

Provenance 16 — The complete solution and its commands: the fermion fable, its canonical quantization in $4 + 4$ dimensions, the coupled system with the primordial gravitational field, the density-functional ground and first excited states, the answers to [1] and [2], and every command that solves, tests, verifies, executes and displays the calculations

Efforts 0, A, B and C of 2026-09-24 and 2026-09-25

2026-09-25

The complete solution of the work of 2026-09-24 and 2026-09-25. This page holds all of it in one place: the foundation that was pushed and completed first (Effort 0, section 2); the fermion fable, a complex 16-component Grassmann spinor canonically quantized in $4+4$ dimensions, with its field equations in the primordial gravitational field, its energy–momentum tensor operator, pressure, density, equations of state and a complete discussion of the canonical spin connection (Effort A, section 3); the fermion fable as the source of the Einstein equations of the primordial gravitational field, the canonical spin connection of the interacting system, the Kohn–Sham source, the density-functional ground and first excited states, the solution from the beginning of the present universe to today, and the answers to the author’s questions [1] and [2] (Effort B, section 4); the adversarial design review with every finding, its verdict and the correction adopted (section 5); and every command that was used to solve the coupled system and to test, verify, execute and display the calculations, with what each command does, why, what it needs, how long it takes, what it prints and how to tell success from failure (Effort C, section 6). Section 7 is the verification of the repository, and section 8 lists what remains open.

This page is complete on its own. Everything a reader needs is restated here: sections 3 and 4 restate the full content of the work of Efforts A and B, every equation, result, number, caveat and answer, so that nothing has to be looked up anywhere else. Code, logs, data files, notebooks, and the TeX and PDF files are named only as the places where things are computed, recorded or produced.

mark	meaning
------	---------

How to read the evidence. Every mathematical or numerical statement on this page carries one of these marks:

mark	meaning
[A], [proved P §n]	an assertion <code>cfAssert[label, test]</code> of the Mathematica notebook, Part P, Section n. The run log prints <code>PASS label</code> ; the label is quoted verbatim , in typewriter type, exactly as the log prints it
[VII], [VIII]	proved in Part VII (Sections 26–29) or Part VIII (Sections 30–33) of the notebook; the labels are quoted verbatim from its run logs
[D], [displayed P §n]	a displayed or printed output of the notebook (a table, a count, a closed form, a <code>Print</code> line of the run log), not an assertion
[P], [prose P §n]	stated only in a <code>Text</code> cell (or a code comment) of the notebook; not checked by an assertion
[nb05 §n], [nb06 §n], [nb07 §n]	a number printed by an executed cell of the fable-cosmology notebook <code>05_fermion_fable_quantum_eos.ipynb</code> , <code>06_fable_primordial_gravity_8d.ipynb</code> or <code>07_fable_dft_states.ipynb</code> , section n, in the executed versions committed in <code>aca5102</code> ; every such cell asserts the claims it prints (a failed assertion stops the notebook). A number quoted from the wall-state report and followed by [nb07 §n] is printed again by notebook 07 (or by a result file it writes in that section), to the digits it prints
[WG n]	a result of the wall-state solver, quoted from its report <code>fable-cosmology/fermion/waveguide_REPORT.txt</code> , section n, which names the log of the run that produced it
[commit h]	stated in the message of commit <i>h</i> of the repository (<code>git log --format=%B -1 h</code> prints it)
[derived here]	a short derivation carried out on this page from marked results, every step shown
[file]	read from a named file of the repository (code, CSV, log), which is named

Numbers are quoted from the files and logs named next to them. Nothing is invented. a_4 is never given a value anywhere: the notebook’s last assertion of Part VIII checks that it is still undefined, PART VIII [control]: `a4 is STILL UNDEFINED -- nothing` in Part VIII gave it a value; the scale factors `scA`, `scB`, `scC` were only ever substituted inside assertions, and none of them entered the canonical frame.

Two kinds of “section”. “Section n” with a capital S and an integer n is a section of the Mathematica notebook (Sections 1–33; Part VI is Sections 23–25, Part VII Sections 26–29, Part VIII Sections 30–33). “section n” in lower case, and any section number with a dot in it, is a section of this page. The mathematics is written in standard notation; where the notebook’s Wolfram names matter they are given in typewriter type. $T_{16}[a]$ (`T16[a]`) are the 16×16 Dirac matrices, σ_{16} (`sigma16`) the spinor metric, $\Psi^\dagger$ (`Psi^ddag`) the classical conjugate, $\bar{\Psi}$ (`Psibar`) the Dirac conjugate, ∂_μ a partial derivative, D_μ the spin-covariant derivative, $\sqrt{g}$ (`Sqrt[g]`) the volume factor $\sqrt{|\det g|}$, $G^\mu{}_\nu$ the Einstein tensor with one index up and one down.

Two records still to come. Section 7.3 is where the results of the final fresh-clone verification of the pushed repository are recorded, and section 7.4 is where the final commit hashes are recorded. Both are filled by the last step of the work.

Contents

Where things live	4
0 The task, in the author’s words, and how the work was organized	6
0.1 The three instructions	6
0.2 The efforts	7
0.3 Who did the work, and how	8
0.4 The order in which it was done	8
0.5 The review loop	9
1 The complete set of ideas	10
2 Effort 0: the foundation of 2026-09-16, pushed and completed	10
2.1 e34d454 (18:52:36): the 2026-09-16 work, as it stood	10
2.2 028b379 (19:24:32) and pull request #1: the reference generator runs clean . . .	11
2.3 2f6d322 (19:57:29): the gaps of 2026-09-16 closed	12
2.4 8459c57 (20:02:04) and e130337 (20:14:25): the solver crate, pushed while it was being built	13
2.5 The last open items of Effort 0, closed in aca5102	14
2.6 The numbers of Effort 0	14
3 Effort A: fermion fable, the complex 16-spinor, canonically quantized in 4+4 dimensions, in the primordial gravitational field	15
3.1 Foundations: the pre-universe, its algebra, the canonical frame, the author’s La and eLa, and the classical fable of Part VI	16
3.2 The refinement: why fable must be a complex 16-spinor	37
3.3 The Lagrangian and its Hermiticity	43
3.4 The field equations in the primordial gravitational field, written out	45
3.5 Canonical quantization in 4+4 dimensions	58
3.6 The energy–momentum tensor operator	70
3.7 Density, pressure and equations of state	73
3.8 The canonical spin connection: a complete discussion	80
3.9 How it was verified	104
4 Effort B: the interacting system — fermion fable as the source of the primordial gravitational field, the solution from the beginning of the present universe to today, and the answers to [1] and [2]	115
4.1 The fermion fable in brief	117
4.2 The Einstein equations of the primordial gravitational field, with fable as their source	126
4.3 The generalized frame, the (0,4) equation, the asymptotic region, and the 8- dimensional Bianchi-I system	131
4.4 The canonical spin connection of the interacting system: a complete discussion .	135
4.5 The fable source in the Kohn–Sham ground state	140
4.6 Density-functional theory: the fable ground and first excited states	150
4.7 Solving the coupled equations from the beginning of the present universe to today	159
4.8 The answers to [1] and [2]	179
5 The design review: four lenses, four skeptics, every finding, and every correc- tion	182
5.1 How the review was organized	182
5.2 The quantization lens (QK)	183

5.3	The field-equation, energy-momentum-tensor and spin-connection lens (EMT) . . .	187
5.4	The Einstein and cosmology lens (E)	188
5.5	The density-functional and numerics lens (DFT)	191
5.6	What the review changed in the answers to [1] and [2]	193
5.7	Corrections to Effort A that the review produced	193
5.8	Corrections found after the review, by the implementation and the checks	193
6	Effort C: the commands that solve the coupled system, and test, verify, execute and display the calculations	195
6.1	The toolchain, and how to set it up	195
6.2	The order in which to run everything	198
6.3	The Mathematica notebook: build, check, map, run, verify, export, reference runs	198
6.4	The Rust solver <code>fable_fermion</code> : build, test, and every run	205
6.5	The independent Python implementation, and comparing results	222
6.6	The wall-state solver, its derivation and its Mathematica checks	224
6.7	The Jupyter notebooks: generate, execute, check, and how they display the results	227
6.8	Reproduce everything with one command	234
6.9	The LaTeX build	236
6.10	Git: commit, push, check the remote, clone afresh	237
7	Verification of the repository: the push, and a fresh clone of it	238
7.1	What was pushed, and what was not yet, when this page was written	238
7.2	What the fresh-clone verification consists of	239
7.3	The results of the fresh-clone verification	239
7.4	The final push	240
8	What remains open	240
8.1	Physics that was not done	240
8.2	Assumptions made, not derived	241
8.3	Where the answers stop	241
8.4	Records that are incomplete	242
8.5	What is still to be done when this page is written	242

Where things live

what	where (relative to the repository root)
the Mathematica notebook (generated, never edited by hand)	<code>claude-fable/claude-fable_Einstein-Rosen-2-Planes.nb</code> (545408 bytes; 384 cells, 217 of them Input cells)
its cell manifests; Part VI (Sections 23–25) is Effort 0, Part VII (Sections 26–29) Effort A, Part VIII (Sections 30–33) Effort B’s symbolic side	<code>claude-fable/cells_part1.wl</code> ... <code>cells_part8.wl</code> ; <code>claude-fable/cells_part6.wl</code> , <code>claude-fable/cells_part7.wl</code> , <code>claude-fable/cells_part8.wl</code>
the builder, the run harness, the runner, the checker, the section map	<code>claude-fable/build_tools.py</code> , <code>claude-fable/runner_header.wl</code> , <code>claude-fable/run_from_nb.wls</code> , <code>claude-fable/verify_nb.wls</code> , <code>claude-fable/nb_section_map.wls</code>
the author’s two helper packages, which must sit next to the notebook	<code>claude-fable/ConvertMapleToMathematicaV2.wl</code> , <code>claude-fable/EtoExp.wl</code>

what	where (relative to the repository root)
the final run log of the whole notebook, Parts I–VIII (217 Input cells, 644/644 assertions, 0 cells with messages, 208.715 s, 46 non-vanishing witnesses; commit 1671155)	<code>claude-fable/run_fermion_fable_final.log</code>
the earlier run logs: Parts I–VI (172 cells, 358/358); through Part VII (198 cells, 528/528); Parts I–VIII before the Rust CSVs existed (217 cells, 642/642)	<code>claude-fable/run_fable51_part6.log</code> ; <code>claude-fable/run_fermion_fable_part7.log</code> ; <code>claude-fable/run_fermion_fable_part8.log</code>
the checker’s logs; the section map	<code>claude-fable/verify_nb_part6.log</code> , <code>verify_nb_part7.log</code> , <code>verify_nb_part8.log</code> ; <code>claude-fable/nb_section_map.log</code>
the TeX/text export of the Part VII results	<code>claude-fable/export_fermion_fable_tex.wls</code> → <code>provenance-latex/generated/*.tex, *.txt</code> (14 files)
the solver of the classical fields of Effort 0 (Rust, pure-Rust SUNDIALS 7.8.0 CVODE)	<code>fable-cosmology/rust/fable_cosmo/</code> (7 unit tests)
the solver of the coupled (fermion fable, primordial gravitational field) system	<code>fable-cosmology/rust/fable_fermion/</code> (<code>src/constants.rs</code> , <code>numerics.rs</code> , <code>potentials.rs</code> , <code>kohn_sham.rs</code> , <code>models.rs</code> , <code>run.rs</code> , <code>cvode_driver.rs</code> , <code>main.rs</code> ; <code>tests/cosmology.rs</code> ; 34 tests)
the engine both solvers depend on by path (made by the setup, not committed)	<code>fable-cosmology/rust/vendor/rustSolveIt/sundials_rs/</code> (a sparse clone of the <code>rustSolveIt</code> repository for the platform)
the independent Python implementation of the coupled system; the development report	<code>fable-cosmology/fermion/crosscheck.py</code> ; <code>fable-cosmology/fermion/report.py</code> (writes into the ignored <code>fable-cosmology/fermion/dev/</code>)
the Mathematica reference runs of the coupled system (Effort B) and of the classical fields (Effort 0)	<code>fable-cosmology/reference/make_reference_fermion.wls</code> → <code>mathematica_fable4d_mass30eV.csv</code> , <code>mathematica_fable4d_power.csv</code> ; <code>fable-cosmology/reference/make_reference.wls</code> → <code>mathematica_scalar_exp.csv</code> , <code>mathematica_spinor_expdamp.csv</code>
the wall-state (x_0) solver, its derivation, its input, its Mathematica check, its report	<code>fable-cosmology/fermion/waveguide.py</code> ; <code>waveguide_derivation.wls</code> → <code>waveguide_derivation.log</code> , <code>waveguide_T16.json</code> ; <code>waveguide_check.wls</code> ; <code>waveguide_REPORT.txt</code>
the seven Jupyter notebooks (generated by <code>notebooks/_build/nbgen.py</code> , checked by <code>notebooks/_build/nbcheck.py</code>); 01–04 are Effort 0, 05–07 Effort B	<code>fable-cosmology/notebooks/01_fableScalar Quintessence.ipynb</code> ... <code>04_dark_matter_dark_energy.ipynb</code> ; <code>05_fermion_fable_quantum_eos.ipynb</code> , <code>06_fable_primordial_gravity_8d.ipynb</code> , <code>07_fable_dft_states.ipynb</code>
every CSV table and PNG figure the notebooks write	<code>fable-cosmology/results/nb01_*</code> ... <code>nb04_*</code> (43 CSV, 16 PNG), <code>nb05_*</code> (11 CSV, 7 PNG), <code>nb06_*</code> (37 CSV, 8 PNG), <code>nb07_*</code> (18 CSV, 9 PNG)
setup and the complete reproduction	<code>fable-cosmology/setup.sh</code> , <code>setup.ps1</code> ; <code>fable-cosmology/run_all.sh</code> , <code>run_all.ps1</code> ; <code>fable-cosmology/requirements.txt</code>
the paper of Effort 0	<code>fable-cosmology/latex/fable_cosmology.tex</code> → <code>.pdf</code> (24 pages), <code>fable-cosmology/latex/figures/</code>

what	where (relative to the repository root)
the LaTeX twins of the three provenance pages of 2026-09-24, their preamble and build scripts	provenance-latex/PROVENANCE-14-FERMION-FABLE-CANONICAL-QUANTIZATION.tex, ...-15-FERMION-FABLE-AND-THE-PRIMORDIAL-GRAVITATIONAL-FIELD.tex, ...-16-THE-COMPLETE-SOLUTION-AND-ITS-COMMANDS.tex → .pdf; provenance-latex/preamble.tex; provenance-latex/build_all.sh, build_all.ps1
the delivered copy of the notebook, outside the repository	C:/Users/nsh/Documents/8-dim/claude-fable_Einstein-Rosen-2-Planes.nb (545408 bytes, byte-identical to the repository's; the two helper packages beside it)

<repo> below is wherever the repository is cloned. Every shell block finds it with `git rev-parse --show-toplevel`.

0 The task, in the author's words, and how the work was organized

0.1 The three instructions

On 2026-09-24 the author gave three instructions. They are quoted here in full as the author's operative words, verbatim (the ellipses [...] are cuts in the quotation, not in the instruction's meaning):

A. “Refine the new 16-component fermi spinor field named fable to be a complex 16-spinor, and quantize it using canonical quantization, extended to 4+4 dimensions. Write out the field equations for the fermion fable field in the presence of a primordial Gravitational Field. Create new provenance markdown file, latex, and pdf files, that contain the exact, correct, fermion fable field equations, energy-momentum tensor operator, pressure, density and equations of state for fermion fable field. Include a complete discussion of the canonical spin connection.”

B. “Include fable as a source in the Einstein Gravitational Field Equations for this primordial Gravitational Field, and solve the coupled field equations over a time interval from the beginning of the present universe until the present time. Try to use some of the relevant ideas from Density Functional Theory to obtain the fable ground and first excited states. Create another new provenance markdown file, latex, and pdf files [...] the interacting (fermion fable field, primordial Gravitational Field) system [...] coupled equations, canonical spin connection for the interacting system, energy-momentum tensor operator, pressure, density and equations of state [...] complete discussion of the canonical spin connections. Answer: [1] Does this system provide a physical mechanism for a time-varying dark energy equation of state (w)? [2] [...] time-varying dark matter equation of state (w)?”

C. “Thoroughly discuss the commands that you use to solve the coupled system, and to test, verify, execute, and display your calculations of the coupled field equations. Create another new provenance markdown file, latex, and pdf files that contain your complete solution. Each of your major efforts should have its own section in the markdown file, latex, and pdf files. Do not ever refer the USER to another markdown file for any reason. The full set of 100% complete ideas must be given its own section in each markdown file.” And: push, check and verify the repository. “DO NOT TAKE SHORTCUTS.”

This page is the third: the complete solution. Instructions A and B each asked for their own provenance files, and they were written (Effort A and Effort B); instruction C asked for this one, which contains the complete solution. Following C’s rule, every major effort has its own section here (Effort 0: section 2; Effort A: section 3; Effort B: section 4; the design review: section 5; Effort C, the commands: section 6; the verification and push: section 7), the content of Efforts A and B is restated in full rather than referred to, and the complete set of ideas of all efforts is section 1.

The author’s choice. Instruction A itself names the field: “a complex 16-spinor”. Section 3.2 shows why that is forced by the author’s own algebra: a real anticommuting 16-spinor with the author’s spinor metric has no dynamics, a Majorana 16-spinor is massless with no potential, and a one-chirality (Weyl or Majorana–Weyl) spinor cannot propagate. When these alternatives were laid out on 2026-09-24, the author confirmed the choice: **a complex 16-spinor**, with Dirac conjugate $\bar{\Psi} = \Psi^\dagger \sigma_{16}$. Every result below is for that field.

The author’s standing rules, which bound all of the work: the notebook is generated from its cell manifests and never edited by hand; the author’s conventions (the metric signs, the τ/T_{16} construction, the 16-spinor split and the canonical frame) are never changed; the free function a_4 of the canonical frame is never given a value; no number is invented; every claim is traced to a file, a log or an assertion; no page sends the reader to another markdown file; nothing is shortcut.

0.2 The efforts

effort	what it is	where on this page	its main commits
0	the foundation: the work of 2026-09-16 on the classical fields fableScalar and fable (Part VI of the notebook, the solver fable_cosmo , notebooks 01–04, the paper), committed, pushed and completed first, with its gaps closed	2	e34d454 , 028b379 (PR #1), 2f6d322 , and the nbgen.py corrections inside aca5102
A	the fermion fable: the complex 16-spinor, its Lagrangian and field equations in the primordial gravitational field, its canonical quantization in 4+4 dimensions, its energy–momentum tensor operator, density, pressure and equations of state, and the canonical spin connection (Part VII of the notebook)	3	4fdfc40 , b7ea18d , fee5603
B	the interacting system: the Einstein equations of the primordial gravitational field with fable as source, the generalized frame and the 8-dimensional Bianchi-I system, the spin connection of the interacting system, the Kohn–Sham source, the density-functional ground and first excited states, the solution from before nucleosynthesis to today, the answers to [1] and [2] (Part VIII of the notebook, the solver fable_fermion , the wall-state solver, notebooks 05–07)	4	8459c57 , e130337 , c5892bd , bc04ed1 , 18a2501 , 2bc936d , aca5102 , 1671155
review	the adversarial review of the design before anything was implemented, and every correction found afterwards	5	(its corrections are in the commits above)
C	the commands: setting up the toolchain, building, running, testing, verifying, executing and displaying every calculation; the three provenance pages; the push and its verification	6, 7	0a789e2 , d0f2a99 , fee5603 , and the final commits recorded in section 7

0.3 Who did the work, and how

The work was done by Claude (Claude Opus 5.5, in Claude Code) as one orchestrating session that planned the work, wrote the design, ran the review, delegated self-contained pieces to subagents running in parallel, checked every delivered piece itself, and made every commit. Every commit of the work carries `Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>` [commit messages]. The agents that appear in the records are these:

- **The orchestrator** (the main session): the design, the review loop, integration, the commits, the pushes. The wall-state report calls its instructions “the orchestrator’s instructions of 2026-09-24” [WG, header] and its §7 confirms or refutes “the orchestrator’s” statements next to the reviewer’s.
- **Four reviewer agents**, one per lens of the design review, each running its own Wolfram and Python probes against the notebook’s own objects (section 5.1).
- **Four skeptic agents**, one per lens, each trying to refute every finding of its reviewer with an independent computation of its own (section 5.1).
- **Implementation agents**, each with one piece: the fermion-fable solver crate (the commit `e130337` calls it “the solver agent”); Part VII of the notebook; Part VIII of the notebook; the wall-state solver (its report records that “another agent works in `claude-fable/`” and that “the notebook was rebuilt by another agent during this session (172 -> 217 Input cells)” [WG 1, WG 8]); and the notebooks 05–07 with their generator and checker.
- **Independent re-runs**. Every notebook part and the solver were re-run by a second process before they were committed: “An independent re-run gave the same result” (Part VII, `4fdffc40`), “An independent re-run confirmed it” (Part VIII, `bc04ed1`), “21 + 8 tests pass (re-run independently)” (the solver, `c5892bd`), “validation 64/64 PASS, re-run independently” (the wall-state solver, `18a2501`) [commit messages].
- **The documentation workflow**: the three provenance pages, each written from the logs, manifests and git history, then fact-checked adversarially against them (“An adversarial fact-check against the logs, manifests and git history corrected 7 errors in both files”, `fee5603`). This page was assembled by a script that copies the sections of Efforts A and B programmatically from the pages of those efforts (so that nothing is retyped), rennumbers their sections and cross-references, and adds the parts that are new here.

Two process incidents are recorded because they bear on what could have been lost: two agents of the design review (the reviewer of the Einstein lens and the skeptic of the DFT lens), while stopping their own stalled runs, each ran `taskkill /F /IM python.exe`, which terminated every Python process on the machine at that moment (three processes the first time, one the second), and each reported it so that anything that had died could be re-run (the design review’s records). No committed result depends on a process that was killed: every committed number was produced or re-produced by a later run whose log or executed notebook is committed.

0.4 The order in which it was done

All times are PDT (UTC−7), from `git log --date=iso` and, for the pull request, from GitHub.

time	step	record
2026-09-24 18:52:36	Effort 0 committed as it stood when the session of 2026-09-16 ended, with its known gaps listed	<code>e34d454</code>
about 19:04	the design of Efforts A and B (revision 1) written, with six facts F1–F6 already computed by probes and eight questions Q1–Q8 for the reviewers	the design’s working file

time	step	record
19:24:32	the reference generator’s underflow messages silenced; the branch <code>fermion-fable-and-primordial-gravity</code> pushed with it	028b379
19:24:50 → 19:32:10	pull request #1, “Fermion fable and the primordial gravitational field (branch opened with a reference-generator fix)”, opened and merged into <code>main</code> (a fast-forward: its merge commit is 028b379 itself)	GitHub, PR #1
19:32 – 20:51	the design review: the reports of the quantization and Einstein lenses (by 19:32), of the field-equation and DFT lenses (by 20:03); the skeptics’ verdicts (20:03–20:04, the DFT skeptic’s by 20:51)	the review’s working files
19:57:29	the gaps of Effort 0 closed	2f6d322
20:02:04, 20:14:25	the author asked for everything to be pushed at once: two work-in-progress snapshots of the solver crate	8459c57, e130337
20:17:18, 20:18:53	the LaTeX build tree of the three new provenance pages	0a789e2, d0f2a99
20:38:49	the finished solver <code>fable_fermion</code> , with its <code>scipy</code> cross-check	c5892bd
20:47:28	Part VII of the notebook (Effort A), 528/528	4fdcf40
20:48:42	a wording correction in an earlier provenance page, found by the fact-check of the spin-connection section	b7ea18d
21:24:04	Part VIII of the notebook (Effort B’s symbolic side), 642/642	bc04ed1
22:13:42	the wall-state solver (the density-functional states along x_0)	18a2501
22:31:20	the page of Effort A (draft) with its LaTeX twin and PDF (110 pages)	fee5603
about 22:31	the design addendum (revision 2, the adopted corrections) in its final form	the addendum’s working file
22:53	notebooks 05 and 06 first executed; they exposed six solver defects	the page of Effort B, as written
23:12:51	the six solver defects fixed, 34/34 tests	2bc936d
23:26	the page of Effort B written (not yet committed)	its section on the push
23:49 – 23:52	notebooks 01–06 executed on the final solver (the cell timestamps of the committed notebooks start at 06:49:12 UTC)	the committed notebooks
2026-09-25 00:06:38	the wall-problem derivation re-run, writing the committed <code>waveguide_T16.json</code> (94/94)	<code>fable-cosmology/fermion/waveguide_derivation.log</code>
00:11 – 00:21	notebook 07 executed (its cells start at 07:11:24 UTC and end 557.6 s later)	the committed notebook 07
00:23:46	notebooks 05–07, their 90 result files and the integration (nbcheck 7/7, <code>run_all.sh</code> passing)	aca5102
00:28:54	the final full run of the notebook: 217 Input cells, 644/644, 0 messages, 208.715 s	1671155
then	this page; the final push and the fresh-clone verification	section 7

0.5 The review loop

The work followed one loop, repeated at every level:

1. **Design before code.** The design (revision 1) stated the physics of Efforts A and B, the facts already computed, the implementation plan (Part VII and Part VIII of the notebook; the Rust solver; the Python cross-checks; the wall-state problem; notebooks 05–07; the three provenance pages; the verification) and eight questions the reviewers had to try to break.

2. **Adversarial review.** Four reviewers, one per lens, each followed by an independent skeptic (section 5). No finding was refuted as a whole; many were confirmed only in part, and then the skeptic’s corrected fix was adopted. The adopted corrections were written into an addendum (revision 2) that overrides the design wherever they differ.
3. **Implementation with assertions.** Every symbolic claim became an assertion of the notebook (Parts VII and VIII), every numerical claim a unit test of the solver, a check of the wall-state solver or an assertion of a notebook cell.
4. **Independent re-runs and cross-implementations.** Parts VII and VIII of the notebook, the solver and the wall-state solver were re-run by a second process before they were committed; every cosmological run was recomputed by an independent Python implementation, and two by Mathematica as well (section 4.7.12); the wall levels were recomputed in Mathematica (section 4.6.4).
5. **Defects found by the checks were fixed and recorded,** never hidden (section 5.8).
6. **Pages from the records.** Every provenance page was written from the logs, the manifests and the git history, and then fact-checked adversarially against them.
7. **Push and verify.** Every commit reached `main` on GitHub (028b379 through pull request #1, all the others directly); the final state is verified from a fresh clone (section 7).

The design and the review records are working documents in the session’s scratch folder. They are not part of the repository. Everything in them that survived review is restated on this page: the findings and verdicts in section 5, the physics in sections 3 and 4.

1 The complete set of ideas

{{COMPLETE-SET-OF-IDEAS}}

2 Effort 0: the foundation of 2026-09-16, pushed and completed

Efforts A and B stand on the work of 2026-09-16: Part VI of the notebook, which put two classical fields on the pre-universe (`fableScalar`, a real scalar, and `fable`, a real commuting 16-component spinor), and the `fable-cosmology/` folder, which solved them numerically. When the session of 2026-09-24 began, that work was finished but not committed. Effort 0 committed it as it stood, pushed it, closed every gap it had, and verified it from a clean clone. Its physics is restated in section 3.1.12 (the classical fable of Part VI and its reference cosmology) and section 3.1.13 (why it has to be refined); this section records what was committed, what each commit closed, and the numbers that verify it.

2.1 e34d454 (18:52:36): the 2026-09-16 work, as it stood

The commit “Add the 2026-09-16 fable-cosmology work: Part VI, the solver, four notebooks” (by `git show --stat: 107 files, 47374 insertions`) contains [commit e34d454]:

- `claude-fable/cells_part6.wl`: Part VI of the notebook (Sections 23–25). The rebuilt notebook runs **172 Input cells, 358/358 assertions, 0 messages** (`claude-fable/run_fable51_part6.log`, `claude-fable/verify_nb_part6.log`).
- `fable-cosmology/`: the design of the numerical work (revision 2); the Rust solver `fable_cosmo` (`src/main.rs`, `models.rs`, `potentials.rs`, `ccode_driver.rs`, with the FMA pin in `.cargo/config.toml`) on the pure-Rust SUNDIALS 7.8.0 CVODE engine of the `rustSolveIt` repositories; four executed Jupyter notebooks (`01_fableScalar_`

quintessence, 02_fable_spinor, 03_pre-universe_dynamics, 04_dark_matter_dark_energy) with their generator `nbgen.py`, their checker `nbcheck.py` and their conventions; the results they write (43 CSV files and 16 PNG figures); the Mathematica reference integrations (`reference/make_reference.wls` $\rightarrow$ `mathematica_scalar_exp.csv`, `mathematica_spinor_expdamp.csv`); `setup.sh`, `setup.ps1`, `run_all.sh`, `run_all.ps1`, `requirements.txt`; and the 16 figures for the paper.

- `.gitignore` (the engine clone, the virtual environment, cargo’s `target/` and the LaTeX build logs are ignored); the repository’s front page (a Part VI row, the fable-cosmology section).

The commit message lists its known gaps “so nobody mistakes this for finished”: three sections of the provenance page of the fable-cosmology work were placeholders; the student instructions of `fable-cosmology/` and the paper `latex/fable_cosmology.tex`, which the front page linked to, had never been written; and the fresh-clone verification had not been run.

2.2 028b379 (19:24:32) and pull request #1: the reference generator runs clean

`fable-cosmology/reference/make_reference.wls` writes the Mathematica `NDSolve` references of Model A (`fableScalar`, `exp` potential) and Model B (the classical spinor fable, `expdamp`). Model B’s H column is evaluated at exact N , and its term $\exp(-s/s_1)$ ($s \approx 1.3 \times 10^9$ at $N = -7$) underflows when `N[]` takes it to machine precision. The value written is unaffected, but a run printed three `General::munfl` messages and a `General::stop`. The formatter became `fmt[x_] := Quiet[N[x], General::munfl];`, so a clean run prints no messages; nothing else changed [commit 028b379]. The first run of the script on 2026-09-24 had taken 427 s and printed, verbatim apart from the messages it no longer prints,

```
mathematica_scalar_exp.csv: 701 rows; w(a=1) = -0.843363803931198
mathematica_spinor_expdamp.csv: 701 rows; w(a=1) = -0.8608456403776481 min w =
↪ -1.3020988149091868
```

and a probe showed that `N[HB[-7]]` is 22740.499428716343 with or without the message. Re-run after the fix (366 s), it printed exactly these two lines and nothing else, and both CSVs were byte-identical to the committed ones.

The commit was pushed on a new branch, `fermion-fable-and-primordial-gravity`, and opened as **pull request #1** of `once-ere/Pre-Universe_with_Claude`, “Fermion fable and the primordial gravitational field (branch opened with a reference-generator fix)”, at 19:24:50. Its description stated what the branch was for (the six tasks of Efforts A and B) and the findings already established that would shape the work: $\sigma_{16}T_{16}[a]$ is antisymmetric for every a , so a real Grassmann fable with σ_{16} has a zero mass term and a total-derivative kinetic term; the canonical anticommutator $\{\Psi, \Psi^\dagger\} \propto -i\sigma_{16}\gamma^4$ has signature (8,8), and the Krein fundamental symmetry $J = -iT_0T_1T_2T_3T_4$ gives a positive Fock space in the sector where the fields do not depend on $x_5..x_7$; the canonical pre-universe metric is not a vacuum solution of Einstein’s equations; in a generalized version of the canonical frame the off-diagonal (0,4) Einstein equation “forces the author’s volume-preserving a_4 structure”. (The review later corrected the last point: the (0,4) equation *singles out* that structure within a family and under stated conditions; it does not *derive* it, section 4.3.2.) The pull request was merged into `main` at 19:32:10; the merge was a fast-forward, so its merge commit is 028b379 itself. (e34d454 had gone straight to `main` before the branch was opened, as the pull request’s description records.) After that the author asked for everything to go to `main` at once, and every later commit of the work was pushed directly to `main`.

2.3 2f6d322 (19:57:29): the gaps of 2026-09-16 closed

The commit that closes the gaps of 2026-09-16 (by `git show --stat`: 9 files, 3032 insertions) contains [commit 2f6d322]:

- **The student instructions** of the numerical cosmology (new, in `fable-cosmology/`): the prerequisites per platform, the setup, the solver’s command-line contract read from `main.rs`, the CSV columns, the notebooks, the tests, the paper and troubleshooting. Their content, extended on 2026-09-25 to the fermion solver and notebooks 05–07, is restated in section 6 of this page.
- **The paper** `fable-cosmology/latex/fable_cosmology.tex` and `.pdf` (new): 24 pages, 16 figures, 20 references. `latexmk` builds it from scratch with no warnings, no undefined references and no over- or underfull boxes (`Output written on fable_cosmology.pdf (24 pages, 2835044 bytes)`).
- **The provenance page of the fable-cosmology work**: its three placeholder sections written out in full, every pointer to another markdown file removed, and its verification recorded.
- **Seven corrections to the fable-cosmology design**, each marked [corr 2026-09-24] there: G_0 is $C^2 \sin^2(6Hx_0)/2$, not $C^2/2$ (as Part VI asserts); the reference `expdamp` minimum is -1.302 at $a = 0.543$ ($z = 0.84$), not “ ≈ -1.27 near $z \approx 1$ ”; the Cai & Wang citation is *Class. Quantum Grav.* 25 (2008) 165014 (arXiv:0806.3890, doi 10.1088/0264-9381/25/16/165014), not “JCAP 2008”; the CSVs carry 16 significant digits (`{:.15e}`), not 15; an unknown model or potential name exits with code 1, not 2; the reference CSVs are written by `make_reference.wls` and checked, not written, by Part VI; and table pipes inside code spans are escaped.
- The same citation corrected in a Text cell of `claude-fable/cells_part6.wl` (text only; no Input cell changed); `setup.sh` now creates the virtual environment with `python3` when that runs and with `python` otherwise (many macOS and Linux systems have no `python`); `run_all.ps1` now stops when `cargo test` fails (PowerShell’s `$ErrorActionPreference = "Stop"` does not apply to native commands, so the script checks `$LASTEXITCODE` and throws). `.gitignore`: comments only.

check	result
-------	--------

The verification recorded with it (run on 2026-09-24 in Git Bash on Windows 11, in the working repository at 028b379, with the engine rustSolveIt_Win11_SUNDIALS_7_8_0 @ a8fdff4):

check	result
cargo test --release in rust/fable_cosmo	test result: ok. 7 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s; fable_cosmo 0.1.0, sundials_rs 7.8.0 (pure Rust), CVODE BDF
bash fable-cosmology/run_all.sh	exit code 0 after 51 s; 4/4 notebooks pass all eight requirements; == all done
nbcheck.py on the committed notebooks	4/4 notebooks pass all eight requirements, exit code 0
reproducibility	git diff --stat -- fable-cosmology/results printed nothing: all 43 CSV files and 16 PNG figures byte-identical to the committed ones; a column-by-column comparison of every CSV against HEAD printed 43 CSV files compared column by column against HEAD; files with any nonzero difference or row-count change: 0; the four re-executed notebooks differed from the committed ones only in the cells' execution timestamps and in how the kernel split printed output into stream chunks (sources, execution counts, stream text, results, images and errors identical)
the paper, from scratch	latexmk exit 0, three pdflatex passes, no errors, warnings, undefined references, overfull or underfull boxes; Output written on fable_cosmology.pdf (24 pages, 2835044 bytes); MiKTeX 26.5, pdfTeX 4.27, latexmk 4.88
a fresh local clone of 028b379 (with the session's three new files copied in, not the PDF)	bash fable-cosmology/setup.sh: exit code 0, 80 s (a new virtual environment, a new engine clone from GitHub, Finished release profile ... in 5.04s); bash fable-cosmology/run_all.sh: exit code 0, 61 s; git diff --stat -- fable-cosmology/results empty; the paper built from source to 24 pages
Part VI, re-run	wolframscript -file run_from_nb.wls: cells evaluated : 172, total seconds : 152.388, cells w/ msgs : 0, Assertions run: 358 passed: 358 failed: 0, 27 non-vanishing witnesses; with the timings masked, identical line for line to the committed run_fable51_part6.log
the classical results table, re-checked against the committed files	every row confirmed (the table is below)

Still open after this commit, and recorded as such: four wording items in the notebook generator nbgen.py, which that session had been told not to touch (section 2.5 closes them).

2.4 8459c57 (20:02:04) and e130337 (20:14:25): the solver crate, pushed while it was being built

“The author asked for everything to be on main now.” So the fermion-fable solver crate `fable-cosmology/rust/fable_fermion` was committed twice at intermediate stages, each commit saying plainly that it was **not finished, tested, reviewed or verified** and would be changed [commits 8459c57, e130337]. The first snapshot held `constants.rs` (ρ_{c0} , Ω_{r0} , Ω_{b0} from physical constants), `kohn_sham.rs` (the $T = 0$ Fermi-sea integrals for $g = 8$ states per momentum, the mean-field potentials $W(\sigma)$ and the gap equation), `models.rs/run.rs` (fable4d with the hidden sheet stabilized, and fable8d, the 8-dimensional Bianchi-I no-go demonstration) and

`cvs_driver.rs` (the path dependency on the engine, as `fable_cosmo` has it); `.gitignore` section 5 ignores the crate’s `target/` and the scratch folder `fable-cosmology/fermion/dev/`. The second recorded the solver agent’s edits to `kohn_sham.rs`, `models.rs`, `potentials.rs` and `run.rs` as they stood, “in the middle of adopting the design review’s corrections”: the per-7-volume normalization, numerically stable Fermi integrals, the stabilized `fable4d` as the physical model, `fable8d` as the no-go demonstration, and the observer-inferred and interacting w definitions. The finished crate is `c5892bd` (section 4.7 and section 6.4).

2.5 The last open items of Effort 0, closed in `aca5102`

The four `nbgen.py` items were fixed on 2026-09-24 with the notebooks of the fermion-fable work, and committed in `aca5102` [commit `aca5102`]. The generator now says:

- notebook 03, §3: $G_0 = C^2 \sin^2(6Hx_0)/2$ (as Part VI asserts), independent of x_4 and so constant for the observer at fixed x_0 ;
- notebooks 01 (§5.8) and 02 (§5.10): the reference CSV “was written by `fable-cosmology/reference/make_reference.wls` (a wolframscript) and is checked by Part VI of the Mathematica notebook, which reads it back”;
- notebook 01, §5: every number is written as `{:.15e}`, “15 decimals, i.e. 16 significant digits”;
- notebooks 01, 02 and 03, §5: exit code 1 is a solver or physics error “and also an unknown model or potential name”; exit code 2 is “a malformed command line (an unknown option, or a missing or non-numeric value)”. This was checked against the binary first: `fable_cosmo nosuchmodel` and `fable_cosmo spinor-flrw --potential nosuch` exit 1, `fable_cosmo spinor-flrw --bogus` and `fable_cosmo spinor-flrw --points x` exit 2.

The four notebooks were regenerated (`nbgen.py 01 02 03 04`) and re-executed in place (7 s, 7 s, 53 s and 6 s). None of the stale phrases occurs in them any more; `nbcheck.py` reported 4/4 notebooks pass all eight requirements; and **all 59 result files of notebooks 01–04 were byte-identical** to the committed ones (`git diff --stat -- fable-cosmology/results` empty; the SHA-256 of each equal to that of its HEAD blob: `checked 59 files, 0 differ`). The corrections are to markdown text only; no code cell and no result changed. `aca5102` records this: “Their 59 result files are byte-identical to before.”

2.6 The numbers of Effort 0

Every one of these is asserted by a notebook, so a re-run that changed any of them would fail [file: the committed notebooks 01–04 and their results; re-checked on 2026-09-24 against the committed CSVs]:

case	quantity	value
Model A, <code>exp λ = 1</code>	$w(a = 1)$; CPL fit (w_0, w_a)	-0.843364 ; $(-0.8440, -0.2172)$
Model A, <code>exp λ = 1.5</code>	CPL fit; distance from Unite	$(-0.6176, -0.5015)$; 0.263
Model A, <code>scan</code>	closest λ ; λ with $w_a = -0.60$	1.4, fit $(-0.6740, -0.4349)$, distance 0.2495; 1.638, with $w_0 = -0.5274$
Model A, every run	$\min w$	-1.000000 (never below -1)
Model A, <code>exp λ = 3</code>	refused; by hand with $v_0 = 10^6$	$\Omega_\phi(1) = 0.3375$ ($3/\lambda^2 = 0.3333$)
Model A', <code>exp λ = 1</code>	CPL fit	$(-0.8371, -0.2257)$
Model B, <code>mass</code>	$\max w $	0 (dust, exactly)
Model B, <code>lambda-mass</code>	$\Omega_\Lambda, \Omega_{\text{dust}}, w(1)$	0.489941, 0.209975, -0.700000
Model B, <code>power n = 0.236</code>	$w(1)$; $w(N = 6)$	-0.38200 ; -0.76399919

case	quantity	value
Model B, <code>expdamp</code> (1.566, 0.839, 2.21)	$w(1)$; crossing a_c ; min w at a ; fit	-0.86085 ; 0.7677195 (analytic 0.7677195); -1.30210 at 0.5434 ; $(-0.9782, -0.2445)$
Model B, <code>expdamp s1</code> $= 3.0$	fit; distance from Unite	$(-0.8513, -0.5692)$; 0.0323 (the closest of all runs)
Model C, quadratic, $x_4 \leq 100$	$w_{\text{avg}}(100)$; max $ \rho $ drift	$+0.004363$; $3.27\text{e-}9$
Model C, quartic, $x_4 \leq 100$	$w_{\text{avg}}(100)$	0.334909 (virial value $1/3$)
Model D, $\varepsilon = 0.05$, $\varepsilon = 0.5$	max $ E_{\text{drift}} $; G_0 fraction at $x_4 = 0$	$4.48\text{e-}9$, $9.05\text{e-}9$; 0.2847 , 0.9780
three integrators, Model A	max $ \Delta w $ CVODE–NDSolve, scipy–NDSolve, CVODE–scipy	$1.149\text{e-}10$, $1.313\text{e-}10$, $9.509\text{e-}11$
three integrators, Model B	the same	$1.861\text{e-}8$, $1.008\text{e-}11$, $1.861\text{e-}8$

(Models A and A' are `fableScalar` in the standard 4-dimensional reference cosmology and on the Unite CPL background; Model B is the classical spinor fable in the reference cosmology; Models C and D put `fableScalar` on the 8-dimensional pre-universe itself. The re-check of 2026-09-24 also printed, from the committed CSVs: `const: max|w+1| = 0.000e+00` `Omega_phi(1) = 0.6999160000`, `mass: max|w| = 0.000e+00` `q_dec(1) = 0.5000420000` (the 0.5 of pure dust, plus the radiation term), Model A `exp1` vs NDSolve: `max|dw| = 1.149e-10` `max|dt| = 5.428e-10`, Model B `expdamp` vs NDSolve: `max|dw| = 1.861e-08` `max|dt| = 5.224e-09`, and the scipy rows 9.509×10^{-11} (`exp` $\lambda = 1$) and $2.201 \times 10^{-8} / 1.006 \times 10^{-9}$ (`lorentz`).)

Why Effort 0 matters for the rest. The classical fable of Part VI has $w = sV'/V - 1$, which crosses $w = -1$ for `expdamp` and `lorentz` (the “phantom crossing” of the table above: min $w = -1.30210$). Effort A refines that field into a fermion, and Effort B shows that the quantized fable **cannot** cross $w = -1$ ($\rho + P_{\text{obs}} = \omega_{Fn} \geq 0$, section 4.5.9); what survives of the crossing is an *apparent* phantom in the dark energy an observer infers (section 4.8.1).

3 Effort A: fermion fable, the complex 16-spinor, canonically quantized in 4+4 dimensions, in the primordial gravitational field

This section is the complete content of Effort A. It is copied section by section, by a script, from the provenance record of Effort A (written on 2026-09-24 and fact-checked against the logs), with its sections renumbered 3.1–3.9 and its cross-references renumbered to match this page; nothing in it was retyped, and nothing was left out. Its subsections:

here	content
3.1	the foundations: the pre-universe, its algebra, the canonical frame, the author’s <code>La</code> and <code>eLa</code> , and the classical fable of Part VI (Effort 0’s physics); these serve all of Efforts A and B
3.2	the refinement: why fable must be a complex 16-spinor
3.3	the Lagrangian and its Hermiticity
3.4	the field equations in the primordial gravitational field, written out: matrix, adjoint, 8+8 split-octonion, all sixteen component equations and their adjoints, the rescaled form, the x_4 -only family, the operator form, the currents
3.5	canonical quantization in 4+4 dimensions: the Dirac bracket, the anticommutator, J , the admissibility theorem, the Krein/ J Fock space, the Dirac sea, the wall condition

here	content
3.6	the energy–momentum tensor operator
3.7	density, pressure and equations of state
3.8	the canonical spin connection: a complete discussion
3.9	how Effort A was verified: Part VII of the notebook, every assertion verbatim, the full runs, the design review’s corrections to Effort A, the Fock check, the TeX export

The commands that build, run, check and export Part VII are in section 6.3; the design review is section 5.

3.1 Foundations: the pre-universe, its algebra, the canonical frame, the author’s L_a and eLa, and the classical fable of Part VI

This section sets out everything that the new work on fermion fable stands on. It restates the author’s pre-universe (a curved $4 + 4$ -dimensional spacetime), the real Clifford algebra and split-octonion spinors built on it, the author’s own wave function, Lagrangian and field equations, the canonical frame field, and the classical field fable of 2026-09-16. It ends with the reason why that classical field has to be refined into a complex 16-component spinor. The reader needs nothing outside this document to follow it.

3.1.1 How the evidence is cited

All the mathematics lives in one Mathematica notebook, `claude-fable/claude-fable_Einstein-Rosen-2-Planes.nb`. That notebook is never edited by hand. Its Parts I–VI, which this section restates, are generated from the cell manifests `claude-fable/cells_part1.wl ... cells_part6.wl`. In a manifest, `Text` cells explain and `Input` cells compute. Every claim the notebook checks is an assertion `cfAssert[label, test]`, which prints `PASS label` or `FAIL label`. The complete run of Parts I–VI is recorded in `claude-fable/run_fable51_part6.log`. It evaluated 172 Input cells in 162.5s with no cell raising a message, and its final tally reads (the witness line is shortened here; in the log it continues with a one-sentence explanation of what a witness is):

```
identities accepted on numerical evidence alone : 0    (stage 3 returned True)
non-vanishing witnesses                       : 27
PASS no identity in this notebook was accepted on numerical evidence alone
Assertions run: 358  passed: 358  failed: 0
```

Each statement below is tagged with one of five kinds of evidence:

- **[A]** an assertion. Its label is quoted exactly as the log prints it after `PASS`.
- **[D]** a displayed result: an Output the notebook prints but does not assert. Where a displayed result was re-read for the provenance pages of 2026-09-24 from the saved kernel state, that is said.
- **[P]** prose only: a statement that appears in a `Text` cell and is not checked by any assertion.
- **[VII]** proved in Part VII, the new part of the notebook written for fermion fable (Sections 26–29). Its labels are quoted verbatim from `claude-fable/run_fermion_fable_part7.log`, which reports `Assertions run: 528 passed: 528 failed: 0` for Parts I–VII together.
- **[VIII]** proved in Part VIII, the notebook’s coupled system (Sections 30–33). Its labels are quoted verbatim from `claude-fable/run_fermion_fable_part8.log`, which reports `Assertions run: 642 passed: 642 failed: 0` for Parts I–VIII together (sections 4.2–4.5 of this document); the final run `claude-fable/run_fermion_fable_final.log` prints the same labels and two more, `Assertions run: 644 passed: 644 failed: 0`.

How an identity is certified (restated from Section 15 of the notebook). `cfZeroQ` and `cfZeroArrayQ` try three stages in turn:

1. exact structural zero;
2. **Simplify** under the geometric assumptions, with a 5-second budget;
3. a 30-digit numerical evaluation at three rational probe points, with test functions substituted for the undetermined a_4 (and for the undetermined rapidity of the notebook's boosted-frame bridge); the result is accepted as zero only if its magnitude is below 10^{-22} .

An identity accepted at stage 3 counts as a *numerical certificate*, and the run reports zero of those: every identity in Parts I–VI is closed symbolically. A claim that something is *not* zero is made with `cfNonZeroWitnessQ`. It passes only if, at some probe point, every entry evaluates to a number and at least one of them has magnitude above 10^{-10} . That is a complete proof of non-vanishing; the run has 27 such witnesses. The probe functions are test inputs only, never definitions.

The session policy is the original notebook's, unchanged: **Simplify** has a 1-second budget and **FullSimplify** a 3-second budget. `DIM8 = 8`, and the model's three constants M (a mass), K and H (a Hubble-like constant) are **Protected**, so no cell can assign them (Section 1: a **Text** cell and the **Input** cells that set this up; no assertion) [P].

3.1.2 Coordinates, the flat 4+4 metric, and which directions are timelike

The coordinates are $X = \{x_0, x_1, \dots, x_7\}$. The flat tangent-space metric is

$$\mathbf{eta4488} = \eta = \text{diag}(+1, +1, +1, +1, -1, -1, -1, -1), \quad \text{signature } (4, 4). \quad (1)$$

coordinate	η	name in the original notebook (Section 2)	reading used in this work
x_0	+1	hidden space	the hidden spacelike coordinate
x_1, x_2, x_3	+1	ordinary 3-space	the observed sheet
x_4	-1	time	the observer's time
x_5, x_6, x_7	-1	superluminal deflating time directions	the hidden timelike sheet

[A] `eta4488 . eta4488 == ID8; eta4488 is symmetric; signature of eta4488 is (4,4).`

Index conventions (restated from Section 2). Latin indices a, b, A, B are flat and run 0..7. Greek indices μ, ν are curved and run 0..7. Wolfram arrays are 1-based, so index A sits at array position $A + 1$. For example, x_4 is entry 5 of X , so Part VI's energy density $\rho = T_{44}$ is the array element `T[[5,5]]`.

The notebook uses two sets of assumptions:

- `ssX = H > 0 && sX0 && 6 H x0 > 0 && 2 la[x4] > 0 && Cot[6 H x0] > 0 && Sin[6 H x0] > 0`, with `sX0 = x0 > 0 && x1 > 0 && ... && x7 > 0` (all eight coordinates positive), for the (x_0, x_4) chart;
- `constraintVars = x0 > 0 && x4 > 0 && z > 0 && t > 0`, for the chart $z = 6Hx_0, t = Hx_4$.

`la` is a free function inherited from the original notebook. It occurs only inside `ssX`. It is never given a value, and nothing here invents one [P, Section 2].

3.1.3 SO(4): the self-dual and anti-self-dual blocks

The six antisymmetric real 4×4 matrices split under the Hodge star into a self-dual 3-space and an anti-self-dual 3-space, the two commuting $\mathfrak{su}(2)$ factors of $\mathfrak{so}(4)$. The original notebook builds both from two elementary tensors (Section 3):

$$Q_a[h, p, q] = \text{Signature}[\{h, p, q, 4\}], \quad Q_b[h, p, q] = \delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}, \quad (2)$$

$$\mathbf{s4by4}[h] = Q_a - Q_b \text{ (self-dual)}, \quad \mathbf{t4by4}[h] = Q_a + Q_b \text{ (anti-self-dual)}, \quad (3)$$

$$h = 1, 2, 3. \quad (4)$$

Explicitly [D]:

$$s_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad s_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad s_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix},$$

$$t_1 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad t_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad t_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

with $s_h = \mathbf{s4by4}[h]$ and $t_h = \mathbf{t4by4}[h]$.

[A] $\mathbf{s4by4}$ is self-dual; $\mathbf{t4by4}$ is anti-self-dual; $\mathbf{s4by4}$ and $\mathbf{t4by4}$ are antisymmetric; the two $\mathfrak{su}(2)$ factors commute; and, for the nine mixed products $\mathbf{st}[J, K] = s_J t_K$, $\mathbf{st}[J, K]$ is symmetric.

3.1.4 The real 8×8 Clifford generators τ , $\bar{\tau}$, and the spinor metric σ

The $O(4,4)$ spinor metric is

$$\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}. \quad (5)$$

[A] $\mathbf{sigma} . \mathbf{sigma} == \text{ID8}$; $\mathbf{sigma}$ is symmetric; $\text{Tr}[\mathbf{sigma}] == 0$. The type-1 and type-2 split-octonion spinors both carry this same σ , because $\sigma = \sigma^{-1}$ [P].

The generators (Section 4) are

$$\tau[0] = I_8, \quad (6)$$

$$\tau[h] = \begin{pmatrix} 0 & s_h \\ s_h & 0 \end{pmatrix}, \quad h = 1, 2, 3, \quad (7)$$

$$\tau[3+k] = \begin{pmatrix} 0 & t_{4-k} \\ -t_{4-k} & 0 \end{pmatrix}, \quad k = 1, 2, 3 \quad (t_3, t_2, t_1), \quad (8)$$

$$\tau[7] = \tau[1]\tau[2]\tau[3]\tau[4]\tau[5]\tau[6] = \text{diag}(-I_4, +I_4) \quad [\text{D}], \quad (9)$$

$$\bar{\tau}[A] = \sigma (\sigma \tau[A])^T, \quad A = 0, \dots, 7 \quad (\text{the } \sigma\text{-adjoint}). \quad (10)$$

The order 3,2,1 is the original's. It is what makes $\tau[7]$ the product of the first six and $\sigma = \tau[1]\tau[2]\tau[3]$ [P]. The original calls the list of the six blocks $\mathbf{sixAntiSymmetric8by8}$, but that name fits only half of it: [A] blocks 1-3 (built from $\mathbf{s4by4}$) are antisymmetric; blocks 4-6 (built from $\mathbf{t4by4}$) are symmetric.

The defining relation is the split Clifford algebra of the $4+4$ quadratic form in its real 8×8 “half-spinor” form:

$$\frac{1}{2}(\tau[A] \bar{\tau}[B] + \tau[B] \bar{\tau}[A]) = \eta_{AB} I_8. \quad (11)$$

[A] $(1/2)(\mathbf{tau}[A].\mathbf{taubar}[B] + \mathbf{tau}[B].\mathbf{taubar}[A]) == \mathbf{eta4488}[[A+1, B+1]] \text{ ID8}$.

The ordering identities [A]:

- $\mathbf{sigma} == \mathbf{tau}[1].\mathbf{tau}[2].\mathbf{tau}[3]$
- $\mathbf{sigma} == \mathbf{tau}[4].\mathbf{tau}[5].\mathbf{tau}[6].\mathbf{tau}[7]$

- `tau[1]...tau[7] == tau[0] == ID8`
- `sigma . taubar[A] == Transpose[sigma . tau[A]]`
- `-eta4488[[A+1,A+1]] tau[A] == Transpose[tau[A]] for A = 1..7`

The last one says that $\tau[1..3]$ are antisymmetric and $\tau[4..7]$ are symmetric. The second invariant of the original is $\Omega = \sigma \tau[7]$: `[A] Omega == tau[4].tau[5].tau[6]`.

A complete basis of the 8×8 algebra (Section 7) is the 63 ordered products of 1, 2 or 3 of $\tau[1..7]$, together with I_8 . `[A] the 8x8 basis has 64 elements; the 64 elements are distinct; (1/8) Tr[bas64[A].bas64[B]] == eta64[[A,B]]`; symmetric 8×8 basis elements number 36; antisymmetric 8×8 basis elements number 28.

3.1.5 The 16×16 Dirac matrices, chirality, σ_{16} , and $\mathfrak{so}(4,4)$

Following E. A. Lord's reduced Brauer–Weyl construction (Math. Proc. Camb. Phil. Soc. 64 (1968) 765–778), the 8×8 generators are doubled off-diagonally (Section 5):

$$T_{16}[A] = \begin{pmatrix} 0 & \bar{\tau}[A] \\ \tau[A] & 0 \end{pmatrix} \quad (A = 0, \dots, 7), \quad (12)$$

$$T_{16}[8] = T_{16}[0]T_{16}[1] \cdots T_{16}[7], \quad \sigma_{16} = T_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]. \quad (13)$$

The $T_{16}[A]$ are real 16×16 matrices, and “ $\gamma[a]$ ” means $T_{16}[a]$. $T_{16}[8]$ is the chirality operator and σ_{16} is the 16×16 spinor metric.

Assertions `[A]`:

- `{T16[A], T16[B]}/2 == eta4488[[A+1,B+1]] ID16`
- `sigma16 == ArrayFlatten[{{-sigma,0},{0,sigma}}]`
- `sigma16 . T16[A] is antisymmetric for A = 0..7`
- `T16[8] == sigma16 . T16[4].T16[5].T16[6].T16[7]`
- `sigma16 is symmetric and sigma16 . sigma16 == ID16`
- `sigma16 . covariantDiffMatrix is symmetric, with covariantDiffMatrix = T16[5]T16[6]T16[7]`

The two chiralities are the two split-octonion spinors. The notebook defines $P_L = \frac{1}{2}(I_{16} - T_{16}[8])$ and $P_R = \frac{1}{2}(I_{16} + T_{16}[8])$. `[A] PL + PR == ID16; PL is idempotent; PR is idempotent; PL . PR == PR . PL == 0; PL projects onto the upper (type-1) components, PR onto the lower (type-2) components.`

The last assertion is equivalent to

$$T_{16}[8] = \text{diag}(-I_8, +I_8). \quad (14)$$

A 16-component spinor is therefore the direct sum $\Psi = (\psi_1, \psi_2)$:

- the upper eight components are the **type-1** split-octonion spinor, of chirality -1 ;
- the lower eight components are the **type-2** spinor, of chirality $+1$.

Every $T_{16}[A]$ exchanges the two halves, since it is block off-diagonal `[P]`.

The $\mathfrak{so}(4,4)$ generators are

$$S^{AB} = \text{SAB}[[A+1,B+1]] = \frac{1}{4} [T_{16}[A], T_{16}[B]]. \quad (15)$$

Assertions `[A]`:

- SAB is antisymmetric in its two flat indices
- sigma16 . SAB is antisymmetric
- T16[8] commutes with every SAB: the chiral halves are each Spin(4,4)-invariant
- [SAB,SAB] closes on the so(4,4) algebra, which is the relation $[S_{AB}, S_{CD}] = -(\eta_{AC}S_{BD} - \eta_{AD}S_{BC} - \eta_{BC}S_{AD} + \eta_{BD}S_{AC})$
- [SAB, T16] reproduces the vector representation, which is $[S_{AB}, \gamma_C] = -\eta_{CA}\gamma_B + \eta_{CB}\gamma_A$

The spin connection of the frame field enters as $\frac{1}{8}\omega_{\mu ab}[\gamma^a, \gamma^b] = \frac{1}{2}\omega_{\mu ab}S^{ab}$.

The 256-element basis (Section 6) is the ordered products of k distinct generators, $k = 1..8$, followed by I_{16} , taken in the original's order. [A] base16[[93]] is sigma16 (the word {0,1,2,3}); base16[[255]] is the chirality operator T16[8]; base16[[256]] is the identity; Tr[M.M]/16 is +1 or -1 for every basis element; 136 basis elements have Tr[M.M] > 0; 120 basis elements have Tr[M.M] < 0; symmetric basis elements number $16*17/2 = 136$; antisymmetric basis elements number $16*15/2 = 120$; Tr[M.M] > 0 exactly when M is symmetric; Tr[M.M] < 0 exactly when M is antisymmetric.

Section 8 also builds the ordinary 4×4 Dirac algebra of the (x_1, x_2, x_3, x_4) block, whose metric is $g_{44} = \text{diag}(1, 1, 1, -1)$. For example [A]: {gamma4[h], gamma4[k]} == 2 g44[[h,k]] ID4 for h,k = 1..4; [S44,S44] closes on the so(3,1) algebra. It plays no role below.

3.1.6 Cartan triality and the split octonions (Section 9): what is proved

The vector, the type-1 spinor and the type-2 spinor are three 8-dimensional representations of Spin(4,4) that triality relates. The original makes the vector-to-spinor bridge concrete with one distinguished spinor, unit. It is row 8 of the eigenvectors of σ , rescaled by $1/\sqrt{2}$. Assertions [A]:

- unit is pinned to the author's spinor {1/Sqrt[2],0,0,0,1/Sqrt[2],0,0,0} - the row order of Eigensystem is not relied on silently
- each uEig row is sigma-normalized to +/-1
- unit is sigma-normalized to +1

The bridge and its inverse are

$$\text{triVecToSpin}[[A+1, a+1]] = (\text{unit} \cdot \sigma \cdot \bar{\tau}[A])[[a+1]], \quad (16)$$

$$\text{triSpinToVec} = \text{triVecToSpin}^{-1}, \quad (17)$$

where the row index A of triVecToSpin is a flat vector index and the column index a is a type-1 spinor index. Assertions [A]:

- triVecToSpin . triSpinToVec == ID8
- triSpinToVec . triVecToSpin == ID8
- triSpinToVec == Transpose[eta_AA tau[A].unit] (the original's closed form)
- etaTri is symmetric; etaTri has signature (4,4)
- triVecToSpin transports etaTri back to eta4488
- THE TRIALITY METRIC IS EXACTLY THE SPINOR METRIC: etaTri == sigma

The split-octonion structure constants are read off from the generators through the bridge. In the type-1 spinor basis they are `mabc`; in the vector basis they are `mABC`, the usual multiplication table. Assertions [A]:

- `eA[1]` is a two-sided identity for the vector-basis product
- `<eA, eB>` composition is compatible with `eta` (basis-unit norm form)
- `m_{ABC}` restricted to imaginary directions is totally antisymmetric

The scope of what is proved: the norm is multiplicative *on the basis units*, and the imaginary structure constants are totally antisymmetric. The general composition law $N(xy) = N(x)N(y)$ for arbitrary elements, and alternativity, are **not** asserted.

3.1.7 The wave function of the un-universe and the author's Lagrangian L_a (Section 11)

The original notebook calls `Psi16` the wave function of the “un-universe”. It is sixteen scalar functions of the hidden coordinate x_0 and the time x_4 :

$$\Psi_{16} = (f_0, \dots, f_{15}), \quad f_k \equiv \text{f16}[k][x_0, x_4], \quad (18)$$

where the upper eight components are the type-1 spinor and the lower eight the type-2 spinor. [A] `Psi16` is the direct sum of the type-1 and type-2 spinors. Two re-parametrizations are recorded:

- `f16[k] -> Z[k][6 H #1, H #2] &`, which is the one used downstream;
- `f16[k] -> nZ[k][6 H #1, H #2]/Sqrt[Sin[6 H #1]] &`, the author's alternative normalization.

The author's Lagrangian, exactly as defined in Section 11:

```
La[] := ((1/H) Transpose[Psi16] . sigma16 .
  Sum[T16[alpha1 - 1] . D[Psi16, X[[alpha1]]], {alpha1, 1, Length[X]}]
  + 2 (M/H) Transpose[Psi16] . sigma16 . Psi16
) // Simplify[#, constraintVars] &
```

that is,

$$L_a = \frac{1}{H} \Psi_{16}^T \sigma_{16} T_{16}[\alpha] \partial_\alpha \Psi_{16} + \frac{2M}{H} \Psi_{16}^T \sigma_{16} \Psi_{16} \quad (19)$$

$$= \frac{1}{H} \Psi_{16}^T \sigma_{16} (T_{16}[0] \partial_{x_0} + T_{16}[4] \partial_{x_4}) \Psi_{16} + \frac{2M}{H} s, \quad s = \Psi_{16}^T \sigma_{16} \Psi_{16}. \quad (20)$$

The second line holds because Ψ_{16} depends only on x_0 and x_4 . It was checked for this document against the evaluated `La[]` (a `Simplify` difference of exactly 0); it is not a notebook assertion. The mass term is $(2M/H)s$: it is $+2M/H$ times the scalar bilinear s .

The evaluated `La[]` [D] is shown below. Write $f_k = \text{f16}[k][x_0, x_4]$, $\partial_0 f_k = \text{Derivative}[1, 0][\text{f16}[k]][x_0, x_4]$ and $\partial_4 f_k = \text{Derivative}[0, 1][\text{f16}[k]][x_0, x_4]$. Then $H L_a$ is exactly

$$\begin{aligned} H L_a = & -4M(f_0 f_4 + f_1 f_5 + f_2 f_6 + f_3 f_7 - f_8 f_{12} - f_9 f_{13} - f_{10} f_{14} - f_{11} f_{15}) \\ & + f_{12}(\partial_4 f_5 + \partial_0 f_0) + f_{13}(-\partial_4 f_4 + \partial_0 f_1) + f_{14}(-\partial_4 f_7 + \partial_0 f_2) \\ & + f_{15}(\partial_4 f_6 + \partial_0 f_3) + f_8(-\partial_4 f_1 + \partial_0 f_4) + f_9(\partial_4 f_0 + \partial_0 f_5) \\ & + f_{10}(\partial_4 f_3 + \partial_0 f_6) + f_{11}(-\partial_4 f_2 + \partial_0 f_7) + f_4(\partial_4 f_{13} - \partial_0 f_8) \\ & - f_5(\partial_4 f_{12} + \partial_0 f_9) - f_6(\partial_4 f_{15} + \partial_0 f_{10}) + f_7(\partial_4 f_{14} - \partial_0 f_{11}) \\ & - f_0(\partial_4 f_9 + \partial_0 f_{12}) + f_1(\partial_4 f_8 - \partial_0 f_{13}) + f_2(\partial_4 f_{11} - \partial_0 f_{14}) \\ & - f_3(\partial_4 f_{10} + \partial_0 f_{15}). \end{aligned}$$

The structural identity behind it. $\sigma_{16}T_{16}[A]$ is antisymmetric (Section 5). An antisymmetric form vanishes on the diagonal, so for commuting components $\Psi_{16}^T\sigma_{16}T_{16}[\alpha]\Psi_{16} = 0$ identically. The kinetic term therefore equals minus its transpose, with no boundary term. [A] Psi16 . sigma16 . T16[alpha] . Psi16 == 0 identically (antisymmetric form); the kinetic term equals minus its transpose, exactly; La[] == its conjugate form when the coefficients match. The last one is $L_a = -\frac{1}{H}(\partial_\alpha\Psi_{16})^T\sigma_{16}T_{16}[\alpha]\Psi_{16} + \frac{2M}{H}\Psi_{16}^T\sigma_{16}\Psi_{16}$. The original's scratch cell compared L_a with a form carrying mismatched factors of H and so found a non-zero difference [P].

3.1.8 The Euler–Lagrange equations eLa (Section 12)

The original's Euler–Lagrange operator, transcribed exactly:

```
eL[Lagrangian_Symbol, detsqrt_] := Module[{L, t},
  L = Lagrangian[];
  t = Table[
    FullSimplify[
      (1/detsqrt) ( D[L, f16[k][x0, x4]]
        - D[D[L, Derivative[1, 0][f16[k]]][x0, x4]], x0]
        - D[D[L, Derivative[0, 1][f16[k]]][x0, x4]], x4) ),
    constraintVars],
  {k, 0, 15}];
  Return[t /. subsDefects];
  (* subsDefects = {} *)
eLa = eL[La, 1]
```

[A] there are sixteen field equations. $eLa[[k]]$ is the Euler–Lagrange expression for the component f_{k-1} , and the field equations are $eLa[[k]] = 0$. **The sixteen expressions, exactly as the notebook computes them** [D], in the notation above:

$$\begin{aligned}
eLa[[1]] &= -\frac{2}{H}(2Mf_4 + \partial_4f_9 + \partial_0f_{12}) & (\text{varied: } f_0) \\
eLa[[2]] &= -\frac{2}{H}(2Mf_5 - \partial_4f_8 + \partial_0f_{13}) & (\text{varied: } f_1) \\
eLa[[3]] &= -\frac{2}{H}(2Mf_6 - \partial_4f_{11} + \partial_0f_{14}) & (\text{varied: } f_2) \\
eLa[[4]] &= -\frac{2}{H}(2Mf_7 + \partial_4f_{10} + \partial_0f_{15}) & (\text{varied: } f_3) \\
eLa[[5]] &= -\frac{2}{H}(2Mf_0 - \partial_4f_{13} + \partial_0f_8) & (\text{varied: } f_4) \\
eLa[[6]] &= -\frac{2}{H}(2Mf_1 + \partial_4f_{12} + \partial_0f_9) & (\text{varied: } f_5) \\
eLa[[7]] &= -\frac{2}{H}(2Mf_2 + \partial_4f_{15} + \partial_0f_{10}) & (\text{varied: } f_6) \\
eLa[[8]] &= -\frac{2}{H}(2Mf_3 - \partial_4f_{14} + \partial_0f_{11}) & (\text{varied: } f_7) \\
eLa[[9]] &= \frac{2}{H}(2Mf_{12} - \partial_4f_1 + \partial_0f_4) & (\text{varied: } f_8) \\
eLa[[10]] &= \frac{2}{H}(2Mf_{13} + \partial_4f_0 + \partial_0f_5) & (\text{varied: } f_9) \\
eLa[[11]] &= \frac{2}{H}(2Mf_{14} + \partial_4f_3 + \partial_0f_6) & (\text{varied: } f_{10}) \\
eLa[[12]] &= \frac{2}{H}(2Mf_{15} - \partial_4f_2 + \partial_0f_7) & (\text{varied: } f_{11}) \\
eLa[[13]] &= \frac{2}{H}(2Mf_8 + \partial_4f_5 + \partial_0f_0) & (\text{varied: } f_{12}) \\
eLa[[14]] &= \frac{2}{H}(2Mf_9 - \partial_4f_4 + \partial_0f_1) & (\text{varied: } f_{13}) \\
eLa[[15]] &= \frac{2}{H}(2Mf_{10} - \partial_4f_7 + \partial_0f_2) & (\text{varied: } f_{14}) \\
eLa[[16]] &= \frac{2}{H}(2Mf_{11} + \partial_4f_6 + \partial_0f_3) & (\text{varied: } f_{15})
\end{aligned}$$

In Wolfram InputForm the first reads, character for character (broken over two lines here),

```
eLa[[1]] = (-2*(2*M*f16[4][x0, x4] + Derivative[0, 1][f16[9]][x0, x4]
  + Derivative[1, 0][f16[12]][x0, x4]))/H
```

and the other fifteen follow the same pattern with the indices and signs above.

These were printed for the provenance pages of 2026-09-24 by loading the notebook's own DumpSave file `claude-fable/claude-fable_Einstein-Rosen-2-Planes-eLa.mx`. That

file is `SameQ` to the author's original `Pre-gravityPre-Big_Bang_M6=3-Generations_of_Einstein-Rosen-2-Planes-eLa.mx` in the repository root; this was re-checked for the provenance pages of 2026-09-24 with `SameQ`, which returned `True`. So they are the author's equations, unchanged. In matrix form they read

$$\mathbf{eLa} = \frac{2}{H} \sigma_{16} (T_{16}[0] \partial_{x_0} + T_{16}[4] \partial_{x_4} + 2M) \Psi_{16}, \quad \text{i.e.} \quad (21)$$

$$(T_{16}[0] \partial_{x_0} + T_{16}[4] \partial_{x_4}) \Psi_{16} = -2M \Psi_{16}. \quad (22)$$

This matrix form was checked for the provenance pages of 2026-09-24 (a `Simplify` difference of exactly 0); it is not a notebook assertion. In Part VI's language (below) it is the flat-frame Dirac equation $\gamma^\mu \partial_\mu \Psi = H V'(s) \Psi$ with $\gamma^\mu = T_{16}[\mu]$ and $V'(s) = -2M/H$. What Part VI asserts is the fidelity anchor below: the Euler–Lagrange expressions of its fable Lagrangian with $V = -(2M/H)s$, on the flat frame and with no volume factor, are exactly `eLa`.

The coupling pattern. Iterating `MergeSetsStep` to a fixed point merges any two sets of component indices that share an index. The result partitions the sixteen components into [D]

$$\mathbf{eLaCouplings} = \{\{0, 5, 8, 13\}, \{1, 4, 9, 12\}, \{2, 7, 10, 15\}, \{3, 6, 11, 14\}\}. \quad (23)$$

[A] the coupled blocks partition all sixteen components; there are four blocks of four. The notebook writes `eLa` and `eLazt` to `.mx` files beside itself, as the original does.

3.1.9 The (z, t) chart, four blocks of four, and the closed-form solutions (Section 13)

The chart. The chart is $z = 6Hx_0$, $t = Hx_4$. Substituting `f16[k] → Z[k] [6 H x0, H x4]` and then $x_0 \rightarrow z/(6H)$, $x_4 \rightarrow t/H$, with the original's normalization $1/(2H)$, gives `eLazt =  $\frac{1}{2H}$  eLa` in the new chart. The first equation, for example, reads [D]

$$\mathbf{eLazt}[[1]] = -\frac{2M Z_4 + H \partial_t Z_9 + 6H \partial_z Z_{12}}{H^2}, \quad Z_k \equiv Z[k] [z, t]. \quad (24)$$

Decoupling. Solving $0 = \mathbf{eLazt}$ for the sixteen t -derivatives gives [A] all sixteen t -derivatives are determined. Relabelling each coupling set as a consecutive block, `sZt0yZ` maps Z_k to yZ_j as follows [D]:

- $Z_0, Z_5, Z_8, Z_{13} \rightarrow yZ_{0..yZ_3}$;
- $Z_1, Z_4, Z_9, Z_{12} \rightarrow yZ_{4..yZ_7}$;
- $Z_2, Z_7, Z_{10}, Z_{15} \rightarrow yZ_{8..yZ_{11}}$;
- $Z_3, Z_6, Z_{11}, Z_{14} \rightarrow yZ_{12..yZ_{15}}$.

Assertions [A]:

- the relabelling is a bijection of the sixteen components
- the system splits into four blocks of four equations
- each 4x4 block is closed in its own four components

The relabelling is a linear map. [A] the `yZ` coefficient matrix is the identity; `ORTHOGONAL: Transpose[S] . S == ID16; ORTHOGONAL: S . Transpose[S] == ID16;` but NOT a `O(8,8)` transformation: it does not preserve `sigma16`; and NOT a direct sum: it mixes type-1 with type-2.

Closed forms. DSolve, wrapped in a 60-second TimeConstrained, returns every block unevaluated. The log prints `block 1: returned unevaluated` through `block 4: returned unevaluated`. The author obtained closed forms with Maple and pasted them into the original as strings. The notebook reproduces them verbatim, parses them with the author's `ConvertMapleToMathematicaV2`, and then substitutes them back into the four blocks. That substitution, not the parser, is what certifies them. [A] all sixteen components have a closed form; there are sixteen pure-function rules; THE MAPLE CLOSED FORMS SOLVE ALL SIXTEEN EQUATIONS.

Block j has one constant C_j and four constants $c_{j1}..c_{j4}$. Every component of block j carries the same exponential

$$\begin{aligned} E_j &= \exp\left(\frac{6(z/6 + t)H^2C_j^2 - 6(-z/6 + t)M^2}{6C_jH^2}\right) \\ &= \exp\left(C_j\left(t + \frac{z}{6}\right) - \frac{M^2}{C_jH^2}\left(t - \frac{z}{6}\right)\right). \end{aligned} \quad (25)$$

Block 1, verbatim from the Maple string:

$$\begin{aligned} yZ_0 &= E_1 \frac{H^2(c_{11}c_{12} - c_{13}c_{14})C_1^2 - M^2(c_{11}c_{12} + c_{13}c_{14})}{2HC_1M}, & yZ_2 &= c_{13}c_{14} E_1, \\ yZ_1 &= -E_1 \frac{H^2(c_{11}c_{12} - c_{13}c_{14})C_1^2 + M^2(c_{11}c_{12} + c_{13}c_{14})}{2HC_1M}, & yZ_3 &= c_{11}c_{12} E_1. \end{aligned}$$

Blocks 2, 3 and 4 have the same shape with their own constants and signs (Section 13 prints all four strings).

3.1.10 Bilinears, the solution branches, and $M_6 = 3$ generations of Einstein–Rosen 2-planes (Section 14)

Five bilinears are formed from the solution written in the relabelled basis. `psi1sol` and `psi2sol` are its first and second halves of eight components **in the relabelled yZ basis**, that is, coupling blocks 1–2 and coupling blocks 3–4. They are not the type-1 and type-2 spinors. [A] the `yZ` split at 8 is NOT the type-1/type-2 split; the first relabelled block holds four type-1 and four type-2 components.

For the full sixteen-component norm, [A]:

- `psi . sigma16 . psi == psi2.sigma.psi2 - psi1.sigma.psi1`
- `sigma16` alone is SYMMETRIC, so `psi . sigma16 . psi` is not forced to vanish
- and for this solution it does NOT vanish: it is not the zero expression

With the light-cone coordinates $z_{\text{tadv}} = 6t + z$ and $z_{\text{tret}} = 6t - z$, the notebook requires the two block norms to be light-cone constants. [A] the **light-cone-constant condition has solutions**. `Solve` returns four branches [D], among them $\{C_2 \rightarrow -C_1, C_4 \rightarrow -C_3\}$, the original's branch. [A] on the branch $C_2 = -C_1, C_4 = -C_3$ the two norms are independent of z_{tadv} and z_{tret} .

The displayed values on that branch [D, re-read for the provenance pages of 2026-09-24 from the saved kernel state] are sharper than the assertion. There, $\psi \cdot \sigma_{16} \cdot \psi$, $\psi_1 \cdot \sigma \cdot \psi_1$ and $\psi_2 \cdot \sigma \cdot \psi_2$ are all identically 0, while $\psi_2 \cdot \sigma \cdot \psi_1$ is not.

The two mass branches [D] are, both taken on the branch $C_2 \rightarrow -C_1, C_4 \rightarrow -C_3$:

- `Psi16ab`: $M \rightarrow H M_{ab} \sqrt{C_1 C_3}$, which keeps a free dimensionless ratio M_{ab} ;
- `Psi16aa`: $M \rightarrow H \sqrt{C_1 C_3}$, the case $M_{ab} = 1$.

In the (x_0, x_4) chart the advanced and retarded coordinates are $\xi_{\text{adv}} = x_0 - x_4$ and $\xi_{\text{ret}} = x_0 + x_4$. Every component of **Psi16aa** is a constant times one of $\exp(\pm H(C_1 \xi_{\text{ret}} + C_3 \xi_{\text{adv}}))$ or $\exp(\pm H(C_3 \xi_{\text{ret}} + C_1 \xi_{\text{adv}}))$ [D, re-read for the provenance pages of 2026-09-24 from the saved kernel state; the notebook prints **Short** forms]. The special case $C_3 \rightarrow C_1$, **Psi16x00ONLY**, depends on x_0 alone. It is exactly [D, re-read in full from the saved kernel state], with $e_{\pm} \equiv e^{\pm 2C_1 H x_0}$:

$$\begin{aligned} \mathbf{Psi16x00ONLY} = & (-c_{13}c_{14}e_+, c_{23}c_{24}e_-, -c_{33}c_{34}e_+, c_{43}c_{44}e_-, \\ & c_{21}c_{22}e_-, -c_{11}c_{12}e_+, c_{41}c_{42}e_-, -c_{31}c_{32}e_+, \\ & c_{13}c_{14}e_+, c_{23}c_{24}e_-, c_{33}c_{34}e_+, c_{43}c_{44}e_-, \\ & c_{21}c_{22}e_-, c_{11}c_{12}e_+, c_{41}c_{42}e_-, c_{31}c_{32}e_+). \end{aligned}$$

Triality turns each spinor of the solution into a vector. The genuine type-1 and type-2 spinors are $\psi_1 = \mathbf{Psi16ab}[[1;;8]]$ and $\psi_2 = \mathbf{Psi16ab}[[9;;16]]$. From them:

$$\mathbf{vectorFormOfType2} = \mathbf{triVecToSpin} \cdot \psi_2, \quad (26)$$

$$\mathbf{vectorFormOfType1} = \eta \cdot (\psi_1^T \cdot \sigma \cdot \mathbf{triSpinToVec}). \quad (27)$$

Assertions [A]:

- both triality vectors have eight components
- $\mathbf{vectorFormOfType2} == \mathbf{psi2}$ contracted on the SPINOR index of $\mathbf{triVecToSpin}$
- $\mathbf{triVecToSpin}$ and $\mathbf{triSpinToVec}$ really are mutually inverse

Each vector is normal to a 7-plane through the origin of the split-octonion algebra, the set of position vectors **Zoct** with $\langle \mathbf{vector}, \mathbf{Zoct} \rangle = 0$. [A] **sevenPlane2** is annihilated by $\mathbf{vectorFormOfType2}$; **sevenPlane1** is annihilated by $\mathbf{vectorFormOfType1}$.

Two non-parallel 7-planes in an 8-space meet in a 6-plane, **M6**. [A] **M6** is a non-empty solution set; **M6** has six free parameters. The intersection determines **zoct**[7] and **zoct**[8] as linear combinations of the free **zoct**[1]..**zoct**[6], with coefficients that depend on x_0, x_4 and the constants C_1, C_3, M_{ab}, c_{jk} [D].

What is proved is that **M6** exists and is a 6-parameter set. The reading of **M6** as the home of “the three generations of Einstein–Rosen 2-plane bridges”, which gives the original notebook its title, is the author’s interpretation and is stated in prose only [P].

Finally [A]: **cfResolution**[unit]: the closed-form bridge equals the matrix inverse; **cfResolution**[unit]: the two bridges are mutually inverse; **cfResolution**[unit] reproduces $\mathbf{triVecToSpin}$.

3.1.11 The canonical frame field (Section 15), restated

At every point of the curved $4 + 4$ spacetime there is a local flat $4 + 4$ Minkowski coordinate system. The frame field installs it; in eight dimensions a vielbein is simply an 8-dimensional vierbein. The frame is stored with the curved index μ as the row and the flat index a as the column. The canonical frame is the one the original notebook records as the value of its symbol **gtrye**:

$$e = \mathbf{frameCanonical} = \text{diag}(\tan 6Hx_0, q, q, q, 1, p, p, p), \quad (28)$$

$$q = \frac{e^{-a_4(Hx_4)}}{(\sin 6Hx_0)^{1/6}}, \quad p = \frac{e^{+a_4(Hx_4)}}{(\sin 6Hx_0)^{1/6}}, \quad g = e\eta e^T, \quad (29)$$

$$ds^2 = \tan^2(6Hx_0) dx_0^2 + q^2(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - p^2(dx_5^2 + dx_6^2 + dx_7^2). \quad (30)$$

Assertions [A]:

- CANONICAL [structural]: frame . coframe == ID8 (certifies only that the frame is invertible)
- CANONICAL [definition]: g == frame . eta4488 . Transpose[frame] (bridge [d]; this IS how g was defined, so it cannot fail)
- CANONICAL [has content]: g == diag(Tan[6 H x0]^2, q^2, q^2, q^2, -1, -p^2, -p^2, -p^2), the stated line element
- CANONICAL [has content]: Diagonal[g] == Diagonal[eta4488] * Diagonal[frame]^2 exactly
- CANONICAL [has content]: g has signature (4,4), GIVEN a4 real and $0 < 6 H x_0 < \pi/2$
- Inverse[g] == Transpose[coframe] . eta4488 . coframe
- g is symmetric
- the metric is non-degenerate
- FABLE-5.1: det g == Sec[6 H x0]^2 (positive, as signature 4+4 requires)

So $\sqrt{|\det g|} = \sec 6Hx_0$ on $0 < 6Hx_0 < \pi/2$. This is the notebook's volume factor `sqrtDetgTele` = Sec[6 H x0], defined in Section 21 and used throughout Part VI (Section 15 itself displays Abs[Sec[6 H x0]]).

Two facts follow from the definitions of q and p . First, $qp = (\sin 6Hx_0)^{-1/3}$: the two exponentials cancel [P, Section 15]. Hence $q^3 p^3 = 1/\sin 6Hx_0$, and the 8-volume element factorizes. [A]:

- FABLESCALAR [has content]: Sqrt[det g_8] == q^3 * (Tan[6 H x0] p^3): observed 3-volume density times hidden 4-volume density
- FABLESCALAR [has content]: the 8-volume element is independent of x4: $d(q^3 p^3)/dx_4 == 0$
- FABLESCALAR [has content]: the hidden 4-volume density scales as $\text{Exp}[+3 a_4[H x_4]]$ and the observed 3-volume as $\text{Exp}[-3 a_4[H x_4]]$: $V_{\text{hid}} \text{Exp}[-3 a_4]$ and $q^3 \text{Exp}[+3 a_4]$ are x4-independent

One Text cell of Section 23 writes $q^3 p^3 = 1/\sqrt{\sin 6Hx_0}$. That is a typo in the prose: from $qp = (\sin 6Hx_0)^{-1/3}$ the cube is $1/\sin 6Hx_0$, which is what makes $\sqrt{\det g} = q^3 \tan(6Hx_0) p^3 = \sec 6Hx_0$. The assertions above are unaffected.

The curved Dirac matrices are $\gamma^\mu = \sum_a \text{coframe}[[a, \mu]] T_{16}[a]$, with $\text{coframe} = e^{-1}$. On the canonical frame they are as follows (read for the provenance pages of 2026-09-24 from the saved kernel state; they follow directly from the diagonal coframe):

$$\begin{aligned} \gamma^0 &= \cot(6Hx_0) T_{16}[0], & \gamma^i &= \frac{1}{q} T_{16}[i] = e^{a_4} (\sin 6Hx_0)^{1/6} T_{16}[i], \\ \gamma^4 &= T_{16}[4], & \gamma^h &= \frac{1}{p} T_{16}[h] = e^{-a_4} (\sin 6Hx_0)^{1/6} T_{16}[h], \end{aligned} \quad (31)$$

for $i = 1, 2, 3$ and $h = 5, 6, 7$. [A] {gammaCurved[mu], gammaCurved[nu]} == 2 gInv[mu, nu] ID16; and, from Part VI, FABLE [has content]: $(\text{gamma}^4)^2 == -\text{ID16}$, so $\text{gamma}^4 d_4 \text{Psi}' == H V'(s) \text{Psi}'$ is solved by $d_4 \text{Psi}' == -H V'(s) \text{gamma}^4 \text{Psi}'$.

The free function a_4 . a_4 is a free, differentiable scalar function of Hx_4 . The original notebook never defines it; it occurs only inside the recorded frame. It is carried symbolically everywhere and is never given a value, and every result holds for arbitrary a_4 . The notebook says so on screen: the log prints NOTE a_4 is an undefined scalar function inherited from the

original notebook together with the explanation, which also names 1a. Near its end Part VI checks [A] PART VI [control]: a4 is STILL UNDEFINED - nothing in Part VI gave it a value; and the FLRW scale factor aF never entered the canonical frame.

The physical reading. The observer is the geodesic observer $u = \partial/\partial x_4$: $g_{44} = -1$, and x_0 is constant along the worldline [P, Section 23]. The cosmological time is $t = x_4$, the proper time of that observer [P, Section 25]. The observed sheet is x_1, x_2, x_3 ; the hidden spacelike coordinate is x_0 ; the hidden timelike sheet is x_5, x_6, x_7 . Part VI makes the kinematic identification $a(t) \sim q \sim e^{-a_4(Ht)}$, $H_{\text{obs}} = -H a'_4(Ht)$. Assertions [A]:

- KINEMATIC [definition]: $\ln q == -a_4[H x_4] + \text{const}(x_0)$: $D[\text{Log}[q], x_4] == -H a_4'[H x_4] == H_{\text{obs}}$
- KINEMATIC [has content]: the second sheet contracts at exactly the opposite rate: $D[\text{Log}[p], x_4] == +H a_4'[H x_4] == -H_{\text{obs}}$
- KINEMATIC [has content]: $H_{\text{obs}} > 0$ exactly when $a_4' < 0$ (and $H_{\text{obs}} < 0$ when $a_4' > 0$) - the sign HYPOTHESIS of this Part, stated, not derived

So which sheet grows is set by the sign of a'_4 , which is never fixed. Section 15's prose describes the 3-space contracting while the deflating directions expand. Part VI reads the same frame the other way round (the observed sheet expands when $a'_4 < 0$) and states that sign as a hypothesis [P].

What the canonical spin connection contributes. The spin connection of this frame follows from the zero-torsion vielbein postulate. Only the facts that the classical fable uses are restated here:

$$\Gamma_{\mu}^{\text{spin}} = \frac{1}{8} \omega_{\mu ab} [T_{16}[a], T_{16}[b]], \quad D_{\mu} \Psi = \partial_{\mu} \Psi + \Gamma_{\mu}^{\text{spin}} \Psi, \quad (32)$$

where each $\Gamma_{\mu}^{\text{spin}}$ is block diagonal: it never mixes type-1 with type-2. Its 24 non-zero components [D, Section 16] are listed below, with $c = \cos 6Hx_0$, $s = \sin 6Hx_0$, $A = a_4(Hx_4)$, $A' = a'_4(Hx_4)$, $j = 1, 2, 3$ and $k = 5, 6, 7$:

component	value
$\omega_{j;0j} = -\omega_{j;j0}$	$Hc^2/(e^A s^{13/6})$
$\omega_{j;4j} = -\omega_{j;j4}$	$HA'/(e^A s^{1/6})$
$\omega_{k;0k} = -\omega_{k;k0}$	$-e^A Hc^2/s^{13/6}$
$\omega_{k;4k} = -\omega_{k;k4}$	$e^A HA'/s^{1/6}$

Assertions [A]:

- CANONICAL [has content]: $\text{omega}[\mu, a, b] == -\text{omega}[\mu, b, a]$ (metric compatibility)
- INDEPENDENT CHECK: the frame-only derivation reproduces omegaCanonical exactly
- [has content] the spinor connection is compatible with omega:
 $[\text{Gamma}^{\text{spin}}_{\mu}, \text{gamma}^a] + \text{omega}_{\mu}^a{}_b \text{gamma}^b == 0$
- $\text{Gamma}^{\text{spin}}[\mu]$ is block diagonal: it never mixes type-1 with type-2
- [has content] and therefore the curved Dirac matrices are covariantly constant: $D_{\mu} \text{gammaCurved}^{\nu} == 0$

From Part V (the fable-5.1 bridge), three results are used below [A]:

- FABLE-5.1 [THE RESULT]: $\gamma^\mu \Gamma_\mu^{\text{spin}} == -(1/2) T[\mu] \gamma^\mu$, one vector term, with the Weitzenböck torsion vector $T_\mu = \partial_\mu \ln \sin 6Hx_0$, i.e. FABLE-5.1 torsion vector has exactly one non-zero component, $6H \cot[6Hx_0]$ along x_0 . Hence $\gamma^\mu \Gamma_\mu^{\text{spin}} = -3H \cot^2(6Hx_0) T_{16}[0]$.
- FABLE-5.1 [THE RESULT]: $\text{DiracCanonical}[\text{Sqrt}[\text{Sin}[6Hx_0]] \text{Psi}'] == \text{Sqrt}[\text{Sin}[6Hx_0]] \text{DiracFable51}[\text{Psi}']$, for a generic Psi'
- FABLE-5.1 [THE RESULT]: $\text{Psi}^T \sigma_{16} \gamma^\mu \Gamma_\mu^{\text{spin}} \text{Psi} == 0$ for EVERY Psi - the spin connection drops out of the Lagrangian, because FABLE-5.1: the reason is Section 11's identity, $\text{Psi}^T \sigma_{16} T_{16}[a] \text{Psi} == 0$ ($\sigma_{16} T_{16}[a]$ antisymmetric). Note that this reason is a statement about **commuting** components.

The Ricci scalar of the canonical metric is [D, Section 16]

$$R = -3H^2 \left((55 + 7 \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0) - 2a'_4(Hx_4)^2 \right). \quad (33)$$

3.1.12 The classical field fable of 2026-09-16 (Part VI, Section 24)

Part VI (Sections 23–25) places two matter fields on the pre-universe. It asks of each the cosmologist's question: what are the energy density ρ , the pressure P , and $w = P/\rho$? The first field, **fableScalar**, is a real scalar with potential $V(\phi)$. The second, **fable**, is the subject of this document.

fableScalar, in brief. It has $\hat{L}_\phi = -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi)$ and the Hilbert tensor $T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi + g_{\mu\nu} \hat{L}_\phi$. Its central theorem is [A] FABLESCALAR [THE RESULT]: the null energy condition $\rho + P_{-1} == 2$ KE holds IDENTICALLY, for every $\phi(x_0, x_4, x_5)$ and every V , so it cannot cross $w = -1$ with positive energy density.

3.1.12.1 The field and its Lagrangian

fable is a **real, commuting** 16-component spinor $\Psi(x)$ that transforms under $\text{Spin}(4,4)$ exactly as Ψ_{16} . It uses $T_{16}[a]$, σ_{16} , the curved γ^μ , the scalar bilinear $s = \Psi^T \sigma_{16} \Psi$, and a general potential $V(s)$:

$$L_\Psi = \sqrt{\det g} \hat{L}_\Psi, \quad \hat{L}_\Psi = \frac{1}{H} \Psi^T \sigma_{16} \gamma^\mu D_\mu \Psi - V(s), \quad (34)$$

$$D_\mu = \partial_\mu + \Gamma_\mu^{\text{spin}}, \quad s = \Psi^T \sigma_{16} \Psi. \quad (35)$$

In Part VI's generic computations $\Psi = \Psi(x_0, x_4)$ has sixteen arbitrary component functions. Assertions [A]:

- FABLE [definition]: $s = \text{Psi}^T \sigma_{16} \text{Psi}$ is a genuine scalar with content: on the constant spinor $e_1 + e_5$, $s == -2$
- FABLE [has content]: Lhat with the covariant derivative $== \text{Lhat}$ with the partial derivative, for the generic $\text{Psi}(x_0, x_4)$ (Part V's theorem, re-asserted here)

The fidelity anchor. The author's Lagrangian is the special case $V(s) = -(2M/H)s$, taken on the flat frame ($e \rightarrow I_8$) and with no volume factor ($\sqrt{g} \rightarrow 1$). With that potential $-V = +(2M/H)s$, which is exactly L_a 's mass term. [A]:

- FABLE [fidelity]: with $V = -(2M/H)s$, frame $\rightarrow \text{ID8}$ and $\text{Sqrt}[g] \rightarrow 1$, on the model's Psi_{16} , the Lagrangian helper IS the author's $\text{La}[]$ of Section 11

- FABLE [fidelity]: and its Euler-Lagrange equations, formed with the author's own eL operator, ARE the sixteen equations eLa of Section 12
- FABLE [solver regression]: the new helper cfEulerLagrange reproduces eLa too
- FABLE-5.1: the covariant Lagrangian with frame $\rightarrow$ ID8 and volume factor $\rightarrow$ 1 IS the author's La of Section 11, term for term

3.1.12.2 The field equation

The equation is obtained by explicit variation of $\sqrt{g} \hat{L}_\Psi$. The kinetic matrix is $A^\mu = \sigma_{16} \gamma^\mu$, which is antisymmetric; the mass matrix σ_{16} is symmetric. [A]:

- FABLE [THE RESULT]: the Euler-Lagrange equations by explicit variation == $\text{Sqrt}[g] \left[(2/H) (A^\mu d_\mu \Psi + (1/2) D \Psi) - 2 V'(s) \sigma_{16} \Psi \right]$, where $D = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} A^\mu)$
- FABLE [has content]: $A^\mu = \sigma_{16} \gamma^\mu$ is antisymmetric and σ_{16} is symmetric - the structure that makes the variation close
- FABLE [has content]: in general $(1/\text{Sqrt}[g]) d_\mu (\text{Sqrt}[g] \gamma^\mu) == [\gamma^\mu, \Gamma^\mu_{\text{spin}}]$ (covariant constancy of γ^μ , Section 17)
- FABLE [has content]: the HYPOTHESIS that makes it a product - the anticommutator $\{\gamma^\mu, \Gamma^\mu_{\text{spin}}\} == 0$ on the canonical frame (no totally antisymmetric part of the connection)
- FABLE [has content]: hence $(1/2)(1/\text{Sqrt}[g]) d_\mu (\text{Sqrt}[g] \gamma^\mu) == \gamma^\mu \Gamma^\mu_{\text{spin}}$, and on this geometry == $-3 H \cot[6 H x_0]^2 T_{16}[0]$
- FABLE [THE RESULT]: $EL == \text{Sqrt}[g] (2/H) \sigma_{16} \cdot (\gamma^\mu D_\mu \Psi - H V'(s) \Psi)$: the field equation of fable is EXACTLY the Levi-Civita covariant Dirac equation $\gamma^\mu D_\mu \Psi == H V'(s) \Psi$
- FABLE [has content]: the connection term does not vanish on an x_0 -independent spinor - witnessed on the constant spinor e1: a spinor homogeneous in everything but x_4 is NOT a solution

So the field equation of the classical fable is

$$\gamma^\mu D_\mu \Psi = H V'(s) \Psi, \quad \text{i.e. on the canonical frame} \quad (36)$$

$$\cot(6 H x_0) T_{16}[0] \partial_0 \Psi + \frac{1}{q} T_{16}[i] \partial_i \Psi + T_{16}[4] \partial_4 \Psi + \frac{1}{p} T_{16}[h] \partial_h \Psi - 3 H \cot^2(6 H x_0) T_{16}[0] \Psi = H V'(s) \Psi \quad (37)$$

(summed over $i = 1, 2, 3$ and $h = 5, 6, 7$; for $\Psi(x_0, x_4)$ only the ∂_0 and ∂_4 terms survive).

The rescaling. [A]:

- FABLE [definition]: under $\Psi = \text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'$, $s == \text{Sin}[6 H x_0]$ $\Psi'^T \sigma_{16} \Psi'$
- FABLE [THE RESULT]: $(\gamma^\mu D_\mu - H V'(s)) [\text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'] == \text{Sqrt}[\text{Sin}[6 H x_0]] (\gamma^\mu d_\mu \Psi' - H V'(s) \Psi')$ for a GENERIC $\Psi'(x_0, x_4)$: the rescaling removes the connection term exactly
- FABLE [has content]: for $\Psi' = \Psi'(x_4)$ the field equation reduces to $\text{Sqrt}[\text{Sin}[6 H x_0]] (\gamma^4 d_4 \Psi' - H V'(s) \Psi')$, $s = \text{Sin}[6 H x_0]$ $\Psi'^T \sigma_{16} \Psi'$

- FABLE [has content]: the reduced equation still depends on x_0 through $V'(\sin[6Hx_0]s)$: for a NONLINEAR V a $\Psi'(x_4)$ alone is not a solution - witnessed with $V = s^2$ on $\Psi' = e_1 + e_5$
- FABLE [has content]: for the mass term $V = -(2M/H)s$, V' is constant and $\Psi'(x_4)$ IS an exact solution

3.1.12.3 The energy-momentum tensor

The tensor is built with the covariant derivative and symmetrized:

$$T_{\mu\nu}^{\text{cov}} = -\frac{1}{2H}(\Psi^T \sigma_{16} \gamma_\mu D_\nu \Psi + \Psi^T \sigma_{16} \gamma_\nu D_\mu \Psi) + g_{\mu\nu} \hat{L}_\Psi, \quad (38)$$

with $\gamma_\mu = g_{\mu\nu} \gamma^\nu$. Assertions [A]:

- FABLE [definition]: T_{cov} is symmetric by construction
- FABLE [has content]: the diagonal of T_{cov} equals the diagonal of the partial-derivative tensor, for the generic $\Psi(x_0, x_4)$: ρ and every pressure are the same in both
- FABLE [has content]: but the two tensors DIFFER off the diagonal - witnessed on the (x_3, x_4) component $T[[4,5]]$, a momentum density along the observed sheet, for the constant spinor $e_1 + e_{13}$ and $V = s^2$
- FABLE [THE RESULT]: $\nabla^\mu T_{\mu\nu} = 0$ ON SHELL in all eight components, for a generic $\Psi(x_0, x_4)$ - the covariant tensor is the conserved one

“On shell” means the following. The field equation is solved for $\partial_4 \Psi = F = -\gamma^4(HV'(s)\Psi - \gamma^0 \partial_0 \Psi - \frac{1}{2} \text{div} G \Psi)$, where $\text{div} G = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu)$ (on this geometry $\frac{1}{2} \text{div} G = -3H \cot^2(6Hx_0) T_{16}[0]$, by the assertion quoted above), and `cfOnShell` replaces every x_4 -derivative of the field by the corresponding derivative of F , repeatedly, until none is left. [A] FABLE [definition]: F solves the field equation for $d_4 \Psi$: $\gamma^4 F + \gamma^0 d_0 \Psi + (1/2) \text{div} G \Psi - H V'(s) \Psi = 0$; FABLE [definition]: `cfOnShell` leaves no x_4 -derivative of the field in ρ . The partial-derivative tensor was checked symbolically while Part VI was designed: its x_0 and x_4 components are conserved and the other six are not. That check takes minutes and is not repeated in the notebook [P].

3.1.12.4 ρ , P , and w on shell

The observer is $u = \partial/\partial x_4$, so $\rho = T_{44}$ ($T[[5,5]]$) and $P_i = T^i_i$ (no sum, $i = 1, 2, 3$). Two kinetic bilinears organize the answer:

$$K_\Psi = \frac{1}{H} \Psi^T \sigma_{16} \gamma^4 \partial_4 \Psi \quad (\text{along the observer's time}), \quad (39)$$

$$K_h = -\frac{1}{H} \Psi^T \sigma_{16} \gamma^0 \partial_0 \Psi \quad (\text{along the hidden coordinate } x_0). \quad (40)$$

Assertions [A]:

- FABLE [THE RESULT]: on shell, $L_{\text{hat}} = s V'(s) - V(s)$
- FABLE [THE RESULT]: on shell, $\rho = V(s) + K_h$, $K_h = -(1/H) \Psi^T \sigma_{16} \gamma^0 d_0 \Psi$
- FABLE [THE RESULT]: on shell, $P_1 = s V'(s) - V(s)$, and $P_2 = P_3 = P_1$

- FABLE [has content]: on shell, $K_{\Psi} == s V'(s) + K_h$ - the x_4 kinetic bilinear is fixed by the potential and the hidden one
- FABLE [has content]: $K_h == 0$ for $\Psi = \text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'(x_4)$ ($d_0 \Psi$ is proportional to Ψ , and $\Psi^T \text{sigma16} \gamma^0 \Psi == 0$)
- FABLE [THE RESULT]: so for those solutions $\rho == V(s)$, hence $w == s V'(s)/V(s) - 1$, and $\rho > 0$ requires $V(s) > 0$

In summary:

$$\rho = V(s) + K_h, \quad P = sV'(s) - V(s), \quad (41)$$

$$\text{and for } \Psi = \sqrt{\sin 6 H x_0} \Psi'(x_4): \quad \rho = V(s), \quad w = \frac{sV'(s)}{V(s)} - 1. \quad (42)$$

3.1.12.5 No dilution on the pre-universe

Assertions [A]:

- FABLE [THE RESULT]: on shell, $ds/dx_4 == 2 \Psi^T \text{sigma16} \gamma^4 \gamma^0 d_0 \Psi$ for a generic $\Psi(x_0, x_4)$
- FABLE [has content]: the two other terms of $d_4 \Psi$ drop out because $\text{sigma16} \gamma^4$ and $\text{sigma16} \gamma^4 \gamma^0$ are ANTISYMMETRIC
- FABLE [control]: that bilinear is NOT identically zero for an x_0 -dependent Ψ - witnessed on $\Psi = e_1 + x_0 e_2$: ' $ds/dx_4 == 0$ for generic $\Psi(x_0, x_4)$ ' would be FALSE
- FABLE [THE RESULT]: for $\Psi = \text{Sqrt}[\text{Sin}[6 H x_0]] \Psi'(x_4)$ it vanishes: s is CONSTANT IN x_4 - no dilution on the pre-universe
- FABLE [THE RESULT]: the same from the reduced equation: $ds'/dx_4 == 0$ on shell for $\Psi'(x_4)$, because $\Psi'^T \text{sigma16} \gamma^4 \Psi' == 0$

So on those solutions s , V , ρ and w are all constant along x_4 . This is the spinor analogue of the scalar's "no Hubble friction" (for the scalar, [A] FABLESCALAR [THE RESULT]: for $\phi = \phi(x_4)$ the field equation is $\phi'' == -V'(\phi)$: NO HUBBLE FRICTION on the pre-universe). For the scalar, Section 23's prose gives the reason as geometric: the factor $\sqrt{g} g^{44} = -\sec 6 H x_0$ does not depend on x_4 , because the 8-volume element is x_4 -independent [P].

3.1.12.6 Dust for the author's mass term

[A]:

- FABLE [THE RESULT]: the author's mass term $V = -(2M/H) s$ is DUST: on shell $P_1 == 0$ identically and $\rho == -(2M/H) s$ ($w == 0$, exactly)
- FABLE [has content]: that density is not zero - witnessed on $\Psi' = e_1 + e_5$ at $M = 1$ ($\rho = (2M/H) * 2 \text{Sin}[6 H x_0]$ there)

The density $\rho = -(2M/H)s$ is positive only on the branch $M s < 0$ [P, Section 24].

3.1.12.7 $w = sV'/V - 1$ for the six potentials, and the classical phantom crossing

With $u \equiv (s/s_1)^2$:

mass:	$V = ms,$	$w = 0,$
lambda-mass:	$V = V_0 + ms,$	$w = -\frac{V_0}{V_0 + ms},$

power:	$V = ms + \lambda s^n,$	$w = \frac{ms + n\lambda s^n}{ms + \lambda s^n} - 1,$
hilltop:	$V = V_0 - \mu(s - s_*)^2,$	$w = \frac{-2\mu s(s - s_*)}{V_0 - \mu(s - s_*)^2} - 1,$
lorentz:	$V = V_0 + \frac{ms}{1 + u},$	$w = \frac{ms(1 - u)}{(1 + u)^2(V_0 + ms/(1 + u))} - 1,$
expdamp:	$V = V_0 + ms e^{-s/s_1},$	$w = \frac{ms e^{-s/s_1}(1 - s/s_1)}{V_0 + ms e^{-s/s_1}} - 1.$

Assertions [A]:

- FABLE [has content]: the closed forms of $w_{\text{Psi}}(s) = s V'(s)/V(s) - 1$ for the six potentials of the numerical work (mass, lambda-mass, power, hilltop, lorentz, expdamp)
- FABLE [THE RESULT]: mass $\rightarrow w == 0$ (dust); $V = \lambda s^n$ alone $\rightarrow w == n - 1$ (accelerating for $n < 2/3$, phantom for $n < 0$); $n = 0.236 \rightarrow w == -0.764$ exactly
- FABLE [THE RESULT]: $w + 1 == s V'/V$ vanishes where V' does, with $V > 0$ there: lorentz and expdamp cross $w = -1$ at $s = s_1$, hilltop at $s = s_*$ - the phantom divide is crossed with no wrong-sign kinetic term
- FABLE [has content]: lorentz and expdamp are bounded below by $V_0 > 0$ for $s \geq 0$ ($\rho > 0$ on the whole history), and $w < -1$ for $s > s_1$ (the past), $w > -1$ for $0 < s < s_1$ (the present)
- FABLE [has content]: the limits - lorentz and expdamp: $w \rightarrow -1$ both as $s \rightarrow \infty$ and as $s \rightarrow 0$; lambda-mass: $w \rightarrow 0$ (dust) as $s \rightarrow \infty$ and $\rightarrow -1$ (Lambda) as $s \rightarrow 0$
- FABLE [has content]: expdamp with $(V_0, m, s_1) = (1.566, 0.839, 2.21)$ at $s = 1$ (today) gives $w == -0.8608$, Unite's $w_0 = -0.861$ to three decimals

The assumptions for the sign statements are $V_0 > 0$, $m > 0$, $s_1 > 0$. This is the **classical phantom crossing**: $w + 1 = sV'/V$ changes sign where V' does, with $V > 0$ there, and no kinetic term has the wrong sign. It is the spinor-quintom mechanism of Cai and Wang, Class. Quantum Grav. 25 (2008) 165014. The stability of perturbations is not examined in Part VI [P]. The V_0 terms of lambda-mass, lorentz and expdamp are a bare cosmological constant and are reported as such [D, the summary table printed in Section 25].

3.1.12.8 The FLRW reference model (Model B)

Because $ds/dx_4 = 0$ on the pre-universe, a history $w(a)$ needs a different background. Part VI builds it as a **separate** frame, $\text{frameFLRW} = \text{diag}(1, a_F(x_4), a_F(x_4), a_F(x_4), 1, 1, 1, 1)$, with its own scale factor a_F , and feeds it to the same spin-connection solver. The canonical frame and a_4 are not touched. Assertions [A]:

- FLRW [definition]: the reference frame does not mention a_4 , and the canonical frame is untouched
- FLRW [solver regression]: vielbein postulate residual is zero
- FLRW [has content]: omega is antisymmetric (metric compatible)
- FLRW [THE RESULT]: $\gamma^\mu \Gamma_{\text{spin}_\mu} == (3/2) (a'/a) \gamma^4$ - the Hubble term of the FLRW Dirac equation
- FLRW [definition]: the rule solves $\gamma^\mu D_\mu \Psi == H V'(s) \Psi$ for $d_4 \Psi$

- FLRW [THE RESULT]: $d(a^3 s)/dx4 == 0$ on shell, for EVERY V : $s = s_0 a^{-3}$, the dust dilution law, and $w_{\Psi}(a) = s V'(s)/V(s) - 1$ on it
- FLRW [control]: without the Hubble term, $d(a^3 s)/dx4 == 3 (a'/a) a^3 s$ instead (i.e. $ds/dx4 == 0$, the pre-universe law): the dilution IS the Hubble term

Model B, solved independently with NDSolve. The units are $8\pi G = 1$, $H_0 = 1$ and $\rho_{\text{crit}} = 3$. The independent variable is $N = \ln a$, and $s = e^{-3N}$ is exact, so only the cosmic time needs integrating:

$$\frac{dt}{dN} = \frac{1}{H}, \quad H^2 = \Omega_m e^{-3N} + \Omega_r e^{-4N} + \frac{V(s)}{3}, \quad \Omega_m = \frac{3}{10}, \quad \Omega_r = \frac{84}{10^6}, \quad (43)$$

$$V = \text{expdamp} \text{ with } (V_0, m, s_1) = \left(\frac{1566}{1000}, \frac{839}{1000}, \frac{221}{100}\right), \quad (44)$$

$$N \in [-7, 0], \quad 701 \text{ rows}, \quad t(N_0) = \frac{1}{2H(N_0)} \text{ added.} \quad (45)$$

All inputs are exact rationals, and `WorkingPrecision -> 24`. Assertions [A]:

- MODEL B [definition]: 701 rows from $N = -7$ to 0 , $s(0) == 1$ and $\rho_{\Psi}(0) == V(1) == 1.566 + 0.839 \text{ Exp}[-1/2.21]$
- MODEL B [THE RESULT]: $w \leq -1$ at every grid point with $s > s_1$ and $w > -1$ at every grid point with $s < s_1$: the crossing is at $s = s_1$, $a = s_1^{-(-1/3)} = 0.7677$
- MODEL B [has content]: a genuine phantom phase: $\min w = -1.302$ (to 0.001) at $a = 0.543$, and $w(a = 1) = -0.8608$
- MODEL B [has content]: $V(s) > 0$ at every row - $\rho_{\Psi} > 0$ on the whole history
- MODEL B [fidelity]: the reference CSV has the expected header and the same 701 values of N
- MODEL B [fidelity]: w agrees with `fable-cosmology/reference/mathematica_spinor_expdamp.csv` at EVERY row to $1e-9$
- MODEL B [fidelity]: so do t , H and ρ_{Ψ} , to $1e-9$ (s differs only by the file's 16-digit rounding of numbers of order $1e9$)

The run prints [D] (one line in the log, broken here):

```
Model B (expdamp):
w(a=1) = -0.86084564037764806560795392619028548152`20.
min w   = -1.30209881490918676547015655743122703056`20. at N = -0.61
          (a = 0.54335086907449978711266408781657974806`20.)
t(a=1) = 0.96433642136311524576147498023448633382`20.
H(a=1) = 0.99998205003431888956547950340514648567`20.
```

Model B applies no normalization: $V(1) = 1.566 + 0.839 e^{-1/2.21} \approx 2.0996 = 3\Omega_{\Psi_0}$ with $\Omega_{\Psi_0} = 0.6999$ [P, Section 24]. That is why the printed $H(a=1)$ is 0.99998 and not exactly 1: $H(a=1)^2 = 3/10 + 84/10^6 + V(1)/3 \approx 0.99996$ (this arithmetic was added for this document; it is not a notebook statement).

3.1.12.9 What is proved on the pre-universe, and what belongs to the reference model

Proved on the 8-manifold, for arbitrary a_4 , with every identity closed symbolically:

- the Lagrangian and its fidelity to L_a and `eLa`;

- the covariant Dirac equation and its rescaling;
- the covariant energy-momentum tensor and its conservation;
- $\rho = V + K_h$ and $P = sV' - V$;
- dust for the author's mass term;
- the constancy of s in x_4 ;
- the volume factorization.

The history $s = s_0 a^{-3}$, the phantom crossing at $a = 0.768$, and all contact with the Unite fits $(w_0, w_a) = (-0.861, -0.60)$ belong to 4-dimensional physics on the separate FLRW frame [P, Section 25].

The volume bookkeeping is proved [A]:

- VOLUME [has content]: the reduced density $\rho_4 = \text{Vhid } \rho_8$ is NOT separately conserved: $d \rho_4 / dx_4 + 3 H_{\text{obs}} (\rho_4 + P_4) == 3 H_{\text{obs}} P_4$, for every w
- VOLUME [has content]: equivalently $d \rho_4 / dx_4 == -3 H_{\text{obs}} \rho_4$: ρ_4 falls as a^{-3} for EVERY w - a volume effect, the same for a cosmological constant as for dust

Which of ρ_4 and ρ_8 gravitates is left undetermined in Part VI, because Part VI has no gravitational sector: no Einstein equations, and a_4 free [P, Section 25].

3.1.13 Why the classical fable must be refined

The fable of Part VI is a field of **real, commuting** numbers. A fermion needs **anticommuting** (Grassmann) components, and the change is not cosmetic. It rests on two algebraic facts about the author's Clifford algebra.

F1, the symmetry of CT . Write $\Gamma_{(r)}$ for an antisymmetrized product of r of the $T_{16}[a]$. With $C_- = \sigma_{16}$:

- σ_{16} is symmetric;
- $\sigma_{16} T_{16}[a]$ is **antisymmetric** (this much is already Part I's `sigma16 . T16[A]` is **antisymmetric** for $A = 0..7$);
- $\sigma_{16} \Gamma_{ab}$ is antisymmetric;
- $\sigma_{16} \Gamma_{abc}$ and $\sigma_{16} \Gamma_{abcd}$ are symmetric;
- $\sigma_{16} T_{16}[8]$ is symmetric;
- $\sigma_{16} T_{16}[a] \sigma_{16}^{-1} = -T_{16}[a]^T$.

The second conjugation is $C_+ = \sigma_{16} T_{16}[8]$:

- it is symmetric, and $C_+ T_{16}[a]$ is **symmetric**;
- $C_+ \Gamma_{ab}$ and $C_+ \Gamma_{abc}$ are antisymmetric;
- $C_+ T_{16}[a] C_+^{-1} = +T_{16}[a]^T$.

The $\text{Spin}(4,4)$ -invariant bilinear forms on the 16-spinor form a two-dimensional space, spanned by σ_{16} and $\sigma_{16} T_{16}[8]$ ($C_+ T_{16}[8] = \sigma_{16}$). Both are symmetric, and both are chirality-diagonal. [VII] REFINEMENT [definition]: $T_{16}[8]^2 == \text{sigma16}^2 == C_+^2 == \text{ID16}$, C_+ is symmetric, and $C_+ . T_{16}[8] == \text{sigma16}$ - the four candidate mass matrices are only two, REFINEMENT [has content]: $C_- T_{16}[a] C_-^{-1} == -T_{16}[a]^T$ and $C_+ T_{16}[a] C_+^{-1} == +T_{16}[a]^T$ for $a = 0..7$, REFINEMENT [THE RESULT]: the symmetry of C Gamma

depends only on the rank r of Γ : C-: S A A S S A A S S, C+: S S A A S S A A S ($r = 0..8$), REFINEMENT [THE RESULT]: the Spin(4,4)-invariant bilinear forms on the 16-spinor form a 2-dimensional space, spanned by C- = σ_{16} and C+ = $\sigma_{16} T_{16}[8]$, REFINEMENT [has content]: every invariant form is SYMMETRIC and chirality-diagonal, and the two chiral pieces PL σ_{16} PL and PR σ_{16} PR are invariant separately.

F2, what anticommuting components do to a bilinear. For Grassmann θ :

- $\theta^T N \theta$ sees only the **antisymmetric** part of N ;
- $\theta^T N d\theta = \frac{1}{2} d(\theta^T N \theta) + \theta^T N_{\text{sym}} d\theta$.

For commuting components the roles are exactly reversed: $\theta^T N \theta$ sees only the symmetric part, and the Euler-Lagrange expression of $\theta^T N \phi$ is $2N_{\text{antisym}}\phi$. That reversal is why Part VI's real commuting fable has both a mass term (σ_{16} symmetric) and a kinetic term ($\sigma_{16} T_{16}[a]$ antisymmetric). [VII] GRASSMANN [THE RESULT]: $\theta^T N \theta == \theta^T N_A \theta$ and $\theta^T N_S \theta == 0$ for a general 16×16 N - an anticommuting bilinear sees only the ANTISYMMETRIC part, GRASSMANN [THE RESULT]: $\theta^T N \phi - (1/2) d(\theta^T N \theta) == \theta^T N_S \phi$ for a general N , GRASSMANN [THE RESULT]: the Euler-Lagrange expression of $L = \theta^T N \phi$ (left derivatives) is $(N + N^T) \phi$: EMPTY field equations exactly when N is antisymmetric, GRASSMANN [control]: for COMMUTING components the roles are reversed - $\theta^T N \theta$ sees only N_S , and the Euler-Lagrange expression of $\theta^T N \phi$ is $2 N_A \phi$.

The consequences:

- A real Grassmann 16-spinor with the author's σ_{16} has no dynamics.** Its mass term $\theta^T \sigma_{16} \theta$ vanishes identically, because σ_{16} is symmetric. Its kinetic term $\theta^T \sigma_{16} T_{16}[a] d\theta$ is a total derivative, because $\sigma_{16} T_{16}[a]$ is antisymmetric. So its Euler-Lagrange expression is identically empty, even though the kinetic term itself is not zero. The statement holds on **every** frame. $\sigma_{16} \Gamma_{abc}$ is symmetric, so the connection term $\theta^T \sigma_{16} \{\gamma^\mu, \Gamma_\mu^{\text{spin}}\} \theta$ vanishes too. [VII] REFINEMENT (a) [THE RESULT]: real Grassmann + σ_{16} : both invariant mass terms vanish, and $\theta^T \sigma_{16} T_{16}[a] \phi == (1/2) d(\theta^T \sigma_{16} T_{16}[a] \theta)$ for every a (a total derivative), REFINEMENT (a) [has content]: its Euler-Lagrange expression is identically empty in every direction - while the kinetic term itself is NOT zero, REFINEMENT (a) [THE RESULT]: FRAME-INDEPENDENT - $\sigma_{16} \{T_{16}[a], S^{bc}\}$ is SYMMETRIC for all a, b, c (so $\theta^T \sigma_{16} \{\gamma^\mu, \Gamma_\mu\} \theta == 0$ on every frame), and $d_\mu(\text{Sqrt}[g] \sigma_{16} \gamma^\mu) == \text{Sqrt}[g] \sigma_{16} [\gamma^\mu, \Gamma_\mu]$ on the canonical frame.
- A Majorana 16-spinor built with C_+ propagates but is massless.** Its kinetic term is genuine, but every invariant bilinear vanishes on it, so there is no s and $V(s)$ is undefined. It has exactly one quartic Lorentz scalar: the quartic scalars $v \cdot v, B \cdot B$ and $C \cdot C$ built from it are all proportional to one quartic. Its commuting shadow has no kinetic term at all. [VII] REFINEMENT (b) [THE RESULT]: Majorana with C+: the kinetic term is genuine - Euler-Lagrange expression $2 C_+ T_{16}[a] \phi$ with C+ $T_{16}[a]$ invertible - but every invariant bilinear vanishes (massless, no $V(s)$), REFINEMENT (b) [has content]: the scope - the quartic Lorentz scalars $v.v, B.B, C.C$ of the Majorana spinor are NOT zero and are all proportional to ONE quartic (same monomials, one constant ratio each), REFINEMENT (b) [has content]: and the commuting shadow of the C+ Majorana spinor has NO kinetic term: for commuting components $\theta^T C_+ T_{16}[a] \phi$ is the total derivative $(1/2) d(\theta^T C_+ T_{16}[a] \theta)$.

(c) **Weyl and Majorana–Weyl 8-spinors (one chirality) cannot propagate.** Both C s are chirality-diagonal and γ^μ is chirality-odd, so $P_L C T_{16}[a] P_L = P_R C T_{16}[a] P_R = 0$. [VII] REFINEMENT (c) [THE RESULT]: Weyl and Majorana–Weyl: $PL C T_{16}[a] PL == PR C T_{16}[a] PR == 0$ for $C = C-, C+$ and every a (C chirality-diagonal, $T_{16}[a]$ chirality-odd) – no kinetic term.

(d) **Fermion fable is the complex 16-component spinor.** It is $\Psi = (\theta_1 + i\theta_2)/\sqrt{2}$, built from two real Grassmann 16-spinors, with the conjugate $\bar{\Psi} = \Psi^\dagger \sigma_{16}$. Here † is the conjugation of the complex Grassmann algebra; the Hilbert-space adjoint is fixed in the quantization, section 3.5.

- It has two scalar bilinears. $s = \bar{\Psi}\Psi = i\theta_1^T \sigma_{16} \theta_2$ is non-zero and Hermitian under † , and so is $p = \bar{\Psi} T_{16}[8] \Psi$. (With the Hilbert-space adjoint that the quantization fixes, p is odd under the fundamental symmetry J and becomes an anti-Hermitian operator, while s stays Hermitian.)
- The symmetrized kinetic term is Hermitian and genuine.
- It reduces **exactly** to Part VI’s real commuting fable: its Lagrangian, its field equation, and its covariant energy–momentum tensor. Through that reduction it reduces to the author’s L_a and eLa .

[VII] REFINEMENT (d) [THE RESULT]: $s = \text{Psibar Psi} == i \text{theta1}^T \text{sigma16 theta2}$: NON-ZERO (16 monomials) and HERMITIAN; $p = \text{Psibar T16}[8] \text{Psi}$ is non-zero and Hermitian too, REFINEMENT (d) [THE RESULT]: its symmetric part is $A_S == (i/2) \Omega$, $\Omega = [[0, K], [-K, 0]]$ real SYMMETRIC and invertible: a genuine kinetic term, REFINEMENT (d) [has content]: the symmetrized kinetic term is HERMITIAN and equals $\text{Psi}^{\text{ddag}} K \text{phi} - (1/2) d(\text{Psi}^{\text{ddag}} K \text{Psi})$; the unsymmetrized one is not Hermitian, FIDELITY [1] [fidelity]: $\text{Psi} \rightarrow \text{psi}$, $\text{Psibar} \rightarrow \text{psi}^T \text{sigma16}$ turns the symmetrized covariant L_{hat} into Part VI’s cfLhatFable EXACTLY (canonical frame, generic V), FIDELITY [2] [fidelity]: on the flat frame with $V = -(2M/H) s$ and no volume factor, the same limit on Psi16 IS the author’s L_a of Section 11, FIDELITY [3] [fidelity]: $\text{sigma16} \cdot \text{EL_Psibar} + \text{EL_Psi}$ in the real limit == Part VI’s explicit variation cfELFable , component for component, FIDELITY [3] [fidelity]: and on the flat frame with $V = -(2M/H) s$ the same combination IS the author’s eLa of Section 12, FIDELITY [4] [fidelity]: the energy-momentum tensor That in the real limit IS Part VI’s covariant tensor cfTCov , all 64 components.

The claim must be stated with its scope. The complex 16-spinor is the **minimal** field that:

1. uses the author’s σ_{16} conjugation;
2. propagates;
3. admits a potential $V(s)$;
4. reduces to Part VI’s classical fable as its real commuting shadow.

It is **not** the unique field with dynamics: a Majorana spinor with C_+ also propagates. The author chose the complex 16-spinor on 2026-09-24.

One more consequence matters for everything that follows. Part VI’s mechanism for the spin connection dropping out of the Lagrangian was $\Psi^T \sigma_{16} T_{16}[a] \Psi = 0$, which is a property of commuting components. It does not carry over to complex Grassmann fields. There, the drop of the connection is a property of the matrix $\{\gamma^\mu, \Gamma_\mu^{\text{spin}}\}$ itself, and it has to be proved again. Whether the classical results of this section, among them w and the phantom crossing, survive

quantization is decided in sections 3.6 and 3.7. The complete refinement, with every assertion of Section 26, is section 3.2.

3.2 The refinement: why fable must be a complex 16-spinor

This is Section 26 of the notebook (manifest `claude-fable/cells_part7.wl`, Input cells 173–179, 38 assertions). It turns the classical, real, commuting fable of section 3.1.12 into a fermion.

Two conjugations, two symbols. $\Psi^\dagger$ is the **classical** conjugate of the field, the conjugation of the Grassmann algebra, the one that appears in the Lagrangian; the Dirac conjugate is $\bar{\Psi} = \Psi^\dagger \sigma_{16}$. After quantization (section 3.5) the Hilbert-space adjoint is a **different** operation, $\Psi^\dagger = \Psi^\dagger J$. The two are never written with the same symbol [**prose VII, introduction of Part VII**].

Two kinds of algebra, and when each is used [**prose VII**]. Statements whose content *is* the anticommutation of the field (that a mass term or a kinetic term vanishes, is Hermitian, or is a total derivative) are proved in a genuine Grassmann (exterior) algebra, built and tested in Section 26 (subsection 3.2.3). Statements about **bilinear** expressions (Lagrangians, Euler–Lagrange equations, currents, energy–momentum tensors, conservation laws) are proved with Ψ and $\bar{\Psi}$ represented by two **independent** vectors of ordinary functions of the coordinates, with $\bar{\Psi}$ always written to the **left** of Ψ . That commuting proxy is exact for such identities: in every term $\bar{\Psi}$ is the leftmost and Ψ the rightmost odd factor, so the left derivative with respect to $\bar{\Psi}$ and the right derivative with respect to Ψ are the ordinary derivatives, and no reordering of odd factors ever occurs. Where reality matters (Hermiticity under †), $\Psi = u + iv$ with real u, v and $\bar{\Psi} = (u - iv)^T \sigma_{16}$.

3.2.1 The Clifford symmetry table

A spinor bilinear $\psi^T C \Gamma \psi$ is built from a “charge conjugation” matrix C and one of the 256 ordered products Γ of the Dirac matrices (Section 6’s basis `base16`, grouped by the rank r , the number of factors). There are two candidates for C :

$$\begin{aligned} C_- &= \sigma_{16}, & C_- T_{16}[a] C_-^{-1} &= -T_{16}[a]^T & \text{(the author’s spinor metric),} \\ C_+ &= \sigma_{16} T_{16}[8], & C_+ T_{16}[a] C_+^{-1} &= +T_{16}[a]^T. \end{aligned}$$

Every product $C\Gamma$ is either symmetric (S) or antisymmetric (A), and the answer depends only on the rank:

rank r	0	1	2	3	4	5	6	7	8
number of products, $\binom{8}{r}$	1	8	28	56	70	56	28	8	1
$C_- \Gamma = \sigma_{16} \Gamma$	S	A	A	S	S	A	A	S	S
$C_+ \Gamma = \sigma_{16} T_{16}[8] \Gamma$	S	S	A	A	S	S	A	A	S

For each C , 136 products are symmetric and 120 antisymmetric, as it must be for 16×16 matrices ($16 \cdot 17/2 = 136$, $16 \cdot 15/2 = 120$). The “four candidate” mass matrices σ_{16} , C_+ , $\sigma_{16} T_{16}[8]$, $C_+ T_{16}[8]$ are only two, because $T_{16}[8]^2 = \text{ID}_{16}$: $\sigma_{16} T_{16}[8] = C_+$ and $C_+ T_{16}[8] = \sigma_{16}$. Why this table decides everything: for **commuting** components $\psi^T N \psi$ sees only the symmetric part of N ; for **anticommuting** (Grassmann) components only the antisymmetric part.

[**proved VII §26**]

- **REFINEMENT** [definition]: `T16[8]^2 == sigma16^2 == C+^2 == ID16`, C_+ is symmetric, and $C_+ \cdot T_{16}[8] == \text{sigma16}$ – the four candidate mass matrices are only two

- REFINEMENT [has content]: $C- T_{16}[a] C^{-1} == -T_{16}[a]^T$ and $C+ T_{16}[a] C^{+1} == +T_{16}[a]^T$ for $a = 0..7$
- REFINEMENT [THE RESULT]: the symmetry of C Gamma depends only on the rank r of Gamma: $C-: S A A S S A A S S$, $C+: S S A A S S A A S$ ($r = 0..8$) (the assertion also checks that the counts per rank are $\binom{8}{r}$)
- REFINEMENT [has content]: for each C , 136 of the 256 products are symmetric and 120 antisymmetric

3.2.2 The Lorentz-invariant bilinears: exactly two, both symmetric

A bilinear $\psi^T N \psi$ (real field) or $\Psi^\dagger N \Psi$ (complex field) is a $\text{Spin}(4,4)$ scalar exactly when $S^T N + N S = 0$ for all 28 generators $S = \text{SAB}$ of Section 5 (the generators are real, so the condition is the same in both cases). Solving these 7168 linear equations (28×256) for the 256 entries of N gives a **two**-dimensional space, spanned by C_- and C_+ , equivalently by the two chiral pieces $P_L \sigma_{16} P_L$ and $P_R \sigma_{16} P_R$, one for each split-octonion spinor type. Every invariant N is **symmetric** and **chirality-diagonal** (it commutes with $T_{16}[8]$). That is the complete list of candidates for a mass term; nothing is left out [**prose VII §26**].

[**proved VII §26**]

- REFINEMENT [THE RESULT]: the $\text{Spin}(4,4)$ -invariant bilinear forms on the 16-spinor form a 2-dimensional space, spanned by $C- = \text{sigma16}$ and $C+ = \text{sigma16 } T_{16}[8]$
- REFINEMENT [has content]: every invariant form is **SYMMETRIC** and **chirality-diagonal**, and the two chiral pieces $PL \text{ sigma16 } PL$ and $PR \text{ sigma16 } PR$ are invariant separately

3.2.3 A Grassmann algebra, built and tested

To prove statements about anticommuting fields the notebook uses an honest exterior algebra, not a sign convention applied by hand. An element is an **Association** from **monomials** to coefficients; a monomial is the sorted list of the generator indices it contains, $\{\}$ being the unit. The product of two monomials is zero if they share a generator; otherwise it is the sorted union with the sign of the permutation that sorts the concatenation (**Signature**). The operations are [**prose VII §26**]:

operation	what it computes
<code>grAdd, grScale, grMul</code>	sums, scalar multiples and products
<code>grSame[x, y]</code>	equality of two elements ($x - y$ has no monomial left)
<code>grBil[xs, N, ys]</code>	the bilinear $\sum_{ab} x_a N_{ab} y_b$ of two lists of elements
<code>grD[x, dmap]</code>	the even derivation that maps generator i to generator <code>dmap[i]</code> (with $\theta_a \mapsto \phi_a$ it is d/dt , with $\phi_a = d\theta_a/dt$)
<code>grDL[x, i]</code>	the left derivative with respect to generator i
<code>grConj[x]</code>	the classical conjugation † : $(xy)^\dagger = y^\dagger x^\dagger$, generators self-conjugate (real), coefficients complex-conjugated
<code>grEL[L, thIdx, phIdx, dmap]</code>	the Euler–Lagrange expression $\partial L / \partial \theta_c - d(\partial L / \partial \phi_c) / dt$, with left derivatives

Before any physics the algebra is tested on facts known independently (random elements drawn with `SeedRandom[20260924]`). [**proved VII §26**]

- GRASSMANN [solver regression]: generators anticommute and square to zero (all pairs of 1..6)

- GRASSMANN [solver regression]: the product is associative, $(x y) z == x (y z)$, on elements of degree 1, 2, 3 (and the products tested are not zero)
- GRASSMANN [solver regression]: graded commutativity $x y == (-1)^{(|x| |y|)} y x$ (degrees 1, 2, 3), and $x^2 == 0$ for x odd
- GRASSMANN [solver regression]: the conjugation is an involutive anti-automorphism, $(x y)^{\text{ddag}} == y^{\text{ddag}} x^{\text{ddag}}$, $(x^{\text{ddag}})^{\text{ddag}} == x$
- GRASSMANN [solver regression]: grD is an even derivation, $d(x y) == d(x) y + x d(y)$; grDL obeys $d_i(x y) == d_i(x) y + (-1)^{|x|} x d_i(y)$

Then the two lemmas that drive the section are proved for a completely general, symbolic 16×16 matrix N (N_A , N_S its antisymmetric and symmetric parts, θ sixteen real Grassmann generators, $\phi = d\theta/dt$ sixteen more):

$$\begin{aligned}\theta^T N \theta &= \theta^T N_A \theta \quad (\text{and } \theta^T N_S \theta = 0), \\ \theta^T N \phi - \frac{1}{2} d(\theta^T N \theta) &= \theta^T N_S \phi,\end{aligned}$$

Euler–Lagrange expression of $L = \theta^T N \phi = (N + N^T) \phi = 2N_S \phi$.

So a first-order Grassmann kinetic term is a **total derivative**, with empty field equations, precisely when its matrix is antisymmetric. For **commuting** variables the roles are exactly reversed (the control). [**proved VII §26**]

- GRASSMANN [THE RESULT]: $\theta^T N \theta == \theta^T N_A \theta$ and $\theta^T N_S \theta == 0$ for a general 16×16 N - an anticommuting bilinear sees only the ANTISYMMETRIC part
- GRASSMANN [THE RESULT]: $\theta^T N \phi - (1/2) d(\theta^T N \theta) == \theta^T N_S \phi$ for a general N
- GRASSMANN [THE RESULT]: the Euler–Lagrange expression of $L = \theta^T N \phi$ (left derivatives) is $(N + N^T) \phi$: EMPTY field equations exactly when N is antisymmetric
- GRASSMANN [control]: for COMMUTING components the roles are reversed - $\theta^T N \theta$ sees only N_S , and the Euler–Lagrange expression of $\theta^T N \phi$ is $2 N_A \phi$

3.2.4 The real options, and why each fails

With the table and the lemmas the verdicts are immediate, and each is proved in the Grassmann algebra.

(a) A real Grassmann 16-spinor with the author’s σ_{16} has no dynamics, on any frame.

Its only invariant mass terms vanish, $\theta^T \sigma_{16} \theta = \theta^T C_+ \theta = 0$ (both matrices are symmetric). In every direction its flat kinetic term $\theta^T \sigma_{16} T_{16}[a] d\theta$ is the total derivative $\frac{1}{2} d(\theta^T \sigma_{16} T_{16}[a] \theta)$ ($\sigma_{16} T_{16}[a]$ is antisymmetric), with an identically empty Euler–Lagrange expression; the total derivative itself is not zero. On a curved frame the covariant kinetic term is

$$\sqrt{g} \theta^T \sigma_{16} \gamma^\mu D_\mu \theta = (\text{total derivative}) + \frac{1}{2} \sqrt{g} \theta^T \sigma_{16} \{\gamma^\mu, \Gamma_\mu\} \theta,$$

because $\partial_\mu(\sqrt{g} \sigma_{16} \gamma^\mu) = \sqrt{g} \sigma_{16} [\gamma^\mu, \Gamma_\mu]$; and the second term vanishes on **every** frame, because $\{T_{16}[a], S^{bc}\}$ is a rank-3 product (or zero) and σ_{16} times a rank-3 product is symmetric. [**proved VII §26**]

- REFINEMENT (a) [THE RESULT]: real Grassmann + sigma16: both invariant mass terms vanish, and $\theta^T \sigma_{16} T_{16}[a] \phi = (1/2) d(\theta^T \sigma_{16} T_{16}[a] \theta)$ for every a (a total derivative)
- REFINEMENT (a) [has content]: its Euler-Lagrange expression is identically empty in every direction - while the kinetic term itself is NOT zero
- REFINEMENT (a) [THE RESULT]: FRAME-INDEPENDENT - sigma16 {T16[a], S^{bc}} is SYMMETRIC for all a, b, c (so $\theta^T \sigma_{16} \{\gamma^\mu, \Gamma_\mu\} \theta = 0$ on every frame), and $d_\mu(\sqrt{g} \sigma_{16} \gamma^\mu) = \sqrt{g} \sigma_{16} [\gamma^\mu, \Gamma_\mu]$ on the canonical frame

(b) **A Majorana 16-spinor built with C_+ propagates, but is massless and has no $V(s)$.** Now $C_+ T_{16}[a]$ is symmetric, so the kinetic term is genuine: its Euler-Lagrange expression is $2C_+ T_{16}[a] \phi$, and $C_+ T_{16}[a]$ is invertible. But both invariant forms are symmetric, so there is **no** Lorentz-scalar bilinear: no mass, no s , no $V(s)$. Its quartic Lorentz scalars are not zero, and they are all **one** scalar: with the vector $v_a = \theta^T \sigma_{16} T_{16}[a] \theta$, the 2-form $B_{ab} = \theta^T C_+ T_{16}[a] T_{16}[b] \theta$ and the 3-form $C_{abc} = \theta^T C_+ T_{16}[a] T_{16}[b] T_{16}[c] \theta$, the contractions $v \cdot v$, $B \cdot B$ and $C \cdot C$ are proportional to each other (same monomials, one constant ratio each). Its commuting shadow has no kinetic term ($C_+ T_{16}[a]$ symmetric is a total derivative for commuting components). [**proved VII §26**]

- REFINEMENT (b) [THE RESULT]: Majorana with C_+ : the kinetic term is genuine - Euler-Lagrange expression $2 C_+ T_{16}[a] \phi$ with $C_+ T_{16}[a]$ invertible - but every invariant bilinear vanishes (massless, no $V(s)$)
- REFINEMENT (b) [has content]: the scope - the quartic Lorentz scalars $v \cdot v$, $B \cdot B$, $C \cdot C$ of the Majorana spinor are NOT zero and are all proportional to ONE quartic (same monomials, one constant ratio each)
- REFINEMENT (b) [has content]: and the commuting shadow of the C_+ Majorana spinor has NO kinetic term: for commuting components $\theta^T C_+ T_{16}[a] \phi$ is the total derivative $(1/2) d(\theta^T C_+ T_{16}[a] \theta)$

(c) **A Weyl or Majorana-Weyl 8-spinor (one chirality, P_L or P_R) cannot propagate.** Both invariant forms are chirality-**diagonal** and every $T_{16}[a]$ is chirality-**off**-diagonal, so $P_L C T_{16}[a] P_L = P_R C T_{16}[a] P_R = 0$: no kinetic term at all. [**proved VII §26**]

- REFINEMENT (c) [THE RESULT]: Weyl and Majorana-Weyl: $P_L C T_{16}[a] P_L = P_R C T_{16}[a] P_R = 0$ for $C = C_-, C_+$ and every a (C chirality-diagonal, $T_{16}[a]$ chirality-odd) - no kinetic term

(d) **Why Part VI's real commuting fable escapes (a), and why it cannot be quantized as a fermion.** For commuting components the symmetric σ_{16} gives a non-zero mass term and the antisymmetric $\sigma_{16} T_{16}[a]$ gives a genuine kinetic term (the control of subsection 3.2.3). That is why Part VI's classical theory is consistent, and why it cannot be quantized as a fermion: a commuting spinor violates the spin-statistics connection [**prose VII §26**].

3.2.5 The complex 16-spinor

Take two real Grassmann 16-spinors θ_1, θ_2 (32 real generators) and

$$\Psi = \frac{\theta_1 + i \theta_2}{\sqrt{2}}, \quad \Psi^\dagger = \frac{(\theta_1 - i \theta_2)^T}{\sqrt{2}}, \quad \bar{\Psi} = \Psi^\dagger \sigma_{16}.$$

Now everything the real options lacked is present:

- $s = \bar{\Psi}\Psi = i\theta_1^T\sigma_{16}\theta_2$ is **non-zero** (16 monomials) and **Hermitian** (real under $\dagger$). The second invariant $p = \bar{\Psi}T_{16}[8]\Psi = \Psi^\dagger C_+\Psi$ is non-zero and Hermitian as well: the complex 16-spinor has **two** scalar bilinears.
- s is a sum of 16 **commuting nilpotent** even elements x_a ($x_a^2 = 0$), so every polynomial $V(s)$ is a finite polynomial: $s^{17} = 0$.
- The kinetic term is genuine: $\Psi^\dagger K\phi = \Theta^T A \Phi$ with $\Theta = (\theta_1, \theta_2)$, $\Phi = d\Theta$, $K = \sigma_{16}T_{16}[a]$, and the symmetric part of the 32×32 matrix A is $A_S = \frac{i}{2} \begin{pmatrix} 0 & K \\ -K & 0 \end{pmatrix} =: \frac{i}{2} \Omega$, which is **not** zero. Ω is real symmetric and invertible (it is the symplectic matrix of section 3.5.3).
- The symmetrized kinetic term $\frac{1}{2}(\Psi^\dagger K\phi - \phi^\dagger K\Psi)$ is **Hermitian**, and it differs from $\Psi^\dagger K\phi$ by the total derivative $-\frac{1}{2}d(\Psi^\dagger K\Psi)$; the unsymmetrized one is **not** Hermitian.

Invariance of s and p under $\text{Spin}(4,4)$ is the statement $S^T N + N S = 0$ of subsection 3.2.2 (the Lorentz generators are real). The self-interaction is taken to depend on s only; section 3.5.7 shows why p cannot enter. **[proved VII §26]**

- REFINEMENT (d) [definition]: `Psi^ddag == (theta1 - i theta2)^T/Sqrt[2]` in the algebra
- REFINEMENT (d) [THE RESULT]: `s = Psibar Psi == i theta1^T sigma16 theta2`: NON-ZERO (16 monomials) and HERMITIAN; `p = Psibar T16[8] Psi` is non-zero and Hermitian too
- REFINEMENT (d) [has content]: `s` is a sum of 16 commuting nilpotent even elements `x_a` (`x_a x_b == x_b x_a`, `x_a^2 == 0`), so `s^17 == 0` and `V(s)` is a finite polynomial
- REFINEMENT (d) [definition]: the 32×32 matrix A read off the algebra reproduces `Psi^ddag K phi` exactly
- REFINEMENT (d) [THE RESULT]: its symmetric part is `A_S == (i/2) Omega`, `Omega = [[0, K], [-K, 0]]` real SYMMETRIC and invertible: a genuine kinetic term
- REFINEMENT (d) [has content]: the symmetrized kinetic term is HERMITIAN and equals `Psi^ddag K phi - (1/2) d(Psi^ddag K Psi)`; the unsymmetrized one is not Hermitian
- REFINEMENT (d) [has content]: the same holds in every direction `a = 0..7`: the symmetrized `Psi^ddag sigma16 T16[a] phi` is Hermitian, with symmetric part `(i/2)[[0, sigma16 T16[a]], [-sigma16 T16[a], 0]] != 0`

3.2.6 Four four-dimensional Dirac fermions

The observer's Clifford algebra is generated by $T_{16}[1], T_{16}[2], T_{16}[3]$ (the observed space) and $T_{16}[4]$ (the observer's time). Its 16 ordered products are linearly independent, so over the complex numbers it is the full matrix algebra $M_4(\mathbb{C})$, the algebra of 4×4 Dirac matrices. Its **commutant** in $M_{16}(\mathbb{C})$, computed by solving $XT_{16}[i] = T_{16}[i]X$ ($i = 1..4$), is again 16-dimensional, generated by the four "hidden" (flavour) gammas

$$h_A = \Omega_4 T_{16}[A], \quad A = 0, 5, 6, 7, \quad \Omega_4 = T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4], \quad \Omega_4^2 = -\text{ID}16, \\ \{h_A, h_B\} = -2\eta_{AB} \text{ID}16,$$

which commute with $T_{16}[1..4]$: a second Clifford algebra, of the hidden directions. The two algebras meet only in the multiples of the identity (the rank of their union is $16 + 16 - 1 = 31$). By the double-commutant theorem

$$\mathbb{C}^{16} = \mathbb{C}^4 \text{ (the observer's Dirac index)} \otimes \mathbb{C}^4 \text{ (the hidden flavour index)}.$$

Under the observer's Lorentz group $\text{Spin}(3,1)$, generated by $T_{16}[i]T_{16}[j]$ with i, j in $\{1, 2, 3, 4\}$, fermion fable is **four 4-dimensional Dirac fermions**. Per momentum that is $4 \times 2 = 8$ particle states (and 8 antiparticle states): the degeneracy $g = 8$ of sections 3.5 and 3.7. [proved VII §26]

- 4D [THE RESULT]: the 16 ordered products of $T_{16}[1..4]$ are linearly independent (the observer's algebra is $M_4(\mathbb{C})$), and its commutant in $M_{16}(\mathbb{C})$ is 16-dimensional
- 4D [has content]: $\Omega^4 = -\text{ID}_{16}$, the hidden gammas $h_A = \Omega^4 T_{16}[A]$ ($A = 0, 5, 6, 7$) commute with $T_{16}[1..4]$ and obey $\{h_A, h_B\} = -2 \eta_{AB}$
- 4D [THE RESULT]: the 16 products of the hidden gammas span the commutant, and the two algebras meet only in the multiples of ID_{16} (rank of the union $16 + 16 - 1 = 31$): $C^{16} = C^4(x) C^4$, four 4D Dirac fermions
- 4D [has content]: the observer's Lorentz generators $T_{16}[i] T_{16}[j]$ ($i < j$ in $1..4$) commute with every hidden gamma

3.2.7 The refined Lagrangian, and the fidelity chain to Part VI, L_a and eLa

The Lagrangian of fermion fable is the **symmetrized covariant** one (section 3.3 shows why no other form is admissible for a complex field):

$$L = \sqrt{\det g} \hat{L}, \quad \hat{L} = \frac{1}{2H} \left[\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi \right] - V(s), \quad s = \bar{\Psi} \Psi,$$

$$D_\mu \Psi = \partial_\mu \Psi + \Gamma_\mu^{\text{spin}} \Psi, \quad D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi} \Gamma_\mu^{\text{spin}},$$

and its energy-momentum tensor (section 3.6) is

$$\hat{T}_{\mu\nu} = -\frac{1}{4H} \left[\bar{\Psi} \gamma_\mu D_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi + (\mu \leftrightarrow \nu) \right] + g_{\mu\nu} \hat{L}.$$

Four helpers implement them for an arbitrary frame (curved gammas and spin connection passed in): `cfLhatFermion`, `cfEquationFermion` ($\gamma^\mu D_\mu \Psi - HV'(s)\Psi$), `cfAdjointEquationFermion` ($(D_\mu \bar{\Psi}) \gamma^\mu + HV'(s)\bar{\Psi}$) and `cfTFermion`. Before they are used for anything new they are anchored to what the notebook already knows. Set $\Psi \rightarrow \psi$ (a real commuting spinor) and $\bar{\Psi} \rightarrow \psi^T \sigma_{16}$ (the “bosonic shadow”). Then:

1. $\hat{L}$ becomes Part VI's `cfLhatFable` **exactly**, on the canonical frame, with generic V ;
2. on the flat frame with $V = -(2M/H)s$ and no volume factor, it becomes the author's `La[]` of Section 11 (section 3.1.7);
3. the Euler-Lagrange expressions of the complex field, combined as $\sigma_{16} \cdot \text{EL}_{\bar{\Psi}} + \text{EL}_\Psi$ (the chain rule for $\bar{\Psi} = \psi^T \sigma_{16}$), become Part VI's explicit variation `cfELFable`, and on the flat frame the author's sixteen equations `eLa` of Section 12 (section 3.1.8);
4. $\hat{T}$ becomes Part VI's covariant tensor `cfTCov` (section 3.1.12.3), all 64 components.

[proved VII §26]

- FIDELITY [1] [fidelity]: `Psi -> psi, Psibar -> psi^T sigma16` turns the symmetrized covariant `Lhat` into Part VI's `cfLhatFable` **EXACTLY** (canonical frame, generic V)
- FIDELITY [2] [fidelity]: on the flat frame with $V = -(2M/H)s$ and no volume factor, the same limit on `Psi16` IS the author's `La[]` of Section 11
- FIDELITY [3] [fidelity]: `sigma16 . EL_Psibar + EL_Psi` in the real limit == Part VI's explicit variation `cfELFable`, component for component

- FIDELITY [3] [fidelity]: and on the flat frame with $V = -(2M/H)$ s the same combination IS the author's eLa of Section 12
- FIDELITY [4] [fidelity]: the energy-momentum tensor That in the real limit IS Part VI's covariant tensor cftCov, all 64 components

Every new result of this effort therefore contains Part VI, and through it the author's L_a and eLa, as its real commuting limit.

3.2.8 The conclusion, with its scope

The complex 16-spinor is the **minimal** field that

1. uses the author's σ_{16} conjugation,
2. propagates,
3. admits a potential $V(s)$ with $s = \bar{\Psi}\Psi$, and
4. reduces to Part VI's classical fable as its real commuting shadow.

It is **not** the unique field with dynamics: the C_+ Majorana spinor propagates too, but it is massless, has no bilinear $V(s)$, and has no commuting shadow with a kinetic term. The statement of minimality is the conclusion of Section 26 as a whole; no single assertion states it. It rests on the assertions of subsections 3.2.1–3.2.7 [**prose VII §26, assembled from the assertions quoted**].

3.3 The Lagrangian and its Hermiticity

This is the first cell of Section 27 (Input cell 180, 10 assertions). From here on Ψ and $\bar{\Psi}$ are two independent 16-component fields of **all eight** coordinates (the commuting proxy of section 3.2, $\bar{\Psi}$ always on the left).

3.3.1 The four candidates

Four candidate Lagrangians differ by where the derivative acts and whether it is covariant:

$$L_{\text{sym}} = \frac{1}{2H} [\bar{\Psi}\gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi})\gamma^\mu \Psi] - V(s) \quad (\text{the Lagrangian of fermion fable}),$$

$$L_{\text{sym},\partial} = \text{the same with } \partial_\mu \text{ in place of } D_\mu,$$

$$L_{\text{un}} = \frac{1}{H} \bar{\Psi}\gamma^\mu D_\mu \Psi - V(s),$$

$$L_{\text{un},\partial} = \frac{1}{H} \bar{\Psi}\gamma^\mu \partial_\mu \Psi - V(s),$$

with $D_\mu \Psi = \partial_\mu \Psi + \Gamma_\mu \Psi$, $D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi}\Gamma_\mu$, and $\Gamma_\mu = \text{GammaSpinCanonical}[[\mu+1]]$, the canonical spin connection of section 3.8. (The notebook's labels call them L_{sym} , L_{sym_d} , L_{un} and L_{un_d} .) The action is $S = \int d^8x \sqrt{\det g} L_{\text{sym}}$, with $\sqrt{\det g} = \sec 6Hx_0$ on the canonical frame. There is **no factor** i in front of the kinetic term, as in the author's L_a : section 3.3.2(b) shows that none is needed.

3.3.2 The facts, each asserted on the canonical frame for generic $\Psi(x_0..x_7)$, $\bar{\Psi}(x_0..x_7)$

(a) **The connection is real and $\sigma_{16}\Gamma_\mu$ is antisymmetric**, i.e. $\Gamma_\mu^T \sigma_{16} = -\sigma_{16} \Gamma_\mu$. So $D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi}\Gamma_\mu$ is exactly the Dirac conjugate of $D_\mu \Psi$. [**proved VII §27**] LAGRANGIAN (a) [definition]: every $\text{Gamma}^\wedge\text{spin}_\mu$ is real and sigma16 Gamma_μ is antisymmetric ($\text{Gamma}_\mu^T \text{sigma16} == -\text{sigma16 Gamma}_\mu$): $D_\mu \text{Psibar}$ is the Dirac conjugate of $D_\mu \text{Psi}$

(b) L_{sym} is **Hermitian**; L_{un} is **not**. With $\Psi = u + iv$ and $\bar{\Psi} = (u - iv)^T \sigma_{16}$, L_{sym} is real. No factor i is needed, because $\sigma_{16} \gamma^\mu$ is real and antisymmetric: $(\Psi^\dagger K d\Psi)^\dagger = -(d\Psi^\dagger) K \Psi$ for real antisymmetric K , so the symmetrized combination is real. L_{un} is not Hermitian: its imaginary part is not zero (a witness). **[proved VII §27]**

- LAGRANGIAN (b) [THE RESULT]: L_{sym} is HERMITIAN - real for $\text{Psi} = u + i v$, $\text{Psibar} = (u - i v)^T \text{sigma16}$, generic u, v of all eight coordinates
- LAGRANGIAN (b) [control]: L_{un} is NOT Hermitian - its imaginary part is not zero, witnessed on the constant spinors $u = e1$, $v = e13$ (with $V = s^2$)

(c) L_{sym} contains the connection only through $\{\gamma^\mu, \Gamma_\mu\}$, and that matrix vanishes for each μ . Expanding D ,

$$L_{\text{sym}} - L_{\text{sym},\partial} = \frac{1}{2H} \sum_{\mu} \bar{\Psi} \{\gamma^\mu, \Gamma_\mu\} \Psi \quad (\text{identically, off shell}).$$

On the canonical frame $\{\gamma^\mu, \Gamma_\mu\} = 0$ for **each** μ separately: every Γ_μ is a combination of $T_{16}[\mu]T_{16}[0]$ and $T_{16}[\mu]T_{16}[4]$, with no totally antisymmetric part (section 3.8.8). So $L_{\text{sym}} = L_{\text{sym},\partial}$ there: the connection drops out of the symmetrized Lagrangian on every frame on which the **matrix** $\{\gamma^\mu, \Gamma_\mu\}$ vanishes (diagonal frames), and **not** in general. **[proved VII §27]** LAGRANGIAN (c) [THE RESULT]: $L_{\text{sym}} - L_{\text{sym}_d} == (1/(2H)) \text{Sum_mu Psibar \{gamma^mu, Gamma_mu\} Psi}$ identically, and $\{\text{gamma}^{\text{mu}}, \text{Gamma_mu}\} == 0$ for EACH μ on the canonical frame: $L_{\text{sym}} == L_{\text{sym}_d}$

(d) L_{un} and L_{sym} differ by a total divergence. **[proved VII §27]** LAGRANGIAN (d) [has content]: $L_{\text{un}} - L_{\text{sym}} == (1/(2H)) (1/\text{Sqrt}[g]) d_mu(\text{Sqrt}[g] \text{Psibar gamma}^{\text{mu}} \text{Psi})$, a total divergence

(e) But L_{un} and $L_{\text{un},\partial}$ differ by a term that is not zero for a complex field.

$$L_{\text{un}} - L_{\text{un},\partial} = \frac{1}{H} \bar{\Psi} \gamma^\mu \Gamma_\mu \Psi = -3 \cot^2(6Hx_0) \bar{\Psi} T_{16}[0] \Psi.$$

This vanishes only in the real commuting limit, which is the scope of the statement “ $\hat{L}$ with $D = \hat{L}$ with ∂ ” of Parts V and VI (section 3.1.12.1). **[proved VII §27]**

- LAGRANGIAN (e) [has content]: $L_{\text{un}} - L_{\text{un}_d} == (1/H) \text{Psibar gamma}^{\text{mu}} \text{Gamma_mu Psi} == -3 \text{Cot}[6 H x_0]^2 \text{Psibar T16}[0] \text{Psi}$
- LAGRANGIAN (e) [control]: that term is NOT zero for a complex field (witness $u = e1$, $v = e13$), and IS zero in the real commuting limit (the scope of Parts V-VI)

(f) So $L_{\text{un},\partial}$ is **inadmissible**. Varying $\bar{\Psi}$ gives $\gamma^\mu \partial_\mu \Psi = HV' \Psi$ with the connection term **missing**. Varying Ψ gives an adjoint equation that **contains** it, $(\partial_\mu \bar{\Psi}) \gamma^\mu + \bar{\Psi} [\gamma^\mu, \Gamma_\mu] = -HV' \bar{\Psi}$. The two are **not** Dirac conjugates of each other: they differ by

$$\bar{\Psi} [\gamma^\mu, \Gamma_\mu] = 2 \bar{\Psi} \gamma^\mu \Gamma_\mu = -6H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0]$$

(a witness). Only a Lagrangian whose two variations are conjugate defines a consistent theory. **[proved VII §27]**

- LAGRANGIAN (f) [THE RESULT]: L_{un_d} varied with respect to Psibar gives $(\text{Sqrt}[g]/H) (\text{gamma}^{\text{mu}} d_mu \text{Psi} - H V' \text{Psi})$: the connection term is MISSING

- LAGRANGIAN (f) [THE RESULT]: varied with respect to Psi it gives $-(\text{Sqrt}[g]/H)((d_\mu \text{Psibar}) \gamma^\mu + \text{Psibar} [\gamma^\mu, \Gamma_\mu] + H V' \text{Psibar})$, with $\text{Psibar} [\gamma^\mu, \Gamma_\mu] == 2 \text{Psibar} \gamma^\mu \Gamma_\mu$: the connection term is PRESENT
- LAGRANGIAN (f) [control]: so Lun_d is INADMISSIBLE - its two field equations are not Dirac conjugates: they differ by $2 \text{Psibar} \gamma^\mu \Gamma_\mu = -6 H \cot^2 \text{Psibar} T_{16}[0]$, non-zero (witness $u = e_1, v = 0, V = s^2$)

3.3.3 Why the connection drops out, and why this is not Part VI's mechanism

For the real commuting field of Part VI the connection dropped out because $\Psi^T \sigma_{16} T_{16}[a] \Psi = 0$ for commuting components ($\sigma_{16} T_{16}[a]$ antisymmetric; section 3.1.11). For a complex field that bilinear identity is false: $\bar{\Psi} T_{16}[0] \Psi = \Psi^\dagger (\sigma_{16} T_{16}[0]) \Psi$ with $\sigma_{16} T_{16}[0]$ real antisymmetric is i times the Hermitian form of $-i \sigma_{16} T_{16}[0]$, which is not zero in general [**derived here**]. The drop-out of the connection from L_{sym} is therefore a property of the **matrix** $\{\gamma^\mu, \Gamma_\mu\}$ (fact (c)), not of a bilinear identity. The general identity behind it is $\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = \frac{1}{2} \omega_{[cab]} \gamma^{cab}$: only the totally antisymmetric part of the connection can enter the symmetrized Lagrangian (section 3.8.8).

Hermiticity rests on two facts: $\sigma_{16} \gamma^\alpha$ and $\sigma_{16} \Gamma_\mu$ are **real antisymmetric**. The first is Part I's `sigma16 . T16[A]` is antisymmetric for $A = 0..7$; the second is fact (a).

3.4 The field equations in the primordial gravitational field, written out

This is Section 27 of the notebook (Input cells 181, 182 and 184) and item (2) of the pre-universe cell of Section 29 (Input cell 197). The **primordial gravitational field** is the author's canonical frame of section 3.1.11:

$$\begin{aligned}
\text{frameCanonical} &= \text{diag}(\tan 6Hx_0, q, q, q, 1, p, p, p), \\
q &= \frac{e^{-a_4(Hx_4)}}{\sin^{1/6}(6Hx_0)}, \quad p = \frac{e^{+a_4(Hx_4)}}{\sin^{1/6}(6Hx_0)} \quad (a_4 \text{ free, never given a value}), \\
ds^2 &= \tan^2(6Hx_0) dx_0^2 + q^2(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - p^2(dx_5^2 + dx_6^2 + dx_7^2), \\
\sqrt{|\det g|} &= \sec 6Hx_0, \\
\gamma^0 &= \cot(6Hx_0) T_{16}[0], \quad \gamma^i = \frac{1}{q} T_{16}[i] \quad (i = 1, 2, 3), \\
\gamma^4 &= T_{16}[4], \quad \gamma^h = \frac{1}{p} T_{16}[h] \quad (h = 5, 6, 7),
\end{aligned}$$

on $0 < 6Hx_0 < \pi/2$. The spin connection enters through $\gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) T_{16}[0]$ (section 3.8.8).

3.4.1 The derivation

Vary $\bar{\Psi}$ and Ψ independently in $L = \sqrt{g} L_{\text{sym}}$ (the left derivative with respect to $\bar{\Psi}$ and the right derivative with respect to Ψ are the ordinary derivatives of the commuting proxy). For generic fields of all eight coordinates:

$$\begin{aligned}
\text{EL}_{\bar{\Psi}} &= \frac{\sqrt{g}}{H} (\gamma^\mu D_\mu \Psi - H V'(s) \Psi), \\
\text{EL}_\Psi &= -\frac{\sqrt{g}}{H} ((D_\mu \bar{\Psi}) \gamma^\mu + H V'(s) \bar{\Psi}),
\end{aligned}$$

so the field equation and the adjoint field equation of fermion fable are

$$\boxed{\gamma^\mu D_\mu \Psi = H V'(s) \Psi, \quad (D_\mu \bar{\Psi}) \gamma^\mu = -H V'(s) \bar{\Psi}}$$

The **sign** of the adjoint equation is derived, not assumed, and it is exactly the one consistency requires: with $\Psi = u + iv$ and $\bar{\Psi} = (u - iv)^T \sigma_{16}$, the adjoint equation is the Dirac conjugate of the field equation, (adjoint residual) = $-(\text{field residual})^* \cdot \sigma_{16}$, because $T_{16}[a]^T \sigma_{16} = -\sigma_{16} T_{16}[a]$. The opposite sign fails that test (control). The derivation uses only the covariant constancy of γ^μ , in the form $\frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu) = [\gamma^\mu, \Gamma_\mu]$ (Section 17); the anticommutator is not needed for the symmetrized Lagrangian. **[proved VII §27]**

- FIELD EQUATION [has content]: the identity the derivation uses - $(1/\text{Sqrt}[g]) \text{d_mu}(\text{Sqrt}[g] \text{gamma}^\mu) == [\text{gamma}^\mu, \text{Gamma_mu}]$ (covariant constancy, Section 17)
- FIELD EQUATION [THE RESULT]: $\text{EL_Psibar} == (\text{Sqrt}[g]/H)(\text{gamma}^\mu \text{D_mu Psi} - H V'(s) \text{Psi})$ for generic Psi, Psibar of all eight coordinates: the field equation is $\text{gamma}^\mu \text{D_mu Psi} == H V'(s) \text{Psi}$
- FIELD EQUATION [THE RESULT]: $\text{EL_Psi} == -(\text{Sqrt}[g]/H)((\text{D_mu Psibar}) \text{gamma}^\mu + H V'(s) \text{Psibar})$: the adjoint equation is $(\text{D_mu Psibar}) \text{gamma}^\mu == -H V'(s) \text{Psibar}$
- FIELD EQUATION [has content]: the adjoint equation IS the Dirac conjugate of the field equation: with $\text{Psi} = u + i v$, $\text{Psibar} = (u - i v)^T \text{sigma16}$, adjoint residual == $-(\text{field residual})^* \cdot \text{sigma16}$
- FIELD EQUATION [control]: the opposite sign, $(\text{D Psibar}) \text{gamma} == +H V' \text{Psibar}$, is NOT the conjugate - witnessed on $u = e1 + e5$, $v = 0$ with $V = s^2$ ($s = -2$ there)

3.4.2 The 16×16 matrix form on the canonical frame

For fields of all eight coordinates, with $\partial_\mu = \partial/\partial x_\mu$, sums over $i = 1, 2, 3$ and $h = 5, 6, 7$:

$$\begin{aligned} & \cot(6Hx_0) T_{16}[0] \partial_0 \Psi + \frac{1}{q} \sum_i T_{16}[i] \partial_i \Psi + T_{16}[4] \partial_4 \Psi + \frac{1}{p} \sum_h T_{16}[h] \partial_h \Psi \\ & - 3H \cot^2(6Hx_0) T_{16}[0] \Psi = H V'(s) \Psi, \\ & \cot(6Hx_0) \partial_0 \bar{\Psi} T_{16}[0] + \frac{1}{q} \sum_i \partial_i \bar{\Psi} T_{16}[i] + \partial_4 \bar{\Psi} T_{16}[4] + \frac{1}{p} \sum_h \partial_h \bar{\Psi} T_{16}[h] \\ & - 3H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0] = -H V'(s) \bar{\Psi}. \end{aligned}$$

In the adjoint equation the connection enters as $-\bar{\Psi} \Gamma_\mu \gamma^\mu = +\bar{\Psi} \gamma^\mu \Gamma_\mu$ (per- μ anticommutator), so it carries the **same** coefficient $-3H \cot^2$ as in the field equation. **[proved VII §27]**

- WRITTEN OUT (i) [THE RESULT]: the field equation on the canonical frame is $\text{Cot } T_{16}[0] \text{d_0 Psi} + (1/q) T_{16}[i] \text{d_i Psi} + T_{16}[4] \text{d_4 Psi} + (1/p) T_{16}[h] \text{d_h Psi} - 3 H \text{Cot}^2 T_{16}[0] \text{Psi} == H V'(s) \text{Psi}$
- WRITTEN OUT (i) [THE RESULT]: the adjoint equation is $\text{Cot d_0 Psibar } T_{16}[0] + \dots - 3 H \text{Cot}^2 \text{Psibar } T_{16}[0] == -H V'(s) \text{Psibar}$ (the connection enters with the sign fixed by $\{\text{gamma}^\mu, \text{Gamma_mu}\} = 0$)

3.4.3 The split-octonion 8+8 form

Write $\Psi = (\psi_1, \psi_2)$: ψ_1 = the upper eight components (type-1 split-octonion spinor, chirality -1), ψ_2 = the lower eight (type-2, chirality $+1$). With $T_{16}[a] = \begin{pmatrix} 0 & \bar{\tau}_a \\ \tau_a & 0 \end{pmatrix}$ and $\Gamma_\mu^{\text{spin}} = \text{diag}(\Gamma_\mu^{(1)}, \Gamma_\mu^{(2)})$ (Section 17; the spin connection never mixes the two types), **covariantly, on any frame**:

$$e_a^\mu \bar{\tau}_a (\partial_\mu + \Gamma_\mu^{(2)}) \psi_2 = H V'(s) \psi_1 \quad (\text{rows 1..8}),$$

$$\begin{aligned}
e_a^\mu \tau_a (\partial_\mu + \Gamma_\mu^{(1)}) \psi_1 &= H V'(s) \psi_2 & (\text{rows } 9..16), \\
s = -\psi_1^\dagger \sigma \psi_1 + \psi_2^\dagger \sigma \psi_2, \quad \bar{\psi}_1 &= -\psi_1^\dagger \sigma, \quad \bar{\psi}_2 = \psi_2^\dagger \sigma, \\
(\partial_\mu \bar{\psi}_2 - \bar{\psi}_2 \Gamma_\mu^{(2)}) \tau_a e_a^\mu &= -H V'(s) \bar{\psi}_1, \\
(\partial_\mu \bar{\psi}_1 - \bar{\psi}_1 \Gamma_\mu^{(1)}) \bar{\tau}_a e_a^\mu &= -H V'(s) \bar{\psi}_2
\end{aligned}$$

($\bar{\Psi} = (\bar{\psi}_1, \bar{\psi}_2) = \Psi^\dagger \sigma_{16}$, $\sigma_{16} = \text{diag}(-\sigma, \sigma)$; $\tau_a = \text{tau}[a]$, $\bar{\tau}_a = \text{taubar}[a]$.) On the canonical frame ($\tau_0 = \bar{\tau}_0 = \text{ID8}$), the connection term is $-3H \cot^2 \psi_2$ in the upper rows and $-3H \cot^2 \psi_1$ in the lower rows:

$$\begin{aligned}
\cot \partial_0 \psi_2 + \frac{1}{q} \sum_i \bar{\tau}_i \partial_i \psi_2 + \bar{\tau}_4 \partial_4 \psi_2 + \frac{1}{p} \sum_h \bar{\tau}_h \partial_h \psi_2 - 3H \cot^2 \psi_2 &= H V'(s) \psi_1, \\
\cot \partial_0 \psi_1 + \frac{1}{q} \sum_i \tau_i \partial_i \psi_1 + \tau_4 \partial_4 \psi_1 + \frac{1}{p} \sum_h \tau_h \partial_h \psi_1 - 3H \cot^2 \psi_1 &= H V'(s) \psi_2,
\end{aligned}$$

with $\cot = \cot(6Hx_0)$. [proved VII §27]

- WRITTEN OUT (ii) [THE RESULT]: the split-octonion 8+8 form - the type-1 rows are the taubar equations for psi2, the type-2 rows the tau equations for psi1 (tau[0] == taubar[0] == ID8)
- WRITTEN OUT (ii) [THE RESULT]: the covariant 8+8 form - rows 1..8 are $e_a^\mu \text{taubar}[a] (d_\mu + \Gamma_\mu^{(2)}) \psi_2 - H V' \psi_1$, rows 9..16 are $e_a^\mu \text{tau}[a] (d_\mu + \Gamma_\mu^{(1)}) \psi_1 - H V' \psi_2$; and the conjugate rows
- WRITTEN OUT (ii) [has content]: with $\text{Psibar} = \Psi^\dagger \text{ddag} \sigma_{16}$ and $\sigma_{16} = \text{diag}(-\sigma, \sigma)$: $s == -\text{psi1}^\dagger \text{ddag} \sigma \text{psi1} + \text{psi2}^\dagger \text{ddag} \sigma \text{psi2}$ ($\text{psibar1} = -\text{psi1}^\dagger \text{ddag} \sigma$, $\text{psibar2} = \text{psi2}^\dagger \text{ddag} \sigma$)

3.4.4 All sixteen component equations

Each $T_{16}[a]$ is a signed permutation matrix, so component equation r contains exactly **one** derivative per coordinate, of the component that $T_{16}[a]$ maps to r ; the connection term multiplies the same component as the x_0 -derivative; and $-HV'$ multiplies ψ_r itself. For fields of all eight coordinates the sixteen equations form **one** coupled block. (The four blocks of four of Section 13, section 3.1.9, belong to the reduction to fields of (x_0, x_4) only.) The term lists `cfFieldEqTerms` and `cfAdjEqTerms` from which the equations below are rendered reproduce all sixteen field equations and all sixteen adjoint equations exactly. [proved VII §27]

- WRITTEN OUT (iii) [definition]: the term lists reproduce all sixteen field equations and all sixteen adjoint equations exactly
- WRITTEN OUT (iii) [has content]: every component equation has exactly one derivative per coordinate $x_0..x_7$, the connection term multiplies the component carrying the x_0 -derivative, and $-H V'$ multiplies ψ_r itself
- WRITTEN OUT (iii) [has content]: for fields of all eight coordinates the sixteen equations form **ONE** coupled block (the four blocks of four of Section 13 belong to the (x_0, x_4) reduction)

How to read the equations below. They are the TeX export written by `claudable/export_fermion_fable_tex.wls` from the notebook's own asserted objects (`provenancelatex/generated/fermion_fable_field_equations.tex`, included here unchanged; the same equations in plain text are the `.txt` file of the same name).

- ψ_k ($k = 0..15$) is component k of Ψ ; $\bar{\psi}_k$ is component k of $\bar{\Psi}$. The index is the notebook's 0-based component index (array position $k + 1$).
- $\partial_\mu \psi_k$ is $\partial \psi_k / \partial x_\mu$, $\mu = 0..7$.
- a_4 is the free function. Hence $e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} = 1/q$ multiplies the observed derivatives $\partial_1, \partial_2, \partial_3$, and $e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} = 1/p$ multiplies the hidden derivatives $\partial_5, \partial_6, \partial_7$.
- $V'(s)$ is evaluated at $s = \bar{\Psi}\Psi$; H is the author's Protected constant.
- Equation E_r is row r of $\gamma^\mu D_\mu \Psi - HV'(s)\Psi = 0$, with the V' term moved to the right.

$$\begin{aligned}
(E_0) \quad & \cot(6Hx_0) \partial_0 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{13} \\
& - \partial_4 \psi_{13} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{14} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{15} - 3H \cot^2(6Hx_0) \psi_8 \\
& = HV'(s) \psi_0 \\
(E_1) \quad & \cot(6Hx_0) \partial_0 \psi_9 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{12} \\
& + \partial_4 \psi_{12} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{15} - 3H \cot^2(6Hx_0) \psi_9 \\
& = HV'(s) \psi_1 \\
(E_2) \quad & \cot(6Hx_0) \partial_0 \psi_{10} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{12} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{13} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{13} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{15} + \partial_4 \psi_{15} - 3H \cot^2(6Hx_0) \psi_{10} \\
& = HV'(s) \psi_2 \\
(E_3) \quad & \cot(6Hx_0) \partial_0 \psi_{11} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{11} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{12} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{13} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{14} - \partial_4 \psi_{14} - 3H \cot^2(6Hx_0) \psi_{11} \\
& = HV'(s) \psi_3 \\
(E_4) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_9 + \partial_4 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{10} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{11} \\
& + \cot(6Hx_0) \partial_0 \psi_{12} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{12} - 3H \cot^2(6Hx_0) \psi_{12} \\
& = HV'(s) \psi_4 \\
(E_5) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_8 - \partial_4 \psi_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{10} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{11} \\
& + \cot(6Hx_0) \partial_0 \psi_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{13} - 3H \cot^2(6Hx_0) \psi_{13} \\
& = HV'(s) \psi_5 \\
(E_6) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_9 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_9 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{11} - \partial_4 \psi_{11} \\
& + \cot(6Hx_0) \partial_0 \psi_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{14} - 3H \cot^2(6Hx_0) \psi_{14} \\
& = HV'(s) \psi_6
\end{aligned}$$

$$\begin{aligned}
(E_7) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_9 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{10} + \partial_4 \psi_{10} \\
& + \cot(6Hx_0) \partial_0 \psi_{15} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{15} - 3H \cot^2(6Hx_0) \psi_{15} \\
& = HV'(s) \psi_7 \\
(E_8) \quad & \cot(6Hx_0) \partial_0 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_5 \\
& + \partial_4 \psi_5 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_6 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_7 - 3H \cot^2(6Hx_0) \psi_0 \\
& = HV'(s) \psi_8 \\
(E_9) \quad & \cot(6Hx_0) \partial_0 \psi_1 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_4 \\
& - \partial_4 \psi_4 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_7 - 3H \cot^2(6Hx_0) \psi_1 \\
& = HV'(s) \psi_9 \\
(E_{10}) \quad & \cot(6Hx_0) \partial_0 \psi_2 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_4 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_5 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_5 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_7 - \partial_4 \psi_7 - 3H \cot^2(6Hx_0) \psi_2 \\
& = HV'(s) \psi_{10} \\
(E_{11}) \quad & \cot(6Hx_0) \partial_0 \psi_3 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_3 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_4 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_5 \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_6 + \partial_4 \psi_6 - 3H \cot^2(6Hx_0) \psi_3 \\
& = HV'(s) \psi_{11} \\
(E_{12}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_1 - \partial_4 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_2 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_4 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_4 - 3H \cot^2(6Hx_0) \psi_4 \\
& = HV'(s) \psi_{12} \\
(E_{13}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_0 + \partial_4 \psi_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_2 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_5 - 3H \cot^2(6Hx_0) \psi_5 \\
& = HV'(s) \psi_{13} \\
(E_{14}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_1 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_1 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_3 + \partial_4 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_6 - 3H \cot^2(6Hx_0) \psi_6 \\
& = HV'(s) \psi_{14} \\
(E_{15}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_1 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_2 - \partial_4 \psi_2 \\
& + \cot(6Hx_0) \partial_0 \psi_7 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_7 - 3H \cot^2(6Hx_0) \psi_7 \\
& = HV'(s) \psi_{15}
\end{aligned}$$

3.4.5 All sixteen adjoint equations

Equation $\bar{E}_r$ is column r of $(D_\mu \bar{\Psi})\gamma^\mu + HV'(s)\bar{\Psi} = 0$, with the V' term moved to the right (provenance-latex/generated/fermion_fable_adjoint_equations.tex, unchanged):

$$\begin{aligned}
(\bar{E}_0) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_8 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{13} \\
& + \partial_4 \bar{\psi}_{13} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{15} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{15} - 3H \cot^2(6Hx_0) \bar{\psi}_8 \\
& = -HV'(s) \bar{\psi}_0 \\
(\bar{E}_1) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_9 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{12} \\
& - \partial_4 \bar{\psi}_{12} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{14} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{15} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{15} - 3H \cot^2(6Hx_0) \bar{\psi}_9 \\
& = -HV'(s) \bar{\psi}_1 \\
(\bar{E}_2) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_{10} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{12} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{13} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{15} - \partial_4 \bar{\psi}_{15} - 3H \cot^2(6Hx_0) \bar{\psi}_{10} \\
& = -HV'(s) \bar{\psi}_2 \\
(\bar{E}_3) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{11} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{12} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{13} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{13} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{14} + \partial_4 \bar{\psi}_{14} - 3H \cot^2(6Hx_0) \bar{\psi}_{11} \\
& = -HV'(s) \bar{\psi}_3 \\
(\bar{E}_4) \quad & -e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_9 - \partial_4 \bar{\psi}_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{10} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{11} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{11} \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_{12} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{12} - 3H \cot^2(6Hx_0) \bar{\psi}_{12} \\
& = -HV'(s) \bar{\psi}_4 \\
(\bar{E}_5) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_8 + \partial_4 \bar{\psi}_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_{10} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_{11} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_{11} \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_{13} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{13} - 3H \cot^2(6Hx_0) \bar{\psi}_{13} \\
& = -HV'(s) \bar{\psi}_5 \\
(\bar{E}_6) \quad & -e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_8 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_9 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_9 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{11} + \partial_4 \bar{\psi}_{11} \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_{14} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{14} - 3H \cot^2(6Hx_0) \bar{\psi}_{14} \\
& = -HV'(s) \bar{\psi}_6 \\
(\bar{E}_7) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_8 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_9 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_{10} - \partial_4 \bar{\psi}_{10} \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_{15} - 3H \cot^2(6Hx_0) \bar{\psi}_{15}
\end{aligned}$$

$$= -HV'(s) \bar{\psi}_7$$

$$\begin{aligned}
(\bar{E}_8) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_0 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_5 \\
& - \partial_4 \bar{\psi}_5 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_7 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_7 - 3H \cot^2(6Hx_0) \bar{\psi}_0 \\
& = -HV'(s) \bar{\psi}_8
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_9) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_1 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_4 \\
& + \partial_4 \bar{\psi}_4 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_6 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_7 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_7 - 3H \cot^2(6Hx_0) \bar{\psi}_1 \\
& = -HV'(s) \bar{\psi}_9
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{10}) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_2 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_4 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_5 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_7 + \partial_4 \bar{\psi}_7 - 3H \cot^2(6Hx_0) \bar{\psi}_2 \\
& = -HV'(s) \bar{\psi}_{10}
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{11}) \quad & \cot(6Hx_0) \partial_0 \bar{\psi}_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_3 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_4 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_5 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_5 \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_6 - \partial_4 \bar{\psi}_6 - 3H \cot^2(6Hx_0) \bar{\psi}_3 \\
& = -HV'(s) \bar{\psi}_{11}
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{12}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_1 + \partial_4 \bar{\psi}_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_2 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_3 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_3 \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_4 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_4 - 3H \cot^2(6Hx_0) \bar{\psi}_4 \\
& = -HV'(s) \bar{\psi}_{12}
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{13}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_0 - \partial_4 \bar{\psi}_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_2 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_3 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_3 \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_5 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_5 - 3H \cot^2(6Hx_0) \bar{\psi}_5 \\
& = -HV'(s) \bar{\psi}_{13}
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{14}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_0 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_1 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_1 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_3 - \partial_4 \bar{\psi}_3 \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_6 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_6 - 3H \cot^2(6Hx_0) \bar{\psi}_6 \\
& = -HV'(s) \bar{\psi}_{14}
\end{aligned}$$

$$\begin{aligned}
(\bar{E}_{15}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \bar{\psi}_0 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \bar{\psi}_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \bar{\psi}_1 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \bar{\psi}_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \bar{\psi}_2 + \partial_4 \bar{\psi}_2 \\
& + \cot(6Hx_0) \partial_0 \bar{\psi}_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \bar{\psi}_7 - 3H \cot^2(6Hx_0) \bar{\psi}_7 \\
& = -HV'(s) \bar{\psi}_{15}
\end{aligned}$$

For every r , E_r and $\bar{E}_r$ contain the same eight (derivative direction, component index) pairs, $\partial_\mu \psi_k$ against $\partial_\mu \bar{\psi}_k$, with the same coefficients. The terms with derivatives along the spacelike directions $x_0..x_3$ have the **same** sign in both; those along the timelike directions $x_4..x_7$ have the **opposite** sign; the connection term $-3H \cot^2$ multiplies the same component with the same sign; and the V' term changes sign. This is what the transposition requires: $T_{16}[a]$ is a real signed permutation matrix with $T_{16}[a]^2 = \eta_{aa}$, so $T_{16}[a]^T = \eta_{aa} T_{16}[a]$ [derived here; the rule was checked for the provenance pages of 2026-09-24 on all 128 derivative terms and 16 connection terms of the two exported files by a short script; it is not a notebook assertion]. The adjoint equations are the Dirac conjugates of the field equations (FIELD EQUATION [has content]: the adjoint equation IS the Dirac conjugate of the field equation: with $\text{Psi} = u + i v$, $\text{Psibar} = (u - i v)^T \text{sigma16}$, $\text{adjoint residual} == -(\text{field residual})^* \cdot \text{sigma16}$, section 3.4.1).

3.4.6 The 8+8 form, component by component

Rows 0–7 are the $\bar{\tau}$ equations for ψ_2 , rows 8–15 the τ equations for ψ_1 ; $\psi_k^{(1)} = \psi_k$ and $\psi_k^{(2)} = \psi_{k+8}$ ($k = 0..7$) (provenance-latex/generated/fermion_fable_field_equations_8plus8.tex, unchanged):

$$\begin{aligned} & \cot(6Hx_0) \partial_0 \psi^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \sum_{i=1}^3 \bar{\tau}_i \partial_i \psi^{(2)} + \bar{\tau}_4 \partial_4 \psi^{(2)} \\ & + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \sum_{h=5}^7 \bar{\tau}_h \partial_h \psi^{(2)} - 3H \cot^2(6Hx_0) \psi^{(2)} = HV'(s) \psi^{(1)}, \\ & \cot(6Hx_0) \partial_0 \psi^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \sum_{i=1}^3 \tau_i \partial_i \psi^{(1)} + \tau_4 \partial_4 \psi^{(1)} \\ & + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \sum_{h=5}^7 \tau_h \partial_h \psi^{(1)} - 3H \cot^2(6Hx_0) \psi^{(1)} = HV'(s) \psi^{(2)}, \\ & s = -\psi^{(1)\dagger} \sigma \psi^{(1)} + \psi^{(2)\dagger} \sigma \psi^{(2)}. \end{aligned}$$

Component form (rows 0–7: the $\bar{\tau}$ equations for $\psi^{(2)}$; rows 8–15: the τ equations for $\psi^{(1)}$):

$$\begin{aligned} (E_0) \quad & \cot(6Hx_0) \partial_0 \psi_0^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_0^{(2)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_5^{(2)} \\ & - \partial_4 \psi_5^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_6^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_6^{(2)} \\ & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_7^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_7^{(2)} - 3H \cot^2(6Hx_0) \psi_0^{(2)} \\ & = HV'(s) \psi_0^{(1)} \\ (E_1) \quad & \cot(6Hx_0) \partial_0 \psi_1^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_1^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_4^{(2)} \\ & + \partial_4 \psi_4^{(2)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_6^{(2)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_6^{(2)} \\ & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_7^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_7^{(2)} - 3H \cot^2(6Hx_0) \psi_1^{(2)} \\ & = HV'(s) \psi_1^{(1)} \\ (E_2) \quad & \cot(6Hx_0) \partial_0 \psi_2^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_2^{(2)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_4^{(2)} \\ & - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_4^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_5^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_5^{(2)} \\ & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_7^{(2)} + \partial_4 \psi_7^{(2)} - 3H \cot^2(6Hx_0) \psi_2^{(2)} \\ & = HV'(s) \psi_2^{(1)} \end{aligned}$$

$$\begin{aligned}
(E_3) \quad & \cot(6Hx_0) \partial_0 \psi_3^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_3^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_4^{(2)} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_4^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_5^{(2)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_5^{(2)} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_6^{(2)} - \partial_4 \psi_6^{(2)} - 3H \cot^2(6Hx_0) \psi_3^{(2)} \\
& = HV'(s) \psi_3^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_4) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_1^{(2)} + \partial_4 \psi_1^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_2^{(2)} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_2^{(2)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_3^{(2)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_3^{(2)} \\
& + \cot(6Hx_0) \partial_0 \psi_4^{(2)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_4^{(2)} - 3H \cot^2(6Hx_0) \psi_4^{(2)} \\
& = HV'(s) \psi_4^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_5) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_0^{(2)} - \partial_4 \psi_0^{(2)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_2^{(2)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_2^{(2)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_3^{(2)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_3^{(2)} \\
& + \cot(6Hx_0) \partial_0 \psi_5^{(2)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_5^{(2)} - 3H \cot^2(6Hx_0) \psi_5^{(2)} \\
& = HV'(s) \psi_5^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_6) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_0^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_0^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_1^{(2)} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_1^{(2)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_3^{(2)} - \partial_4 \psi_3^{(2)} \\
& + \cot(6Hx_0) \partial_0 \psi_6^{(2)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_6^{(2)} - 3H \cot^2(6Hx_0) \psi_6^{(2)} \\
& = HV'(s) \psi_6^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_7) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_0^{(2)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_0^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_1^{(2)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_1^{(2)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_2^{(2)} + \partial_4 \psi_2^{(2)} \\
& + \cot(6Hx_0) \partial_0 \psi_7^{(2)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_7^{(2)} - 3H \cot^2(6Hx_0) \psi_7^{(2)} \\
& = HV'(s) \psi_7^{(1)}
\end{aligned}$$

$$\begin{aligned}
(E_8) \quad & \cot(6Hx_0) \partial_0 \psi_0^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_0^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_5^{(1)} \\
& + \partial_4 \psi_5^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_6^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_6^{(1)} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_7^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_7^{(1)} - 3H \cot^2(6Hx_0) \psi_0^{(1)} \\
& = HV'(s) \psi_0^{(2)}
\end{aligned}$$

$$\begin{aligned}
(E_9) \quad & \cot(6Hx_0) \partial_0 \psi_1^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_1^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_4^{(1)} \\
& - \partial_4 \psi_4^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_6^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_6^{(1)} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_7^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_7^{(1)} - 3H \cot^2(6Hx_0) \psi_1^{(1)} \\
& = HV'(s) \psi_1^{(2)}
\end{aligned}$$

$$\begin{aligned}
(E_{10}) \quad & \cot(6Hx_0) \partial_0 \psi_2^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_2^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_4^{(1)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_4^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_5^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_5^{(1)} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_7^{(1)} - \partial_4 \psi_7^{(1)} - 3H \cot^2(6Hx_0) \psi_2^{(1)} \\
& = HV'(s) \psi_2^{(2)}
\end{aligned}$$

$$(E_{11}) \quad \cot(6Hx_0) \partial_0 \psi_3^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_3^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_4^{(1)}$$

$$\begin{aligned}
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_4^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_5^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_5^{(1)} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_6^{(1)} + \partial_4 \psi_6^{(1)} - 3H \cot^2(6Hx_0) \psi_3^{(1)} \\
& = HV'(s) \psi_3^{(2)} \\
(E_{12}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_1^{(1)} - \partial_4 \psi_1^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_2^{(1)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_2^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_3^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_3^{(1)} \\
& + \cot(6Hx_0) \partial_0 \psi_4^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_4^{(1)} - 3H \cot^2(6Hx_0) \psi_4^{(1)} \\
& = HV'(s) \psi_4^{(2)} \\
(E_{13}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_0^{(1)} + \partial_4 \psi_0^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_2^{(1)} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_2^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_3^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_3^{(1)} \\
& + \cot(6Hx_0) \partial_0 \psi_5^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_5^{(1)} - 3H \cot^2(6Hx_0) \psi_5^{(1)} \\
& = HV'(s) \psi_5^{(2)} \\
(E_{14}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_0^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_0^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_1^{(1)} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_1^{(1)} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_3^{(1)} + \partial_4 \psi_3^{(1)} \\
& + \cot(6Hx_0) \partial_0 \psi_6^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_6^{(1)} - 3H \cot^2(6Hx_0) \psi_6^{(1)} \\
& = HV'(s) \psi_6^{(2)} \\
(E_{15}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_0^{(1)} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_0^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_1^{(1)} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_1^{(1)} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_2^{(1)} - \partial_4 \psi_2^{(1)} \\
& + \cot(6Hx_0) \partial_0 \psi_7^{(1)} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_7^{(1)} - 3H \cot^2(6Hx_0) \psi_7^{(1)} \\
& = HV'(s) \psi_7^{(2)}
\end{aligned}$$

3.4.7 The rescaled form: the spin connection disappears

Put $\Psi = \sqrt{\sin 6Hx_0} \Psi'$ and $\bar{\Psi} = \sqrt{\sin 6Hx_0} \bar{\Psi}'$. Then $s = \sin(6Hx_0) s'$ with $s' = \bar{\Psi}' \Psi'$, and

$$\begin{aligned}
\gamma^\mu D_\mu \Psi - HV'(s) \Psi &= \sqrt{\sin 6Hx_0} \left(\cot(6Hx_0) T_{16}[0] \partial_0 \Psi' + \frac{1}{q} \sum_i T_{16}[i] \partial_i \Psi' + T_{16}[4] \partial_4 \Psi' \right. \\
&\quad \left. + \frac{1}{p} \sum_h T_{16}[h] \partial_h \Psi' - HV'(\sin(6Hx_0) s') \Psi' \right)
\end{aligned}$$

with **no connection term at all**, and the same for the adjoint equation. This works because the connection term is a gradient: $\gamma^\mu \Gamma_\mu = -\frac{1}{2} T_\mu \gamma^\mu$ with the Weitzenböck torsion vector $T_\mu = \partial_\mu \ln \sin(6Hx_0)$ (section 3.8.10), and multiplying by $\exp(\frac{1}{2} \ln \sin) = \sqrt{\sin}$ removes it. **[proved VII §27]**

- WRITTEN OUT (iv) [THE RESULT]: `Psi = Sqrt[Sin[6Hx0]] Psi'` turns the field equation into `Sqrt[Sin] (Cot T16[0] d_0 Psi' + (1/q) T16[i] d_i Psi' + T16[4] d_4 Psi' + (1/p) T16[h] d_h Psi' - H V'(Sin s') Psi')`: NO connection term
- WRITTEN OUT (iv) [THE RESULT]: and the adjoint equation into `Sqrt[Sin] (Cot d_0 Psi'bar T16[0] + ... + H V'(Sin s') Psi'bar)`, with no connection term either

- WRITTEN OUT (iv) [definition]: under the rescaling $s == \text{Sin}[6 H x_0] s'$, $s' = \text{Psi}'\text{bar Psi}'$

The sixteen rescaled equations (ψ'_k = component k of Ψ' ; V' evaluated at $s = \text{sin}(6Hx_0) \bar{\Psi}'\Psi'$; provenance-latex/generated/fermion_fable_field_equations_rescaled.tex, unchanged):

$$\begin{aligned}
(E'_0) \quad & \cot(6Hx_0) \partial_0 \psi'_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{13} \\
& - \partial_4 \psi'_{13} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{14} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{15} \\
& = HV'(s) \psi'_0 \\
(E'_1) \quad & \cot(6Hx_0) \partial_0 \psi'_9 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{12} \\
& + \partial_4 \psi'_{12} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{15} \\
& = HV'(s) \psi'_1 \\
(E'_2) \quad & \cot(6Hx_0) \partial_0 \psi'_{10} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{12} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{13} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{13} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{15} + \partial_4 \psi'_{15} \\
& = HV'(s) \psi'_2 \\
(E'_3) \quad & \cot(6Hx_0) \partial_0 \psi'_{11} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{11} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{12} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{13} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{14} - \partial_4 \psi'_{14} \\
& = HV'(s) \psi'_3 \\
(E'_4) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_9 + \partial_4 \psi'_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{10} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{11} \\
& + \cot(6Hx_0) \partial_0 \psi'_{12} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{12} \\
& = HV'(s) \psi'_4 \\
(E'_5) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_8 - \partial_4 \psi'_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_{10} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_{11} \\
& + \cot(6Hx_0) \partial_0 \psi'_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{13} \\
& = HV'(s) \psi'_5 \\
(E'_6) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_9 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_9 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{11} - \partial_4 \psi'_{11} \\
& + \cot(6Hx_0) \partial_0 \psi'_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{14} \\
& = HV'(s) \psi'_6 \\
(E'_7) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_9 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_{10} + \partial_4 \psi'_{10}
\end{aligned}$$

$$\begin{aligned}
& + \cot(6Hx_0) \partial_0 \psi'_{15} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_{15} \\
& = HV'(s) \psi'_7
\end{aligned}$$

$$\begin{aligned}
(E'_8) \quad & \cot(6Hx_0) \partial_0 \psi'_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_5 \\
& + \partial_4 \psi'_5 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_6 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_7 \\
& = HV'(s) \psi'_8
\end{aligned}$$

$$\begin{aligned}
(E'_9) \quad & \cot(6Hx_0) \partial_0 \psi'_1 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_4 \\
& - \partial_4 \psi'_4 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_7 \\
& = HV'(s) \psi'_9
\end{aligned}$$

$$\begin{aligned}
(E'_{10}) \quad & \cot(6Hx_0) \partial_0 \psi'_2 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_4 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_5 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_5 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_7 - \partial_4 \psi'_7 \\
& = HV'(s) \psi'_{10}
\end{aligned}$$

$$\begin{aligned}
(E'_{11}) \quad & \cot(6Hx_0) \partial_0 \psi'_3 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_3 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_4 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_5 \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_6 + \partial_4 \psi'_6 \\
& = HV'(s) \psi'_{11}
\end{aligned}$$

$$\begin{aligned}
(E'_{12}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_1 - \partial_4 \psi'_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_2 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_3 \\
& + \cot(6Hx_0) \partial_0 \psi'_4 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_4 \\
& = HV'(s) \psi'_{12}
\end{aligned}$$

$$\begin{aligned}
(E'_{13}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_0 + \partial_4 \psi'_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_2 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_3 \\
& + \cot(6Hx_0) \partial_0 \psi'_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_5 \\
& = HV'(s) \psi'_{13}
\end{aligned}$$

$$\begin{aligned}
(E'_{14}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_1 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_1 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_3 + \partial_4 \psi'_3 \\
& + \cot(6Hx_0) \partial_0 \psi'_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_6 \\
& = HV'(s) \psi'_{14}
\end{aligned}$$

$$\begin{aligned}
(E'_{15}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi'_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi'_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi'_1 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi'_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi'_2 - \partial_4 \psi'_2 \\
& + \cot(6Hx_0) \partial_0 \psi'_7 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi'_7 \\
& = HV'(s) \psi'_{15}
\end{aligned}$$

3.4.8 The x_4 -only family and the condition $V''s' = 0$

The rescaled field with no x_0 -dependence, $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$, satisfies the field equation at every x_0 **if and only if** $\partial_{x_0} V'(\sin(6Hx_0) s') = 0$, and $\partial_{x_0} V'(\sin s') = 6H \cos(6Hx_0) s' V''(\sin s')$. So it is a solution exactly when $V''s' = 0$:

- either V is **linear** (the author's mass term $V = -(2M/H)s$, $V' = -2M/H$ constant), or
- the data are **null**, $s' = \bar{\Psi}'\Psi' = 0$.

For the complex field s' is **conserved** by the x_4 -only equations $\partial_4 \Psi' = -HV' T_{16}[4] \Psi'$, $\partial_4 \bar{\Psi}' = HV' \bar{\Psi}' T_{16}[4]$, for **any** V . So null data give x_4 -only solutions for any V , with $\rho = V(0)$. For a nonlinear V and $s' \neq 0$ the family fails (witness: $V = s^2$). [**proved VII §29**]

- PRE-UNIVERSE (2) [THE RESULT]: $\Psi = \text{Sqrt}[\text{Sin}] \Psi'(x_4)$ satisfies the field equation iff $d_{x_0} V'(\text{Sin}[6Hx_0] s') == 0$, and $d_{x_0} V'(\text{Sin} s') == 6 H \text{Cos}[6 H x_0] s' V''(\text{Sin} s')$: V linear OR $s' == 0$
- PRE-UNIVERSE (2) [has content]: $s' = \Psi' \bar{\Psi}'$ is conserved by the x_4 -only equations ($d_4 \Psi' = -H V' T_{16}[4] \Psi'$, $d_4 \bar{\Psi}' = H V' \bar{\Psi}' T_{16}[4]$) for ANY V ; so null data $s' = 0$ are solutions for any V , with $\rho == V(0)$; and for $V = s^2$ with $s' \neq 0$ the x_0 -dependence is real (witness)

3.4.9 The operator form of the field equation, and its ordering

These are definitions, not identities [**prose VII §27**]. The Hamiltonian (section 3.5.11) contains $V(:s:)$, defined by the spectral calculus of the Hermitian operator $s(x) = \Psi^\dagger \beta \Psi$ of section 3.5.5 (regularized and normal-ordered). No ordering prescription is needed there, and a non-polynomial V is covered. The operator field equation is

$$\gamma^\mu D_\mu \Psi = H :V'(s) \Psi:$$

where $V'(s)\Psi$ is the **Weyl-symmetrized** product, the average over the positions at which Ψ can be inserted in $V'(s)$.

- It is **exact** for linear V (the author's mass term).
- For nonlinear V it differs from the naive product by Hartree and Fock contractions with the coincident J -vacuum propagator. That propagator is a c-number **matrix** with two parts: a **scalar** part, which is divergent and renormalizes V' ; and a γ^4 part, the charge density of the Dirac sea, which is a constant chemical-potential shift. The γ^4 part is removed by charge-symmetric ordering or by a phase $\Psi \rightarrow e^{i\mu x_4} \Psi$, **not** by renormalizing V .
- In the Kohn–Sham mean field of section 3.7, $V'(s)$ is replaced by $V'(\langle s \rangle)$ and the equation is linear.

3.4.10 The U(1) current and the particle-number current

L_{sym} is invariant under $\Psi \rightarrow e^{i\alpha} \Psi$, $\bar{\Psi} \rightarrow e^{-i\alpha} \bar{\Psi}$. The Noether current is

$$J^\mu = \frac{\partial L}{\partial(\partial_\mu \Psi)} (i\Psi) + (-i\bar{\Psi}) \frac{\partial L}{\partial(\partial_\mu \bar{\Psi})} = \frac{\sqrt{g}}{H} i \bar{\Psi} \gamma^\mu \Psi,$$

so $j^\mu = i \bar{\Psi} \gamma^\mu \Psi$ is the U(1) current. It is real, because $i\sigma_{16}\gamma^\mu$ is Hermitian. **Off shell**, for generic fields of all eight coordinates,

$$\frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} j^\mu) = i (\bar{E} \Psi + \bar{\Psi} E), \quad E, \bar{E} = \text{the two field-equation residuals},$$

so the current is conserved on shell. [**proved VII §27**]

- CURRENT [THE RESULT]: the Noether current of the U(1) phase is $(\text{Sqrt}[g]/H) i \text{Psibar } \gamma^\mu \Psi$
- CURRENT [has content]: $i \text{sigma16 } \gamma^\mu$ is Hermitian for every μ , so j^μ is real
- CURRENT [THE RESULT]: OFF SHELL $(1/\text{Sqrt}[g]) d_\mu(\text{Sqrt}[g] j^\mu) == i (\text{Ebar } \Psi + \text{Psibar } E)$ for generic fields of all eight coordinates - conserved on shell

The **particle-number current** is $n^\mu := -\frac{i}{H} \bar{\Psi} \gamma^\mu \Psi = -j^\mu/H$. As section 3.5.15 shows, $N = \int \sqrt{g} n^4 d^7x$ is the normal-ordered number of particles minus antiparticles, so the charge of j carries a conventional minus sign: $\int \sqrt{g} j^4 d^7x = -HN$ [**prose VII §27**]. So one particle (one occupied positive-frequency J -mode, $N = +1$) carries j -charge $-H$, and one antiparticle $+H$ [derived here from the two definitions; the design review, QK-10, states the same sign as “a positive-frequency J -mode carries j -charge -1 ”, that is, in units of H].

3.4.11 The only internal symmetry is the vector phase

The commutant of all eight $T_{16}[a]$ in $M_{16}(\mathbb{C})$ is one-dimensional (multiples of ID16), so the only internal U(1) is the vector phase. In particular the **axial** phase $\Psi \rightarrow \exp(i\alpha T_{16}[8]) \Psi$ leaves $s = \bar{\Psi} \Psi$ invariant but **not** the kinetic term ($T_{16}[8]$ anticommutes with every gamma): $\exp(-i\alpha T_8) T_{16}[a] \exp(i\alpha T_8) = T_{16}[a] \exp(2i\alpha T_8)$, so $\bar{\Psi} \gamma^\mu \partial_\mu \Psi$ changes at first order by $2i\alpha \bar{\Psi} \gamma^\mu T_{16}[8] \partial_\mu \Psi$, which is not zero. This is the reverse of the four-dimensional intuition, where the axial phase preserves the kinetic term and breaks the mass term. [**proved VII §27**]

- SYMMETRY [THE RESULT]: the only internal U(1) is the vector phase - the commutant of all eight $T_{16}[a]$ in $M_{16}(\mathbb{C})$ is one-dimensional (multiples of ID16)
- SYMMETRY [has content]: the axial phase $\exp(i \alpha T_{16}[8])$ leaves s invariant but not the kinetic term: $\exp(-i \alpha T_8) T_{16}[a] \exp(i \alpha T_8) == T_{16}[a] \exp(2 i \alpha T_8)$, so $\text{Psibar } \gamma^\mu d_\mu \Psi$ changes at first order by $2 i \alpha \text{Psibar } \gamma^\mu T_{16}[8] d_\mu \Psi$, which is not zero (witness)

3.5 Canonical quantization in 4+4 dimensions

This is Section 28 of the notebook (Input cells 185–191, 56 assertions). “Extended to 4+4 dimensions” means: the time is one of the four timelike directions, x_4 , and the slice of constant time is seven-dimensional and contains the other three timelike directions. Everything below follows from the symmetrized Lagrangian of section 3.3; the result is summarized first.

The result in one paragraph. The Dirac bracket gives the anticommutator $\{\Psi_a(x), \Psi_b^\dagger(y)\} = \frac{H}{\sqrt{|g|}} (-i\sigma_{16}\gamma^4)_{ab} \delta^7(x-y)$. Its matrix is $J = -iT_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4]$, which is Hermitian with square 1 and has signature (8, 8): the naive Fock space would have negative-norm states. J acts only on a hidden flavour index, $\sigma_{16} = J\beta$ with $\beta = -iT_{16}[4]$, and the Hilbert adjoint $\Psi^\dagger := \Psi^\dagger J$ makes the anticommutator positive, $\{\Psi, \Psi^\dagger\} = \frac{H}{\sqrt{|g|}} \delta^7$. This works exactly when the field carries no momentum along the three hidden timelike directions x_5, x_6, x_7 (the admissibility theorem), and there J is unique. The admissible theory is a dimensional reduction with a finite hidden coordinate volume V_{hid} . Its Fock space is built on J -eigenmodes: per momentum, 8 positive-frequency and 8 negative-frequency modes; the ground state is the filled Dirac sea; the first excitations are 8 particles and 8 antiparticles. A finite Jordan–Wigner Fock space checks all of this exactly.

3.5.1 Time, slices, and why the slice is not a Cauchy surface

Time is x_4 , the observer's proper time. On the canonical frame $g_{44} = -1$ (lapse 1) and $g_{4\mu} = 0$ for $\mu \neq 4$ (shift 0). A slice $x_4 = \text{const}$ is **seven**-dimensional with signature (4,3): x_0, x_1, x_2, x_3 spacelike and x_5, x_6, x_7 **timelike**. It is therefore not a Cauchy surface, and the quantization is carried out where it is well posed (the admissible sector, section 3.5.10). [proved VII §28] QUANTIZATION [definition]: lapse 1 and shift 0 ($g_{44} == -1$, $g_{4\mu} == 0$), and the slice $x_4 = \text{const}$ has signature (4,3) - $x_0..x_3$ spacelike, $x_5..x_7$ timelike (given a_4 real and $0 < 6 H x_0 < \text{Pi}/2$)

3.5.2 The momenta, the kinetic form G , and the signature (8,8)

From L_{sym} the canonical momenta are

$$\Pi_\Psi = \frac{\partial L}{\partial(\partial_4 \Psi)} = \frac{\sqrt{g}}{2H} \bar{\Psi} \gamma^4, \quad \Pi_{\bar{\Psi}} = \frac{\partial L}{\partial(\partial_4 \bar{\Psi})} = -\frac{\sqrt{g}}{2H} \gamma^4 \Psi.$$

The symplectic potential $\Pi_\Psi d\Psi + d\bar{\Psi} \Pi_{\bar{\Psi}}$ differs from that of the unsymmetrized Lagrangian, $\frac{\sqrt{g}}{H} \bar{\Psi} \gamma^4 d\Psi$, by the exact variation $\delta[\frac{\sqrt{g}}{2H} \bar{\Psi} \gamma^4 \Psi]$: the two define the **same** symplectic form, hence the same brackets. With $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ the time-derivative part of the Lagrangian is

$$\Psi^\dagger K \partial_4 \Psi, \quad K = \frac{\sqrt{g}}{H} \sigma_{16} \gamma^4 = i \tilde{G}, \quad \tilde{G} = \frac{\sqrt{g}}{H} G, \quad G = -i \sigma_{16} \gamma^4.$$

On the canonical frame $\gamma^4 = T_{16}[4]$, so

$$G = -i \sigma_{16} T_{16}[4] = -i T_{16}[0] T_{16}[1] T_{16}[2] T_{16}[3] T_{16}[4] =: J.$$

G is Hermitian, $G^2 = \text{ID}_{16}$, with eigenvalues $+1$ (8 times) and -1 (8 times): the kinetic form is **indefinite**, signature (8,8). The canonical frame has $\Gamma_4 = 0$, so the time-derivative part, which alone determines the anticommutator, contains no spin connection. [proved VII §28]

- QUANTIZATION [THE RESULT]: the momenta of L_{sym} are $\text{Pi}_\Psi == (\text{Sqrt}[g]/(2H)) \text{Psibar} \gamma^4$ and $\text{Pi}_{\text{Psibar}} == -(\text{Sqrt}[g]/(2H)) \gamma^4 \Psi$
- QUANTIZATION [has content]: $\text{theta_un} - \text{theta_sym} == \text{delta}[(\text{Sqrt}[g]/(2H)) \text{Psibar} \gamma^4 \Psi]$ exactly: the same symplectic form, the same brackets
- QUANTIZATION [THE RESULT]: $G = -i \sigma_{16} \gamma^4 == J = -i T_{16}[0].T_{16}[1].T_{16}[2].T_{16}[3].T_{16}[4]$ on the canonical frame; J is Hermitian, $J^2 == \text{ID}_{16}$, eigenvalues $+1$ (8) and -1 (8): signature (8,8)
- QUANTIZATION [has content]: $\text{Gamma}_4 == 0$ - the time-derivative part of the Lagrangian, which fixes the anticommutator, contains no spin connection

3.5.3 The Dirac bracket and the canonical anticommutator

Write $\Psi = (\theta_1 + i\theta_2)/\sqrt{2}$ with real Grassmann θ_1, θ_2 and $\Theta = (\theta_1, \theta_2)$. In the Grassmann algebra of subsection 3.2.3, for $K = c \sigma_{16} T_{16}[4]$ with any scalar c ,

$$\Psi^\dagger K d\Psi = \frac{i}{2} \Theta^T \Omega_c d\Theta + \frac{1}{4} d(\theta_1^T K \theta_1 + \theta_2^T K \theta_2),$$

$$\Omega_c = \begin{pmatrix} 0 & K \\ -K & 0 \end{pmatrix} \quad (\text{real, symmetric, invertible; } \Omega_c^{-1} = \begin{pmatrix} 0 & -K^{-1} \\ K^{-1} & 0 \end{pmatrix}).$$

So up to a total time derivative the Lagrangian is the first-order system $\frac{i}{2} \Theta^T \Omega \Theta' - h$. Its momenta are proportional to Θ ; the constraints $\pi -$ (momentum as a function of Θ) are second

class with constraint matrix proportional to Ω ; and the Dirac bracket of the Θ 's is proportional to Ω^{-1} . The constant is fixed by the elementary case $\Omega = 1$, $\frac{i}{2} \xi \xi'$, whose quantization is $\{\xi, \xi\} = 1$. Hence

$$\begin{aligned}\{\Theta_i, \Theta_j\} &= (\Omega^{-1})_{ij}, \\ \{\Psi_a, \Psi_b^\dagger\} &= \frac{1}{2}[(\Omega^{-1})_{11} + (\Omega^{-1})_{22} - i(\Omega^{-1})_{12} + i(\Omega^{-1})_{21}]_{ab} = i(K^{-1})_{ab} = (\tilde{G}^{-1})_{ab}, \\ \{\Psi_a, \Psi_b\} &= 0.\end{aligned}$$

The same sign and normalization are confirmed independently, with no convention for graded Poisson brackets, by the Fock space of section 3.5.14: with this anticommutator the Heisenberg equation generated by the canonical Hamiltonian reproduces the field equation, and with the opposite sign it gives the time-reversed one. On the canonical frame $J^{-1} = J$ and $\sqrt{g} = \sec 6Hx_0$, so

$$\boxed{\{\Psi_a(x), \Psi_b^\dagger(y)\}_{x_4=y_4} = \frac{H}{\sqrt{g}} J_{ab} \delta^7(x-y) = H \cos(6Hx_0) J_{ab} \delta^7(x-y), \quad \{\Psi_a, \Psi_b\} = 0}$$

which is $\frac{H}{\sqrt{|g|}} (-i\sigma_{16}\gamma^4)_{ab}$, because $(-i\sigma_{16}T_{16}[4])^{-1} = -i\sigma_{16}T_{16}[4]$. The spin connection does not appear: the anticommutator comes from the time-derivative term alone, and $\Gamma_4 = 0$. [**proved VII §28**]

- DIRAC BRACKET [THE RESULT]: in the Grassmann algebra $\Psi_i \text{ddag } K \text{ phi} == (i/2) \text{Theta}^T \Omega_{\text{c}} \text{Phi} + (1/4) d(\text{theta1}^T K \text{theta1} + \text{theta2}^T K \text{theta2})$, for $K = c \text{ sigma16 } T_{16}[4]$ with a free scalar c
- DIRAC BRACKET [has content]: Ω_{c} is real symmetric and invertible, $\Omega_{\text{c}}^{-1} == [[0, -K^{-1}], [K^{-1}, 0]]$
- DIRAC BRACKET [THE RESULT]: $\{\text{Theta}_i, \text{Theta}_j\} = (\Omega_{\text{c}}^{-1})_{ij}$ gives $\{\Psi_i, \Psi_j \text{ddag}\} == i (K^{-1})_{ij} == (G_{\text{ilde}}^{-1})_{ij}$ and $\{\Psi_i, \Psi_j\} == 0$
- ANTICOMMUTATOR [THE RESULT]: on the canonical frame ($c = \text{Sqrt}[g]/H$) $\{\Psi_i, \Psi_j \text{ddag}\} == (H/\text{Sqrt}[g]) J == H \text{Cos}[6 H x_0] J$ ($J^{-1} == J$), which IS the design's $(H/\text{Sqrt}[g])(-i \text{sigma16 } \gamma^4)$

The exported anticommutator, with its matrix (provenance-latex/generated/fermion_fable_anticommutator.tex, unchanged):

$$\begin{aligned}\{\Psi_a(x), \Psi_b^\dagger(y)\}_{x_4=y_4} &= \frac{H}{\sqrt{|g|}} J_{ab} \delta^7(x-y) = H \cos(6Hx_0) J_{ab} \delta^7(x-y), \quad \{\Psi_a, \Psi_b\} = 0, \\ \{\Psi_a(x), \Psi_b^\dagger(y)\}_{x_4=y_4} &= \frac{H}{\sqrt{|g|}} \delta_{ab} \delta^7(x-y)\end{aligned}$$

with $\Psi^\dagger = \Psi^\dagger J$; the anticommutator matrix is $\frac{H}{\sqrt{|g|}} J = H \cos(6Hx_0) J$ with

$$J = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i \\ 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

3.5.4 The admissible truncation, V_{hid} , and the rescaled field

The admissible theory (section 3.5.10) is a **truncation**: a dimensional reduction to fields $\Psi = \Psi(x_0, x_1, x_2, x_3, x_4)$, independent of x_5, x_6, x_7 . It needs a **finite** hidden coordinate volume $V_{\text{hid}} = \int dx_5 dx_6 dx_7$. That is an assumption, and it has a consequence that must be stated: compact timelike directions contain closed timelike curves. In the truncation the delta function becomes $\delta^4(x_{0..3} - y_{0..3})$ and the factor $H/\sqrt{g}$ becomes $H/(V_{\text{hid}}\sqrt{g})$:

$$\{\Psi_a(x), \Psi_b^\dagger(y)\}_{x_4=y_4} = \frac{H}{V_{\text{hid}}\sqrt{|g|}} G_{ab} \delta^4(x_{0..3} - y_{0..3}), \quad G = -i \sigma_{16} \gamma^4.$$

For the rescaled field $\Psi' = \Psi/\sqrt{\sin 6Hx_0}$ of section 3.4.7 the factor is $H \cos(6Hx_0)/\sin(6Hx_0) = H \cot(6Hx_0)$. On a frame whose $\sqrt{g}$ depends on x_4 one quantizes $\chi = (\sqrt{g}/H)^{1/2} \Psi$, with $\{\chi, \chi^\dagger\} = G \delta^7$; on the canonical frame $\sqrt{g}$ does not depend on x_4 and the two coincide [**prose VII §28**]. [**proved VII §28**] ANTICOMMUTATOR [has content]: the rescaled field $\Psi' = \Psi/\text{Sqrt}[\text{Sin}[6 H x_0]]$ has $\{\Psi', \Psi'^{\text{ddag}}\} == H \text{Cot}[6 H x_0] J \delta^7$, and $\text{Sqrt}[g]$ is x_4 -independent (so quantizing $\chi = (\text{Sqrt}[g]/H)^{(1/2)} \Psi$ changes nothing on the canonical frame)

(For the time-dependent warped frames of the interacting system, Part VIII of the notebook, Section 32, proves that the rescaled field has an x_4 -independent anticommutator: RESCALING [THE RESULT]: $\{\Psi, \Psi^{\text{ddag}}\} == (H/\text{Sqrt}[|\det g|]) G$ on the warped frame (Section 28's `cfAntiPsiPsiDd` at `c = \text{Sqrt}[|\det g|]/H`), and for $\chi = \Psi/f$, $f = \text{Sqrt}[\text{Sin} (A^3 B^3 C)^{(-1/2)}]$: $\{\chi, \chi^{\text{ddag}}\} == H \text{Cot}[6 H x_0] G$ - the $\text{Sqrt}[\det g]$ cancels, independent of x_4 and of A, B, C and RESCALING [has content]: on the warped frame $G = -i \sigma_{16} \gamma^4$ is Section 28's `cfGK` ($= J$) - the same matrix on every member of the family ($\gamma^4 == T_{16}[4]$, `lapse 1`).

3.5.5 What J is, and the Hilbert adjoint

$J = -i T_{16}[0] \cdots T_{16}[4] = -i \Omega_4 T_{16}[0] = -i h_0$, with $\Omega_4 = T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4]$ of section 3.2.6. So J is one of the hidden **flavour** gammas: it lies in the commutant of the observer's Clifford algebra, and on $\mathbb{C}^{16} = \mathbb{C}^4(\text{observer}) \otimes \mathbb{C}^4(\text{flavour})$ it acts on the **flavour factor only**, where it has signature $(2, 2)$ (verified on the image of a rank-4 primitive idempotent of the observer's algebra). With $\beta := -i T_{16}[4]$ (Hermitian, $\beta^2 = 1$):

$$\sigma_{16} = J\beta, \quad \beta T_{16}[j] \beta = -T_{16}[j] \text{ and } T_{16}[j] \text{ Hermitian } (j = 0..3),$$

so β is the observer's Dirac-adjoint matrix. Define the **Hilbert adjoint** by the J -involution, $\Psi^\dagger := \Psi^\dagger J$. Then

$$\bar{\Psi} = \Psi^\dagger \sigma_{16} = \Psi^\dagger \beta \quad (\text{the standard 4D Dirac adjoint}),$$

$$s = \Psi^\dagger \beta \Psi,$$

$$\Psi^\dagger G \partial_4 \Psi = \Psi^\dagger \partial_4 \Psi \quad (\text{the time-kinetic term becomes } \frac{\sqrt{g}}{H} i \Psi^\dagger \partial_4 \Psi),$$

$$\{\Psi_a, \Psi_b^\dagger\} = \frac{H}{\sqrt{g}} \delta_{ab} \delta^7 \quad (\text{POSITIVE}).$$

So fermion fable is four flavours times a 4D Dirac field, and J -quantization replaces the classical flavour metric of signature $(2, 2)$ by the identity. The adjoint is **mode-independent**, and it must be defined this way: a mode-by-mode prescription is basis-dependent unless every mode is a J -eigenvector (section 3.5.13). [**proved VII §28**]

- J [THE RESULT]: $J == -i \Omega_4 T_{16}[0] == -i h_0$ is a hidden flavour gamma: it lies in the commutant of $T_{16}[1..4]$ (in the span of the hidden-gamma products of Section 26)

- J [has content]: on the flavour factor J has signature (2,2): cfIdemObs is a rank-4 projector commuting with J, and J restricted to its image has eigenvalues +1, +1, -1, -1
- J [THE RESULT]: sigma16 == J beta with beta = -i T16[4] Hermitian, beta^2 == 1; T16[0..3] Hermitian with beta T16[j] beta == -T16[j]: beta is the observer's Dirac adjoint
- J [THE RESULT]: with Psi^dagger := Psi^ddag J: Psibar = Psi^dagger beta (J sigma16 == beta), Psi^ddag G = Psi^dagger (J G == ID16), {Psi, Psi^dagger} == (H/Sqrt[g]) ID16 (G^-1 J == ID16): POSITIVE

J and β , as exported (provenance-latex/generated/fermion_fable_krein_J.tex, unchanged):

$$J = -i T_{16}[0] T_{16}[1] T_{16}[2] T_{16}[3] T_{16}[4] = -i \sigma_{16} T_{16}[4] = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\beta = -i T_{16}[4] = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -i \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & i & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \sigma_{16} = J\beta, \quad J^2 = \beta^2 = 1, \quad \bar{\Psi} = \Psi^\dagger \beta.$$

3.5.6 Why this is legitimate for fermions, and why it would fail for bosons

For Grassmann fields the *-structure is a free choice: the Berezin integral treats Ψ and $\bar{\Psi}$ as independent, and the J -involution is that choice. The filled Dirac sea makes the Hamiltonian bounded below. For a **commuting** 16-component field the same involution gives $[b, b^\dagger] = 1$, but the Hamiltonian $\sum_u \omega_u b_u^\dagger b_u$ then has 8 negative-frequency modes per momentum and no exclusion principle to fill them: it is unbounded below. The only local involutions compatible with a Hermitian s and a Hermitian Hamiltonian are $+J$ and $-J$ (the uniqueness theorem, section 3.5.10), and $-J$ gives every mode negative norm; making the bosonic Hamiltonian bounded below would turn 4 modes into negative-norm modes. For fermions the negative-frequency modes are filled and the obstruction disappears [prose VII §28; design review QK-5].

The J -odd bilinears ($\hat{T}_{\mu h}, j^h, \bar{\Psi} T_{16}[8] \Psi$) become **anti-Hermitian** operators (section 3.5.8). So the classical real theory of Part VI is the classical limit of the J -**even** sector only. (A reflection-positivity claim that appeared in the design was dropped by the review: it was not verified.)

3.5.7 Chirality, and why V depends on s only

J anticommutes with $T_{16}[8]$: J exchanges the two split-octonion types. Each chirality subspace is G -null ($P_L G P_L = P_R G P_R = 0$: the Krein form pairs type-1 with type-2 only), so the J -vacuum

is not chirality-graded; but the two chirality subspaces **are** orthogonal in the Hilbert product ($P_L P_R = 0$, with P_L, P_R Hermitian). Neither $p = \bar{\Psi} T_{16}[8] \Psi$ (anti-Hermitian under J) nor ip (anti-Hermitian under the classical conjugation $\dagger$, so it would make the Lagrangian complex) can enter V consistently: $V = V(s)$. [proved VII §28] J [has content]: chirality - $\{J, T_{16}[8]\} = 0$, each chirality subspace is G-NULL (PL G PL == PR G PR == 0), and the two chiralities are orthogonal in the Hilbert product (PL, PR Hermitian, PL PR == 0)

3.5.8 The Hermiticity rule and the J table

Under the J -adjoint, a bilinear $\Psi^\dagger M \Psi$ with M classically Hermitian has $(\Psi^\dagger M \Psi)^\dagger = \Psi^\dagger (J M J) \Psi$: it is **Hermitian** iff $[J, M] = 0$ and **anti-Hermitian** iff $\{J, M\} = 0$. The table records J against every matrix that occurs [proved VII §28]:

matrix	relation to J	consequence
σ_{16}	commutes	s is Hermitian
$T_{16}[0], \dots, T_{16}[4]$	commute	
$T_{16}[5], T_{16}[6], T_{16}[7]$	anticommute	
$\sigma_{16} \gamma^\mu, \mu = 0..4$	commute	$j^\mu, \mu = 0..4$, Hermitian
$\sigma_{16} \gamma^h, h = 5, 6, 7$	anticommute	j^h anti-Hermitian
each $\gamma^\mu \Gamma_\mu$ (no sum), $\mu = 0..7$	commute	the connection term of the Dirac operator is J -even
Γ_0, Γ_4 (both zero), $\Gamma_1, \Gamma_2, \Gamma_3$	commute	
$\Gamma_5, \Gamma_6, \Gamma_7$	anticommute	boosts mixing x_0 or x_4 with the hidden timelike sheet
$T_{16}[8]$ and $C_+ = \sigma_{16} T_{16}[8]$	anticommute	$p = \bar{\Psi} T_{16}[8] \Psi$ is anti-Hermitian

- J TABLE [THE RESULT]: J commutes with σ_{16} , $T_{16}[0..4]$, $\sigma_{16} \gamma^\mu$ ($\mu = 0..4$), every $\gamma^\mu \Gamma_\mu$ and $\Gamma_{1..3}$; it ANTICOMMUTES with $T_{16}[5..7]$, $\sigma_{16} \gamma^h$ ($h = 5..7$), $\Gamma_{5..7}$, $T_{16}[8]$ and C_+
- J [has content]: the Hermiticity rule - for classically Hermitian M , $(J M)^\dagger = J M$ iff $[J, M] = 0$ and $= -J M$ iff $\{J, M\} = 0$: so $s = \Psi^\dagger \sigma_{16} \Psi$ is Hermitian and $p = \Psi^\dagger C_+ \Psi$ is ANTI-Hermitian

Consequences: s is Hermitian and p anti-Hermitian (so V may depend on s but not on p); the hidden current components j^h and the energy-momentum components $\hat{T}_{\mu h}$ ($\mu = 0..4$) are built from anticommuting matrices and are anti-Hermitian: their expectation values are purely imaginary in every state and vanish in every state invariant under the hidden rotations (the J -vacuum and the Kohn-Sham states of section 3.7), consistent with $G_{\mu h} = 0$ on the canonical frame. $\hat{T}_{hh'}$ ($h \neq h'$) vanishes identically for fields independent of $x_5..x_7$, and $\hat{T}_{hh} = g_{hh} \hat{L}$ is Hermitian (section 3.6.5).

(The same J table holds on the dynamical frames of the interacting system; Part VIII, Section 32, proves it: J [THE RESULT]: $[J, \gamma^\mu \Gamma_\mu] = 0$ for EACH μ separately ($h = 5, 6, 7$ included), on the warped and on the Bianchi-I frame and J [THE RESULT]: $[J, \Gamma_\mu] = 0$ for $\mu = 0..4$ and $\{J, \Gamma_h\} = 0$ for $h = 5, 6, 7$, on the warped and on the Bianchi-I frame.)

3.5.9 The symmetry that survives, and what is lost

Of the 28 Lorentz generators S_{ab} , **13 commute** with J : Spin(4,1) of $x_0..x_4$ (10 generators) and Spin(3) of $x_5..x_7$ (3). The other **15 anticommute**: the 12 boosts between $x_0..x_3$ and $x_5..x_7$, and the 3 **rotations** between x_4 and $x_5..x_7$ ($x_4..x_7$ are all timelike, so those planes are rotations). The Krein form G itself is invariant ($S^T G + G S = 0$) under exactly the 21 generators with

$a, b \neq 4$: $\text{Spin}(4,3)$, the group of the slice. [proved VII §28] (The J structure is defined in the diagonal gauge of the canonical frame and transforms as $J \rightarrow S^{-1}JS$ under a change of frame gauge S ; design review QK-6.)

- J [THE RESULT]: 13 of the 28 Lorentz generators commute with J (a, b both in $\{0..4\}$: $\text{Spin}(4,1)$, or both in $\{5,6,7\}$: $\text{Spin}(3)$); the other 15 anticommute - 12 boosts $x_0..x_3 \leftrightarrow x_5..x_7$ and 3 rotations $x_4 \leftrightarrow x_5..x_7$
- J [has content]: the Krein form G is invariant ($S^T G + G S = 0$) under exactly the 21 generators with $a, b \neq 4$ - $\text{Spin}(4,3)$, the group of the slice

3.5.10 The admissible sector: a theorem

Take a plane wave at momentum k (flat local analysis, the frame factors frozen). The one-particle Hamiltonian is $h_k = \sigma_{16} (m - i \gamma \cdot k)$, with $\gamma \cdot k = \sum_{a \neq 4} k_a T_{16}[a]$, and the evolution generator is $E_k = G^{-1}h_k = G h_k$, $i \partial_4 \Psi = E_k \Psi$. Write $k_s^2 = k_0^2 + k_1^2 + k_2^2 + k_3^2$ and $k_h^2 = k_5^2 + k_6^2 + k_7^2$.

1. For **every** k , $E_k^2 = (k_s^2 - k_h^2 + m^2)$ ID16: the dispersion relation is $\omega^2 = k_s^2 - k_h^2 + m^2$.
2. **Admissible** k ($k_h = 0$): $[J, E_k] = 0$ and J itself gives the positive quantization ($GJ = \text{ID16}$).
3. Hidden momentum **below** threshold ($k_h^2 < k_s^2 + m^2$): a momentum-dependent, boosted $J' = S^{-1}JS$ (S the spin boost in the plane of k_s and k_h that removes k_h) commutes with E_k and with β , $J'^2 = 1$, and $GJ' = S^2$ is positive definite: each such momentum can be quantized positively on its own (checked at $k = 5e_1 + 3e_5$, $m = 3$, $\omega^2 = 25 - 9 + 9 = 25$, boost with $\tanh \theta = 3/5$).
4. **At** threshold E_k is not zero but $E_k^2 = 0$: E_k is nilpotent and cannot be diagonalized (Jordan blocks of size 2, rank 8); there is no mode expansion (checked at $k = 4e_1 + 5e_5$, $m = 3$).
5. **Above** threshold $\omega^2 < 0$: the frequencies are imaginary (growth along x_4 , the ill-posed ultrahyperbolic Cauchy problem) and the modes are **G -neutral** ($u^\dagger G u = 0$), so they carry no norm at all. Example: $k = (0, \frac{3}{4}, 1, 0 \mid k_5 = 2)$, $m = 1$: $\omega^2 = -23/16$, $\omega = \pm i\sqrt{23}/4 = \pm 1.19896 i$.
6. **No** single momentum-independent positive J' exists once a hidden momentum is present. If $V(s)$ and the local energy-momentum tensor are to be Hermitian, J' must commute with β (because $s = \Psi^\dagger G \beta \Psi$) and with E_k for all k . Certificate: for the four admissible unit momenta and one hidden momentum, **every** X commuting with β and all five E_k has $\text{Tr}(GX) = 0$, so GX is never positive definite. Analytic proof for a purely hidden momentum: $[E_k, \beta] = 2i k_5 T_{16}[5]$, so J' commutes with $T_{16}[5]$; $T_{16}[5]$ anticommutes with G , hence $T_{16}[5]^\dagger (GJ') T_{16}[5] = -(GJ')$: a positive matrix would be congruent to a negative one.
7. J is **unique**: the joint commutant of β and all admissible E_k is 8-dimensional and every one of its elements commutes with J ; if X is in it with $X^2 = 1$ and $GX = JX$ definite, then $(JX)^2 = 1$, and a definite Hermitian matrix with square 1 is $\pm \text{ID16}$: $X = J$ (positive metric) or $X = -J$ (every mode of negative norm).

The general identity behind (6) is $[E_k, \beta] = 2i T_k$, $T_k = \sum_a k_a T_{16}[a]$, for every k .

THEOREM (admissibility). Let W be the linear span of the momenta carried by the modes, a subspace of the slice $\mathbb{R}^{(4,3)}$. A positive (Hilbert-space) Fock quantization in which the local $s(x)$, the local energy-momentum tensor and the free Hamiltonian are all Hermitian exists **if and only if** W is **spacelike-definite**. Then $J' = S^{-1}JS$ with S a slice boost carrying W into $\text{span}(x_0..x_3)$, and J' is unique (with $J'^2 = 1$) when $W = \text{span}(x_0..x_3)$. On the canonical frame

the fields depend on all of $x_0 \dots x_3$, so the condition is: **no momentum along** x_5, x_6, x_7 , and $J = -i T_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4]$ is unique.

The admissible theory is a **truncation** (a dimensional reduction $\Psi = \Psi(x_0, \dots, x_4)$), not a superselection sector: $V(s)$ contains terms like $b_{k_h}^\dagger b_{-k_h}^\dagger b_0 b_0$ that would connect it to hidden momenta. What survives of the multi-time (Tomonaga–Schwinger) picture is the isometry fact: the metric, the frame and the connection do not depend on x_5, x_6, x_7 , so the hidden translations commute with the x_4 -evolution, $[P_4, P_h] = 0$ [prose VII §28]. [proved VII §28]

- ADMISSIBILITY (1) [THE RESULT]: $E_k^2 == (k_s^2 - k_h^2 + m^2)$ ID16 for every k : $\omega^2 = k_s^2 - k_h^2 + m^2$
- ADMISSIBILITY (2) [THE RESULT]: for admissible k ($k_h = 0$), $[J, E_k] == 0$, E_k is J-Hermitian, and $G J == \text{ID16}$ is positive: J quantizes every admissible momentum positively
- ADMISSIBILITY (3) [THE RESULT]: below threshold ($k = 5 \text{ e1} + 3 \text{ e5}$, $m = 3$) the boosted $J' = S^{-1} J S$ commutes with E_k and β , $J'^2 == 1$, and $G J' == S^2$ is Hermitian positive definite
- ADMISSIBILITY (4) [THE RESULT]: AT threshold ($k = 4 \text{ e1} + 5 \text{ e5}$, $m = 3$) $E_k \neq 0$ but $E_k^2 == 0$: nilpotent, not diagonalizable (rank 8: Jordan blocks of size 2)
- ADMISSIBILITY (5) [THE RESULT]: ABOVE threshold ($k = (0, 3/4, 1, 0 \mid k_5 = 2)$, $m = 1$) $\omega^2 == -23/16$: every eigenvalue of E_k is $\pm i \sqrt{23}/4 = \pm 1.19896 i$ and every eigenvector is G-NEUTRAL ($u^\dagger G u == 0$)
- ADMISSIBILITY [has content]: the general identity $[E_k, \beta] == 2 i T_k$, $T_k = \sum_a k_a T_{16}[a]$, for every k (admissible and hidden components alike)
- ADMISSIBILITY (6) [THE RESULT]: traceless certificate - every X commuting with β , the four admissible E_k and one hidden-momentum E_k has $\text{Tr}(G X) == 0$, so no positive $G J'$ exists; J itself is not in that commutant
- ADMISSIBILITY (6) [THE RESULT]: the analytic proof for a purely hidden momentum - $[E_k, \beta] == 2 i k_5 T_{16}[5]$; $T_{16}[5]$ anticommutes with G and $T_{16}[5]^\dagger G T_{16}[5] == -G$ (so a J' commuting with $T_{16}[5]$ has $T_5^\dagger (G J') T_5 == -(G J')$)
- ADMISSIBILITY (7) [THE RESULT]: J is UNIQUE - the joint commutant of β and the admissible E_k is 8-dimensional, contains J , and every element of it commutes with J (so $X^2 == 1$ with $J X > 0$ forces $X == J$)
- ADMISSIBILITY [has content]: $[P_4, P_h] == 0$ - the metric, the frame and the spin connection do not depend on x_5, x_6, x_7

The theorem statement combines items (1)–(7); the items are asserted individually, the “if and only if” is their combination [prose VII §28].

3.5.11 The Hamiltonian on the canonical frame, and the spin connection

The Hamiltonian **density** of the symmetrized Lagrangian, $\Pi_\Psi \partial_4 \Psi + \partial_4 \bar{\Psi} \Pi_\Psi - L$, is

$$\sqrt{g} \left[-\frac{1}{2H} \sum_{k \neq 4} (\bar{\Psi} \gamma^k \partial_k \Psi - \partial_k \bar{\Psi} \gamma^k \Psi) + V(s) \right]$$

with **no spin connection** in it, for fields of all eight coordinates. (This corrects design revision 1, which said that the connection “enters the Hamiltonian”.) Varying it gives the one-particle

operator. In the admissible sector ($\Psi = \Psi(x_0, \dots, x_4)$) the field equation is $i \tilde{G} \partial_4 \Psi = h \Psi$ with

$$h \Psi = \sqrt{g} \sigma_{16} \left[V'(s) \Psi - \frac{1}{H} \left(\sum_{j=0}^3 \gamma^j \partial_j \Psi + \sum_{\mu} \gamma^{\mu} \Gamma_{\mu} \Psi \right) \right]$$

($V'(s)$ is the mean-field $V'(\langle s \rangle)$ here). The connection appears here, through the equation of motion, and it has a job: since $\frac{1}{2} \sum_{\mu \neq 4} \partial_{\mu} (\sqrt{g} \sigma_{16} \gamma^{\mu}) = \sqrt{g} \sigma_{16} \gamma^{\mu} \Gamma_{\mu}$ and $\partial_4 (\sqrt{g} \sigma_{16} \gamma^4) = 0$, the derivative part of h is $-\frac{1}{H} \sum_j (A_j \partial_j + \frac{1}{2} \partial_j A_j)$ with $A_j = \sqrt{g} \sigma_{16} \gamma^j$ real **antisymmetric**, and for such an operator $\langle f, h g \rangle - \langle h f, g \rangle$ is a total derivative. The connection term is exactly the term that makes the one-particle Hamiltonian Hermitian. [proved VII §28]

- HAMILTONIAN [THE RESULT]: the Hamiltonian density $\text{Pi_Psi } d_4 \text{ Psi} + d_4 \text{ Psibar Pi_Psibar} - L$ of the symmetrized Lagrangian is $\text{Sqrt}[g] [-(1/(2H)) \text{Sum}_{\{k \neq 4\}} (\text{Psibar } \gamma^k d_k \text{ Psi} - d_k \text{ Psibar } \gamma^k \text{ Psi}) + V]$: NO spin connection (fields of all eight coordinates)
- HAMILTONIAN [THE RESULT]: $(\text{Sqrt}[g]/H) \text{sigma16} (\gamma^{\mu} D_{\mu} \text{ Psi} - H V' \text{ Psi}) == i \text{Gtilde } d_4 \text{ Psi} - h \text{ Psi}$ for admissible $\text{Psi}(x_0..x_4)$: the field equation is $i \text{Gtilde } d_4 \text{ Psi} = h \text{ Psi}$
- HAMILTONIAN [THE RESULT]: $(1/2) \text{Sum}_{\{\mu \neq 4\}} d_{\mu} (\text{Sqrt}[g] \text{sigma16 } \gamma^{\mu}) == \text{Sqrt}[g] \text{sigma16 } \text{Sum } \gamma^{\mu} \Gamma_{\mu}$, and $d_4 (\text{Sqrt}[g] \text{sigma16 } \gamma^4) == 0$: the connection term is the one that makes h Hermitian
- HAMILTONIAN [has content]: $A_0 = \text{Sqrt}[g] \text{sigma16 } \gamma^0$ is real antisymmetric, and for $h_0 = -(A_0 d_0 + (1/2) A_0')$: $\langle f, h_0 g \rangle - \langle h_0 f, g \rangle == -d_0 (f^{\dagger} g)$, a pure boundary term

J commutes with h : every matrix in h (σ_{16} , $\sigma_{16} \gamma^j$ for $j = 0..3$, $\sigma_{16} \gamma^{\mu} \Gamma_{\mu}$) commutes with J , for every a_4 , every x_0 -profile and every V . A hidden derivative term would anticommute (control). The x_4 -dependence of h enters only through the observed frame factor $1/q$; it mixes positive and negative frequencies (a **Bogoliubov** transformation, particle production by the x_4 -dependence of a_4) **within** the admissible sector, and since it commutes with J that mixing is unitary on the Fock space. [proved VII §28]

- HAMILTONIAN [THE RESULT]: $[J, h] == 0$ on the admissible sector, for every a_4 , every x_0 -profile and every V' - including the spin-connection term
- HAMILTONIAN [control]: a hidden derivative term breaks it - $[J, h] \neq 0$ once Psi depends on x_5 (witnessed on $\text{Psi} = x_5 e_1$)
- HAMILTONIAN [has content]: the x_4 -dependence of h enters only through $1/q$: $d_4 h \text{ Psi} == (H a_4' [H x_4]) \times (\text{the observed-direction part of } h)$, which commutes with J

3.5.12 The x_0 direction and the boundary condition at the wall

For the rescaled field $\Psi' = \Psi / \sqrt{\sin 6Hx_0}$ the measure becomes flat in the proper distance $z = -\ln(\cos 6Hx_0)/(6H)$ along x_0 :

$$\sqrt{g} dx_0 |\Psi|^2 = |\Psi'|^2 dz \quad (\sec \cdot \sin = \tan = dz/dx_0).$$

z runs over $[0, \infty)$ as $6Hx_0$ runs over $[0, \pi/2)$; the x_0 part of the rescaled Dirac operator is exactly $T_{16}[0] \partial_z$, and the frame factors are bounded, $\sin(6Hx_0) = \sqrt{1 - e^{-12Hz}}$. So the one-particle space is $L^2(dz)$. The operator is **regular** at the wall $z = 0$, so a boundary condition is needed there, and **limit-point** at $z \rightarrow \infty$ (by the standard theorem that a one-dimensional Dirac system with locally integrable coefficients is always limit-point at an infinite endpoint), so no condition

is needed or allowed there. The condition at the wall, in this real-Clifford convention, is the bag-type

$$T_{16}[0] \Psi' = +\Psi' \quad \text{or} \quad T_{16}[0] \Psi' = -\Psi' \quad \text{at } z = 0 \quad (\text{no factor } i, \text{ because } T_{16}[0]^2 = +1).$$

It kills the normal flux $\Psi'^\dagger \beta T_{16}[0] \Psi'$ (because $\{\beta, T_{16}[0]\} = 0$) and commutes with J . It is part of the canonical quantization. [proved VII §28] It also gives $\bar{\Psi}\Psi = 0$ at the wall [derived here: $\sigma_{16} = J\beta$ anticommutes with $T_{16}[0]$, because J commutes and β anticommutes with it, so $P\sigma_{16}P = 0$ on either eigenspace P of $T_{16}[0]$; design review QK-9].

- BOUNDARY [has content]: the x0 direction in proper distance $z = -\text{Log}[\text{Cos}[6 H x0]]/(6 H)$: $dz/dx0 == \text{Tan}[6Hx0] == \text{Sec Sin}$ (so $\text{Sqrt}[g] dx0 |\Psi|^2 = |\Psi'|^2 dz$), $z(0) == 0$ and $z \rightarrow \text{infinity}$ at $6 H x0 \rightarrow \text{Pi}/2$, $\gamma^0 d_0 == T_{16}[0] d_z$, and $\text{Sin}[6Hx0] == \text{Sqrt}[1 - \text{Exp}[-12 H z]]$ (bounded frame factors)
- BOUNDARY [THE RESULT]: the bag condition $T_{16}[0] \Psi' = \pm \Psi'$ at the wall - $T_{16}[0]^2 == 1$, $\{\beta, T_{16}[0]\} == 0$, the normal-flux form vanishes on both eigenspaces ($P \beta T_{16}[0] P == 0$), and $[J, T_{16}[0]] == 0$

(The full family of self-adjoint, Lorentz-covariant, J -sector-preserving wall conditions is $P = T_{16}[0] Q$, with Q any Hermitian involution in the 8-dimensional commutant of $T_{16}[0..4]$, which contains J ; the uniform choices $Q = \pm 1$ are the two above. That refinement belongs to the treatment of the states along the hidden coordinate in the coupled system; it was adopted by the design review, DFT-8, and is not asserted in Part VII.)

3.5.13 Krein modes, and why the mode basis must consist of J -eigenvectors

For an admissible momentum $k = (k_0, k_1, k_2, k_3)$ and mass m (flat local analysis; on the canonical frame k is the physical momentum, the coordinate momentum along $x_1..x_3$ divided by q), put $A = Gh = E_k$ and $w = \sqrt{k \cdot k + m^2}$. Then $A^2 = w^2$, $[A, J] = 0$, A is Hermitian, and the four Hermitian projectors

$$P_{s,\eta} = \frac{1}{4} (\text{ID16} + s A/w) (\text{ID16} + \eta J), \quad s, \eta = +1, -1,$$

have rank 4 each (trace 4), are mutually orthogonal, sum to ID16, and $\sum \eta P_{s,\eta} = J = G^{-1}$: the **completeness relation** $\sum_u \eta_u u u^\dagger = G^{-1}$ of any J -orthonormal mode basis. So each frequency $\pm w$ is 8-fold degenerate, and the G -form restricted to each frequency eigenspace has signature (4,4). For **every** mode u in the image of $P_{s,\eta}$, normalized by $u^\dagger G u = \eta$, the projector identities $PXP = cP$ give, exactly and for arbitrary k and m ,

$$\begin{aligned} \eta u^\dagger h u &= s w = \omega_u & (\text{energy} = \text{frequency}), \\ \eta u^\dagger \sigma_{16} u &= m/\omega_u & (\text{scalar density}), \\ \eta u^\dagger (\partial h / \partial k_j) u &= k_j/\omega_u = \partial \omega_u / \partial k_j & (\text{velocity: Hellmann-Feynman}), \end{aligned}$$

with $\partial h / \partial k_j = -i \sigma_{16} T_{16}[j]$. The identities are proved with w a symbol and w^2 reduced to $k \cdot k + m^2$ (exact polynomial remainder); nothing is numerical. [proved VII §28]

- KREIN MODES [has content]: $A = G h$ satisfies $A^2 == w^2 \text{ID16}$ and $[A, J] == 0$, and A is Hermitian (G and h commute) for every admissible k and m
- KREIN MODES [THE RESULT]: the four $P_{\{s,\eta\}}$ are projectors of trace (rank) 4, mutually orthogonal, summing to ID16, and $\sum \eta P_{\{s,\eta\}} == J == G^{-1}$ (completeness)
- KREIN MODES [THE RESULT]: for every mode, $\eta u^\dagger h u == s w$ (energy = frequency), $\eta u^\dagger \sigma_{16} u == m/\omega_u$, $\eta u^\dagger (\partial h / \partial k_j) u == k_j/\omega_u$: $PXP == cP$ with $c = \eta s w$, $\eta s m/w$, $\eta s k_j/w$

- KREIN MODES [has content]: $\text{dh/dk}_j == -i \text{ sigma16 T16[j]}$, and $\text{d omega/d k}_j == \text{k}_j/\text{omega}$ for $\text{omega} = +-w$
- KREIN MODES [THE RESULT]: every mode of $P_{\{s,\text{eta}\}}$ is a J -eigenvector: $J P_{\{s,\text{eta}\}} == \text{eta} P_{\{s,\text{eta}\}}$

The mode basis must consist of J -eigenvectors. The Krein prescription $b_u^\dagger := \eta_u b_u^\dagger$ depends on the basis. It reproduces the J -involution $\Psi^\dagger = \Psi^\dagger J$ only if every mode is a J -eigenvector, $Ju = \eta_u u$, which is what the $P_{s,\eta}$ deliver ($JP_{s,\eta} = \eta P_{s,\eta}$), and which is possible because $[J, Gh_k] = 0$ for momenta along $x_0..x_3$. A G -orthonormal basis mixed by a Krein boost inside one frequency eigenspace still diagonalizes the evolution, but it induces the involution $J_U = G U U^\dagger \neq J$, and under it s is **not** Hermitian (control, exact, at $k = 3e_1$, $m = 4$, with the boost $\cosh = 5/4$, $\sinh = 3/4$). A generic generalized-eigenvector routine does not choose J -eigenvectors; the Fock construction of section 3.5.14 does. [proved VII §28] KREIN MODES [control]: a Krein-boosted pair $u_1 = (5/4)u_+ + (3/4)u_-$, $u_2 = (3/4)u_+ + (5/4)u_-$ in the $\text{omega} = +5$ eigenspace is still G -orthonormal and still consists of modes, but not of J -eigenvectors; the induced $J_U = G U U^\dagger$ differs from J , and under it s is NOT Hermitian

3.5.14 A finite Fock space at one momentum (Jordan–Wigner): the Dirac sea, 8 particles and 8 antiparticles

At the momentum $k = 3e_1$ with $m = 4$ (so $\omega = 5$ exactly) the sixteen modes of one momentum are represented on $\mathbb{C}^{2^{16}} = \mathbb{C}^{65536}$ by Jordan–Wigner operators $f_1..f_{16}$ (`SparseArray`, exact integers), which obey the positive CAR $\{f_i, f_j^\dagger\} = \delta_{ij}$, $\{f_i, f_j\} = 0$. Two realizations are checked.

Field level. $\Psi_a := f_a$ (the factor $H/\sqrt{g}$ set to 1), $\Psi^\dagger = f^\dagger$, $\Psi^\ddagger := \Psi^\dagger J$. Then:

- $\{\Psi_a, \Psi_b^\ddagger\} = J_{ab} = (G^{-1})_{ab}$: exactly the Dirac-bracket anticommutator, realized on a positive Fock space;
- the free Hamiltonian $H_{\text{free}} = \Psi^\ddagger h \Psi = \Psi^\dagger (Jh) \Psi$ and $s = \Psi^\ddagger \sigma_{16} \Psi = \Psi^\dagger \beta \Psi$ are Hermitian operators, and so is the interacting $H_{\text{free}} + \frac{3}{7} s^2$; $p = \Psi^\ddagger C_+ \Psi$ is anti-Hermitian (control); the one-body matrix Jh has eigenvalues $+5$ and -5 , eight times each;
- the **Heisenberg equation** holds: $[H_{\text{free}}, \Psi_a] = -(G^{-1}h\Psi)_a$ for all 16 components, i.e. $i \partial_4 \Psi = i [H, \Psi]$ reproduces $iG \partial_4 \Psi = h\Psi$;
- with the **opposite** sign of the anticommutator ($\Psi^\ddagger := -\Psi^\dagger J$) the same construction gives $H' = -H_{\text{free}}$ and the time-**reversed** equation (control). This is the convention-free confirmation of the sign derived with the Dirac bracket.

Mode level. $b_u := f_u$ for the sixteen J -eigenmodes of section 3.5.13 ($u = 1..8$: $\omega = +5$; $u = 9..16$: $\omega = -5$; $Ju = \eta_u u$ with $\eta = +1$ for four of each and -1 for the other four):

- the Krein conjugate $b_u^\ddagger := \eta_u b_u^\dagger$ obeys $\{b_u, b_v^\ddagger\} = \eta_u \delta_{uv}$, the indefinite CAR that the field anticommutator demands through $\sum_u \eta_u u u^\dagger = G^{-1}$, and the J -involution $b^\ddagger = \eta b^\dagger$ restores $\{b, b^\dagger\} = 1$;
- the naive Krein “norm” $\langle 0 | b_u b_u^\ddagger | 0 \rangle = \eta_u$ is **negative** for $\eta = -1$: that is the indefinite metric the J -involution removes (the Fock norm $\langle 0 | b_u b_u^\dagger | 0 \rangle = 1$);
- $\text{Ham} = \sum_u \omega_u \eta_u b_u^\ddagger b_u = \sum_u \omega_u b_u^\dagger b_u$, and $[\text{Ham}, b_u] = -\omega_u b_u$ ($b_u(x_4) = b_u e^{-i\omega_u x_4}$, the time dependence of the classical mode);
- the spectrum on the 65536 states: a **unique ground state** at $-8\omega = -40$, the **Dirac sea** (all eight negative-frequency modes filled); the normal-ordered $:\text{Ham}: = \text{Ham} + 40 \geq 0$;

- the first excited level, $\omega = 5$ above the sea (energy -35), is **16-fold: 8 particle states** (one positive-frequency mode added, $N = +1$) and **8 antiparticle states** (one negative-frequency mode emptied, a hole, $N = -1$), with $N = \sum_u b_u^\dagger b_u - 8$;
- in every Fock basis state tested (the sea and the 16 first excited states) the expectation of $b_u^\dagger b_v$ is $\eta_u n_u \delta_{uv}$: the rule $\langle \Psi^\dagger M \Psi \rangle = \sum_{\text{occupied}} \eta_u u^\dagger M u$ used in section 3.7.

[proved VII §28]

- FOCK [solver regression]: the Jordan-Wigner operators obey the positive CAR $\{f_i, f_j^\dagger\} == \delta_{ij}$, $\{f_i, f_j\} == 0$ on C⁶⁵⁵³⁶
- FOCK (field level) [THE RESULT]: $\{\Psi_a, \Psi_{ddag_b}\} == J_{ab} == (G^{-1})_{ab}$ with $\Psi = f$, $\Psi_{ddag} = f^\dagger J$ - the Dirac-bracket anticommutator, realized on a positive Fock space
- FOCK (field level) [THE RESULT]: J h is Hermitian with eigenvalues $+5$ (8) and -5 (8); H_{free} and s are Hermitian operators, so is $H_{\text{free}} + (3/7) s^2$; $p = \Psi_{ddag} C \Psi$ is ANTI-Hermitian
- FOCK (field level) [THE RESULT]: Heisenberg - $[H_{\text{free}}, \Psi_a] == -(G^{-1} h \Psi)_a$ for all 16 components, so $i d_4 \Psi = i [H, \Psi]$ IS the field equation $i G d_4 \Psi = h \Psi$
- FOCK (field level) [control]: with the opposite sign of the anticommutator the Hamiltonian is $-H_{\text{free}}$ and $[H', \Psi_a] == +(G^{-1} h \Psi)_a$: the time-REVERSED equation
- FOCK (mode level) [THE RESULT]: $\{b_u, b_v^\dagger\} == \eta_u \delta_{uv}$ (the indefinite CAR demanded by $\sum \eta_u u u^\dagger = G^{-1}$), and the J-involution $b^\dagger = \eta b_{ddag}$ restores $\{b, b^\dagger\} == 1$
- FOCK (mode level) [has content]: the naive Krein 'norm' $\langle 0 | b_u b_u^\dagger | 0 \rangle$ equals η_u - NEGATIVE for the four $\eta = -1$ modes of each frequency - while the Fock norm $\langle 0 | b_u b_u^\dagger | 0 \rangle == 1$
- FOCK (mode level) [THE RESULT]: $\sum \omega_u \eta_u b_u^\dagger b_u == \sum \omega_u b_u^\dagger b_u$, and $[Ham, b_u] == -\omega_u b_u$ for every u ($b_u(x_4) = b_u e^{-i \omega_u x_4}$)
- FOCK (mode level) [THE RESULT]: the spectrum - a UNIQUE ground state at -8 $\omega = -40$, the Dirac sea (modes 9..16 filled, 1..8 empty); $:Ham: = Ham + 40 \geq 0$
- FOCK (mode level) [THE RESULT]: the first excited level, $\omega = 5$ above the sea, is 16-fold: 8 particle states ($N = +1$) and 8 antiparticle states ($N = -1$), with $N = \sum b^\dagger b - 8$
- FOCK (mode level) [THE RESULT]: in every Fock basis state tested (the sea and the 16 first excited states) $\langle n | b_u^\dagger b_v | n \rangle == \eta_u n_u \delta_{uv}$ - the expectation rule of Section 29

The Jordan-Wigner CAR check on the 65536-dimensional space took 0.749 s and the Heisenberg check 1.868 s; the whole Fock cell (Input cell 191) took 9.858 s (claude-fable/run_fermion_fable_part7.log, lines [0.749 s], [1.868 s], CELL 191 t=9.858 s).

3.5.15 The particle number

The particle-number density is

$$n^4 = -\frac{i}{H} \bar{\Psi} \gamma^4 \Psi = \frac{1}{H} \Psi^\dagger G \Psi = \frac{1}{H} \Psi^\dagger \Psi \geq 0$$

and its mode form $\sum_u \eta_u b_u^\dagger b_u = \sum_u b_u^\dagger b_u$ is the particle number (normal-ordered: particles minus antiparticles). [proved VII §28] CHARGE [THE RESULT]: $n^4 = -(i/H) \bar{\Psi} \gamma^4 \Psi = (1/H) \bar{\Psi} \gamma^4 \Psi$, whose mode form $\sum_u \eta_u b_u^\dagger b_u = \sum_u b_u^\dagger b_u$ is the particle number (normal-ordered: particles - antiparticles)

3.5.16 The dynamical-frame field χ

On a frame whose volume factor depends on x_4 the field to quantize is $\chi = (\sqrt{g}/H)^{1/2} \Psi$, with $\{\chi, \chi^\dagger\} = G \delta^7$ (section 3.5.4). On the canonical frame $\sqrt{g} = \sec 6Hx_0$ does not depend on x_4 , and quantizing χ changes nothing (ANTICOMMUTATOR [has content]: the rescaled field $\Psi' = \Psi/\sqrt{\sin[6Hx_0]}$ has $\{\Psi', \Psi'^\dagger\} = H \cot[6Hx_0] \delta^7$, and $\sqrt{g}$ is x_4 -independent (so quantizing $\chi = (\sqrt{g}/H)^{1/2} \Psi$ changes nothing on the canonical frame)). On the time-dependent warped frames of the interacting system (Part VIII), the rescaled $\chi = \Psi/(\sqrt{\sin 6Hx_0} (A^3 B^3 C)^{-1/2})$ has $\{\chi, \chi^\dagger\} = H \cot(6Hx_0) G$, independent of x_4 and of the scale factors (RESCALING [THE RESULT]: $\{\Psi, \Psi^\dagger\} = (H/\sqrt{|\det g|}) G$ on the warped frame (Section 28's cfAntiPsiPsiDd at $c = \sqrt{|\det g|}/H$), and for $\chi = \Psi/f$, $f = \sqrt{\sin(A^3 B^3 C)^{-1/2}}$: $\{\chi, \chi^\dagger\} = H \cot[6Hx_0] G$ - the $\sqrt{|\det g|}$ cancels, independent of x_4 and of A, B, C).

3.6 The energy-momentum tensor operator

This is the first half of Section 29 of the notebook (Input cells 192–194, 16 assertions).

3.6.1 The definition, and where it comes from

The energy-momentum tensor of fermion fable is

$$\hat{T}_{\mu\nu} = -\frac{1}{4H} \left[\bar{\Psi} \gamma_\mu D_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi + (\mu \leftrightarrow \nu) \right] + g_{\mu\nu} \hat{L}, \quad \gamma_\mu = g_{\mu\nu} \gamma^\nu$$

normal-ordered with respect to the J -vacuum of section 3.5 when it is an operator. It is the **Hilbert** (symmetric-tetrad) tensor of the symmetrized action, $T^{\mu\nu} = \frac{2}{\sqrt{g}} \frac{\delta S}{\delta g_{\mu\nu}}$, and it equals the covariant expression above **off shell**. The notebook checks this the direct way: it perturbs the canonical frame by a symmetric first-order tetrad perturbation $h(x_0, x_4)$ in the (μ, ν) slot, recomputes to first order everything that depends on the frame (the metric, the Christoffel symbols, the spin connection from the vielbein postulate, the curved gammas, $\sqrt{g}$), and takes the Euler-Lagrange derivative with respect to h . Three facts come out:

1. The result is the tensor above, checked on the diagonal $(x_1, x_1), (x_4, x_4)$ and off the diagonal $(x_1, x_4), (x_0, x_5), (x_5, x_4)$.
2. The variation of the spin connection contributes **exactly nothing**. Its totally antisymmetric part does not change under a symmetric tetrad variation, and the symmetrized Lagrangian sees the connection only through that part, $\frac{1}{2} \omega_{[cab]} \gamma^{cab}$. The covariant-derivative terms of the tensor come instead from the frame dependence of $\{\gamma^\mu, \Gamma_\mu\}$, which vanishes on the background while its first variation does not.
3. The Hilbert tensor is **not** the partial-derivative tensor T_{par} (control, at (x_1, x_4)): one never varies a partial-derivative Lagrangian with respect to the frame.

[proved VII §29]

- EMT [THE RESULT]: the Hilbert (symmetric-tetrad) tensor of the symmetrized action IS That, OFF shell - components $(x_1, x_1), (x_4, x_4), (x_1, x_4), (x_0, x_5), (x_5, x_4)$ (the five components took 0.693s)

- EMT [has content]: the spin-connection variation contributes NOTHING - dropping it leaves the Hilbert tensor unchanged at (x1,x4), (x0,x5), (x5,x4)
- EMT [control]: the Hilbert tensor is NOT the partial-derivative tensor: at (x1,x4) they differ - witnessed on constant spinors (Psibar = e1, Psi = e11: (T16[4] T16[1] T16[0])_{1,11} != 0)

3.6.2 Symmetric and Hermitian

$\hat{T}$ is symmetric, and it is Hermitian under the classical conjugation: real for $\Psi = u + iv$, $\bar{\Psi} = (u - iv)^T \sigma_{16}$, in all 64 components. [proved VII §29]

- EMT [definition]: That is symmetric
- EMT [THE RESULT]: That is Hermitian under the classical conjugation: real for Psi = u + i v, Psibar = (u - i v)^T sigma16, all 64 components

3.6.3 Conservation on shell, and why the commuting proxy proves it for the fermion

Solve the two field equations for the x_4 -derivatives:

$$\begin{aligned}\partial_4 \Psi &= F = -\gamma^4 (H V'(s) \Psi - \gamma^0 \partial_0 \Psi - (\gamma^\mu \Gamma_\mu) \Psi), \\ \partial_4 \bar{\Psi} &= \bar{F} = (H V'(s) \bar{\Psi} + \partial_0 \bar{\Psi} \gamma^0 - \bar{\Psi} (\Gamma_\mu \gamma^\mu)) \gamma^4,\end{aligned}$$

and let cfOnShellC replace every x_4 -derivative of either field (of any order) by the corresponding derivative of F or $\bar{F}$, to a fixed point. Then $\nabla^\mu \hat{T}_{\mu\nu} = 0$ in all eight components, for generic $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$ and generic V .

Why a commuting computation proves the Grassmann statement. After the substitution, every term of the divergence is a bilinear with $\bar{\Psi}$ (or a derivative of it) on the left and Ψ (or a derivative) on the right, multiplied by functions of the **even** composites s and ∂s (through V' , V'' and their derivatives). No two odd factors are ever exchanged, so the same algebra holds verbatim for anticommuting components. (The argument is not valid for arbitrary quartic identities; it is valid here because of that structure.) A configuration that is not a solution has a non-zero divergence (control).

The connection does not drop out of the tensor. $\hat{T}$ differs from the partial-derivative tensor by

$$\hat{T}_{\mu\nu} - T_{\mu\nu}^{\text{par}} = -\frac{1}{4H} \bar{\Psi} (\{\gamma_\mu, \Gamma_\nu\} + \{\gamma_\nu, \Gamma_\mu\}) \Psi,$$

which vanishes on the diagonal (so ρ and every pressure are the same in both) but not off it, and only $\hat{T}$ is conserved. (The symbolic divergence of T_{par} is not attempted: it takes minutes, and the off-diagonal difference already shows the two tensors are different.) [proved VII §29]

- EMT CONSERVATION [definition]: F and Fbar solve the field equation and the adjoint equation for the x4-derivatives
- EMT CONSERVATION [THE RESULT]: nabla^mu That_{mu nu} == 0 ON SHELL in all eight components, for generic Psi(x0, x4), Psibar(x0, x4) and generic V (the divergence took 0.767s and its on-shell reduction 2.599s)
- EMT CONSERVATION [control]: OFF shell the divergence is not zero - witnessed on Psi = (x4 + x0^2) e1 + e13, Psibar = e1 + x4 e5 with V = s^2, which is not a solution

- EMT [THE RESULT]: That $-T_{\text{par}} == -(1/(4H)) \text{Psibar}(\{\gamma_{\mu}, \Gamma_{\mu}\} + \{\gamma_{\nu}, \Gamma_{\nu}\}) \text{Psi}$: equal on the diagonal (every $\{\gamma_{\mu}, \Gamma_{\mu}\}$ vanishes), different off it (the (x_1, x_4) witness above)

The statement is proved for fields of (x_0, x_4) . That is the family in which ρ , P and the pre-universe results of section 3.7 are evaluated.

3.6.4 On shell: the Lagrangian, the density, the pressures and the trace

With `cfOnShellC`, for generic $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$ and generic V :

$$\begin{aligned} \frac{1}{2H} [\bar{\Psi} \gamma^{\mu} D_{\mu} \Psi - (D_{\mu} \bar{\Psi}) \gamma^{\mu} \Psi] &= s V'(s) \quad (\text{the kinetic bilinear; NOT zero}), \\ \hat{L} &= s V'(s) - V(s), \\ \rho = \hat{T}_{44} &= V(s) + K_h, \quad K_h = -\frac{1}{2H} (\bar{\Psi} \gamma^0 \partial_0 \Psi - \partial_0 \bar{\Psi} \gamma^0 \Psi), \\ T^i_i = T^h_h &= s V'(s) - V(s) \\ &\quad (i = 1, 2, 3 \text{ observed; } h = 5, 6, 7 \text{ hidden; no sum}), \\ T^0_0 &= s V'(s) - V(s) + K_h, \\ T^{\mu}_{\mu} &= 7 s V'(s) - 8 V(s) \quad (\text{off shell: } 7L_{\text{kin}} - 8V). \end{aligned}$$

The observer is $u = \partial/\partial x_4$, so $\rho = T_{44}$ (the array element `[[5,5]]`) and the pressure along a direction j is $P_j = T^j_j$ (no sum). $P_i = P_h$ holds because $\{\gamma^{\mu}, \Gamma_{\mu}\} = 0$ for **each** direction separately: the kinetic parts of T^i_i and T^h_h vanish for fields of (x_0, x_4) , and both reduce to $\hat{L}$. The trace is consistent with the list: $T^4_4 = -\rho$, so $-(V+K_h) + (sV'-V+K_h) + 6(sV'-V) = 7sV'-8V$ [**derived here**]. In the real commuting limit K_h is Part VI's `cfKh` (section 3.1.12.4), and for the rescaled solutions $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$ (and the same for $\bar{\Psi}$) $K_h = 0$ and $\rho = V(s)$: Part VI's formulas, now for the complex field. [**proved VII §29**]

- EMT ON SHELL [THE RESULT]: the kinetic bilinear == $s V'(s)$ (not zero) and $L_{\text{hat}} == s V'(s) - V(s)$
- EMT ON SHELL [THE RESULT]: $\rho = T_{44} == V(s) + K_h$
- EMT ON SHELL [THE RESULT]: $T^i_i == T^h_h == s V'(s) - V(s)$ for the three observed and the three hidden directions, and $T^0_0 == s V'(s) - V(s) + K_h$
- EMT [has content]: the trace $T^{\mu}_{\mu} == 7 L_{\text{kin}} - 8 V$ off shell, == $7 s V'(s) - 8 V(s)$ on shell
- EMT [fidelity]: K_h in the real commuting limit IS Part VI's `cfKh`, and $K_h == 0$ on the rescaled solutions `Sqrt[Sin] Psi'(x4)`, `Sqrt[Sin] Psi'bar(x4)`, where $\rho == V(s)$

3.6.5 Which components are Hermitian operators: the J -parity table

For fields independent of x_5, x_6, x_7 (the admissible sector, here $\Psi(x_0, \dots, x_4)$, $\bar{\Psi}(x_0, \dots, x_4)$), replace $\Psi \rightarrow J\Psi$ and $\bar{\Psi} \rightarrow \bar{\Psi}J$ ($J^2 = 1$, s is unchanged). A component that goes into itself is built from J -even matrices and is a **Hermitian** operator; one that goes into minus itself is **anti-Hermitian** (section 3.5.8). The notebook computes the parity of all 64 components (3.561 s) and asserts the table:

component	J -parity	operator
$\hat{T}_{\mu\nu}, \mu, \nu \text{ in } \{0..4\}$	even	Hermitian

component	J -parity	operator
$\hat{T}_{\mu h}, \mu \text{ in } \{0..4\}, h \text{ in } \{5, 6, 7\}$	odd	anti-Hermitian: purely imaginary expectation values, zero in every state invariant under the hidden rotations ($\hat{T}_{\mu h}$ is a vector under them)
$\hat{T}_{hh'}, h \neq h'$	zero identically	—
$\hat{T}_{hh}$	even	Hermitian; $\hat{T}_{hh} = g_{hh}\hat{L}$ exactly, because $\{\gamma_h, \Gamma_h\} = 0$

The hidden momenta $P_h = \int \sqrt{g} \hat{T}^4_h$ are J -odd operators with zero expectation in the J -eigenmode Fock states of section 3.5 [**prose VII §29**]. The anti-Hermitian components are consistent with $G_{\mu h} = 0$ on the canonical frame. [**proved VII §29**]

- EMT HERMITICITY [THE RESULT]: J-parity table - (mu, nu) in {0..4}: EVEN (Hermitian); (mu in 0..4, h in 5..7): ODD (anti-Hermitian); (h, h'), h != h': ZERO identically; (h, h): even
- EMT HERMITICITY [has content]: for x5..x7-independent fields That_{hh} == g_hh Lhat exactly, because {gamma_h, Gamma_h} == 0

3.6.6 The real commuting limit

Set $\Psi \rightarrow \psi$ (real, commuting) and $\bar{\Psi} \rightarrow \psi^T \sigma_{16}$. Then $\hat{T}$ becomes Part VI's covariant tensor T_{cov} of section 3.1.12.3 exactly, all 64 components: FIDELITY [4] [fidelity]: the energy-momentum tensor That in the real limit IS Part VI's covariant tensor cftCov, all 64 components. The classical tensor of 2026-09-16 is therefore the commuting shadow of the operator derived here.

3.7 Density, pressure and equations of state

This is the second half of Section 29 of the notebook (Input cells 195–197, 23 assertions), with the numerically stable forms used by the solver and the vacuum-energy caveat added from the design review.

3.7.1 The operators

The observer is $u = \partial/\partial x_4$. The energy-density and pressure **operators** are the normal-ordered components of section 3.6:

$$\begin{aligned} \rho_{\text{op}} &= :\hat{T}_{44}: && \text{(the energy density measured by the observer),} \\ P_{j,\text{op}} &= :\hat{T}^j_j: \quad (\text{no sum}) \quad j = 1, 2, 3: \text{ the observed sheet; } j = 0: \text{ the hidden spacelike direction;} \\ & && h = 5, 6, 7: \text{ the hidden timelike sheet,} \\ w &= \langle P_{\text{obs}} \rangle / \langle \rho \rangle && P_{\text{obs}} = \text{the pressure along the observed sheet.} \end{aligned}$$

On shell, for fields of (x_0, x_4) : $\rho = V(s) + K_h$, $P_i = P_h = s V'(s) - V(s)$, $P_0 = s V'(s) - V(s) + K_h$ (section 3.6.4). These are operator identities in the sense of section 3.4.9 (ordering by the Weyl rule; exact for linear V).

3.7.2 Expectation values: the mode sums, and one particle

Use the **canonical normalization** $\chi = \Psi/\sqrt{H}$. Then $V(s) = V(H\sigma) =: W(\sigma)$ with $\sigma = \langle \bar{\chi} \chi \rangle$, and $m = W'(\sigma)$ is the mass in the mode equation. On a flat-frame plane wave $\chi = u e^{i(k \cdot x - \omega x_4)}$, $\bar{\chi} = \bar{u} e^{-i(k \cdot x - \omega x_4)}$ ($\bar{u} = u^\dagger \sigma_{16}$), the kinetic parts of the tensor are

$$T_{44} + \hat{L} = \omega \bar{u} (-i T_{16}[4]) u = \omega u^\dagger G u,$$

$$T^j_j - \hat{L} = k_j \bar{u} (-i T_{16}[j]) u = k_j u^\dagger (\partial h / \partial k_j) u \quad (j = 0..3, \text{ flat frame}).$$

By the expectation rule $\langle \Psi^\dagger N \Psi \rangle = \sum_{\text{occupied}} \eta_u u^\dagger N u$ (section 3.5.14) and the exact mode identities of section 3.5.13, each occupied positive-frequency mode contributes ω to the energy density, k_j^2/ω to the pressure along j , and m/ω to σ ; each hole in the sea contributes the same with $\omega \rightarrow |\omega|$. So for a state of N modes in a coordinate volume Vol:

$$\rho = \sum \frac{\omega}{\text{Vol}} + (W - \sigma W'), \quad P_j = \sum \frac{k_j^2}{\omega \text{Vol}} + (\sigma W' - W), \quad \sigma = \sum \frac{m}{\omega \text{Vol}}.$$

For **one particle**, $w = (k^2/3)/\omega^2$ (isotropic average) on top of the background $W - \sigma W'$. **[proved VII §29]**

- **MODE SUMS [THE RESULT]:** on a flat-frame plane wave (canonical normalization), `T_44 + Lhat == omega ubar (-i T16[4]) u` and `T^j_j - Lhat == k_j ubar (-i T16[j]) u` ($j = 0..3$): energy and pressure are the frequency and $k_j dh/dk_j$ bilinears
- **MODE SUMS [has content]:** with `ubar = u^ddag sigma16` these are `omega u^ddag G u` and `k_j u^ddag (dh/dk_j) u`, so by Section 28 each occupied mode contributes `omega`, `k_j^2/omega` and (to `sigma`) `m/omega`

3.7.3 The vacuum: normal ordering, the sea energy, and the cosmological-constant problem

$\langle 0 | \hat{T}_{\mu\nu} | 0 \rangle = 0$ by normal ordering with respect to the J -vacuum. That statement is clean only for a **static** background ($a_4 = \text{const}$) or in the adiabatic limit. On an x_4 -dependent background normal ordering is ambiguous: with respect to the instantaneous vacuum it breaks covariant conservation, and with respect to a fixed in-vacuum it leaves a divergence. A conserved renormalized $\langle T \rangle$ needs adiabatic subtraction or point-splitting **[prose VII §29]**.

The **unrenormalized sea energy** per unit volume, $-g \int^\Lambda \frac{d^3 k}{(2\pi)^3} \omega$, is minus the Fermi-sea energy at $k_F = \Lambda$:

$$-\varepsilon_{\text{KS}}(\Lambda) = -\frac{g}{16\pi^2} \left[2\Lambda^4 + 2m^2\Lambda^2 + \frac{m^4}{4} - m^4 \ln \frac{2\Lambda}{m} \right] + o(1) \quad (\Lambda \rightarrow \infty).$$

Its finite, mass-dependent part contains $-\frac{g}{32\pi^2} m^4 \ln m^2$. It is absorbed into V (the renormalized, “no-sea” functional). **The cosmological-constant problem is not solved here.** **[proved VII §29]** **VACUUM [THE RESULT]:** the unrenormalized sea energy `-eps_KS(Lambda) == -(g/(16 pi^2))(2 Lambda^4 + 2 m^2 Lambda^2 + m^4/4 - m^4 Log[2 Lambda/m]) + o(1)` as `Lambda -> infinity`; its m -dependent finite part contains `-(g/(32 pi^2)) m^4 Log[m^2]`

For a mass that varies (the coupled system), the sea energy cannot be absorbed silently (design review DFT-4, adopted). The renormalized one-loop Dirac-sea energy that the no-sea functional drops (relativistic Hartree approximation, Chin 1977), per 3-volume, with reference mass M and counterterms through m^4 , is

$$\Delta E_{\text{vac}}(m) = -\frac{g}{16\pi^2} \left[m^4 \ln \frac{m}{M} + M^3(M-m) - \frac{7}{2} M^2(M-m)^2 + \frac{13}{3} M(M-m)^3 - \frac{25}{12} (M-m)^4 \right]$$

[file: `fable-cosmology/rust/fable_fermion/src/kohn_sham.rs`, function `vacuum_energy`; its unit test `vacuum_energy_vanishes_to_fifth_order_at_the_reference_mass` checks that it vanishes like $(m - M)^5$ at $m = M$ and is even in m]. It is neither zero nor absorbable into W without changing the gap equation; the solver reports it (column `vac_over_rho`) and

does not include it in the dynamics. Reading W as the fully renormalized effective potential absorbs it formally, but that is a fine-tuning. Its size **[derived here]**: at $m = 0$ the bracket is $M^4(1 - \frac{7}{2} + \frac{13}{3} - \frac{25}{12}) = -M^4/4$, so $\Delta E_{\text{vac}}(0) - \Delta E_{\text{vac}}(M) = gM^4/(64\pi^2)$. With $g = 8$ and the unit $E_c = \rho_{c0}^{1/4} = 2.46261 \times 10^{-3} \text{ eV}$ (asserted in Part VIII: fable4d RUNS **[definition]**: the units computed from CODATA 2018 - $E_c = \rho_{c0}^{1/4} == 2.46261\text{e-}3 \text{ eV}$, $\Omega_{r0} == 9.2096\text{e-}5$, $\Omega_{b0} = 0.02237/h^2 == 0.049243$ (to the quoted digits)), a mass variation of order m changes the vacuum energy by $\frac{gm^4}{64\pi^2}/\rho_{c0} = 3.44, 3.44 \times 10^4, 3.44 \times 10^8, 3.44 \times 10^{12}$ for $m = 0.01, 0.1, 1, 10 \text{ eV}$ (it scales as m^4). So for $m \gtrsim 10 \text{ meV}$ it exceeds the critical density. (The design review quoted 3.44×10^{-4} and 3.44 at 0.01 and 0.1 eV ; those two figures are off by 10^4 and are corrected here.)

3.7.4 The homogeneous Fermi sea (the Kohn–Sham ground state) in closed form

Fill the positive-frequency modes with $|k| < k_F$ along the observed sheet (zero modes along x_0 — the $\sqrt{\sin}$ profile — and along x_5, x_6, x_7). The degeneracy is $\mathbf{g} = \mathbf{8} = 4 \text{ flavours} \times 2 \text{ spins}$ (sections 3.2.6 and 3.5: eight positive-frequency modes per momentum). With $\omega_F = \sqrt{k_F^2 + m^2}$ and $\mathcal{L} = \ln((k_F + \omega_F)/m)$ ($m > 0$; everything depends on m^2 except σ , which has the sign of m):

$$\begin{aligned} n &= \frac{g k_F^3}{6\pi^2} &= g \int \frac{d^3 k}{(2\pi)^3} 1, \\ \sigma_{\text{KS}} &= \frac{g m}{4\pi^2} [k_F \omega_F - m^2 \mathcal{L}] &= g \int \frac{d^3 k}{(2\pi)^3} \frac{m}{\omega}, \\ \varepsilon_{\text{KS}} &= \frac{g}{16\pi^2} [k_F \omega_F (2k_F^2 + m^2) - m^4 \mathcal{L}] &= g \int \frac{d^3 k}{(2\pi)^3} \omega, \\ P_{\text{KS}} &= \frac{g}{48\pi^2} [k_F \omega_F (2k_F^2 - 3m^2) + 3m^4 \mathcal{L}] &= g \int \frac{d^3 k}{(2\pi)^3} \frac{k^2}{3\omega}. \end{aligned}$$

Each closed form is proved equal to its integral: its k_F -derivative is the integrand times $gk_F^2/(2\pi^2)$ and it vanishes at $k_F = 0$; independently, Mathematica's **Integrate** returns the same function (with $\text{artanh}(k_F/\omega_F)$, which equals $\mathcal{L}$); and a numerical quadrature at $k_F = 3/2, m = 1, g = 8$ agrees to 10^{-12} (a cross-check; the proof is the symbolic one). **[proved VII §29]**

- KOHN-SHAM **[definition]**: the degeneracy $\mathbf{g} = \mathbf{8}$ - per momentum the positive-frequency eigenspace has rank 8 (Section 28's $P_{\{+,+\}} + P_{\{+,-\}}$), four flavours times two spins
- KOHN-SHAM **[THE RESULT]**: each closed form has k_F -derivative $(g/(2\pi^2)) k_F^{-2} f(k_F)$ with $f = 1, m/\omega, \omega, k^2/(3\omega)$, and vanishes at $k_F = 0$ - so it IS the integral $g \int d^3 k / (2\pi)^3 f$
- KOHN-SHAM **[has content]**: **Integrate** returns the same functions, with $\text{ArcTanh}[k_F/w_F]$ in place of $\mathcal{L}$ - and $\text{ArcTanh}[k_F/w_F] == \mathcal{L}$ (equal derivatives, both 0 at $k_F = 0$)
- KOHN-SHAM **[control]**: numerical quadrature agrees at $k_F = 3/2, m = 1, g = 8$ to $1\text{e-}12$ (a cross-check; the proof is the symbolic one above)

The exported closed forms at $g = 8$ ($k_F = k_F$; provenance-latex/generated/fermion_fable_eos.tex, unchanged):

$$\begin{aligned} n &= \frac{4k_F^3}{3\pi^2} \\ \sigma_{\text{KS}} &= \frac{2m \left(k_F \sqrt{k_F^2 + m^2} - m^2 \log \left(\frac{\sqrt{k_F^2 + m^2} + k_F}{m} \right) \right)}{\pi^2} \end{aligned}$$

$$\varepsilon_{KS} = \frac{k_F \sqrt{k_F^2 + m^2} (2k_F^2 + m^2) - m^4 \log \left(\frac{\sqrt{k_F^2 + m^2} + k_F}{m} \right)}{2\pi^2}$$

$$P_{KS} = \frac{k_F \sqrt{k_F^2 + m^2} (2k_F^2 - 3m^2) + 3m^4 \log \left(\frac{\sqrt{k_F^2 + m^2} + k_F}{m} \right)}{6\pi^2}$$

$$\rho = \varepsilon_{KS} + W - \sigma W', \quad P_{obs} = P_{KS} + \sigma W' - W, \quad P_{hid} = \sigma W' - W, \quad \rho + P_{obs} = w_F n, \quad w_F = \sqrt{k_F^2 + m^2}$$

$$w = P_{KS}/\varepsilon_{KS} \in [0, 1/3] \text{ for the mass term } W = m\sigma \quad (w \rightarrow k_F^2/(5m^2) \text{ as } k_F/m \rightarrow 0, \quad w \rightarrow 1/3 \text{ as } k_F/m \rightarrow \infty)$$

The identities that organize them:

$$\varepsilon_{KS} + P_{KS} = \omega_F n \quad (\text{the zero-temperature Gibbs relation, chemical potential } \mu = \omega_F),$$

$$\varepsilon_{KS} - 3P_{KS} = m \sigma_{KS} \quad (\text{the trace identity}),$$

$$\partial \varepsilon_{KS} / \partial m = \sigma_{KS}, \quad \partial \varepsilon_{KS} / \partial k_F = \omega_F dn / dk_F.$$

[proved VII §29] KOHN-SHAM [THE RESULT]: `eps + P == w_F n` (Gibbs), `eps - 3 P == m sigma` (trace), `d eps/dm == sigma`, `d eps/dk_F == w_F dn/dk_F`

3.7.5 The numerically stable forms

The closed forms cancel catastrophically for $x = k_F/|m| \ll 1$: $\varepsilon \sim mn$, $\sigma \sim n$ and $P \sim nk_F^2/(5m)$ are small differences of large terms. The solver therefore uses three regimes, all written with the integral representations

$$S(x) = 2 \int_0^x \frac{t^2}{\sqrt{1+t^2}} dt, \quad E(x) = 8 \int_0^x t^2 \sqrt{1+t^2} dt, \quad Q(x) = 8 \int_0^x \frac{t^4}{\sqrt{1+t^2}} dt,$$

with $\sigma = g m^3 S/(4\pi^2)$, $\varepsilon = g m^4 E/(16\pi^2)$, $P = g m^4 Q/(48\pi^2)$ [file: `fable-cosmology/rust/fable_fermion/src/kohn_sham.rs`, module documentation and constants `X_LO = 0.6`, `X_HI = 100.0`]:

regime	form used
$x < X_{LO} = 0.6$	the convergent binomial series (radius 1) in x^2 , e.g. $\varepsilon = m n [1 + \frac{3}{10}x^2 - \frac{3}{56}x^4 + \dots]$, $\sigma = \text{sgn}(m) n [1 - \frac{3}{10}x^2 + \dots]$
$0.6 \leq x \leq X_{HI} = 100$	the closed forms (with $\text{asinh } x = \mathcal{L}$), losing at most about 1.3 digits at $x = 0.6$
$x > 100$ $m = 0$ exactly	the ultra-relativistic series in $y = m /k_F$, with the logarithm kept exact $\sigma = 0$, $\varepsilon = 3P = g k_F^4/(8\pi^2)$

The series coefficients are $\binom{-1/2}{j}$ for S and Q and $\binom{1/2}{j}$ for E :

$$S = \sum_j 2 \binom{-1/2}{j} \frac{x^{2j+3}}{2j+3}, \quad E = \sum_j 8 \binom{1/2}{j} \frac{x^{2j+3}}{2j+3}, \quad Q = \sum_j 8 \binom{-1/2}{j} \frac{x^{2j+5}}{2j+5}$$

(the `series_nr` function of the same file). The commit that finished the solver (c5892bd) records that these agree with quadrature to 10^{-12} from $x = 10^{-8}$ to 10^6 , and that `cargo test-release` passes 21 + 8 tests, among them `kohn_sham::tests::fermi_integrals_match_quadrature` and `kohn_sham::tests::series_and_closed_forms_are_continuous_at_the_switch_points`. (The design review had proposed a switch at $x = 0.25$; the implemented switch is 0.6.)

3.7.6 The mean field

With $W(\sigma) := V(H\sigma)$, the gap equation $m = W'(\sigma)$ and self-consistency $\sigma = \sigma_{\text{KS}}(k_F, m)$:

$$\begin{aligned}\rho &= \varepsilon_{\text{KS}} + W - \sigma W', \\ P_{\text{obs}} &= P_{\text{KS}} + \sigma W' - W, \\ P_{\text{hid}} &= P_{x_0} = \sigma W' - W \quad (\text{along } x_0 \text{ and along } x_5, x_6, x_7: \text{ zero modes} \\ &\quad \text{carry no kinetic pressure}), \\ \rho + P_{\text{obs}} &= \omega_F n \quad (\text{exact}), \\ d\rho/dn &= \omega_F + (\sigma_{\text{KS}} - \sigma) W'' \sigma'(n), \quad \text{i.e.} = \omega_F \text{ at the self-consistent point.}\end{aligned}$$

[proved VII §29] MEAN FIELD [THE RESULT]: $\rho = \varepsilon_{\text{KS}} + W - \sigma W'$, $P_{\text{obs}} = P_{\text{KS}} + \sigma W' - W$ give $\rho + P_{\text{obs}} = \omega_F n$ exactly; and $d\rho/dn = \omega_F + (\sigma_{\text{KS}} - \sigma) W'' \sigma'(n)$, i.e. $= \omega_F$ at the self-consistent point $\sigma = \sigma_{\text{KS}}$ ($m = W'(\sigma)$ throughout)

The split $\rho = \varepsilon_{\text{KS}} + U$ with $U := W - \sigma W'$ separates a **quasiparticle** part (ε_{KS} , P_{KS} , with $0 \leq P_{\text{KS}}/\varepsilon_{\text{KS}} \leq 1/3$) from a **mean-field condensate** U , which has $P = -U$ in the observed and in the hidden directions alike: an 8-dimensional vacuum-like energy that varies with σ [derived here from the formulas above].

3.7.7 The classical limit is the non-relativistic limit, and the sign rule

At the self-consistent point the trace identity gives, **exactly**,

$$\rho = W(\sigma) + 3P_{\text{KS}}, \quad P_{\text{obs}} = (\sigma W'(\sigma) - W(\sigma)) + P_{\text{KS}},$$

that is, Part VI's $\rho = V(s)$, $P = sV' - V$ (with $s = H\sigma$) plus the degeneracy pressure and three times it. Part VI is recovered exactly in the limit $P_{\text{KS}}/\varepsilon_{\text{KS}} \rightarrow 0$, which is $k_F/|m| \rightarrow 0$: the cold, **non-relativistic** limit of the Fermi sea. (It is not " $k_F \rightarrow 0$ at fixed n ", which is impossible, and not a coherent zero mode, which the Pauli principle forbids.) [proved VII §29] CLASSICAL LIMIT [THE RESULT]: at the self-consistent point $\rho = W + 3P_{\text{KS}}$ and $P_{\text{obs}} = (\sigma W' - W) + P_{\text{KS}}$ exactly ($\sigma = \sigma_{\text{KS}}$, $m = W'$); Part VI's $\rho = V(s)$, $P = sV' - V$ is the limit $P_{\text{KS}}/\varepsilon_{\text{KS}} \rightarrow 0$, i.e. $k_F/m \rightarrow 0$

Expanded in $x = k_F/|m|$ with the series of section 3.7.5 and $W' = m$ [derived here]:

$$\begin{aligned}\sigma &= \text{sgn}(m) n [1 - \frac{3}{10}x^2 + \dots], \\ \rho &= W(\text{sgn}(m) n) + \frac{3}{10} n k_F^2/|m| + \dots, \\ P_{\text{obs}} &= [\sigma W' - W] + n k_F^2/(5|m|) + \dots, \\ P_{\text{hid}} &= \sigma W' - W\end{aligned}$$

($\rho = |m| n (1 + \frac{3}{10}x^2) + W(\sigma) - |m| n (1 - \frac{3}{10}x^2)$ with $W(\sigma) = W(\text{sgn}(m) n) - \frac{3}{10}|m| n x^2 + \dots$). **The sign rule:** self-consistency forces $\text{sign}(\sigma) = \text{sign}(m) = \text{sign}(W')$. This removes the Part VI branch $M s > 0$. For the author's mass term $W = -2M\sigma$ ($V = -(2M/H)s$, $m = -2M$), $\rho \rightarrow 2|M|n > 0$ in the classical limit.

3.7.8 The author's mass term: w falls from 1/3 to 0

For $W = m\sigma$ ($V = -(2M/H)s$ gives $m = -2M$), $W - \sigma W' = 0$, so

$$\rho = \varepsilon_{\text{KS}} > 0, \quad P_{\text{obs}} = P_{\text{KS}} \geq 0, \quad 0 \leq w = P_{\text{KS}}/\varepsilon_{\text{KS}} \leq \frac{1}{3},$$

and w depends only on $x = k_F/|m|$: $w = I_2/(3I_1)$ with the $m = 1$ integrals

$$I_1 = \frac{1}{8}(x w_x(2x^2 + 1) - \ln(x + w_x)), \quad I_2 = \frac{1}{8}(x w_x(2x^2 - 3) + 3 \ln(x + w_x)),$$

$$I_3 = \frac{1}{2}(x w_x - \ln(x + w_x)), \quad w_x = \sqrt{x^2 + 1}.$$

w rises **monotonically**, since $dw/dx = x^2 \Delta / (3 w_x I_1^2)$ with $\Delta = x^2 I_1 - w_x^2 I_2$, $\Delta(0) = 0$ and $\Delta' = 2x I_3 > 0$, from $w \rightarrow 0$ ($x \rightarrow 0$, $w = x^2/5 + O(x^4)$: dust) to $w \rightarrow 1/3$ ($x \rightarrow \infty$: radiation). As a gas of fixed particle number cools, w falls from $1/3$ to 0 .

The classical sign problem disappears. Part VI needed $Ms < 0$ for $\rho = -(2M/H) s > 0$. Quantum mechanically $\sigma_{\text{KS}} = g \int m/\omega$ is **odd** in m , and $m \sigma_{\text{KS}} = \frac{g m^4}{2\pi^2} I_3(k_F/|m|) \geq 0$ for either sign of m . With $m = -2M$, $M \sigma_{\text{KS}} \leq 0$: exactly the branch $Ms < 0$ that Part VI had to assume. [proved VII §29]

- AUTHOR'S MASS TERM [THE RESULT]: $W = m \sigma$ gives $W - \sigma W' == 0$, so $\rho == \text{eps_KS}$, $P == P_{\text{KS}}$, and $w = P_{\text{KS}}/\text{eps_KS}$ depends only on $x = k_F/m$: $w == I_2/(3 I_1)$, I_n the $m = 1$ integrals
- AUTHOR'S MASS TERM [THE RESULT]: $0 \leq w \leq 1/3 - P_{\text{KS}} \geq 0$ ($I_2' = x^4/w_x > 0$, $I_2(0) = 0$) and $\text{eps} - 3P = m \sigma = (g m^4/(2 \pi^2)) I_3 \geq 0$ ($I_3' = x^2/w_x > 0$, $I_3(0) = 0$)
- AUTHOR'S MASS TERM [THE RESULT]: w rises MONOTONICALLY - $dw/dx == x^2 \Delta / (3 w_x I_1^2)$ with $\Delta = x^2 I_1 - w_x^2 I_2$, $\Delta(0) = 0$ and $\Delta' = 2x I_3 > 0$ - from $w \rightarrow 0$ ($x \rightarrow 0$, $w = x^2/5 + O(x^4)$: dust) to $w \rightarrow 1/3$ ($x \rightarrow \infty$: radiation)
- AUTHOR'S MASS TERM [has content]: the classical sign problem disappears - $\sigma_{\text{KS}} = g \int m/\omega$ is ODD in m and $m \sigma_{\text{KS}} = (g m^4/(2 \pi^2)) I_3(k_F/|m|) \geq 0$ for either sign, so with $m = -2M$: $M \sigma_{\text{KS}} \leq 0$, the branch $M s < 0$ that Part VI had to assume

3.7.9 $w \geq -1$ always: no phantom crossing for the quantized fable

From $\rho + P_{\text{obs}} = \omega_F n$ (section 3.7.6), which is exact, and $\omega_F n \geq 0$:

$$w_f + 1 = \frac{\rho + P_{\text{obs}}}{\rho} = \frac{\omega_F n}{\rho} \geq 0 \quad \text{wherever } \rho > 0.$$

(w_f is the fable's equation of state $w = \langle P_{\text{obs}} \rangle / \langle \rho \rangle$ of section 3.7.1; $\omega_F = \sqrt{k_F^2 + m^2}$ is the Fermi energy.) The quantized fermion fable **cannot cross the phantom divide**, for any potential W [derived here from MEAN FIELD [THE RESULT]: $\rho = \text{eps} + W - \sigma W'$, $P_{\text{obs}} = P_{\text{KS}} + \sigma W' - W$ give $\rho + P_{\text{obs}} == w_F n$ exactly; and $d\rho/dn == w_F + (\sigma_{\text{KS}} - \sigma) W'' \sigma'(n)$, i.e. $== w_F$ at the self-consistent point $\sigma = \sigma_{\text{KS}}$ ($m = W'(\sigma)$ throughout)]. The classical Part VI crossing at $V'(s_1) = 0$ (section 3.1.12.7) is unreachable. In the quantum theory $\rho + P_{\text{obs}} = \omega_F n > 0$ at every non-zero density, whatever the value of m . And for the **expdamp** potential the gap equation keeps $0 < \sigma < \min(n, s_1)$ strictly, so $m = W'(\sigma)$ never reaches the classical crossing point at finite density: $\sigma \rightarrow s_1$ and $m \rightarrow 0$ only as $n \rightarrow \infty$ (design review DFT-6; solver tests `kohn_sham::tests::expdamp_pins_sigma_below_s1_and_mass_goes_to_zero_at_high_density` and `kohn_sham::tests::no_phantom_crossing_rho_plus_p_obs_is_nonnegative`). The classical phantom crossing of 2026-09-16 therefore does **not** survive quantization. (What an observer who fits a cold-dark-matter-plus-dark-energy model to the coupled system can infer — an apparent crossing, when the dust is subtracted at today's dark-matter density and the effective mass grows — is a property of that fit, not of the fluid; it is part of the coupled system, Effort B, and is stated in section 1.)

3.7.10 On the pre-universe: the quantum gas cools, the classical field does not

Four facts, each proved on the canonical frame with a_4 free.

(1) Charge. The $U(1)$ current is conserved on shell for generic $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$. For the x_0 -profile $\Psi = \sqrt{\sin 6Hx_0} \Psi'$ with Ψ' independent of x_0 , $\sqrt{g} j^0$ is independent of x_0 ($\sec \cdot \sin \cdot \cot = 1$), so $\partial_4(\sqrt{g} j^4) = 0$: the charge per coordinate volume does not depend on x_4 , and since $\sqrt{g}$ does not either, neither does the proper 8-density. The 7-volume is constant because the observed sheet grows exactly as the hidden sheet shrinks ($q^3 p^3$ is x_4 -independent). [**proved VII §29**]

- PRE-UNIVERSE (1) [THE RESULT]: the $U(1)$ current is conserved on shell, $(1/\text{Sqrt}[g]) \text{d_mu}(\text{Sqrt}[g] \text{ j}^\mu) == 0$, for generic $\text{Psi}(x_0, x_4)$, $\text{Psibar}(x_0, x_4)$
- PRE-UNIVERSE (1) [THE RESULT]: for $\text{Psi} = \text{Sqrt}[\text{Sin}] \text{Psi}'(x_4)$, $\text{Sqrt}[g] \text{ j}^0$ is x_0 -independent ($\sec \sin \cot == 1$), so $\text{d_4}(\text{Sqrt}[g] \text{ j}^4) == 0$ on shell; and $\text{Sqrt}[g]$ and $q^3 p^3$ are x_4 -independent

(2) Which x_4 -only fields solve the equations: $V''s' = 0$ (section 3.4.8).

(3) Redshift. A mode $\Psi = \sqrt{\sin} u(x_4) e^{ikx_1}$ (coordinate momentum k along the observed sheet) obeys $T_{16}[4] u' + i(k/q) T_{16}[1] u = HV'u$ exactly (mass term, V' constant): its **physical** momentum is k/q , and $d \ln(k/q)/dx_4 = H a'_4(Hx_4) = -H_{\text{obs}}$. The observed-sheet momenta redshift as $e^{a_4} \sim 1/a$, and the instantaneous frequency is $\sqrt{k^2/q^2 + m^2}$ ($E^2 = \omega^2$ at $k_{\text{phys}} = k/q$, section 3.5.10). [**proved VII §29**]

- PRE-UNIVERSE (3) [THE RESULT]: for $\text{Psi} = \text{Sqrt}[\text{Sin}] u(x_4) \text{Exp}[i k x_1]$ the field equation (mass term, $V' = \text{constant}$) is $\text{Sqrt}[\text{Sin}] \text{Exp}[i k x_1] (T_{16}[4] u' + i(k/q) T_{16}[1] u - H V' u)$: the physical momentum is k/q
- PRE-UNIVERSE (3) [THE RESULT]: $d \text{Log}[k/q]/dx_4 == H a_4'[H x_4] == -H_{\text{obs}}$ (observed momenta redshift as $\text{Exp}[a_4]$), and $E^2 == (k^2/q^2 + m^2)$ ID16 at the physical momentum

(4) Cooling. The occupied coordinate momenta are fixed (the mode labels are conserved; the x_4 -dependence of q only mixes $\pm\omega$, a Bogoliubov effect that vanishes adiabatically). So $x = k_{F,\text{phys}}/m$ falls as e^{a_4} : $d \ln x/dx_4 = H a'_4$, and since $w(x)$ is increasing, $\text{sign}(dw/dx_4) = \text{sign}(a'_4)$. In the regime $a'_4 < 0$ (the observed sheet expanding, the sign hypothesis of Part VI, section 3.1.11) the quantum fable gas **cools**, $w : 1/3 \rightarrow 0$, even though its 8-density does not dilute. The classical Part VI fable, by contrast, has $ds/dx_4 = 0$ on the rescaled solutions and $w = sV'/V - 1$ **frozen**. [**proved VII §29**]

- PRE-UNIVERSE (4) [THE RESULT]: $x = k_{F,\text{phys}}/m = x_c/(q m)$ obeys $d \text{Log}[x]/dx_4 == H a_4'[H x_4]$; since $dw/dx > 0$, $dw/dx_4 = w'(x) x H a_4' < 0$ exactly when $a_4' < 0$ (the observed sheet expanding): the quantum gas COOLS
- PRE-UNIVERSE (4) [control]: the classical Part VI fable does NOT cool - $ds/dx_4 == 0$ on its rescaled solutions (Part VI's result, re-asserted), and its $w = s V'/V - 1$ is a function of s alone: frozen

quantity	classical fable (Part VI)	fermion fable (Part VII, Kohn–Sham, mass term)
ρ	$V(s)$	$\varepsilon_{\text{KS}}(k_F, m) > 0$
P (observed)	$sV' - V$ ($= 0$: dust)	P_{KS} in $[0, \rho/3]$
w	$sV'/V - 1$, frozen ($ds/dx_4 = 0$)	$P_{\text{KS}}/\varepsilon_{\text{KS}}$: $1/3 \rightarrow 0$ as the observed sheet expands
8-density	constant	constant (charge per coordinate volume conserved)

(This table is the notebook’s displayed summary at the end of Section 29 [displayed VII §29].)

Finally, a_4 is still undefined after Part VII: PART VII [control]: a4 is STILL UNDEFINED - nothing in Part VII gave it a value

3.8 The canonical spin connection: a complete discussion

This section covers the canonical spin connection of the 4+4 pre-universe in full: the frame it comes from, how it is computed, every non-zero component, its 16×16 spinor form, the gauge-covariant derivative of the 16-component spinor, how it relates to the three bridges of Part IV and to the fable-5.1 (Weitzenböck) bridge of Part V, and what it does and does not do for the complex fermion fable. It stands on its own. Every object is defined here and every result is restated here, with the place where it is proved.

Throughout, “Section n” (capital S) means a section of the notebook. The numbered parts of this discussion are called “subsections” 3.8.1–3.8.13.

3.8.1 Sources, and how the status of each statement is marked

Where the proofs live. The proofs are in the Mathematica notebook `claude-fable/claude-fable_Einstein-Rosen-2-Planes.nb`. That notebook is generated from cell manifests and is never edited by hand:

manifest	Part	Sections	subject
<code>claude-fable/cells_part1.wl</code>	I	0–4	session, coordinates, η_{4488} , τ , σ
<code>claude-fable/cells_part2.wl</code>	I	5–9	T_{16} , σ_{16} , SAB, matrix bases, Cartan triality
<code>claude-fable/cells_part3.wl</code>	II	10–14	the author’s physics reproduced (Ψ_{16} , La, eLa)
<code>claude-fable/cells_part4.wl</code>	III, IV	15–17, 18–20	frame, Christoffel symbols, canonical spin connection, spinor covariant derivative; the three bridges
<code>claude-fable/cells_part5.wl</code>	V	21–22	the fable-5.1 (Weitzenböck) bridge; the master comparison
<code>claude-fable/cells_part6.wl</code>	VI	23–25	fableScalar and the real (classical) fable spinor
<code>claude-fable/cells_part7.wl</code>	VII	26–29	the complex fermion fable (new): refinement, Lagrangian and field equations, canonical quantization, energy–momentum tensor
<code>claude-fable/cells_part8.wl</code>	VIII	30–33	fable as a source of the Einstein equations of the primordial field (the coupled system; its spin connection is Section 32)

The run of Parts I–VI, `claude-fable/run_fable51_part6.log`, evaluates 172 Input cells straight out of the `.nb`. It reports Assertions run: 358 passed: 358 failed: 0 and cells w/ msgs : 0, with identities accepted on numerical evidence alone : 0 and non-vanishing witnesses : 27. The Part VII run is recorded in `claude-fable/run_fermion_fable_part7.log`. It evaluates 198 Input cells and reports Assertions run: 528 passed: 528 failed: 0 and cells w/ msgs : 0, with identities accepted on numerical evidence alone : 0 and non-vanishing witnesses : 38 (Part VII adds 170 assertions and 11 witnesses to Parts I–VI). The full run of Parts I–VIII,

claude-fable/run_fermion_fable_part8.log, reports Assertions run: 642 passed: 642 failed: 0.

How an identity is certified. Each identity is certified in up to three stages:

1. exact structural equality (the entry is literally 0);
2. Simplify under the geometric assumptions, with a 5-second budget;
3. a numerical check at three rational probe points to 30 significant digits.

Stage 3 has two uses, and the notebook counts them separately. The first is to accept an identity. The notebook reports *zero* of those, and its last assertion fails if that count is ever non-zero, so every identity quoted below is closed symbolically. The second is to witness that something is *not* zero, through `cfNonZeroWitnessQ`. A witness succeeds only if every entry evaluates to an actual number at some probe point and at least one of them exceeds 10^{-10} in magnitude. That is a complete proof of non-vanishing.

The probe functions used in stage 3 are test inputs only, never definitions: `a4 -> Function[u, 3/7 + u/5 + Sin[u]/11]`, a probe for the boost rapidity of Bridge 1, and probe values of the Bridge 3 parameter `lambdaOct`.

Status marks used below.

mark	meaning
[proved P §n]	“label” or label a <code>cfAssert</code> of Part P, Section n, that prints PASS in <code>run_fable51_part6.log</code> (Parts I–VI), <code>run_fermion_fable_part7.log</code> (Part VII) or <code>run_fermion_fable_part8.log</code> (Part VIII); the label is quoted verbatim
[displayed P §n]	the printed output of a cell of Part P, Section n (a table, a count or a closed form); an output, not an assertion
[prose P §n]	stated only in prose in the notebook (a Text cell, or a comment inside an Input cell), not asserted
[derived here]	a short derivation carried out in this section from results marked above, with every step shown

The notebook grades its own assertion labels. [definition] means true by construction. [structural] means true for any input of the right shape. [solver regression] is a check of the code, not of the geometry. [has content] could have failed. [THE RESULT] marks a headline result. [control] shows that the wrong alternative fails.

3.8.2 Conventions

- Coordinates $X = \{x_0, x_1, \dots, x_7\}$. The Wolfram array position is the coordinate index plus one, so `x4` is entry 5. Greek indices are curved and Latin indices are flat; both run 0..7. x_4 is the observer’s time ($g_{44} = -1$), x_1, x_2, x_3 are the observed 3-space, x_0 is the hidden spacelike direction, and x_5, x_6, x_7 are the hidden timelike sheet. The notebook’s own words for $x_5..x_7$ are “the superluminal deflating directions”.
- Flat metric: `eta4488 = DiagonalMatrix[{1,1,1,1,-1,-1,-1,-1}]`. [proved I §2] “signature of eta4488 is (4,4)”, “eta4488 . eta4488 == ID8”.
- The 8×8 generators have `tau[0] = ID8`. The conjugates are `taubar[A] = sigma . Transpose[sigma . tau[A]]`, with `sigma = ArrayFlatten[{{0,ID4},{ID4,0}}]` ([proved I §4] “sigma . sigma == ID8”, “sigma is symmetric”). It follows that `taubar[0] = ID8`.

- The 16×16 Dirac matrices are E. A. Lord's reduced Brauer–Weyl doubling, $T16[A] = \text{ArrayFlatten}[\{\{0, \text{taubar}[A]\}, \{\text{tau}[A], 0\}\}]$ for $A = 0..7$. They are real, and $T16[0] = \text{ArrayFlatten}[\{\{0, \text{ID8}\}, \{\text{ID8}, 0\}\}]$. [proved I §5] “ $\{T16[A], T16[B]\}/2 == \text{eta4488}[[A+1, B+1]] \text{ ID16}$ ”. Throughout, $T_{16}[a]$ is γ^a , with the flat index *up*. Since eta4488 is its own inverse, $\gamma_a = \eta_{ab} T_{16}[b]$ differs from $T_{16}[a]$ by the sign η_{aa} .
- Spinor metric: $\text{sigma16} = T16[0].T16[1].T16[2].T16[3]$. [proved I §5] “ $\text{sigma16} == \text{ArrayFlatten}[\{\{-\text{sigma}, 0\}, \{0, \text{sigma}\}\}]$ ”, “ $\text{sigma16} . T16[A]$ is antisymmetric for $A = 0..7$ ”, “ sigma16 is symmetric and $\text{sigma16} . \text{sigma16} == \text{ID16}$ ”.
- Chirality: $T16[8] = T16[0].T16[1]...T16[7] = \text{diag}(-\text{ID8}, +\text{ID8})$. [proved I §5] “PL projects onto the upper (type-1) components, PR onto the lower (type-2) components”. A 16-spinor is $\Psi = (\psi_1, \psi_2)$. The upper 8 components ψ_1 are a split-octonion type-1 spinor, and the lower 8 components ψ_2 are a type-2 spinor.
- $\text{so}(4,4)$ generators: $\text{SAB}[[A+1, B+1]] = (1/4) (T16[A].T16[B] - T16[B].T16[A])$. [proved I §5] “SAB is antisymmetric in its two flat indices”, “ $\text{sigma16} . \text{SAB}$ is antisymmetric”, “ $T16[8]$ commutes with every SAB: the chiral halves are each $\text{Spin}(4,4)$ -invariant”, “[SAB, T16] reproduces the vector representation”.
- H , K and M are Protected constants of the original notebook.

3.8.3 The canonical frame (vielbein), the metric, and the volume factor

A vielbein here is simply an 8-dimensional vierbein, that is, an 8-dimensional frame field. The canonical frame is the diagonal matrix that the original notebook records as the value of the symbol it writes “gtrye with indices alpha and (A)” [prose III §15]:

$$e = \text{diag}(\tan 6Hx_0, q, q, q, 1, p, p, p), \quad (46)$$

$$q = \frac{e^{-a_4(Hx_4)}}{\sin^{1/6} 6Hx_0}, \quad p = \frac{e^{+a_4(Hx_4)}}{\sin^{1/6} 6Hx_0}. \quad (47)$$

In the notebook: $\text{frameCanonical} = \text{DiagonalMatrix}[\{\text{Tan}[6 H x_0], q, q, q, 1, p, p, p\}]$, $q = \text{cfQminus}$ (observed directions x_1, x_2, x_3), $p = \text{cfQplus}$ (hidden directions x_5, x_6, x_7).

Rows curved, columns flat. $\text{frameCanonical}[[\mu+1, a+1]] = e_\mu^a$. The coframe is the matrix inverse, stored with the flat index on the row: $\text{coframeCanonical} = \text{Inverse}[\text{frameCanonical}]$, $\text{coframeCanonical}[[a+1, \mu+1]] = e_a^\mu$. [proved III §15] “CANONICAL [structural]: $\text{frame} . \text{coframe} == \text{ID8}$ (certifies only that the frame is invertible)”.

The free function a_4 . The original notebook never defines a_4 . It appears only inside the recorded frame. It is carried symbolically everywhere and is *never given a value*. The notebook prints a notice to this effect when Section 15 runs [prose III §15], and Part VI ends with [proved VI §25] “PART VI [control]: a_4 is STILL UNDEFINED - nothing in Part VI gave it a value; and the FLRW scale factor a_F never entered the canonical frame”. Every result in this section holds for an arbitrary differentiable a_4 . Below, a'_4 means $\text{Derivative}[1][a_4][H x_4]$.

Reciprocity. $qp = \sin^{-1/3} 6Hx_0$ does not depend on x_4 : the observed directions contract exactly as fast as the hidden directions expand [prose III §15].

The metric. The bridge between the curved and flat metrics is

$$[d] \quad g_{\mu\nu} = e_{\mu}^a \eta_{ab} e_{\nu}^b, \quad (48)$$

that is, `gCanonical = frameCanonical . eta4488 . Transpose[frameCanonical]`, and the line element is

$$ds^2 = \tan^2(6Hx_0) dx_0^2 + q^2 (dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - p^2 (dx_5^2 + dx_6^2 + dx_7^2). \quad (49)$$

- [proved III §15] “CANONICAL [definition]: `g == frame . eta4488 . Transpose[frame] (bridge [d]; this IS how g was defined, so it cannot fail)`”.
- [proved III §15] “CANONICAL [has content]: `g == diag(Tan[6 H x0]^2, q^2, q^2, q^2, -1, -p^2, -p^2, -p^2)`, the stated line element”.
- [proved III §15] “CANONICAL [has content]: `Diagonal[g] == Diagonal[eta4488] * Diagonal[frame]^2` exactly” (Sylvester’s law made explicit).
- [proved III §15] “CANONICAL [has content]: `g` has signature (4,4), GIVEN a4 real and $0 < 6 H x_0 < \text{Pi}/2$ ”.
- [proved III §15] “Inverse[g] == Transpose[coframe] . eta4488 . coframe”, “g is symmetric”, “the metric is non-degenerate”.

The volume factor. Section 15 displays `Det[g] = Sec[6 H x0]^2` and `Sqrt[Abs[Det[g]]] = Abs[Sec[6 H x0]]` [displayed III §15]. [proved V §21] “FABLE-5.1: `det g == Sec[6 H x0]^2` (positive, as signature 4+4 requires)”. On the domain $0 < 6Hx_0 < \pi/2$ the root is `sec6Hx0`. From Part V on, the notebook uses `sqrtdetgTele = Sec[6 H x0]` rather than `Abs[Sec[...]]`, so that `Abs` is never differentiated [prose V §21]. The same result follows from the frame directly [derived here]:

$$\det e = \tan(6Hx_0) q^3 p^3 = \frac{\tan 6Hx_0}{\sin 6Hx_0} = \sec 6Hx_0 = \sqrt{|\det g|}. \quad (50)$$

The volume factor does not depend on x_4 .

The curved Dirac matrices. `gammaCurvedCanonical[[mu+1]] = Sum_a coframe[[a,mu]] T16[a]`. The frame is diagonal, so [derived here]

$$\begin{aligned} \gamma^0 &= \cot(6Hx_0) T_{16}[0], \\ \gamma^j &= \frac{1}{q} T_{16}[j] = e^{a_4} \sin^{1/6}(6Hx_0) T_{16}[j] \quad (j = 1, 2, 3), \\ \gamma^4 &= T_{16}[4], \\ \gamma^k &= \frac{1}{p} T_{16}[k] = e^{-a_4} \sin^{1/6}(6Hx_0) T_{16}[k] \quad (k = 5, 6, 7). \end{aligned} \quad (51)$$

- [proved III §17] “{`gammaCurved[mu]`, `gammaCurved[nu]`} == 2 `gInv[mu,nu]` ID16”.
- [proved VI §24] “FABLE [has content]: `(gamma^4)^2 == -ID16`, so `gamma^4 d_4 Psi' == H V'(s) Psi'` is solved by `d_4 Psi' == -H V'(s) gamma^4 Psi'`”.

3.8.4 The generalized Christoffel symbols

These are the Levi-Civita symbols of g :

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho s} (\partial_{\mu} g_{s\nu} + \partial_{\nu} g_{s\mu} - \partial_s g_{\mu\nu}). \quad (52)$$

cfChristoffel[g, ginv, coords] computes the first derivatives $dg[[k,m,p]] = D[g[[m,p]], \text{coords}[[k]]]$ once, reuses them in the triple loop, and simplifies each entry under the notebook's assumptions. The result is GammaCanonical[[rho+1, mu+1, nu+1]]. Section 16 lists its 37 non-zero entries (out of 512) [displayed III §16]; here Gamma[r;m,n] stands for Γ^r_{mn} :

```
Gamma[0;0,0] = 12*H*Csc[12*H*x0]
Gamma[0;j,j] = (H*Cos[6*H*x0]^3)/(E^(2*a4[H*x4])*Sin[6*H*x0]^(10/3))  j = 1,2,3
Gamma[0;k,k] = -(E^(2*a4[H*x4])*H*Cos[6*H*x0]^3)/Sin[6*H*x0]^(10/3)    k = 5,6,7
Gamma[j;0,j] = Gamma[j;j,0] = -H*Cot[6*H*x0]                             j = 1,2,3
Gamma[j;j,4] = Gamma[j;4,j] = -H*a4'[H*x4]                               j = 1,2,3
Gamma[4;j,j] = -(H*a4'[H*x4])/(E^(2*a4[H*x4])*Sin[6*H*x0]^(1/3))        j = 1,2,3
Gamma[4;k,k] = -(E^(2*a4[H*x4])*H*a4'[H*x4])/Sin[6*H*x0]^(1/3)          k = 5,6,7
Gamma[k;0,k] = Gamma[k;k,0] = -H*Cot[6*H*x0]                             k = 5,6,7
Gamma[k;k,4] = Gamma[k;4,k] = H*a4'[H*x4]                                k = 5,6,7
```

The count is $1 + 3 + 3 + 6 + 6 + 3 + 3 + 6 + 6 = 37$.

- [proved III §16] “Gamma is symmetric in its two lower indices (torsion-free affine connection)”. This check is structural: swapping the two lower indices in the formula reproduces the formula whenever g is symmetric.
- [proved VI §23] “HELPER [solver regression]: cfCovariantDivergence of the metric itself vanishes (metric compatibility of Section 16’s Gamma; a wrong index slot in the helper fails this)”.

3.8.5 The vielbein postulate, and how cfSpinConnection solves it

The spin connection is fixed by the zero-torsion (Levi-Civita) condition, written as the *vielbein postulate*. The total covariant derivative of the frame vanishes when Γ acts on the curved index and ω acts on the flat index [prose III §16]:

$$\partial_\mu e_\nu^a - \Gamma^\rho_{\mu\nu} e_\rho^a + \omega_\mu^a{}_b e_\nu^b = 0. \quad (53)$$

In the notebook: $D[e[\text{nu},a], x[\text{mu}]] - \text{Gamma}[\text{rho},\text{mu},\text{nu}] e[\text{rho},a] + \text{omegaMixed}[\text{mu},a,b] e[\text{nu},b] == 0$, with $\text{omegaMixed}[\text{mu},a,b] = \omega_\mu^a{}_b$. Contracting with the coframe frees ν :

$$\omega_\mu^a{}_c = \sum_\nu \left(\sum_\rho \Gamma^\rho_{\mu\nu} e_\rho^a - \partial_\mu e_\nu^a \right) e_c^\nu. \quad (54)$$

The notebook’s solver is exactly this formula (Section 16):

```
cfSpinConnection[frame_, coords_, etaFlat_, gamma_] :=
Module[{n = Length[coords], cofr, dE},
  cofr = cfSimp[Inverse[frame]]; (* cofr[[a,nu]] = e_a^nu *)
  dE = Table[D[frame[[nu, a]], coords[[mu]]], {mu, n}, {nu, n}, {a, n}];
  Table[
    cfSimp[Sum[(Sum[gamma[[r, mu, nu]] frame[[r, a]], {r, n})
      - dE[[mu, nu, a]] cofr[[c, nu]], {nu, n}]],
    {mu, n}, {a, n}, {c, n}];

cfLowerFirstFlat[om_, etaFlat_] := Module[{n = Length[etaFlat]},
  Table[Sum[etaFlat[[a, c]] om[[mu, c, b]], {c, n}], {mu, n}, {a, n}, {b, n}];

(* omega_mu^a_b *)
omegaCanonicalMixed =
  cfSpinConnection[frameCanonical, X, eta4488, GammaCanonical];
(* omega_{mu a b} *)
omegaCanonical = cfLowerFirstFlat[omegaCanonicalMixed, eta4488];
```

The same routine produces every connection of the notebook that comes from the vielbein postulate: the canonical one, those of Bridges 1 and 2, the Levi-Civita part omegaTriLC of Bridge 3 (to which Section 20 then adds the contortion), and the fable-5.1 connection. The Christoffel symbols are computed once and passed in. So the comparisons below compare results, not different pieces of code [prose III §16, IV, V §21].

What each certificate is worth. The notebook grades its certificates as follows [prose III §16]:

- [proved III §16] “CANONICAL [solver regression]: vielbein postulate residual is zero”. This is an algebraic identity of the solver. `cfSpinConnection` contracts the postulate with `Inverse[frame]` and `cfVielbeinResidual` contracts it back with `frame`, and the two cancel for any affine connection and any invertible frame.
- [proved III §16] “CANONICAL [structural]: torsion vanishes”. Once the residual is zero, this reduces to the symmetry of Γ in its lower indices, which is itself structural. The torsion 2-form used is $T^a_{\mu\nu} = \partial_\mu e_\nu^a - \partial_\nu e_\mu^a + \omega_\mu^a{}_b e_\nu^b - \omega_\nu^a{}_b e_\mu^b$ (`cfTorsion`).
- [proved III §16] “CANONICAL [has content]: `omega[mu,a,b] == -omega[mu,b,a]` (metric compatibility)”. This is not imposed anywhere. It holds only because the affine connection fed in is the Levi-Civita connection of the metric the frame builds, so a wrong Christoffel symbol would break it. The connection is therefore $\mathfrak{so}(4,4)$ -valued.
- [proved III §16] “INDEPENDENT CHECK: the frame-only derivation reproduces `omegaCanonical` exactly” and “INDEPENDENT CHECK: and it is antisymmetric in its two raised flat indices”. The connection is re-derived from the frame alone by the anholonomy formula, which never mentions Γ or g (`cfSpinConnectionFromFrame`, with $e^{a\nu} = (\eta^{-1} \cdot \text{coframe})^{a\nu}$):

$$\begin{aligned} \omega_\mu^{ab} = & \frac{1}{2} e^{a\nu} (\partial_\mu e_\nu^b - \partial_\nu e_\mu^b) - \frac{1}{2} e^{b\nu} (\partial_\mu e_\nu^a - \partial_\nu e_\mu^a) \\ & - \frac{1}{2} e^{a\rho} e^{b\sigma} (\partial_\rho e_\sigma^c - \partial_\sigma e_\rho^c) \eta_{cd} e_\mu^d. \end{aligned} \quad (55)$$

- The canonical frame is diagonal and so equals its own transpose. Section 16 therefore cannot detect a transposition of the frame’s two slots. Section 18 closes that gap on a frame that is not symmetric (the canonical frame times a constant boost of rapidity $1/3$): [proved IV §18] “the slot-test frame is genuinely not symmetric”, “INDEPENDENT CHECK on a non-symmetric frame: the two solvers agree, so the curved/flat slots are placed correctly in both”, “and the result is the canonical connection conjugated by the constant boost, as the gauge law demands”.

3.8.6 The canonical spin connection: all its non-zero components

`omegaCanonical` has **24 non-zero components out of 512** [displayed III §16]. They are listed by `cfShowConnection[omegaCanonical, "omega"]`. In this subsection and the next only, write $c = \cos 6Hx_0$, $s = \sin 6Hx_0$ and $A = a_4(Hx_4)$. For $j = 1, 2, 3$ and $k = 5, 6, 7$, with all indices down, $\omega_{\mu ab} = \text{omega}[\mu; a, b] = \text{omegaCanonical}[[\mu+1, a+1, b+1]]$:

component (both flat indices down)	value
$\omega_{j0j} = -\omega_{jj0}$	$Hc^2/(e^A s^{13/6})$
$\omega_{j4j} = -\omega_{jj4}$	$Ha'_4/(e^A s^{1/6})$
$\omega_{k0k} = -\omega_{kk0}$	$-e^A Hc^2/s^{13/6}$
$\omega_{k4k} = -\omega_{kk4}$	$e^A Ha'_4/s^{1/6}$

That gives 6 directions $\times$ 2 planes $\times$ 2 orderings = 24 components. Every other component is zero. Part VII re-asserts the count and the pattern: [proved VII §27] `SPIN CONNECTION [has content]: metric compatible (omega_mu^{ab} antisymmetric)` and every non-zero component is `omega_mu^{mu 0}` or `omega_mu^{mu 4}` (up to antisymmetry): `omega_0 == omega_4 == 0`.

Structure [derived here], from the table.

- $\omega_0 = \omega_4 = 0$: the connection vanishes along the hidden direction x_0 and along the observer's time x_4 .
- In each remaining direction μ (1, 2, 3, 5, 6, 7) only the two flat planes $(0, \mu)$ and $(4, \mu)$ are excited.
- Whether the generator of a plane is a rotation or a boost depends on the signs of `eta4488` in that plane. For $j = 1, 2, 3$, $(0, j)$ is a rotation (both directions spacelike) and $(4, j)$ is a boost. For $k = 5, 6, 7$, $(0, k)$ is a boost and $(4, k)$ is a rotation (both directions timelike).
- The x_0 -dependent components (planes $(0, \mu)$) do not involve a'_4 . The a_4 -dependent components (planes $(4, \mu)$) are proportional to a'_4 , so they all vanish when a_4 is constant.
- No component depends on $x_1, x_2, x_3, x_5, x_6, x_7$. The metric, the frame and the connection are invariant under translations along the observed and hidden sheets.
- The *mixed* components $\omega_\mu{}^a{}_b = \text{omegaCanonicalMixed}$ follow by raising the first flat index with `eta4488`. For example, $\omega_j{}^4{}_j = -\omega_{j4j}$ because $\eta_{44} = -1$.

Curvature. The curvature 2-form is $R_{\mu\nu}{}^a{}_b = \partial_\mu \omega_\nu{}^a{}_b - \partial_\nu \omega_\mu{}^a{}_b + [\omega_\mu, \omega_\nu]{}^a{}_b$ (`cfCurvature`). It has **156 non-zero components** [`displayed III §16`]. [`proved III §16`] “CANONICAL: curvature is antisymmetric in the two spacetime indices”. The Ricci scalar is [`displayed III §16`]

$$R = -3H^2 \left((55 + 7 \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0) - 2a_4'^2 \right) \quad (56)$$

In the notebook:

```
RicciScalarCanonical = -3*H^2*((55 + 7*Cos[12*H*x0])*Cot[6*H*x0]^2*Csc[6*H*x0]^2
- 2*a4'[H*x4]^2)
```

This is equivalent to $-6H^2(24 \cot^2 6Hx_0 + 31 \cot^4 6Hx_0 - a_4'^2)$ [`derived here`], using $\cos 12Hx_0 = 1 - 2 \sin^2 6Hx_0$ and $\csc^2 = 1 + \cot^2$. It depends on a_4 only through $a_4'^2$. It is the frame-independent invariant that every bridge below is checked against.

Summary of the certified properties of the canonical connection. It is metric compatible (antisymmetric in its flat indices, a statement with content), it is torsion-free (structural), it is reproduced exactly by an independent frame-only derivation, and it is not flat.

3.8.7 The spinor connection Γ_μ and its exact normalization

The 16×16 spin-connection matrix is built by `cfSpinMatrix` in Section 17:

```
cfSpinMatrix[om_, mu_Integer] :=
  (1/8) Sum[om[[mu, a, b]] (T16[a - 1] . T16[b - 1] - T16[b - 1] . T16[a - 1]),
    {a, 8}, {b, 8}];
GammaSpinCanonical =
  Table[cfSimpArray[cfSpinMatrix[omegaCanonical, mu]], {mu, 8}];
```

It is fed `omegaCanonical`, the connection with *both flat indices down*. Since $T_{16}[a] = \gamma^a$ has its flat index *up*, the exact convention is

$$\begin{aligned} \Gamma_\mu &= \frac{1}{8} \omega_{\mu ab} [\gamma^a, \gamma^b] = \frac{1}{4} \sum_{a,b=0}^7 \omega_{\mu ab} T_{16}[a] T_{16}[b] \\ &= \frac{1}{2} \sum_{a < b} \omega_{\mu ab} T_{16}[a] T_{16}[b] = \frac{1}{2} \omega_{\mu ab} \text{SAB}[[a+1, b+1]]. \end{aligned} \quad (57)$$

The second line uses the antisymmetry of $\omega_{\mu ab}$ and $T_{16}[a] T_{16}[b] = -T_{16}[b] T_{16}[a]$ for $a \neq b$ [`derived here`]. The equivalent form with raised connection indices is $\frac{1}{4} \omega_\mu{}^{ab} \gamma_a \gamma_b$, with $\gamma_a = \eta_{ab} T_{16}[b]$.

Index warning. The expression $\frac{1}{4}\omega_\mu^{ab}T_{16}[a]T_{16}[b]$, with the raised connection contracted against the unlowered **T16**, is *not* the notebook's Γ_μ . It would multiply each plane by $\eta_{aa}\eta_{bb}$, and so flip the sign of the $(4,j)$ and $(0,k)$ terms [derived here].

The coefficient and the sign are not conventions left open. They are fixed against the vielbein postulate:

- [proved III §17] “[definition] $(1/8)\ \text{omega}\ [\text{gamma}^a, \text{gamma}^b] == (1/2)\ \text{omega}\ \text{SAB}$ (SAB is defined as $(1/4)[\text{gamma}, \text{gamma}]$, so this cannot fail)”.
- [proved III §17] “[has content] the spinor connection is compatible with omega: $[\text{Gamma}^{\text{spin_mu}}, \text{gamma}^a] + \text{omega_mu}^a{}_b \text{gamma}^b == 0$ ”. The spinor connection rotates the Dirac matrices exactly as ω rotates a vector. With $1/4$ in place of $1/8$, or with the opposite sign, this check fails.
- [proved III §17] “[control] with the opposite sign the compatibility check FAILS (witnessed): the sign is fixed by the vielbein postulate, not free”.
- [proved VII §27] the normalization used throughout Part VII, **SPIN CONNECTION** [definition]: $\text{Gamma}^{\text{spin_mu}} == \text{Sum}_{\{a<b\}} \text{omega_mu,ab}\ (1/2)\ \text{T16}[a]\ \text{T16}[b]$.

The explicit matrices [derived here], from the table of subsection 3.8.6 and the formula above. Only the planes $(0,\mu)$ and $(4,\mu)$ contribute. For $j = 1, 2, 3$ the ordered pair $(j, 4)$ gives $\omega_{jj4}T_{16}[j]T_{16}[4] = \omega_{j4j}T_{16}[4]T_{16}[j]$, so

$$\begin{aligned}\Gamma_0 &= \Gamma_4 = 0, \\ \Gamma_j &= \frac{1}{2} \left[\frac{Hc^2}{e^A s^{13/6}} T_{16}[0]T_{16}[j] + \frac{Ha'_4}{e^A s^{1/6}} T_{16}[4]T_{16}[j] \right] \quad (j = 1, 2, 3), \\ \Gamma_k &= \frac{1}{2} \left[-\frac{e^A Hc^2}{s^{13/6}} T_{16}[0]T_{16}[k] + \frac{e^A Ha'_4}{s^{1/6}} T_{16}[4]T_{16}[k] \right] \quad (k = 5, 6, 7).\end{aligned}\tag{58}$$

The provenance script `claude-fable/prov03_covariant_derivative.wls` (log `claude-fable/prov03.log`) prints the non-zero directions of $\text{Gamma}^{\text{spin}}$ as $\{1, 2, 3, 5, 6, 7\}$, each with 32 non-zero entries. That matches the two terms above: each $T_{16}[a]T_{16}[b]$ with $a \neq b$ is a signed permutation matrix with 16 entries, and the two terms occupy different positions.

The direct-sum structure. Every $T_{16}[a]$ is off-diagonal in the type-1/type-2 split, so every product of two of them is block-diagonal: `T16[A].T16[B] = ArrayFlatten[{{taubar[A].tau[B], 0}, {0, tau[A].taubar[B]}}`. The spin connection therefore never mixes the two split-octonion spinors.

- [proved III §17] “ $\text{Gamma}^{\text{spin}}[\mu]$ is block diagonal: it never mixes type-1 with type-2”.
- [proved III §17] “the type-1 block is $(1/8)\ \text{omega}\ (\text{taubar}^a \tau^b - \text{taubar}^b \tau^a)$ ”.
- [proved III §17] “the type-2 block is $(1/8)\ \text{omega}\ (\tau^a \text{taubar}^b - \tau^b \text{taubar}^a)$ ”.

Below, the upper (type-1) block is written $\Gamma_\mu^{(1)}$ and the lower (type-2) block $\Gamma_\mu^{(2)}$. In the notebook these are `GammaSpinType1` and `GammaSpinType2`.

3.8.8 The gauge-covariant derivative of the 16-component spinor

$$\begin{aligned}D_\mu \Psi &= \partial_\mu \Psi + \Gamma_\mu \Psi, & D_\mu \psi_1 &= \partial_\mu \psi_1 + \Gamma_\mu^{(1)} \psi_1, \\ D_\mu \psi_2 &= \partial_\mu \psi_2 + \Gamma_\mu^{(2)} \psi_2, & \text{Dirac}[\Psi] &= \gamma^\mu D_\mu \Psi.\end{aligned}\tag{59}$$

In the notebook these are `cfDcov16[psi, mu] = D[psi, X[[mu]]] + GammaSpinCanonical[[mu]] . psi, cfDcovType1, cfDcovType2` and `cfDiracOperator[psi] = Sum[gammaCurvedCanonical[[mu]] . cfDcov16[psi, mu], {mu, 8}]`.

- **[proved III §17]** “Dcov16 on the direct sum is the direct sum of Dcov1 and Dcov2” (on a generic spinor of all eight coordinates).

Covariant constancy of γ^μ . The Christoffel symbols act on the curved index, the spinor connection acts by commutator, and the frame connects the two. If any of the three had the wrong sign or index order, the curved Dirac matrices would not be covariantly constant:

$$D_\mu \gamma^\nu = \partial_\mu \gamma^\nu + \Gamma^\nu_{\mu\rho} \gamma^\rho + [\Gamma_\mu, \gamma^\nu] = 0. \quad (60)$$

- **[proved III §17]** “[has content] and therefore the curved Dirac matrices are covariantly constant: `D_mu gammaCurved^nu == 0`”.

The commutator identity. Contract $\mu = \nu$ and use $\Gamma^\mu_{\mu\rho} = \partial_\rho \ln \sqrt{g}$ **[derived here]**:

$$\frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu) = [\gamma^\mu, \Gamma_\mu]. \quad (61)$$

- **[proved VI §24]** “FABLE [has content]: in general $(1/\text{Sqrt}[g]) \text{d_mu}(\text{Sqrt}[g] \text{gamma}^\mu) == [\text{gamma}^\mu, \text{Gamma}^\text{spin_mu}]$ (covariant constancy of gamma^μ , Section 17)”.

The connection term of the Dirac operator, direction by direction [derived here]. From the explicit Γ_μ of subsection 3.8.7 and the curved gammas of subsection 3.8.3, using $T_{16}[j]^2 = +\text{ID16}$, $T_{16}[k]^2 = -\text{ID16}$ and the anticommutation of distinct **T16**:

$$\begin{aligned} \gamma^j \Gamma_j &= \frac{1}{2q} T_{16}[j] (w_0 T_{16}[0] T_{16}[j] + w_4 T_{16}[4] T_{16}[j]) \\ &= -\frac{1}{2q} (w_0 T_{16}[0] + w_4 T_{16}[4]), \end{aligned} \quad (62)$$

$$\begin{aligned} \gamma^k \Gamma_k &= \frac{1}{2p} T_{16}[k] (v_0 T_{16}[0] T_{16}[k] + v_4 T_{16}[4] T_{16}[k]) \\ &= +\frac{1}{2p} (v_0 T_{16}[0] + v_4 T_{16}[4]), \end{aligned} \quad (63)$$

where w_0, w_4 and v_0, v_4 are the coefficients in Γ_j and Γ_k . Since $w_0/q = Hc^2/s^2 = H \cot^2 6Hx_0$, $w_4/q = Ha'_4$, $v_0/p = -H \cot^2 6Hx_0$ and $v_4/p = Ha'_4$:

μ	$\gamma^\mu \Gamma_\mu$ (no sum)	$\{\gamma^\mu, \Gamma_\mu\}$ (no sum)
0	0	0
$j = 1, 2, 3$	$-\frac{1}{2} H \cot^2(6Hx_0) T_{16}[0] - \frac{1}{2} Ha'_4 T_{16}[4]$	0
4	0	0
$k = 5, 6, 7$	$-\frac{1}{2} H \cot^2(6Hx_0) T_{16}[0] + \frac{1}{2} Ha'_4 T_{16}[4]$	0

The anticommutator column follows because $\Gamma_j \gamma^j$ equals $+\frac{1}{2q}(w_0 T_{16}[0] + w_4 T_{16}[4])$, the negative of $\gamma^j \Gamma_j$, and similarly for k .

The sum. Summing the table **[derived here]** gives $3(-\frac{1}{2}) + 3(-\frac{1}{2}) = -3$ for the $T_{16}[0]$ terms and $3(-\frac{1}{2}) + 3(+\frac{1}{2}) = 0$ for the $T_{16}[4]$ terms. The observed and hidden a'_4 terms cancel:

$$\gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) T_{16}[0], \quad \sum_\mu \{\gamma^\mu, \Gamma_\mu\} = 0. \quad (64)$$

- [proved VI §24] “FABLE [has content]: the HYPOTHESIS that makes it a product - the anticommutator $\{\gamma^\mu, \Gamma^{\text{spin}_\mu}\} == 0$ on the canonical frame (no totally antisymmetric part of the connection)” (the sum over μ).
- [proved VI §24] “FABLE [has content]: hence $(1/2)(1/\text{Sqrt}[g]) \text{d}_\mu(\text{Sqrt}[g] \gamma^\mu) == \gamma^\mu \Gamma^{\text{spin}_\mu}$, and on this geometry $== -3 H \text{Cot}[6 H x_0]^2 T_{16}[0]$ ”.
- [proved VII §27] the per-direction form: every $\gamma^\mu \Gamma_\mu$ is $\alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$, with $\sum \alpha_\mu = -3H \cot^2$ and $\sum \beta_\mu = 0$, SPIN CONNECTION [THE RESULT]: direction by direction $\gamma^\mu \Gamma_\mu == \alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$; Sum $\alpha_\mu == -3 H \text{Cot}[6 H x_0]^2$ and Sum $\beta_\mu == 0$ (the observed and hidden a4 terms cancel); and the anticommutator vanishes for *each* μ separately, LAGRANGIAN (c) [THE RESULT]: $\text{Lsym} - \text{Lsym}_d == (1/(2H)) \sum_\mu \text{Psibar} \{\gamma^\mu, \Gamma_\mu\} \text{Psi}$ identically, and $\{\gamma^\mu, \Gamma_\mu\} == 0$ for EACH μ on the canonical frame: $\text{Lsym} == \text{Lsym}_d$.

Why the anticommutator vanishes [derived here]. For $a \neq b$, $\{\gamma^c, \gamma^a \gamma^b\} = 2\gamma^a \gamma^b \gamma^c + 2\eta^{ca} \gamma^b - 2\eta^{cb} \gamma^a$. This is $2\gamma^{cab}$ (the antisymmetrized product) when c differs from both a and b , and 0 when $c = a$ or $c = b$. Hence

$$\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = \frac{1}{2} \omega_{[cab]} \gamma^{cab}, \quad \omega_{cab} = e_c^\mu \omega_{\mu ab}, \quad (65)$$

and only the totally antisymmetric part of ω_{cab} survives. For the diagonal canonical frame, c equals the direction μ , and every non-zero $\omega_{\mu ab}$ has μ equal to a or b (the planes $(0, \mu)$, $(4, \mu)$). So the totally antisymmetric part is zero, and it is zero direction by direction.

Contrast with an expanding frame. On the separate 4-dimensional FLRW reference frame of Part VI, the same computation leaves a Hubble term: [proved VI §24] “FLRW [THE RESULT]: $\gamma^\mu \Gamma_\mu == (3/2) (a'/a) \gamma^4$ - the Hubble term of the FLRW Dirac equation”. On the canonical frame the analogous $T_{16}[4]$ terms cancel between the observed sheet and the hidden sheet ($\sum \beta_\mu = 0$), because qp does not depend on x_4 [derived here]: the per-direction coefficient $-\frac{1}{2} H a'_4$ of the table equals $\frac{1}{2} \partial_4 \ln q$, and $+\frac{1}{2} H a'_4$ equals $\frac{1}{2} \partial_4 \ln p$, so the sum is $\frac{3}{2} \partial_4 \ln(qp) = 0$.

On the model’s own spinor Psi16 of Section 11.

- [proved III §17] “[definition] $\text{Dcov}[\mu] \text{Psi16} - \text{D}[\text{Psi16}, x_\mu] == \Gamma^{\text{spin}[\mu]} \cdot \text{Psi16}$ ”.
- [proved III §17] “in the six directions on which Psi16 does not depend, Dcov Psi16 is purely the connection term”.
- [proved III §17] “the connection term does not vanish on Psi16: the connection matrices themselves are non-zero (witnessed)”.
- [proved VI §24] “FABLE [has content]: the connection term does not vanish on an x_0 -independent spinor - witnessed on the constant spinor e_1 : a spinor homogeneous in everything but x_4 is NOT a solution”.

Section 17 constructs the operator and applies it. It does not re-derive the flat field equations of Part II from a covariant Lagrangian [prose III §17].

3.8.9 The three bridges of Part IV

The gauge law [prose IV], verified per bridge. Suppose a new frame is `frameNew = frameOld . L` with $L(x)$ invertible on the flat indices. Substituting into the vielbein postulate, with $M = \text{Transpose}[L]$, gives

$$\omega_\mu^{\text{new}} = M \omega_\mu^{\text{old}} M^{-1} - (\partial_\mu M) M^{-1} \quad (\text{mixed components}). \quad (66)$$

The second term is the inhomogeneous gauge term, and it vanishes exactly when L is constant. The law is not assumed anywhere. Each new connection is derived independently by `cfSpinConnection`, and the law is then verified against the result. A frame `frameCanonical . L` with L preserving the flat metric describes the *same geometry*, and its Levi-Civita connection is the canonical one in a different gauge, *not* a new connection [prose IV].

3.8.9.1 Bridge 1: a locally boosted frame, the same connection in a local gauge (Section 18)

A new object, not in the original notebook, is introduced here: the rapidity $\theta = \text{cfBoostRapidity}[x_0, x_4]$, an arbitrary function. It generates a boost in the flat (0,4) plane:

$$\Lambda[\theta] = \text{ID8} + (\cosh \theta - 1)(E_{00} + E_{44}) + \sinh \theta (E_{04} + E_{40}) = e^{\theta K}, \quad (67)$$

where $K = E_{04} + E_{40}$ is the (0,4) boost generator on flat indices, and `frameBoost = frameCanonical . Lambda`.

- [proved IV §18] “Lambda is in O(4,4): `Lambda . eta . Transpose[Lambda] == eta`”, “Lambda is NOT in O(8): it is a genuine boost, not a rotation”, “`Lambda == MatrixExp[theta K]` with K the (0,4) boost generator”.
- [proved IV §18] “BRIDGE 1 reproduces the SAME curved metric g ”, “BRIDGE 1 frame really is different from the canonical frame”.
- [proved IV §18] “BRIDGE 1: vielbein postulate holds identically”, “BRIDGE 1: `omega[mu,a,b] == -omega[mu,b,a]`”, “BRIDGE 1: torsion still vanishes”.
- [proved IV §18] “COMPARISON 1: `omegaBoost == M omegaCanonical Inverse[M] - D[M,x_mu] Inverse[M]`”, “COMPARISON 1: the WHOLE difference is the inhomogeneous gauge term”, “COMPARISON 1: the difference equals `-D[theta, x_mu] * (boost generator)`”.
- [proved IV §18] “COMPARISON 1: curvature transforms covariantly, `R' == M R Inverse[M]`”, “COMPARISON 1: BRIDGE 1 has the SAME Ricci scalar as the canonical frame”.
- [displayed IV §18] `omegaBoost` has 28 non-zero components. The difference `deltaBoost` has 4: $-\partial_\mu \theta K$ for $\mu = 0$ and $\mu = 4$, in the two orderings of the plane (0,4).

At the spinor level, the spin lift is `S = MatrixExp[theta SAB[[1,5]]]`, with no extra factor 1/2, because `SAB` already carries 1/4 [prose IV §18]:

- [proved IV §18] “BRIDGE 1 spin lift: `Inverse[S] . S == ID16`”, “BRIDGE 1 spin lift: `S` preserves the spinor metric, `Transpose[S] . sigma16 . S == sigma16`”, “BRIDGE 1 spin lift [has content]: `Inverse[S] . gamma^a . S == Lambda^a_b gamma^b` (S is the spin lift of Λ)”.
- [proved IV §18] “COMPARISON 1 at the spinor level [THE GAUGE LAW]: `Gamma'_mu == S Gamma_mu Inverse[S] - D[S,x_mu] Inverse[S]`”, “COMPARISON 1 at the spinor level: the inhomogeneous term is `-D[theta,x_mu] SAB[[1,5]]`, the spin image of `-D[theta,x_mu] K`”, “COMPARISON 1 at the spinor level: and that is exactly `cfSpinMatrix` applied to the vector difference `deltaBoost`”.

- [proved IV §18] “BRIDGE 1 [has content]: $D'_{\mu}(S \psi) == S D_{\mu} \psi$ for a generic spinor - one covariant derivative, two gauges”, “BRIDGE 1: the curved Dirac matrices of the boosted frame are $S \gamma^{\mu} \text{Inverse}[S]$ ”, “BRIDGE 1: so the Dirac operator is one operator in two gauges, $\text{Dirac}'[S \psi] == S \text{Dirac}[\psi]$ ”.

A solution of the Dirac equation in one gauge is therefore a solution in the other, after $\psi \rightarrow S\psi$. Part V identifies the inhomogeneous term as the fable-5.1 connection of the boosted frame (subsection 3.8.10).

3.8.9.2 Bridge 2: the null (light-cone) frame, the same connection in a constant gauge (Section 19)

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} \text{ID4} & \text{ID4} \\ \text{ID4} & -\text{ID4} \end{pmatrix}, \quad U = U^T, \quad U \cdot U = \text{ID8}, \quad (68)$$

with $\text{etaNull} = \text{Transpose}[U] \cdot \text{eta4488} \cdot U$ and $\text{frameNull} = \text{frameCanonical} \cdot U$. The eight flat directions are paired as (0, 4), (1, 5), (2, 6) and (3, 7), one spacelike and one timelike in each pair, and each pair is replaced by its two null combinations [prose IV §19].

- [proved IV §19] “U is an involution: $U \cdot U == \text{ID8}$ ”, “U is symmetric”, “THE NULL TANGENT METRIC IS EXACTLY THE SPINOR METRIC σ ”, “etaNull is symmetric and squares to ID8”, “all eight frame directions are null: etaNull has zero diagonal”. The identity $\text{etaNull} == \sigma$ is a statement about the flat metric, not about the connection.
- [proved IV §19] “BRIDGE 2 reproduces the SAME curved metric g ”, “BRIDGE 2: vielbein postulate holds identically”, “BRIDGE 2: $\omega_{[\mu,a,b]} == -\omega_{[\mu,b,a]}$ with indices lowered by etaNull”, “BRIDGE 2: torsion vanishes”.
- [proved IV §19] “COMPARISON 2: $\omega_{\text{Null}} == U \cdot \omega_{\text{Canonical}} \cdot U$ exactly”, “COMPARISON 2: the inhomogeneous gauge term is ZERO because U is constant”, “COMPARISON 2: $R_{\text{null}} == U \cdot R_{\text{canonical}} \cdot U$ ”, “COMPARISON 2: BRIDGE 2 has the SAME Ricci scalar as the canonical frame”.
- [displayed IV §19] ω_{Null} has 48 non-zero components. A constant change of basis spreads the canonical 24 over more entries, and the difference from $U\omega U$ is zero.

3.8.9.3 Bridge 3: triality frame with totally antisymmetric split-octonion torsion, a different connection (Section 20)

The flat index is a *type-1 split-octonion spinor index*, reached through the constant triality bridge triVecToSpin of Section 9. [proved I §9] “THE TRIALITY METRIC IS EXACTLY THE SPINOR METRIC: $\text{etaTri} == \sigma$ ”. The frame is $\text{frameTri} = \text{frameCanonical} \cdot \text{triVecToSpin}$, with flat tangent metric σ , and the connection is

$$\omega_{\mu ab}^{\text{Oct}} = \omega_{\mu ab}^{\text{TriLC}} + \lambda m_{abc}^{\text{Skew}} (e_{\text{Tri}})_{\mu}^c, \quad (e_{\text{Tri}})_{\mu}^c = \text{frameTri}[[\mu, c]]. \quad (69)$$

Here ω_{TriLC} is the Levi-Civita connection in the triality frame. $m^{\text{Skew}} = m^{\text{SkewSpin}}$ is the totally antisymmetrized split-octonion structure constant m^{Lower} , transported to the spinor basis. $\lambda = \text{lambdaOct}$ is a free real torsion-strength parameter introduced in Section 20, not in the original notebook.

- [proved IV §20] “the triality bridge is Euclidean-orthogonal: $\text{Transpose}[P] == \text{Inverse}[P]$ ”, “but it is NOT in $O(4,4)$: $P \cdot \text{eta4488} \cdot \text{Transpose}[P] != \text{eta4488}$ (it carries eta4488 to σ)”, “BRIDGE 3 reproduces the SAME curved metric g ”, “the triality tangent metric is σ ”.

- [proved IV §20] “the antisymmetrized structure constants are totally antisymmetric”, “they agree with mLower on the seven imaginary directions”, “mSkewSpin is totally antisymmetric” (48 non-zero components [displayed IV §20]).
- [proved IV §20] “BRIDGE 3 (lambda=0): vielbein postulate holds identically”, “BRIDGE 3 (lambda=0): omega[mu,a,b] == -omega[mu,b,a]”, “BRIDGE 3 (lambda=0) is the constant triality conjugate of the canonical connection”.
- [proved IV §20] “BRIDGE 3: omegaOct[mu,a,b] == -omegaOct[mu,b,a] (still metric compatible)”, “BRIDGE 3: at lambda = 0 it reduces to the Levi-Civita connection”.
- [proved IV §20] “BRIDGE 3: torsion vanishes at lambda = 0”, “BRIDGE 3: torsion is NOT zero for lambda != 0”, “BRIDGE 3: the flat torsion T[a,b,c] is TOTALLY ANTISYMMETRIC”, “BRIDGE 3: T[a,b,c] == -2 lambda mSkewSpin[a,b,c]”.
- [proved IV §20] “COMPARISON 3: omegaOct - (triality conjugate of omegaCanonical) == contortion”, “COMPARISON 3: the difference is linear in lambda and vanishes at lambda = 0”. The difference is a *tensor* (a contortion), so this is a different connection.

Its curvature is quadratic in λ . With K the contortion, $R(\omega^{\text{Oct}}) = R(\omega^{\text{TriLC}}) + \lambda(dK + [\omega^{\text{TriLC}}, K]) + \lambda^2[K, K]$:

- [proved IV §20] “BRIDGE 3: at lambda = 0 the curvature is the triality conjugate of the canonical curvature”, “BRIDGE 3 [has content]: the curvature DOES depend on lambda (witnessed)”, “BRIDGE 3: the lambda-dependence is a polynomial of degree exactly two”.
- [proved IV §20] “BRIDGE 3: the lambda¹ term is the Levi-Civita covariant exterior derivative of the contortion, $dK + [\omega^{\text{TriLC}}, K]$ ”. Written out, it is $\partial_\mu K_\nu - \partial_\nu K_\mu + [\omega_\mu^{\text{TriLC}}, K_\nu] - [\omega_\nu^{\text{TriLC}}, K_\mu]$.
- [proved IV §20] “BRIDGE 3: the lambda² term is $[K_\mu, K_\nu]$, the contortion commuted with itself”.

The Ricci shift. $-42\lambda^2 = -\frac{1}{4}T_{abc}T^{abc}$:

- [proved IV §20] “BRIDGE 3: at lambda = 0 the Ricci scalar is the canonical one”.
- [proved IV §20] “BRIDGE 3 [THE RESULT]: $R(\omega^{\text{Oct}}) - R(\text{canonical}) == -(1/4) T_{\{abc\}} T^{\{abc\}}$, a constant shift set by the torsion alone”.
- [proved IV §20] “BRIDGE 3: equivalently $-\lambda^2 (m\text{Skew} \cdot m\text{Skew})$, with $m\text{Skew} \cdot m\text{Skew} == 42$, so the shift is exactly $-42 \lambda^2$ ”. Check [derived here]: $T = -2\lambda m^{\text{Skew}}$ gives $T \cdot T = 4\lambda^2 \cdot 42 = 168\lambda^2$, and $\frac{1}{4} \cdot 168 = 42$.

This connection resembles the Cartan–Schouten connections of a Lie group, but it differs in two ways that must not be assumed away [prose IV §20]. It is not flat. And the octonion structure constants violate the Jacobi identity, so they cannot be the anholonomy of any frame.

Spinor level. cfSpinMatrix is correct only for eta4488 vector indices. Bridge 3’s flat index is a triality-spinor index, so the matching Dirac matrices are $T16\text{Tri}[a] = \text{Sum_A triSpinToVec}[[a,A]] T16[A-1]$, and they are contracted by cfSpinMatrixTri [prose IV §20]:

- [proved IV §20] “THE TRIALITY DIRAC MATRICES OBEY THE sigma CLIFFORD RELATION: {T16Tri[a],T16Tri[b]} == 2 sigma[[a,b]] ID16”, “they are NOT the vector-frame Dirac matrices, so the distinction is real”.
- [proved IV §20] “at lambda = 0 the Bridge 3 spinor connection equals the canonical one”, “the extra spinor coupling vanishes at lambda = 0”, “the extra spinor coupling is non-zero for lambda != 0”, “the extra spinor coupling is still block diagonal (type-1 + type-2 preserved)”.
- The extra coupling (lambda/8) mSkew[a,b,c] frameTri[[mu,c]] [T16Tri[a], T16Tri[b]] is *quadratic* in the gammas. The third index is contracted with the frame, not with a third gamma. The cubic Einstein–Cartan axial term would appear only after contracting with γ^μ in the Dirac operator, and the notebook does *not* carry that contraction out [prose IV §20]. The extra coupling has 160 non-zero entries [displayed IV §20].

3.8.10 The fable-5.1 bridge of Part V: the Weitzenböck (teleparallel) connection

The fable-5.1 bridge keeps the metric, the frame, eta4488 and the bridge [d] unchanged, and changes only the rule for parallel transport. It declares the canonical frame itself to be parallel [prose V §21]:

$$(\Gamma^W)^\rho{}_{\mu\nu} = e_a{}^\rho \partial_\mu e_\nu{}^a, \quad (70)$$

in the notebook `GammaWeitzenboeck[[rho, mu, nu]] = Sum_a coframe[[a, rho]] D[frame[[nu, a]], x[mu]]`.

- [proved V §21] “FABLE-5.1 [definition]: the frame is parallel - nabla^W e == 0 for all eight frame vectors”, “FABLE-5.1 [has content]: Gamma^W is metric compatible, nabla^W g == 0”, “FABLE-5.1 [has content]: Gamma^W is NOT symmetric in its lower indices - it has torsion”, “FABLE-5.1 [has content]: Gamma^W is NOT the Levi-Civita connection”.
- Γ^W has 13 non-zero components [displayed V §22].

The spin connection is zero, and so is the curvature. Γ^W is fed to the same solver. Because the first two terms of the vielbein postulate then cancel by construction, the solver returns $\omega = 0$:

- [proved V §21] “FABLE-5.1 [solver regression]: vielbein postulate residual is zero”, “FABLE-5.1 [THE RESULT]: the spin connection in the canonical frame is IDENTICALLY ZERO”, “FABLE-5.1: trivially antisymmetric, so trivially metric compatible”.
- [proved V §21] “FABLE-5.1 [has content]: the torsion is NOT zero”, “FABLE-5.1: $T^{\rho}{}_{\mu\nu} == \Gamma^{\rho W}{}_{\mu\nu} - \Gamma^{\rho W}{}_{\nu\mu}$ (torsion is the antisymmetric part of $\Gamma^{\rho W}$)”. With $\omega = 0$ the torsion is $T^a{}_{\mu\nu} = \partial_\mu e_\nu{}^a - \partial_\nu e_\mu{}^a$.
- [proved V §21] “FABLE-5.1 [THE RESULT]: the curvature is IDENTICALLY ZERO - a flat connection on a curved metric”.

The canonical connection is minus the contortion of this torsion. The contortion is $K = \Gamma^W - \Gamma^{LC}$, and in terms of the torsion $K_{\rho\mu\nu} = \frac{1}{2}(T_{\rho\mu\nu} + T_{\mu\rho\nu} + T_{\nu\rho\mu})$ (all indices down, first index then raised). Substituting $\Gamma^{LC} = \Gamma^W - K$ into the vielbein postulate leaves `omegaCanonical[mu,a,b] = -K[rho,mu,nu] e[rho,a] coframe[[b,nu]]` [prose V §21].

- [proved V §21] “COMPARISON F [has content]: $\Gamma^W - \Gamma^{LC} == (1/2)(T_{\{rmn\}} + T_{\{mrn\}} + T_{\{nrm\}})$ with the index raised, the contortion formula”.
- [proved V §21] “COMPARISON F [THE RESULT]: $\omega\text{Canonical} == -\text{contortion}(\text{fable-5.1 torsion})$, pulled back to flat indices, exactly”.
- [proved V §21] “COMPARISON F: so $\omega\text{Canonical} - \omega\text{Fable51}$ is a TENSOR, not a gauge term ($\omega\text{Fable51} == 0$)”.
- contortionFable51 has 24 non-zero components, the same 24 as $\omega\text{Canonical}$. $\text{torsionFable51Flat}$ has 24, and RiemannFable51 has 0 [displayed V §22].

The kind of torsion. A torsion tensor splits into a vector part, an axial part and a tensor part:

- [proved V §21] “FABLE-5.1 torsion vector $T[\mu] = T[\text{nu}, \text{nu}, \mu]$ is a GRADIENT: $T[\mu] == D[\text{Log}[\text{Sin}[6 H x_0]], x[\mu]]$ ”.
- [proved V §21] “FABLE-5.1 torsion vector has exactly one non-zero component, $6 H \text{Cot}[6 H x_0]$ along x_0 ”.
- [proved V §21] “FABLE-5.1: the AXIAL (totally antisymmetric) part of the torsion vanishes - the complement of Bridge 3”, “FABLE-5.1: the torsion itself does not vanish, so the tensor part carries the rest”.

For any diagonal frame the torsion vector is, component by component (no sum) [derived here],

$$T_\mu = \sum_\nu e_a{}^\nu (\partial_\nu e_\mu{}^a - \partial_\mu e_\nu{}^a) = \partial_\mu \ln \frac{e_\mu{}^\mu}{\det e}. \quad (71)$$

On the canonical frame this is $\partial_0 \ln(\tan 6Hx_0 / \sec 6Hx_0) = \partial_0 \ln \sin 6Hx_0 = 6H \cot 6Hx_0$ for $\mu = 0$. It is zero for $\mu = 4$, because $e_4{}^4 = 1$ and $\det e = \sec 6Hx_0$ do not depend on x_4 . It is zero for $\mu = 1, 2, 3, 5, 6, 7$, because nothing depends on those coordinates. This agrees with the assertion.

$R = -T + B$ in closed form. With the torsion scalar $T = \frac{1}{4}T^{\rho\mu\nu}T_{\rho\mu\nu} + \frac{1}{2}T^{\rho\mu\nu}T_{\nu\mu\rho} - T^\mu T_\mu$ and the boundary term $B = \frac{2}{\sqrt{\det g}} \partial_\mu (\sqrt{\det g} T^\mu)$, where $\sqrt{\det g} = \sec 6Hx_0$:

$$R = -3H^2 \left((55 + 7 \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0) - 2a_4'^2 \right), \quad (72)$$

$$T = -6H^2 (5 \cot^4(6Hx_0) + a_4'^2), \quad (73)$$

$$B = -36H^2 (5 + \cos 12Hx_0) \cot^2(6Hx_0) \csc^2(6Hx_0). \quad (74)$$

These closed forms are [displayed V §21] ($\text{RicciScalarCanonical}$, $\text{torsionScalarFable51}$, $\text{boundaryTermFable51}$). The following are [proved V §21]: “COMPARISON F [THE RESULT]: $R == -T + B$ - the teleparallel equivalent of GR, in closed form for this geometry”, “FABLE-5.1: T is not zero and B is not zero - neither side is trivial”. The Einstein–Hilbert and teleparallel actions therefore differ by a boundary term on this geometry.

The spinor: one vector term, removed by a rescaling. In the fable-5.1 gauge the spinor connection is zero, so the Dirac operator is $\gamma^\mu \partial_\mu$. The canonical Dirac operator carries in addition $\gamma^\mu \Gamma_\mu$, and this whole term is one vector term:

$$\gamma^\mu \Gamma_\mu = -\frac{1}{2} T_\mu \gamma^\mu = -\frac{1}{2} \partial_\mu (\ln \sin 6Hx_0) \gamma^\mu$$

$$= -3H \cot^2(6Hx_0) T_{16}[0], \quad (75)$$

$$\text{Dirac}_{LC}[\sqrt{\sin 6Hx_0} \Psi'] = \sqrt{\sin 6Hx_0} \gamma^\mu \partial_\mu \Psi' \quad \text{for every } \Psi'. \quad (76)$$

The first line checks against subsection 3.8.8 **[derived here]**: $-\frac{1}{2}(6H \cot)(\cot T_{16}[0]) = -3H \cot^2 T_{16}[0]$, because $\gamma^0 = \cot T_{16}[0]$. The matrix $\gamma^\mu \Gamma_\mu$ is $-3H \cot^2$ times the permutation matrix $T_{16}[0]$, so it has 16 non-zero entries.

- **[proved V §21]** “FABLE-5.1 [THE RESULT]: $\gamma^\mu \Gamma_\mu \Psi == -(1/2) T[\mu] \gamma^\mu \Psi$, one vector term”, “FABLE-5.1: equivalently $-(1/2) D[\text{Log}[\sin 6Hx_0], x_\mu] \gamma^\mu \Psi$ ”.
- **[proved V §21]** “FABLE-5.1 [THE RESULT]: $\text{DiracCanonical}[\sqrt{\sin 6Hx_0} \Psi'] == \sqrt{\sin 6Hx_0} \text{DiracFable51}[\Psi']$, for a generic Ψ' ”, “FABLE-5.1: the rescaling is NOT a no-op - the two Dirac operators differ on the constant spinor”.
- **[proved VI §24]** with a potential: “FABLE [THE RESULT]: $(\gamma^\mu D_\mu - H V'(s))[\sqrt{\sin 6Hx_0} \Psi'] == \sqrt{\sin 6Hx_0} (\gamma^\mu d_\mu \Psi' - H V'(s) \Psi')$ for a GENERIC $\Psi'(x_0, x_4)$: the rescaling removes the connection term exactly”.

The rescaling works because the torsion vector is a gradient. A vector-torsion term that is a gradient is removed by multiplying the spinor by $\exp(\frac{1}{2} \ln \sin 6Hx_0) = \sqrt{\sin 6Hx_0}$ **[derived here]**.

For the real spinor of Parts II–VI the connection drops out of the Lagrangian.

σ_{16} . $T_{16}[a]$ is antisymmetric, so $\Psi^T \sigma_{16} T_{16}[a] \Psi = 0$ for real commuting Ψ . Since $\gamma^\mu \Gamma_\mu$ is a multiple of γ^μ , the connection contributes nothing to that Lagrangian:

- **[proved V §21]** “FABLE-5.1 [THE RESULT]: $\Psi^T \sigma_{16} \gamma^\mu \Gamma_\mu \Psi == 0$ for EVERY Ψ - the spin connection drops out of the Lagrangian”, “FABLE-5.1: the reason is Section 11’s identity, $\Psi^T \sigma_{16} T_{16}[a] \Psi == 0$ ($\sigma_{16} T_{16}[a]$ antisymmetric)”, “FABLE-5.1: but the connection term does NOT drop out of the Dirac OPERATOR (it is $-(1/2) T[\mu] \gamma^\mu$, non-zero)”.
- **[proved V §21]** “FABLE-5.1: the covariant Lagrangian with frame $\rightarrow$ ID8 and volume factor $\rightarrow$ 1 IS the author’s La of Section 11, term for term”, “FABLE-5.1: and the covariant Lagrangian has NO connection term to omit - its kinetic term is already the fable-5.1 one”. The author’s flat La omits only the volume factor $\sec 6Hx_0$ and the coframe factors inside γ^μ **[prose V §21]**.

This mechanism is specific to a *real* field. subsection 3.8.12 shows that it does not carry over to the complex fermion fable.

Frame covariance: Bridge 1 explained. In Bridge 1’s boosted frame, the same Γ^W fed to the same solver gives a fable-5.1 spin connection that is pure gauge:

- **[proved V §21]** “FABLE-5.1 in the boosted frame: vielbein postulate residual is zero”, “FABLE-5.1 in the boosted frame [THE RESULT]: $\omega == -D[M, x_\mu] \text{Inverse}[M]$, pure gauge and nothing else”, “FABLE-5.1 in the boosted frame: the lemma’s prediction itself satisfies the vielbein postulate with Γ^W ”, “FABLE-5.1 in the boosted frame: the curvature is STILL identically zero”.
- **[proved V §21]** “and Bridge 1’s inhomogeneous term of Section 18 IS this object: $\text{deltaBoost} == \omega \text{Fable51Boost}$ ”, “so $\omega \text{Boost} == (\text{canonical},$

conjugated) + (fable-5.1 in the boosted frame)".

- `omegaFable51BoostMixed` has 4 non-zero components, the same 4 as `deltaBoost` [displayed V §21].

3.8.11 Master comparison of the five connections (Section 22)

Section 22 prints two tables: the summary `cfConnectionSummary`, and a grid of the comparisons side by side [displayed V §22]. The table below is `cfConnectionSummary`, with three entries of the side-by-side grid added to the column “difference from canonical”: Bridge 1’s spinor-level law, and Bridge 3’s curvature and Ricci shift. Part VII restates the bridge results it relies on, so that it can be read alone: [proved VII §27] SPIN CONNECTION [has content]: Parts IV–V restated – Bridge 1’s difference is the pure gauge term, Bridge 2 is a constant conjugate, Bridge 3 has totally antisymmetric torsion that is not zero and SPIN CONNECTION [has content]: Part V restated – $\omega^W = 0$ in the canonical frame, $\omega^{LC} = -\text{contortion}$, $T_\mu = d_\mu \text{Log}[\text{Sin}[6 H x_0]]$, $R = -T + B$, and $\gamma^\mu \Gamma_\mu = -(1/2) T_\mu \gamma^\mu$. For all five connections the curved metric g is the same (“yes (reference)” for the canonical one), and the connection is metric compatible; those two columns of the notebook’s table read “yes” throughout and are not repeated below.

connection; frame field; flat metric	torsion zero? curvature zero?	difference from canonical	verdict
<code>omegaCanonical;</code> <code>frameCanonical</code> (diagonal); <code>eta4488</code>	torsion: yes; curvature: no	— (reference)	Levi-Civita
<code>omegaBoost</code> (Bridge 1); <code>frameCanonical . Lambda(x);</code> <code>eta4488</code>	torsion: yes; curvature: no	gauge term $-\partial_\mu \theta K$, which is the fable-5.1 connection of the boosted frame; at spinor level $\Gamma' = S \Gamma S^{-1} - (\partial S) S^{-1}$ with the explicit spin lift S	SAME connection, local gauge
<code>omegaNull</code> (Bridge 2); <code>frameCanonical . U; etaNull == sigma</code>	torsion: yes; curvature: no	constant similarity $U \omega U$	SAME connection, constant gauge
<code>omegaOct</code> (Bridge 3); <code>frameCanonical . triVecToSpin; etaTri == sigma</code>	torsion: no , axial torsion $-2\lambda m^{\text{Skew}}$; curvature: no	contortion $\lambda m_{abc}^{\text{Skew}} (e_{\text{Tri}})_\mu{}^c$ (with $e_{\text{Tri}} = \text{frameTri}$), a tensor; curvature quadratic in λ ; Ricci shift $-42\lambda^2$	DIFFERENT connection
<code>omegaFable51</code> (fable-5.1); <code>frameCanonical</code> (parallel); <code>eta4488</code>	torsion: no , vector + tensor torsion, axial part zero; curvature: yes , flat	$\omega^{\text{canonical}} = -\text{contortion}(T_{\text{fable51}})$, a tensor; $\omega^{\text{fable51}} = 0$	DIFFERENT connection

Non-zero component counts (Section 22 fingerprint table and the counts printed in Sections 16–21, [displayed]):

object	non-zero components
GammaCanonical (Christoffel)	37
GammaWeitzenboeck	13
omegaCanonical	24
omegaBoost / its difference deltaBoost	28 / 4
omegaNull / its difference	48 / 0
omegaTriLC (Bridge 3, $\lambda = 0$)	84
contortionOct (Bridge 3)	78
torsionOctFlat (Bridge 3)	48
omegaFable51 (canonical frame)	0
contortionFable51 / torsionFable51Flat	24 / 24
RiemannCanonical	156
RiemannFable51	0

3.8.12 What the canonical connection does for the complex fermion fable

Everything in this subsection concerns the *complex* 16-component fermion fable of Part VII. Two kinds of statement appear. Some are consequences of results already proved in Parts I–VI, and the derivation is shown here. The others are assertions of the new Part VII, marked [**proved VII §n**], with the label quoted verbatim from `claude-fable/run_fermion_fable_part7.log`.

3.8.12.1 The field and the symmetrized Lagrangian

Fermion fable is a complex 16-component Grassmann spinor Ψ . It is the minimal field that uses the author's σ_{16} conjugation, propagates, admits a potential $V(s)$, and reduces to Part VI's real field as its real commuting shadow. It is not the unique field with dynamics. This is the conclusion of Section 26 as a whole (section 3.2.8); no single assertion states it. It rests on [**proved VII §26**] REFINEMENT (d) [THE RESULT]: `s = Psibar Psi == i theta1^T sigma16 theta2`: NON-ZERO (16 monomials) and HERMITIAN; `p = Psibar T16[8] Psi` is non-zero and Hermitian too, REFINEMENT (b) [THE RESULT]: Majorana with C+: the kinetic term is genuine - Euler-Lagrange expression `2 C+ T16[a] phi` with C+ T16[a] invertible - but every invariant bilinear vanishes (massless, no $V(s)$), REFINEMENT (c) [THE RESULT]: Weyl and Majorana-Weyl: `PL C T16[a] PL == PR C T16[a] PR == 0` for $C = C-, C+$ and every a (C chirality-diagonal, T16[a] chirality-odd) - no kinetic term and FIDELITY [1] [fidelity]: `Psi -> psi, Psibar -> psi^T sigma16` turns the symmetrized covariant Lhat into Part VI's `cflhatFable` EXACTLY (canonical frame, generic V), among the others quoted in section 3.2. Its Dirac conjugate and scalar bilinear are

$$\bar{\Psi} = \Psi^\dagger \sigma_{16}, \quad s = \bar{\Psi} \Psi = -\psi_1^\dagger \sigma \psi_1 + \psi_2^\dagger \sigma \psi_2, \quad (77)$$

where † is the classical conjugate transpose. The second form follows from $\sigma_{16} = \text{diag}(-\sigma, \sigma)$. In the quantized theory the Hilbert adjoint is $\Psi^\dagger := \Psi^\dagger J$ (subsection 3.8.12.6).

The only Lagrangian used is the *symmetrized covariant* one:

$$L = \sqrt{g} \hat{L}, \quad \hat{L} = \frac{1}{2H} \left[\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi \right] - V(s), \quad (78)$$

$$D_\mu \Psi = \partial_\mu \Psi + \Gamma_\mu \Psi, \quad D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi} \Gamma_\mu.$$

$D_\mu \bar{\Psi}$ is the Dirac conjugate of $D_\mu \Psi$, and $\hat{L}$ is real, because $\sigma_{16} \gamma^a$ and $\sigma_{16} \Gamma_\mu$ are *real antisymmetric* matrices. That rests on [**proved I §5**] “`sigma16 . T16[A]` is antisymmetric for $A = 0..7$ ” and “`sigma16 . SAB` is antisymmetric”, together with $\Gamma_\mu = \frac{1}{2} \omega_{\mu ab} \text{SAB}$ with real ω (subsection 3.8.7). [**proved VII §27**] LAGRANGIAN (a) [definition]: every `Gamma^spin_mu` is real and `sigma16 Gamma_mu` is antisymmetric (`Gamma_mu^T sigma16 == -sigma16 Gamma_mu`): `D_mu Psibar` is the Dirac conjugate of `D_mu Psi`.

3.8.12.2 It drops out of the symmetrized Lagrangian because the matrix $\{\gamma^\mu, \Gamma_\mu\}$ vanishes

Expanding D in (78) gives [derived here]

$$\hat{L}(D) = \hat{L}(\partial) + \frac{1}{2H} \bar{\Psi} \left(\sum_\mu \{\gamma^\mu, \Gamma_\mu\} \right) \Psi. \quad (79)$$

The connection terms are $\bar{\Psi} \gamma^\mu \Gamma_\mu \Psi$ from the first product and $+\bar{\Psi} \Gamma_\mu \gamma^\mu \Psi$ from $-(D_\mu \bar{\Psi}) \gamma^\mu \Psi$. On the canonical frame $\{\gamma^\mu, \Gamma_\mu\} = 0$ for each μ (subsection 3.8.8), so $\hat{L}(D) = \hat{L}(\partial)$ identically. That is, “covariant = partial” holds for the symmetrized complex Lagrangian on this frame. It holds on any frame where the totally antisymmetric part $\omega_{[cab]}$ vanishes (65), and fails in general.

- [proved VII §27] LAGRANGIAN (c) [THE RESULT]: Lsym - Lsym_d == (1/(2H)) Sum_mu Psibar {gamma^mu, Gamma_mu} Psi identically, and {gamma^mu, Gamma_mu} == 0 for EACH mu on the canonical frame: Lsym == Lsym_d.

This is *not* Part VI’s mechanism. For the real commuting field, $\Psi^T \sigma_{16} \gamma^\mu \Gamma_\mu \Psi$ vanishes because $\sigma_{16} T_{16}[0]$ is antisymmetric (subsection 3.8.10). For a complex field, $\bar{\Psi} T_{16}[0] \Psi = \Psi^\dagger (\sigma_{16} T_{16}[0]) \Psi$ with $\sigma_{16} T_{16}[0]$ real antisymmetric is i times the Hermitian form of $-i \sigma_{16} T_{16}[0]$, which is not zero in general. So the drop-out for the complex field is a property of the *matrix* $\{\gamma^\mu, \Gamma_\mu\}$, not of a bilinear identity [derived here]. The other Lagrangians behave differently:

- The *unsymmetrized covariant* Lagrangian $\frac{1}{H} \bar{\Psi} \gamma^\mu D_\mu \Psi - V$ equals the symmetrized one plus $\frac{1}{2H} \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \bar{\Psi} \gamma^\mu \Psi)$, a total divergence. Its connection term is $\frac{1}{H} \bar{\Psi} \gamma^\mu \Gamma_\mu \Psi = -3 \cot^2(6Hx_0) \bar{\Psi} T_{16}[0] \Psi$, which is not zero. [proved VII §27] LAGRANGIAN (d) [has content]: Lun - Lsym == (1/(2H)) (1/Sqrt[g]) d_mu(Sqrt[g] Psibar gamma^mu Psi), a total divergence, LAGRANGIAN (e) [has content]: Lun - Lun_d == (1/H) Psibar gamma^mu Gamma_mu Psi == -3 Cot[6 H x0]^2 Psibar T16[0] Psi, LAGRANGIAN (e) [control]: that term is NOT zero for a complex field (witness u = e1, v = e13), and IS zero in the real commuting limit (the scope of Parts V-VI).
- The *unsymmetrized partial* Lagrangian $\frac{1}{H} \bar{\Psi} \gamma^\mu \partial_\mu \Psi - V$ is *inadmissible*. Its $\bar{\Psi}$ and Ψ equations are not Dirac conjugates of each other: they differ by $\bar{\Psi} \sum_\mu [\gamma^\mu, \Gamma_\mu] = 2 \bar{\Psi} \gamma^\mu \Gamma_\mu = -6H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0]$. [proved VII §27] LAGRANGIAN (f) [THE RESULT]: Lun_d varied with respect to Psibar gives (Sqrt[g]/H)(gamma^mu d_mu Psi - H V' Psi): the connection term is MISSING, LAGRANGIAN (f) [THE RESULT]: varied with respect to Psi it gives -(Sqrt[g]/H)((d_mu Psibar) gamma^mu + Psibar [gamma^mu, Gamma_mu] + H V' Psibar), with Psibar [gamma^mu, Gamma_mu] == 2 Psibar gamma^mu Gamma_mu: the connection term is PRESENT, LAGRANGIAN (f) [control]: so Lun_d is INADMISSIBLE - its two field equations are not Dirac conjugates: they differ by 2 Psibar gamma^mu Gamma_mu = -6 H Cot^2 Psibar T16[0], non-zero (witness u = e1, v = 0, V = s^2).

3.8.12.3 It enters the field equation as $-3H \cot^2 T_{16}[0]$, and the rescaling removes it

Varying $\bar{\Psi}$ and Ψ independently in the symmetrized action gives the field equation and its conjugate:

$$\gamma^\mu D_\mu \Psi = H V'(s) \Psi, \quad (D_\mu \bar{\Psi}) \gamma^\mu = -H V'(s) \bar{\Psi}. \quad (80)$$

[proved VII §27] FIELD EQUATION [THE RESULT]: EL_Psibar == (Sqrt[g]/H)(gamma^mu D_mu Psi - H V'(s) Psi) for generic Psi, Psibar of all eight coordinates: the field equation is gamma^mu D_mu Psi == H V'(s) Psi, FIELD EQUATION [THE RESULT]: EL_Psi == -(Sqrt[g]/H)((D_mu Psibar) gamma^mu + H V'(s) Psibar): the adjoint equation is (D_mu Psibar) gamma^mu == -H V'(s) Psibar. For the real field of Part VI the

same equation is [proved VI §24] “FABLE [THE RESULT]: $EL == \text{Sqrt}[g] (2/H) \text{sigma16}$. ($\gamma^\mu D_\mu \Psi - H V'(s) \Psi$): the field equation of fable is EXACTLY the Levi-Civita covariant Dirac equation $\gamma^\mu D_\mu \Psi == H V'(s) \Psi$ ”.

On the canonical frame the connection enters as follows [derived here], from (64). In the equation it is $\gamma^\mu \Gamma_\mu \Psi = -3H \cot^2(6Hx_0) T_{16}[0] \Psi$. In the conjugate it is $-\bar{\Psi} \Gamma_\mu \gamma^\mu = +\bar{\Psi} \gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) \bar{\Psi} T_{16}[0]$, using the per- μ anticommutator. Because $T_{16}[0] = \begin{pmatrix} 0 & \text{ID}_8 \\ \text{ID}_8 & 0 \end{pmatrix}$, in the 8+8 split-octonion form $\Psi = (\psi_1, \psi_2)$:

$$e_a^\mu \bar{\tau}_a (\partial_\mu + \Gamma_\mu^{(2)}) \psi_2 = H V'(s) \psi_1 \quad (\text{upper 8 rows}), \quad (81)$$

$$e_a^\mu \tau_a (\partial_\mu + \Gamma_\mu^{(1)}) \psi_1 = H V'(s) \psi_2 \quad (\text{lower 8 rows}), \quad (82)$$

where the connection term is $-3H \cot^2(6Hx_0) \psi_2$ in the upper rows and $-3H \cot^2(6Hx_0) \psi_1$ in the lower rows, and $\tau_a = \text{tau}[a]$, $\bar{\tau}_a = \text{taubar}[a]$, $s = -\psi_1^\dagger \sigma \psi_1 + \psi_2^\dagger \sigma \psi_2$, $\bar{\psi}_1 = -\psi_1^\dagger \sigma$ and $\bar{\psi}_2 = \psi_2^\dagger \sigma$. The conjugate equations are $(\partial \bar{\psi}_2 - \bar{\psi}_2 \Gamma^{(2)}) \tau_a e_a^\mu = -H V' \bar{\psi}_1$ and $(\partial \bar{\psi}_1 - \bar{\psi}_1 \Gamma^{(1)}) \bar{\tau}_a e_a^\mu = -H V' \bar{\psi}_2$. [proved VII §27] WRITTEN OUT (ii) [THE RESULT]: the split-octonion 8+8 form - the type-1 rows are the taubar equations for psi2, the type-2 rows the tau equations for psi1 (tau[0] == taubar[0] == ID8), WRITTEN OUT (ii) [THE RESULT]: the covariant 8+8 form - rows 1..8 are $e_a^\mu \text{taubar}[a] (d_\mu + \Gamma_\mu^{(2)}) \psi_2 - H V' \psi_1$, rows 9..16 are $e_a^\mu \text{tau}[a] (d_\mu + \Gamma_\mu^{(1)}) \psi_1 - H V' \psi_2$; and the conjugate rows, WRITTEN OUT (ii) [has content]: with $\text{Psibar} = \Psi^\dagger \text{sigma16}$ and $\text{sigma16} = \text{diag}(-\text{sigma}, \text{sigma})$: $s == -\text{psi1}^\dagger \text{sigma} \text{psi1} + \text{psi2}^\dagger \text{sigma} \text{psi2}$ ($\text{psibar1} = -\text{psi1}^\dagger \text{sigma}$, $\text{psibar2} = \text{psi2}^\dagger \text{sigma}$).

The rescaling. The rescaling of Part V applies unchanged, because it is a statement about the operator. $\Psi = \sqrt{\sin 6Hx_0} \Psi'$ turns the equation into $\sqrt{\sin 6Hx_0} (\gamma^\mu \partial_\mu \Psi' - H V'(s) \Psi') = 0$ with $s = \sin(6Hx_0) \bar{\Psi}' \Psi'$. The connection term is gone. [proved VII §27] WRITTEN OUT (iv) [THE RESULT]: $\Psi = \text{Sqrt}[\text{Sin}[6Hx_0]] \Psi'$ turns the field equation into $\text{Sqrt}[\text{Sin}] (\text{Cot } T_{16}[0] d_0 \Psi' + (1/q) T_{16}[i] d_i \Psi' + T_{16}[4] d_4 \Psi' + (1/p) T_{16}[h] d_h \Psi' - H V'(\text{Sin } s') \Psi')$: NO connection term, WRITTEN OUT (iv) [THE RESULT]: and the adjoint equation into $\text{Sqrt}[\text{Sin}] (\text{Cot } d_0 \bar{\Psi}' T_{16}[0] + \dots + H V'(\text{Sin } s') \bar{\Psi}')$, with no connection term either. The x_4 -only family $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$ solves the equation if and only if $V'' s' = 0$. That holds for linear V (the author's mass term), or for null data $s' = 0$. For the complex field s' is conserved on the x_4 -only equation, so null data give x_4 -only solutions for any V . [proved VII §29] PRE-UNIVERSE (2) [THE RESULT]: $\Psi = \text{Sqrt}[\text{Sin}] \Psi'(x_4)$ satisfies the field equation iff $d_{x0} V'(\text{Sin}[6Hx_0] s') == 0$, and $d_{x0} V'(\text{Sin } s') == 6 H \text{Cos}[6 H x_0] s' V''(\text{Sin } s')$: V linear OR $s' == 0$, PRE-UNIVERSE (2) [has content]: $s' = \bar{\Psi}' \Psi'$ is conserved by the x_4 -only equations ($d_4 \Psi' = -H V' T_{16}[4] \Psi'$, $d_4 \bar{\Psi}' = H V' \bar{\Psi}' T_{16}[4]$) for ANY V ; so null data $s' = 0$ are solutions for any V , with $\rho == V(0)$; and for $V = s'^2$ with $s' != 0$ the x_0 -dependence is real (witness). The real-field version is [proved VI §24] “FABLE [has content]: for the mass term $V = -(2M/H) s$, V' is constant and $\Psi'(x_4)$ IS an exact solution”.

The frame-independent form. Covariant constancy of γ^μ (61) together with a vanishing $\sum_\mu \{\gamma^\mu, \Gamma_\mu\}$ gives, for any frame on which that anticommutator vanishes,

$$\gamma^\mu \Gamma_\mu = \frac{1}{2} \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu) = -\frac{1}{2} T_\mu \gamma^\mu, \quad T_\mu = \partial_\mu \ln \frac{e_\mu^\mu}{\det e} \quad (\text{no sum}), \quad (83)$$

where T_μ is the Weitzenböck torsion vector of a diagonal frame (71). The second equality is a one-line computation for a diagonal frame, since $\sqrt{g} = \det e$ and $\gamma^\mu = T_{16}[\mu]/e_\mu^\mu$ [derived here]. This links the canonical connection directly to the fable-5.1 bridge: the connection term of the

Dirac operator is the Weitzenböck torsion vector. On the canonical frame $T_\mu = \partial_\mu \ln \sin 6Hx_0$ (subsection 3.8.10), and the rescaling by $\exp(\frac{1}{2} \ln \sin) = \sqrt{\sin 6Hx_0}$ removes it. [proved VII §27] SPIN CONNECTION [has content]: Part V restated - $\omega^W = 0$ in the canonical frame, $\omega^{LC} = -\text{contortion}$, $T_\mu = d_\mu \text{Log}[\text{Sin}[6 H x_0]]$, $R = -T + B$, and $\gamma^\mu \Gamma_\mu = -(1/2) T_\mu \gamma^\mu$ (on the canonical frame); and on the time-dependent warped frames of the coupled system, Part VIII, Section 32: WEITZENBOECK [THE RESULT]: the torsion vector of the warped frame is a GRADIENT, $T_\mu = d_\mu \text{Log}[\text{Sin}[6 H x_0]/(A^3 B^3 C)]$; on the canonical member it is Part V's torsionVectorFable51, WEITZENBOECK [THE RESULT]: $\gamma^\mu \Gamma_\mu = -(1/2) T_\mu \gamma^\mu$ on the warped frame - one vector term, a gradient.

3.8.12.4 It enters neither the canonical anticommutator nor the symmetrized Hamiltonian density

The admissible theory is a dimensional reduction $\Psi = \Psi(x_0, x_1, x_2, x_3, x_4)$. It uses a finite hidden coordinate volume $V_{\text{hid}} = \int dx_5 dx_6 dx_7$, which is an assumption: compact timelike directions contain closed timelike curves. Time is x_4 . The canonical anticommutator on a slice $x_4 = y_4$ is

$$\{\Psi_a(x), \Psi_b^\dagger(y)\} = \frac{H}{V_{\text{hid}} \sqrt{|g|}} G_{ab} \delta^4(x_{0..3} - y_{0..3}), \quad G = -i \sigma_{16} \gamma^4. \quad (84)$$

[proved VII §28] DIRAC BRACKET [THE RESULT]: $\{\Theta_i, \Theta_j\} = (\Omega^{-1})_{ij}$ gives $\{\Psi_a, \Psi_b^\dagger\} = i (K^{-1})_{ab} = (G^{-1})_{ab}$ and $\{\Psi_a, \Psi_b\} = 0$, ANTICOMMUTATOR [THE RESULT]: on the canonical frame ($c = \sqrt{g}/H$) $\{\Psi, \Psi^\dagger\} = (H/\sqrt{g}) J = H \cos[6 H x_0] J$ ($J^{-1} = J$), which IS the design's $(H/\sqrt{g})(-i \sigma_{16} \gamma^4)$. No spin connection appears in it. It is fixed by the coefficient of $\partial_4 \Psi$ in the Lagrangian, and $\Gamma_4 = 0$ in any case. On the canonical frame $\gamma^4 = T_{16}[4]$, so $G = -i T_{16}[0] T_{16}[1] T_{16}[2] T_{16}[3] T_{16}[4] = J$ [derived here]. J is the fundamental symmetry of subsection 3.8.12.6.

The Hamiltonian density of the symmetrized Lagrangian is

$$\mathcal{H} = \sqrt{g} \left[-\frac{1}{2H} \sum_{k \neq 4} (\bar{\Psi} \gamma^k \partial_k \Psi - \partial_k \bar{\Psi} \gamma^k \Psi) + V(s) \right], \quad (85)$$

and it contains *no spin connection* [proved VII §28] HAMILTONIAN [THE RESULT]: the Hamiltonian density $\Pi_\Psi d_4 \Psi + d_4 \bar{\Psi} \Pi_{\bar{\Psi}} - \mathcal{L}$ of the symmetrized Lagrangian is $\sqrt{g} [-(1/(2H)) \sum_{k \neq 4} (\bar{\Psi} \gamma^k d_k \Psi - d_k \bar{\Psi} \gamma^k \Psi) + V]$: NO spin connection (fields of all eight coordinates). This follows from subsection 3.8.12.2: the symmetrized $\hat{L}$ does not depend on the connection. The connection re-emerges in the equation of motion. Integrating the symmetrized spatial derivative terms by parts produces $\gamma^\mu \Gamma_\mu = \frac{1}{2} \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} \gamma^\mu)$, because the volume factor and the curved gammas depend on x_0 . [proved VII §28] HAMILTONIAN [THE RESULT]: $(\sqrt{g}/H) \sigma_{16} (\gamma^\mu d_\mu \Psi - H V' \Psi) = i G^{-1} d_4 \Psi - h \Psi$ for admissible $\Psi(x_0..x_4)$: the field equation is $i G^{-1} d_4 \Psi = h \Psi$, HAMILTONIAN [THE RESULT]: $(1/2) \sum_{\mu \neq 4} d_\mu (\sqrt{g} \sigma_{16} \gamma^\mu) = \sqrt{g} \sigma_{16} \sum \gamma^\mu \Gamma_\mu$, and $d_4 (\sqrt{g} \sigma_{16} \gamma^4) = 0$: the connection term is the one that makes h Hermitian.

The same rescaling organizes the one-particle space. $\sqrt{g} dx_0 |\Psi|^2 = |\Psi'|^2 dz$ for $\Psi' = \Psi/\sqrt{\sin 6Hx_0}$, where $z = -\ln \cos(6Hx_0)/(6H)$ is proper distance along x_0 . So the one-particle space of Ψ' is $L^2(dz)$. The x_0 -operator is regular at the wall $z = 0$ and needs a boundary condition there. The Lorentz-covariant, J -compatible ones are $T_{16}[0] \Psi' = \pm \Psi'$ at $z = 0$. It is limit-point at $z \rightarrow \infty$, where no condition is needed. [proved VII §28] BOUNDARY [has content]: the x_0 direction in proper distance $z = -\text{Log}[\text{Cos}[6 H x_0]]/(6 H)$:

$dz/dx_0 == \tan[6Hx_0] == \sec \sin$ (so $\sqrt{g} dx_0 |\Psi|^2 = |\Psi'|^2 dz$), $z(0) == 0$ and $z \rightarrow \infty$ at $6Hx_0 \rightarrow \pi/2$, $\gamma^0 d_0 == T_{16}[0] d_z$, and $\sin[6Hx_0] == \sqrt{1 - \exp[-12Hz]}$ (bounded frame factors), **BOUNDARY [THE RESULT]**: the bag condition $T_{16}[0] \Psi' = \pm \Psi'$ at the wall - $T_{16}[0]^2 == 1$, $\{\beta, T_{16}[0]\} == 0$, the normal-flux form vanishes on both eigenspaces ($P \beta T_{16}[0] P == 0$), and $[J, T_{16}[0]] == 0$.

3.8.12.5 It does not drop out of the energy-momentum tensor

The energy-momentum tensor operator is

$$\hat{T}_{\mu\nu} = -\frac{1}{4H} \left[\bar{\Psi} \gamma_\mu D_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi + (\mu \leftrightarrow \nu) \right] + g_{\mu\nu} \hat{L}, \quad \gamma_\mu = g_{\mu\nu} \gamma^\nu. \quad (86)$$

This is the Hilbert (symmetric-tetrad) tensor of the symmetrized action, and it equals the covariant tensor T_{cov} off shell. It is *built with D , not ∂* . The difference from the tensor T_{par} built with partial derivatives is **[derived here]**

$$T_{\mu\nu}^{\text{cov}} - T_{\mu\nu}^{\text{par}} = -\frac{1}{4H} \bar{\Psi} (\{\gamma_\mu, \Gamma_\nu\} + \{\gamma_\nu, \Gamma_\mu\}) \Psi. \quad (87)$$

The $g_{\mu\nu} \hat{L}$ terms agree by subsection 3.8.12.2. For $\mu = \nu$ the difference is $-\frac{1}{2H} \bar{\Psi} \{\gamma_\mu, \Gamma_\mu\} \Psi = 0$ by the per- μ anticommutator, because $\gamma_\mu = g_{\mu\mu} \gamma^\mu$ on the diagonal frame. So the energy density and all the pressures are the same with D or ∂ . Not every off-diagonal component differs. The (0,4) component coincides, because $\Gamma_0 = \Gamma_4 = 0$. The (5,6) component coincides too, because there the two anticommutators cancel **[derived here]**: $\gamma_5 = -p T_{16}[5]$, $\gamma_6 = -p T_{16}[6]$, the coefficients of Γ_5 and Γ_6 are equal, and $T_{16}[5] T_{16}[a] T_{16}[6] = -T_{16}[6] T_{16}[a] T_{16}[5]$ for $a = 0, 4$. The pair at which the notebook shows the two tensors to differ is given in the third item below.

- **[proved VII §29]** Hilbert = T_{cov} , via a first-order symmetric perturbation of the frame, **EMT [THE RESULT]**: the Hilbert (symmetric-tetrad) tensor of the symmetrized action IS That, OFF shell - components (x1,x1), (x4,x4), (x1,x4), (x0,x5), (x5,x4).
- **[proved VII §29]** the piece coming from the variation of the spin connection is zero, **EMT [has content]**: the spin-connection variation contributes NOTHING - dropping it leaves the Hilbert tensor unchanged at (x1,x4), (x0,x5), (x5,x4). The Lagrangian depends on the connection only through $\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = \frac{1}{2} \omega_{[cab]} \gamma^{cab}$ (65), and under a symmetric variation of the tetrad $\delta \omega_{[cab]} = 0$. So the spin-connection variation contributes *exactly nothing*. The covariant terms of $\hat{T}$ come from the dependence of $\{\gamma^\mu, \Gamma_\mu\}$ on the frame. A partial-derivative Lagrangian must never be varied with respect to the frame. (Part VI's Text cell described this contribution as "a term antisymmetric in μ , ν that drops from the symmetrised tensor". The statement above is the sharper one.)
- **[proved VII §29]** Hilbert $\neq T_{\text{par}}$, witnessed on the non-vacuous pair (x_1, x_4) (the pairs (0,4) and (5,6) coincide on the canonical frame, as shown above), **EMT [control]**: the Hilbert tensor is NOT the partial-derivative tensor: at (x1,x4) they differ - witnessed on constant spinors ($\bar{\Psi} = e_1$, $\Psi = e_{11}$: $(T_{16}[4] T_{16}[1] T_{16}[0])_{\{1,11\}} \neq 0$).
- **[proved VII §29]** T_{cov} is conserved on shell, **EMT CONSERVATION [THE RESULT]**: $\nabla^\mu T_{\mu\nu} == 0$ ON SHELL in all eight components, for generic $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$ and generic V .

The real-field analogues are already proved. **[proved VI §24]** "FABLE [definition]: T_{cov} is symmetric by construction". **[proved VI §24]** "FABLE [has content]: the diagonal of T_{cov} equals the diagonal of the partial-derivative tensor, for the generic

Psi(x0, x4): rho and every pressure are the same in both". [proved VI §24]
 "FABLE [has content]: but the two tensors DIFFER off the diagonal - witnessed
 on the (x3, x4) component T[[4,5]], a momentum density along the observed sheet,
 for the constant spinor e1 + e13 and $V = s^2$ ". [proved VI §24] "FABLE [THE
 RESULT]: $\nabla^\mu T_{\mu\nu} == 0$ ON SHELL in all eight components, for a generic
 Psi(x0, x4) - the covariant tensor is the conserved one". That only the covariant
 tensor is conserved, and T_{par} is not, is [prose VI §24]: the divergence of T_{par} is not computed
 symbolically.

3.8.12.6 The connection and the fundamental symmetry J

$$J = -i T_{16}[0] T_{16}[1] T_{16}[2] T_{16}[3] T_{16}[4], \quad J^2 = \text{ID16}. \quad (88)$$

J is the fundamental symmetry that gives the positive quantization of the admissible sector.
 With $\beta = -i T_{16}[4]$, $\sigma_{16} = J\beta$ holds, and the Hilbert adjoint is $\Psi^\dagger = \Psi^\dagger J$. [proved VII §28]
 J [THE RESULT]: sigma16 == J beta with beta = -i T16[4] Hermitian, beta^2
 == 1; T16[0..3] Hermitian with beta T16[j] beta == -T16[j]: beta is the
 observer's Dirac adjoint, J [THE RESULT]: with Psi^dagger := Psi^ddag J: Psibar
 = Psi^dagger beta (J sigma16 == beta), Psi^ddag G = Psi^dagger (J G == ID16),
 {Psi, Psi^dagger} == (H/Sqrt[g]) ID16 (G^-1 J == ID16): POSITIVE.

$J^2 = \text{ID16}$ follows directly [derived here]. The product $P = T_{16}[0] \cdots T_{16}[4]$ of five mutually
 anticommuting matrices satisfies $P^2 = (-1)^{5 \cdot 4/2} T_{16}[0]^2 \cdots T_{16}[4]^2 = (+1)(1)(1)(1)(-1) =$
 $-\text{ID16}$, so $(-iP)^2 = +\text{ID16}$.

The commutation rules also follow by counting sign changes [derived here]. A single $T_{16}[a]$
 anticommutes with the four factors of P other than itself and commutes with itself, if $a \in \{0..4\}$.
 It anticommutes with all five if $a \in \{5, 6, 7\}$. Hence J commutes with $T_{16}[0..4]$ and anticommutes
 with $T_{16}[5], T_{16}[6], T_{16}[7]$. Applying this to the explicit Γ_μ of subsection 3.8.7:

object	relation to J	reason
$\Gamma_0 = \Gamma_4 = 0$	commute	zero
Γ_j ($j = 1, 2, 3$)	commute	made of $T_{16}[0]T_{16}[j]$, $T_{16}[4]T_{16}[j]$, all factors in $\{0..4\}$
Γ_k ($k = 5, 6, 7$)	anticommute	made of $T_{16}[0]T_{16}[k]$, $T_{16}[4]T_{16}[k]$: one factor in $\{0..4\}$, one in $\{5, 6, 7\}$
each $\gamma^\mu \Gamma_\mu$ (no sum), and their sum	commute	each is $\alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$ (subsection 3.8.8)

The hidden components Γ_5, Γ_6 and Γ_7 are J -odd. They enter the Dirac operator only through
 $\gamma^k \Gamma_k$, a product of two J -odd matrices, which is J -even. So every term of the admissible-sector
 Hamiltonian, the connection term included, is J -even. The J -odd objects, namely the $(0, h)$, (i, h)
 and $(4, h)$ components of T_{cov} and the hidden momenta P_h ($h = 5, 6, 7$), are nonzero operators
 with zero expectation in J -eigenmode Fock states. [proved VII §28] J TABLE [THE RESULT]:
 J commutes with sigma16, T16[0..4], sigma16 gamma^mu (mu = 0..4), every gamma^mu
 Gamma_mu and Gamma_1..Gamma_3; it ANTICommutes with T16[5..7], sigma16 gamma^h
 (h = 5..7), Gamma_5..Gamma_7, T16[8] and C+. The same J table holds on the dynamical
 (warped and Bianchi-I) frames of the coupled system. Part VIII, Section 32, proves it: J [THE
 RESULT]: [J, gamma^mu Gamma_mu] == 0 for EACH mu separately (h = 5,6,7 included),
 on the warped and on the Bianchi-I frame, J [THE RESULT]: [J, Gamma_mu] == 0
 for mu = 0..4 and {J, Gamma_h} == 0 for h = 5, 6, 7, on the warped and on
 the Bianchi-I frame and J [has content]: the anticommuting Gamma_h are not zero
 (witnessed on $A = 1 + x4/2$, $B = 1 + x4/3$, $C = 1 + x4^2/3$), so {J, Gamma_h} == 0
 is not vacuous.

3.8.13 Status ledger

statement	status	where (notebook Part and Section; assertion tag)
frame e , $g = e \eta e^T$, stated line element, signature (4,4)	proved	III §15
a_4 is free and never given a value	proved	VI §25, VII §29 and VIII (controls); notice printed in III §15
$\det g = \sec^2 6Hx_0$; $\sqrt{ \det g } = \sec 6Hx_0$ on $0 < 6Hx_0 < \pi/2$	displayed; proved	displayed III §15; proved V §21
37 Christoffel symbols; symmetric in the lower indices; metric compatible	displayed; proved	III §16; VI §23
the vielbein postulate is solved by <code>cfSpinConnection</code> (residual zero)	proved (solver regression)	III §16
the 24 non-zero components of <code>omegaCanonical</code>	displayed; independent frame-only derivation proved; count 24 proved	III §16; VII §27 (SPIN CONNECTION [has content])
antisymmetry in the flat indices (metric compatibility)	proved (has content)	III §16; VII §27
torsion-free	proved (structural)	III §16
curvature: 156 components; Ricci scalar	displayed	III §16
$\Gamma_\mu = \frac{1}{8}\omega_{\mu ab}[T_{16}[a], T_{16}[b]]$; sign and coefficient fixed by compatibility	proved (with control)	III §17; VII §27 (SPIN CONNECTION [definition])
block-diagonal; type-1 and type-2 blocks; <code>Dcov16</code> = direct sum	proved	III §17
covariant constancy of γ^μ	proved	III §17
$\frac{1}{\sqrt{g}}\partial_\mu(\sqrt{g}\gamma^\mu) = [\gamma^\mu, \Gamma_\mu]$	proved	VI §24; VII §27 (FIELD EQUATION [has content])
$\sum_\mu \{\gamma^\mu, \Gamma_\mu\} = 0$; $\gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) T_{16}[0]$	proved	VI §24; VII §27
per-direction $\gamma^\mu \Gamma_\mu = \alpha_\mu T_{16}[0] + \beta_\mu T_{16}[4]$, $\sum \alpha_\mu = -3H \cot^2(6Hx_0)$, $\sum \beta_\mu = 0$;	derived here; proved	VII §27 (SPIN CONNECTION [THE RESULT], LAGRANGIAN (c) [THE RESULT])
anticommutator zero for each μ		
Bridges 1 and 2: the same connection in a local / constant gauge, at vector and spinor level	proved	IV §18, IV §19; restated VII §27
Bridge 3: a different connection; torsion $-2\lambda m^{\text{Skew}}$; curvature of degree 2 in λ ; Ricci shift	proved	IV §20; restated VII §27
$-42\lambda^2 = -\frac{1}{4}T \cdot T$		
fable-5.1: $\omega = 0$, zero curvature, $\omega^{LC} = -\text{contortion}(T)$, $T_\mu = \partial_\mu \ln \sin 6Hx_0$, axial part zero, $R = -T + B$	proved	V §21; restated VII §27
$\gamma^\mu \Gamma_\mu = -\frac{1}{2}T_\mu \gamma^\mu$; the rescaling $\Psi = \sqrt{\sin 6Hx_0} \Psi'$	proved	V §21; VI §24; VII §27 (SPIN CONNECTION [has content], WRITTEN OUT (iv)); VIII §32 on the warped frames (WEITZENBOECK, RESCALING)
Bridge 1's gauge term is the fable-5.1 connection of the boosted frame	proved	V §21
master table of the five connections	displayed	V §22
complex fable: the connection drops out of the symmetrized L because the matrix $\{\gamma^\mu, \Gamma_\mu\}$ vanishes	proved	VII §27 (LAGRANGIAN (c) [THE RESULT])
unsymmetrized covariant L = symmetrized + divergence; unsymmetrized partial L inadmissible	proved	VII §27 (LAGRANGIAN (d), (e), (f))
field equations; connection term	proved	VII §27 (FIELD EQUATION, WRITTEN OUT (i)-(iv), SPIN CONNECTION); VIII §32
$-3H \cot^2(6Hx_0) T_{16}[0]$; 8+8 form; rescaling; frame-independent form		
no connection in the canonical anticommutator or in the symmetrized Hamiltonian density; the connection makes h Hermitian	proved	VII §28 (QUANTIZATION [has content]: <code>Gamma_4 == 0</code> , ANTICOMMUTATOR, HAMILTONIAN)

statement	status	where (notebook Part and Section; assertion tag)
EMT: Hilbert = T_{cov} ; spin-connection variation zero; Hilbert $\neq T_{\text{par}}$; T_{cov} conserved	proved	VII §29 (EMT, EMT CONSERVATION)
J commutes with each $\gamma^\mu \Gamma_\mu$ and with $\Gamma_0 \dots \Gamma_4$, and anticommutes with $\Gamma_5 \dots \Gamma_7$	derived here; proved	VII §28 (J TABLE [THE RESULT]); VIII §32 on the warped and Bianchi-I frames (J [THE RESULT])

3.9 How it was verified

3.9.1 Part VII of the notebook: sections, cells, assertions

Part VII is the manifest `claude-fable/cells_part7.wl` (1777 lines; 170 `cfAssert` calls). Built into the notebook by `claude-fable/build_tools.py` (section 6.3.1), it adds 26 Input cells, 173–198, to the 172 of Parts I–VI. The section-to-cell map is printed by `claude-fable/nb_section_map.wls` (`claude-fable/nb_section_map.log`, section 6.3.3):

notebook Section	subject	Input cells	assertions	seconds (sum of the cells' times)	this document
26	the refinement: why fermion fable is a complex 16-spinor	173–179	38	1.957	3.2
27	the Lagrangian, the field equations in the primordial gravitational field, and the canonical spin connection	180–184	36	2.291	3.3, 3.4, 3.8
28	canonical quantization in 4+4 dimensions: the Dirac bracket, the Krein structure J , and the admissible sector	185–191	56	12.311	3.5
29	the energy–momentum tensor operator: definition, conservation, and ρ , P and w	192–198	40	17.874	3.6, 3.7
Part VII		26 cells	170	34.433	

The counts and times per cell, read from `claude-fable/run_fermion_fable_part7.log` (each cell's `PASS` lines are printed before its `CELL n t=...` line):

cell	173	174	175	176	177	178	179	180	181	182	183	184	185
assertions	4	2	9	7	7	4	5	10	5	3	5	13	5
seconds	0.016	0.541	0.117	0.592	0.053	0.064	0.574	1.005	0.403	0.395	0.209	0.279	0.062
cell	186	187	188	189	190	191	192	193	194	195	196	197	198
assertions	5	9	10	9	6	12	5	4	7	3	11	9	1
seconds	0.017	0.344	0.129	0.221	1.680	9.858	2.248	4.431	4.080	2.098	4.945	0.072	0.000

The 170 assertions by tag (the text of a label before its first `[`), counted from the log:

tag	count	tag	count
REFINEMENT	6	ADMISSIBILITY (all items)	10
GRASSMANN	9	HAMILTONIAN	7
REFINEMENT (a), (b), (c), (d)	3, 3, 1, 7	BOUNDARY	2
4D	4	KREIN MODES	6

tag	count	tag	count
FIDELITY	5	FOCK, FOCK (field level), FOCK (mode level)	1, 4, 6
LAGRANGIAN (a)–(f)	1, 2, 1, 1, 2, 3	CHARGE	1
FIELD EQUATION	5	EMT	8
CURRENT	3	EMT CONSERVATION, EMT ON SHELL, EMT HERMITICITY	3, 3, 2
SPIN CONNECTION	5	MODE SUMS, VACUUM	2, 1
WRITTEN OUT (i)–(iv)	2, 3, 3, 3	KOHN-SHAM, MEAN FIELD	5, 1
SYMMETRY	2	AUTHOR’S MASS TERM, CLASSICAL LIMIT	4, 1
QUANTIZATION	5	PRE-UNIVERSE (1)–(4)	2, 2, 2, 2
DIRAC BRACKET, ANTICOMMUTATOR	3, 2	PART VII	1
J, J TABLE	8, 1	“no identity in this notebook was accepted on numerical evidence alone”	1

By grade: the labels carry [THE RESULT] (a headline result), [has content] (a statement that could have failed), [definition], [structural], [solver regression] (a check of the code), [control] (the wrong alternative is shown to fail) and [fidelity] (agreement with the author’s or an earlier Part’s objects).

The final tally and the run summary, verbatim from the end of `claude-fable/run_fermion_fable_part7.log`:

```

identities accepted on numerical evidence alone : 0    (stage 3 returned True)
non-vanishing witnesses                        : 38    (cfNonZeroWitnessQ found a probe point at which
↪ every entry is a                               number and one of them is non-zero: a complete
                                                    ↪ proof of non-vanishing)

PASS no identity in this notebook was accepted on numerical evidence alone
Assertions run: 528   passed: 528   failed: 0
CELL 198  t=0.000 s
===== RUN SUMMARY =====
cells evaluated : 198
total seconds   : 232.146
cells w/ msgs   : 0
---- slowest cells ----
cell 80  89.913 s
cell 88  27.790 s
cell 86  11.167 s
cell 87  10.303 s
cell 191  9.858 s
cell 137  8.740 s
cell 160  6.445 s
cell 121  5.401 s
cell 196  4.945 s
cell 193  4.431 s
cell 107  4.158 s
cell 194  4.080 s
===== ASSERTIONS =====
assertions run : 528
passed         : 528
FAILED         : 0
=====
RUN-DONE

```

So through Part VII: 198 Input cells, **528 assertions, 528 passed, 0 failed, 0 cells with messages, 0 identities accepted on numerical evidence alone** (every identity is closed symbolically; the notebook’s last assertion is `no identity in this notebook was accepted on numerical evidence alone`), and **38 non-vanishing witnesses** (19 from Parts I–V, 8 more from Part VI, 11 more from Part VII: the Part V and Part VI tallies in the same log print 19 and 27). Parts I–VI account for 358 of the assertions; Part VII adds 170. The whole run took

232.146 s; the slowest cells are Part II's (cell 80, the bilinears of the author's closed-form solution, 89.913 s). The last line, RUN-DONE, is not printed by `run_from_nb.wls`: it was appended by the shell wrapper that ran it.

3.9.2 Every Part VII assertion, verbatim

This is the complete output of Input cells 173–198 in `claude-fable/run_fermion_fable_part7.log` (lines 634–844), from the line that closes cell 172 to the line that closes cell 198. Every label is printed after PASS; a bracketed `[t s]` line is a timing printed by `cfTimed`. Lines longer than the page are broken, and the break is marked with a hook arrow.

```
CELL 172  t=0.000 s
PASS REFINEMENT [definition]: T16[8]^2 == sigma16^2 == C+^2 == ID16, C+ is symmetric, and C+ . T16[8]
  ↪ == sigma16 -- the four candidate mass matrices are only two
PASS REFINEMENT [has content]: C- T16[a] C-^-1 == -T16[a]^T and C+ T16[a] C+^-1 == +T16[a]^T for a =
  ↪ 0..7
PASS REFINEMENT [THE RESULT]: the symmetry of C Gamma depends only on the rank r of Gamma: C-: S A A
  ↪ S S A A S S, C+: S S A A S S A A S (r = 0..8)
PASS REFINEMENT [has content]: for each C, 136 of the 256 products are symmetric and 120
  ↪ antisymmetric
CELL 173  t=0.016 s
PASS REFINEMENT [THE RESULT]: the Spin(4,4)-invariant bilinear forms on the 16-spinor form a
  ↪ 2-dimensional space, spanned by C- = sigma16 and C+ = sigma16 T16[8]
PASS REFINEMENT [has content]: every invariant form is SYMMETRIC and chirality-diagonal, and the two
  ↪ chiral pieces PL sigma16 PL and PR sigma16 PR are invariant separately
CELL 174  t=0.541 s
PASS GRASSMANN [solver regression]: generators anticommute and square to zero (all pairs of 1..6)
PASS GRASSMANN [solver regression]: the product is associative, (x y) z == x (y z), on elements of
  ↪ degree 1, 2, 3 (and the products tested are not zero)
PASS GRASSMANN [solver regression]: graded commutativity x y == (-1)^(|x| |y|) y x (degrees 1, 2,
  ↪ 3), and x^2 == 0 for x odd
PASS GRASSMANN [solver regression]: the conjugation is an involutive anti-automorphism, (x y)^ddag
  ↪ == y^ddag x^ddag, (x^ddag)^ddag == x
PASS GRASSMANN [solver regression]: grD is an even derivation, d(x y) == d(x) y + x d(y); grDL obeys
  ↪ d_i(x y) == d_i(x) y + (-1)^|x| x d_i(y)
PASS GRASSMANN [THE RESULT]: theta^T N theta == theta^T N_A theta and theta^T N_S theta == 0 for a
  ↪ general 16x16 N -- an anticommuting bilinear sees only the ANTISYMMETRIC part
PASS GRASSMANN [THE RESULT]: theta^T N phi - (1/2) d(theta^T N theta) == theta^T N_S phi for a
  ↪ general N
PASS GRASSMANN [THE RESULT]: the Euler-Lagrange expression of L = theta^T N phi (left derivatives) is
  ↪ (N + N^T) phi: EMPTY field equations exactly when N is antisymmetric
PASS GRASSMANN [control]: for COMMUTING components the roles are reversed -- theta^T N theta sees
  ↪ only N_S, and the Euler-Lagrange expression of theta^T N phi is 2 N_A phi
CELL 175  t=0.117 s
PASS REFINEMENT (a) [THE RESULT]: real Grassmann + sigma16: both invariant mass terms vanish, and
  ↪ theta^T sigma16 T16[a] phi == (1/2) d(theta^T sigma16 T16[a] theta) for every a (a total
  ↪ derivative)
PASS REFINEMENT (a) [has content]: its Euler-Lagrange expression is identically empty in every
  ↪ direction -- while the kinetic term itself is NOT zero
PASS REFINEMENT (a) [THE RESULT]: FRAME-INDEPENDENT -- sigma16 {T16[a], S^bc} is SYMMETRIC for all a,
  ↪ b, c (so theta^T sigma16 {gamma^mu, Gamma_mu} theta == 0 on every frame), and d_mu(Sqrt[g] sigma16
  ↪ gamma^mu) == Sqrt[g] sigma16 [gamma^mu, Gamma_mu] on the canonical frame
PASS REFINEMENT (b) [THE RESULT]: Majorana with C+: the kinetic term is genuine -- Euler-Lagrange
  ↪ expression 2 C+ T16[a] phi with C+ T16[a] invertible -- but every invariant bilinear vanishes
  ↪ (massless, no V(s))
PASS REFINEMENT (b) [has content]: the scope -- the quartic Lorentz scalars v.v, B.B, C.C of the
  ↪ Majorana spinor are NOT zero and are all proportional to ONE quartic (same monomials, one constant
  ↪ ratio each)
PASS REFINEMENT (b) [has content]: and the commuting shadow of the C+ Majorana spinor has NO kinetic
  ↪ term: for commuting components theta^T C+ T16[a] phi is the total derivative (1/2) d(theta^T C+
  ↪ T16[a] theta)
PASS REFINEMENT (c) [THE RESULT]: Weyl and Majorana-Weyl: PL C T16[a] PL == PR C T16[a] PR == 0 for C
  ↪ = C-, C+ and every a (C chirality-diagonal, T16[a] chirality-odd) -- no kinetic term
CELL 176  t=0.592 s
PASS REFINEMENT (d) [definition]: Psi^ddag == (theta1 - i theta2)^T/Sqrt[2] in the algebra
PASS REFINEMENT (d) [THE RESULT]: s = Psibar Psi == i theta1^T sigma16 theta2: NON-ZERO (16
  ↪ monomials) and HERMITIAN; p = Psibar T16[8] Psi is non-zero and Hermitian too
PASS REFINEMENT (d) [has content]: s is a sum of 16 commuting nilpotent even elements x_a (x_a x_b ==
  ↪ x_b x_a, x_a^2 == 0), so s^17 == 0 and V(s) is a finite polynomial
PASS REFINEMENT (d) [definition]: the 32x32 matrix A read off the algebra reproduces Psi^ddag K phi
  ↪ exactly
```

PASS REFINEMENT (d) [THE RESULT]: its symmetric part is $A_S == (i/2) \Omega$, $\Omega = [[0, K], [-K, 0]]$
 $\hookrightarrow$ real SYMMETRIC and invertible: a genuine kinetic term
PASS REFINEMENT (d) [has content]: the symmetrized kinetic term is HERMITIAN and equals $\Psi^\dagger \ddag K \Psi$
 $\hookrightarrow \phi - (1/2) d(\Psi^\dagger \ddag K \Psi)$; the unsymmetrized one is not Hermitian
PASS REFINEMENT (d) [has content]: the same holds in every direction $a = 0..7$: the symmetrized
 $\hookrightarrow \Psi^\dagger \ddag \sigma_{16} T_{16}[a] \phi$ is Hermitian, with symmetric part $(i/2)[[0, \sigma_{16} T_{16}[a]], [-\sigma_{16} T_{16}[a], 0]] \neq 0$

CELL 177 $t=0.053$ s
PASS 4D [THE RESULT]: the 16 ordered products of $T_{16}[1..4]$ are linearly independent (the observer's
 $\hookrightarrow$ algebra is $M_4(C)$), and its commutant in $M_{16}(C)$ is 16-dimensional
PASS 4D [has content]: $\Omega^4 = -ID_{16}$, the hidden gammas $h_A = \Omega^4 T_{16}[A]$ ($A = 0,5,6,7$) commute
 $\hookrightarrow$ with $T_{16}[1..4]$ and obey $\{h_A, h_B\} = -2 \eta_{AB}$
PASS 4D [THE RESULT]: the 16 products of the hidden gammas span the commutant, and the two algebras
 $\hookrightarrow$ meet only in the multiples of ID_{16} (rank of the union $16 + 16 - 1 = 31$): $C^{16} = C^4(x) C^4$, four
 $\hookrightarrow$ 4D Dirac fermions
PASS 4D [has content]: the observer's Lorentz generators $T_{16}[i] T_{16}[j]$ ($i < j$ in $1..4$) commute with
 $\hookrightarrow$ every hidden gamma

CELL 178 $t=0.064$ s
[0.024 s] Euler-Lagrange expressions of the complex fable with respect to Ψ_{ibar} , generic (x_0, x_4)
[0.023 s] Euler-Lagrange expressions of the complex fable with respect to Ψ , generic (x_0, x_4)
PASS FIDELITY [1] [fidelity]: $\Psi \rightarrow \psi$, $\Psi_{\text{ibar}} \rightarrow \psi^T \sigma_{16}$ turns the symmetrized covariant
 $\hookrightarrow$ Lhat into Part VI's cfLhatFable EXACTLY (canonical frame, generic V)
PASS FIDELITY [2] [fidelity]: on the flat frame with $V = -(2M/H)$ s and no volume factor, the same
 $\hookrightarrow$ limit on Ψ_{16} IS the author's $\text{La}[]$ of Section 11
PASS FIDELITY [3] [fidelity]: $\sigma_{16} \cdot \text{EL}_{\Psi_{\text{ibar}}} + \text{EL}_{\Psi}$ in the real limit == Part VI's explicit
 $\hookrightarrow$ variation cfELFable , component for component
PASS FIDELITY [3] [fidelity]: and on the flat frame with $V = -(2M/H)$ s the same combination IS the
 $\hookrightarrow$ author's eLa of Section 12
PASS FIDELITY [4] [fidelity]: the energy-momentum tensor That in the real limit IS Part VI's
 $\hookrightarrow$ covariant tensor cfTCov , all 64 components

CELL 179 $t=0.574$ s
PASS LAGRANGIAN (a) [definition]: every Γ^{spin_μ} is real and $\sigma_{16} \Gamma_\mu$ is antisymmetric
 $\hookrightarrow$ ($\Gamma_\mu^T \sigma_{16} = -\sigma_{16} \Gamma_\mu$): $D_\mu \Psi_{\text{ibar}}$ is the Dirac conjugate of $D_\mu \Psi$
PASS LAGRANGIAN (b) [THE RESULT]: L_{sym} is HERMITIAN -- real for $\Psi = u + i v$, $\Psi_{\text{ibar}} = (u - i v)^T$
 $\hookrightarrow$ σ_{16} , generic u, v of all eight coordinates
PASS LAGRANGIAN (b) [control]: L_{un} is NOT Hermitian -- its imaginary part is not zero, witnessed on
 $\hookrightarrow$ the constant spinors $u = e_1, v = e_{13}$ (with $V = s^2$)
PASS LAGRANGIAN (c) [THE RESULT]: $L_{\text{sym}} - L_{\text{sym}_d} = (1/(2H)) \sum_\mu \Psi_{\text{ibar}} \{\Gamma^\mu, \Gamma_\mu\} \Psi$
 $\hookrightarrow$ identically, and $\{\Gamma^\mu, \Gamma_\mu\} = 0$ for EACH μ on the canonical frame: $L_{\text{sym}} = L_{\text{sym}_d}$
PASS LAGRANGIAN (d) [has content]: $L_{\text{un}} - L_{\text{sym}} = (1/(2H)) (1/\sqrt{g}) d_\mu (\sqrt{g} \Psi_{\text{ibar}} \Gamma^\mu$
 $\hookrightarrow \Psi)$, a total divergence
PASS LAGRANGIAN (e) [has content]: $L_{\text{un}} - L_{\text{un}_d} = (1/H) \Psi_{\text{ibar}} \Gamma^\mu \Gamma_\mu \Psi = -3 \text{Cot}[6 H$
 $\hookrightarrow x_0]^2 \Psi_{\text{ibar}} T_{16}[0] \Psi$
PASS LAGRANGIAN (e) [control]: that term is NOT zero for a complex field (witness $u = e_1, v = e_{13}$),
 $\hookrightarrow$ and IS zero in the real commuting limit (the scope of Parts V-VI)
PASS LAGRANGIAN (f) [THE RESULT]: L_{un_d} varied with respect to Ψ_{ibar} gives $(\sqrt{g}/H)(\Gamma^\mu d_\mu \Psi_{\text{ibar}}$
 $\hookrightarrow \Psi - H V' \Psi)$: the connection term is MISSING
PASS LAGRANGIAN (f) [THE RESULT]: varied with respect to Ψ it gives $-(\sqrt{g}/H)((d_\mu \Psi_{\text{ibar}})$
 $\hookrightarrow \Gamma^\mu + \Psi_{\text{ibar}} [\Gamma^\mu, \Gamma_\mu] + H V' \Psi_{\text{ibar}}$, with $\Psi_{\text{ibar}} [\Gamma^\mu, \Gamma_\mu] = 2$
 $\hookrightarrow \Psi_{\text{ibar}} \Gamma^\mu \Gamma_\mu$: the connection term is PRESENT
PASS LAGRANGIAN (f) [control]: so L_{un_d} is INADMISSIBLE -- its two field equations are not Dirac
 $\hookrightarrow$ conjugates: they differ by $2 \Psi_{\text{ibar}} \Gamma^\mu \Gamma_\mu = -6 H \text{Cot}^2 \Psi_{\text{ibar}} T_{16}[0]$, non-zero
 $\hookrightarrow$ (witness $u = e_1, v = 0, V = s^2$)

CELL 180 $t=1.005$ s
[0.134 s] Euler-Lagrange expressions with respect to Ψ_{ibar} , generic fields of all eight coordinates
[0.118 s] Euler-Lagrange expressions with respect to Ψ , generic fields of all eight coordinates
PASS FIELD EQUATION [has content]: the identity the derivation uses -- $(1/\sqrt{g}) d_\mu (\sqrt{g}$
 $\hookrightarrow \Gamma^\mu) = [\Gamma^\mu, \Gamma_\mu]$ (covariant constancy, Section 17)
PASS FIELD EQUATION [THE RESULT]: $\text{EL}_{\Psi_{\text{ibar}}} = (\sqrt{g}/H)(\Gamma^\mu D_\mu \Psi - H V'(s) \Psi)$ for
 $\hookrightarrow$ generic Ψ, Ψ_{ibar} of all eight coordinates: the field equation is $\Gamma^\mu D_\mu \Psi = H V'(s)$
 $\hookrightarrow \Psi$
PASS FIELD EQUATION [THE RESULT]: $\text{EL}_{\Psi} = -(\sqrt{g}/H)((D_\mu \Psi_{\text{ibar}}) \Gamma^\mu + H V'(s) \Psi_{\text{ibar}})$:
 $\hookrightarrow$ the adjoint equation is $(D_\mu \Psi_{\text{ibar}}) \Gamma^\mu = -H V'(s) \Psi_{\text{ibar}}$
PASS FIELD EQUATION [has content]: the adjoint equation IS the Dirac conjugate of the field equation:
 $\hookrightarrow$ with $\Psi = u + i v, \Psi_{\text{ibar}} = (u - i v)^T \sigma_{16}$, adjoint residual == $-(\text{field residual})^* \cdot$
 $\hookrightarrow \sigma_{16}$
PASS FIELD EQUATION [control]: the opposite sign, $(D \Psi_{\text{ibar}}) \Gamma = +H V' \Psi_{\text{ibar}}$, is NOT the
 $\hookrightarrow$ conjugate -- witnessed on $u = e_1 + e_5, v = 0$ with $V = s^2$ ($s = -2$ there)

CELL 181 $t=0.403$ s
PASS CURRENT [THE RESULT]: the Noether current of the $U(1)$ phase is $(\sqrt{g}/H) i \Psi_{\text{ibar}} \Gamma^\mu \Psi$
PASS CURRENT [has content]: $i \sigma_{16} \Gamma^\mu$ is Hermitian for every μ , so j^μ is real
PASS CURRENT [THE RESULT]: OFF SHELL $(1/\sqrt{g}) d_\mu (\sqrt{g} j^\mu) = i (E_{\text{bar}} \Psi + \Psi_{\text{ibar}} E)$ for
 $\hookrightarrow$ generic fields of all eight coordinates -- conserved on shell

CELL 182 $t=0.395$ s

PASS SPIN CONNECTION [has content]: metric compatible ($\omega_{\mu}^{\{ab\}}$ antisymmetric) and every
 $\hookrightarrow$ non-zero component is $\omega_{\mu}^{\{0\}}$ or $\omega_{\mu}^{\{4\}}$ (up to antisymmetry): $\omega_{\mu}^{\{0\}} = \omega_{\mu}^{\{4\}}$
 $\hookrightarrow = 0$

PASS SPIN CONNECTION [definition]: $\Gamma^{\text{spin}}_{\mu} = \sum_{\{a,b\}} \omega_{\mu,ab} (1/2) T_{16}[a] T_{16}[b]$

PASS SPIN CONNECTION [THE RESULT]: direction by direction $\gamma_{\mu} \Gamma_{\mu} = \alpha_{\mu} T_{16}[0] +$
 $\hookrightarrow \beta_{\mu} T_{16}[4]$; $\sum \alpha_{\mu} = -3 H \cot[6 H x_0]^2$ and $\sum \beta_{\mu} = 0$ (the observed and hidden
 $\hookrightarrow$ a4 terms cancel)

PASS SPIN CONNECTION [has content]: Parts IV-V restated -- Bridge 1's difference is the pure gauge
 $\hookrightarrow$ term, Bridge 2 is a constant conjugate, Bridge 3 has totally antisymmetric torsion that is not
 $\hookrightarrow$ zero

PASS SPIN CONNECTION [has content]: Part V restated -- $\omega^W = 0$ in the canonical frame, ω^{LC}
 $\hookrightarrow = -\text{contortion}$, $T_{\mu} = d_{\mu} \log[\sin[6 H x_0]]$, $R = -T + B$, and $\gamma_{\mu} \Gamma_{\mu} = -(1/2) T_{\mu}$
 $\hookrightarrow \gamma_{\mu}$

CELL 183 $t=0.209$ s

PASS WRITTEN OUT (i) [THE RESULT]: the field equation on the canonical frame is $\cot T_{16}[0] d_0 \Psi +$
 $\hookrightarrow (1/q) T_{16}[i] d_i \Psi + T_{16}[4] d_4 \Psi + (1/p) T_{16}[h] d_h \Psi - 3 H \cot^2 T_{16}[0] \Psi = H V'(s) \Psi$

PASS WRITTEN OUT (i) [THE RESULT]: the adjoint equation is $\cot d_0 \bar{\Psi} T_{16}[0] + \dots - 3 H \cot^2$
 $\hookrightarrow \bar{\Psi} T_{16}[0] = -H V'(s) \bar{\Psi}$ (the connection enters with the sign fixed by $\{\gamma_{\mu},$
 $\hookrightarrow \Gamma_{\mu}\} = 0$)

PASS WRITTEN OUT (ii) [THE RESULT]: the split-octonion 8+8 form -- the type-1 rows are the taubar
 $\hookrightarrow$ equations for ψ_2 , the type-2 rows the tau equations for ψ_1 ($\tau[0] = \text{taubar}[0] = \text{ID8}$)

PASS WRITTEN OUT (ii) [THE RESULT]: the covariant 8+8 form -- rows 1..8 are $e_a^{\mu} \text{taubar}[a] (d_{\mu} +$
 $\hookrightarrow \Gamma_{\mu}^{(2)}) \psi_2 - H V' \psi_1$, rows 9..16 are $e_a^{\mu} \tau[a] (d_{\mu} + \Gamma_{\mu}^{(1)}) \psi_1 - H V'$
 $\hookrightarrow \psi_2$; and the conjugate rows

PASS WRITTEN OUT (ii) [has content]: with $\bar{\Psi} = \Psi^{\dagger} \sigma_{16}$ and $\sigma_{16} = \text{diag}(-\sigma,$
 $\hookrightarrow \sigma)$: $s = -\psi_1^{\dagger} \sigma_{16} \psi_2 + \psi_2^{\dagger} \sigma_{16} \psi_1$ ($\bar{\psi}_1 = -\psi_1^{\dagger} \sigma_{16}$, $\bar{\psi}_2 =$
 $\hookrightarrow \psi_2^{\dagger} \sigma_{16}$)

PASS WRITTEN OUT (iii) [definition]: the term lists reproduce all sixteen field equations and all
 $\hookrightarrow$ sixteen adjoint equations exactly

PASS WRITTEN OUT (iii) [has content]: every component equation has exactly one derivative per
 $\hookrightarrow$ coordinate $x_0..x_7$, the connection term multiplies the component carrying the x_0 -derivative, and $-H$
 $\hookrightarrow V'$ multiplies ψ_r itself

PASS SYMMETRY [THE RESULT]: the only internal $U(1)$ is the vector phase -- the commutant of all eight
 $\hookrightarrow T_{16}[a]$ in $M_{16}(C)$ is one-dimensional (multiples of ID_{16})

PASS SYMMETRY [has content]: the axial phase $\exp(i \alpha T_{16}[8])$ leaves s invariant but not the
 $\hookrightarrow$ kinetic term: $\exp(-i \alpha T_8) T_{16}[a] \exp(i \alpha T_8) = T_{16}[a] \exp(2 i \alpha T_8)$, so $\bar{\Psi}$
 $\hookrightarrow \gamma_{\mu} d_{\mu} \Psi$ changes at first order by $2 i \alpha \bar{\Psi} \gamma_{\mu} T_{16}[8] d_{\mu} \Psi$, which is
 $\hookrightarrow$ not zero (witness)

PASS WRITTEN OUT (iii) [has content]: for fields of all eight coordinates the sixteen equations form
 $\hookrightarrow$ ONE coupled block (the four blocks of four of Section 13 belong to the (x_0, x_4) reduction)

PASS WRITTEN OUT (iv) [THE RESULT]: $\Psi = \sqrt{\sin[6 H x_0]} \Psi'$ turns the field equation into
 $\hookrightarrow \sqrt{\sin} (\cot T_{16}[0] d_0 \Psi' + (1/q) T_{16}[i] d_i \Psi' + T_{16}[4] d_4 \Psi' + (1/p) T_{16}[h] d_h \Psi' -$
 $\hookrightarrow H V'(\sin s') \Psi')$: NO connection term

PASS WRITTEN OUT (iv) [THE RESULT]: and the adjoint equation into $\sqrt{\sin} (\cot d_0 \bar{\Psi}' T_{16}[0] +$
 $\hookrightarrow \dots + H V'(\sin s') \bar{\Psi}')$, with no connection term either

PASS WRITTEN OUT (iv) [definition]: under the rescaling $s = \sin[6 H x_0] s'$, $s' = \bar{\Psi}' \Psi'$

CELL 184 $t=0.279$ s

PASS QUANTIZATION [definition]: lapse 1 and shift 0 ($g_{44} = -1$, $g_{4\mu} = 0$), and the slice $x_4 =$
 $\hookrightarrow$ const has signature (4,3) -- $x_0..x_3$ spacelike, $x_5..x_7$ timelike (given a4 real and $0 < 6 H x_0 <$
 $\hookrightarrow \pi/2$)

PASS QUANTIZATION [THE RESULT]: the momenta of L_{sym} are $\Pi_{\Psi} = (\sqrt{g}/(2H)) \bar{\Psi} \gamma^4$ and
 $\hookrightarrow \Pi_{\bar{\Psi}} = -(\sqrt{g}/(2H)) \gamma^4 \Psi$

PASS QUANTIZATION [has content]: $\theta_{\text{un}} - \theta_{\text{sym}} = \delta[(\sqrt{g}/(2H)) \bar{\Psi} \gamma^4 \Psi]$
 $\hookrightarrow$ exactly: the same symplectic form, the same brackets

PASS QUANTIZATION [THE RESULT]: $G = -i \sigma_{16} \gamma^4 = J = -i T_{16}[0].T_{16}[1].T_{16}[2].T_{16}[3].T_{16}[4]$
 $\hookrightarrow$ on the canonical frame; J is Hermitian, $J^2 = \text{ID}_{16}$, eigenvalues $+1$ (8) and -1 (8): signature
 $\hookrightarrow (8,8)$

PASS QUANTIZATION [has content]: $\Gamma_4 = 0$ -- the time-derivative part of the Lagrangian, which
 $\hookrightarrow$ fixes the anticommutator, contains no spin connection

CELL 185 $t=0.062$ s

PASS DIRAC BRACKET [THE RESULT]: in the Grassmann algebra $\Psi^{\dagger} K \Psi = (i/2) \theta^T \Omega_c \Phi$
 $\hookrightarrow + (1/4) d(\theta^T K \theta + \theta^T K \theta^2)$, for $K = c \sigma_{16} T_{16}[4]$ with a free scalar c

PASS DIRAC BRACKET [has content]: Ω_c is real symmetric and invertible, $\Omega_c^{-1} = [[0,$
 $\hookrightarrow -K^{-1}], [K^{-1}, 0]]$

PASS DIRAC BRACKET [THE RESULT]: $\{\theta_i, \theta_j\} = (\Omega_c^{-1})_{ij}$ gives $\{\Psi_a, \Psi^{\dagger} b\} = i$
 $\hookrightarrow (K^{-1})_{ab} = (G^{-1})_{ab}$ and $\{\Psi_a, \Psi_b\} = 0$

PASS ANTICOMMUTATOR [THE RESULT]: on the canonical frame ($c = \sqrt{g}/H$) $\{\Psi, \Psi^{\dagger}\} =$
 $\hookrightarrow (H/\sqrt{g}) J = H \cos[6 H x_0] J$ ($J^{-1} = J$), which IS the design's $(H/\sqrt{g})(-i \sigma_{16}$
 $\hookrightarrow \gamma^4)$

PASS ANTICOMMUTATOR [has content]: the rescaled field $\Psi' = \Psi/\sqrt{\sin[6 H x_0]}$ has $\{\Psi',$
 $\hookrightarrow \Psi'^{\dagger}\} = H \cot[6 H x_0] J \delta^7$, and $\sqrt{g}$ is x_4 -independent (so quantizing $\chi =$
 $\hookrightarrow (\sqrt{g}/H)^{(1/2)} \Psi$ changes nothing on the canonical frame)

CELL 186 $t=0.017$ s

PASS J [THE RESULT]: $J = -i \Omega_4 T_{16}[0] = -i h_0$ is a hidden flavour gamma: it lies in the
 $\hookrightarrow$ commutant of $T_{16}[1..4]$ (in the span of the hidden-gamma products of Section 26)

PASS J [has content]: on the flavour factor J has signature (2,2): cfIdemObs is a rank-4 projector
 $\hookrightarrow$ commuting with J, and J restricted to its image has eigenvalues +1, +1, -1, -1

PASS J [THE RESULT]: $\sigma_{16} = J \beta$ with $\beta = -i T_{16}[4]$ Hermitian, $\beta^2 = 1$; $T_{16}[0..3]$
 $\hookrightarrow$ Hermitian with $\beta T_{16}[j] \beta = -T_{16}[j]$: β is the observer's Dirac adjoint

PASS J [THE RESULT]: with $\Psi^\dagger := \Psi^\ddagger$ J: $\bar{\Psi} = \Psi^\dagger \beta$ ($J \sigma_{16} = \beta$),
 $\hookrightarrow \Psi^\ddagger G = \Psi^\dagger (J G = ID_{16})$, $\{\Psi, \Psi^\dagger\} = (H/\sqrt{g}) ID_{16}$ ($G^{-1} J = ID_{16}$):
 $\hookrightarrow$ POSITIVE

PASS J TABLE [THE RESULT]: J commutes with σ_{16} , $T_{16}[0..4]$, $\sigma_{16} \gamma^\mu$ ($\mu = 0..4$), every
 $\hookrightarrow \gamma^\mu \Gamma_\mu$ and Γ_3 ; it ANTICOMMUTES with $T_{16}[5..7]$, $\sigma_{16} \gamma^h$ ($h =$
 $\hookrightarrow 5..7$), $\Gamma_5..7$, $T_{16}[8]$ and C^+

PASS J [has content]: the Hermiticity rule -- for classically Hermitian M, $(J M)^\dagger = J M$ iff
 $\hookrightarrow [J, M] = 0$ and $= -J M$ iff $\{J, M\} = 0$: so $s = \Psi^\ddagger \sigma_{16} \Psi$ is Hermitian and $p = \Psi^\ddagger$
 $\hookrightarrow C^+ \Psi$ is ANTI-Hermitian

PASS J [THE RESULT]: 13 of the 28 Lorentz generators commute with J (a, b both in $\{0..4\}$: Spin(4,1),
 $\hookrightarrow$ or both in $\{5,6,7\}$: Spin(3)); the other 15 anticommute -- 12 boosts $x_0..x_3 \leftrightarrow x_5..x_7$ and 3
 $\hookrightarrow$ rotations $x_4 \leftrightarrow x_5..x_7$

PASS J [has content]: the Krein form G is invariant ($S^T G + G S = 0$) under exactly the 21
 $\hookrightarrow$ generators with a, b $\neq 4$ -- Spin(4,3), the group of the slice

PASS J [has content]: chirality -- $\{J, T_{16}[8]\} = 0$, each chirality subspace is G-NULL ($PL G PL = PR$
 $\hookrightarrow G PR = 0$), and the two chiralities are orthogonal in the Hilbert product (PL, PR Hermitian, $PL PR$
 $\hookrightarrow = 0$)

CELL 187 t=0.344 s

PASS ADMISSIBILITY (1) [THE RESULT]: $E_k^2 = (k_s^2 - k_h^2 + m^2) ID_{16}$ for every k: $\omega^2 = k_s^2$
 $\hookrightarrow -k_h^2 + m^2$

PASS ADMISSIBILITY (2) [THE RESULT]: for admissible k ($k_h = 0$), $[J, E_k] = 0$, E_k is J-Hermitian,
 $\hookrightarrow$ and $G J = ID_{16}$ is positive: J quantizes every admissible momentum positively

PASS ADMISSIBILITY (3) [THE RESULT]: below threshold ($k = 5 e_1 + 3 e_5$, $m = 3$) the boosted $J' = S^{-1} J$
 $\hookrightarrow S$ commutes with E_k and β , $J'^2 = 1$, and $G J' = S^2$ is Hermitian positive definite

PASS ADMISSIBILITY (4) [THE RESULT]: AT threshold ($k = 4 e_1 + 5 e_5$, $m = 3$) $E_k \neq 0$ but $E_k^2 = 0$:
 $\hookrightarrow$ nilpotent, not diagonalizable (rank 8: Jordan blocks of size 2)

PASS ADMISSIBILITY (5) [THE RESULT]: ABOVE threshold ($k = (0, 3/4, 1, 0 \mid k_5 = 2)$, $m = 1$) $\omega^2 =$
 $\hookrightarrow -23/16$: every eigenvalue of E_k is $\pm i \sqrt{23}/4 = \pm 1.19896 i$ and every eigenvector is G-NEUTRAL
 $\hookrightarrow (u^\dagger G u = 0)$

PASS ADMISSIBILITY [has content]: the general identity $[E_k, \beta] = 2 i T_k$, $T_k = \sum_a k_a$
 $\hookrightarrow T_{16}[a]$, for every k (admissible and hidden components alike)

PASS ADMISSIBILITY (6) [THE RESULT]: traceless certificate -- every X commuting with β , the four
 $\hookrightarrow$ admissible E_k and one hidden-momentum E_k has $\text{Tr}(G X) = 0$, so no positive $G J'$ exists; J itself
 $\hookrightarrow$ is not in that commutant

PASS ADMISSIBILITY (6) [THE RESULT]: the analytic proof for a purely hidden momentum -- $[E_k, \beta]$
 $\hookrightarrow = 2 i k_5 T_{16}[5]$; $T_{16}[5]$ anticommutes with G and $T_{16}[5]^\dagger G T_{16}[5] = -G$ (so a J' commuting
 $\hookrightarrow$ with $T_{16}[5]$ has $T_{16}[5]^\dagger (G J') T_{16}[5] = -(G J')$)

PASS ADMISSIBILITY (7) [THE RESULT]: J is UNIQUE -- the joint commutant of β and the admissible
 $\hookrightarrow E_k$ is 8-dimensional, contains J, and every element of it commutes with J (so $X^2 = 1$ with $J X >$
 $\hookrightarrow 0$ forces $X = J$)

PASS ADMISSIBILITY [has content]: $[P_4, P_h] = 0$ -- the metric, the frame and the spin connection do
 $\hookrightarrow$ not depend on x_5, x_6, x_7

CELL 188 t=0.129 s

PASS HAMILTONIAN [THE RESULT]: the Hamiltonian density $\Pi_\Psi d_4 \Psi + d_4 \bar{\Psi} \Pi_\Psi - L$ of
 $\hookrightarrow$ the symmetrized Lagrangian is $\sqrt{g}[-(1/(2H)) \sum_{k \neq 4} (\bar{\Psi} \gamma^k d_k \Psi - d_k \bar{\Psi} \gamma^k \Psi) + V]$: NO spin connection (fields of all eight coordinates)

PASS HAMILTONIAN [THE RESULT]: $(\sqrt{g}/H) \sigma_{16} (\gamma^\mu D_\mu \Psi - H V' \Psi) = i \tilde{G} d_4 \Psi$
 $\hookrightarrow -h \Psi$ for admissible $\Psi(x_0..x_4)$: the field equation is $i \tilde{G} d_4 \Psi = h \Psi$

PASS HAMILTONIAN [THE RESULT]: $(1/2) \sum_{\mu \neq 4} d_\mu (\sqrt{g} \sigma_{16} \gamma^\mu) = \sqrt{g} \sigma_{16}$
 $\hookrightarrow \sum \gamma^\mu \Gamma_\mu$, and $d_4 (\sqrt{g} \sigma_{16} \gamma^4) = 0$: the connection term is the one that
 $\hookrightarrow$ makes h Hermitian

PASS HAMILTONIAN [has content]: $A_0 = \sqrt{g} \sigma_{16} \gamma^0$ is real antisymmetric, and for $h_0 =$
 $\hookrightarrow -(A_0 d_0 + (1/2) A_0')$: $\langle f, h_0 g \rangle - \langle h_0 f, g \rangle = -d_0(f^\dagger G A_0 g)$, a pure boundary term

PASS BOUNDARY [has content]: the x_0 direction in proper distance $z = -\text{Log}[\text{Cos}[6 H x_0]]/(6 H)$: dz/dx_0
 $\hookrightarrow = \text{Tan}[6 H x_0] = \text{Sec Sin}$ (so $\sqrt{g} dx_0 |\Psi|^2 = |\Psi'|^2 dz$), $z(0) = 0$ and $z \rightarrow \infty$ at $6 H$
 $\hookrightarrow x_0 \rightarrow \Pi/2$, $\gamma^0 d_0 = T_{16}[0] d_z$, and $\text{Sin}[6 H x_0] = \sqrt{1 - \text{Exp}[-12 H z]}$ (bounded frame
 $\hookrightarrow$ factors)

PASS BOUNDARY [THE RESULT]: the bag condition $T_{16}[0] \Psi' = \pm \Psi'$ at the wall -- $T_{16}[0]^2 = 1$,
 $\hookrightarrow \{\beta, T_{16}[0]\} = 0$, the normal-flux form vanishes on both eigenspaces ($P \beta T_{16}[0] P = 0$), and
 $\hookrightarrow [J, T_{16}[0]] = 0$

PASS HAMILTONIAN [THE RESULT]: $[J, h] = 0$ on the admissible sector, for every a_4 , every x_0 -profile
 $\hookrightarrow$ and every V' -- including the spin-connection term

PASS HAMILTONIAN [control]: a hidden derivative term breaks it -- $[J, h] \neq 0$ once Ψ depends on x_5
 $\hookrightarrow$ (witnessed on $\Psi = x_5 e_1$)

PASS HAMILTONIAN [has content]: the x_4 -dependence of h enters only through $1/q$: $d_4 h \Psi = (H a_4' [H$
 $\hookrightarrow x_4]) x$ (the observed-direction part of h), which commutes with J

CELL 189 t=0.221 s

PASS KREIN MODES [has content]: $A = G h$ satisfies $A^2 = w^2 \text{ID}_{16}$ and $[A, J] = 0$, and A is Hermitian
 $\hookrightarrow$ (G and h commute) for every admissible k and m
PASS KREIN MODES [THE RESULT]: the four $P_{\{s, \eta\}}$ are projectors of trace (rank) 4, mutually
 $\hookrightarrow$ orthogonal, summing to ID_{16} , and $\sum \eta P_{\{s, \eta\}} = J = G^{-1}$ (completeness)
PASS KREIN MODES [THE RESULT]: for every mode, $\eta u^\dagger h u = s w$ (energy = frequency), η
 $\hookrightarrow u^\dagger \sigma_{16} u = m/\omega_u$, $\eta u^\dagger (dh/dk_j) u = k_j/\omega_u$: $P X P = c P$ with $c =$
 $\hookrightarrow \eta s w$, $\eta s m/w$, $\eta s k_j/w$
PASS KREIN MODES [has content]: $dh/dk_j = -i \sigma_{16} T_{16}[j]$, and $d\omega/dk_j = k_j/\omega$ for
 $\hookrightarrow \omega = +w$
PASS KREIN MODES [THE RESULT]: every mode of $P_{\{s, \eta\}}$ is a J -eigenvector: $J P_{\{s, \eta\}} = \eta$
 $\hookrightarrow P_{\{s, \eta\}}$
PASS KREIN MODES [control]: a Krein-boosted pair $u_1 = (5/4)u_+ + (3/4)u_-$, $u_2 = (3/4)u_+ + (5/4)u_-$ in
 $\hookrightarrow$ the $\omega = +5$ eigenspace is still G -orthonormal and still consists of modes, but not of
 $\hookrightarrow J$ -eigenvectors; the induced $J_U = G U U^\dagger$ differs from J , and under it s is NOT Hermitian
CELL 190 $t=1.680$ s
[0.749 s] the CAR of 16 Jordan-Wigner operators
PASS FOCK [solver regression]: the Jordan-Wigner operators obey the positive CAR $\{f_i, f_j^\dagger\} =$
 $\hookrightarrow \delta_{ij}$, $\{f_i, f_j\} = 0$ on C^{65536}
PASS FOCK (field level) [THE RESULT]: $\{\Psi_a, \Psi^\dagger_b\} = J_{ab} = (G^{-1})_{ab}$ with $\Psi = f$, $\Psi^\dagger = f^\dagger J$ -- the Dirac-bracket anticommutator, realized on a positive Fock space
PASS FOCK (field level) [THE RESULT]: $J h$ is Hermitian with eigenvalues $+5$ (8) and -5 (8); H_{free} and
 $\hookrightarrow s$ are Hermitian operators, so is $H_{\text{free}} + (3/7) s^2$; $p = \Psi^\dagger G \Psi$ is ANTI-Hermitian
[1.868 s] the Heisenberg equation on the Fock space
PASS FOCK (field level) [THE RESULT]: Heisenberg -- $[H_{\text{free}}, \Psi_a] = -(G^{-1} h \Psi)_a$ for all 16
 $\hookrightarrow$ components, so $i d_4 \Psi = i [H, \Psi]$ IS the field equation $i G d_4 \Psi = h \Psi$
PASS FOCK (field level) [control]: with the opposite sign of the anticommutator the Hamiltonian is
 $\hookrightarrow -H_{\text{free}}$ and $[H, \Psi_a] = +(G^{-1} h \Psi)_a$: the time-REVERSED equation
PASS FOCK (mode level) [THE RESULT]: $\{b_u, b_v^\dagger\} = \eta_u \delta_{uv}$ (the indefinite CAR demanded
 $\hookrightarrow$ by $\sum \eta_u u^\dagger u = G^{-1}$), and the J -involution $b^\dagger = \eta b^\dagger$ restores $\{b, b^\dagger\}$
 $\hookrightarrow = 1$
PASS FOCK (mode level) [has content]: the naive Krein 'norm' $\langle 0 | b_u b_u^\dagger | 0 \rangle$ equals η_u --
 $\hookrightarrow$ NEGATIVE for the four $\eta = -1$ modes of each frequency -- while the Fock norm $\langle 0 | b_u b_u^\dagger$
 $\hookrightarrow | 0 \rangle = 1$
PASS FOCK (mode level) [THE RESULT]: $\sum \omega_u \eta_u b_u^\dagger b_u = \sum \omega_u b_u^\dagger b_u$,
 $\hookrightarrow$ and $[H_{\text{m}}, b_u] = -\omega_u b_u$ for every u ($b_u(x_4) = b_u e^{-i \omega_u x_4}$)
PASS FOCK (mode level) [THE RESULT]: the spectrum -- a UNIQUE ground state at $-8 \omega = -40$, the
 $\hookrightarrow$ Dirac sea (modes 9..16 filled, 1..8 empty); $:H_{\text{m}} = H_{\text{m}} + 40 \geq 0$
PASS FOCK (mode level) [THE RESULT]: the first excited level, $\omega = 5$ above the sea, is 16-fold: 8
 $\hookrightarrow$ particle states ($N = +1$) and 8 antiparticle states ($N = -1$), with $N = \sum b^\dagger b - 8$
PASS FOCK (mode level) [THE RESULT]: in every Fock basis state tested (the sea and the 16 first
 $\hookrightarrow$ excited states) $\langle n | b_u^\dagger b_v | n \rangle = \eta_u \eta_v \delta_{uv}$ -- the expectation rule of Section 29
PASS CHARGE [THE RESULT]: $n^4 = -(i/H) \text{Psib} \gamma^4 \Psi = (1/H) \Psi^\dagger G \Psi = (1/H) \Psi^\dagger$
 $\hookrightarrow \Psi$, whose mode form $\sum_u \eta_u b_u^\dagger b_u = \sum_u b_u^\dagger b_u$ is the particle number
 $\hookrightarrow$ (normal-ordered: particles - antiparticles)
CELL 191 $t=9.858$ s
[0.693 s] Hilbert tensor from first-order symmetric tetrad perturbations, five components
PASS EMT [THE RESULT]: the Hilbert (symmetric-tetrad) tensor of the symmetrized action IS That, OFF
 $\hookrightarrow$ shell -- components (x_1, x_1) , (x_4, x_4) , (x_1, x_4) , (x_0, x_5) , (x_5, x_4)
PASS EMT [has content]: the spin-connection variation contributes NOTHING -- dropping it leaves the
 $\hookrightarrow$ Hilbert tensor unchanged at (x_1, x_4) , (x_0, x_5) , (x_5, x_4)
PASS EMT [control]: the Hilbert tensor is NOT the partial-derivative tensor: at (x_1, x_4) they differ
 $\hookrightarrow$ -- witnessed on constant spinors ($\text{Psib} = e_1$, $\Psi = e_{11}$: $T_{16}[4] T_{16}[1] T_{16}[0]_{\{1,11\}} \neq 0$)
PASS EMT [definition]: That is symmetric
PASS EMT [THE RESULT]: That is Hermitian under the classical conjugation: real for $\Psi = u + i v$,
 $\hookrightarrow \text{Psib} = (u - i v)^T \sigma_{16}$, all 64 components
CELL 192 $t=2.248$ s
PASS EMT CONSERVATION [definition]: F and $\bar{F}$ solve the field equation and the adjoint equation for
 $\hookrightarrow$ the x_4 -derivatives
[0.767 s] $\nabla_\mu \text{That}_{\{\mu\}}$, generic $\Psi(x_0, x_4)$, $\text{Psib}(x_0, x_4)$
[2.599 s] the on-shell divergence, eight components
PASS EMT CONSERVATION [THE RESULT]: $\nabla_\mu \text{That}_{\{\mu\}} = 0$ ON SHELL in all eight components, for
 $\hookrightarrow$ generic $\Psi(x_0, x_4)$, $\text{Psib}(x_0, x_4)$ and generic V
PASS EMT CONSERVATION [control]: OFF shell the divergence is not zero -- witnessed on $\Psi = (x_4 +$
 $\hookrightarrow x_0^2) e_1 + e_{13}$, $\text{Psib} = e_1 + x_4 e_5$ with $V = s^2$, which is not a solution
PASS EMT [THE RESULT]: $\text{That} - T_{\text{par}} = -(1/(4H)) \text{Psib} (\{\gamma_\mu, \Gamma_\mu\} + \{\gamma_\mu, \Gamma_\mu\})$
 $\hookrightarrow \Psi$: equal on the diagonal (every $\{\gamma_\mu, \Gamma_\mu\}$ vanishes), different off it (the (x_1, x_4)
 $\hookrightarrow$ witness above)
CELL 193 $t=4.431$ s
PASS EMT ON SHELL [THE RESULT]: the kinetic bilinear $= s V'(s)$ (not zero) and $L_{\text{hat}} = s V'(s) - V(s)$
PASS EMT ON SHELL [THE RESULT]: $\rho = \text{That}_{44} = V(s) + K_h$
PASS EMT ON SHELL [THE RESULT]: $T^i_i = T^h_h = s V'(s) - V(s)$ for the three observed and the three
 $\hookrightarrow$ hidden directions, and $T^0_0 = s V'(s) - V(s) + K_h$
PASS EMT [has content]: the trace $T^\mu_\mu = 7 L_{\text{kin}} - 8 V$ off shell, $= 7 s V'(s) - 8 V(s)$ on shell
PASS EMT [fidelity]: K_h in the real commuting limit IS Part VI's $\text{cf} K_h$, and $K_h = 0$ on the rescaled
 $\hookrightarrow$ solutions $\text{Sqrt}[\text{Sin}] \Psi'(x_4)$, $\text{Sqrt}[\text{Sin}] \text{Psib}'(x_4)$, where $\rho = V(s)$

[3.561 s] the J-parity of all 64 components of That, admissible fields
PASS EMT HERMITICITY [THE RESULT]: J-parity table -- (mu, nu) in {0..4}: EVEN (Hermitian); (mu in
→ 0..4, h in 5..7): ODD (anti-Hermitian); (h, h'), h != h': ZERO identically; (h, h): even
PASS EMT HERMITICITY [has content]: for x5..x7-independent fields That_{hh} == g_hh Lhat exactly,
→ because {gamma_h, Gamma_h} == 0
CELL 194 t=4.080 s
PASS MODE SUMS [THE RESULT]: on a flat-frame plane wave (canonical normalization), T_44 + Lhat ==
→ omega ubar (-i T16[4]) u and T_j_j - Lhat == k_j ubar (-i T16[j]) u (j = 0..3): energy and
→ pressure are the frequency and k_j dh/dk_j bilinears
PASS MODE SUMS [has content]: with ubar = u^ddag sigma16 these are omega u^ddag G u and k_j u^ddag
→ (dh/dk_j) u, so by Section 28 each occupied mode contributes omega, k_j^2/omega and (to sigma)
→ m/omega
PASS VACUUM [THE RESULT]: the unrenormalized sea energy -eps_KS(Lambda) == -(g/(16 pi^2))(2 Lambda^4
→ + 2 m^2 Lambda^2 + m^4/4 - m^4 Log[2 Lambda/m]) + o(1) as Lambda -> infinity; its m-dependent
→ finite part contains -(g/(32 pi^2)) m^4 Log[m^2]
CELL 195 t=2.098 s
PASS KOHN-SHAM [definition]: the degeneracy g = 8 -- per momentum the positive-frequency eigenspace
→ has rank 8 (Section 28's P_{+,+} + P_{+,-}), four flavours times two spins
PASS KOHN-SHAM [THE RESULT]: each closed form has k_F-derivative (g/(2 pi^2)) k_F^2 f(k_F) with f =
→ 1, m/omega, omega, k^2/(3 omega), and vanishes at k_F = 0 -- so it IS the integral g Int d^3k/(2
→ pi)^3 f
PASS KOHN-SHAM [has content]: Integrate returns the same functions, with ArcTanh[k_F/w_F] in place of
→ L -- and ArcTanh[k_F/w_F] == L (equal derivatives, both 0 at k_F = 0)
PASS KOHN-SHAM [control]: numerical quadrature agrees at k_F = 3/2, m = 1, g = 8 to 1e-12 (a
→ cross-check; the proof is the symbolic one above)
PASS KOHN-SHAM [THE RESULT]: eps + P == w_F n (Gibbs), eps - 3 P == m sigma (trace), d eps/dm ==
→ sigma, d eps/dk_F == w_F dn/dk_F
PASS MEAN FIELD [THE RESULT]: rho = eps + W - sigma W', P_obs = P_KS + sigma W' - W give rho + P_obs
→ == w_F n exactly; and d rho/dn == w_F + (sigma_KS - sigma) W' sigma'(n), i.e. == w_F at the
→ self-consistent point sigma = sigma_KS (m = W'(sigma) throughout)
PASS AUTHOR'S MASS TERM [THE RESULT]: W = m sigma gives W - sigma W' == 0, so rho == eps_KS, P ==
→ P_KS, and w = P_KS/eps_KS depends only on x = k_F/m: w == I2/(3 I1), I_n the m = 1 integrals
PASS AUTHOR'S MASS TERM [THE RESULT]: 0 <= w <= 1/3 -- P_KS >= 0 (I2' = x^4/w_x > 0, I2(0) = 0) and
→ eps - 3P = m sigma = (g m^4/(2 pi^2)) I3 >= 0 (I3' = x^2/w_x > 0, I3(0) = 0)
PASS AUTHOR'S MASS TERM [THE RESULT]: w rises MONOTONICALLY -- dw/dx == x^2 Delta/(3 w_x I1^2) with
→ Delta = x^2 I1 - w_x^2 I2, Delta(0) = 0 and Delta' = 2 x I3 > 0 -- from w -> 0 (x -> 0, w = x^2/5
→ + 0(x^4): dust) to w -> 1/3 (x -> infinity: radiation)
PASS AUTHOR'S MASS TERM [has content]: the classical sign problem disappears -- sigma_KS = g Int
→ m/omega is ODD in m and m sigma_KS = (g m^4/(2 pi^2)) I3(k_F/|m|) >= 0 for either sign, so with m
→ = -2M: M sigma_KS <= 0, the branch M s < 0 that Part VI had to assume
PASS CLASSICAL LIMIT [THE RESULT]: at the self-consistent point rho == W + 3 P_KS and P_obs == (sigma
→ W' - W) + P_KS exactly (sigma = sigma_KS, m = W'); Part VI's rho = V(s), P = s V' - V is the limit
→ P_KS/eps_KS -> 0, i.e. k_F/m -> 0
CELL 196 t=4.945 s
PASS PRE-UNIVERSE (1) [THE RESULT]: the U(1) current is conserved on shell, (1/Sqrt[g]) d_mu(Sqrt[g]
→ j^mu) == 0, for generic Psi(x0, x4), Psi_bar(x0, x4)
PASS PRE-UNIVERSE (1) [THE RESULT]: for Psi = Sqrt[Sin] Psi'(x4), Sqrt[g] j^0 is x0-independent (Sec
→ Sin Cot == 1), so d_4(Sqrt[g] j^4) == 0 on shell; and Sqrt[g] and q^3 p^3 are x4-independent
PASS PRE-UNIVERSE (2) [THE RESULT]: Psi = Sqrt[Sin] Psi'(x4) satisfies the field equation iff d_x0
→ V'(Sin[6Hx0] s') == 0, and d_x0 V'(Sin s') == 6 H Cos[6 H x0] s' V''(Sin s'): V linear OR s' == 0
PASS PRE-UNIVERSE (2) [has content]: s' = Psi_bar Psi' is conserved by the x4-only equations (d_4
→ Psi' = -H V' T16[4] Psi', d_4 Psi_bar = H V' Psi_bar T16[4]) for ANY V; so null data s' = 0 are
→ solutions for any V, with rho == V(0); and for V = s^2 with s' != 0 the x0-dependence is real
→ (witness)
PASS PRE-UNIVERSE (3) [THE RESULT]: for Psi = Sqrt[Sin] u(x4) Exp[i k x1] the field equation (mass
→ term, V' = constant) is Sqrt[Sin] Exp[i k x1] (T16[4] u' + i (k/q) T16[1] u - H V' u): the
→ physical momentum is k/q
PASS PRE-UNIVERSE (3) [THE RESULT]: d Log[k/q]/dx4 == H a4'[H x4] == -H_obs (observed momenta
→ redshift as Exp[a4]), and E^2 == (k^2/q^2 + m^2) ID16 at the physical momentum
PASS PRE-UNIVERSE (4) [THE RESULT]: x = k_F,phys/m = x_c/(q m) obeys d Log[x]/dx4 == H a4'[H x4];
→ since dw/dx > 0, dw/dx4 = w'(x) x H a4' < 0 exactly when a4' < 0 (the observed sheet expanding):
→ the quantum gas COOLS
PASS PRE-UNIVERSE (4) [control]: the classical Part VI fable does NOT cool -- ds/dx4 == 0 on its
→ rescaled solutions (Part VI's result, re-asserted), and its w = s V'/V - 1 is a function of s
→ alone: frozen
PASS PART VII [control]: a4 is STILL UNDEFINED -- nothing in Part VII gave it a value
CELL 197 t=0.072 s
identities accepted on numerical evidence alone : 0 (stage 3 returned True)
non-vanishing witnesses : 38 (cfNonZeroWitnessQ found a probe point at which
→ every entry is a
number and one of them is non-zero: a complete
→ proof of non-vanishing)
PASS no identity in this notebook was accepted on numerical evidence alone
Assertions run: 528 passed: 528 failed: 0

CELL 198 t=0.000 s

3.9.3 The same assertions in the full run of Parts I–VIII

After Part VIII (the coupled system, notebook Sections 30–33) was added, the whole notebook was run again (`claude-fable/run_fermion_fable_part8.log`). The Part VII lines of that log are identical to those of the Part VII log apart from the timing lines [checked for the provenance pages of 2026-09-24 with `diff` on the two logs, the `CELL` and `[t s]` lines removed]. The Part VII cells took 29.706 s in that run. Its summary:

```
===== RUN SUMMARY =====
cells evaluated : 217
total seconds   : 233.289
cells w/ msgs   : 0
---- slowest cells ----
cell 80  80.343 s
cell 88  23.494 s
cell 204 10.364 s
cell 86   9.821 s
cell 137   9.306 s
cell 191   9.225 s
cell 87   8.387 s
cell 121   6.069 s
cell 160   4.250 s
cell 196   4.185 s
cell 107   3.836 s
cell 193   3.140 s
===== ASSERTIONS =====
assertions run  : 642
passed         : 642
FAILED         : 0
=====
RUN-DONE
```

The last Part VII line of that log is still `Assertions run: 528 passed: 528 failed: 0 (line 843)`, and the run ends with `Assertions run: 642 passed: 642 failed: 0` and 46 non-vanishing witnesses.

The final run (commit 1671155, `claude-fable/run_fermion_fable_final.log`) was made after the `fable-cosmology` notebook 06 had written the Rust solver's CSVs `results/nb06_fable4d_mass30eV.csv` and `nb06_fable4d_power.csv`, so that Part VIII's two conditional comparisons with them ran (in the Part VIII log they had been skipped, not failed). Its Part VII lines are identical to those of the Part VII log apart from the cells' timing lines [checked while writing this document, with `diff` on the two logs, from the line that closes cell 172 to the line that closes cell 198]; the Part VII cells took 28.435 s and the Part VIII cells 22.932 s [summed from its `CELL n t=...` lines]. Its last Part VII tally is still `Assertions run: 528 passed: 528 failed: 0 (line 843)`, and it ends with `identities accepted on numerical evidence alone : 0 (stage 3 returned True), non-vanishing witnesses : 46, Assertions run: 644 passed: 644 failed: 0` and the summary

```
===== RUN SUMMARY =====
cells evaluated : 217
total seconds   : 208.715
cells w/ msgs   : 0
---- slowest cells ----
cell 80  69.797 s
cell 88  22.871 s
cell 204   9.939 s
cell 191   8.828 s
cell 86   8.339 s
cell 87   7.361 s
cell 137   7.359 s
cell 121   4.840 s
cell 196   3.848 s
```

```

cell 160  3.824 s
cell 107  3.789 s
cell 193  3.062 s
===== ASSERTIONS =====
assertions run   : 644
passed          : 644
FAILED          : 0
=====

```

It has no RUN-DONE line (that line of the earlier logs was appended by the wrapper that ran them). Its 644 PASS labels are the 642 of the Part VIII log, identical, plus the two comparisons with the Rust CSVs (section 4.7.12).

3.9.4 The design review, and every correction it made to Effort A

The design (revision 1) was reviewed by four adversarial reviewers — quantization (findings QK-*); field equations, energy-momentum tensor and spin connection (EMT-*); Einstein equations and cosmology (E-*); density-functional theory and numerics (DFT-*) — each followed by an independent skeptic who tried to refute each finding. **No finding was refuted.** Several were confirmed only in part, and then the corrected fix was adopted. The review records are working documents and are not part of the repository; every adopted correction that concerns Effort A is restated here, with where it now lives. Section 5 of this document lists every finding of all four lenses, Efforts A and B together, with its verdict and the correction adopted.

design item	findings	what revision 1 said, or lacked	what was adopted	where in this document (notebook)
refinement	QK-7	“fermion fable must be complex” without scope; F2(a) only on the flat frame	the invariant forms are exactly $\text{span}\{\sigma_{16}, \sigma_{16}T_{16}[8]\}$; Majorana has one quartic scalar and no commuting kinetic shadow; Weyl/Majorana–Weyl cannot propagate; the complex spinor is the minimal , not the unique, field; F2(a) holds on every frame	3.2.2, 3.2.4, 3.2.8 (REFINEMENT (a), (b))
Lagrangian	QK-4 (corrected), EMT-5	several Lagrangians used interchangeably; Part VI’s drop-out mechanism assumed to carry over	only the symmetrized covariant Lagrangian; the unsymmetrized partial one is inadmissible (its two equations are not conjugate); the drop-out is a property of the matrix $\{\gamma^\mu, \Gamma_\mu\}$, per μ , on diagonal frames only	3.3 (LAGRANGIAN (a)–(f))
field equations	EMT-9, EMT-11	8+8 form sketched; operator ordering “normal-ordered or Weyl”; x_4 -only solutions assumed	the 8+8 form with the conjugate rows; the axial phase is not a symmetry; the ordering rule with its scalar and γ^4 contraction parts; the x_4 -only family needs $V''s' = 0$, and s' is conserved	3.4.3, 3.4.8, 3.4.9, 3.4.11
quantization: truncation	QK-2	“physical states obey $P_5 = P_6 = P_7 = 0$ ” (a superselection rule) and Tomonaga–Schwinger integrability	a truncation with finite V_{hid} (closed timelike curves stated); anticommutator with $H/(V_{\text{hid}}\sqrt{ g })$ and δ^4 ; only $[P_4, P_h] = 0$ kept	3.5.4, 3.5.10
quantization: theorem	QK-1 (corrected)	admissibility asserted from one example	the theorem: positive quantization iff the momentum span is spacelike-definite, with items (1)–(7) (threshold Jordan block, G -neutral modes above threshold, traceless certificate, uniqueness)	3.5.10 (ADMISSIBILITY)

design item	findings	what revision 1 said, or lacked	what was adopted	where in this document (notebook)
quantization: J	QK-5 (corrected), plus the missed basis point	Krein prescription $b^\dagger := \eta b^\ddagger$, basis unspecified; reflection positivity claimed	$J = -i\Omega_4 T_{16}[0]$, $\sigma_{16} = J\beta$, flavour metric $(2, 2)$; the adjoint $\Psi^\dagger := \Psi^\ddagger J$ is mode-independent and requires a J-eigenmode basis (control: a Krein boost breaks Hermiticity of s); why it fails for bosons; reflection positivity dropped	3.5.5, 3.5.6, 3.5.13 (J, KREIN MODES [control])
quantization: other	QK-6, QK-8, QK-9, QK-10, QK-11, EMT-2	symmetry count, dynamical frames, wall condition, current sign and J -connection relation incomplete or wrong	13 kept / 15 lost generators, G invariant under $\text{Spin}(4,3)$; $\chi = (\sqrt{g}/H)^{1/2}\Psi$; one-particle space $L^2(dz)$, regular at the wall, limit-point at infinity, bag condition $T_{16}[0]\Psi' = \pm\Psi'$; $n^\mu = -\frac{i}{H}\bar{\Psi}\gamma^\mu\Psi$; the J table with $\Gamma_{5..7}$ anticommuting	3.5.9, 3.5.12, 3.5.15, 3.5.16, 3.5.8
EMT	EMT-1, EMT-3, EMT-4, EMT-8, QK-3	EMT stated without derivation; conservation “for independent $\Psi, \bar{\Psi}$ ” without scope; “ $\langle:T:\rangle = 0$ in the vacuum”	Hilbert = T_{cov} off shell; spin-connection variation exactly zero; Hilbert $\neq T_{\text{par}}$ (witness at a non-vacuous pair); why the commuting proxy proves conservation; the J -parity table; the vacuum statement only for static backgrounds; do not run the symbolic divergence of T_{par}	3.6 (EMT, EMT CONSERVATION, EMT HERMITICITY), 3.7.3
expectation values	DFT-5, DFT-6, DFT-12, DFT-15, DFT-4	classical limit “ $k_F \rightarrow 0$ at fixed n ” (impossible) or a coherent zero mode; unstable closed forms; phantom crossing not examined; sea energy only for constant m	the classical limit is the non-relativistic limit, with the sign rule; the numerically stable forms; $w_f \geq -1$ (no phantom crossing); the pair-excitation thresholds ($\omega_F + m $ at $q = k_F$, $2\omega_F$ at $q = 0$; not asserted in Part VII); ΔE_{vac} for mass-varying W	3.7.5, 3.7.7, 3.7.9, 3.7.3
spin connection	EMT-6, EMT-10	“the connection enters the Hamiltonian”	it enters neither the anticommutator nor the Hamiltonian density; it re-emerges in the equation of motion as the term that makes h Hermitian; $\gamma^\mu\Gamma_\mu = -\frac{1}{2}T_\mu\gamma^\mu$ (the Weitzenböck torsion vector)	3.5.11, 3.8.12.3, 3.8.12.4

Two of these adopted corrections are also stated in Part VII’s own Text cells: the Hamiltonian-density correction (EMT-6; “This corrects the design’s ‘it enters the Hamiltonian’”, Section 28), and the identification of the classical limit with the non-relativistic one (DFT-5; Section 29). One numerical slip in the review’s vacuum-energy figures is corrected in this document (section 3.7.3).

3.9.5 The Fock check

The finite Fock space of section 3.5.14 is the independent check of the quantization: it realizes the Dirac-bracket anticommutator on a positive Hilbert space of dimension $2^{16} = 65536$, with exact integer sparse matrices, and confirms the sign of the anticommutator through the Heisenberg equation (the opposite sign gives the time-reversed equation). It shows the unique Dirac-sea ground state at -40 , the 16-fold first excited level (8 particles, 8 antiparticles), the negative Krein “norms” that the J -involution removes, and the expectation rule used for ρ , P and σ . Twelve assertions (FOCK, CHARGE), Input cell 191, 9.858 s.

3.9.6 The TeX export

claude-fable/export_fermion_fable_tex.wls evaluates the notebook’s Input cells (output suppressed), checks that the Part VII objects exist, and writes fourteen files into provenance-

latex/generated/:

file (.tex and .txt)	content	taken from the notebook object
fermion_fable_field_	the 16 component field equations on	cfFieldEqTerms
equations	the canonical frame	
fermion_fable_adjoint_	the 16 component adjoint equations	cfAdjEqTerms
equations		
fermion_fable_field_	the split-octonion 8+8 form, compact	cfFieldOp16, cfUpper8,
equations_8plus8	and by component	cfLower8, cfFieldEqTerms
fermion_fable_field_	the 16 equations for	the rescaled operator of
equations_rescaled	$\Psi' = \Psi / \sqrt{\sin 6Hx_0}$	Section 27 (iv), asserted
		equal to cfE8 under the
		rescaling
fermion_fable_	$\{\Psi_a, \Psi_b^\dagger\} = H \cos(6Hx_0) J_{ab} \delta^7$, with	cfAntiPsiPsiDd at
anticommutator	the matrix	$c = \sqrt{g}/H$
fermion_fable_krein_J	$J, \beta, \sigma_{16} = J\beta$	cfJK, cfBeta
fermion_fable_eos	the Kohn–Sham closed forms n, σ, ε ,	cfNKS, cfSigKS, cfEpsKS,
	P, w at $g = 8$	cfPKS

Every expression comes from an object that Part VII asserts; the script does no mathematics of its own beyond formatting (its header says so, and its code only renders term lists and matrices). The .tex files are \input by this document in sections 3.4, 3.5 and 3.7; the .txt files hold the same content in plain text. The export was run on 2026-09-24 at 20:41 (PDT) on the 198-cell notebook, and it printed:

```
evaluating 198 Input cells of C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\claude-fable\claude-
↪ fable_Einstein-Rosen-2-Planes.nb (output suppressed)
assertions: 528   failed: 0   cells with messages: 0
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_f
↪ ield_equations.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_a
↪ djoint_equations.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_f
↪ ield_equations_8plus8.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_f
↪ ield_equations_rescaled.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_a
↪ nticommutator.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_k
↪ rein_J.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\fermion_fable_e
↪ os.tex / .txt
done: 14 files in C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated
```

(That console output was captured outside the repository; the fourteen files it wrote were committed in 4fd4c40.) A test document that \inputs all seven .tex fragments (with amsmath) compiled with pdfLaTeX (MiKTeX) to 9 pages with no errors, warnings or overfull boxes (a scratch test on 2026-09-24, outside the repository).

4 Effort B: the interacting system — fermion fable as the source of the primordial gravitational field, the solution from the beginning of the present universe to today, and the answers to [1] and [2]

The quantized fermion fable of section 3 is put into the Einstein equations of the author’s primordial gravitational field as their source. The Einstein tensor of the author’s canonical

frame is computed in full, with the free function a_4 never given a value, and what it demands of any source is proved. The frame is generalized to a family on which fable can act back; its asymptotic region is an 8-dimensional Bianchi-I universe. The canonical spin connection of that interacting system is computed and discussed completely. The fable’s energy–momentum tensor operator is evaluated in its Kohn–Sham ground state, and the coupled equations are solved from before nucleosynthesis to today, three times independently (Rust/CVODE, Python/scipy, Mathematica/NDSolve). Ideas of density-functional theory give the fable’s ground and first excited states: in the homogeneous gas and in the discrete states bound along the hidden coordinate x_0 . The author’s two questions — [1] a time-varying dark-energy equation of state, [2] a time-varying dark-matter equation of state — are answered directly and quantitatively, with every caveat (section 4.8).

How this section was made. It is copied section by section, by a script, from the provenance record of Effort B (written on 2026-09-24 at 23:26 PDT and fact-checked against the logs), with its sections renumbered 4.1–4.8 and its cross-references renumbered to match this page. Two of that record’s sections are not repeated because they are word for word the same as sections of Effort A already on this page: its foundations are section 3.1, and the first thirteen subsections of its spin-connection discussion (the canonical spin connection) are sections 3.8.1–3.8.13; its cross-references to them point there. Three things were brought up to date, because they changed after that record was written, and each change is marked where it is made: (i) notebook 07 has since been executed and committed (`aca5102`), so every wall-state number it prints again now carries the mark `[nb07 §n]`, and the few it does not print are named; (ii) the derivation of the wall problem was re-run when its output `waveguide_T16.json` was committed, so its log’s timing changed; (iii) notebook 06 was re-executed on the final solver of `2bc936d` before it was committed, so its refusal message for the repulsive quadratic is now the bisected one. Every other number is as it was quoted, and every one of them was checked again, for this page, against the committed notebooks 05–07, their result files, the wall-state report and the notebook’s run logs.

The work of Effort B, in the order it was done.

1. A design (revision 1) was written for the whole of 2026-09-24’s work. Four adversarial reviewers examined it — quantization; field equations, energy–momentum tensor and spin connection; Einstein equations and cosmology (findings `E-*`); density-functional theory and numerics (findings `DFT-*`) — each followed by an independent skeptic. No finding was refuted as a whole; several were confirmed only in part, and the corrected fix was adopted. Every adopted correction that concerns Effort B is stated where it applies below, and all of them are listed in section 5.
2. The solver `fable-cosmology/rust/fable_fermion` was written on the vendored pure-Rust SUNDIALS 7.8.0 CVODE of the `rustSolveIt` repositories, with the independent Python check `fable-cosmology/fermion/crosscheck.py` (commit `c5892bd`).
3. Part VIII of the Mathematica notebook (Sections 30–33, manifest `claud-fable/cells_part8.wl`) proves the symbolic side: the Einstein tensor, the three theorems, the generalized frame, the Bianchi-I system, the spin connection of the interacting system, and the coupled equations for the Kohn–Sham fluid, and it solves two reference cases with NDSolve (commit `bc04ed1`).
4. The wall-state solver `fable-cosmology/fermion/waveguide.py`, its symbolic derivation and its Mathematica check `compute` the discrete states along x_0 and their density-functional treatment (commit `18a2501`).
5. Notebooks 05 and 06 ran the solver on every case and exposed six solver defects, which were fixed (commit `2bc936d`, section 5.8, item 4).
6. The `fable-cosmology` notebooks `05_fermion_fable_quantum_eos.ipynb`, `06_fable_`

primordial_gravity_8d.ipynb and 07_fable_dft_states.ipynb were executed on the final solver; they run every case, re-compute every run independently, reproduce the wall states, and state the answers (commit aca5102).

7. The whole notebook was run once more, now with the Rust solver's CSVs present, so that Part VIII's comparison with the Rust solver also ran: 644/644 (commit 1671155).

The evidence marks of this section are those of the head of this page. The marks [VIII] and [proved VIII §n] quote Part VIII's labels verbatim from `claude-fable/run_fermion_fable_part8.log`, which reports `Assertions run: 642 passed: 642 failed: 0` for Parts I–VIII together; the final run `claude-fable/run_fermion_fable_final.log` prints the same 642 labels identically and two more (the comparison with the Rust CSVs, section 4.7.12), `Assertions run: 644 passed: 644 failed: 0`. The Part VIII labels were spliced into the record of Effort B by a script from that log, not retyped.

4.1 The fermion fable in brief

This section restates, complete enough to be read alone, the quantized fermion fable that is the source of the Einstein equations in the rest of section 4: the field, its Lagrangian, its field equations in the primordial gravitational field, its canonical anticommutator, its energy–momentum tensor operator, and its density, pressures and equation of state. Everything here is proved in Part VII of the notebook (Sections 26–29, manifest `claude-fable/cells_part7.wl`, Input cells 173–198, 170 assertions); the labels are quoted verbatim from `claude-fable/run_fermion_fable_part7.log`, whose tally reads `Assertions run: 528 passed: 528 failed: 0` for Parts I–VII together. Section 3.1.13 gave the algebraic reasons (F1, F2) for the refinement; this section states the results.

4.1.1 The field

Fermion fable is a **complex 16-component Grassmann spinor** Ψ , two real Grassmann 16-spinors θ_1, θ_2 combined as $\Psi = (\theta_1 + i\theta_2)/\sqrt{2}$, with the classical conjugate $\Psi^\dagger = (\theta_1 - i\theta_2)^T/\sqrt{2}$ and the Dirac conjugate

$$\bar{\Psi} = \Psi^\dagger \sigma_{16}, \quad \sigma_{16} = T_{16}[0] T_{16}[1] T_{16}[2] T_{16}[3] = \text{diag}(-\sigma, \sigma).$$

It has two Lorentz-scalar bilinears, $s = \bar{\Psi}\Psi$ and $p = \bar{\Psi} T_{16}[8] \Psi$; the self-interaction depends on s only (section 4.1.4). Written as $\Psi = (\psi_1, \psi_2)$ (upper eight components the type-1 split-octonion spinor of chirality -1 , lower eight the type-2 spinor of chirality $+1$), $s = -\psi_1^\dagger \sigma \psi_1 + \psi_2^\dagger \sigma \psi_2$.

Why complex, and with what scope. The $\text{Spin}(4,4)$ -invariant bilinear forms on the 16-spinor are exactly $\text{span}\{\sigma_{16}, \sigma_{16} T_{16}[8]\}$, both symmetric and chirality-diagonal. Therefore a real Grassmann 16-spinor with the author's σ_{16} has no dynamics on any frame (its mass term vanishes and its kinetic term is a total derivative), a Majorana 16-spinor built with $C_+ = \sigma_{16} T_{16}[8]$ propagates but is massless and has no $V(s)$, and a one-chirality (Weyl or Majorana–Weyl) spinor cannot propagate. The complex 16-spinor is the **minimal** field that uses the author's σ_{16} conjugation, propagates, admits $V(s)$, and reduces to Part VI's classical fable as its real commuting shadow; it is **not** the unique field with dynamics. The author chose it on 2026-09-24. [proved VII §26]

- REFINEMENT [THE RESULT]: the $\text{Spin}(4,4)$ -invariant bilinear forms on the 16-spinor form a 2-dimensional space, spanned by $C_- = \sigma_{16}$ and $C_+ = \sigma_{16} T_{16}[8]$

- REFINEMENT (a) [THE RESULT]: real Grassmann + sigma16: both invariant mass terms vanish, and $\theta^T \sigma_{16} T_{16}[a] \psi = (1/2) d(\theta^T \sigma_{16} T_{16}[a] \theta)$ for every a (a total derivative)
- REFINEMENT (b) [THE RESULT]: Majorana with C+: the kinetic term is genuine - Euler-Lagrange expression $2 C^+ T_{16}[a] \psi$ with $C^+ T_{16}[a]$ invertible - but every invariant bilinear vanishes (massless, no $V(s)$)
- REFINEMENT (c) [THE RESULT]: Weyl and Majorana-Weyl: $PL C T_{16}[a] PL = PR C T_{16}[a] PR = 0$ for $C = C^-, C^+$ and every a (C chirality-diagonal, $T_{16}[a]$ chirality-odd) - no kinetic term
- REFINEMENT (d) [THE RESULT]: $s = \bar{\psi} \psi = i \theta_1^T \sigma_{16} \theta_2$: NON-ZERO (16 monomials) and HERMITIAN; $p = \bar{\psi} T_{16}[8] \psi$ is non-zero and Hermitian too

Four four-dimensional Dirac fermions. The observer's Clifford algebra $T_{16}[1..4]$ is $M_4(\mathbb{C})$; its commutant in $M_{16}(\mathbb{C})$ is generated by the hidden gammas $h_A = \Omega_4 T_{16}[A]$ ($A = 0, 5, 6, 7$, $\Omega_4 = T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4]$). So $\mathbb{C}^{16} = \mathbb{C}^4$ (observer Dirac index) $\otimes \mathbb{C}^4$ (hidden flavour index): under the observer's $\text{Spin}(3,1)$ of x_1, x_2, x_3, x_4 fermion fable is four Dirac fermions, $g = 4 \times 2 = 8$ particle states (and 8 antiparticle states) per momentum. [proved VII §26] 4D [THE RESULT]: the 16 products of the hidden gammas span the commutant, and the two algebras meet only in the multiples of ID_{16} (rank of the union $16 + 16 - 1 = 31$): $\mathbb{C}^{16} = \mathbb{C}^4 \otimes \mathbb{C}^4$, four 4D Dirac fermions.

Fidelity. In the real commuting limit ($\Psi \rightarrow \psi, \bar{\Psi} \rightarrow \psi^T \sigma_{16}$) the Lagrangian, the field equations and the energy-momentum tensor below become Part VI's classical fable exactly (section 3.1.12), and through it the author's L_a and eL_a (sections 3.1.7, 3.1.8). [proved VII §26]

- FIDELITY [1] [fidelity]: $\psi \rightarrow \psi, \bar{\psi} \rightarrow \psi^T \sigma_{16}$ turns the symmetrized covariant L_{hat} into Part VI's cfL_{hatFable} EXACTLY (canonical frame, generic V)
- FIDELITY [2] [fidelity]: on the flat frame with $V = -(2M/H) s$ and no volume factor, the same limit on ψ_{16} IS the author's L_a of Section 11
- FIDELITY [4] [fidelity]: the energy-momentum tensor T_{hat} in the real limit IS Part VI's covariant tensor $cfTCov$, all 64 components

4.1.2 The Lagrangian

The only admissible Lagrangian is the **symmetrized covariant** one:

$$L = \sqrt{g} \hat{L}, \quad \hat{L} = \frac{1}{2H} \left[\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi \right] - V(s), \quad s = \bar{\Psi} \Psi,$$

$$D_\mu \Psi = \partial_\mu \Psi + \Gamma_\mu \Psi, \quad D_\mu \bar{\Psi} = \partial_\mu \bar{\Psi} - \bar{\Psi} \Gamma_\mu,$$

with Γ_μ the canonical spin connection of section 4.4 (on the dynamical frames of this document, the connection of those frames, section 4.4.1 onwards). There is no factor i , as in the author's L_a : the Lagrangian is Hermitian because $\sigma_{16} \gamma^a$ and $\sigma_{16} \Gamma_\mu$ are real antisymmetric matrices. It contains the connection only through the matrix $\sum_\mu \{\gamma^\mu, \Gamma_\mu\}$, which vanishes for **each** μ on the canonical frame (and on the warped and Bianchi-I frames of this document, section 4.4.4), so there it equals the partial-derivative symmetrized Lagrangian. The unsymmetrized covariant Lagrangian differs from it by a total divergence; the unsymmetrized *partial* Lagrangian is inadmissible (its two field equations are not Dirac conjugates). [proved VII §27]

- LAGRANGIAN (b) [THE RESULT]: Lsym is HERMITIAN - real for Psi = u + i v, Psibar = (u - i v)^T sigma16, generic u, v of all eight coordinates
- LAGRANGIAN (c) [THE RESULT]: Lsym - Lsym_d == (1/(2H)) Sum_mu Psibar {gamma^mu, Gamma_mu} Psi identically, and {gamma^mu, Gamma_mu} == 0 for EACH mu on the canonical frame: Lsym == Lsym_d
- LAGRANGIAN (f) [control]: so Lun_d is INADMISSIBLE - its two field equations are not Dirac conjugates: they differ by 2 Psibar gamma^mu Gamma_mu = -6 H Cot^2 Psibar T16[0], non-zero (witness u = e1, v = 0, V = s^2)

4.1.3 The field equations in the primordial gravitational field

Varying $\bar{\Psi}$ and Ψ independently:

$$\gamma^\mu D_\mu \Psi = H V'(s) \Psi, \quad (D_\mu \bar{\Psi}) \gamma^\mu = -H V'(s) \bar{\Psi}.$$

[proved VII §27]

- FIELD EQUATION [THE RESULT]: EL_Psibar == (Sqrt[g]/H)(gamma^mu D_mu Psi - H V'(s) Psi) for generic Psi, Psibar of all eight coordinates: the field equation is gamma^mu D_mu Psi == H V'(s) Psi
- FIELD EQUATION [THE RESULT]: EL_Psi == -(Sqrt[g]/H)((D_mu Psibar) gamma^mu + H V'(s) Psibar): the adjoint equation is (D_mu Psibar) gamma^mu == -H V'(s) Psibar
- FIELD EQUATION [has content]: the adjoint equation IS the Dirac conjugate of the field equation: with Psi = u + i v, Psibar = (u - i v)^T sigma16, adjoint residual == -(field residual)^* . sigma16

On the author's canonical frame (section 3.1.11; $q = e^{-a_4(Hx_4)}/\sin^{1/6}(6Hx_0)$, $p = e^{+a_4(Hx_4)}/\sin^{1/6}(6Hx_0)$, a_4 free; this p is the frame factor, not the bilinear of section 4.1.1), with $\gamma^\mu \Gamma_\mu = -3H \cot^2(6Hx_0) T_{16}[0]$ (section 3.8.8), summed over $i = 1, 2, 3$ and $h = 5, 6, 7$:

$$\cot(6Hx_0) T_{16}[0] \partial_0 \Psi + \frac{1}{q} \sum_i T_{16}[i] \partial_i \Psi + T_{16}[4] \partial_4 \Psi + \frac{1}{p} \sum_h T_{16}[h] \partial_h \Psi - 3H \cot^2(6Hx_0) T_{16}[0] \Psi = H V'(s) \Psi.$$

[proved VII §27] WRITTEN OUT (i) [THE RESULT]: the field equation on the canonical frame is Cot T16[0] d_0 Psi + (1/q) T16[i] d_i Psi + T16[4] d_4 Psi + (1/p) T16[h] d_h Psi - 3 H Cot^2 T16[0] Psi == H V'(s) Psi.

In the split-octonion 8 + 8 form, covariantly on any frame ($\Gamma_\mu = \text{diag}(\Gamma_\mu^{(1)}, \Gamma_\mu^{(2)})$), the connection never mixes the two types; $\tau_a = \text{tau}[a]$, $\bar{\tau}_a = \text{taubar}[a]$):

$$\begin{aligned} e_a^\mu \bar{\tau}_a (\partial_\mu + \Gamma_\mu^{(2)}) \psi_2 &= H V'(s) \psi_1, & e_a^\mu \tau_a (\partial_\mu + \Gamma_\mu^{(1)}) \psi_1 &= H V'(s) \psi_2, \\ (\partial_\mu \bar{\psi}_2 - \bar{\psi}_2 \Gamma_\mu^{(2)}) \tau_a e_a^\mu &= -H V'(s) \bar{\psi}_1, & (\partial_\mu \bar{\psi}_1 - \bar{\psi}_1 \Gamma_\mu^{(1)}) \bar{\tau}_a e_a^\mu &= -H V'(s) \bar{\psi}_2, \\ \bar{\psi}_1 &= -\psi_1^\dagger \sigma, & \bar{\psi}_2 &= \psi_2^\dagger \sigma, \end{aligned}$$

and on the canonical frame the connection term is $-3H \cot^2 \psi_2$ in the upper and $-3H \cot^2 \psi_1$ in the lower rows. [proved VII §27] WRITTEN OUT (ii) [THE RESULT]: the covariant 8+8 form - rows 1..8 are $e_a^\mu \text{taubar}[a] (d_\mu + \text{Gamma}^{(2)}_\mu) \text{psi2} - H V' \text{psi1}$, rows 9..16 are $e_a^\mu \text{tau}[a] (d_\mu + \text{Gamma}^{(1)}_\mu) \text{psi1} - H V' \text{psi2}$; and the conjugate rows.

All sixteen component equations on the canonical frame. They are the notebook's export provenance-latex/generated/fermion_fable_field_equations.tex, included here unchanged (the same equations in plain text are the file provenance-latex/generated/fermion_fable_field_equations.txt; both are written by claude-fable/export_fermion_fable.tex.wls from the asserted term lists). ψ_k is component k of Ψ (the notebook's 0-based component index), $\partial_\mu \psi_k = \partial \psi_k / \partial x_\mu$, $e^{a_4(Hx_4)}$ is the exponential of the free function a_4 at Hx_4 , and $V'(s)$ is evaluated at $s = \bar{\Psi}\Psi$:

$$\begin{aligned}
(E_0) \quad & \cot(6Hx_0) \partial_0 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{13} \\
& - \partial_4 \psi_{13} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{14} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{15} - 3H \cot^2(6Hx_0) \psi_8 \\
& = HV'(s) \psi_0 \\
(E_1) \quad & \cot(6Hx_0) \partial_0 \psi_9 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{12} \\
& + \partial_4 \psi_{12} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{14} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{15} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{15} - 3H \cot^2(6Hx_0) \psi_9 \\
& = HV'(s) \psi_1 \\
(E_2) \quad & \cot(6Hx_0) \partial_0 \psi_{10} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{12} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{13} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{13} \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{15} + \partial_4 \psi_{15} - 3H \cot^2(6Hx_0) \psi_{10} \\
& = HV'(s) \psi_2 \\
(E_3) \quad & \cot(6Hx_0) \partial_0 \psi_{11} + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{11} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{12} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{12} + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{13} \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{14} - \partial_4 \psi_{14} - 3H \cot^2(6Hx_0) \psi_{11} \\
& = HV'(s) \psi_3 \\
(E_4) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_9 + \partial_4 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{10} \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{11} \\
& + \cot(6Hx_0) \partial_0 \psi_{12} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{12} - 3H \cot^2(6Hx_0) \psi_{12} \\
& = HV'(s) \psi_4 \\
(E_5) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_8 - \partial_4 \psi_8 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_{10} \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_{10} - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_{11} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_{11} \\
& + \cot(6Hx_0) \partial_0 \psi_{13} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{13} - 3H \cot^2(6Hx_0) \psi_{13} \\
& = HV'(s) \psi_5 \\
(E_6) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_9 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_9 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{11} - \partial_4 \psi_{11} \\
& + \cot(6Hx_0) \partial_0 \psi_{14} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{14} - 3H \cot^2(6Hx_0) \psi_{14} \\
& = HV'(s) \psi_6 \\
(E_7) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_8 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_8 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_9
\end{aligned}$$

$$\begin{aligned}
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_9 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_{10} + \partial_4 \psi_{10} \\
& + \cot(6Hx_0) \partial_0 \psi_{15} - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_{15} - 3H \cot^2(6Hx_0) \psi_{15} \\
& = HV'(s) \psi_7
\end{aligned}$$

$$\begin{aligned}
(E_8) \quad & \cot(6Hx_0) \partial_0 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_5 \\
& + \partial_4 \psi_5 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_6 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_7 - 3H \cot^2(6Hx_0) \psi_0 \\
& = HV'(s) \psi_8
\end{aligned}$$

$$\begin{aligned}
(E_9) \quad & \cot(6Hx_0) \partial_0 \psi_1 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_4 \\
& - \partial_4 \psi_4 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_6 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_7 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_7 - 3H \cot^2(6Hx_0) \psi_1 \\
& = HV'(s) \psi_9
\end{aligned}$$

$$\begin{aligned}
(E_{10}) \quad & \cot(6Hx_0) \partial_0 \psi_2 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_4 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_5 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_5 \\
& + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_7 - \partial_4 \psi_7 - 3H \cot^2(6Hx_0) \psi_2 \\
& = HV'(s) \psi_{10}
\end{aligned}$$

$$\begin{aligned}
(E_{11}) \quad & \cot(6Hx_0) \partial_0 \psi_3 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_3 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_4 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_4 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_5 \\
& - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_6 + \partial_4 \psi_6 - 3H \cot^2(6Hx_0) \psi_3 \\
& = HV'(s) \psi_{11}
\end{aligned}$$

$$\begin{aligned}
(E_{12}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_1 - \partial_4 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_2 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_4 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_4 - 3H \cot^2(6Hx_0) \psi_4 \\
& = HV'(s) \psi_{12}
\end{aligned}$$

$$\begin{aligned}
(E_{13}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_0 + \partial_4 \psi_0 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_2 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_2 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_3 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_5 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_5 - 3H \cot^2(6Hx_0) \psi_5 \\
& = HV'(s) \psi_{13}
\end{aligned}$$

$$\begin{aligned}
(E_{14}) \quad & e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_1 \\
& + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_1 + e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_3 + \partial_4 \psi_3 \\
& + \cot(6Hx_0) \partial_0 \psi_6 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_6 - 3H \cot^2(6Hx_0) \psi_6 \\
& = HV'(s) \psi_{14}
\end{aligned}$$

$$\begin{aligned}
(E_{15}) \quad & - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_1 \psi_0 - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_6 \psi_0 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_2 \psi_1 \\
& - e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_5 \psi_1 - e^{a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_3 \psi_2 - \partial_4 \psi_2 \\
& + \cot(6Hx_0) \partial_0 \psi_7 + e^{-a_4(Hx_4)} \sqrt[6]{\sin(6Hx_0)} \partial_7 \psi_7 - 3H \cot^2(6Hx_0) \psi_7
\end{aligned}$$

$$= HV'(s) \psi_{15}$$

The sixteen adjoint equations $\bar{E}_r$ (provenance-latex/generated/fermion_fable_adjoint_equations.txt) contain, for each r , the same eight (direction, component) pairs with $\bar{\psi}_k$ in place of ψ_k : the terms along the spacelike directions $x_0..x_3$ have the same sign, those along the timelike directions $x_4..x_7$ the opposite sign, the connection term $-3H \cot^2$ is the same, and the V' term changes sign ($T_{16}[a]^T = \eta_{aa} T_{16}[a]$). [proved VII §27] WRITTEN OUT (iii) [definition]: the term lists reproduce all sixteen field equations and all sixteen adjoint equations exactly.

The rescaled form. $\Psi = \sqrt{\sin 6Hx_0} \Psi'$ removes the connection term exactly, because $\gamma^\mu \Gamma_\mu = -\frac{1}{2} T_\mu \gamma^\mu$ with the gradient Weitzenböck torsion vector $T_\mu = \partial_\mu \ln \sin(6Hx_0)$:

$$\gamma^\mu D_\mu \Psi - H V'(s) \Psi = \sqrt{\sin 6Hx_0} (\gamma^\mu \partial_\mu \Psi' - H V'(\sin(6Hx_0) s') \Psi'), \quad s = \sin(6Hx_0) s'.$$

[proved VII §27] WRITTEN OUT (iv) [THE RESULT]: Psi = Sqrt[Sin[6Hx0]] Psi' turns the field equation into Sqrt[Sin] (Cot T16[0] d_0 Psi' + (1/q) T16[i] d_i Psi' + T16[4] d_4 Psi' + (1/p) T16[h] d_h Psi' - H V'(Sin s') Psi'): NO connection term. The x_4 -only family $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$ solves the equation if and only if $V''s' = 0$: for linear V (the author's mass term) or for null data. [proved VII §29] PRE-UNIVERSE (2) [THE RESULT]: Psi = Sqrt[Sin] Psi'(x4) satisfies the field equation iff d_x0 V'(Sin[6Hx0] s') == 0, and d_x0 V'(Sin s') == 6 H Cos[6 H x0] s' V''(Sin s'): V linear OR s' == 0.

The operator form. As an operator equation the right-hand side is $H:V'(s)\Psi$: with the Weyl-symmetrized product: exact for linear V ; for nonlinear V it differs by Hartree and Fock contractions with the coincident J -vacuum propagator (a scalar part that renormalizes V' , and a γ^4 part, the sea charge, removed by charge-symmetric ordering). In the Kohn–Sham mean field $V'(s) \rightarrow V'(\langle s \rangle)$ and the equation is linear [prose VII §27].

The currents. The only internal U(1) is the vector phase; its current $j^\mu = i \bar{\Psi} \gamma^\mu \Psi$ is real and conserved on shell. The particle-number current is $n^\mu = -\frac{i}{H} \bar{\Psi} \gamma^\mu \Psi$, with $n^4 = \Psi^\dagger \Psi / H \geq 0$ and $N = \int \sqrt{g} n^4 d^7x$ the normal-ordered number of particles minus antiparticles. [proved VII §27, §28]

- CURRENT [THE RESULT]: OFF SHELL (1/Sqrt[g]) d_mu(Sqrt[g] j^mu) == i (Ebar Psi + Psibar E) for generic fields of all eight coordinates - conserved on shell
- CHARGE [THE RESULT]: n^4 = -(i/H) Psibar gamma^4 Psi == (1/H) Psi^ddag G Psi == (1/H) Psi^dagger Psi, whose mode form Sum_u eta_u b_u^ddag b_u == Sum_u b_u^dagger b_u is the particle number (normal-ordered: particles - antiparticles)

4.1.4 The canonical anticommutator in 4+4 dimensions

Time is x_4 (lapse 1, shift 0 on every frame of this document). The slice $x_4 = \text{const}$ is seven-dimensional with signature (4,3) — $x_0..x_3$ spacelike, x_5, x_6, x_7 timelike — and is not a Cauchy surface. The Dirac bracket of the symmetrized Lagrangian gives

$$\{\Psi_a(x), \Psi_b^\dagger(y)\}_{x_4=y_4} = \frac{H}{\sqrt{g}} G_{ab} \delta^7(x-y), \quad G = -i \sigma_{16} \gamma^4, \quad \{\Psi_a, \Psi_b\} = 0.$$

On the canonical frame $\gamma^4 = T_{16}[4]$ and $\sqrt{g} = \sec 6Hx_0$, so the matrix is

$$G = J := -iT_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4], \quad J \text{ Hermitian}, \quad J^2 = \text{ID}_{16},$$

eigenvalues $+1$ (8), -1 (8): signature (8,8),

$$\{\Psi_a(x), \Psi_b^\dagger(y)\} = H \cos(6Hx_0) J_{ab} \delta^7(x - y).$$

[proved VII §28]

- QUANTIZATION [THE RESULT]: $G = -i \text{ sigma16 } \gamma^4 == J = -i T_{16}[0].T_{16}[1].T_{16}[2].T_{16}[3].T_{16}[4]$ on the canonical frame; J is Hermitian, $J^2 == \text{ID}_{16}$, eigenvalues $+1$ (8) and -1 (8): signature (8,8)
- ANTICOMMUTATOR [THE RESULT]: on the canonical frame ($c = \text{Sqrt}[g]/H$) $\{\Psi, \Psi^\dagger\} == (H/\text{Sqrt}[g]) J == H \cos[6 H x_0] J$ ($J^{-1} == J$), which IS the design's $(H/\text{Sqrt}[g])(-i \text{ sigma16 } \gamma^4)$

The spin connection does not enter it ($\Gamma_4 = 0$; the anticommutator comes from the time-derivative term alone).

The fundamental symmetry and the Hilbert adjoint. $J = -i\Omega_4 T_{16}[0]$ is a hidden flavour gamma, of signature (2,2) on the flavour factor; $\sigma_{16} = J\beta$ with $\beta = -iT_{16}[4]$ the observer's Dirac-adjoint matrix. The Hilbert adjoint is defined by the J -involution, $\Psi^\dagger := \Psi^\dagger J$; then $\bar{\Psi} = \Psi^\dagger \beta$, $s = \Psi^\dagger \beta \Psi$, and $\{\Psi_a, \Psi_b^\dagger\} = (H/\sqrt{g}) \delta_{ab} \delta^7$: **positive**. The adjoint is mode-independent, and a mode basis must consist of J -eigenvectors. For Grassmann fields the $*$ -structure is a free choice and the filled Dirac sea makes the Hamiltonian bounded below; for a commuting field the same construction fails. Under this adjoint a bilinear $\Psi^\dagger M \Psi$ (with M classically Hermitian) is a Hermitian operator iff $[J, M] = 0$ and anti-Hermitian iff $\{J, M\} = 0$: $s = \bar{\Psi} \Psi$ is Hermitian and $p = \bar{\Psi} T_{16}[8] \Psi$ anti-Hermitian, which is why the self-interaction depends on s only. [proved VII §28]

- J [THE RESULT]: with $\Psi^\dagger := \Psi^\dagger J$: $\text{Psibar} = \Psi^\dagger \beta$ ($J \text{ sigma16} == \beta$), $\Psi^\dagger G = \Psi^\dagger (J G == \text{ID}_{16})$, $\{\Psi, \Psi^\dagger\} == (H/\text{Sqrt}[g]) \text{ID}_{16}$ ($G^{-1} J == \text{ID}_{16}$): **POSITIVE**
- J [has content]: the Hermiticity rule - for classically Hermitian M , $(J M)^\dagger == J M$ iff $[J, M] == 0$ and $== -J M$ iff $\{J, M\} == 0$: so $s = \Psi^\dagger \text{sigma16 } \Psi$ is Hermitian and $p = \Psi^\dagger C^+ \Psi$ is ANTI-Hermitian
- KREIN MODES [control]: a Krein-boosted pair $u_1 = (5/4)u^+ + (3/4)u^-$, $u_2 = (3/4)u^+ + (5/4)u^-$ in the $\omega = +5$ eigenspace is still G -orthonormal and still consists of modes, but not of J -eigenvectors; the induced $J_U = G U U^\dagger$ differs from J , and under it s is NOT Hermitian

The admissible sector (a theorem). A positive Fock quantization in which the local $s(x)$, the local energy-momentum tensor and the free Hamiltonian are Hermitian exists if and only if the span of the momenta carried by the modes is spacelike-definite. On the canonical frame: no momentum along x_5, x_6, x_7 , and J is then unique. With a hidden momentum above threshold the frequencies are imaginary ($\omega^2 = k_s^2 - k_h^2 + m^2$; at $k = (0, \frac{3}{4}, 1, 0 | k_5 = 2)$, $m = 1: \pm 1.19896i$), at threshold the evolution is nilpotent. The admissible theory is a truncation (dimensional reduction to $\Psi(x_0, \dots, x_4)$) with a finite hidden coordinate volume $V_{\text{hid}} = \int dx_5 dx_6 dx_7$ — an assumption, and compact timelike directions contain closed timelike curves — and its anticommutator is

$$\{\Psi_a(x), \Psi_b^\dagger(y)\} = \frac{H}{V_{\text{hid}} \sqrt{|g|}} G_{ab} \delta^4(x_{0..3} - y_{0..3}).$$

[proved VII §28]

- ADMISSIBILITY (1) [THE RESULT]: $E_k^2 == (k_s^2 - k_h^2 + m^2)$ ID16 for every k : $\omega^2 = k_s^2 - k_h^2 + m^2$
- ADMISSIBILITY (5) [THE RESULT]: ABOVE threshold ($k = (0, 3/4, 1, 0 \mid k_5 = 2)$, $m = 1$) $\omega^2 == -23/16$: every eigenvalue of E_k is $\pm i \sqrt{23}/4 = \pm 1.19896 i$ and every eigenvector is G-NEUTRAL ($u^\dagger G u == 0$)
- ADMISSIBILITY (7) [THE RESULT]: J is UNIQUE - the joint commutant of β and the admissible E_k is 8-dimensional, contains J , and every element of it commutes with J (so $X^2 == 1$ with $J X > 0$ forces $X == J$)

The Fock space. Per spatial momentum the sixteen J -eigenmodes are 8 of frequency $+\omega$ and 8 of frequency $-\omega$, $\omega = \sqrt{k \cdot k + m^2}$; every mode has energy equal to its frequency, the negative-frequency modes are filled (the Dirac sea, the unique ground state), and the first excitations are 8 particles and 8 antiparticles. A finite Jordan–Wigner Fock space at one momentum ($k = 3e_1$, $m = 4$, dimension $2^{16} = 65536$) checks all of it exactly, including the sign of the anticommutator through the Heisenberg equation. [proved VII §28]

- FOCK (mode level) [THE RESULT]: the spectrum - a UNIQUE ground state at -8 $\omega = -40$, the Dirac sea (modes 9..16 filled, 1..8 empty); $:Ham: = Ham + 40 \geq 0$
- FOCK (mode level) [THE RESULT]: the first excited level, $\omega = 5$ above the sea, is 16-fold: 8 particle states ($N = +1$) and 8 antiparticle states ($N = -1$), with $N = \sum b^\dagger b - 8$
- FOCK (field level) [THE RESULT]: Heisenberg - $[H_{free}, \Psi_a] == -(G^{-1} h \Psi)_a$ for all 16 components, so $i d_4 \Psi = i [H, \Psi]$ IS the field equation $i G d_4 \Psi = h \Psi$

What survives. 13 of the 28 Lorentz generators commute with J (Spin(4,1) of $x_0..x_4$ and Spin(3) of $x_5..x_7$); the 12 boosts between $x_0..x_3$ and $x_5..x_7$ and the 3 rotations between x_4 and $x_5..x_7$ are lost. The wall $x_0 = 0$ of the canonical frame is at finite proper distance $z = -\ln \cos(6Hx_0)/(6H)$; the one-particle space of $\Psi' = \Psi/\sqrt{\sin}$ is $L^2(dz)$, the x_0 -operator is regular at the wall and limit-point at infinity, and the Lorentz-covariant, J -compatible wall conditions are $T_{16}[0] \Psi' = \pm \Psi'$. [proved VII §28]

- J [THE RESULT]: 13 of the 28 Lorentz generators commute with J (a, b both in $\{0..4\}$: Spin(4,1), or both in $\{5,6,7\}$: Spin(3)); the other 15 anticommute - 12 boosts $x_0..x_3 \leftrightarrow x_5..x_7$ and 3 rotations $x_4 \leftrightarrow x_5..x_7$
- BOUNDARY [THE RESULT]: the bag condition $T_{16}[0] \Psi' = \pm \Psi'$ at the wall - $T_{16}[0]^2 == 1$, $\{\beta, T_{16}[0]\} == 0$, the normal-flux form vanishes on both eigenspaces ($P \beta T_{16}[0] P == 0$), and $[J, T_{16}[0]] == 0$

The anticommutator on the time-dependent frames of the coupled system is section 4.4.6.

4.1.5 The energy–momentum tensor operator

$$\hat{T}_{\mu\nu} = -\frac{1}{4H} \left[\bar{\Psi} \gamma_\mu D_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi + (\mu \leftrightarrow \nu) \right] + g_{\mu\nu} \hat{L}, \quad \gamma_\mu = g_{\mu\nu} \gamma^\nu,$$

normal-ordered with respect to the J -vacuum when it is an operator. It is the **Hilbert** (symmetric-tetrad) tensor of the symmetrized action, equal to this covariant expression off shell; the variation of the spin connection contributes exactly nothing to it (the Lagrangian sees the connection only through its totally antisymmetric part, which a symmetric tetrad variation does not change); it is **not** the partial-derivative tensor. It is symmetric, Hermitian under the classical conjugation, and conserved on shell. [proved VII §29]

- EMT [THE RESULT]: the Hilbert (symmetric-tetrad) tensor of the symmetrized action IS That, OFF shell - components (x1,x1), (x4,x4), (x1,x4), (x0,x5), (x5,x4)
- EMT [has content]: the spin-connection variation contributes NOTHING - dropping it leaves the Hilbert tensor unchanged at (x1,x4), (x0,x5), (x5,x4)
- EMT CONSERVATION [THE RESULT]: $\text{nabla}^\mu \text{That}_{\{\mu \nu\}} == 0$ ON SHELL in all eight components, for generic $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$ and generic V

On shell, for fields of (x_0, x_4) and generic V :

$$\begin{aligned}\hat{L} &= s V'(s) - V(s) && \text{(the kinetic bilinear is } s V'(s)\text{),} \\ \rho &= \hat{T}_{44} = V(s) + K_h, && K_h = -\frac{1}{2H}(\bar{\Psi}\gamma^0\partial_0\Psi - \partial_0\bar{\Psi}\gamma^0\Psi), \\ T^i_i &= T^h_h = s V'(s) - V(s) && \text{(observed } i = 1, 2, 3 \text{ and hidden } h = 5, 6, 7), \\ T^0_0 &= s V'(s) - V(s) + K_h, && \text{trace} = 7 s V'(s) - 8 V(s).\end{aligned}$$

[proved VII §29]

- EMT ON SHELL [THE RESULT]: $\rho = \text{That}_{44} == V(s) + K_h$
- EMT ON SHELL [THE RESULT]: $T^i_i == T^h_h == s V'(s) - V(s)$ for the three observed and the three hidden directions, and $T^0_0 == s V'(s) - V(s) + K_h$

$\hat{T}_{\mu\nu}$ is a Hermitian operator for μ, ν in $\{0..4\}$; its components that mix 0..4 with the hidden sheet are anti-Hermitian, with zero expectation value in every state invariant under the hidden rotations. [proved VII §29] EMT HERMITICITY [THE RESULT]: J-parity table - (mu, nu) in $\{0..4\}$: EVEN (Hermitian); (mu in 0..4, h in 5..7): ODD (anti-Hermitian); (h, h'), $h \neq h'$: ZERO identically; (h, h): even.

4.1.6 Density, pressures and the equation of state

The observer is $u = \partial/\partial x_4$: $\rho = \langle \hat{T}_{44} \rangle$, $P_j = \langle \hat{T}^j_j \rangle$ (no sum), $w = P_{\text{obs}}/\rho$ with P_{obs} the pressure along the observed sheet. With the canonical normalization $\chi = \Psi/\sqrt{H}$ and $W(\sigma) := V(H\sigma)$, $\sigma = \langle \bar{\chi}\chi \rangle$, each occupied positive-frequency mode contributes ω to the energy density, k_j^2/ω to the pressure along j , and m/ω to σ . [proved VII §29] MODE SUMS [has content]: with $\text{ubar} = \text{u}^\dagger \text{sigma16}$ these are $\omega \text{u}^\dagger G u$ and $k_j \text{u}^\dagger (dh/dk_j) u$, so by Section 28 each occupied mode contributes ω , k_j^2/ω and (to sigma) m/ω .

The Kohn-Sham Fermi sea. Fill the positive-frequency modes with $|k| < k_F$ along the observed sheet (zero modes along x_0 and x_5, x_6, x_7), $g = 8$, mass m , $\omega_F = \sqrt{k_F^2 + m^2}$, $\mathcal{L} = \ln((k_F + \omega_F)/m)$:

$$\begin{aligned}n &= \frac{g k_F^3}{6\pi^2}, \\ \sigma_{\text{KS}} &= \frac{g m}{4\pi^2} [k_F \omega_F - m^2 \mathcal{L}], \\ \varepsilon_{\text{KS}} &= \frac{g}{16\pi^2} [k_F \omega_F (2k_F^2 + m^2) - m^4 \mathcal{L}], \\ P_{\text{KS}} &= \frac{g}{48\pi^2} [k_F \omega_F (2k_F^2 - 3m^2) + 3m^4 \mathcal{L}],\end{aligned}$$

$$\varepsilon_{\text{KS}} + P_{\text{KS}} = \omega_F n, \quad \varepsilon_{\text{KS}} - 3P_{\text{KS}} = m \sigma_{\text{KS}}, \quad \partial \varepsilon_{\text{KS}} / \partial m = \sigma_{\text{KS}}, \quad \partial \varepsilon_{\text{KS}} / \partial k_F = \omega_F dn / dk_F.$$

[proved VII §29]

- KOHN-SHAM [THE RESULT]: each closed form has k_F -derivative $(g/(2\pi^2)) k_F^2 f(k_F)$ with $f = 1, m/\omega, \omega, k_F^2/(3\omega)$, and vanishes at $k_F = 0$ - so it IS the integral $\int d^3k/(2\pi)^3 f$
- KOHN-SHAM [THE RESULT]: $\epsilon + P = w_F n$ (Gibbs), $\epsilon - 3P = m \sigma$ (trace), $d\epsilon/dm = \sigma$, $d\epsilon/dk_F = w_F dn/dk_F$

In the mean field $m = W'(\sigma)$:

$$\rho = \epsilon_{KS} + W - \sigma W', \quad P_{obs} = P_{KS} + \sigma W' - W, \quad P_{hid} = P_{x_0} = \sigma W' - W, \quad \rho + P_{obs} = \omega_F n.$$

[proved VII §29] MEAN FIELD [THE RESULT]: $\rho = \epsilon + W - \sigma W'$, $P_{obs} = P_{KS} + \sigma W' - W$ give $\rho + P_{obs} = w_F n$ exactly; and $d\rho/dn = w_F + (\sigma_{KS} - \sigma) W'' \sigma'(n)$, i.e. $= w_F$ at the self-consistent point $\sigma = \sigma_{KS}$ ($m = W'(\sigma)$ throughout). Hence $w_f + 1 = \omega_F n / \rho \geq 0$ wherever $\rho > 0$: **the quantized fable cannot cross the phantom divide**. For the author's mass term $W = m\sigma$ the condensate $W - \sigma W'$ vanishes, $\rho = \epsilon_{KS} > 0$, $P_{obs} = P_{KS} \geq 0$, and $w = P_{KS}/\epsilon_{KS}$ rises monotonically with $x = k_F/|m|$ from 0 (dust) to $1/3$ (radiation); the classical limit is the non-relativistic limit $k_F/|m| \rightarrow 0$, which reproduces Part VI's $\rho = V(s)$, $P = sV' - V$, with the sign rule $\text{sign}(\sigma) = \text{sign}(m) = \text{sign}(W')$. [proved VII §29]

- AUTHOR'S MASS TERM [THE RESULT]: w rises MONOTONICALLY - $dw/dx = x^2 \Delta/(3 w_x I_1^2)$ with $\Delta = x^2 I_1 - w_x^2 I_2$, $\Delta(0) = 0$ and $\Delta' = 2 x I_3 > 0$ - from $w \rightarrow 0$ ($x \rightarrow 0$, $w = x^2/5 + O(x^4)$: dust) to $w \rightarrow 1/3$ ($x \rightarrow \infty$: radiation)
- CLASSICAL LIMIT [THE RESULT]: at the self-consistent point $\rho = W + 3 P_{KS}$ and $P_{obs} = (\sigma W' - W) + P_{KS}$ exactly ($\sigma = \sigma_{KS}$, $m = W'$); Part VI's $\rho = V(s)$, $P = sV' - V$ is the limit $P_{KS}/\epsilon_{KS} \rightarrow 0$, i.e. $k_F/m \rightarrow 0$

On the pre-universe (the canonical frame, a_4 free) the charge per coordinate volume is conserved, the observed-sheet momenta redshift as e^{a_4} , and the quantum gas **cools** ($w : 1/3 \rightarrow 0$) when $a_4' < 0$, while the classical Part VI fable's w is frozen. [proved VII §29] PRE-UNIVERSE (4) [THE RESULT]: $x = k_{F,phys}/m = x_c/(q m)$ obeys $d \text{Log}[x]/dx^4 = H a_4' [H x^4]$; since $dw/dx > 0$, $dw/dx^4 = w'(x) x H a_4' < 0$ exactly when $a_4' < 0$ (the observed sheet expanding): the quantum gas **COOLS**.

Section 4.5 below turns all of this into the per-7-volume source of the coupled system and proves the properties the Einstein equations need.

4.2 The Einstein equations of the primordial gravitational field, with fable as their source

This is Section 30 of the notebook (manifest `claude-fable/cells_part8.w1`, Input cells 199–203, 28 assertions, 4.119s in `claude-fable/run_fermion_fable_part8.log`). Part VIII opens with a statement of purpose: Parts III–V built the $4+4$ geometry of the pre-universe but never asked what **sources** it; Part VI put fields on it without letting them act back; Part VII made fable a quantized fermion with an energy–momentum tensor operator; Part VIII closes the loop [prose VIII, introduction]. It never redefines anything of Parts I–VII, and it never gives a_4 a value.

4.2.1 The semiclassical system and its conventions

The metric stays classical; fable is quantized. The coupled equations are

$$G^\mu{}_\nu = \kappa \langle \hat{T}^\mu{}_\nu \rangle \quad (\text{semiclassical gravity}),$$

$$\kappa = 8\pi G_8 > 0 \quad (\text{the eight-dimensional gravitational coupling, a plain symbol}),$$

with the expectation value taken in a state of the fable Fock space of section 4.1.4 — in section 4.5 and after, the Kohn–Sham ground state. The Einstein tensor is used with one index up and one down, $G^\mu{}_\nu$, because then the answer does not depend on how the metric signs are distributed. For a source at rest in the observer’s frame the energy density and the pressures are the diagonal of the mixed tensor [**prose VIII, introduction**]:

$$\rho = -T^4{}_4, \quad P_{\text{obs}} = T^i{}_i \quad (i = 1, 2, 3), \quad P_{\text{hid}} = T^h{}_h \quad (h = 5, 6, 7), \quad P_C = T^0{}_0,$$

so $G^4{}_4 = -\kappa\rho$, $G^i{}_i = \kappa P_{\text{obs}}$, $G^h{}_h = \kappa P_{\text{hid}}$, $G^0{}_0 = \kappa P_C$. The dynamical frames of Sections 31–33 carry their own scale factors **scA**, **scB**, **scC** (functions of x_4 , separate symbols); the canonical frame is recovered from them only by an explicit, labelled substitution inside an assertion.

4.2.2 The tool, and the three tests it must pass

One helper, `cfEinsteinData`, computes every Einstein tensor of Part VIII from the Christoffel symbols of Section 16’s solver:

$$\begin{aligned} R_{\mu\nu} &= \partial_\lambda \Gamma^\lambda_{\mu\nu} - \partial_\nu \Gamma^\lambda_{\mu\lambda} + \Gamma^\lambda_{\lambda\sigma} \Gamma^\sigma_{\mu\nu} - \Gamma^\lambda_{\nu\sigma} \Gamma^\sigma_{\mu\lambda}, \\ R &= g^{\mu\nu} R_{\mu\nu}, \quad G^\mu{}_\nu = g^{\mu\kappa} R_{\kappa\nu} - \tfrac{1}{2} \delta^\mu{}_\nu R. \end{aligned}$$

It is not trusted until it passes three tests, each of which a wrong index or sign would fail [**proved VIII §30**]:

1. on the separate FLRW frame of Part VI it returns the Friedmann equations with the textbook signs: `EINSTEIN [solver regression]:` on the FLRW frame of Section 24, `G^4_4 == -3 (a'/a)^2` and `G^i_i == -(2 a''/a + (a'/a)^2)`: the Friedmann equations with the textbook signs; `EINSTEIN [solver regression]:` ... and `G == -3 (a''/a + (a'/a)^2)` along the four static directions `x0`, `x5`, `x6`, `x7`, with no off-diagonal component;
2. on the canonical frame its Ricci scalar equals the one Section 16 obtained from the curvature 2-form of the spin connection: `EINSTEIN [fidelity]:` the Ricci scalar from the Christoffel symbols == `RicciScalarCanonical` of Section 16 (from the curvature 2-form of the spin connection): two routes, one answer;
3. its output obeys the contracted Bianchi identity: `EINSTEIN [has content]:` the contracted Bianchi identity `nabla_mu G^mu_nu == 0` holds for the canonical Einstein tensor, all eight components, `a4` arbitrary.

And the canonical metric is not a vacuum solution: `EINSTEIN [control]:` the canonical metric is NOT a vacuum solution - `G^0_0` is not zero (witnessed).

4.2.3 The Einstein tensor of the canonical metric: all sixty-four components

With $\cot = \cot(6Hx_0)$, $\csc = \csc(6Hx_0)$, $a'_4 = a'_4(Hx_4)$, $a''_4 = a''_4(Hx_4)$ (derivatives with respect to the argument of a_4), the mixed Einstein tensor of the canonical metric

$$ds^2 = \tan^2(6Hx_0) dx_0^2 + q^2 dx_{\text{obs}}^2 - dx_4^2 - p^2 dx_{\text{hid}}^2$$

is **diagonal** — all 56 off-diagonal components vanish identically — and

$$\begin{aligned} G^0{}_0 &= 3H^2(5\cot^4 - a_4'^2), \\ G^i{}_i &= \frac{H^2}{2} \left(15(9 + \cos(12Hx_0)) \cot^2 \csc^2 - 6a_4'^2 - 2a_4'' \right) \quad i = 1, 2, 3, \end{aligned}$$

$$\begin{aligned}
G^4_4 &= \frac{3H^2}{2} \left((55 + 7 \cos(12Hx_0)) \cot^2 \csc^2 + 2 a_4'^2 \right), \\
G^h_h &= \frac{H^2}{2} \left(15 (9 + \cos(12Hx_0)) \cot^2 \csc^2 - 6 a_4'^2 + 2 a_4'' \right) \quad h = 5, 6, 7, \\
R &= -3H^2 \left((55 + 7 \cos(12Hx_0)) \cot^2 \csc^2 - 2 a_4'^2 \right)
\end{aligned}$$

for every a_4 [prose VIII §30], and [proved VIII §30]

- EINSTEIN [THE RESULT]: all 64 components of G^{μ}_{ν} of the canonical metric, a_4 arbitrary: 8 diagonal ones as stated, 56 off-diagonal ones identically zero
- EINSTEIN [THE RESULT]: the Ricci scalar $R == -3 H^2 ((55 + 7 \cos[12 H x_0]) \cot^2 \csc^2 - 2 a_4'^2)$

The split, with no cross terms. Every component is a function of x_0 alone plus a function of x_4 alone. The statement is made without giving a_4 a value: the x_0 -part is the Einstein tensor of the **separate** frame $\text{diag}(\tan 6Hx_0, S^{-1/6}, S^{-1/6}, S^{-1/6}, 1, S^{-1/6}, S^{-1/6}, S^{-1/6})$ ($S = \sin 6Hx_0$), which does not contain a_4 , and the a_4 -part is the Einstein tensor of the **separate** frame $\text{diag}(1, e^{-a_4}, e^{-a_4}, e^{-a_4}, 1, e^{+a_4}, e^{+a_4}, e^{+a_4})$, which does not contain x_0 ; the two frames multiply to the canonical frame. [proved VIII §30]

- EINSTEIN [definition]: the two factor frames multiply to the canonical frame, and neither is it: the x_0 -only frame contains no a_4 , the a_4 -only frame contains no x_0
- EINSTEIN [THE RESULT]: NO CROSS TERMS - $G^{\mu}_{\nu}(\text{canonical}) == G^{\mu}_{\nu}(x_0\text{-only frame}) + G^{\mu}_{\nu}(a_4\text{-only frame})$, all 64 components
- EINSTEIN [has content]: the x_0 -part is the same on the observed and the hidden sheet ($G^i_i == G^h_h$), and the a_4 -part is x_0 -independent

The two parts, component by component [derived here: the x_0 -part is the canonical formula at $a_4' = a_4'' = 0$, since the a_4 -part depends on a_4 only through the rates a_4' and a_4'' (it is a Bianchi-I tensor, section 4.3.1) and vanishes for constant a_4 , where the a_4 -only frame is constant; the a_4 -part is the difference]:

component	x_0 -part G_{x_0} (the x_0 -only frame)	a_4 -part G_{a_4} (the a_4 -only frame)
G^0_0	$15H^2 \cot^4$	$-3H^2 a_4'^2$
G^i_i ($i = 1, 2, 3$)	$\frac{15H^2}{2} (9 + \cos(12Hx_0)) \cot^2 \csc^2$	$-3H^2 a_4'^2 - H^2 a_4''$
G^4_4	$\frac{3H^2}{2} (55 + 7 \cos(12Hx_0)) \cot^2 \csc^2$	$+3H^2 a_4'^2$
G^h_h ($h = 5, 6, 7$)	$\frac{15H^2}{2} (9 + \cos(12Hx_0)) \cot^2 \csc^2$	$-3H^2 a_4'^2 + H^2 a_4''$

The x_0 -part does not distinguish the observed from the hidden sheet; only a_4'' does.

4.2.4 Theorem (a): energy — the canonical metric demands negative total energy

$G^4_4 = -\kappa \rho_{\text{total}}$, and the minimum of $55 + 7 \cos(12Hx_0)$ is 48. Therefore, for every a_4 and everywhere in $0 < 6Hx_0 < \pi/2$,

$$\begin{aligned}
\rho_{\text{total}} &= -G^4_4 / \kappa \leq -(72H^2 \cot^2 \csc^2 + 3H^2 a_4'^2) / \kappa < 0, \\
G^4_4 - (72H^2 \cot^2 \csc^2 + 3H^2 a_4'^2) &= \frac{21}{2} H^2 (1 + \cos(12Hx_0)) \cot^2 \csc^2 \geq 0 \quad \text{exactly.}
\end{aligned}$$

The total energy density that sources the canonical metric is **negative everywhere**. No source with $\rho \geq 0$ can be the sole source of this geometry — in particular not the Kohn–Sham fable with the author’s mass term, whose $\rho = \varepsilon_{\text{KS}}$ is positive. [proved VIII §30]

- ENERGY [has content]: min over x_0 of $55 + 7 \cos[12 H x_0]$ is 48 (attained where $\cos[12 H x_0] = -1$)
- ENERGY [THE RESULT]: $G^4_4 - (72 H^2 \cot^2 \text{Csc}^2 + 3 H^2 a_4'^2) == (21/2) H^2 (1 + \cos[12 H x_0]) \cot^2 \text{Csc}^2$, which is ≥ 0
- ENERGY [THE RESULT]: $72 H^2 \cot^2 \text{Csc}^2 + 3 H^2 a_4'^2 > 0$ in $0 < 6 H x_0 < \pi/2$ for real a_4 , hence $\rho_{\text{total}} = -G^4_4/\kappa < 0$: the canonical metric demands **NEGATIVE** total energy density
- ENERGY [has content]: Section 29’s eps_{KS} vanishes at $k_F = 0$ and $d \text{eps}_{\text{KS}}/d k_F == g k_F^2 w_F/(2 \pi^2) > 0$, so the KS fable with the author’s mass term ($\rho = \text{eps}_{\text{KS}}$) cannot be the sole source

4.2.5 The background-stress reading (an illustration; the splitting is a choice)

One may split the source as $T = T_{\text{bg}} + T_{\text{rest}}$ with $T_{\text{bg}} := G_{x_0}/\kappa$, the x_0 -part held as a fixed background. Its energy density is negative everywhere:

$$\rho_{\text{bg}} = -\frac{3H^2}{2\kappa} (55 + 7 \cos(12Hx_0)) \cot^2 \text{csc}^2 < 0.$$

The remaining equations $G_{a_4} = \kappa T_{\text{rest}}$ then demand a stress that does not depend on x_0 , because G_{a_4} does not; the fable solutions of Part VI do — the rescaled solution $\Psi = \sqrt{\sin 6Hx_0} \Psi'(x_4)$ has $\rho = V(\sin(6Hx_0) s')$, which depends on x_0 for the author’s mass term. So fable would have to deform the x_0 -profile. This is an illustration of the splitting, not a theorem about the full coupled problem, and the notebook states it as such [prose VIII §30]. [proved VIII §30]

- BACKGROUND [has content]: with the splitting $T_{\text{bg}} := G_{x_0}/\kappa$, $\rho_{\text{bg}} == -(3 H^2/(2 \kappa)) (55 + 7 \cos[12 H x_0]) \cot^2 \text{Csc}^2 < 0$ everywhere
- BACKGROUND [has content]: the illustration - G_{a_4} is x_0 -independent, while the fable density of Section 24, $\rho = V(\sin[6 H x_0] s')$ on $\Psi = \text{Sqrt}[\sin] \Psi'(x_4)$, depends on x_0 for the mass term (witnessed on $\Psi' = e_1 + e_5$, $M = 1$)

4.2.6 Theorem (b): anisotropy

Subtracting the hidden-sheet equation from the observed-sheet one, the x_0 -parts cancel:

$$G^i_i - G^h_h = -2H^2 a_4'' \quad \text{so} \quad a_4'' = -\frac{\kappa (P_{\text{obs}} - P_{\text{hid}})}{2H^2} \quad \text{for ANY source.}$$

Take fable as the source, as a c-number configuration $\Psi(x_0, x_4), \bar{\Psi}(x_0, x_4)$ (independent functions) of the symmetrized Lagrangian. Along a direction on which the fields do not depend, $D_i \Psi = \Gamma_i \Psi$ and $D_i \bar{\Psi} = -\bar{\Psi} \Gamma_i$, so the kinetic part of $\hat{T}_{ii}$ is $-\frac{1}{2H} \bar{\Psi} \{\gamma_i, \Gamma_i\} \Psi$, which vanishes because $\{\gamma_\mu, \Gamma_\mu\} = 0$ for each μ separately. Hence, **off shell** and for every V ,

$$P_{\text{obs}} = T^i_i = \hat{L} = T^h_h = P_{\text{hid}},$$

and such configurations force $a_4'' = 0$: the anisotropy of the canonical frame cannot be sourced by them. The Kohn–Sham Fermi gas is different: its quasiparticles move along the observed sheet only, so $P_{\text{obs}} - P_{\text{hid}} = P_{\text{KS}}/v > 0$ (with v the hidden 4-volume factor of section 4.5) and it **can** source $a_4'' < 0$. [proved VIII §30]

- ANISOTROPY [THE RESULT]: $G^i_i - G^h_h == -2 H^2 a_4'' [H x_4]$ for every observed i and hidden h , hence $a_4'' == -\kappa (P_{\text{obs}} - P_{\text{hid}})/(2 H^2)$ for any source
- ANISOTROPY [has content]: the premise, re-checked with the index lowered - $\{\gamma_\mu, \Gamma_\mu\} == 0$ for EACH μ separately on the canonical frame ($\gamma_\mu = g_{\mu\nu} \gamma^\nu$; Section 27 (c))
- ANISOTROPY [THE RESULT]: OFF SHELL, for generic independent $\Psi(x_0, x_4)$, $\bar{\Psi}(x_0, x_4)$ and every V : $T^i_i == \hat{L} == T^h_h$ ($i = 1, 2, 3$; $h = 5, 6, 7$), so $P_{\text{obs}} == P_{\text{hid}}$ and such fable configurations force $a_4'' == 0$
- ANISOTROPY [control]: T^0_0 is NOT $\hat{L}$ off shell - the x_0 -derivatives enter it (witnessed on $\Psi = x_0 e_1$, $\bar{\Psi}$ = the first column of $T_{16}[0]$, $V = s^2$): the statement is special to the directions on which the fields do not depend
- ANISOTROPY [has content]: Section 29's P_{KS} vanishes at $k_F = 0$ and $dP_{\text{KS}}/dk_F == g k_F^4/(6 \pi^2 w_F) > 0$, so $P_{\text{KS}} > 0$ for $k_F > 0$
- ANISOTROPY [THE RESULT]: for the KS fluid $P_{\text{obs}} - P_{\text{hid}} == P_{\text{KS}}/v$ identically, for every W : the Fermi gas CAN source $a_4'' < 0$

4.2.7 Theorem (c): momentum

$G^0_4 = G^4_0 = 0$ on the canonical frame, so the source must carry no momentum along the hidden coordinate, $T^0_4 = 0$. That is a real condition on fable: its T_{04} is not identically zero. [**proved VIII §30**]

- MOMENTUM [THE RESULT]: $G^0_4 == G^4_0 == 0$ on the canonical frame, so $T^0_4 == 0$ is required of the source
- MOMENTUM [has content]: $T^0_4 == 0$ is a genuine condition - fable's $T_{\{04\}}$ is not identically zero (witnessed on $\Psi = x_4 e_1$, $\bar{\Psi}$ = $T_{16}[0] e_1$, $V = s^2$)

4.2.8 The asymptotic canonical frame is sourced by a ghost stiff fluid

The a_4 -part of the Einstein tensor is the Einstein tensor of $\text{diag}(1, e^{-a_4}, e^{-a_4}, e^{-a_4}, 1, e^{a_4}, e^{a_4}, e^{a_4})$, which section 4.3.3 shows to be the asymptotic ($6Hx_0 \rightarrow \pi/2$) form of the canonical frame. With $h = H a_4'(Hx_4)$ (so that the observer's Hubble rate is $H_{\text{obs}} = -h$) and $h' = dh/dx_4 = H^2 a_4''$, the stress it demands is

$$\rho = P_C = -\frac{3h^2}{\kappa}, \quad P_{\text{obs}} = \rho - \frac{h'}{\kappa}, \quad P_{\text{hid}} = \rho + \frac{h'}{\kappa}.$$

Its x_0 -pressure equals its energy density (stiff, $w_C = +1$), and both are negative. When $h' = 0$ all seven pressures equal $\rho = -3h^2/\kappa$: exactly the energy-momentum tensor of a **ghost** scalar $\phi(x_4)$ (a scalar with the wrong sign of its kinetic term, $L = +\frac{1}{2} g^{mn} \partial_m \phi \partial_n \phi$, $T_{mn} = -\partial_m \phi \partial_n \phi + g_{mn} L$) with $\phi'^2 = 6h^2/\kappa$. In general the demanded stress is the ghost tensor plus the traceless anisotropic stress (h'/κ) $\text{diag}(0, -1, -1, -1, 0, +1, +1, +1)$. The volume-preserving expansion of the observed sheet ($h < 0$, $H_{\text{obs}} > 0$) is sourced by negative energy. [**proved VIII §30**]

- GHOST [THE RESULT]: the a_4 -part demands $\rho == P_C == -3 h^2/\kappa$, $P_{\text{obs}} == \rho - h'/\kappa$, $P_{\text{hid}} == \rho + h'/\kappa$, with $h = H a_4' [H x_4]$, $h' = dh/dx_4$

- GHOST [THE RESULT]: demanded stress == ghost-scalar tensor with $\phi'^2 = 6 h^2/\kappa + (h'/\kappa) \text{diag}(0, -1, -1, -1, 0, 1, 1, 1)$: negative-energy, stiff ($P_C = \rho$), plus a traceless anisotropic part
- GHOST [has content]: the ghost scalar's energy density is $-\phi'^2/2 < 0$ and all seven of its pressures equal it

and PART VIII [control]: a_4 is STILL UNDEFINED after Section 30.

4.2.9 What Section 30 proves

Proved symbolically, with a_4 arbitrary: the Einstein tensor of the canonical metric (all 64 components, the $x_0 + a_4$ split with no cross terms, the Ricci scalar two ways, the Bianchi identity); negative total energy for every source; the anisotropy law, with $P_{\text{obs}} = P_{\text{hid}}$ for every c-number fable configuration of (x_0, x_4) and $P_{\text{obs}} - P_{\text{hid}} = P_{\text{KS}}/v$ for the Kohn–Sham gas; zero x_0 -momentum; and the ghost-scalar reading of the asymptotic canonical frame. The conclusion that drives the rest of this document: **fable, whose energy is positive, cannot source the canonical frame alone; the frame must be generalized** (section 4.3).

4.3 The generalized frame, the (0,4) equation, the asymptotic region, and the 8-dimensional Bianchi-I system

This is Section 31 of the notebook (Input cells 204–207, 21 assertions, 12.672s). Section 30 showed that the canonical frame, with its single free function a_4 , cannot absorb a positive energy density. Section 31 builds the family of frames on which fable can act back.

4.3.1 The generalized warped frame and its Einstein tensor

The generalization keeps the x_0 -profile of the canonical frame and gives each block its own scale factor, a function of the observer's time x_4 :

$$\begin{aligned}
 e &= \text{diag}(\tan(6Hx_0)C, AS^{-1/6}, AS^{-1/6}, AS^{-1/6}, 1, BS^{-1/6}, BS^{-1/6}, BS^{-1/6}), \\
 S &= \sin 6Hx_0, \\
 A &= \text{scA}[x_4] \text{ (observed sheet)}, \quad B = \text{scB}[x_4] \text{ (hidden timelike sheet)}, \\
 C &= \text{scC}[x_4] \text{ (hidden } x_0 \text{ direction)}, \\
 H_A &= A'/A, \quad H_B = B'/B, \quad H_C = C'/C, \quad \Theta = 3H_A + 3H_B + H_C \quad (' = d/dx_4).
 \end{aligned}$$

The canonical frame is the member $A = e^{-a_4}$, $B = e^{+a_4}$, $C = 1$ — a labelled substitution, used only inside assertions. The theorem:

$$\begin{aligned}
 \text{diag } G^\mu{}_\nu &= \text{diag } G_{x_0}/C^2 + \text{diag } G_{\text{BI}}(A, B, C) \quad \text{EXACTLY,} \\
 G^4{}_0 &= 3H \cot(6Hx_0) (2H_C - H_A - H_B), \quad G^0{}_4 = -3H \cot^3(6Hx_0) (2H_C - H_A - H_B)/C^2 \\
 &\quad \text{(the only off-diagonal components),}
 \end{aligned}$$

where G_{x_0} is the x_0 -part of section 4.2.3 and G_{BI} is the Einstein tensor of the unwarped 8-dimensional Bianchi-I frame $\text{diag}(C, A, A, A, 1, B, B, B)$. The contracted Bianchi identity holds for the whole family, and on the canonical member everything reduces to Section 30. **[proved VIII §31]**

- WARPED [definition]: the canonical frame IS the member $A = \text{Exp}[-a_4]$, $B = \text{Exp}[+a_4]$, $C = 1$ of the family (labelled substitution)
- WARPED [has content]: the contracted Bianchi identity $\nabla_\mu G^\mu{}_\nu = 0$ holds for the warped family, A, B, C arbitrary

- WARPED [THE RESULT]: $\text{diag } G^{\mu}_{\mu} == \text{diag } G_{x0}/C^2 + \text{diag } G_{\text{BI}}(A, B, C)$ EXACTLY (the x_0 -part of Section 30 and the unwarped Bianchi-I tensor)
- WARPED [THE RESULT]: the only off-diagonal components are $G^4_0 == 3 H \text{Cot}(2 H_C - H_A - H_B)$ and $G^0_4 == -3 H \text{Cot}(2 H_C - H_A - H_B)/C^2$
- WARPED [fidelity]: on the canonical member the warped Einstein tensor IS Section 30's, all 64 components, and the Bianchi-I part IS Section 30's a_4 -part

(the two geometries took 9.584s and 0.709s, `claude-fable/run_fermion_fable_part8.log`).

The Bianchi-I part, component by component [derived here from the Ricci components asserted in section 4.3.4, $R^i_i = H'_i + H_i\Theta$ and $R^4_4 = \sum_7 (H'_i + H_i^2)$, with $R = R^4_4 + 3R^A + 3R^B + R^C$ and $G^\mu_\mu = R^\mu_\mu - R/2$; $\Sigma_2 = 3H_A^2 + 3H_B^2 + H_C^2$, $\dot{S} = 3H'_A + 3H'_B + H'_C$]:

$$\begin{aligned} G_{\text{BI}}^4_4 &= -\frac{1}{2}(\Theta^2 - \Sigma_2) = -(3H_A^2 + 3H_B^2 + 9H_AH_B + 3H_AH_C + 3H_BH_C), \\ G_{\text{BI}}^i_i &= H'_A + H_A\Theta - \dot{S} - \frac{1}{2}(\Sigma_2 + \Theta^2) \quad (i = 1, 2, 3), \\ G_{\text{BI}}^h_h &= H'_B + H_B\Theta - \dot{S} - \frac{1}{2}(\Sigma_2 + \Theta^2) \quad (h = 5, 6, 7), \\ G_{\text{BI}}^0_0 &= H'_C + H_C\Theta - \dot{S} - \frac{1}{2}(\Sigma_2 + \Theta^2). \end{aligned}$$

Checks [derived here]: on the canonical member ($H_A = -h$, $H_B = h$, $H_C = 0$, $h = Ha'_4$, $\Theta = 0$) these give $G^4_4 = 3h^2$, $G^i_i = -h' - 3h^2$, $G^h_h = h' - 3h^2$, $G^0_0 = -3h^2$ — the a_4 -part of section 4.2.3 ($h' = H^2a''_4$); on the FLRW frame ($A = a$, $B = C = 1$, $H_A = a'/a$) they give $G^4_4 = -3H_A^2$, $G^i_i = -(2a''/a + H_A^2)$, $G^0_0 = G^h_h = -3(a''/a + H_A^2)$ — the solver regression of section 4.2.2.

4.3.2 The (0,4) equation singles out the author's volume-preserving structure; it does not derive it

The Einstein equation $G^4_0 = \kappa T^4_0$ with a source that carries no momentum along x_0 ($T^4_0 = 0$: required on the canonical frame by theorem (c), and true of the Kohn–Sham fluid, whose quasiparticles move along the observed sheet only) gives, since $\cot(6Hx_0)$ is not zero,

$$2H_C = H_A + H_B.$$

With the x_0 direction **static** ($C = 1$, a labelled substitution) this is $H_A + H_B = 0$, i.e. $AB = \text{const}$: exactly the author's volume-preserving pairing $A = e^{-a_4}$, $B = e^{+a_4}$. So the author's ansatz is singled out by the momentum constraint once $C = 1$ is imposed; it is **not** derived, because $C = 1$ is itself an assumption. With C dynamical the constraint is $H_C = (H_A + H_B)/2$, and the energy bill changes sign. The Bianchi-I part of $\kappa\rho = -G^4_4$ is the sum over the 21 pairs of the seven directions:

$$\kappa\rho = -G_{x_0}^4_4/C^2 + 3H_A^2 + 3H_B^2 + 9H_AH_B + 3H_AH_C + 3H_BH_C,$$

pair sum = $-3H_A^2 < 0$	on the volume-preserving branch ($H_B = -H_A$, $H_C = 0$): section 4.2.4's negative energy,
pair sum = $\frac{9}{2}(H_A^2 + H_B^2) + 12H_AH_B$	on the branch $H_C = (H_A + H_B)/2$
$= +\frac{9}{2}H_A^2 > 0$	when the hidden sheet is static ($H_B = 0$).

With C dynamical a **positive-energy** expansion of the observed sheet is allowed (the x_0 -term $-G_{x_0}^4_4/C^2$ is still negative; section 4.3.3 shows how fast it decays). [proved VIII §31]

- (0,4) [THE RESULT]: with $T^4_0 = 0$ the (0,4) Einstein equation is $2 H_C == H_A + H_B$ ($\text{Cot}[6 H x_0]$ is not zero for $0 < 6 H x_0 < \text{Pi}/2$)

- (0,4) [THE RESULT]: with $C = 1$ (labelled substitution) it reads $H_A + H_B == 0$, i.e. $d(A B)/dx_4 == 0$: the volume-preserving pairing is SINGLED OUT, and the canonical member satisfies it
- (0,4) [has content]: $\kappa \rho == -G^4_4 == -G_{x_0^4_4}/C^2 + (\text{sum over the 21 pairs of the seven directions of } H_i H_j)$
- (0,4) [THE RESULT]: the pair sum is $-3 H_A^2$ on the volume-preserving branch (negative energy) and $(9/2)(H_A^2 + H_B^2) + 12 H_A H_B$ on the branch $H_C = (H_A + H_B)/2$, i.e. $+(9/2) H_A^2 > 0$ for a static hidden sheet

4.3.3 The asymptotic region, and the decay rates

The proper distance along x_0 at fixed x_4 ($C = 1$) is

$$z = \int_0^{x_0} \tan(6Hy) dy = -\frac{\ln \cos(6Hx_0)}{6H}, \quad x_0(z) = \frac{\arccos(e^{-6Hz})}{6H},$$

and $6Hx_0 \rightarrow \pi/2$ is $z \rightarrow \infty$. In the coordinate z the warped metric is **exactly**

$$ds^2 = C^2 dz^2 + A^2(1 - e^{-12Hz})^{-1/6} dx_{\text{obs}}^2 - dx_4^2 - B^2(1 - e^{-12Hz})^{-1/6} dx_{\text{hid}}^2,$$

so it tends to the unwarped Bianchi-I metric, and every x_0 -term of the Einstein tensor decays with $\cot^2(6Hx_0) = e^{-12Hz}/(1 - e^{-12Hz})$. As $6Hx_0 \rightarrow \pi/2$:

$$G_{x_0^0_0} \sim 15H^2 \cot^4, \quad G_{x_0^i_i} = G_{x_0^h_h} \sim 60H^2 \cot^2, \quad G_{x_0^4_4} \sim 72H^2 \cot^2, \\ G^4_0 \sim \cot, \quad G^0_4 \sim \cot^3.$$

Read as a background stress $T_{\text{bg}} = G_{x_0}/(\kappa C^2)$, in units of $H^2 \cot^2/(\kappa C^2)$: $\rho_{\text{bg}} \rightarrow -72$, $P_{\text{obs}} = P_{\text{hid}} \rightarrow 60$, $P_C \rightarrow 15 \cot^2 \rightarrow 0$. If T_{bg} is **not** counted as a source, the x_0 -terms act on the Bianchi-I evolution equations (section 4.3.4) as residual drivers $\kappa (P_i - T/6)|_{\text{bg}} \rightarrow -12H^2 \cot^2/C^2$ for A and for B , and $-72H^2 \cot^2/C^2$ for C . The present universe may be treated as the 8-dimensional Bianchi-I region when the background is negligible against the observer's critical density:

$$\frac{\kappa |\rho_{\text{bg}}|}{3H_A^2} \rightarrow 24 \left(\frac{H}{H_A} \right)^2 \frac{\cot^2}{C^2} \ll 1 \quad (\text{the validity inequality of the models of section 4.7}).$$

[proved VIII §31]

- ASYMPTOTIC [definition]: $z = \text{Int}_0^{x_0} \tan[6 H y] dy == -\text{Log}[\cos[6 H x_0]]/(6 H)$, and $x_0(z) = \text{ArcCos}[\exp[-6 H z]]/(6 H)$ inverts it
- ASYMPTOTIC [THE RESULT]: in the coordinate z , $g_{zz} == C^2$ exactly and $g_{ii} = A^2 (1 - \exp[-12 H z])^{(-1/6)}$, $g_{hh} = -B^2 (1 - \exp[-12 H z])^{(-1/6)}$: the metric tends to the unwarped Bianchi-I metric
- ASYMPTOTIC [has content]: $\cot[6 H x_0]^2 == \exp[-12 H z]/(1 - \exp[-12 H z])$: every x_0 -term decays like $\exp[-12 H z]$
- ASYMPTOTIC [THE RESULT]: decay rates as $6 H x_0 \rightarrow \pi/2$: $G_{x_0^0_0}/\cot^4 \rightarrow 15 H^2$, $G_{x_0^i_i}/\cot^2 \rightarrow 60 H^2$, $G_{x_0^h_h}/\cot^2 \rightarrow 60 H^2$, $G_{x_0^4_4}/\cot^2 \rightarrow 72 H^2$; $G^4_0/\cot$ and $G^0_4/\cot^3$ are free of x_0
- ASYMPTOTIC [THE RESULT]: in units of $H^2 \cot^2/(\kappa C^2)$: $\rho_{\text{bg}} \rightarrow -72$, $P_{\text{obs}} = P_{\text{hid}} \rightarrow 60$, $P_C/\cot^2 \rightarrow 15$

- ASYMPTOTIC [THE RESULT]: residual drivers $\kappa(P_i - T/6)|_{bg} \rightarrow -12 H^2 \cot^2/C^2$ for A and B, and $-72 H^2 \cot^2/C^2$ for C (if T_{bg} is not counted as a source)
- ASYMPTOTIC [THE RESULT]: validity of 'present universe = Bianchi-I': $\kappa |\rho_{bg}|/(3 H_A^2)$ divided by 24 $(H/H_A)^2 \cot^2/C^2 \rightarrow 1$

What the asymptotic reading neglects [derived here]. The Bianchi-I system below treats the rates H_A, H_B, H_C as free; the (0,4) equation of section 4.3.2 is then satisfied only to the extent that its coefficient $3H \cot(6Hx_0) = 3H e^{-6Hz}/\sqrt{1 - e^{-12Hz}}$ is negligible. In the stabilized model of section 4.7 ($H_B = H_C = 0, H_A > 0$) the bracket is $2H_C - H_A - H_B = -H_A$, so at finite z the equation would demand an x_0 -momentum $\kappa T^4_0 = -3H \cot H_A$, of order $\cot$ — one power of $\cot$ less suppressed than the diagonal terms of order $\cot^2$ in the validity inequality. The author's constant H and the observer's position z along x_0 are not fixed by anything in the notebook, so the size of these terms is a parameter of the reading, not a computed number.

4.3.4 The 8-dimensional Bianchi-I system

In the asymptotic region the primordial field is

$$ds^2 = C^2 dz^2 + A^2(dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 - B^2(dx_5^2 + dx_6^2 + dx_7^2),$$

and its mixed Einstein tensor does **not** depend on whether the hidden directions are timelike or spacelike: flipping the three hidden signs of η leaves every G^μ_ν unchanged. With the source $T^\mu_\nu = \text{diag}(P_C, P_{\text{obs}}, P_{\text{obs}}, P_{\text{obs}}, -\rho, P_{\text{hid}}, P_{\text{hid}}, P_{\text{hid}})$ and $T = -\rho + 3P_{\text{obs}} + 3P_{\text{hid}} + P_C$, the Einstein equations are equivalent to

$$\begin{aligned} \text{constraint: } & 3H_A^2 + 3H_B^2 + 9H_A H_B + 3H_A H_C + 3H_B H_C = \kappa \rho \\ & (-G^4_4 = \tfrac{1}{2}(\Theta^2 - \sum_i H_i^2), \text{ the 21 pairs}), \\ \text{evolution: } & H'_i + H_i \Theta = \kappa(P_i - T/6), \quad i = A, B, C, \quad \Theta = 3H_A + 3H_B + H_C, \end{aligned}$$

the evolution form being the Ricci form $R^i_i = \kappa(T^i_i - T/(D-2))$ with $D-2=6$, and $R^4_4 =$ the sum over the seven directions of $(H'_i + H_i^2)$. [proved VIII §31]

- BIANCHI-I [THE RESULT]: the mixed Einstein tensor does NOT depend on the signs of the hidden directions (timelike x5..x7 or spacelike: identical G^μ_ν)
- BIANCHI-I [THE RESULT]: $-G^4_4 == 3 H_A^2 + 3 H_B^2 + 9 H_A H_B + 3 H_A H_C + 3 H_B H_C == (\Theta^2 - \text{Sum}_i H_i^2)/2$, the sum over the 21 pairs of the seven directions
- BIANCHI-I [has content]: $R^i_i == H'_i + H_i \Theta$ for $i = A$ (x1), B (x5), C (x0), and $R^4_4 == \text{Sum over the seven directions of } (H'_i + H_i^2)$
- BIANCHI-I [THE RESULT]: $G^\mu_\nu == \kappa T^\mu_\nu$ is EQUIVALENT to the constraint sum-over-pairs == $\kappa \rho$ plus the evolution equations $H'_i + H_i \Theta == \kappa (P_i - T/6)$, $i = A, B, C$

(the sign-flipped geometry took 0.680 s).

The same Bianchi-I Einstein tensor was re-computed numerically, from the metric, in notebook 06 (section 4, for test functions A, B, C), and every statement agreed to machine precision [nb06 §4]:

off-diagonal $G^\mu_\nu = 0$	0.00e+00
$-G^4_4 - (3H_A^2 + 3H_B^2 + 9H_A H_B + 3H_A H_C + 3H_B H_C)$	1.63e-16
$R^0_0 - (dH_C/dt + H_C \Theta)$	2.78e-17

<code>R^1_1 - (dH_A/dt + H_A Theta)</code>	<code>5.55e-17</code>
<code>R^5_5 - (dH_B/dt + H_B Theta)</code>	<code>2.43e-17</code>
<code>G^mu_nu(hidden timelike) - G^mu_nu(hidden spacelike)</code>	<code>0.00e+00</code>
<code>div T (nu = 4) + (drho/dt + 3H_A(rho+P_obs) + 3H_B(rho+P_hid) + H_C(rho+P_x0))</code>	<code>2.22e-16</code>

the Bianchi-I Einstein tensor, computed from the metric, is the one the solver integrates

Constraint propagation. With S the pair sum, the evolution equations and the conservation law give $\frac{d}{dt}(S - \kappa\rho) = -2\Theta(S - \kappa\rho)$: the constraint decays forward in time (for $\Theta > 0$) and grows backward [file: `fable-cosmology/rust/fable_fermion/src/models.rs`, module documentation; prose nb06 §4]. The solver therefore monitors the constraint and does not impose it, and it integrates the unstabilized model forward only (section 4.7.11).

4.3.5 The 8-dimensional conservation law for anisotropic pressures

The source must obey the $\nu = 4$ component of $\nabla_\mu T^\mu{}_\nu = 0$:

$$\rho' + 3H_A(\rho + P_{\text{obs}}) + 3H_B(\rho + P_{\text{hid}}) + H_C(\rho + P_C) = 0,$$

and the other seven components vanish identically for a diagonal, x_4 -dependent source. [**proved VIII §31**] BIANCHI-I [THE RESULT]: 8D conservation for anisotropic pressures - `nabla_mu T^mu_4 == -(rho' + 3 H_A (rho + P_obs) + 3 H_B (rho + P_hid) + H_C (rho + P_C))`, and the other seven components vanish identically. Radiation (on the observed sheet: $P_{\text{obs}} = \rho_r/3$, $P_{\text{hid}} = P_C = 0$) obeys it with $\rho_r \propto A^{-4}(B^3C)^{-1}$, and baryons (dust in every direction) with $\rho_b \propto A^{-3}(B^3C)^{-1}$ [file: `fable-cosmology/rust/fable_fermion/src/models.rs`, module documentation]; section 4.5.5 proves it for the Kohn–Sham fable.

4.4 The canonical spin connection of the interacting system: a complete discussion

The spin connection is discussed in full in two parts. The first part is the **canonical** spin connection of the 4+4 pre-universe: the frame it comes from, how it is computed, every non-zero component, its 16×16 spinor form, the gauge-covariant derivative of the 16-component spinor, how it relates to the three bridges of Part IV and to the fable-5.1 (Weitzenböck) bridge of Part V, and what it does and does not do for the complex fermion fable. It is identical for Efforts A and B and is section 3.8 of this document (subsections 3.8.1–3.8.13), which Effort B uses as it stands. The second part, subsections 4.4.1–4.4.9 below, is the spin connection of the **interacting system**: the connection of the generalized warped frame and of the 8-dimensional Bianchi-I frame on which fable acts back on the primordial field (section 4.3), with every non-zero component, its action on the Dirac operator, its Weitzenböck form, the rescaling that removes it, the anticommutator of the rescaled field, its relation to the fundamental symmetry J , and the recovery of the canonical connection. Every object is defined here or in section 3.8, and every result is restated with the place where it is proved.

Throughout, “Section n ” (capital S) means a section of the notebook.

4.4.1 The interacting system: the same solver on two new frames

The spin connection of the frames on which fable acts back (section 4.3) is computed exactly as every connection of Parts III–VI was: the vielbein postulate of subsection 3.8.5, solved by `cfSpinConnection` with the Christoffel symbols of each frame, lowered with `eta4488` by `cfLowerFirstFlat`, checked by the postulate residual (a regression test of the solver) and by antisymmetry (metric compatibility, which has content), and turned into the 16×16 matrices $\Gamma_\mu = \frac{1}{8} \omega_{\mu ab} [T_{16}[a], T_{16}[b]]$ by Section 17’s `cfSpinMatrix` [**prose VIII §32**]. This is Section 32 of the notebook (Input cells 208–210, 26 assertions, 4.694s). The two frames are the warped

frame of section 4.3.1,

$$e_{\text{warped}} = \text{diag}(\tan(6Hx_0)C, AS^{-1/6}, AS^{-1/6}, AS^{-1/6}, 1, BS^{-1/6}, BS^{-1/6}, BS^{-1/6}),$$

$$S = \sin 6Hx_0,$$

and the 8-dimensional Bianchi-I frame $e_{\text{BI}} = \text{diag}(C, A, A, A, 1, B, B, B)$ of section 4.3.4 (the connections took 0.709 s and 0.509 s). **[proved VIII §32]**

- SPIN CONNECTION [solver regression]: the vielbein-postulate residual is zero on the warped and on the Bianchi-I frame
- SPIN CONNECTION [has content]: $\omega_{\mu}[a,b] == -\omega_{\mu}[b,a]$ on both frames (metric compatibility)

4.4.2 Every non-zero component of the connection of the interacting system

On the **warped** frame, with $c = \cos 6Hx_0$, $S = \sin 6Hx_0$ and $' = d/dx_4$, the complete list is thirteen independent components and their antisymmetric partners:

component (both flat indices down, as the array the notebook compares)	value
$\omega[0; 0, 4] = -\omega[0; 4, 0]$	$\tan(6Hx_0)C'$
$\omega[i; 0, i] = -\omega[i; i, 0], i = 1, 2, 3$	$Hc^2A/(CS^{13/6})$
$\omega[i; i, 4] = -\omega[i; 4, i], i = 1, 2, 3$	$A'/S^{1/6}$
$\omega[h; 0, h] = -\omega[h; h, 0], h = 5, 6, 7$	$-Hc^2B/(CS^{13/6})$
$\omega[h; 4, h] = -\omega[h; h, 4], h = 5, 6, 7$	$B'/S^{1/6}$

$1 + 3 + 3 + 3 + 3 = 13$ independent components, 26 non-zero entries of the 512. The first is **new**: the canonical frame ($C = 1$) has no connection along x_0 at all. On the **Bianchi-I** frame only seven survive:

$$\omega[0; 0, 4] = C', \quad \omega[i; i, 4] = A' \quad (i = 1, 2, 3), \quad \omega[h; 4, h] = B' \quad (h = 5, 6, 7).$$

[proved VIII §32]

- SPIN CONNECTION [THE RESULT]: the complete list of non-zero ω_{μ}^{ab} of the WARPED frame is the thirteen stated components (and their antisymmetric partners) - nothing else
- SPIN CONNECTION [THE RESULT]: the complete list for the Bianchi-I frame is $\omega_{\mu 0} = C'$, $\omega_{\mu i} = A'$, $\omega_{\mu h} = B'$ - nothing else
- SPIN CONNECTION [fidelity]: on the canonical member (labelled substitution) the warped connection IS $\omega_{\mu}^{\text{Canonical}}$ of Section 16, and the new component $\omega_{\mu 0}$ vanishes there

Index position. The notebook's Text cell and the two labels write these components as ω_{μ}^{ab} ; the assertion compares them with `cfOmegaWarp = cfLowerFirstFlat[...]`, whose two flat indices are both **down**, and the table above is that array [file: `claude-fable/cells_part8.wl`, the lists `cfOmegaWarpList`, `cfOmegaBIList` and the assertion that uses them]. Raising both flat indices with `eta4488` leaves the $(0, i)$ and $(4, h)$ planes unchanged and flips the sign of the $(i, 4)$, $(0, 4)$ and $(0, h)$ planes [derived here: one index of each of those planes is timelike]. **Recovery** [derived here, and asserted by the fidelity label above]: at $A = e^{-a_4}$, $B = e^{+a_4}$, $C = 1$, $A' = -Ha'_4 e^{-a_4}$ and $B' = +Ha'_4 e^{a_4}$, so $\omega[i; 4, i] = Ha'_4/(e^{\mathcal{A}}s^{1/6})$ and $\omega[h; 4, h] = e^{\mathcal{A}}Ha'_4/s^{1/6}$ with $\mathcal{A} = a_4(Hx_4)$: the 24 non-zero components of subsection 3.8.6.

4.4.3 The spinor connection of the interacting system

With $\Gamma_\mu = \frac{1}{2} \sum_{a<b} \omega_{\mu ab} T_{16}[a] T_{16}[b]$ (subsection 3.8.7), the table gives [derived here]:

$$\begin{aligned}\Gamma_0 &= \frac{1}{2} \tan(6Hx_0) C' T_{16}[0] T_{16}[4], \\ \Gamma_i &= \frac{1}{2} \left[\frac{H c^2 A}{C S^{13/6}} T_{16}[0] T_{16}[i] + \frac{A'}{S^{1/6}} T_{16}[i] T_{16}[4] \right] \quad i = 1, 2, 3, \\ \Gamma_4 &= 0, \\ \Gamma_h &= \frac{1}{2} \left[-\frac{H c^2 B}{C S^{13/6}} T_{16}[0] T_{16}[h] + \frac{B'}{S^{1/6}} T_{16}[4] T_{16}[h] \right] \quad h = 5, 6, 7.\end{aligned}$$

The first and last lines are asserted: $\Gamma_{x_4} = 0$, and Γ_{x_0} is not zero as soon as the x_0 direction is dynamical. Γ_μ rotates the Dirac matrices as ω rotates a vector, and the curved γ^μ are covariantly constant on the warped frame. [proved VIII §32]

- DIRAC [THE RESULT]: `Gamma_x4 == 0`, and `Gamma_x0 == (1/2) Tan[6 H x0] C' (x4) T16[0].T16[4]`, which is NOT zero when `C` varies (witnessed on `C = 1 + x4^2/3`)
- SPIN CONNECTION [has content]: `Gamma_mu` rotates the Dirac matrices as `omega` rotates a vector, `[Gamma_mu, T16[a]] == -omega_mu^a_b T16[b]`, on the warped frame
- SPIN CONNECTION [has content]: the curved `gamma^mu` are covariantly constant on the warped frame, `D_mu gamma^nu == 0`

4.4.4 What the connection of the interacting system does to the Dirac operator

The whole connection enters the Dirac operator as **one** matrix:

$$\begin{aligned}\gamma^\mu \Gamma_\mu &= -\frac{3H \cot^2(6Hx_0)}{C} T_{16}[0] + \frac{1}{2} (3H_A + 3H_B + H_C) \gamma^4, \quad \gamma^4 = T_{16}[4] \quad (\text{warped frame}), \\ \gamma^\mu \Gamma_\mu &= \frac{1}{2} (3H_A + 3H_B + H_C) T_{16}[4] \quad (\text{Bianchi-I frame}).\end{aligned}$$

The first term is the canonical term of subsection 3.8.8 divided by C ; the second is the Hubble (dilution) term of an expanding 7-volume — with the **plus** sign; the opposite sign fails. And $\{\gamma_\mu, \Gamma_\mu\} = 0$ for **each** μ separately on both frames, so the symmetrized Lagrangian of section 4.1.2 is the same with D_μ as with ∂_μ there, for fields of all eight coordinates and every V . [proved VIII §32]

- DIRAC [THE RESULT]: `gamma^mu Gamma_mu == -(3 H Cot^2/C) T16[0] + (1/2) (3 H_A + 3 H_B + H_C) gamma^4` on the warped frame, and `gamma^4 == T16[4]`
- DIRAC [control]: the opposite sign of the Hubble term FAILS (witnessed on `A = 1 + x4/2`, `B = 1 + x4/3`, `C = 1 + x4^2/3`)
- DIRAC [THE RESULT]: on the Bianchi-I frame `gamma^mu Gamma_mu == (1/2) (3 H_A + 3 H_B + H_C) T16[4]`: the pure Hubble term of an expanding 7-volume
- DIRAC [THE RESULT]: `{gamma_mu, Gamma_mu} == 0` for EACH μ separately, on the warped and on the Bianchi-I frame
- DIRAC [THE RESULT]: hence Section 26's symmetrized Lagrangian is the same with `D_mu` as with `d_mu` on the warped frame, for `Psi`, `Psibar` of all eight coordinates (`cfPsiC8`, `cfPsiB8`) and every V

Direction by direction [derived here, from subsection 4.4.3 and the coframe $\gamma^0 = \cot T_{16}[0]/C$, $\gamma^i = S^{1/6} T_{16}[i]/A$, $\gamma^4 = T_{16}[4]$, $\gamma^h = S^{1/6} T_{16}[h]/B$, using $T_{16}[i]^2 = +1$, $T_{16}[h]^2 = -1$ and the anticommutation of distinct T_{16}]:

μ	$\gamma^\mu \Gamma_\mu$ (no sum), warped frame	on the canonical member
0	$\frac{1}{2} H_C T_{16}[4]$	0
$i = 1, 2, 3$	$-\frac{1}{2} (H \cot^2 / C) T_{16}[0] + \frac{1}{2} H_A T_{16}[4]$	$-\frac{1}{2} H \cot^2 T_{16}[0] - \frac{1}{2} H a'_4 T_{16}[4]$
4	0	0
$h = 5, 6, 7$	$-\frac{1}{2} (H \cot^2 / C) T_{16}[0] + \frac{1}{2} H_B T_{16}[4]$	$-\frac{1}{2} H \cot^2 T_{16}[0] + \frac{1}{2} H a'_4 T_{16}[4]$

The sum over the seven directions is the asserted $-(3H \cot^2 / C) T_{16}[0] + \frac{1}{2} \Theta T_{16}[4]$, and the last column is subsection 3.8.8's table.

4.4.5 The Weitzenböck form, and the rescaling that removes the connection

Computed as in Part V — the torsion of the connection in which the frame is parallel, $T^a_{\mu\nu} = \partial_\mu e_\nu^a - \partial_\nu e_\mu^a$, pulled to curved indices and traced, $T_\mu = T^\nu_{\nu\mu}$ — the torsion vector of the warped frame is a **gradient**, and it carries the whole connection term of the Dirac operator:

$$T_\mu = \partial_\mu \ln \frac{\sin(6Hx_0)}{A^3 B^3 C}, \quad \gamma^\mu \Gamma_\mu = -\frac{1}{2} T_\mu \gamma^\mu$$

(on the canonical member it is Part V's torsion vector $\partial_\mu \ln \sin(6Hx_0)$). Hence the rescaling

$$\Psi = \sqrt{\sin 6Hx_0} (A^3 B^3 C)^{-1/2} \chi, \quad \gamma^\mu D_\mu \Psi = \sqrt{\sin 6Hx_0} (A^3 B^3 C)^{-1/2} \gamma^\mu \partial_\mu \chi$$

removes the connection from the Dirac operator **exactly**, for a generic $\chi(x_0, \dots, x_7)$. [**proved VIII §32**]

- WEITZENBOECK [THE RESULT]: the torsion vector of the warped frame is a GRADIENT, $T_\mu == d_\mu \text{Log}[\text{Sin}[6 H x_0]/(A^3 B^3 C)]$; on the canonical member it is Part V's torsionVectorFable51
- WEITZENBOECK [THE RESULT]: $\gamma^\mu \Gamma_\mu == -(1/2) T_\mu \gamma^\mu$ on the warped frame - one vector term, a gradient
- RESCALING [THE RESULT]: $\gamma^\mu D_\mu [\text{Sqrt}[\text{Sin}[6 H x_0]] (A^3 B^3 C)^{(-1/2)} \chi] == \text{Sqrt}[\text{Sin}[6 H x_0]] (A^3 B^3 C)^{(-1/2)} \gamma^\mu d_\mu \chi$ for a GENERIC $\chi(x_0, \dots, x_7)$: the connection is removed exactly

(the torsion took 0.078s). The frame-independent form of subsection 3.8.12.3, $T_\mu = \partial_\mu \ln(e_\mu^\mu / \det e)$ for a diagonal frame, gives the same vector [derived here: $\det e = \tan(6Hx_0) C A^3 B^3 / S$; for $\mu = 0$, $e_0^0 / \det e = S / (A^3 B^3)$, whose x_0 -derivative of the logarithm is $6H \cot(6Hx_0)$; for $\mu = 4$, $e_4^4 = 1$ and $-\partial_4 \ln \det e = -\Theta$, so $-\frac{1}{2} T_4 \gamma^4 = \frac{1}{2} \Theta T_{16}[4]$]. The factor $(A^3 B^3 C)^{-1/2}$ is the square root of the comoving 7-volume: the dilution of the density of an expanding 7-volume.

4.4.6 The anticommutator of the rescaled field is independent of x_4

The anticommutator of section 4.1.4 carries over to the warped family because its derivation is local in time and the lapse is still 1 with $\gamma^4 = T_{16}[4]$: the time-derivative part is $\Psi^\dagger K \partial_4 \Psi$ with $K = (\sqrt{|\det g|} / H) \sigma_{16} T_{16}[4]$, and Section 28's Dirac-bracket result for $K = c \sigma_{16} T_{16}[4]$ with $c = \sqrt{|\det g|} / H$ gives

$$\{\Psi_a(x), \Psi_b^\dagger(y)\} = \frac{H}{\sqrt{|\det g|}} G_{ab} \delta^7(x - y),$$

$$G = -i \sigma_{16} T_{16}[4] = J \quad (\text{the same matrix on every member}),$$

$$\sqrt{|\det g|} = \tan(6Hx_0) C A^3 B^3 / \sin(6Hx_0),$$

$$\{\chi_a(x), \chi_b^\dagger(y)\} = H \cot(6Hx_0) G_{ab} \delta^7(x-y) \quad (\text{independent of } x_4 \text{ and of } A, B, C)$$

— the $\sqrt{|\det g|}$ cancels against the rescaling; on the canonical member this is Section 28's $H \cot(6Hx_0) J$. So the field to quantize on the dynamical frames is χ , whose canonical structure does not change as the universe expands. **[proved VIII §32]**

- RESCALING [has content]: $\det g == (\det e)^2$ and $\det e == \tan[6 H x_0] C A^3 B^3 / \sin[6 H x_0] > 0$, so $\text{Sqrt}[|\det g|] == \tan[6 H x_0] C A^3 B^3 / \sin[6 H x_0]$ on the warped frame
- RESCALING [has content]: on the warped frame $G = -i \sigma_{16} \gamma^4$ is Section 28's cfGK ($= J$) – the same matrix on every member of the family ($\gamma^4 == T_{16}[4]$, lapse 1)
- RESCALING [THE RESULT]: $\{\Psi, \Psi^{\dagger}\} == (H/\text{Sqrt}[|\det g|]) G$ on the warped frame (Section 28's cfAntiPsiPsiDd at $c = \text{Sqrt}[|\det g|]/H$), and for $\chi = \Psi/f$, $f = \text{Sqrt}[\sin] (A^3 B^3 C)^{-1/2}$: $\{\chi, \chi^{\dagger}\} == H \cot[6 H x_0] G$ – the $\text{Sqrt}[\det g]$ cancels, independent of x_4 and of A, B, C

(Two uses of the letter χ must be kept apart in this document: here χ is the rescaled field of the dynamical frame; in sections 4.1.6 and 4.5 $\chi = \Psi/\sqrt{H}$ is the canonically normalized field of the Kohn–Sham gas.)

4.4.7 The fundamental symmetry J and the connection of the interacting system

$J = -i T_{16}[0]T_{16}[1]T_{16}[2]T_{16}[3]T_{16}[4]$ (Hermitian, $J^2 = 1$) commutes with $T_{16}[0..4]$ and anticommutes with $T_{16}[5..7]$. The connection of the interacting system respects this split, direction by direction, on the warped **and** on the Bianchi-I frame:

object	relation to J	on both frames
Γ_0 (now $\frac{1}{2} \tan C' T_{16}[0]T_{16}[4]$, non-zero when C varies)	commutes	both factors in $\{0..4\}$
$\Gamma_1, \Gamma_2, \Gamma_3$	commute	made of $T_{16}[0]T_{16}[i]$, $T_{16}[i]T_{16}[4]$
Γ_4	commutes	zero
$\Gamma_5, \Gamma_6, \Gamma_7$	anticommute , and they are not zero	made of $T_{16}[0]T_{16}[h]$, $T_{16}[4]T_{16}[h]$
each $\gamma^\mu \Gamma_\mu$ (no sum), $h = 5, 6, 7$ included	commutes	each is a combination of $T_{16}[0]$ and $T_{16}[4]$

So every connection term of the admissible-sector Hamiltonian is J -even on the frames of the interacting system, as on the canonical one. **[proved VIII §32]**

- J [definition]: Section 28's $J = -i T_{16}[0] \dots T_{16}[4]$ is Hermitian with $J^2 == 1$; it commutes with $T_{16}[0..4]$ and anticommutes with $T_{16}[5..7]$
- J [THE RESULT]: $[J, \gamma^\mu \Gamma_\mu] == 0$ for EACH μ separately ($h = 5, 6, 7$ included), on the warped and on the Bianchi-I frame
- J [THE RESULT]: $[J, \Gamma_\mu] == 0$ for $\mu = 0..4$ and $\{J, \Gamma_h\} == 0$ for $h = 5, 6, 7$, on the warped and on the Bianchi-I frame
- J [has content]: the anticommuting Γ_h are not zero (witnessed on $A = 1 + x_4/2$, $B = 1 + x_4/3$, $C = 1 + x_4^2/3$), so $\{J, \Gamma_h\} == 0$ is not vacuous

4.4.8 Recovery of the canonical connection, and why Part VI had no dilution

On the canonical member (labelled substitution) Γ_μ is Section 17's `GammaSpinCanonical` in every direction, and $\gamma^\mu \Gamma_\mu$ is $-3H \cot^2 T_{16}[0]$. The Hubble coefficient vanishes there: $3H_A + 3H_B + H_C = -3Ha'_4 + 3Ha'_4 + 0 = 0$ — the 7-volume of the canonical frame is constant in x_4 . That is why Part VI found no dilution ($ds/dx_4 = 0$) on the pre-universe: the dilution term is present in the general frame and cancels exactly on the author's. [**proved VIII §32**]

- RECOVERY [fidelity]: on the canonical member (labelled substitution) `Gamma_mu` IS Section 17's `GammaSpinCanonical` for every `mu`, and `gamma^mu Gamma_mu` IS Section 24's $-3 H \cot^2 T_{16}[0]$
- RECOVERY [THE RESULT]: the Hubble coefficient $3 H_A + 3 H_B + H_C$ vanishes on the canonical member — the canonical 7-volume is constant in x_4 , which is why Part VI had no dilution
- PART VIII [control]: `a4` is STILL UNDEFINED after Section 32, and the scale factors `scA`, `scB`, `scC` have no values either

4.4.9 Status ledger of the connection of the interacting system

statement	status	where
warped and Bianchi-I connections from the vielbein postulate; residual zero; antisymmetric	proved	VIII §32 (SPIN CONNECTION [solver regression], [has content])
the complete lists: 13 independent components (warped), 7 (Bianchi-I); the new $\omega[0;0,4] = \tan C'$	proved	VIII §32 (SPIN CONNECTION [THE RESULT])
the explicit Γ_μ of 4.4.3	derived here; Γ_0 , Γ_4 proved	VIII §32 (the DIRAC assertion on Γ_{x_4} and Γ_{x_0})
$\gamma^\mu \Gamma_\mu = -(3H \cot^2 / C) T_{16}[0] + \frac{1}{2} \Theta \gamma^4$; the sign of the Hubble term	proved (with control)	VIII §32 (DIRAC)
per- μ $\{\gamma_\mu, \Gamma_\mu\} = 0$; symmetrized Lagrangian $D = \partial$	proved	VIII §32 (DIRAC)
the per-direction table of 4.4.4	derived here	—
Weitzenböck gradient form; exact rescaling by $\sqrt{\sin(A^3 B^3 C)}^{-1/2}$	proved	VIII §32 (WEITZENBOECK, RESCALING)
x_4 -independent anticommutator $H \cot(6Hx_0) G$ of χ	proved	VIII §32 (RESCALING [THE RESULT])
the J table on both frames	proved	VIII §32 (J)
recovery of the canonical connection; the Hubble coefficient vanishes on it	proved	VIII §32 (RECOVERY)

4.5 The fable source in the Kohn–Sham ground state

This is Section 33 of the notebook, first half (Input cells 211–214, 25 assertions), with the numerical checks of notebook `05_fermion_fable_quantum_eos.ipynb` and the solver's own statements [file: `fable-cosmology/rust/fable_fermion/src/kohn_sham.rs` and `potentials.rs`, module documentation]. The expectation value $\langle \hat{T}^\mu{}_\nu \rangle$ of section 4.2.1 is taken in the Kohn–Sham ground state of section 4.1.6: a homogeneous Fermi sea at zero temperature, $g = 8$ states per spatial momentum, momenta along the observed sheet only (zero modes along x_0, x_5, x_6, x_7 — the admissible sector of section 4.1.4), in the mean field $m = W'(\sigma)$ of the potential $W(\sigma) := V(H\sigma)$, $\sigma = \langle \bar{\chi} \chi \rangle$, $\chi = \Psi/\sqrt{H}$ [**prose VIII §33**]. In the asymptotic region of section 4.3.3 the metric is 8-dimensional Bianchi-I and this source is diagonal.

4.5.1 Densities per 7-volume, and the gap equation

The closed forms of section 4.1.6 are densities per unit **observed** 3-volume. Spread over the hidden 4-volume they become densities per 7-volume. With the hidden volume factor

$$v = \frac{B^3 C}{(B^3 C)_{\text{today}}} \quad (v = 1 \text{ today in every run; the solver's normalization),$$

the fluid that sources the Einstein equations is **[prose VIII §33]**

$$\begin{aligned} \sigma_7 &= \sigma_{\text{KS}}/v && \text{(the solver calls it `sigma8`:} \\ & && \text{the scalar density per 7-volume),} \\ \text{gap: } m &= W'(\sigma_7), \\ \rho &= \varepsilon_{\text{KS}}/v + W(\sigma_7) - \sigma_7 W'(\sigma_7) && (= 3P_{\text{KS}}/v + W(\sigma_7) \text{ on the gap),} \\ P_{\text{obs}} &= P_{\text{KS}}/v + \sigma_7 W' - W && (\text{along } x_1, x_2, x_3), \\ P_{\text{hid}} &= P_C = \sigma_7 W' - W && (\text{along } x_5, x_6, x_7 \text{ and } x_0: \\ & && \text{the gas has no momentum there).} \end{aligned}$$

At $v = 1$ this is Section 29's mean field; the massless limit is $\sigma \rightarrow 0$, $\varepsilon = 3P = gk_{\text{F}}^4/(8\pi^2)$. The couplings of W run as 4-dimensional couplings, $\lambda_4 = \lambda_8/V_{\text{hid}}$, and $\kappa_{4,0} = \kappa_8/V_{h,0}$ [file: `models.rs`, module documentation]. **[proved VIII §33]**

- KS FLUID [fidelity]: at $v = 1$ the fluid IS Section 29's mean field: $\rho = \text{eps_KS} + W - \sigma W'$, $P_{\text{obs}} = P_{\text{KS}} + \sigma W' - W$, $P_{\text{hid}} = \sigma W' - W$
- KS FLUID [fidelity]: Section 29's identities, re-checked where they are used - $\text{eps} + P == wF \ n$, $\text{eps} - 3 P == m \ \sigma$, $T_s = \text{eps} - m \ \sigma == 3 P$, $d \ \text{eps}/d m == \sigma$, $d \ \text{eps}/d k_F == wF \ dn/d k_F$
- KS FLUID [THE RESULT]: ON THE GAP $m = W'(\sigma_7)$, $\rho == 3 P_{\text{KS}}/v + W(\sigma_7)$ (Section 29's trace identity, per 7-volume), and $P_{\text{obs}} - P_{\text{hid}} == P_{\text{KS}}/v$, for every W
- KS FLUID [has content]: the massless limit $m \rightarrow 0+$: $\sigma_{\text{KS}} \rightarrow 0$ and $\text{eps_KS} = 3 P_{\text{KS}} \rightarrow g \ k_F^4/(8 \ \pi^2)$

Negative m . σ_{KS} is odd in m and ε_{KS} , P_{KS} are even. Under $m \rightarrow -m$, $\sigma \rightarrow -\sigma$ and $W(x) \rightarrow \tilde{W}(x) := W(-x)$ the fluid is unchanged and the gap $m = W'(\sigma_7)$ goes into $-m = \tilde{W}'(-\sigma_7)$; so every statement proved for $m > 0$ and an arbitrary W holds for $m < 0$ as well. **[proved VIII §33]** KS FLUID [THE RESULT]: NEGATIVE m - the fluid is invariant under $\sigma \rightarrow -\sigma$, $W(x) \rightarrow \tilde{W}(x) = W(-x)$, and the gap $m = W'(\sigma_7)$ becomes $-m = \tilde{W}'(-\sigma_7)$; with σ_{KS} odd and eps_KS , P_{KS} even in m (Section 29) the case $m < 0$ reduces to $m > 0$.

Why per 7-volume, and not per 3-volume. The solver's unit test `kohn_sham::tests::per_3_volume_mean_field_violates_8d_conservation` checks that the per-3-volume prescription violates the 8-dimensional conservation law of section 4.3.5 [file: `fable-cosmology/rust/fable_fermion/src/kohn_sham.rs`]; the per-7-volume one obeys it exactly (section 4.5.5).

4.5.2 The Fermi-sea integrals, their stable forms, and their accuracy

The closed forms of section 4.1.6 cancel catastrophically for $x = k_{\text{F}}/|m| \ll 1$ ($\varepsilon \sim mn$, $\sigma \sim n$, $P \sim nk_{\text{F}}^2/(5m)$ are small differences of large terms). The solver uses three regimes, written with

$$S(x) = 2 \int_0^x \frac{t^2}{\sqrt{1+t^2}} dt, \quad E(x) = 8 \int_0^x t^2 \sqrt{1+t^2} dt, \quad Q(x) = 8 \int_0^x \frac{t^4}{\sqrt{1+t^2}} dt,$$

$\sigma = g m^3 S / (4\pi^2)$, $\varepsilon = g m^4 E / (16\pi^2)$, $P = g m^4 Q / (48\pi^2)$ [file: `kohn_sham.rs`, module documentation]:

regime	form used
$x < X_{\text{LO}} = 0.6$	the convergent binomial series (radius 1) in x^2 ; e.g. $\varepsilon = m n \left[1 + \frac{3}{10}x^2 - \frac{3}{56}x^4 + \dots \right]$, $\sigma = \text{sgn}(m) n \left[1 - \frac{3}{10}x^2 + \dots \right]$
$0.6 \leq x \leq X_{\text{HI}} = 100$	the closed forms ($\text{asinh } x = \mathcal{L}$), losing at most about 1.3 digits at $x = 0.6$
$x > 100$	the ultra-relativistic series in $y = m /k_{\text{F}}$, with the logarithm kept exact
$m = 0$ exactly	$\sigma = 0$, $\varepsilon = 3P = g k_{\text{F}}^4 / (8\pi^2)$, $\chi = d\sigma/dm = g k_{\text{F}}^2 / (4\pi^2)$

Notebook 05 compared them with adaptive quadrature from $x = 10^{-8}$ to 10^6 and with the Rust solver [nb05 §5.1]:

```
max relative error of the stable forms against quadrature over 1e-8 <= x <= 1e6: 9.88e-14 (at x = 0.251)
x = 1e-08: stable rel. errors (sigma, eps, P) = 0.0e+00 0.0e+00 0.0e+00 closed forms: 1.5e+00
↳ 2.4e-01 4.1e+16
x = 1e-06: stable rel. errors (sigma, eps, P) = 1.1e-16 1.1e-16 4.4e-16 closed forms: 7.8e-05
↳ 1.8e-06 2.6e+08
x = 0.0001: stable rel. errors (sigma, eps, P) = 0.0e+00 2.2e-16 2.2e-16 closed forms: 7.8e-09
↳ 1.5e-09 1.0e+00
x = 0.01: stable rel. errors (sigma, eps, P) = 2.2e-16 2.2e-16 3.3e-16 closed forms: 4.4e-13
↳ 1.1e-13 5.4e-09
x = 0.251: stable rel. errors (sigma, eps, P) = 4.3e-15 1.8e-15 9.9e-14 closed forms: 4.3e-15
↳ 1.8e-15 9.9e-14
x = 1: stable rel. errors (sigma, eps, P) = 0.0e+00 0.0e+00 2.2e-16 closed forms: 0.0e+00
↳ 0.0e+00 2.2e-16
x = 100: stable rel. errors (sigma, eps, P) = 0.0e+00 1.1e-16 2.2e-16 closed forms: 0.0e+00
↳ 1.1e-16 2.2e-16
x = 1e+06: stable rel. errors (sigma, eps, P) = 1.1e-16 2.2e-16 2.2e-16 closed forms: 1.1e-16
↳ 2.2e-16 2.2e-16
```

and against the Rust solver's `gap` command at fifteen points from $x = 10^{-8}$ to 10^6 : largest relative difference Rust - Python over these 15 points: $3.11e-14$.

(the closed forms are wrong by a factor 4.1×10^{16} in P at $x = 10^{-8}$; the stable forms are exact there). The thermodynamic identities hold to rounding [nb05 §5.3, file: `fable-cosmology/results/nb05_identities.csv`]:

identity	largest residual
$\varepsilon - 3P - m\sigma$ (relative to ε)	1.532e-15
$\varepsilon + P - \omega_{\text{F}} n$ (relative)	5.551e-16
$d\varepsilon/dn _m - \omega_{\text{F}}$ (relative, central differences)	7.439e-11
$d\varepsilon/dm _{k_{\text{F}}} - \sigma$ (relative, central differences)	3.610e-08
$\chi - d\sigma/dm$ (relative)	2.633e-07
$\max \chi/\chi_0$	1.000e+00

and $w_{\text{KS}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ rises from $2.000000e-17$ at $x = 10^{-8}$ ($x^2/5 = 2.000000e-17$) through 0.121969 at $x = 1$ to 0.333333333332667 at $x = 10^6$ ($1/3 - 2/(3x^2)$) [nb05 §5.2]; figures `fable-cosmology/results/nb05_ks_accuracy.png`, `nb05_w_ks.png` (figures 1 and 2).

4.5.3 The gap equation: uniqueness, branch selection, parameter domains

$f(m) = m - W'(\sigma_{\text{KS}}(m, k_{\text{F}})/v)$. Since σ_{KS} is odd and increasing in m with slope $\chi = d\sigma_{\text{KS}}/dm \leq \chi_0 = g k_{\text{F}}^2 / (4\pi^2)$, $f' = 1 - W''\chi/v$: the root is **unique** when $W'' \leq 0$ (attractive, concave W) or when $W''\chi_0/v < 1$; otherwise there can be several roots [prose nb05 §4; file: `kohn_sham.rs`, module documentation]. Every potential states where its roots can lie and whether the root is provably unique there (`Potential::gap_plan` in `potentials.rs`). The adopted parameter

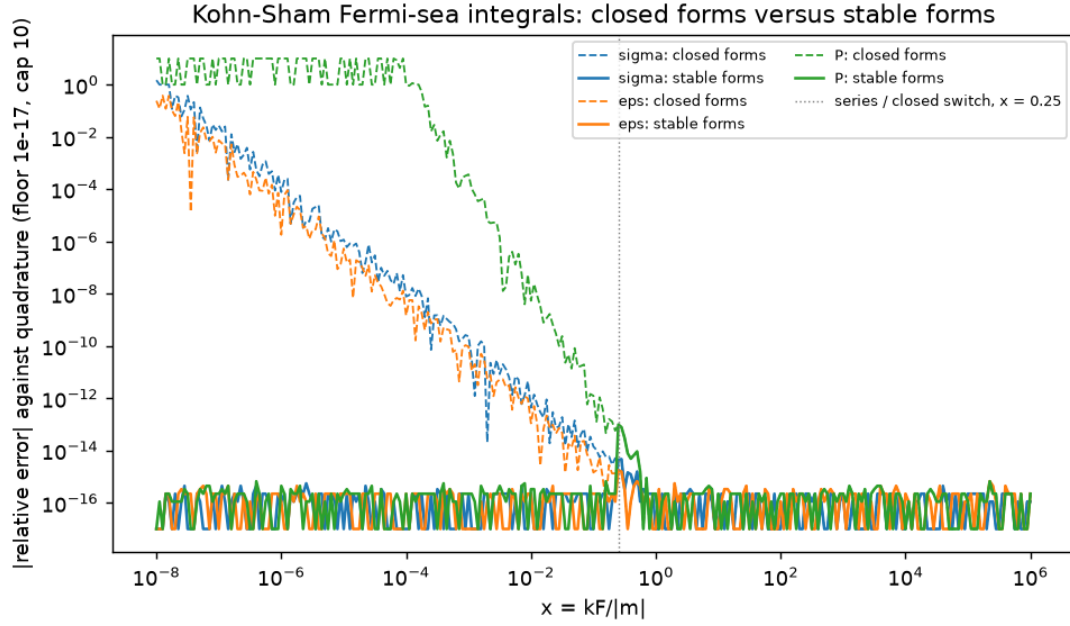


Figure 1: fable-cosmology/results/nb05_ks_accuracy.png (notebook 05, section 5.1): the accuracy of the stable forms of the Fermi-sea integrals against adaptive quadrature, and of the closed forms.

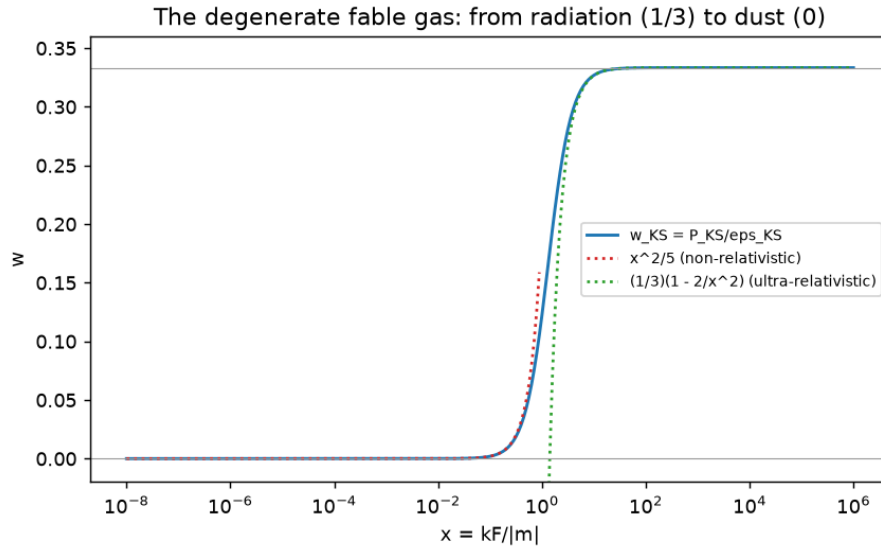


Figure 2: fable-cosmology/results/nb05_w_ks.png (notebook 05, section 5.2): $w_{KS}(x) = P_{KS}/\epsilon_{KS}$ against $x = k_F/|m|$, from 0 (dust) to $1/3$ (radiation).

domains (design review DFT-16): **expdamp** $m_0 > 0$, $s_1 > 0$ (the gap then pins $0 < \sigma < \min(n, s_1)$ strictly); **power** $\lambda > 0$, $0 < \nu < 1$, σ in $(0, n)$ (the bare mass m_0 may come out negative: section 4.7.8); **quadratic** $\lambda < 0$ unique, $\lambda > 0$ all roots enumerated.

Branch selection. When several roots exist, the solver takes the one of **lowest** energy density ρ — not the minimum of the Kohn–Sham energy $E_n[\sigma]$ over σ in $(-n, n)$. The no-sea functional is a minimax: its kinetic part $T_s = \varepsilon_{\text{KS}}(m(\sigma)) - m(\sigma)\sigma$ is **concave** ($d^2T_s/d\sigma^2 = -1/\chi < 0$), so the physical root is a **maximum** of E_n over σ , and its infimum is the variational collapse $E_n \rightarrow -|m_0|n$ as $\sigma \rightarrow -n$ (every particle in the negative-mass band, i.e. in states of the Dirac sea). For attractive W the auxiliary-field **Walecka functional** $\Omega_n(m) = \varepsilon_{\text{KS}}(m) + W(\sigma_m) - m\sigma_m$, $W'(\sigma_m) = m$, is convex and minimized at the gap root, with $\Omega_n(m^*) = \rho$. The numbers are in section 4.6.1.

4.5.4 The energy–momentum tensor’s expectation: ρ , P_{obs} , P_{hid} , and w

On the gap:

$$\begin{aligned}\rho &= 3P_{\text{KS}}/v + W(\sigma_7) && \text{(Section 29’s trace identity } \varepsilon - 3P = m\sigma, \text{ per 7-volume),} \\ P_{\text{obs}} &= P_{\text{KS}}/v + \sigma_7 W' - W, \\ P_{\text{hid}} &= P_C = \sigma_7 W' - W, \\ P_{\text{obs}} - P_{\text{hid}} &= P_{\text{KS}}/v \geq 0 && \text{(for every } W: \text{ the Fermi gas can source} \\ &&& a_4'' < 0, \text{ section 4.2.6),} \\ w_f &= P_{\text{obs}}/\rho.\end{aligned}$$

The split $\rho = \rho_{\text{qp}} + \rho_U$ with $\rho_{\text{qp}} := \varepsilon_{\text{KS}}/v$ (quasiparticles, $w_{\text{qp}} = w_{\text{DM}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ in $[0, 1/3]$) and $\rho_U := W - \sigma_7 W'$ (the mean-field condensate, with $P = -\rho_U$ along **all seven** spatial directions: an 8-dimensional vacuum energy that varies with σ) is a **convention** [prose nb06 §3; file: `kohn_sham.rs`]. Section 4.5.10 defines the dark-matter and dark-energy equations of state built on it.

4.5.5 Theorem: the Kohn–Sham fluid is exactly conserved in 8-dimensional Bianchi-I

The fable number in a comoving 7-volume, $(n_{\text{KS}}/v) A^3 v = n_{\text{KS}} A^3$, is conserved, so $k_F = k_{F0}/A$. Take A , B , C and the mass $m(x_4) > 0$ as **arbitrary** functions and W arbitrary. Then, identically,

$$\rho' + 3H_A(\rho + P_{\text{obs}}) + (3H_B + H_C)(\rho + P_{\text{hid}}) = \sigma_7 \frac{d}{dx_4} [m - W'(\sigma_7)].$$

The right-hand side is the time derivative of the gap equation; on the self-consistent solution it vanishes at every x_4 , so the Kohn–Sham fluid obeys section 4.3.5’s conservation law (with $P_C = P_{\text{hid}}$) **exactly**, for every W — the Einstein equations with this source are consistent. The mechanism: $\rho + P_{\text{obs}} = \omega_F n_{\text{KS}}/v$ and $\rho + P_{\text{hid}} = \varepsilon_{\text{KS}}/v$ hold identically, $d\varepsilon = \omega_F dn + \sigma dm$, and the gap cancels the σdm term against the variation of W . Off the gap the fluid is **not** conserved. [proved VIII §33]

- CONSERVATION [definition]: the fable number per comoving 7-volume, $(n_{\text{KS}}/v) A^3 B^3 C$, is constant when $k_F = k_{F0}/A$
- CONSERVATION [has content]: $\rho + P_{\text{obs}} == \omega_F n_{\text{KS}}/v$ and $\rho + P_{\text{hid}} == \varepsilon_{\text{KS}}/v$ identically (no gap needed)
- CONSERVATION [THE RESULT]: $\rho' + 3 H_A (\rho + P_{\text{obs}}) + (3 H_B + H_C)(\rho + P_{\text{hid}}) == \sigma_7 d/dx_4[m - W'(\sigma_7)]$ IDENTICALLY, for arbitrary A , B , C , $m(x_4) > 0$ and W : the KS fluid is exactly conserved on the gap

- CONSERVATION [control]: without the gap the fluid is NOT conserved - the left side is not identically zero (witnessed on $A = 1 + x^4/2$, $B = 1 + x^4/3$, $C = 1 + x^4/2/3$, $m = 2 + x^4$, $W = \sigma^2$, $k_F = 1$, $g = 8$)

The solver's form of the same law: $d\rho_8 = \omega_F dn/v - (\varepsilon/v) d \ln v$, so at fixed v , $d\rho_8/dn_8 = \omega_F$ [file: `kohn_sham.rs`, module documentation]; its unit test `kohn_sham::tests::thermodynamic_identities_at_the_self_consistent_point` and the end-to-end test `covariant_conservation_along_real_fable8d_runs` check it [file: `kohn_sham.rs`, `tests/cosmology.rs`].

4.5.6 The hidden-sheet driver

With $P_C = P_{\text{hid}}$, section 4.3.4's evolution equations for B and C both read

$$H'_B + H_B \Theta = H'_C + H_C \Theta = \kappa F/6, \quad F = \rho - 3P_{\text{obs}} + 2P_{\text{hid}}.$$

F is the driver of the hidden sheet:

source	F
radiation ($P_{\text{obs}} = \rho/3$, $P_{\text{hid}} = 0$)	0
dust	ρ
an 8-dimensional vacuum energy (all seven pressures $-\rho$; a bare V_0 , the condensate ρ_U)	2ρ
the Kohn–Sham fable, on the gap	$2W - \sigma_7 W'$ (off the gap: $\sigma_7(m - W') + 2W - \sigma_7 W'$)
the author's mass term $W = m_0 \sigma$	$m_0 \sigma_7$ (like dust)
$W = V_0 + m_0 \sigma$	$2V_0 + m_0 \sigma_7$

[proved VIII §33]

- DRIVER [THE RESULT]: with $P_C = P_{\text{hid}}$, $\kappa(P_{\text{hid}} - T/6) == \kappa(P_C - T/6) == \kappa F/6$, $F = \rho - 3P_{\text{obs}} + 2P_{\text{hid}}$: the right-hand side of the B and C equations
- DRIVER [has content]: $F == 0$ for radiation, $F == \rho$ for dust, $F == 2\rho$ for an 8D vacuum
- DRIVER [THE RESULT]: for the KS fluid $F == \sigma_7(m - W') + 2W - \sigma_7 W'$ identically, i.e. $F == 2W - \sigma_7 W'$ on the gap; the mass term gives $F == m_0 \sigma_7$ (dust-like), $W = V_0 + m_0 \sigma$ gives $2V_0 + m_0 \sigma_7$

4.5.7 Theorem: a frozen hidden sheet makes the fable radiation

Given $H_B = H_C = 0$ at one time, the hidden sheet stays **frozen** for all time iff $F = 0$ along the history; for the Kohn–Sham fluid, for all σ_7 the history sweeps, iff $\sigma W' = 2W$, i.e. $W = c\sigma^2$. The gap is then $m = 2c\sigma_7$, and $m = 0$ is always a root. For $c < 0$ it is the only one ($m\sigma_{\text{KS}} > 0$ for $m \neq 0$, so a root $m \neq 0$ would give $m^2 = 2c(m\sigma_{\text{KS}})/v < 0$). For $c > 0$ a non-trivial root exists iff $cgk_F^2/(2\pi^2v) > 1$ (σ_{KS}/m decreases strictly, from $gk_F^2/(4\pi^2)$ at $m \rightarrow 0$ to 0 at $m \rightarrow \infty$), and it has **higher** energy: on it $\rho = (\varepsilon_{\text{KS}}(m) - m\sigma_{\text{KS}}(m)/2)/v$, and $\Delta(m) = \varepsilon_{\text{KS}}(m) - m\sigma_{\text{KS}}/2 - \varepsilon_{\text{KS}}(0)$ vanishes at $m = 0^+$ with

$$\frac{d\Delta}{dm} = \frac{gm^3}{4\pi^2} \left(\mathcal{L} - \frac{k_F}{\omega_F} \right) > 0$$

($\mathcal{L} - k_F/\omega_F = \varphi(k_F/m)$, $\varphi(x) = \ln(x + \sqrt{1+x^2}) - x/\sqrt{1+x^2}$, which vanishes at 0 and has derivative $x^2/(1+x^2)^{3/2} > 0$). So the ground state of a frozen-compatible fable is the $m = 0$

branch, on which $\rho = \varepsilon_{\text{KS}}(k_F, 0)/v$, $P_{\text{obs}} = \rho/3$, $P_{\text{hid}} = 0$: **radiation**. A Kohn–Sham fable compatible with a frozen hidden sheet is neither dark matter nor dark energy. [proved VIII §33]

- FROZEN [THE RESULT]: $2W - \sigma W' = 0$ for all σ iff $W = c\sigma^2$ (the general solution of $\sigma W' = 2W$)
- FROZEN [has content]: $m\sigma_{\text{KS}} > 0$ for $m > 0$ (zero at $k_F = 0$, k_F -derivative $g k_F^2 m^2/(2\pi^2 w_F) > 0$; even in m , so also for $m < 0$); for $c < 0$ a root $m \neq 0$ of $m = 2c\sigma_{\text{KS}}/v$ would give $m^2 = 2c(m\sigma_{\text{KS}})/v < 0$: only $m = 0$
- FROZEN [has content]: $\phi(x) = \text{Log}[x + \text{Sqrt}[1 + x^2]] - x/\text{Sqrt}[1 + x^2]$ is 0 at $x = 0$ and increasing (derivative $x^2/(1 + x^2)^{(3/2)}$), hence positive for $x > 0$; and $L - k_F/w_F = \phi(k_F/m)$
- FROZEN [THE RESULT]: for $c > 0$, σ_{KS}/m decreases strictly ($d/dm = -(g m/(2\pi^2))(L - k_F/w_F)$) from $g k_F^2/(4\pi^2)$ at $m \rightarrow 0$ to 0 at $m \rightarrow \infty$: a non-trivial root of $1 = 2c(\sigma_{\text{KS}}/m)/v$ exists iff $c g k_F^2/(2\pi^2 v) > 1$
- FROZEN [THE RESULT]: on the non-trivial root ($m = 2c\sigma_{\text{KS}}$) $\rho = (\varepsilon_{\text{KS}} - m\sigma_{\text{KS}}/2)/v$, and $\Delta = \varepsilon_{\text{KS}}(m) - m\sigma_{\text{KS}}/2 - \varepsilon_{\text{KS}}(0)$ has $\Delta(0+) = 0$ and $d\Delta/dm = (g m^3/(4\pi^2))(L - k_F/w_F) > 0$: the $m = 0$ branch has LOWER ρ
- FROZEN [has content]: witness numbers - at $g = 8$, $v = 1$, $c = 15$, $k_F = 1$ the non-trivial root m^* exists (gap residual $< 10^{-20}$) and $\rho(m = 0) < \rho(m^*)$
- FROZEN [THE RESULT]: on the $m = 0$ branch the frozen-compatible fable is RADIATION: $\rho = \varepsilon_{\text{KS}}(k_F, 0)/v$, $P_{\text{obs}} = \rho/3$, $P_{\text{hid}} = 0$

The witness the run prints [displayed VIII §33], `claude-fable/run_fermion_fable_part8.log`:

```
frozen-compatible witness (g = 8, v = 1, c = 15, kF = 1): non-trivial root m* =
↪ 3.97853200077825223420767254928521681893`12.; rho(m = 0) =
↪ 0.10132118364233777144387946320972763891`12. < rho(m*) =
↪ 0.28374208489508058306307225309709278221`12.
```

4.5.8 The classical limit, and the sign rule

The classical limit is the **non-relativistic** limit $k_F \ll |m|$ (not a coherent zero mode: the Pauli principle forbids that). Exactly, on the gap, with $s = H\sigma_7$ and $V(s) := W(s/H)$:

$$\rho = V(s) + 3P_{\text{KS}}/v, \quad P_{\text{obs}} = sV'(s) - V(s) + P_{\text{KS}}/v, \quad P_{\text{hid}} = sV'(s) - V(s),$$

and as $k_F/m \rightarrow 0$ ($m > 0$), $\sigma_{\text{KS}}/n_{\text{KS}} = 1 - \frac{3}{10} k_F^2/m^2 + \dots$, $(\varepsilon_{\text{KS}} - n_{\text{KS}}m)/(n_{\text{KS}}k_F^2/m) \rightarrow 3/10$, $P_{\text{KS}}/(n_{\text{KS}}k_F^2/m) \rightarrow 1/5$; $P_{\text{KS}}/\rho \rightarrow 0$ and the quantized fluid reduces to Part VI's $\rho = V(s)$, $P = sV'(s) - V(s)$ (its $K_h = 0$ branch); P_{hid} is exactly Part VI's on-shell pressure. With $s_0 = n_{\text{KS}}/v$, exactly on the gap,

$$\rho - W(s_0) = (\varepsilon - nm)/v - [W(s_0) - W(\sigma_7) - W'(\sigma_7)(s_0 - \sigma_7)] :$$

a second-order Taylor remainder in $s_0 - \sigma_7$. **The sign rule:** $m\sigma_{\text{KS}} \geq 0$, so on every self-consistent solution $\text{sign}(\sigma_7) = \text{sign}(m) = \text{sign}(W'(\sigma_7))$; for Part VI's mass term $W = -2M\sigma$ the gap is $m = -2M$ and $\rho \rightarrow 2|M|n/v > 0$ — the Part VI branch $Ms > 0$, of negative energy, is not a self-consistent state. [proved VIII §33]

- CLASSICAL LIMIT [THE RESULT]: exactly, on the gap and with $s = H \sigma_7$, $V(s) = W(s/H)$: $\rho == V(s) + 3 P_{KS}/v$, $P_{obs} == s V'(s) - V(s) + P_{KS}/v$, $P_{hid} == s V'(s) - V(s)$
- CLASSICAL LIMIT [fidelity]: Section 24's on-shell P_1 (cfP1Fable, re-checked here) is $s V'(s) - V(s)$; evaluated at $s = H \sigma_7$ with $V(s) = W(s/H)$ it IS the KS $P_{hid} = \sigma_7 W' - W$
- CLASSICAL LIMIT [THE RESULT]: NR series ($kF/m \rightarrow 0$, $m > 0$): $\sigma_{KS}/n_{KS} = 1 - (3/10) kF^2/m^2 + \dots$, $(\epsilon_{KS} - n_{KS} m)/(n_{KS} kF^2/m) \rightarrow 3/10$, $P_{KS}/(n_{KS} kF^2/m) \rightarrow 1/5$
- CLASSICAL LIMIT [has content]: the design's form, exactly: on the gap ($m = W'(\sigma_7) > 0$) $\rho - W(s_0) == (\epsilon - n m)/v - [W(s_0) - W(\sigma_7) - W'(\sigma_7)(s_0 - \sigma_7)]$, $s_0 = n/v$ - a second-order Taylor remainder in $s_0 - \sigma_7$
- SIGN RULE [THE RESULT]: $m \sigma_{KS} \geq 0$ gives $\text{sign}(\sigma_7) == \text{sign}(m) == \text{sign}(W'(\sigma_7))$; for Section 24's mass term $W = -2 M \sigma$ the gap is $m = -2 M$ and $W(\text{sign}(m) n/v) = -2 M \text{sign}(-2 M) n/v == 2 |M| n/v > 0$

Notebook 05 checks the series at $x = 0.001$ and the sign at $m = -2$ [nb05 §5.4]: $(\sigma/n - 1)/x^2 = -0.300000$ ($-3/10$), $(\epsilon/(|m| n) - 1)/x^2 = +0.300000$ ($+3/10$), $P/(n kF^2/|m|) = 0.200000$ ($1/5$); $m = -2$: $\sigma/n = -0.999999700$: $\text{sgn}(\sigma) = \text{sgn}(m)$.

4.5.9 $w_f \geq -1$: no phantom crossing

$$\rho + P_{obs} = \omega_F n_{KS}/v \geq 0 \quad \text{so} \quad w_f \geq -1 \quad \text{wherever } \rho > 0.$$

The quantized fermion fable **cannot cross the phantom divide**; the classical crossing of Part VI (section 3.1.12.7, at $V'(s^*) = 0$) does not survive quantization. [proved VIII §33] PHANTOM [THE RESULT]: $\rho + P_{obs} == w_F n_{KS}/v \geq 0$, so $w_f \geq -1$ wherever $\rho > 0$ - the quantized fable cannot cross the phantom divide. Notebook 05 compared the classical $w_{cl} = s V'/V - 1$ of five of Part VI's potential shapes with the quantum $w = P_{obs}/\rho$ of the lowest-energy gap root, over densities $10^{-3} \leq n \leq 30$ (in units where the classical s equals n), for $W = \mathcal{A} V$ with $\mathcal{A} = 10^4, 10^2, 1$ — a large $\mathcal{A}$ makes the quasiparticles heavy ($m = \mathcal{A} V'$), which is the classical limit (915 points) [nb05 §5.4, file: fable-cosmology/results/nb05_classical_vs_quantum.csv, figure nb05_classical_vs_quantum.png] (figure 3):

potential ($V(s)$)	classical min w_{cl}	quantum min w ($\mathcal{A} = 10^4, 100, 1$)
mass (s)	+0.0000, never crosses -1	+0.00000, +0.00000, +0.00741
lambda-mass ($0.7 + 0.3 s$)	-0.9996, never crosses -1	-0.99957, -0.99957, -0.99949
power ($s + s^{0.236}$)	-0.7601, never crosses -1	-0.76012, -0.76012, -0.76012
expdamp ($1.566 + 0.839 s e^{-s/2.21}$)	-1.3020, crosses -1 at $n = 2.2047$	-0.99952, -0.99947, -0.99945
lorentz ($1 + s/(1 + (s/1.5)^2)$)	-1.2481, crosses -1 at $n = 1.4999$	-0.99900, -0.99900, -0.99898

and it printed the quantum $w \geq -1$ at every density, every $\mathcal{A}$ and every potential; the classical expdamp and lorentz cross -1 . For expdamp the gap keeps

$0 < \sigma < \min(n, s_1)$ strictly, with $\sigma \rightarrow s_1$ and $m \rightarrow 0$ only as $n \rightarrow \infty$ (solver test `kohn_sham::tests::expdamp_pins_sigma_below_s1_and_mass_goes_to_zero_at_high_density`).

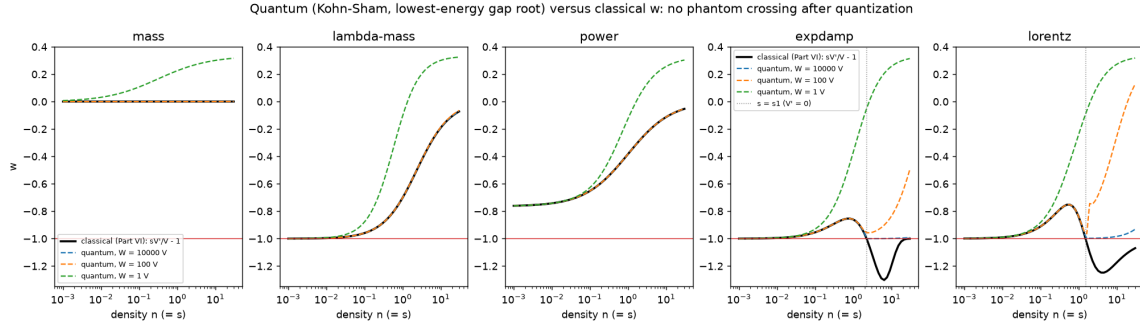


Figure 3: fable-cosmology/results/nb05_classical_vs_quantum.png (notebook 05, section 5.4): the classical $w_{\text{cl}} = s V'/V - 1$ against the quantum $w = P_{\text{obs}}/\rho$ of the lowest-energy gap root.

4.5.10 The equations of state an observer can define

Four definitions are used, and they must not be confused [file: `models.rs`, documentation of the output columns; prose nb06 §5]:

1. $w_f = P_{\text{obs},f}/\rho_f$ — the intrinsic equation of state of the whole fable fluid;
2. $w_{\text{DM}} = w_{\text{qp}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ — the intrinsic equation of state of the quasiparticles, in $[0, 1/3]$;
3. $w_{\text{DM,eff}} = w_{\text{DM}} - \sigma_{\text{KS}}(dm/dN)/(3\varepsilon_{\text{KS}})$ — the effective equation of state of the quasiparticles when the mass varies, which exchange energy with the condensate at the rate $Q = \sigma_8 dm/dt$ (the exact exchange; $n dm/dt$ is its non-relativistic limit);
4. $w_{\text{DE,inf}}$ — the dark energy an observer **infers** by fitting cold dark matter plus dark energy: $\rho_{\text{DE,inf}} = H_A^2 - \Omega_{r0}A^{-4} - \Omega_{b0}A^{-3} - m_{\text{today}} n_{\text{today}} A^{-3}$ (radiation, baryons, and cold dark matter with **today's** mass $m_{\text{eff}}(A = 1)$ subtracted), $w_{\text{DE,inf}} = -1 - \frac{1}{3} d \ln \rho_{\text{DE,inf}} / d \ln A$.

The adiabatic sound speed of the fable is $c_s^2 = (dP_{\text{obs},f}/dN)/(d\rho_f/dN)$; $c_s^2 < 0$ is an adiabatic instability. The split into dark matter and dark energy is a convention (section 4.5.4): the condensate is an 8-dimensional vacuum energy.

4.5.11 The vacuum energy of mass-varying potentials

For a constant mass the Dirac-sea energy is a constant; it is absorbed into V and appears as a bare cosmological constant. **The cosmological-constant problem is not solved here.** For a mass that varies along the history the sea energy cannot be absorbed silently (design review DFT-4, adopted). The renormalized one-loop Dirac-sea energy that the no-sea functional drops (relativistic Hartree approximation, Chin 1977), per 3-volume, with reference mass M and counterterms through m^4 , is

$$\Delta E_{\text{vac}}(m) = -\frac{g}{16\pi^2} \left[m^4 \ln \frac{m}{M} + M^3(M-m) - \frac{7}{2} M^2(M-m)^2 + \frac{13}{3} M(M-m)^3 - \frac{25}{12} (M-m)^4 \right]$$

[file: `kohn_sham.rs`, function `vacuum_energy`; its test `kohn_sham::tests::vacuum_energy_vanishes_to_fifth_order_at_the_reference_mass`]. At $m = 0$ the bracket is $M^4(1 - \frac{7}{2} + \frac{13}{3} - \frac{25}{12}) = -M^4/4$, so $\Delta E_{\text{vac}}(0) - \Delta E_{\text{vac}}(M) = g M^4/(64\pi^2)$ [derived here]. With $g = 8$ and $E_c = \rho_{c0}^{1/4} = 2.46261 \times 10^{-3} \text{ eV}$ (asserted: fable4d RUNS [definition]: the units computed from CODATA 2018 - $E_c = \rho_{c0}^{1/4} == 2.46261\text{e-3 eV}$, $\Omega_{r0} ==$

9.2096e-5, $\Omega_{\text{b0}} = 0.02237/h^2 = 0.049243$ (to the quoted digits)), a mass variation of order m changes the vacuum energy by

$$\frac{g m^4 / (64 \pi^2)}{\rho_{c0}} = 3.44, \quad 3.44 \times 10^4, \quad 3.44 \times 10^8, \quad 3.44 \times 10^{12}$$

at $m = 0.01, 0.1, 1, 10 \text{ eV}$ (it scales as m^4)

[derived here: $g/(64\pi^2) = 0.012665$, $(m/E_c)^4 = 271.9$ at $m = 0.01 \text{ eV}$]. So for $m \gtrsim 10 \text{ meV}$ it exceeds the critical density; notebook 05 prints the exact threshold [nb05 §5.8]: $\Delta E_{\text{vac}}(0; M) = g M^4 / (64 \pi^2)$: check 1.000000000000003; $g M^4 / (64 \pi^2)$ exceeds ρ_{c0} for $M > (64 \pi^2 / g)^{1/4}$ $E_c = 7.341 \text{ meV}$, and at $M = 1 \text{ eV}$ it gives $|\Delta E_{\text{vac}}|/\rho_{c0} = 2.760\text{e-}02, 2.802\text{e+}03, 9.454\text{e+}06, 3.444\text{e+}08$ at $m/M = 0.99, 0.9, 0.5, 0$ (the last is the 3.44×10^8 above); at $M = 1000 \text{ eV}$, $3.444\text{e+}20$ at $m = 0$; the fifth-order zero at $m = M$ is confirmed by the ratio 31.9470 ($2^5 = 32$).

The design review's refinement (DFT-4, adopted): the sea energy **can** be absorbed formally by reading W as the fully renormalized effective potential, $\Omega(m) = \varepsilon_{\text{KS}} + \Delta E_{\text{vac}} + F = \text{no-sea}$ with $\tilde{F} = F + \Delta E_{\text{vac}}$; that reading is a **fine-tuning** of the bare potential against the sea energy. The vacuum also adds $\sigma_{\text{vac}} = d\Delta E_{\text{vac}}/dm$ to $\langle \bar{\Psi}\Psi \rangle$. The two options are (a) to include ΔE_{vac} in ρ and in the gap equation, or (b) to declare W the renormalized potential. **The runs use (b)**, and they print $\Delta E_{\text{vac}}(m(t)) - \Delta E_{\text{vac}}(m_{\text{today}})$ in units of ρ_{c0} (the solver's column `vac_over_rho_c`, $\Delta E_{\text{vac}}(m_{\text{eff}}; M = m_{\text{eff}}(A = 1))/v$). Along the power run $\nu = 0.236$, $m_{\text{today}} = 30 \text{ eV}$ notebook 05 found [nb05 §5.8] (figure 4):

```
power nu = 0.236, m_today = 30 eV: m_eff/m_today runs from 0.2007 (a = 1e-12) to 1; max |solver
↪ vac_over_rho_c - formula| / (g M^4/(64 pi^2)) = 5.4e-15
|Delta E_vac(m(t)) - Delta E_vac(m_today)| / rho_total: max over a >= 0.3 = 1.314e+13, at a = 0.5:
↪ 1.149e+13
```

Section 4.7.6 gives the ratio for every run.

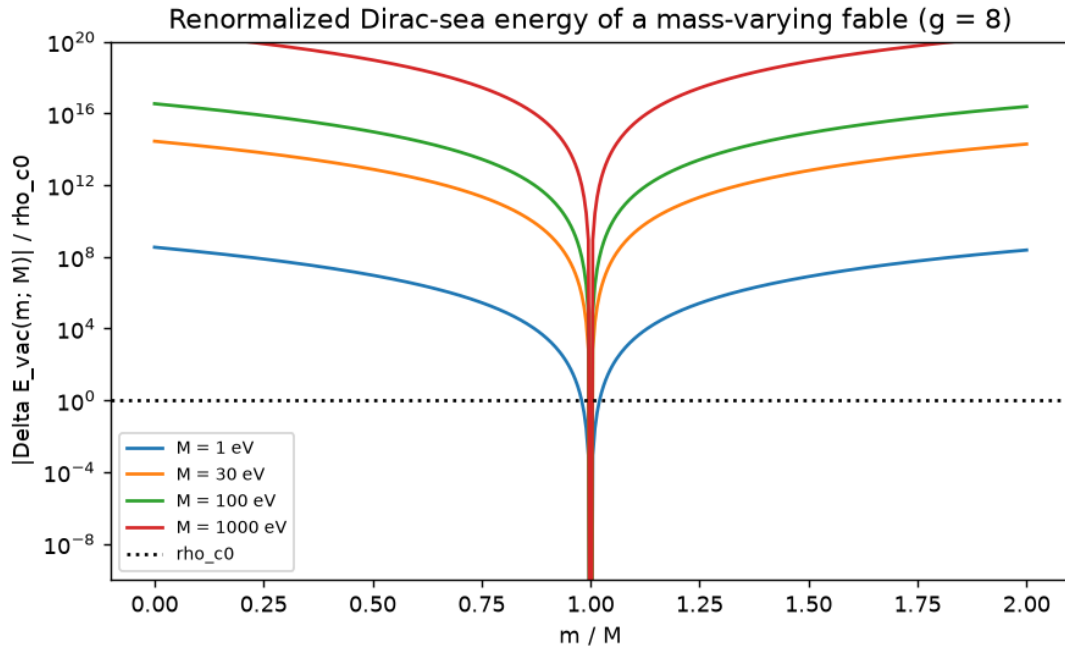


Figure 4: `fable-cosmology/results/nb05_vacuum_energy.png` (notebook 05, section 5.8): the renormalized Dirac-sea energy of a mass-varying fable.

4.5.12 Exchange and correlation

The Kohn–Sham functional of section 4.6.1 has E_{xc} neglected (the Hartree mean field). For the author’s mass term (W linear) the action is bilinear, the theory is free, and the mean field is **exact** ($E_{xc} = 0$). For a nonlinear W the design review (DFT-14) estimated that the Fock exchange of the contact interaction is suppressed relative to the Hartree term by $\sim 1/g = 1/8$ only in the non-relativistic regime $k_F \ll m$, and is of order one or larger for $k_F \gtrsim m$; in the relativistic era its neglect is controlled only by the smallness of the interaction energy relative to $3P_{KS}$. That ratio was not computed in the runs; the neglect of E_{xc} is a stated approximation of every mass-varying result in this document.

4.6 Density-functional theory: the fable ground and first excited states

The author asked to “use some of the relevant ideas from Density Functional Theory to obtain the fable ground and first excited states”. Two systems are treated: the **homogeneous** fable gas of the present universe (sections 4.6.1–4.6.2), and the fable **along the hidden coordinate** x_0 of the primordial field, where the wall $x_0 = 0$ binds discrete states (sections 4.6.3–4.6.7). The homogeneous numbers are printed by notebook 05 [nb05 §5.5, §5.6]; the wall-state numbers are quoted from `fable-cosmology/fermion/waveguide_REPORT.txt` [WG n] and its logs, and notebook 07_fable_dft_states.ipynb, executed and committed in aca5102, reproduces them (a number it prints again carries the mark [nb07 §n], with the notebook’s section). The development logs that the report names are files under `fable-cosmology/fermion/dev/`, a folder the repository ignores; the report, which is tracked, quotes them.

4.6.1 The functional, and the homogeneous Kohn–Sham ground state

The functional. For the conserved charge n and the scalar density σ (the relativistic Hohenberg–Kohn–Sham scheme; the Serot–Walecka mean field is its Hartree limit), the energy of the gas at fixed n is

$$E_n[\sigma] = T_s(\sigma; n) + W(\sigma) + E_{xc}, \quad T_s = \varepsilon_{KS}(m(\sigma)) - m(\sigma)\sigma, \quad \sigma_{KS}(m(\sigma), k_F) = \sigma, \\ \frac{dE_n}{d\sigma} = W'(\sigma) - m(\sigma) \quad (\text{stationary exactly at the gap roots}), \quad \frac{d^2T_s}{d\sigma^2} = -\frac{1}{\chi} < 0,$$

with E_{xc} neglected (section 4.5.12) and the Dirac sea excluded (the no-sea functional; its energy is section 4.5.11). The Kohn–Sham Slater determinant is the filled Dirac sea (renormalized away) plus the Fermi sea $|k| < k_F$ of the lowest positive band. **The ground state is the stationary point of this minimax functional** — a maximum in σ and a minimum in the occupations — and **never** its minimum over σ [prose nb05 §5.6; solver: design departure D1 of the development report, and the unit tests `kohn_sham::tests::ks_functional_stationary_point_is_a_maximum_for_the_mass_term`, `walecka_functional_is_minimized_at_the_gap_root`, `multiple_gap_roots_are_found_and_the_lowest_energy_one_is_taken` in `kohn_sham.rs`]. For linear W the theory is free, $m = m_0$ exactly, and the Kohn–Sham state is the exact ground state.

The minimax, in numbers [nb05 §5.6, file: `fable-cosmology/results/nb05_minimax.csv`, figure `nb05_minimax.png`] (figure 5):

```
mass term W = 2 sigma, kF = 1.5: n = 0.4559453264, the gap root sigma* = 0.3961298703 (sigma*/n =
↪ 0.868810)
E_n(sigma*) = 1.052962422118 = rho = eps_KS(m0) = 1.052962422118; max of E_n over the grid =
↪ 1.052959281278
E_n/n at sigma = -(1 - 1e-7) n: -1.999480 (the collapse value -|m0| = -2.0)

attractive W = sigma - 10 sigma^2, kF = 1: unique gap root m* = 0.2112489356, rho = 0.1211932853;
↪ Omega_n(m*) = 0.121193285306
```

```
Omega_n on the grid: minimum 0.121193451922 at m = 0.2100; second differences all positive (convex):
→ True
```

The root of the mass term is the **maximum** of E_n (the grid maximum lies just below it), and E_n/n falls to the collapse value $-|m_0|$ as $\sigma \rightarrow -n$. For the attractive W the convex Walecka functional $\Omega_n(m) = \varepsilon_{\text{KS}}(m) + W(\sigma_m) - m\sigma_m$ has its minimum at the gap root, equal to ρ .

Several roots, and a first-order transition. For the repulsive $W = m_0\sigma + (\lambda/2)\sigma^2$ with $(m_0, \lambda) = (0.05, 20)$ at $k_F = 1$, $W''\chi_0 = \lambda g k_F^2/(4\pi^2) = 4.0528 > 1$ and there are three roots; the Rust solver's `gap` command and the Python solver agree [nb05 §5.5]:

```
# potential Quadratic { v0: 0.0, m0: 0.05, lam: 20.0 }, kF = 1e0, v = 1e0, n8 = 1.3509491152311703e-1,
→ plan = Scan { lo: -2.651898230462341, hi: 2.7518982304623405 }
m,sigma8,rho8,P_obs,P_hid,gap_residual
-2.535353916022531e0,-1.292676958011266e-1,1.909782056219713e-1,1.772147779888931e-1,1.671013717773260e-1
→ -1,8.506e-17
-1.640336879334347e-2,-3.320168439667173e-3,1.012381942988485e-1,3.387489103006236e-2,1.102351846776195-1
→ e-4,-1.898e-17
2.644093986040535e0,1.297046993020268e-1,2.039272771420881e-1,1.779694075325236e-1,1.682330902102918e-1-1
→ ,0.000e0
# selected (Lowest): m = -1.640336879334347e-2, rho8 = 1.012381942988485e-1, roots = 3
```

with $(\rho + P)/(\omega_F n) = 1.000000000000$ on every root (so $w \geq -1$ on every branch), and uniqueness criterion $\text{lam } \chi_0 < 1$ holds for $k_F < 0.4967$; three roots first appear at $k_F = 0.625$ on this grid; the ground state jumps from $m = +0.4756$ ($k_F = 0.600$) to $m = -0.0998$ ($k_F = 0.625$) [nb05 §5.5, figure nb05_gap_roots.png] (figure 6): the ground state jumps between branches — a first-order transition. That is why the solver refuses a repulsive quadratic run (section 4.7.11): energy conservation across the jump would need a Maxwell construction, which it does not implement.

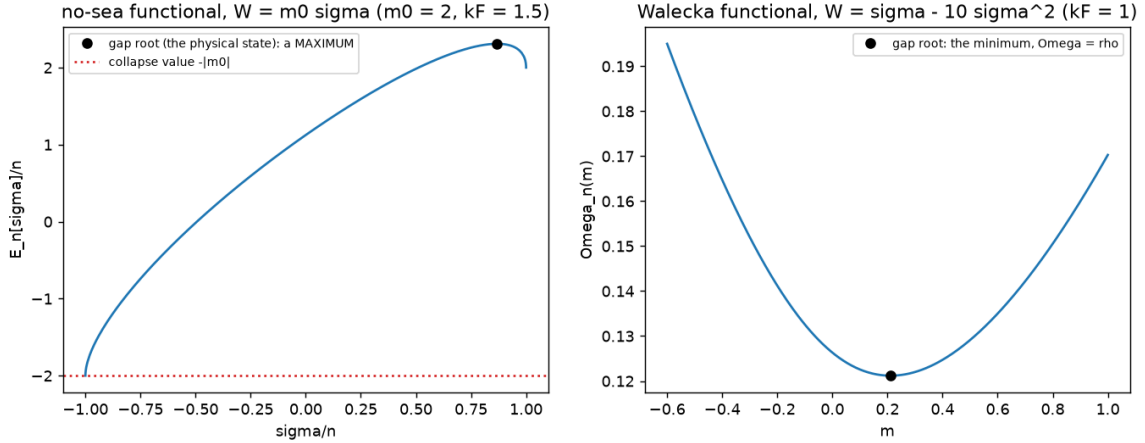


Figure 5: fable-cosmology/results/nb05_minimax.png (notebook 05, section 5.6): the no-sea Kohn–Sham functional $E_n[\sigma]$ of the mass term (its gap root is a maximum) and the Walecka functional $\Omega_n(m)$ of an attractive W (its gap root is a minimum).

4.6.2 The first excited states of the homogeneous gas

The design review (DFT-15, adopted) states them; they are not assertions of the notebook, and the derivation is short [derived here]:

- **Particle–hole excitations** (a quasiparticle moved from just inside to just outside the Fermi surface) are **gapless**: $\Delta E \rightarrow 0$ as the momentum transfer goes to zero. The lowest excited state of the gas is degenerate with the ground state in the thermodynamic limit, and it has no effect on w .

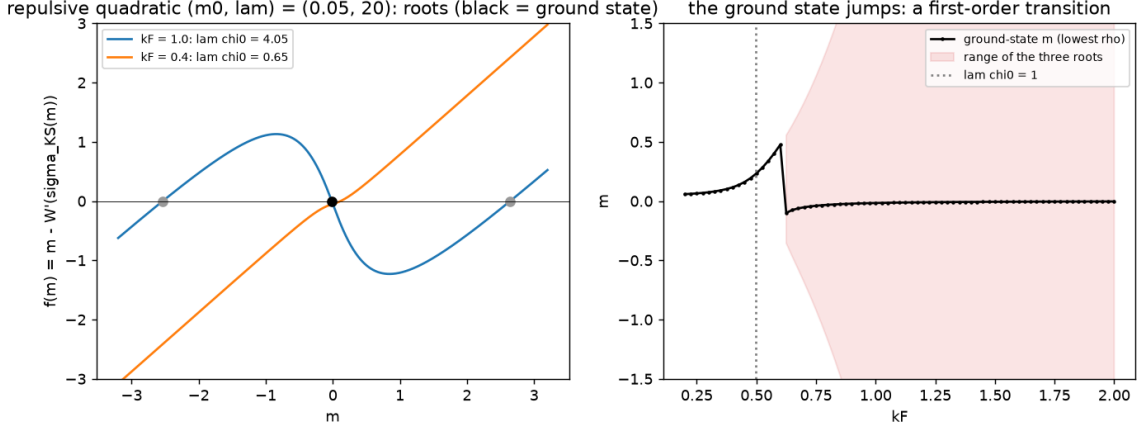


Figure 6: fable-cosmology/results/nb05_gap_roots.png (notebook 05, section 5.5): the roots of the gap equation of the repulsive quadratic W against k_F , and the jump of the ground state between branches.

- **Pair excitations** create a particle above the Fermi sea (energy $\geq \omega_F$, momentum $|k_1| \geq k_F$) and a hole in the Dirac sea, an antiparticle (energy $\sqrt{k_2^2 + m^2} \geq |m|$), with total momentum $q = k_1 + k_2$. At $q = 0$, $k_2 = -k_1$, and the threshold is $2\omega_F$ (Pauli blocking pushes it from $2|m|$ to $2\omega_F$). The absolute threshold is $\omega_F + |m|$, at $|q| = k_F$ ($|k_1| = k_F$, $k_2 = 0$).
- **Delta-SCF** (moving one quantum and re-solving the gap): in the homogeneous gas one quantum changes σ by $O(1/\text{volume})$, so the self-consistent excitation energy of one quantum equals the single-particle one. A finite effect needs a finite fraction of the particles, which is how the wall problem below defines its first excited state (design review DFT-9).

4.6.3 The states along the hidden coordinate: the problem [WG 2]

The wall-state solver works at fixed observer's time on the canonical frame, with a_4 frozen to a constant (the adiabatic approximation). The spectrum depends on the transverse momentum k and the frozen a_4 only through $K_\infty = k e^{a_4}$, so every number below is a function of K_∞ ; the code works with K_∞ directly (internally it sets the frozen constant to zero, which only identifies k with K_∞ ; nothing here or in the notebook gives a_4 a value). Units: $H = 1$, $\hbar = c = 1$. Its derivation, fable-cosmology/fermion/waveguide_derivation.wls, loads the notebook's own Input cells 1–117 (with DumpSave blocked) and uses only the notebook's objects (T16, sigma16, gammaCurvedCanonical, GammaSpinCanonical, cfDcov16, gCanonical, RicciScalarCanonical, eta4488); its log records 94 checks, 94 PASS, 0 FAIL in 162.8 s (total time 162.8 s, with cells 1–117 evaluated in 125. s, in the run of 2026-09-25 00:06:38 that wrote the committed waveguide_T16.json; the first committed run, in 18a2501, took 171.4 s, with 132.3 s of loading), and that the notebook's own 142 assertions of cells 1–117 passed while loading [file: fable-cosmology/fermion/waveguide_derivation.log].

The reduction. With $\Psi = e^{ikx_1 - i\omega x_4} \sqrt{\sin 6Hx_0} \Psi'(x_0)$ the Dirac equation $\gamma^\mu D_\mu \Psi = m\Psi$ ($m = H V'(s)$; $m = -2M$ for the author's mass term) becomes **exactly**

$$(\gamma^\mu D_\mu - m) \Psi = \sqrt{\sin} e^{(\dots)} \left[\cot T_0 \partial_0 + i k e^{A_4} \sin^{1/6} T_1 - i \omega T_4 - m \right] \Psi',$$

$$T_0 \partial_z \Psi' + i K(z) T_1 \Psi' - i \omega T_4 \Psi' = m \Psi', \quad K(z) = K_\infty (1 - e^{-12Hz})^{1/12}, \quad K_\infty = k e^{a_4}$$

in the proper distance $z = -\ln \cos(6Hx_0)/(6H)$ ($\cot d/dx_0 = d/dz$; $z = 0$ at the wall, a finite proper distance away; $z \rightarrow \infty$ as $6Hx_0 \rightarrow \pi/2$). The measure is $\sqrt{|g|} |\Psi|^2 dx_0 = |\Psi'|^2 dz$, so the one-particle space is $L^2(dz)$. The transverse term rises from $K(0) = 0$ at the wall

($K = K_\infty(12Hz)^{1/12}(1 - Hz/2 + \dots)$, K' integrable) to K_∞ : a “pocket” of width $\sim 1/(12H)$ ($K^2/K_\infty^2 = 0.875776, 0.942018, 0.984275$ at $z = 0.05, 0.1, 0.2/H$). The wall is a curvature singularity at finite proper distance: $Rz^2 \rightarrow -31/24$ as $z \rightarrow 0$.

Hamiltonian form and the 2×2 blocks. $\omega\phi = h\phi$, $h = -i\alpha_z\partial_z + K(z)\alpha_x + m\beta$ with $\alpha_z = -T_4T_0$, $\alpha_x = -T_4T_1$, $\beta = -iT_4$ (Hermitian involutions, pairwise anticommuting); $J = -iT_0T_1T_2T_3T_4$ commutes with h . $w_3 = \alpha_z\alpha_x\beta$ ($w_3^2 = -1$) is central; the algebra $\langle\alpha_z, \alpha_x, \beta, J\rangle$ has dimension 16, and $\mathbb{C}^{16}$ splits into **8 invariant 2-dimensional blocks**: 4 with $w_3 = -i$ ($J = +1$ twice, $J = -1$ twice) and 4 with $w_3 = +i$. In an explicit block basis every block has $\alpha_z = \sigma_y$, $\beta = \sigma_z$, $T_0 = \sigma_x$, and with $\phi = (f, g)$:

$$\begin{aligned} \text{irrep } w_3 = -i: \quad & f' = -K(z)f + (m + \omega)g, \quad g' = K(z)g + (m - \omega)f, \\ \text{irrep } w_3 = +i: \quad & \text{the same with } K \rightarrow -K. \end{aligned}$$

The wall conditions. The self-adjoint conditions of a block are exactly the real lines $f(0)\cos\vartheta + g(0)\sin\vartheta = 0$, ϑ in $[0, \pi)$, and all of them have zero normal current. Only $\vartheta = \pm\pi/4$ give $\bar{\Psi}\Psi = 0$ at the wall, and only they are covariant under the wall’s $\text{Spin}(1,3)$ of $x_1\dots x_4$: the MIT (bag) conditions $T_{16}[0]\Psi = \pm\Psi$ (MIT+: $f = g$; MIT–: $f = -g$ in the basis above). The complete covariant, J -preserving family is $P\Psi(0) = \Psi(0)$ with $P = T_{16}[0]Q$, Q any Hermitian involution in the 8-dimensional commutant of the Clifford algebra of $T_{16}[0..4]$ (which contains J); all of them have zero normal current and $\bar{\Psi}\Psi = 0$. Flavour symmetry (Q central: $\pm 1, \pm J$) together with charge conjugation (P real) select $Q = \pm 1$, the uniform MIT walls; a mixed Q gives one MIT+ sector and one MIT– sector with halved multiplicities. Without J -preservation there are more, e.g. $P = T_0\exp(iv_t T_0T_5)$. A **same-sign** wall is $T_{16}[0]\Psi = \text{sign}(m)\Psi$ (the limit of an outside mass of the same sign); an **opposite-sign** wall is $T_{16}[0]\Psi = -\text{sign}(m)\Psi$ (a domain wall); for the author’s mass term $m = -2M$, so both are physically possible.

The structure of the spectrum. With MIT walls the 16-component spectrum at transverse momentum K is $S(K) = A(K) \cup (-A(K))$, $A(K)$ the levels of the $w_3 = -i$ block equation, symmetric under $\omega \rightarrow -\omega$; a positive level is **4-fold** per transverse momentum vector (two blocks with $J = +1$, two with $J = -1$, of one irrep), and 8-fold only when ω and $-\omega$ both lie in $A(K)$. The 8 states per momentum of the homogeneous gas split at the wall into two generally non-degenerate 4-fold branches. The bulk continuum is $|\omega| \geq R = \sqrt{m^2 + K_\infty^2}$; $z = 0$ is a regular (limit-circle) endpoint and $z \rightarrow \infty$ limit point. Second-order form: $f'' = (K^2 - K' + m^2 - \omega^2)f$, a supersymmetric pair with superpotential K ; for $K > 0$ the f -channel has the attractive pocket $-K'$.

Exact solutions (used for validation): constant K , MIT–: $\omega = -K$, $\phi = e^{-mz}(1, -1)$; the true profile with $g(0) = 0$: $\omega = m$ for **every** $K_\infty > 0$ (a flat band); constant K , any ϑ : a level $\omega = -(m\cos 2\vartheta + K\sin 2\vartheta)$ exists iff $\kappa = m\sin 2\vartheta - K\cos 2\vartheta > 0$; constant $K > 0$, MIT+: **no** level.

4.6.4 Numerical method and its validation [WG 3, WG 5]

The block problem is solved by the Prüfer angle $(f, g) = r(\cos p, \sin p)$, $p' = m\cos 2p + K\sin 2p - \omega$, $p(0) = \vartheta + \pi/2$, matched to the exact decaying direction of the constant-coefficient problem beyond $z_a = 3/H$ (where $K(z)/K_\infty - 1 < 2 \times 10^{-17}$). The mismatch $\Delta(\omega)$ is strictly decreasing in ω , so the number of levels in the gap is exactly $\#\{j : \Delta(+R) < j\pi < \Delta(-R)\}$ — a rigorous count that catches shallow levels. Roots are found on a logarithmic grid of the decay rate κ and refined by Brent’s method; two integrators (adaptive DOP853 at 10^{-12} , and a 4th-order Magnus propagator) are compared. `python waveguide.py validate` runs 64 checks, all

PASS (log dev/waveguide_run_validate.log, quoted in waveguide_REPORT.txt §3), among them: the exact solutions above (errors $\leq 6 \times 10^{-16}$); Hellmann–Feynman for the scalar density ($d\omega/dm = \int (f^2 - g^2) dz$, agreement 3×10^{-8}); the full 16×16 system with the notebook’s T16, in which every level has multiplicity exactly 4; and the small- K slope of the edge band against its closed form $\omega/K_\infty \rightarrow -\frac{m}{6H} B(\frac{m}{6H}, \frac{13}{12})$ (B the Euler beta function): -0.980822728 vs -0.980822733 at $m = H$.

The independent Mathematica check fable-cosmology/fermion/waveguide_check.wls (a different language, NDSolve at 30 digits, the normalized Wronskian with FindRoot, NullSpace for the asymptotic data) reports 11 checks, 11 PASS, 0 FAIL and agreement with the Python levels to at most $4.68\text{e-}11$ [WG 5]. Its level table:

case	waveguide.py	Mathematica	difference
same-sign $K = 80, n = 0$ (irrep $-i$)	70.985307229908 [nb07 §5.5]	70.98530722995475448906	$4.68\text{e-}11$
same-sign $K = 80, n = 1$ (irrep $+i$)	78.1442493324274 [nb07 §5.5]	78.14424933243057439331	$3.17\text{e-}12$
same-sign $K = 80, n = 2$ (irrep $-i$)	79.3714957790936 [nb07 §5.5]	79.37149577909911904082	$5.51\text{e-}12$
same-sign $K = 80, n = 3$ (irrep $+i$)	79.9873085520475 [nb07 §5.5]	79.98730855204863978335	$1.15\text{e-}12$
same-sign $K = 8$, shallow (irrep $-i$)	8.05383822789729 [nb07 §5.7]	8.053838227898042799445	$7.53\text{e-}13$
same-sign $K = 40, n = 1$ (irrep $+i$)	39.9665602657784 [nb07 §5.7]	39.96656026577953705636	$1.14\text{e-}12$
opposite-sign $K = 3$, edge (irrep $-i$)	-2.93849067695725 [nb07 §5.3]	-2.938490676960405133909	$3.16\text{e-}12$
opposite-sign $K = 3$, edge (irrep $+i$)	2.9384906769572 [nb07 §5.8]	2.938490676960405133909	$3.21\text{e-}12$
$g(0) = 0, K = 3$, flat band (irrep $-i$)	1. [nb07 §5.3]	1.0000000000000000000000	0
$\vartheta = 1.0, K = 3$ (irrep $-i$)	-2.21043470304386	-2.210434703050103980776	$6.24\text{e-}12$
same-sign $m = 4, K = 20$ (irrep $-i$)	20.1427411133772	20.14274111338139405878	$4.20\text{e-}12$

It also confirms the 4-fold multiplicities in the full 16-component system (four singular values below 10^{-3} of the fifth, rising at $\omega \pm d\omega$) and the thresholds $K_c(1) = 6.70624065181$ and $K_c(2) = 33.9189585083$ (the Prüfer mismatch crosses an integer multiple of π between $(1 - 10^{-6})K_c$ and $(1 + 10^{-6})K_c$). At $K = 80$ NDSolve on the 16×8 matrix system did not finish within 30 minutes, so there the 16×16 check uses an independently written Magnus/MatrixExp propagator.

4.6.5 The wall levels: ground and first excited states, in tables [WG 4]

All at $m = H = 1$ unless stated; the levels are the **positive** levels of the full 16-component problem, each 4-fold per transverse momentum vector.

Thresholds of the same-sign wall (the transverse momentum $K_c(n)$ at which the n -th positive level appears; exact count, bisection to 10^{-9} ; dev/waveguide_run_threshold.log):

m/H	$K_c(1)/H$	irrep	$K_c(1)/m$	$K_c(2)/H$	irrep	$K_c(2)/m$
0.25	3.39259952 [nb07 §5.6]	$-i$	13.57040	33.87815035 [nb07 §5.6]	$+i$	135.51260
0.50	4.77916520 [nb07 §5.6]	$-i$	9.55833	33.89177761 [nb07 §5.6]	$+i$	67.78356
1.00	6.70624065 [nb07 §5.6]	$-i$	6.70624	33.91895851 [nb07 §5.6]	$+i$	33.91896

m/H	$K_c(1)/H$	irrep	$K_c(1)/m$	$K_c(2)/H$	irrep	$K_c(2)/m$
2.00	9.33759991 [nb07 §5.6]	$-i$	4.66880	33.97299731 [nb07 §5.6]	$+i$	16.98650
4.00	12.80358979 [nb07 §5.6]	$-i$	3.20090	34.07956537 [nb07 §5.6]	$+i$	8.51989
8.00	17.04604924 [nb07 §5.6]	$-i$	2.13076	34.28513351 [nb07 §5.6]	$+i$	4.28564

The first level always appears in irrep $-i$, with $K_c(1)$ growing roughly like $6.7(m/H)^{1/2}H$; the second always in the other irrep, at $K_c(2) = 33.9\text{--}34.3H$, almost independent of m . Within irrep $-i$ alone the second level appears only at $K = 49.540216755$ [nb07 §5.6] ($m = H$). **For $K < K_c(1)$ there is no bound level at all** — the design’s first guess (“for $k \neq 0$ the wall region is a potential pocket: bound states below $\sqrt{m^2 + k^2 e^{2a_4}}$ ”) was refuted by the review (DFT-7), and the refutation is confirmed.

Dispersion $\omega_n(K)$ of the same-sign wall ($[R - \omega] = \text{binding below the continuum } R = \sqrt{m^2 + K^2}$; dev/waveguide_dispersion_same_m1.csv):

K	R	$n = 0$ (irrep $-i$)	$n = 1$ (irrep $+i$)	$n = 2$ (irrep $-i$)	$n = 3$ (irrep $+i$)
7	7.07106781	7.07061457 [nb07 §5.7] [4.53e-4]			
8	8.06225775	8.05383823 [nb07 §5.7] [8.42e-3]			
10	10.04987562	9.99920505 [nb07 §5.7] [0.0507]			
20	20.02498439	19.39490065 [nb07 §5.7] [0.630]			
34	34.01470270	31.96226139 [nb07 §5.7] [2.05]	34.01469386 [nb07 §5.7] [8.8e-6]		
40	40.01249805	37.21339774 [nb07 §5.7] [2.80]	39.96656027 [nb07 §5.7] [0.0459]		
50	50.00999900	45.83408535 [nb07 §5.7] [4.18]	49.72238768 [nb07 §5.7] [0.288]	50.00982259 [nb07 §5.7] [1.8e-4]	
60	60.00833275	54.32136437 [nb07 §5.7] [5.69]	59.31765921 [nb07 §5.7] [0.691]	59.92357565 [nb07 §5.7] [0.0848]	
80	80.00624976	70.98530723 [nb07 §5.7] [9.02]	78.14424933 [nb07 §5.7] [1.86]	79.37149578 [nb07 §5.7] [0.635]	79.98730855 [nb07 §5.7] [0.0189]
100	100.00499988	87.32733251 [nb07 §5.7] [12.7]	96.59932036 [nb07 §5.7] [3.41]	98.43212058 [nb07 §5.7] [1.57]	99.66483153 [nb07 §5.7] [0.340]

(at $K = 100$ a fifth level 99.96879653 [nb07 §5.7] [0.0362], irrep $-i$). The band gap at fixed K ($\omega_1 - \omega_0$) of the bare wall: 2.05243246 [nb07 §5.14] ($K = 34$), 2.75316253 [nb07 §5.14] (40), 4.99629484 [nb07 §5.14] (60), 7.15894210 [nb07 §5.14] (80), 9.27198785 (100).

The ground and first excited level at $K = 80$ (the case the review examined, DFT-7): in the full 16-component problem the positive levels are 70.98531 [nb07 §5.5] ($n = 0$, irrep $-i$), 78.14425 [nb07 §5.5] ($n = 1$, from the **other** irrep, $+i$), 79.37150 [nb07 §5.5] ($n = 2$, irrep $-i$) and 79.98731 [nb07 §5.5] ($n = 3$, irrep $+i$), each 4-fold. The review’s “ $n = 1$ at 79.37150” labelled the second level of one irrep; it is $n = 2$ of the full problem, and the first excited level is 78.144249332 [nb07 §5.5]. The levels command printed, for both irreps, irrep $-i$: -79.987308552047, -78.144249332428, +70.985307229908, +79.371495779094; irrep $+i$: -79.371495779091, -70.985307229895, +78.144249332427, +79.987308552047 [nb07 §5.5] — the $\omega \rightarrow -\omega$ symmetry between the irreps to 10^{-11} [WG 1].

Wavefunctions (normalized in $L^2(dz)$): at $K = 80$, $n = 0$ has $\int(f^2 - g^2) = 0.24150473$ [nb07 §5.10] with f and g nodeless; $n = 1$ ($+i$) 0.01572127 [nb07 §5.10], g with one node; $n = 2$ ($-i$) 0.02345024, f and g one node each; they live within $z < 0.4/H$ (decay rates $\kappa = 36.9, 17.2, 10.1$) [WG 4.5].

The opposite-sign wall: a gapless edge band. For every $K > 0$ there is one edge level per block, deep inside the gap: $|\omega|/K \rightarrow \frac{m}{6H} B(\frac{m}{6H}, \frac{13}{12}) = 0.9808227327$ [nb07 §5.8] as $K \rightarrow 0$ (exact), 2.93849068 [nb07 §5.8] at $K = 3$, and the ratio falls for larger K [WG 4.3]:

K	R	edge $ \omega $	$ \omega /K$
0.10	1.004988	0.09808212 [nb07 §5.8]	0.980821
1.00	1.414214	0.98067175 [nb07 §5.8]	0.980672
3.00	3.162278	2.93849068 [nb07 §5.8]	0.979497
10.00	10.049876	9.69024420 [nb07 §5.8]	0.969024
20.00	20.024984	18.99601285 [nb07 §5.8]	0.949801
40.00	40.012498	36.76546650 [nb07 §5.8]	0.919137
80.00	80.006250	70.51329445 [nb07 §5.8]	0.881416
100.00	100.005000	86.85053394 [nb07 §5.8]	0.868505

The edge band lies inside the bulk gap $|\omega| < m$ for $K < \sim 1.02m$: massless fermions bound to the wall, 4 positive states per transverse momentum, with a small **negative** scalar density ($\int(f^2 - g^2) = -0.0498$ at $K = 3$). Further positive levels of the opposite-sign wall appear at $K = 33.80968834$ [nb07 §5.6] (irrep $-i$) and 48.79382020 [nb07 §5.6] ($+i$).

The dependence on the wall condition (the uniform family $f(0) \cos \vartheta + g(0) \sin \vartheta = 0$, symmetry breaking except $\vartheta = \pm\pi/4$): at $K = 3$ the levels sweep the **whole** gap $(-R, R)$ as ϑ varies; at $K = 80$ the lowest positive level moves from $\omega = m = 1$ [nb07 §5.9] ($\vartheta = -\pi/2$, $g(0) = 0$) to 70.985 [nb07 §5.9] (MIT+) and 70.513 [nb07 §5.9] (MIT−). The wall condition, not the pocket, sets the low end of the spectrum [WG 4.4].

Convergence: the reference levels are accurate to about 10^{-10} (tolerance-limited; matching point, `rtol` from 10^{-9} to 10^{-13} , z_a from 2 to 6, the κ grid and the integrator each change ω at $K = 80$, $n = 0$ by at most 1.0×10^{-8}) [WG 4.6].

4.6.6 The Kohn–Sham (local-density) treatment of the wall [WG 6]

Setting. The functional per 8-volume is $W(\sigma) = m_0\sigma + (\lambda/2)\sigma^2$ with the local mean field $m(z) = W'(\sigma(z)) = m_0 + \lambda\sigma(z)$, $m_0 = 1/H$; $\lambda < 0$ is attractive. The admissible modes are zero modes along $x_5 \dots x_7$, normalized over the hidden coordinate volume, chosen as $V_{\text{hid}} = 1/H^3$ (so λ also stands for the effective 5-dimensional coupling λ/V_{hid}); N_s is the number per unit coordinate observed 3-volume. Per transverse momentum each positive level of either irrep contributes 4 degenerate states; a “band” is one level family $\omega_b(k)$. The scalar density of a block orbital is the Krein-weighted $|f|^2 - |g|^2$, the number density $|f|^2 + |g|^2$, and since the transverse proper volume is $1/\text{Sin}(z)$:

$$\sigma(z) = \text{Sin}(z) \sum_b 4 \int \frac{d^3k}{(2\pi)^3} \theta(\text{occupied}) (f^2 - g^2)_{b,k}(z), \quad n(z) \text{ likewise with } (f^2 + g^2),$$

$$E/A = \sum_{\text{occ}} \omega + \int (W - \sigma W') \frac{dz}{\text{Sin}} = \sum_{\text{occ}} \omega + \frac{|\lambda|}{2} \int \frac{\sigma^2}{\text{Sin}} dz,$$

no-sea, E_{xc} neglected, a common Fermi level μ , Anderson mixing; the particle number is reproduced to 10^{-8} in every iteration.

(a) The author’s mass term: exact. For $W = m_0\sigma$ ($\lambda = 0$) the action is bilinear, the theory is **free**, the single-particle levels of section 4.6.5 are the exact spectrum, and every many-body eigenstate is a Slater determinant of exact modes; no self-consistency is needed. Consequences: on the **same-sign** wall there is **no** bound positive level for $K < K_c(1) = 6.70624$ [nb07

§5.11]; every wall level lies above m_0 by more than $\sqrt{m_0^2 + K_c^2} - m_0 = 5.78 m_0$ [**nb07 §5.11: 5.780388**], while the bulk continuum starts at m_0 . The N_s -fermion ground state of the free theory therefore has **no wall-localized fermions**; the wall levels are excited states. On the **opposite-sign** wall the gapless edge band lies below m_0 for $K < \sim 1.02 m_0$; the free ground state of $N_s \leq \frac{4}{6\pi^2} (1.02)^3 = 0.072 H^3$ [**nb07 §5.8, §5.11: 0.071622, with $K^* = 1.01971543$**] fermions fills it — a genuine wall-bound massless $(3+1)$ -dimensional Fermi sea — and beyond that the particles spill into the bulk.

(b) Same-sign wall, attractive W : the wall-band sector ($\lambda = -0.01 H^{-3}$, $N_s = 100 H^3$, $m_0 = 1$). The self-consistency **converges** (9 iterations, $\max |m_{\text{new}} - m| = 1.5 \times 10^{-10}$). At the box size $Z = 30/H$:

```
E/A = 937.2277360325 H^4    (sum of occupied omega = 936.0540086871, interaction Int(W - sigma W') =
↪ 1.1737273454)
E/N = 9.37227736 H          mu = 11.6958993131 H
band n = 0 (irrep -i) holds all N_s: k in [5.45898483, 11.80021359]; band bottom omega = 5.54982120 at
↪ the self-consistent threshold k_lo = 5.45898 (bare 6.70624)
```

(every number in this block [**nb07 §5.17**]); the density profile has $\sigma = n = 0$ at the wall, a maximum $\sigma = 23.845616$ [**nb07 §5.17**] (mass minimum $m = 0.761543845$ [**nb07 §5.17**]) at $z = 0.167$, and an inverse-square tail. The truncation at Z converges slowly (a self-consistent tail $m - m_0 = -0.032/z^2$); extrapolated, $E/A \rightarrow 936.46\text{--}936.53$, $\mu \rightarrow 11.692$, $k_{\text{lo}} \rightarrow 5.439$; grid, quadrature and k -grid are converged to $\leq 4 \times 10^{-5}$ in E/A [**WG 6.5**]. **But this converged state is not the ground state.** (i) $\mu - m_0 = 10.696 > 0$: bulk continuum states lie below the Fermi level, so the global Kohn–Sham ground state has no wall layer at this coupling. (ii) Even within the wall-band sector the Walecka functional $\Omega[m] = \sum_{\text{occ}} \omega[m] + \int (m - m_0)^2 / (2|\lambda|) dz / \text{Sin}$ is **not stationary**: along $m = m_0 + s(m_{\text{sc}} - m_0)$, $\Omega = 938.1963767831, 937.2277359473, 936.2641446367$ [**nb07 §5.17**] at $s = 0.98, 1.00, 1.02$ and $d\Omega/ds = -48.305804$ [**nb07 §5.17**] — the threshold term $\frac{4}{2\pi^2} k_{\text{lo}}^2 (-dk_{\text{lo}}/ds) (R(k_{\text{lo}}) - \mu) = -48.306186$ ($dk_{\text{lo}}/ds = -1.301519$, $R(k_{\text{lo}}) = 5.549821$; agreement to 8×10^{-6} relative): a deeper well pulls new bound states out of the continuum at the band bottom (energy ~ 5.55) and they replace Fermi-surface states (energy $\mu = 11.70$). So the result is reported as the **self-consistent wall-band configuration**, not as a variational ground state.

The first excited state (Delta-SCF) of that configuration: a fraction x of N_s in band $n = 1$ (irrep $+i$), the rest in $n = 0$, each band filled from its bottom; all converged to 10^{-9} :

x	E/A	$\Delta E/A$	ΔE per promoted fermion
0.01	957.9790646375 [nb07 §5.17]	20.7513286050	20.75132861 [nb07 §5.17]
0.05	1041.8217858764 [nb07 §5.17]	104.5940498439	20.91880997 [nb07 §5.17]
0.10	1148.4782948320 [nb07 §5.17]	211.2505587995	21.12505588 [nb07 §5.17]
0.20	1367.8210883907 [nb07 §5.17]	430.5933523582	21.52966762 [nb07 §5.17]

The lowest Delta-SCF excitation is the $x \rightarrow 0^+$ limit, about $20.75 H$ per promoted fermion, close to the bottom of band $n = 1$ ($\omega = 31.99$ at $k_{\text{lo}} = 31.975$) minus the top of $n = 0$ (11.63), $= 20.36$, the remaining 0.4 being the mean-field rearrangement. Intra-band particle–hole excitations are gapless. The self-consistent band gaps at fixed k ($\omega_1 - \omega_0$): 2.08513398 [**nb07 §5.17**] ($k = 34$), 2.78024477 [**nb07 §5.17**] (40), 5.02368345 [**nb07 §5.17**] (60), 7.18485799 [**nb07 §5.17**] (80).

(c) Opposite-sign wall: the edge band is a true ground state ($m_0 = 1$, $N_s = 0.05 H^3$). The edge band starts at the Dirac point ($\omega = 0$, $k = 0$); $N_s = 0.05$ fills it to $k_F = 0.90459$ [**nb07 §5.12**] with $\mu < m_0$: every state below μ is a bound wall state, so this is a genuine ground state of N_s fermions.

	$\lambda = -0.01 H^{-3}$	$\lambda = -10 H^{-3}$
iterations	3 ($\max dm < 10^{-10}$)	6 ($\max dm < 10^{-10}$)
E/A	0.033268937058 H^4 [nb07 §5.12]	0.033267529405 H^4 [nb07 §5.12]
E/N	0.66537874 H [nb07 §5.12]	0.66535059 H [nb07 §5.12]
μ	0.8871343842 H [nb07 §5.12]	0.8870591549 H [nb07 §5.12]
$\sigma(z)$	in $[-7.964\text{e-}4, 0]$ (negative: $f \sim -g$)	in $[-7.986\text{e-}4, 0]$
$m(z)$	in $[1, 1.00000796]$	in $[1, 1.00798557]$ (the attraction raises m near the wall)
Walecka check	slope $-1.2\text{e-}14$, curvature $+5.6\text{e-}13$	slope $+2.0\text{e-}13$, curvature $+5.6\text{e-}10$

Stationary and a **minimum**: a variationally consistent Kohn–Sham ground state. $E/N = 0.6654$ is the massless value $\frac{3}{4} v k_F$ with $v = 0.9807$: the wall fermions are essentially massless. It is fully converged ($Z = 45$, a finer grid and more quadrature points change E/A by at most 1.6×10^{-12}). **Its first excited state**: there is no bound band $n = 1$ at the occupied momenta (the next positive bands of the opposite-sign wall appear only at $K = 33.81$ and 48.79); the lowest excitations are gapless intra-band particle–hole pairs and promotion into the bulk continuum, which costs at least $m_0 - \mu = 0.113 H$ [nb07 §5.12: $m_0 - \mu$ **from its** μ] and unbinds the fermion. A Delta-SCF with a finite fraction in a bound band $n = 1$ would require $k \geq 33.81$, at $\omega - \mu \geq \sim 32.9 H$.

(d) Strong attraction on the same-sign wall (exploratory; $\lambda = -100 H^{-3}$, $N_s = 0.03 H^3$): did not converge. Homogeneous bulk matter is self-bound only for $|\lambda|$ between 60 and 80 H^{-3} or more (min $E/n - m_0 = +0.000033$ for $\lambda = -0.01 \dots -60$, -0.038261 for -80 , -0.072213 for -100). At $\lambda = -100$ the iteration **failed** (not converged after 60 iterations; $\max |m_{\text{new}} - m|$ wandered between 6.6×10^{-3} and 0.71). What was obtained: the layer **is** bound ($\mu = 0.8796 H$, $\mu - m_0 = -0.1204 H$), the two irreps are occupied almost equally (the two 4-fold branches become degenerate away from the wall: 8 states per momentum, as in the homogeneous gas), and the layer drifts **outward** from $z_c = 2.07$ to $3.4\text{--}3.8$. A rigid-shift diagnostic shows why: the Walecka energy of the layer decreases monotonically with its distance from the wall (0.0318696 at $z_c = 0.6$ to 0.0286926 at 6.0): the same-sign wall **repels** the layer, which has no stationary position at finite distance. There is no wall-bound Kohn–Sham ground state on the same-sign wall at this coupling either.

4.6.7 What the density-functional treatment establishes, and what it does not

- **Homogeneous gas.** The Kohn–Sham ground state is the gap root of lowest ρ (a maximum of the no-sea $E_n[\sigma]$, a minimum of the Walecka functional for attractive W); for the author’s mass term it is the exact free Fermi sea. The first excitations are gapless particle–hole pairs; pair creation starts at $\omega_F + |m|$ ($q = k_F$) and at $2\omega_F$ for $q = 0$.
- **Along x_0 .** The geometry binds states only above a transverse-momentum threshold on the same-sign wall ($K_c(1) = 6.70624 H$ at $m = H$ [nb07 §5.6]), so the **ground state of the free fable has no wall-bound fermions there**; the wall levels (ground 70.98531 [nb07 §5.5], first excited 78.14425 [nb07 §5.5] at $K = 80$) are excited states of the many-body system. On the opposite-sign wall a gapless edge band of massless wall fermions exists at every K , and it carries a variationally consistent Kohn–Sham ground state; its first excitations are gapless.
- **What did not work** [WG 8]: on the same-sign wall the self-consistent wall-band configuration converges but is neither the global ground state nor a variational minimum within its sector; the strong-coupling layer does not converge (the wall repels it). A grand-canonical treatment with a bulk fable density and a position-constrained iteration were not done. Two runs of the derivation, before **DumpSave** was blocked, rewrote the notebook’s two

Part II .mx files; their content was verified identical to the committed version (the final script blocks `DumpSave`).

- **Caveats** [WG 8]: a_4 frozen (adiabatic; with a_4 varying the momentum redshifts as $K_\infty = k e^{a_4}$); zero modes along $x_5..x_7$ and along the transverse directions other than the momentum; the wall condition is a choice of self-adjoint extension and the results depend strongly on it; no-sea, E_{xc} neglected, local functional $m(z) = W'(\sigma(z))$; the Delta-SCF states are stationary configurations with constrained occupations, not eigenstates of an exact many-body Hamiltonian.

Notebook 07. `fable-cosmology/notebooks/07_fable_dft_states.ipynb` re-runs these computations with `waveguide.py` (its same-sign Kohn–Sham runs at a reduced resolution, compared in its §5.17 with the solver’s full-resolution run, which it starts in the background), writes their tables and figures under `fable-cosmology/results/nb07_*`, and asserts every number it prints. Executed and committed in `aca5102`, it prints again every wall-state number above that carries the mark `[nb07 §n]`. The numbers of this section that it does **not** print again, and which are therefore quoted from the report alone [WG n], are: the levels -2.21043470304386 ($\vartheta = 1.0$, $K = 3$) and 20.1427411133772 ($m = 4$, $K = 20$) of the Mathematica comparison; the bare band gap 9.27198785 at $K = 100$; $\int(f^2 - g^2) = 0.02345024$ of the level $n = 2$ at $K = 80$ and -0.0498 of the opposite-sign edge level at $K = 3$; and the extrapolation of the same-sign Kohn–Sham state to $Z \rightarrow \infty$ ($E/A \rightarrow 936.46\text{--}936.53$, $\mu \rightarrow 11.692$, $k_{lo} \rightarrow 5.439$). For that last item notebook 07 prints instead the larger box $Z = 45$: $E/A = 937.06107832$ ($Z = 30$: 937.22904016 ; difference -0.167962 , $-1.79e-04$ relative); $\mu = 11.69499658$ ($-9.07e-04$); $k_{lo} = 5.454492$ and the tail $\sigma(20)/\sigma(10) = 0.2529$ (an inverse-square tail gives 0.25) [nb07 §5.13], consistent with the slow convergence in Z described above.

4.7 Solving the coupled equations from the beginning of the present universe to today

The coupled system of sections 4.2–4.5 is solved three times independently: by the Rust solver `fable_fermion` on the pure-Rust SUNDIALS 7.8.0 CVODE of the `rustSolveIt` repositories (`fable_fermion` 0.1.0, `sundials_rs` 7.8.0 (pure Rust), CVODE BDF, its `-version` line), by the independent Python implementation `fable-cosmology/fermion/crosscheck.py` (adaptive quadrature, `brentq`, `DOP853`), and, for two reference cases, by Mathematica (Part VIII, Section 33, and `fable-cosmology/reference/make_reference_fermion.wls`). The runs, their tables and their figures are those of notebook 06 [nb06], which writes every file named below under `fable-cosmology/results/`.

4.7.1 The two models, and the system handed to CVODE

fable8d — the unstabilized 8-dimensional system. The asymptotic region of the primordial field is the Bianchi-I metric of section 4.3.4; the sources are radiation (on the observed sheet, $P_{\text{obs}} = \rho_r/3$, $P_{\text{hid}} = P_{x_0} = 0$, $\rho_r = \Omega_{r0}A^{-4}/v$), baryons (dust in every direction, $\rho_b = \Omega_{b0}A^{-3}/v$) and the Kohn–Sham fable of section 4.5 ($k_F = k_{F0}/A$, per 7-volume, the gap solved at every step). The equations are the constraint (monitored, not imposed) and the evolution equations of section 4.3.4 with $\kappa_8\rho \rightarrow 3\hat{\rho}$ (densities per 7-volume in ρ_{c0}), so that $\kappa_8(P_i - T/6)$ becomes $F/2 + 3(P_i - P_{\text{hid}})$:

$$\frac{dH_A}{dt} = -H_A\Theta + \frac{F}{2} + 3(P_{\text{obs}} - P_{\text{hid}}), \quad \frac{dH_B}{dt} = -H_B\Theta + \frac{F}{2}, \quad \frac{dH_C}{dt} = -H_C\Theta + \frac{F}{2},$$

with the hidden driver F and $P_{\text{obs}} - P_{\text{hid}}$ formed per source analytically — radiation 0 and $\rho_r/3$, baryons ρ_b and 0, the fable $2W - \sigma_8 W'$ and P_{KS}/v — never by subtracting large numbers.

The independent variable is $N = \ln A$; since the physical rates fall like A^{-2} by many orders of magnitude while H_B and H_C start at exactly zero, CVODE integrates the scaled variables $h_i = H_i A^2$ and $\tau = t/A^2$, state $y = (\ln B, \ln C, h_A, h_B, h_C, \tau)$:

$$\frac{d \ln B}{dN} = \frac{H_B}{H_A}, \quad \frac{d \ln C}{dN} = \frac{H_C}{H_A}, \quad \frac{dh_i}{dN} = 2h_i + \frac{A^2 (dH_i/dt)}{H_A}, \quad \frac{d\tau}{dN} = -2\tau + \frac{A^{-2}}{H_A}.$$

Initial state: $H_B = H_C = 0$, H_A from the constraint, $t(a_i) = 1/(2H_A(a_i))$. The forward run is a **boundary-value problem** ($H_B = H_C = 0$ at a_i ; $H_A = B = C = 1$ today), solved by an outer shooting (bracket and Brent) on the fable's amplitude and an inner secant iteration on $\ln v(a_i)$. It runs forward only (section 4.7.11). [prose nb06 §4, §4.1; file: `fable-cosmology/rust/fable_fermion/src/models.rs` and `run.rs`, module documentation.] (The module summary of `models.rs` still lists the unscaled state $(\ln B, \ln C, H_A, H_B, H_C, t)$; its right-hand side and the notebook integrate the scaled one above.)

fable4d — the stabilized model, the physical one. `fable8d` plus a zero-energy stabilizing stress $P_{\text{stab}} = -F_{\text{total}}/2$ added to the hidden pressures of B and C . Then $F_{\text{total}} + 2P_{\text{stab}} = 0$, $H_B = H_C = 0$ is preserved exactly, $v = 1$, and the observer's equations are **exactly** the Friedmann equations $3H_A^2 = \kappa\rho$, $H'_A = -\frac{\kappa}{2}(\rho + P_{\text{obs}})$, with the hidden equations satisfied identically. The stabilizer is a **Lagrange multiplier** that enforces the frozen hidden sheet, not a stress derived from any field; it has zero energy density, drops out of the conservation law at $H_B = H_C = 0$, and it **violates the null energy condition along** x_0 whenever dust is present: for the null vector $k = e_0/C + e_4$ in the (x_0, x_4) plane $T_{\text{stab}}(k, k) = P_{\text{stab}} = -F/2$, which is $-\rho_{\text{dust}}/2 < 0$ for dust alone. [proved VIII §33]

- `fable4d` [THE RESULT]: with $P_{\text{stab}} = -F/2$ on B and C , the hidden drivers vanish, $F + 2 P_{\text{stab}} == 0$, so $H_B = H_C = 0$ is preserved exactly
- `fable4d` [THE RESULT]: with $B = C = 1$ (labelled substitution) the Einstein equations are EXACTLY $3 H_A^2 == \text{kappa } \rho$ and $H_A' == -(\text{kappa}/2)(\rho + P_{\text{obs}})$; the x_0 and hidden equations then hold with $P_{\text{hid}} + P_{\text{stab}}$
- `fable4d` [has content]: the Friedmann pair is self-consistent - $d/dx_4 (3 H_A^2 - \text{kappa } \rho) == 0$ when $H_A' = -(\text{kappa}/2)(\rho + P_{\text{obs}})$ and $\rho' = -3 H_A (\rho + P_{\text{obs}})$ (the conservation law at $H_B = H_C = 0$)
- `fable4d` [THE RESULT]: the stabilizer VIOLATES the null energy condition along x_0 : $T_{\text{stab}}(k, k) == P_{\text{stab}} == -F/2$ for the null k in the (x_0, x_4) plane, $== -\rho_{\text{dust}}/2 < 0$ for dust alone
- `fable4d` [has content]: it is a Lagrange multiplier with zero energy - $\rho_{\text{stab}} == 0$, and at $H_B = H_C = 0$ it drops out of the conservation law (its term is $(3 H_B + H_C) P_{\text{stab}}$)

(Without the stabilizer, dust would decelerate as $dH/dt = -\frac{5}{2}H^2$ instead of $-\frac{3}{2}H^2$ [file: `models.rs`, module documentation].) `fable4d` is **algebraic** in a : $k_F = k_{F0}/a$, the gap is local, $H_A(a)^2 = \Omega_{r0}a^{-4} + \Omega_{b0}a^{-3} + \rho_f(a)$, and only the age $t(a)$ is integrated ($d\tau/dN = -2\tau + A^{-2}/H_A(N)$). `fable8d -freeze-hidden` integrates the `fable8d` system with the stabilizer ($dH_A/dt = -3H_A^2 + \frac{3}{2}(\rho - P_{\text{obs}})$, H_A evolved, not algebraic) and must reproduce `fable4d`; it does (section 4.7.10).

CVODE settings: BDF, Newton iteration, a dense Jacobian, $\text{rtol} = 1\text{e-}10$, $\text{atol} = 1\text{e-}12$ [prose nb06 §4.1]. **Normalization:** `fable4d` by a closure today, $\rho_r + \rho_b + \rho_f = 1$ at $A = 1$, with one shooting parameter per potential (section 4.7.3). **The branch:** the gap root of lowest ρ (section 4.5.3).

Units (fable_fermion -constants, printed by notebooks 05 and 06; $\hbar = c = k_B = 1$):

```

h                = 0.674
HO               = 2.1842852411e-18 1/s = 6.8930799924e-11 1/yr = 1.4377226622e-33 eV
1/HO            = 1.4507302992e10 yr
M_pl (reduced, from G)= 2.4353234593e27 eV = 2.4353234593e18 GeV
rho_c0 = 3 H0^2 M_pl^-2 = 3.6777719498e-11 eV^4
E_c = rho_c0^(1/4)    = 2.4626131773e-3 eV    (unit of m, kF; sigma, n in E_c^3; W, rho in E_c^4 =
  ↪ rho_c0)
1 eV            = 4.0607270732e2 E_c
T_CMB           = 2.7255 K = 2.3486541806e-4 eV; T_nu0 = 1.6763891605e-4 eV
Omega_gamma0    = 5.4437728013e-5 (Omega_gamma0 h^2 = 2.4729753331e-5)
Omega_nu(1 species)0 = 1.2363206389e-5
Omega_r0 (N_eff=3.046) = 9.2096054673e-5 (Omega_r0 h^2 = 4.1837027333e-5)
Omega_b0 = 0.02237/h^2 = 4.9243191364e-2
(g_*(T) is not followed: radiation is Omega_r0 A^-4 at all times, which misstates rho_r A^4 by up to a
  ↪ factor ~0.39 before e+e- annihilation and the QCD transition)
g (fable states per momentum) = 8
a_BBN (T = 1 MeV) = 1.6763891605e-10 (= T_nu0 / 1 MeV)
a_rec (z = 1090) = 9.1659028414e-4

```

and the notebooks recompute it in Python from CODATA and agree to 10^{-9} . Part VIII computes the same constants in Mathematica and asserts them [displayed and proved VIII §33]: **units:** $1/H_0 = 1.4507303e10$ yr; $M_{\text{pl}} = 2.4353235e18$ GeV; $\rho_{\text{c}0} = 3.6777719e-11$ eV⁴; $E_{\text{c}} = \rho_{\text{c}0}^{(1/4)} = 0.0024626132$ eV, $\Omega_{\text{gamma}0} = 0.000054437728$, $\Omega_{\text{gamma}r0} = 0.000092096055$, $\Omega_{\text{gamma}b0} = 0.049243191$; fable4d RUNS [definition]: the units computed from CODATA 2018 - $E_{\text{c}} = \rho_{\text{c}0}^{(1/4)} == 2.46261e-3$ eV, $\Omega_{\text{gamma}r0} == 9.2096e-5$, $\Omega_{\text{gamma}b0} = 0.02237/h^2 == 0.049243$ (to the quoted digits). The coupling is $\kappa_{4,0} = \kappa_8/V_{h,0}$ with $v = 1$ today, and $\Omega_i := \kappa_8 \rho_{7,i}/(3H_A^2)$.

Assumptions stated with every run [prose nb06 §4]: x_0 is compactified in the asymptotic region with a Kaluza–Klein gap far above the temperatures considered; the hidden timelike sheet is a formal comoving volume (compactifying it gives closed timelike curves); every field is truncated to its zero modes along the hidden directions; gravity is semiclassical ($G = \kappa \langle \hat{T} \rangle$); the exchange–correlation energy is neglected; the Dirac-sea energy is absorbed into a renormalized W (section 4.5.11); the terms of order $\cot$ and $\cot^2$ of the asymptotic reading are neglected (section 4.3.3).

4.7.2 The interval, and why it is the right one

“From the beginning of the present universe.” The present universe is the asymptotic region of the primordial field (section 4.3.3), in which the primordial field is the 8-dimensional Bianchi-I metric. Its beginning, for this system, is the earliest time at which the model applies with ordinary matter: deep in the radiation era, before nucleosynthesis, with the fable ultra-relativistic. The runs start at

$$\begin{aligned}
 a_i &\leq \min(10^{-10}, 0.01 a_{\text{nr}}(m)) \\
 (\text{default } a_i &= 10^{-12}: T = T_{\text{CMB}}/a_i = 234.9 \text{ MeV;} \\
 a_{\text{BBN}} &= 1.6763891605 \times 10^{-10} \text{ at } T = 1 \text{ MeV})
 \end{aligned}$$

and end at $A = 1$, defined by $T_{\text{CMB}} = 2.7255$ K. The criterion is checked for every run [prose nb06 §4, §5.1]. Three facts justify it: (i) before a_{nr} the fable is radiation-like ($w = 1/3$ to 10^{-8} , asserted for the reference case in Part VIII) and its history is fixed by its conserved number; (ii) the pre-universe itself does **not** fix a_i — nothing in the notebook ties the author’s constant H or the position along x_0 to a time in the present universe (section 4.3.3); (iii) the results do not depend on a_i [nb06 §5.1, file: fable-cosmology/results/nb06_ai_insensitivity.csv]:

model	a_i	$T(a_i)$ [MeV]	$H_B(1)$	$\rho_U(1)$	q_0	t_0 [1/ H_0]	$w_f(0.5)$	$G(a_{\text{BBN}})$ / $G(1)$
fable4d	1e-12	234.9	0.0000e+00	0.685664712581	-0.528451020843	0.951246512174	-0.244385837789	1.000000e+00
fable4d	1e-11	23.5	0.0000e+00	0.685664712581	-0.528451020843	0.951246510304	-0.244385846968	1.000000e+00
fable4d	1e-10	2.3	0.0000e+00	0.685664712581	-0.528451020843	0.951246512298	-0.244385836589	1.000000e+00
fable8d	1e-12	234.9	9.9990e-01	6.684880971343	-0.842486541055	0.652899266076	-0.164884834974	1.046057e+12
fable8d	1e-11	23.5	9.9990e-01	6.684880972924	-0.842486540972	0.652899265712	-0.164884977537	1.046057e+12
fable8d	1e-10	2.3	9.9990e-01	6.684880970384	-0.842486544662	0.652899266556	-0.164884818972	1.046057e+12

(**lambda-mass**, $m_0 = 30$ eV.) The largest relative change between $a_i = 10^{-12}$ and 10^{-10} is 2.0×10^{-9} (t_0) and 3.8×10^{-8} ($w_f(0.5)$) in **fable4d**, 8.6×10^{-7} ($w_f(0.5)$) in **fable8d**. The one approximation the early start exposes is $g_*(T)$, which is not followed (the solver's constants line above).

4.7.3 The potentials, and what each run fixes

For the potentials that supply dark energy the split is imposed today: the quasiparticles carry $\Omega_{\text{qp}0} = 0.265$ (the dark matter) at the mass $m_{\text{today}} = W'(\sigma_t)$, and the amplitude closes the budget [prose nb06 §3; file: `run.rs`, module documentation]:

name	$W(\sigma)$	given	shot (to close the budget today)
mass (the author's)	$m_0\sigma$	m_0	In k_{F0} ; the fable closes the budget alone, $\Omega_{f0} = 1 - \Omega_{r0} - \Omega_{b0}$ (Einstein–de Sitter-like)
lambda-mass	$V_0 + m_0\sigma$	m_0 , $\Omega_{\text{qp}0} = 0.265$	V_0 (a bare Lambda, reported as such); k_{F0} from $\varepsilon_{\text{KS}}(m_0, k_{F0}) = \Omega_{\text{qp}0}$
power	$m_0\sigma + \lambda\sigma^\nu$	m_{today} , ν , $\Omega_{\text{qp}0}$	the condensate energy today $U_t = (1 - \nu) \lambda \sigma_t^\nu$; then $\lambda = U_t / ((1 - \nu) \sigma_t^\nu)$, $m_0 = m_{\text{today}} - \nu \lambda \sigma_t^{\nu-1}$ (may be negative)
expdamp	$V_0 + m_0 \sigma e^{-\sigma/s_1}$	m_{today} , $x_t = \sigma_t/s_1$ (Part VI's 1/2.21), $\Omega_{\text{qp}0}$	V_0 ; $s_1 = \sigma_t/x_t$, $m_0 = m_{\text{today}} e^{x_t} / (1 - x_t)$
quadratic	$V_0 + m_0\sigma + (\lambda/2) \sigma^2$	m_{today} , $g_q = \lambda \sigma_t / m_{\text{today}}$, $\Omega_{\text{qp}0}$	V_0 ; $m_0 = m_{\text{today}} (1 - g_q)$
lorentz	$V_0 + m_0\sigma / (1 + (\sigma/s_1)^2)$	m_{today} , x_t (Part VI's 1/1.5), $\Omega_{\text{qp}0}$	V_0 ; $m_0 = m_{\text{today}} (1 + x_t^2)^2 / (1 - x_t^2)$

(The solver's parameter names: x_t is `xt`, g_q is `gq`, λ is `lam`, ν is `nu`.)

4.7.4 Runs (a): the author's mass term alone — an Einstein–de Sitter universe with a fable that cools

$W = m_0\sigma$, **fable4d**, $m_0 = 1, 30, 100, 1000, 2000$ eV. There is no condensate ($U = 0$): the fable is a free degenerate gas and, with no Lambda, it closes the budget. Its $w_f = w_{\text{DM}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ runs from 1/3 to 0 through a_{nr} , the scale factor at which $k_F = m$ [nb06 §5.2, file: `results/nb06_runs.csv`, figure `results/nb06_a_mass_wdm.png`] (figure 7):

run	k_{F0} [eV]	a_{nr}	a_i ok	ΔN_{eff} BBN	ΔN_{eff} rec	$3P/\rho_\nu$ rec	q_0	t_0 [Gyr]	Ω_{f0}
mass 1 eV (EdS)	6.3727e-04	6.3727e-04	True	36.75	79.96	17.67	0.50005	9.6706	0.95066
mass 30 eV (EdS)	2.0509e-04	6.8364e-06	True	0.3943	70.49	0.002353	0.50005	9.6707	0.95066
mass 100 eV (EdS)	1.3730e-04	1.3730e-06	True	0.07918	70.48	9.489e-05	0.50005	9.6707	0.95066
mass 1000 eV (EdS)	6.3727e-05	6.3727e-08	True	0.003675	70.48	2.044e-07	0.50005	9.6707	0.95066

run	k_{F0} [eV]	a_{nr}	a_i ok	ΔN_{eff} BBN	ΔN_{eff} rec	$3P/\rho_\nu$ rec	q_0	t_0 [Gyr]	Ω_{f0}
mass 2000 eV (EdS)	5.0580e-05	2.5290e-08	True	0.001459	70.48	3.22e-08	0.50005	9.6707	0.95066

(ΔN_{eff} is the whole fable energy in units of one massless neutrino species; at recombination it is large because the fable is the dark matter there, and $3P/\rho_\nu$ is its radiation-like part.) The solver's own log of the 30 eV run (fable_fermion fable4d -potential mass -param m0_ev=30, printed in notebook 05): # NOTE: W = m0 sigma alone (no V0): the closure forces $\Omega_{\text{f}0} = 1 - \Omega_{\text{b}0} - \Omega_{\text{r}0} = 0.950665$: an Einstein-de Sitter-like universe ($q_0 = 0.5000$, +1/2 for pure dust), and a_{nr} ($k_F = |m_{\text{eff}}|$) = 6.837070e-6 ($z_{nr} = 1.4626e5$), $a_{eq} = 9.2105e-5$: WARM / not hot (a_{nr} between 1e-6 and a_{eq}) [nb05 §5.7]. This model has no dark energy; it is the author's mass term as the only content of the dark sector.

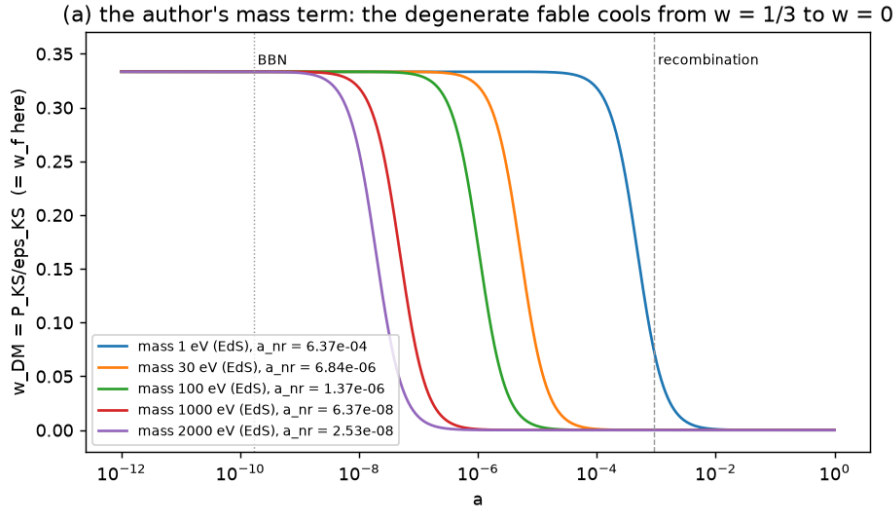


Figure 7: fable-cosmology/results/nb06_a_mass_wdm.png (notebook 06, section 5.2), runs (a): the author's mass term, $w_{\text{DM}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ ($= w_f$ there) against a : the degenerate fable cools from 1/3 to 0.

4.7.5 Runs (b): a mass term plus a bare cosmological constant — Λ CDM with fable dark matter

$W = V_0 + m_0\sigma$, fable4d, $\Omega_{\text{qp}0} = 0.265$, V_0 (a bare Lambda) closing the budget [nb06 §5.3, figure results/nb06_b_lambda_mass.png] (figure 8):

run	a_{nr}	$\Delta N_{\text{eff}} \text{ BBN}$	$\Delta N_{\text{eff}} \text{ rec}$	CPL $w_f (w_0, w_a)$	CPL $w_{\text{DE,inf}} (w_0, w_a)$	$\max w_{\text{DE,inf}} + 1 , a \geq 0.3$	q_0	$t_0 \text{ [Gyr]}$	$V_0 = \rho_U$
lambda-mass 30 eV	4.4658e-06	0.07179	19.65	(-0.7573, +1.0100)	(-1.0000000003, +1.33e-09)	1.47e-09	-0.52845	13.8000	0.685665
lambda-mass 100 eV	8.9687e-07	0.01442	19.65	(-0.7573, +1.0100)	(-1.0000000000, +5.34e-11)	5.91e-11	-0.52845	13.8000	0.685665
lambda-mass 1000 eV	4.1629e-08	0.0006692	19.65	(-0.7573, +1.0100)	(-1.0000000000, +1.16e-13)	1.31e-13	-0.52845	13.8000	0.685665
lambda-mass 2000 eV	1.6521e-08	0.0002656	19.65	(-0.7573, +1.0100)	(-1.0000000000, +1.72e-14)	2.40e-14	-0.52845	13.8000	0.685665
lambda-mass 30 eV (reference)	4.4658e-06	0.07179	19.65	(-0.7569, +1.0092)	(-1.0000000003, +1.31e-09)	1.44e-09	-0.52845	13.8000	0.685665

The mass does not vary ($w_{\text{DM,eff}} = w_{\text{DM}}$, $Q = 0$) and the condensate is the constant V_0 , so the dark energy an observer infers **is a cosmological constant**, to the precision of the quasiparticles' kinetic energy that the observer counts as dark energy. The intrinsic w_f today is -0.72124767 [file: `nb06_runs.csv`] — it mixes the gas and V_0 and is not a dark-energy equation of state. The fifth run ($a_i = 10^{-10}$, 701 rows, `results/nb06_fable4d_mass30eV.csv`) is Part VIII's reference case (section 4.7.12).

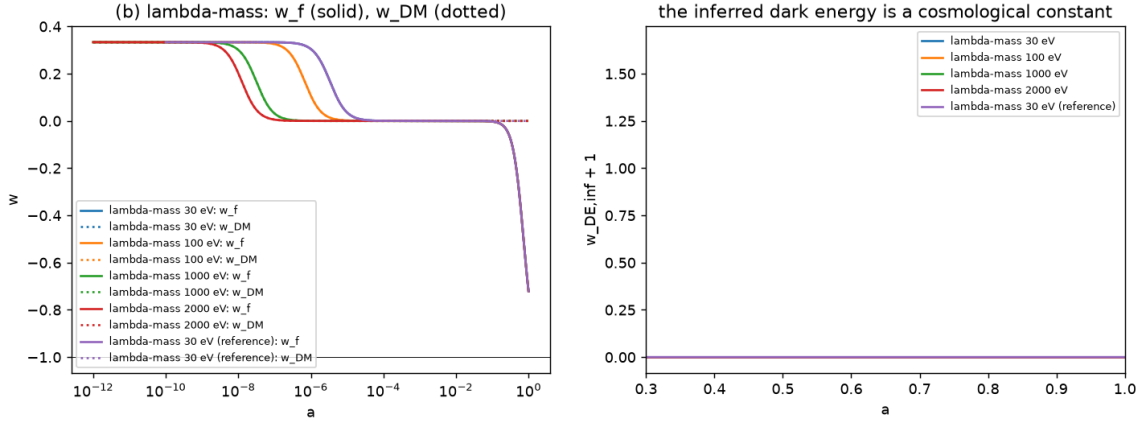


Figure 8: `fable-cosmology/results/nb06_b_lambda_mass.png` (notebook 06, section 5.3), runs (b): w_f (solid) and w_{DM} (dotted); and $w_{\text{DE,inf}} + 1$ on $0.3 \leq a \leq 1$: the inferred dark energy is a cosmological constant.

4.7.6 Runs (c): potentials with which the fable supplies the dark energy itself

power ($\nu = 0.236, 0.5$), `expdamp` (Part VI's $x_t = 1/2.21$), attractive quadratic ($g_q = -0.5$), each at $m_{\text{today}} = 30$ and 100 eV, `fable4d`, and Part VIII's mass-varying reference case (`power`, $\nu = 0.5$, $m_{\text{today}} = 100$ eV, $\Omega_{\text{qp}0} = 0.55$, 701 rows, `results/nb06_fable4d_power.csv`) [**nb06 §5.4**, figures `results/nb06_c_de_potentials.png`, `results/nb06_c_exchange_vacuum_stabilizer.png`] (figures 9 and 10):

run	a_{nr}	min m/m_{today} ($10^{-3} \leq a \leq 0.3$)	$w_f(1)$	min w_f	CPL w_f	poles of $w_{\text{DE},\text{inf}}$	$w = -1$ crossings	phantom in $0.3 \leq a \leq 1$?	CPL $w_{\text{DE},\text{inf}}$	fit from a	min c_s^2
power nu=0.236, 30 eV	2.22e-05	2.01e-01	-0.7212	-0.7212	(-0.783, +0.445)	3.43e-06, 0.533	4.55966e-06, 1	True	(-0.082, -7.986)	0.559	-0.611
power nu=0.236, 100 eV	4.47e-06	2.01e-01	-0.7212	-0.7212	(-0.783, +0.445)	6.89e-07, 0.533	9.15884e-07, 1	True	(-0.082, -7.986)	0.559	-0.611
power nu=0.5, 30 eV	0.669	7.91e-12	-0.7212	-1.0000	(-0.804, -0.358)	3.35e-06, 0.619	4.46748e-06, 1	True	(-0.249, -10.191)	0.650	-27
power nu=0.5, 100 eV	0.669	1.59e-12	-0.7212	-1.0000	(-0.804, -0.359)	6.73e-07, 0.619	8.97015e-07, 1	True	(-0.249, -10.191)	0.650	-27
expdamp xt=1/2.21, 30 eV	0.712	6.58e-12	-0.7212	-1.0000	(-0.829, -0.317)	3.35e-06, 0.625	4.46748e-06, 1	True	(-0.128, -11.778)	0.656	-21.8
expdamp xt=1/2.21, 100 eV	0.712	1.32e-12	-0.7212	-1.0000	(-0.829, -0.317)	6.73e-07, 0.625	8.97015e-07, 1	True	(-0.128, -11.779)	0.656	-21.8
quadratic gq=-0.5, 30 eV	0.643	8.93e-12	-0.7212	-1.0000	(-0.730, -0.483)	3.35e-06, 0.602	4.46748e-06, 1	True	(-0.180, -9.321)	0.632	-47.1
quadratic gq=-0.5, 100 eV	0.643	1.79e-12	-0.7212	-1.0000	(-0.730, -0.483)	6.73e-07, 0.602	8.97015e-07, 1	True	(-0.180, -9.321)	0.632	-47.1
power nu=0.5, 100 eV, $\Omega_{\text{qp}0} = 0.55$	4.21e-06	2.72e-01	-0.4215	-0.4215	(-0.448, +0.264)	8.96e-07, 0.63	1.18957e-06, 1	True	(-0.327, -9.682)	0.662	-0.364

and, for the same runs [nb06 §5.4]:

run	$c_s^2 < 0$ on ($a \geq 10^{-3}$)	$w_{\text{DM},\text{eff}}$ range ($a \geq 10^{-3}$)	$ \Delta E_{\text{vac}} /\rho_{\text{tot}}, a \geq 0.3$	$P_{\text{stab}}(1) [\rho_{c0}]$	ΔN_{eff} BBN
power nu=0.236, 30 eV	[0.00409, 1]	(-0.611, +9.85e-05)	1.314e+13	-0.8428	0.0718
power nu=0.236, 100 eV	[0.00194, 1]	(-0.611, +3.59e-06)	1.622e+15	-0.8428	0.0144
power nu=0.5, 30 eV	[0.608, 1]	(-27.6, +0.333)	2.229e+14	-0.8428	0.0718
power nu=0.5, 100 eV	[0.608, 1]	(-27.7, +0.333)	2.752e+16	-0.8428	0.0144
expdamp xt=1/2.21, 30 eV	[0.643, 1]	(-21.8, +0.333)	2.327e+14	-0.8428	0.0718
expdamp xt=1/2.21, 100 eV	[0.643, 1]	(-21.8, +0.333)	2.873e+16	-0.8428	0.0144
quadratic gq=-0.5, 30 eV	[0.592, 1]	(-47.1, +0.333)	2.031e+14	-0.8428	0.0718
quadratic gq=-0.5, 100 eV	[0.592, 1]	(-47.1, +0.333)	2.508e+16	-0.8428	0.0144
power nu=0.5, 100 eV, $\Omega_{\text{qp}0} = 0.55$	[0.001, 1]	(-0.364, -3.89e-05)	2.350e+14	-0.7003	0.0382

and smallest w_f over all 9 runs: -0.9999987921 (never below -1); every run has $c_s^2 < 0$ somewhere after $a = 1e-3$: True. Reading the tables:

- Only the two $\nu = 0.236$ runs and the reference run stay **massive** through the matter era. The $\nu = 0.5$, expdamp and quadratic runs with $\Omega_{qp0} = 0.265$ have $\min m/m_{\text{today}} \sim 10^{-12}$: the gap pins their mass near zero and the fable is a **massless gas** (radiation, $w = 1/3$) until $a_{\text{nr}} = 0.64\text{--}0.71$ — it supplies no dark matter at recombination (ΔN_{eff} at recombination equals its BBN value, e.g. 0.071792425 for power $\nu=0.5$, 30 eV [file: nb06_runs.csv]). Their CPL fits of w_f look like the supernova fits $(-0.861, -0.60)$, but they are not viable cosmologies.
- The observer's inferred dark energy $w_{\text{DE,inf}}$ has **poles** (its $\rho_{\text{DE,inf}}$ changes sign) inside $0.3 \leq a \leq 1$ in every run, and it is phantom and crosses -1 in every run; a CPL line is not a faithful description (the fits above the pole are in results/nb06_cpl_fits.csv).
- $c_s^2 < 0$ in every run: an **adiabatic instability**.
- The Dirac-sea energy that the no-sea functional drops changes along every run by 1.31×10^{13} to 2.87×10^{16} times the total density at $a \geq 0.3$.

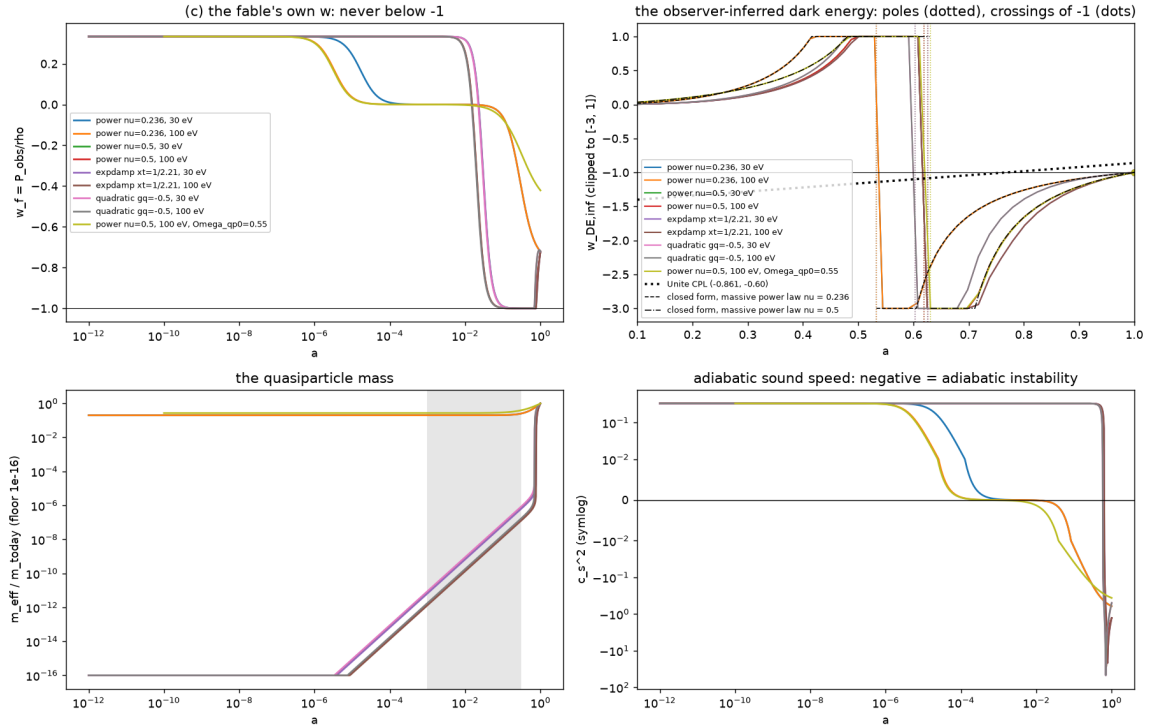


Figure 9: fable-cosmology/results/nb06_c_de_potentials.png (notebook 06, section 5.4), runs (c): w_f (never below -1), the observer's $w_{\text{DE,inf}}$ (clipped to $[-3, 1]$, poles and crossings marked), $m_{\text{eff}}/m_{\text{today}}$, and c_s^2 (negative below the line).

4.7.7 The inferred dark energy of a massive power law, in closed form

Once the gas is non-relativistic, $\sigma_8 \approx n \propto a^{-3}$, the condensate is $U = U_t a^{-3\nu}$ and the mass $m(a) = m_0 + \nu U_t a^{3(1-\nu)} / ((1-\nu) \sigma_t)$. Subtracting cold dark matter with today's mass leaves $\rho_{\text{DE,inf}} = U + (m(a) - m_{\text{today}}) n = U_t [a^{-3\nu} - \nu a^{-3}] / (1-\nu)$ and $P_{\text{DE}} = -U$, hence

$$w_{\text{DE,inf}}(a) = -\frac{1-\nu}{1-\nu a^{-3(1-\nu)}},$$

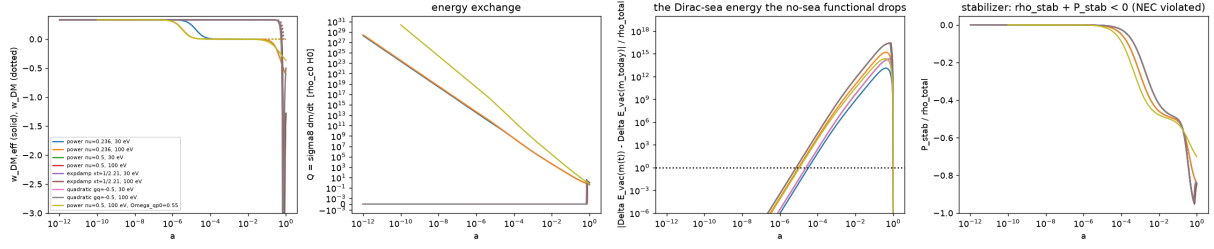


Figure 10: fable-cosmology/results/nb06_c_exchange_vacuum_stabilizer.png (notebook 06, section 5.4), runs (c): w_{DM} and $w_{\text{DM,eff}}$, the exchange Q , the Dirac-sea energy relative to the total density, and P_{stab}/ρ .

a pole at $a_{\text{pole}} = \nu^{1/(3(1-\nu))}$, $w > 0$ before it, phantom ($w < -1$) between it and today, and $w = -1$ at $a = 1$ up to the quasiparticles' kinetic energy, which makes it cross -1 just before today. It depends on ν alone [prose nb06 §5.4]. The runs agree [nb06 §5.4]:

```
power nu=0.236, 30 eV          w_DE,inf against -(1-nu)/(1 - nu a^(-3(1-nu))) on a >= 0.3: max
↪ deviation 1.2e-09; pole: formula 0.532600, run 0.532723; crossing of -1: 1.00000000
power nu=0.236, 100 eV        w_DE,inf against -(1-nu)/(1 - nu a^(-3(1-nu))) on a >= 0.3: max
↪ deviation 4.9e-11; pole: formula 0.532600, run 0.532723; crossing of -1: 1.00000000
power nu=0.5, 100 eV, Omega_qp0=0.55 w_DE,inf against -(1-nu)/(1 - nu a^(-3(1-nu))) on a >= 0.3: max
↪ deviation 4.2e-11; pole: formula 0.629961, run 0.630028; crossing of -1: 1.00000000
```

The mechanism is that of Das, Corasaniti and Khoury, Phys. Rev. D **73**, 083509 (2006): subtracting cold dark matter with today's mass when the mass was different leaves $\rho_{\text{DE}} + P_{\text{DE}} \approx (m_{\text{eff}}(a) - m_{\text{today}})n$, which is negative whenever m_{eff} grows; an apparent phantom with no phantom field.

4.7.8 The scans: where does a dark-energy fable stay massive through the matter era?

Notebook 06 scanned the shape parameters — power $\nu = 0.05 \dots 0.65$, quadratic $g_q = -0.95 \dots -0.05$, expdamp $x_t = 0.1 \dots 0.9$, and for power $\nu = 0.3$ and 0.5 the quasiparticle share $\Omega_{\text{qp0}} = 0.265 \dots 0.8$ — at $m_{\text{today}} = 1000 \text{ eV}$ and 100 eV (fable4d, 401 output rows, 45 + 45 runs), and re-computed every scan run independently as it ran [nb06 §5.6, files results/nb06_scan.csv, figures results/nb06_scan.png, results/nb06_scan_cpl.png] (figures 11 and 12). Its conclusions, verbatim:

```
scan points whose fable stays massive through 1e-3 <= a <= 0.3 (min m/m_today > 1e-3): 28 of 90: power,
↪ nu=0.05 (1000 eV), power, nu=0.1 (1000 eV), power, nu=0.15 (1000 eV), power, nu=0.2 (1000 eV),
↪ power, nu=0.25 (1000 eV), power nu=0.3, Omega_qp0=0.35 (1000 eV), power nu=0.3, Omega_qp0=0.45 (1000
↪ eV), power nu=0.3, Omega_qp0=0.55 (1000 eV), power nu=0.3, Omega_qp0=0.65 (1000 eV), power nu=0.3,
↪ Omega_qp0=0.8 (1000 eV), power nu=0.5, Omega_qp0=0.5 (1000 eV), power nu=0.5, Omega_qp0=0.55 (1000
↪ eV), power nu=0.5, Omega_qp0=0.65 (1000 eV), power nu=0.5, Omega_qp0=0.8 (1000 eV), power, nu=0.05
↪ (100 eV), power, nu=0.1 (100 eV), power, nu=0.15 (100 eV), power, nu=0.2 (100 eV), power, nu=0.25
↪ (100 eV), power nu=0.3, Omega_qp0=0.35 (100 eV), power nu=0.3, Omega_qp0=0.45 (100 eV), power
↪ nu=0.3, Omega_qp0=0.55 (100 eV), power nu=0.3, Omega_qp0=0.65 (100 eV), power nu=0.3, Omega_qp0=0.8
↪ (100 eV), power nu=0.5, Omega_qp0=0.5 (100 eV), power nu=0.5, Omega_qp0=0.55 (100 eV), power nu=0.5,
↪ Omega_qp0=0.65 (100 eV), power nu=0.5, Omega_qp0=0.8 (100 eV)
of these, with c_s^2 >= 0 after a = 1e-3: 0; smallest c_s^2 among them: -0.6469; largest: -0.0565
power law: 'massive through the matter era' <=> 'bare m0 > 0' at all 52 power scan points: True
the estimate Omega_qp0/U_t > nu/(1 - nu) agrees with the sign of m0 at 52 of 52 points; with Omega_qp0
↪ = 0.265 it allows nu < Omega_qp0/(1 - Omega_b0 - Omega_r0) = 0.2788
```

So: every quadratic and expdamp point is massless in the matter era; a power law stays massive exactly when its bare mass $m_0 = m_{\text{today}} - \nu \lambda \sigma_t^{\nu-1}$ is positive, i.e. when $\Omega_{\text{qp0}}/U_t > \nu/(1 - \nu)$; with the observed split $\Omega_{\text{qp0}} = 0.265$ that requires $\nu < 0.2788$ (massive up to $\nu = 0.25$, massless from $\nu = 0.3$ in the scan); a larger ν needs a larger quasiparticle share ($\Omega_{\text{qp0}} \geq 0.35$ for $\nu = 0.3$, ≥ 0.5 for $\nu = 0.5$). And every massive point has $c_s^2 < 0$ after $a = 10^{-3}$. (At high density $\nu \lambda \sigma^{\nu-1} \rightarrow 0$, so if $m_0 < 0$ the gap can only be met with $m \rightarrow 0^+$;

with $\sigma_t \approx n_t$ and $\varepsilon_{\text{KS}} \approx m_{\text{today}} n_t$ today this happens exactly when $\Omega_{\text{qp0}}/U_t < \nu/(1-\nu)$ [prose nb06 §5.6].) In the massless cases the smallest c_s^2 sits at the sharp release of the pinned gap, and its magnitude depends on how finely the output grid samples that release; its sign does not [prose nb06 §5.6]. The largest cross-check difference over the 90 scan runs is $4.43\text{e-}09$ [nb06 §5.10].

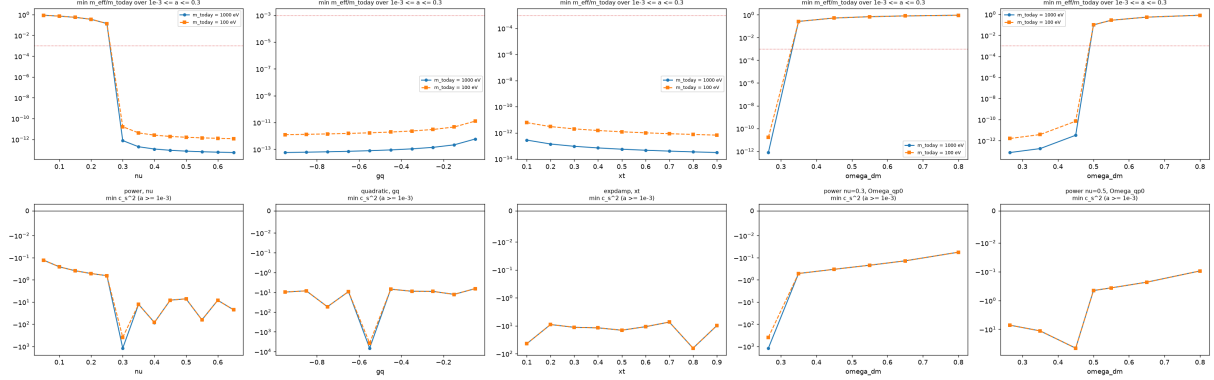


Figure 11: fable-cosmology/results/nb06_scan.png (notebook 06, section 5.6), the scans: per family, the smallest $m_{\text{eff}}/m_{\text{today}}$ over $10^{-3} \leq a \leq 0.3$ and the smallest c_s^2 ($a \geq 10^{-3}$) against the scanned parameter.

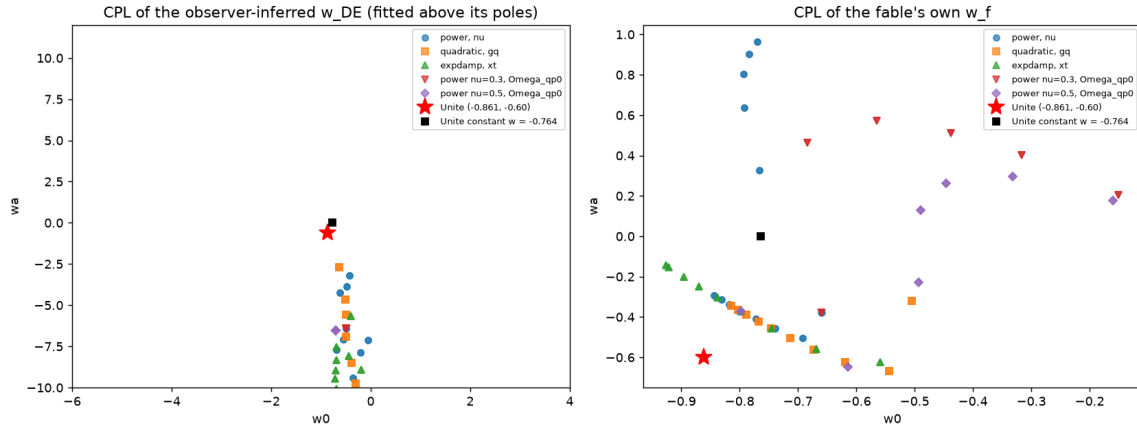


Figure 12: fable-cosmology/results/nb06_scan_cpl.png (notebook 06, section 5.6), the scans: the CPL points (w_0, w_a) of $w_{\text{DE,inf}}$ (fitted above its poles) and of w_f .

4.7.9 How heavy must a fable that is all of the dark matter be?

[nb06 §5.7, file results/nb06_dm_mass_bounds.csv, figure results/nb06_dm_mass_bounds.png] (figure 13)

bound	fable mass	detail
$\Delta N_{\text{eff}}(\text{BBN}) < 0.3$	$m > 10.2645 \text{ eV}$	fit over 17 runs: $\Delta N_{\text{eff}} = 6.6922 (m/\text{eV})^{-1.33333}$, max $ \ln \text{residual} = 7.1\text{e-}05$
$\Delta N_{\text{eff}}(\text{BBN}) < 0.2$	$m > 13.9126 \text{ eV}$	same fit
free streaming: v_{rms0} matched to thermal WDM of 3.3 keV (Viel et al. 2013)	$m \approx 1124.92 \text{ eV}$	$v_{\text{rms0}} = 8.2629\text{e-}03 \text{ km/s}$

bound	fable mass	detail
free streaming: $v_{\text{rms}0}$ matched to thermal WDM of 5.3 keV (Iršič et al. 2017)	$m \approx 1806.69 \text{ eV}$	$v_{\text{rms}0} = 4.3932\text{e-}03 \text{ km/s}$
Tremaine–Gunn, $g = 8$, $\sigma = 10 \text{ km/s}$, $r_c = 0.3 \text{ kpc}$	$m > 231.315 \text{ eV}$	$m^4 > 9 h^3 / (2 (2\pi)^{5/2} g G \sigma r_c^2)$
Tremaine–Gunn, $g = 8$, $\sigma = 7 \text{ km/s}$, $r_c = 0.15 \text{ kpc}$	$m > 357.639 \text{ eV}$	same

$\Delta N_{\text{eff}} \propto m^{-4/3}$ because $\Delta N_{\text{eff}}(\text{BBN})$ depends only on k_{F0} , which $\varepsilon_{\text{KS}}(m, k_{F0}) = \Omega_{\text{qp}0}$ fixes, and $k_{F0} \propto m^{-1/3}$ for a non-relativistic gas today; at 1, 30, 100, 1000, 2000 eV it is 6.69227, 0.0717924, 0.0144181, 0.000669238, 0.00026561. A degenerate gas has the velocity dispersion of a filled Fermi sphere, $v_{\text{rms}} = \sqrt{3/5} k_F/m$: 4.4870e+00, 2.6132e+00, 2.0827e-01, 9.6670e-03, 3.8364e-03 km/s at 10, 15, 100, 1000, 2000 eV. The Lyman-alpha-allowed thermal-relic warm dark matter ($m_{\text{WDM}} \gtrsim 3.3 \text{ keV}$ and 5.3 keV) has $v_{\text{rms}0} = 8.2629\text{e-}03$ and 4.3932e-03 km/s; a degenerate fable with the same $v_{\text{rms}0}$ needs $m \approx 1.1\text{--}1.8 \text{ keV}$ — a velocity-matching estimate, not a transfer-function computation. The Tremaine–Gunn bound uses illustrative dwarf-spheroidal values (inputs, not results). The free-streaming bound is far stronger than ΔN_{eff} . At such masses the change of w_{DM} from 1/3 to 0 happens around $a_{\text{nr}} = 4.16 \times 10^{-8}$ (1 keV) or 1.65×10^{-8} (2 keV).

4.7.10 Runs (d): without the stabilizer, the hidden sheet moves and the Newton constant varies

`fable8d` for the mass term at 1 and 100 eV and `lambda-mass` at 30 eV and 1 keV, and two controls: radiation alone (`-no-fable -omega-b 0`), which must keep the hidden sheet exactly frozen ($F = 0$), and radiation plus baryons (`-no-fable`), in which dust alone drives it. $G(a)/G(1) = 1/v$ (the observed Newton constant is $G_4 = G_8/V_{\text{hid}} \propto 1/v$) [nb06 §5.8, file `results/nb06_nogo_8d.csv`, figure `results/nb06_d_nogo_8d.png`] (figure 14):

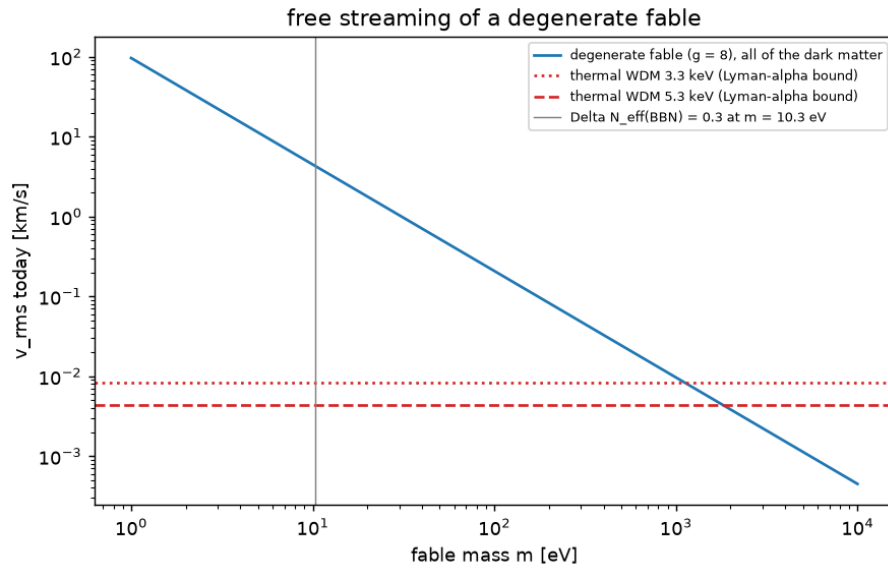


Figure 13: fable-cosmology/results/nb06_dm_mass_bounds.png (notebook 06, section 5.7): the free streaming of a degenerate fable: v_{rms} today against the fable mass, with the warm-dark-matter velocities.

run	$B(a_i) = C(a_i)$	$G(a_i)/G(1)$	$G(\text{BBN})/G(1)$	$\Delta G/G$ since BBN	$ d \ln G / dt $ today [/yr]	$H_B/H_A(1) = H_C/H_A(1)$	max constraint residual	$\Omega_{\text{sum}}(1)$	q_0	t_0 [Gyr]
mass 1 eV, fable8d	2.3224e-03	3.4376e+10	3.4376e+10	-1.0000e+00	2.7571e-10	0.999959	1.16e-08	6.99967	2.5000	4.145
mass 100 eV, fable8d	4.8207e-05	1.8516e+17	1.8516e+17	-1.0000e+00	2.7571e-10	0.999963	2.12e-08	6.99971	2.5000	4.145
lambda-mass 30 eV, fable8d	9.8881e-04	1.0461e+12	1.0461e+12	-1.0000e+00	2.7570e-10	0.999902	2.47e-08	6.99922	-0.8425	9.472
lambda-mass 1 keV, fable8d	9.8389e-04	1.0671e+12	1.0671e+12	-1.0000e+00	2.7570e-10	0.999902	2.07e-08	6.99922	-0.8425	9.472
radiation only, fable8d	1.0000e+00	1.0000e+00	1.0000e+00	0.0000e+00	0.0000e+00	0.000000	9.89e-16	1.00000	1.0000	755.851
radiation + baryons, fable8d	6.2532e-03	6.5402e+08	6.5402e+08	-1.0000e+00	2.3096e-11	0.994782	1.04e-08	6.95831	2.4935	49.354

Radiation alone keeps $H_B = H_C = 0$ exactly ($\max |H_B/H_A| = 0.0e+00$; $F = 0$); dust or vacuum energy drives the hidden sheet to the 7-dimensional isotropic attractor ($H_B/H_A \rightarrow 1$), and the observed Newton constant changes by 6.5×10^8 to 1.9×10^{17} between the start and today, with $|d \ln G/dt|$ today 2.3×10^{-11} to 2.76×10^{-10} /yr against the lunar-laser-ranging bound 10^{-13} /yr, and $\Delta G/G$ since nucleosynthesis -1 against the bound of about 0.1. The closure today then needs $\Omega_{\text{sum}}(1) \approx 7$ ($\sum \Omega_i = 1 + (3H_B^2 + 9H_A H_B + 3H_A H_C + 3H_B H_C)/(3H_A^2)$, which is 7 on the attractor). The two controls have nothing to shoot, so their H_A today is not 1 and their ages are not those of our universe. **fable8d is a no-go for observers; that is why the physical model is the stabilized fable4d.** The last run, fable8d -freeze-hidden (lambda-mass 30 eV), agrees with the algebraic fable4d: $\max |dH_A|/H_A = 4.1e-09$, $\max |dt|/t = 9.8e-09$, $\max |dw_f| = 0.0e+00$, $\max |H_B| = 0.0e+00$.

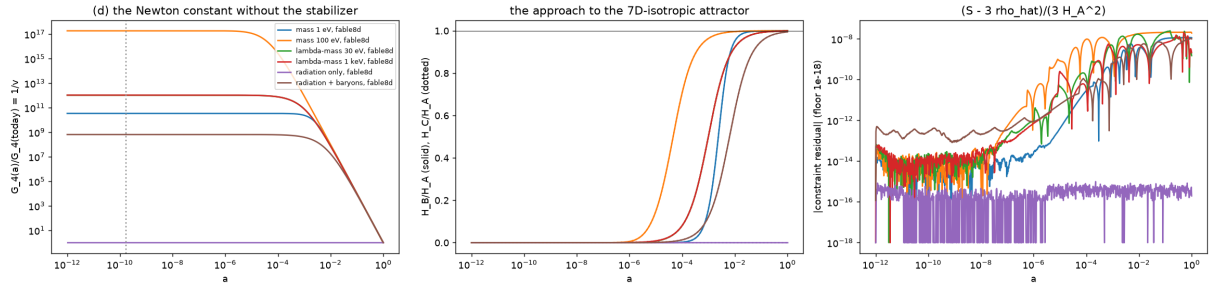


Figure 14: fable-cosmology/results/nb06_d_nogo_8d.png (notebook 06, section 5.8), runs (d): $G_4(a)/G_4(\text{today}) = 1/v$, H_B/H_A and H_C/H_A (the approach to the 7-dimensional isotropic attractor), and the constraint residual $(S - 3\hat{\rho})/(3H_A^2)$.

4.7.11 The cases the solver refuses, and why

Each exits with code 1 and one line of physical reason [nb06 §5.9]:

```
$ fable_fermion fable4d --potential lorentz --points 1001
exit code 1: fable_fermion: rho_hat = H_A^2 <= 0 from N = -0.7597808064 (a = 0.4677689478) on: the
↳ Kohn-Sham ground state has negative energy there (m_eff = -3.802078e-2, sigma8 = -9.789374e-4,
↳ rho_f = -4.830421e-1 (U = -4.847891e-1) against rho_r + rho_b = 4.830421e-1); H_A^2 = rho_hat is
↳ impossible (the onset located past the last accepted CVODE step and bisected)
$ fable_fermion fable4d --potential quadratic --param gq=0.5 --points 1001
exit code 1: fable_fermion: gap-branch jump at N = -0.0037262796 (a = 0.9962806544): a first-order
↳ transition of the Kohn-Sham ground state, the lowest-energy gap root jumps from the branch m_eff =
↳ -1.518260e0 (rho_f = 6.856759e-1) to the branch m_eff = 4.083552e2 (rho_f = 9.536521e-1); the
↳ right-hand side is discontinuous there, and energy conservation across it would need the Maxwell
↳ construction, which this solver does not implement (the transition detected inside the right-hand
↳ side and bisected in N to adjacent doubles)
$ fable_fermion fable8d --potential mass --direction backward
exit code 1: fable_fermion: fable8d runs forward only (design review E4): backward in time the shear
↳ modes grow as A^-3 v^-1 relative to H_A ~ A^-2, the solution runs to the 8D Kasner point H_B/H_A =
↳ -0.2929 and the constraint drifts to O(1); use fable4d --direction backward
all three refused with exit code 1 and the expected reason
```

lorentz: the Kohn–Sham ground state moves to the negative- σ branch, whose energy is negative (a well of W on $\sigma < 0$ that the quantum gas reaches and the classical $s \geq 0$ field never did). **Repulsive quadratic:** a first-order transition (section 4.6.1). **fable8d backward:** the constraint grows backward (section 4.3.4). (These lines are the output of the committed notebook 06, executed on the final solver of 2bc936d. That commit detects both the negative-energy onset and the gap-branch jump inside the right-hand side, on every internal CVODE step, and bisects them to adjacent doubles, independently of the output grid: **lorentz** at $a = 0.4677689478$ “whatever -points is”, the repulsive-quadratic jump at $a = 0.9962806544$ “for any grid”, with **scipy** agreeing to 8×10^{-12} [commit 2bc936d]. The first execution of the notebook, at 22:53 PDT on the solver before that commit, had bracketed the jump only between two output rows and printed gap-branch jump (sign of m_{eff}) between $N = -0.027631021115929855$ and

$N = 0$: $m = -3.1667192351800344e-1 \rightarrow 4.060727073199006e2$; the lorentz line was the same then as now.)

4.7.12 Three implementations, one answer: CVODE, scipy and Mathematica

Part VIII's own solution. Section 33 solves two `fable4d` cases in Mathematica, independently of the reference script: every tenth row (71 of 701) by exact algebra from Section 29's closed forms (with `FindRoot` for the gap), the age of `mass30eV` by `NIntegrate` between the rows, and the age of power by `NDSolve` of the pair $(m(N), \ln t(N))$ with the **differentiated** gap equation

$$\frac{dm}{dN} = -\frac{W''(\sigma) (\partial\sigma/\partial k_F) k_F}{1 - W''(\sigma) \partial\sigma/\partial m}$$

instead of solving it [**prose VIII §33**]. The run printed [**displayed VIII §33**], `claude-fable/run_fermion_fable_part8.log`:

```
[0.242 s] fable4d mass30eV: 71 rows by exact algebra, the age by NIntegrate between the rows
[0.603 s] fable4d power: NDSolve of (m(N), ln t(N)) with the differentiated gap equation
[0.047 s] fable4d power: 71 rows by exact algebra (FindRoot of the gap at every row)
mass30eV: m0 = 30 eV = 12182.18122 E_c; kF0 = 0.0544035545089 E_c = 0.00013397491 eV; V0 =
↪ 0.6856647126; a_nr = kF0/m0 = 4.46583e-6; t0 = 0.951246513195 / H0
power: m_today = 100 eV; kF0 = 0.0464558024014 E_c; lam = 217.736704888; m0 = 27.15187044 eV;
↪ U_today = 0.4006647126; w_f(1) = -0.421457436338; t0 = 0.9218031611 / H0
P_stab(a) [units of rho_c0] at a = 1e-10, 1e-6, 1e-3, 1: mass30eV {-2.4626e28, -6.53147e16,
↪ -1.57121e8, -0.842786}; power {-2.46243e28, -4.87299e16, -9.93078e7, -0.700286}
```

and asserted [**proved VIII §33**]

- `fable4d RUNS [THE RESULT]: mass30eV - flat today ( $H_A(1) == 1$ ),  $\text{eps}_{KS}(1) == 0.265$ , ultra-relativistic at the start ( $w_f(a_i) = 1/3$  to  $1e-8$ ), non-relativistic at  $a_{nr} = kF0/m0 == 4.16e-4$   $(m/eV)^{-4/3}$  (1%), and  $w_f \geq -1$  at every row`
- `fable4d RUNS [THE RESULT]: power - flat today,  $m_{eff}(1) == 100$  eV,  $m0 == 27.15$  eV  $> 0$ ,  $m_{eff}$  INCREASES monotonically from  $m0$  (early) to  $m_{today}$ , and  $w_f \geq -1$  at every row`
- `fable4d RUNS [has content]:` the differentiated gap equation carries $m(N)$ along the history - `NDSolve`'s m agrees with the per-row `FindRoot` to $1e-15$ at all 71 rows
- `fable4d RUNS [THE RESULT]:` the stabilizer $P_{stab} = -F_{total}/2$ is NEGATIVE at every row of both runs - the null-energy violation along x_0 is present along the whole history
- `fable4d RUNS [fidelity]:` the reference CSVs have the expected header and 701 rows
- `fable4d RUNS [fidelity]:` at every tenth row ALL TEN columns agree with `fable-cosmology/reference/mathematica_fable4d_mass30eV.csv` and `_power.csv` to $1e-10$ (relative; P_{obs_f} relative to ρ_f ; N and w_f absolute)

The comparison with the reference CSVs printed max difference per column `{0., 0., 0., 0., 0., 0., 0., 0.}` for both files (three-digit formatting). When this log was recorded the Rust CSVs `results/nb06_fable4d_mass30eV.csv` and `nb06_fable4d_power.csv` did not yet exist, and the conditional comparison with them was skipped, not failed (NOTE Rust CSV not found; that comparison was skipped, not failed); notebook 06 now writes them, and the three-way comparison below includes them. The final run of the notebook (`claude-fable/run_fermion_fable_final.log`, commit 1671155), made after notebook 06 had written them, ran both comparisons and passed (the absolute path before `fable-cosmology` shortened to ...):

```

Rust ...fable-cosmology\results\nb06_fable4d_mass30eV.csv: 71 rows compared; max difference
↪ per column {a, t, H_A, rho_f, P_obs_f, w_f, m_eff, sigma, kF_over_m}: {4.9e-14, 2.35e-9,
↪ 3.35e-11, 1.87e-13, 6.22e-15, 6.8e-15, 2.22e-15, 1.39e-13, 4.84e-14}
PASS fable4d RUNS [fidelity]: the Rust solver's mass30eV run agrees with this cell to 1e-6
↪ in every column, at every one of the rows its grid covers
Rust ...fable-cosmology\results\nb06_fable4d_power.csv: 71 rows compared; max difference per
↪ column {a, t, H_A, rho_f, P_obs_f, w_f, m_eff, sigma, kF_over_m}: {4.9e-14, 1.39e-9,
↪ 3.36e-11, 1.92e-13, 1.34e-14, 1.39e-14, 5.04e-14, 1.41e-13, 8.82e-14}
PASS fable4d RUNS [fidelity]: the Rust solver's power run agrees with this cell to 1e-6 in
↪ every column, at every one of the rows its grid covers

```

The reference script `fable-cosmology/reference/make_reference_fermion.wls` writes `mathematica_fable4d_mass30eV.csv` and `mathematica_fable4d_power.csv`: 701 values of N from $\ln 10^{-10}$ to 0, columns N , a , t , H_A , ρ_f , $P_{\text{obs},f}$, w_f , m_{eff} , σ , k_F/m ; every row is exact algebra at 40 significant digits (the power-law gap by Brent's method at 45 digits), the age by `NDSolve` at working precision 30 with the stable series of the Fermi-sea integrals for $k_F/|m| < 1/4$ [file: its header].

Three-way [nb06 §5.5, file `results/nb06_threeway.csv`] — `CVODE` (Rust), `scipy` (the independent implementation), and `Mathematica` (the reference CSVs), column by column, on both reference cases (the table of the notebook, transposed: one row per compared column):

column	nb06_fable4d_mass30eV	nb06_fable4d_power
<i>Rust – Mathematica:</i>		
t	2.352e-09	1.500e-09
H_A	3.338e-11	3.353e-11
ρ_f	1.271e-14	5.540e-14
$P_{\text{obs},f}$	1.603e-13	6.990e-13
m_{eff}	1.792e-15	5.032e-14
σ	9.938e-15	5.242e-14
k_F/m	6.506e-15	5.128e-14
w_f (abs)	3.608e-15	8.160e-15
<i>scipy – Mathematica:</i>		
H_A	3.347e-11	3.362e-11
ρ_f	1.419e-12	1.362e-12
$P_{\text{obs},f}$	1.172e-12	1.170e-12
m_{eff}	2.986e-16	4.586e-14
$t(A=1)$	2.665e-14	4.241e-14

written: `results/nb06_threeway.csv`; `CVODE`, `scipy` and `Mathematica` agree to better than $1e-6$ on both reference cases; in the power case m_{eff} runs from 27.1519 eV ($a = 1e-10$) to 100.0000 eV (today).

Every run re-computed independently [nb06 §5.10, file `results/nb06_crosscheck.csv`]: the independent implementation takes the normalized parameters from the `# params`: line of each CSV and re-computes the run — every Fermi-sea integral by adaptive quadrature, every gap root by `brentq`, `fable8d` by `DOP853` at `rto1 = 1e-11`, `fable4d` algebraically with $t(A=1)$ by adaptive quadrature — and checks the Friedmann consistency of the solver's columns and the covariant conservation of the independently computed fluid (Richardson-extrapolated central differences). The notebook fails if any difference exceeds 10^{-6} . It printed **largest difference over the 29 kept runs**: $1.08e-08$; **over the 90 scan runs**: $4.43e-09$; **largest conservation residual of the independent fluid**: $1.24e-07$.

4.7.13 The consolidated table of every kept run

[nb06 §5.11, files `results/nb06_runs.csv` (29 rows, 34 columns), `results/nb06_cpl_fits.csv`], verbatim (table 37, on the next landscape page):

Table 37: The consolidated table of every kept run, as notebook 06 printed it (section 5.11).

run	m [eV]	a_nr	dNeff	BBN	CPL w_f	CPL w_DE,inf	min cs2	vac/rho	a>=.3 G(a_i)/G(1)	Gdot/G /yr
mass 1 eV (EdS)	1	0.000637	36.75	(-0.000, +0.000)	(+nan, +nan)	1.35e-07	0.000e+00	1.000e+00	0.000e+00	0.000e+00
mass 30 eV (EdS)	30	6.84e-06	0.3943	(-0.000, +0.000)	(+nan, +nan)	1.56e-11	0.000e+00	1.000e+00	0.000e+00	0.000e+00
mass 100 eV (EdS)	100	1.37e-06	0.07918	(-0.000, +0.000)	(+nan, +nan)	6.28e-13	0.000e+00	1.000e+00	0.000e+00	0.000e+00
mass 1000 eV (EdS)	1000	6.37e-08	0.003675	(-0.000, +0.000)	(+nan, +nan)	1.35e-15	0.000e+00	1.000e+00	0.000e+00	0.000e+00
mass 2000 eV (EdS)	2000	2.53e-08	0.001459	(-0.000, +0.000)	(+nan, +nan)	2.13e-16	0.000e+00	1.000e+00	0.000e+00	0.000e+00
lambda-mass 30 eV	30	4.47e-06	0.07179	(-0.757, +1.010)	(-1.000, +0.000)	6.65e-12	0.000e+00	1.000e+00	0.000e+00	0.000e+00
lambda-mass 100 eV	100	8.97e-07	0.01442	(-0.757, +1.010)	(-1.000, +0.000)	2.68e-13	0.000e+00	1.000e+00	0.000e+00	0.000e+00
lambda-mass 1000 eV	1000	4.16e-08	0.0006692	(-0.757, +1.010)	(-1.000, +0.000)	5.78e-16	0.000e+00	1.000e+00	0.000e+00	0.000e+00
lambda-mass 2000 eV	2000	1.65e-08	0.0002656	(-0.757, +1.010)	(-1.000, +0.000)	9.1e-17	0.000e+00	1.000e+00	0.000e+00	0.000e+00
lambda-mass 30 eV (reference)	30	4.47e-06	0.07179	(-0.757, +1.009)	(-1.000, +0.000)	6.65e-12	0.000e+00	1.000e+00	0.000e+00	0.000e+00
power nu=0.236, 30 eV	30	2.22e-05	0.07179	(-0.783, +0.445)	(-0.082, -7.986)	-0.611	1.314e+13	1.000e+00	0.000e+00	0.000e+00
power nu=0.236, 100 eV	100	4.47e-06	0.01442	(-0.783, +0.445)	(-0.082, -7.986)	-0.611	1.622e+15	1.000e+00	0.000e+00	0.000e+00
power nu=0.5, 30 eV	30	0.669	0.07179	(-0.804, -0.358)	(-0.249, -10.191)	-27	2.229e+14	1.000e+00	0.000e+00	0.000e+00
power nu=0.5, 100 eV	100	0.669	0.01442	(-0.804, -0.359)	(-0.249, -10.191)	-27	2.752e+16	1.000e+00	0.000e+00	0.000e+00
expdamp xt=1/2.21, 30 eV	30	0.712	0.07179	(-0.829, -0.317)	(-0.128, -11.778)	-21.8	2.327e+14	1.000e+00	0.000e+00	0.000e+00
expdamp xt=1/2.21, 100 eV	100	0.712	0.01442	(-0.829, -0.317)	(-0.128, -11.779)	-21.8	2.873e+16	1.000e+00	0.000e+00	0.000e+00
quadratic gq=-0.5, 30 eV	30	0.643	0.07179	(-0.730, -0.483)	(-0.180, -9.321)	-47.1	2.031e+14	1.000e+00	0.000e+00	0.000e+00
quadratic gq=-0.5, 100 eV	100	0.643	0.01442	(-0.730, -0.483)	(-0.180, -9.321)	-47.1	2.508e+16	1.000e+00	0.000e+00	0.000e+00
power nu=0.5, 100 eV, Omega_qp0=0.55	100	4.21e-06	0.03817	(-0.448, +0.264)	(-0.327, -9.682)	-0.364	2.350e+14	1.000e+00	0.000e+00	0.000e+00
power nu=0.236, 1 eV (Delta N_eff)	1	0.00207	6.692	(-0.783, +0.445)	(-0.082, -7.985)	-0.611	1.622e+07	1.000e+00	0.000e+00	0.000e+00
power nu=0.5, 1 eV (Delta N_eff)	1	0.67	6.692	(-0.807, -0.347)	(-0.251, -10.172)	-21.7	2.720e+08	1.000e+00	0.000e+00	0.000e+00
expdamp xt=1/2.21, 1 eV (Delta N_eff)	1	0.712	6.692	(-0.832, -0.305)	(-0.130, -11.753)	-21.7	2.850e+08	1.000e+00	0.000e+00	0.000e+00
quadratic gq=-0.5, 1 eV (Delta N_eff)	1	0.643	6.692	(-0.732, -0.473)	(-0.182, -9.306)	-46.4	2.490e+08	1.000e+00	0.000e+00	0.000e+00
mass 1 eV, fable8d	1	0.00124	521.5	(-0.000, +0.000)	(+0.394, -8.104)	3.93e-07	0.000e+00	3.438e+10	2.757e-10	2.757e-10
mass 100 eV, fable8d	100	2.66e-06	1.124	(-0.000, +0.000)	(+0.394, -8.104)	1.83e-12	0.000e+00	1.852e+17	2.757e-10	2.757e-10
lambda-mass 30 eV, fable8d	30	4.47e-06	0.07179	(-1.103, +1.715)	(-1.716, +5.459)	5.13e-12	0.000e+00	1.046e+12	2.757e-10	2.757e-10
lambda-mass 1 keV, fable8d	1000	4.16e-08	0.0006692	(-1.103, +1.715)	(-1.716, +5.459)	4.46e-16	0.000e+00	1.067e+12	2.757e-10	2.757e-10
radiation only, fable8d	nan	nan	0	(+nan, +nan)	(+nan, +nan)	nan	0.000e+00	1.000e+00	0.000e+00	0.000e+00
radiation + baryons, fable8d	nan	nan	0	(+nan, +nan)	(+0.386, -8.020)	nan	0.000e+00	6.540e+08	2.310e-11	2.310e-11

(The `vac/rho` column is 0 for constant-mass runs, where the sea energy is a constant absorbed into V_0 .)

4.7.14 The figures

All written by notebook 06 under `fable-cosmology/results/` (and those of notebook 05, which show the fluid itself):

figure	what it shows
<code>nb06_a_mass_wdm.png</code> (figure 7)	runs (a): the author's mass term, $w_{\text{DM}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$ ($= w_f$ there) against a : the degenerate fable cools from $1/3$ to 0
<code>nb06_b_lambda_mass.png</code> (figure 8)	runs (b): w_f (solid) and w_{DM} (dotted); and $w_{\text{DE,inf}} + 1$ on $0.3 \leq a \leq 1$: the inferred dark energy is a cosmological constant
<code>nb06_c_de_potentials.png</code> (figure 9)	runs (c): w_f (never below -1), the observer's $w_{\text{DE,inf}}$ (clipped to $[-3, 1]$, poles and crossings marked), $m_{\text{eff}}/m_{\text{today}}$, and c_s^2 (negative below the line)
<code>nb06_c_exchange_vacuum_stabilizer.png</code> (figure 10)	runs (c): w_{DM} and $w_{\text{DM,eff}}$, the exchange Q , the Dirac-sea energy relative to the total density, and P_{stab}/ρ
<code>nb06_scan.png</code> , <code>nb06_scan_cpl.png</code> (figures 11, 12)	the scans: per family, the smallest $m_{\text{eff}}/m_{\text{today}}$ over $10^{-3} \leq a \leq 0.3$ and the smallest c_s^2 ($a \geq 10^{-3}$) against the scanned parameter; the CPL points (w_0, w_a) of $w_{\text{DE,inf}}$ (fitted above its poles) and of w_f
<code>nb06_dm_mass_bounds.png</code> (figure 13)	the free streaming of a degenerate fable: v_{rms} today against the fable mass, with the warm-dark-matter velocities
<code>nb06_d_nogo_8d.png</code> (figure 14)	runs (d): $G_4(a)/G_4(\text{today}) = 1/v$, H_B/H_A and H_C/H_A (the approach to the 7-dimensional isotropic attractor), and the constraint residual $(S - 3\hat{\rho})/(3H_A^2)$
<code>nb05_w_ks.png</code> , <code>nb05_ks_accuracy.png</code> , <code>nb05_classical_vs_quantum.png</code> , <code>nb05_gap_roots.png</code> , <code>nb05_minimax.png</code> , <code>nb05_preuniverse_cooling.png</code> , <code>nb05_vacuum_energy.png</code> (figures 2, 1, 3, 6, 5, 15, 4)	the Kohn–Sham fluid: $w_{\text{KS}}(x)$, the accuracy of the stable forms, classical vs quantum w , the gap roots, the minimax and Walecka functionals, the cooling on the pre-universe, the Dirac-sea energy

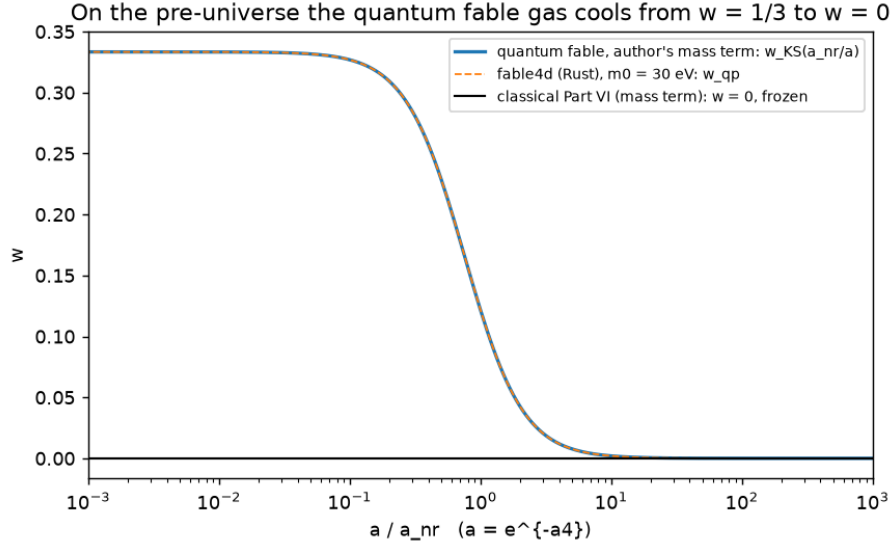


Figure 15: fable-cosmology/results/nb05_preuniverse_cooling.png (notebook 05): the cooling on the pre-universe — the quantum fable gas of the author’s mass term goes from $w = 1/3$ to $w = 0$ as the observed sheet expands, while the classical Part VI fable’s $w = 0$ is frozen (section 4.1.6).

4.8 The answers to [1] and [2]

The author asked: [1] “Does this system provide a physical mechanism for a time-varying dark energy equation of state (w)?” [2] “[...] time-varying dark matter equation of state (w)?” The answers below rest on the runs of section 4.7 and the theorems of sections 4.2–4.5. Notebook 06 composes its own statement of the answers from its tables and asserts every qualitative claim in it; its headlines read [nb06 §7]: “**Formally yes, in the stabilized model, but in none of the runs is it a viable one.**” for [1], and “**Yes, intrinsically, but for an allowed mass the variation is over long before recombination.**” for [2].

4.8.1 Answer [1]: a time-varying dark-energy equation of state

Yes, as a mechanism — in the stabilized model, and only with a mass-varying potential. None of the runs is a viable cosmology, and the mechanism carries every one of the following caveats.

What the mechanism is. In the Kohn–Sham ground state the fable’s energy is $\rho = \rho_{qp} + \rho_U$, and the mean-field condensate $\rho_U = W(\sigma_7) - \sigma_7 W'(\sigma_7)$ has $P = -\rho_U$ along all seven spatial directions: a vacuum energy. When W is not linear, ρ_U changes as the density falls, the quasiparticle mass $m_{\text{eff}} = W'(\sigma_7)$ changes with it, and the two parts exchange energy at the rate $Q = \sigma_8 dm/dt$ (section 4.5.10). The dark energy that an observer infers therefore varies in time. For a massive power law $W = m_0 \sigma + \lambda \sigma^\nu$ it is, in closed form and confirmed by the runs to 1.2×10^{-9} ,

$$w_{\text{DE,inf}}(a) = -\frac{1 - \nu}{1 - \nu a^{-3(1-\nu)}},$$

with a pole at $a = \nu^{1/(3(1-\nu))}$ (0.5326 for $\nu = 0.236$) (section 4.7.7). The observer’s CPL fits of the eight dark-energy runs of section 4.7.6 lie at distances from the supernova fit $(w_0, w_a) = (-0.861, -0.60)$ of 7.43 (power $\nu = 0.236$: $(-0.082, -7.986)$) to 11.2 (expdamp: $(-0.128, -11.778)$) [file: fable-cosmology/results/nb06_cpl_fits.csv], and the fits are not faithful, because $w_{\text{DE,inf}}$ has a pole inside $0.3 \leq a \leq 1$ in all of them.

What does not give it. With the author's mass term alone there is no dark energy at all: the fable is a free gas that closes the budget, an Einstein–de Sitter-like universe with $\Omega_{f0} = 0.95066$ and $q_0 = 0.50005$ (section 4.7.4). With a mass term plus a bare V_0 (**lambda-mass**), the observer infers **exactly a cosmological constant**: CPL $(w_0, w_a) = (-1.000000, +1.3 \times 10^{-9})$, $(-1.000000, +5.3 \times 10^{-11})$, $(-1.000000, +1.2 \times 10^{-13})$, $(-1.000000, +1.7 \times 10^{-14})$ at 30, 100, 1000, 2000 eV — no time variation (section 4.7.5). V_0 is a bare cosmological constant and is reported as such.

The caveats, each quantified:

1. **No true phantom — but an apparent phantom and phantom crossing in the observer's fit.** The fable's own equation of state never goes below -1 : $\rho + P_{\text{obs}} = \omega_F n/v \geq 0$ is an identity (section 4.5.9), and the smallest w_f over every run is -0.9999987921 . The classical Part VI crossing of -1 (at $V'(s^*) = 0$, section 3.1.12.7) does **not** survive quantization. The **observer-inferred** dark energy, however, is phantom or crosses -1 inside $0.3 \leq a \leq 1$ in 9 of 9 runs of section 4.7.6 and in 90 of 90 scan points: subtracting cold dark matter with today's mass when the mass was different leaves $\rho_{\text{DE}} + P_{\text{DE}} \approx (m_{\text{eff}}(a) - m_{\text{today}}) n$, negative whenever m_{eff} grows (the mechanism of Das, Corasaniti and Khoury, Phys. Rev. D **73**, 083509 (2006)), with no phantom field. For the massive power law the crossing lies in the last output interval before today ($a > 0.9727$), where the quasiparticles' kinetic energy, of relative size $\frac{3}{10}(k_F/m)^2$, finally exceeds $(m_{\text{today}} - m_{\text{eff}}(a)) n$ [**nb06 §7**].
2. **Mass-varying dark energy with massive dark matter needs a power law with $\nu < 0.279$.** In the scans 28 of 90 points keep the fable massive over $10^{-3} \leq a \leq 0.3$, all of them power laws; every **quadratic** and **expdamp** point is a massless gas (radiation) until late times and supplies no dark matter at recombination. A power law stays massive exactly when its bare mass m_0 is positive, i.e. when $\Omega_{\text{qp0}}/U_t > \nu/(1-\nu)$; with the observed split $\Omega_{\text{qp0}} = 0.265$ that is $\nu < 0.2788$ (section 4.7.8). The massive mass-varying example with a larger quasiparticle share (Part VIII's case: **power**, $\nu = 0.5$, $m_{\text{today}} = 100$ eV, $\Omega_{\text{qp0}} = 0.55$, no bare Λ) has m_{eff} rising from 27.15 eV to 100.00 eV, w_f today -0.4215 , $q_0 = -0.1010$, and matter today $\Omega_{\text{qp0}} + \Omega_{b0} = 0.599$, about twice the measured $\Omega_m \approx 0.315$ (Planck 2018): it is massive because its dark matter outweighs its dark energy, which the observed universe does not allow [**nb06 §7**].
3. **It is adiabatically unstable.** Every scan point that stays massive has $c_s^2 < 0$ after $a = 10^{-3}$ (smallest c_s^2 between -0.647 and -0.056), and so do all 9 runs of section 4.7.6 (-0.611 for $\nu = 0.236$, -0.364 for Part VIII's case, down to -47.1 for **quadratic**). This is the adiabatic instability of mass-varying fermions (as found by Afshordi, Zaldarriaga and Kohri for mass-varying neutrinos).
4. **The Dirac-sea vacuum energy demands a fine-tuning.** The renormalized sea energy that the no-sea functional drops changes, for a mass variation of order m , by $g m^4/(64\pi^2)$: 3.44 , 3.44×10^4 , 3.44×10^8 , 3.44×10^{12} times ρ_{c0} at $m = 0.01, 0.1, 1, 10$ eV, more than the critical density for $m > 7.341$ meV (section 4.5.11). Along every mass-varying run it changes by **1.31×10^{13} to 2.87×10^{16} times the total density at $a \geq 0.3$** (over the scans at least 3.79×10^9) [**nb06 §7**]. The runs read W as the fully renormalized effective potential, which absorbs the sea energy only by fine-tuning the bare potential against it to that precision. The cosmological-constant problem is not solved.
5. **It needs the stabilizer, which violates the null energy condition.** All of this holds with the hidden sheet held by $P_{\text{stab}} = -F/2$ (today $-0.8428 \rho_{c0}$ in the Λ CDM-like runs, -0.7003 in Part VIII's power case), a Lagrange multiplier, not a stress derived from any field, with zero energy density; it violates the null energy condition along x_0 on 100.0 % of

the output rows of every stabilized run (section 4.7.1).

6. **Without the stabilizer it is excluded (the fable8d no-go).** Left free, the hidden sheet is driven by dust and vacuum energy to the 7-dimensional isotropic attractor, and the observed Newton constant $G_4 = G_8/V_{\text{hid}}$ varies: $G(a_i)/G(1) = 3.44 \times 10^{10}$, 1.85×10^{17} , 1.05×10^{12} , 1.07×10^{12} in the four fable runs and 6.54×10^8 for radiation plus baryons alone; $\Delta G/G$ since nucleosynthesis is -1 (against about 0.1), and $|d \ln G/dt|$ today $\geq 2.31 \times 10^{-11}/\text{yr}$ (against the lunar-laser-ranging bound $10^{-13}/\text{yr}$) (section 4.7.10). The only frozen-compatible Kohn–Sham fable is radiation (section 4.5.7).
7. **The split into dark matter and dark energy is a convention.** The condensate ρ_U is an 8-dimensional vacuum energy, with $P = -\rho_U$ along the hidden directions as well; calling it dark energy and the quasiparticles dark matter is a choice of bookkeeping (section 4.5.4).
8. **The approximations under which all of this holds:** semiclassical gravity ($G = \kappa \langle \hat{T} \rangle$); the Kohn–Sham mean field with exchange–correlation neglected (section 4.5.12); zero modes along x_0 and the hidden timelike sheet, with x_0 compactified (a Kaluza–Klein gap above the temperatures considered) and the hidden timelike sheet a formal comoving volume (compactifying it gives closed timelike curves); the asymptotic Bianchi-I reading of the primordial field, which neglects terms of order $\cot(6Hx_0)$ and $\cot^2(6Hx_0)$ (section 4.3.3); $g_*(T)$ not followed.

4.8.2 Answer [2]: a time-varying dark-matter equation of state

Yes, intrinsically: the quantized fable’s dark-matter equation of state runs from 1/3 to 0. For an allowed mass the variation is over long before recombination.

1. **The mechanism.** The fable’s quasiparticles are a degenerate Fermi gas ($g = 8$ states per momentum), and their equation of state is $w_{\text{DM}} = P_{\text{KS}}/\varepsilon_{\text{KS}}$, a function of $x = k_{\text{F}}/|m|$ alone that rises monotonically from 0 (dust) to 1/3 (radiation) (section 4.1.6). As the universe expands, $k_{\text{F}} = k_{\text{F}0}/a$ falls, and w_{DM} falls from 1/3 to 0: **a time-varying dark-matter equation of state**, which is a property of the quantum gas and has no counterpart in the classical Part VI fable (whose w was frozen on the pre-universe and 0 for the mass term). Every run starts at $w_{\text{DM}} = 0.333333$ and ends today at most 8.1×10^{-8} [nb06 §7]; the transition is at $a_{\text{nr}} = k_{\text{F}0}/m$, 4.466×10^{-6} for 30 eV and 4.163×10^{-8} for 1 keV, scaling as $m^{-4/3}$ (sections 4.7.4–4.7.5). The same cooling happens on the pre-universe itself, on the canonical frame with a_4 free, whenever the observed sheet expands ($a'_4 < 0$, section 4.1.6).
2. **How heavy it must be, and hence when the variation happens.** A fable that is all of the dark matter is radiation before a_{nr} : $\Delta N_{\text{eff}}(\text{BBN}) = 6.692 (m/\text{eV})^{-4/3} < 0.3$ requires $m > 10.3 \text{ eV}$ (< 0.2 : $m > 13.9 \text{ eV}$). Free streaming is far more restrictive: matching the rms velocity of the Lyman-alpha-allowed thermal warm dark matter requires $m \approx 1125\text{--}1807 \text{ eV}$ ($\sim 1.1\text{--}1.8 \text{ keV}$, a velocity-matching estimate, not a transfer-function computation), and the Tremaine–Gunn phase-space bound for illustrative dwarf spheroidals gives $m > 231\text{--}358 \text{ eV}$ (section 4.7.9). At such masses the change of w_{DM} from 1/3 to 0 happens around $a_{\text{nr}} = 4.16 \times 10^{-8}$ (1 keV) or 1.65×10^{-8} (2 keV): **indistinguishable from cold dark matter in the late universe.**
3. **With a mass-varying potential the effective equation of state varies late.** The quasiparticles then exchange energy with the condensate ($Q = \sigma_8 dm/dt$), and their effective equation of state $w_{\text{DM,eff}} = w_{\text{DM}} - \sigma_{\text{KS}} (dm/dN)/(3\varepsilon_{\text{KS}})$ ranges, after $a = 10^{-3}$, over $[-0.611, +9.85 \times 10^{-5}]$ (power $\nu = 0.236$, 30 eV), $[-0.611, +3.59 \times 10^{-6}]$ (100 eV), $[-27.6, +0.333]$ and $[-27.7, +0.333]$ (power $\nu = 0.5$), $[-21.8, +0.333]$ (expdamp), $[-47.1, +0.333]$ (quadratic), and $[-0.364, -3.89 \times 10^{-5}]$ (Part VIII’s case) (section 4.7.6). This late variation is the same exchange that makes the inferred dark energy vary in [1],

and it carries all of [1]’s caveats (adiabatic instability, the sea energy, the viability window $\nu < 0.2788$).

4. **The author’s mass term alone is dark matter without dark energy.** $W = m_0\sigma$ gives an Einstein–de Sitter-like universe ($q_0 = +0.50005$, age 9.67 Gyr with $H_0 = 67.4$ km/s/Mpc), not ours; with a bare cosmological constant (`lambda-mass`) it is Λ CDM with fable dark matter ($q_0 = -0.52845$, age 13.8000 Gyr) (sections 4.7.4–4.7.5).
5. **The same caveats as [1]** apply: the stabilized model with its NEC-violating stabilizer (without it the Newton constant varies by 6.5×10^8 to 1.9×10^{17}); the split into dark matter and dark energy is a convention; semiclassical gravity, the Kohn–Sham mean field with E_{xc} neglected (exact for the author’s mass term, which is free), zero modes along the hidden directions, the asymptotic reading of the primordial field.

5 The design review: four lenses, four skeptics, every finding, and every correction

Nothing of Efforts A and B was implemented before its design had been attacked. This section records the attack: how it was organized, every finding with its severity and its verdict, what was adopted, the sub-claims of the reviewers that the skeptics refuted, the points the skeptics found that the reviewers had missed, and the corrections that were found only later, by the implementation and the checks. Each adopted correction is named with the section of this page where its content now stands.

5.1 How the review was organized

The design (revision 1), written on 2026-09-24 before any code, fixed the conventions of the notebook (never to be changed), recorded six facts already computed by probes (F1 the Clifford symmetry table; F2 the Grassmann consequences; F3 the Einstein tensor of the canonical metric, which is not a vacuum solution; F4 the generalized warped frame and its (0,4) equation; F5 the 8-dimensional Bianchi-I system; F6 the quantization algebra: $G = -i\sigma_{16}\gamma^4$, signature (8,8), J , the plane-wave spectrum), stated the physics of Efforts A (A.1 refinement, A.2 Lagrangian, A.3 field equations, A.4 quantization, A.5 EMT, A.6 expectation values, A.7 spin connection) and B (B.1–B.6: the semiclassical system, the canonical ansatz, the present-universe models, the DFT states, the spin connection of the dynamical frames, the answers), gave the implementation plan, and asked the reviewers eight questions to try to break:

	question
Q1	Is the Dirac-bracket anticommutator (factor, sign, $H/\sqrt{g}$, $-i\sigma_{16}\gamma^4$) right, including for the symmetrized Lagrangian and on the curved slice?
Q2	Is the Krein/ J quantization legitimate (Hermiticity of the interacting H under the J -involution; what exactly is lost; does J commute with every term of the admissible-sector Hamiltonian, the spin connection included)?
Q3	Is the impossibility of a Majorana mass / scalar bilinear right for every candidate bilinear, and is “fermion fable must be complex” stated with the right scope?
Q4	The EMT operator: sign and normalization against Part VI; symmetrization; the spin-connection-variation argument; conservation with independent Ψ , $\bar{\Psi}$.
Q5	Kohn–Sham thermodynamics: the closed forms, $g = 8$, the classical limit, the no-sea statement, the mass-term conclusions.
Q6	The 8D Einstein equations: F3–F5, the frozen-hidden theorem and its corollary, the conservation law, the units (κ_8 against κ_4), the Bianchi-I claim, the negative background energy.

	question
Q7	DFT: the functional $E_n[\sigma]$ on $(-n, n)$, uniqueness, the Δ SCF / wall-state construction, the MIT condition in this real-Clifford convention, whether the wall problem is well posed and bound states exist, whether <code>expdamp</code> pins $\sigma \leq s_1$.
Q8	Cosmology: are the runs meaningful (Ω values, time interval, shooting), the definitions of $w_{\text{DE,eff}}$ and w_{DM} , adiabatic instability, the bounds on G ; are answers [1] and [2] honest?

The four lenses. Each reviewer ran its own probes on the notebook’s own state (the Input cells loaded headlessly, or the notebook’s T_{16} and σ_{16} exported to Python) and wrote findings, each with an ID, a severity (critical, major, minor), a kind (wrong, incomplete, ill-posed, hidden assumption, implementation risk, better formulation), the statement and a proposed fix:

lens	covers	findings
quantization: canonical quantization in 4+4 signature, Grassmann and Majorana algebra, Krein spaces	F1, F2, F6; A.1, A.2, A.4; Q1–Q3	QK-1 ... QK-11
field equations, energy–momentum tensor operator, canonical spin connection	A.2, A.3, A.5, A.7, B.5; Q4	EMT-1 ... EMT-11
the 8D Einstein equations with fable as source, and the cosmological runs	F3, F4, F5; B.1, B.2, B.3, B.6; Q6, Q8	E1 ... E12
density-functional theory and the numerics	A.6, B.3 sources, B.4; Q5, Q7; the Rust and Python plan	DFT-1 ... DFT-16

The four skeptics. Each lens was followed by an independent skeptic, who received the findings and tried to **refute** each one with computations of its own (its own Riemann-tensor and spin-connection code, its own Python on the exported matrices, its own numerical runs), and returned a verdict per finding — *confirmed*, *partly confirmed* or *refuted* — with a corrected fix wherever it disagreed, and a list of what the reviewer had **missed**.

The outcome. 50 findings: **30 confirmed, 20 partly confirmed, 0 refuted**. Where a finding was only partly confirmed, the skeptic’s corrected fix was adopted. All adopted corrections were written into an addendum to the design (revision 2), which overrides revision 1 wherever they differ, and then implemented. The DFT skeptic’s final verdicts came last and override the addendum where they differ.

lens	confirmed	partly confirmed	refuted
quantization (QK)	3 (QK-2, QK-7, QK-8)	8 (QK-1, 3, 4, 5, 6, 9, 10, 11)	0
field equations / EMT / spin connection (EMT)	9 (EMT-1, 3–10)	2 (EMT-2, EMT-11)	0
Einstein / cosmology (E)	6 (E2, E4, E6, E8, E10, E11)	6 (E1, E3, E5, E7, E9, E12)	0
DFT / numerics (DFT)	12 (DFT-1–3, 5, 9–16)	4 (DFT-4, 6, 7, 8)	0

5.2 The quantization lens (QK)

ID	severity, kind	the finding	verdict and the skeptic's correction	adopted, and where it stands now
QK-1	major, wrong	The admissibility argument had the wrong mechanism: the free theory below threshold can be quantized with a momentum-dependent J_k ; the real obstruction is locality — one momentum-independent J' must commute with $\beta = -iT_{16}[4]$ and every one-particle generator $E_k = Gh_k$; no positive J' exists once a hidden momentum is present; at threshold E_k is nilpotent (Jordan blocks); above it the modes are G -neutral.	Partly confirmed. Computed with the notebook's matrices: the boosted $J' = S^{-1}JS$ works for one mixed momentum ($[J', E] = 10^{-15}$, GJ' positive, minimum eigenvalue 0.655); the theorem is about the span V of the mode momenta: positive quantization iff V is spacelike-definite; J' unique (with $J'^2 = 1$) when $V = \text{span}(x_0..x_3)$. The reviewer's above-threshold frequencies $\pm 1.705872i$ were wrong: for $k = (0, .75, 1, 0 k_5 = 2)$, $m = 1$ they are $\pm 1.19896i$. Below threshold the free field is quantizable mode by mode but not uniformly: $\text{cond}(GJ_k) = (m + k_5)/(m - k_5) = 3, 19, 199, 1999$ at $k_5 = .5, .9, .99, .999$.	The theorem as corrected, with its proof ingredients as assertions ($[E_k, \beta] = 2iT_k$, the T_5 anti-congruence, the traceless certificate, the positive boosted J' , the rank-8 Jordan block with $E^2 = 0$): section 3.5.10 (ADMISSIBILITY).
QK-2	major, ill-posed	"Physical states obey $P_5 = P_6 = P_7 = 0$ " is not a superselection rule: pairs $(k_h, -k_h)$ have $P_h = 0$, and $V(s)$ contains $b_{k_h}^\dagger b_{-k_h}^\dagger b_0 b_0$; the admissible theory is a truncation needing a finite hidden volume.	Confirmed (derivation; normalization by projection gives $(H/(V_{\text{hid}}\sqrt{g}))G\delta^4$). V_{hid} is a coordinate volume; the truncated theory is a (4,1)-dimensional theory whose Cauchy surface is $x_0..x_3$.	The truncation $\Psi = \Psi(x_0..x_4)$ with finite V_{hid} (closed timelike curves stated); the anticommutator $(H/(V_{\text{hid}}\sqrt{ g }))G\delta^4$; the Tomonaga–Schwinger remark kept only as $[P_4, P_h] = 0$: sections 3.5.1, 3.5.4.
QK-3	major, in- complete	Under the J -involution, $\Psi^\dagger M \Psi$ is Hermitian iff $[J, M] = 0$ and anti-Hermitian iff $\{J, M\} = 0$; $\hat{T}_{\mu h}$, $\hat{T}_{hh'}$ and j^h are anti-Hermitian.	Partly confirmed. Refuted sub-claim: $\hat{T}_{hh'}$ ($h \neq h'$) is not anti-Hermitian — its connection pieces commute with J , and for $x_5..x_7$ -independent Ψ it vanishes identically on all three frames. The mixed $\hat{T}_{\mu h}$ are nonzero and anti-Hermitian; an anti-Hermitian operator has a purely imaginary expectation in every state, so $G_{\mu h} = \kappa \langle \hat{T}_{\mu h} \rangle$ is consistent only where the expectation is 0, which hidden-SO(3) invariance ensures.	The Hermiticity table with $\hat{T}_{hh'} \equiv 0$, $\hat{T}_{hh} = g_{hh} \hat{L}$, anti-Hermitian $\hat{T}_{\mu h}$ and j^h with zero expectation in hidden-SO(3)-invariant states: sections 3.5.8, 3.6.5.
QK-4	major, implemen- tation risk	Part V/VI's " $\hat{L}$ with $D = \hat{L}$ with ∂ " holds only for real commuting Ψ ; for Grassmann fields the unsymmetrized connection term $-3H \cot^2 \Psi^\dagger \sigma_{16} T_{16}[0] \Psi$ is nonzero and anti-Hermitian, and the unsymmetrized partial-derivative Lagrangian gives inconsistent equations.	Partly confirmed. Overstated part: "never assert covariant = partial" is too strong — for the symmetrized Lagrangian $\hat{L}_{\text{sym}}(D) - \hat{L}_{\text{sym}}(\partial) = \frac{1}{2H} \bar{\Psi} \sum_\mu \{\gamma^\mu, \Gamma_\mu\} \Psi$, which is identically 0 on the canonical, warped and Bianchi-I frames, so the identity holds there in symmetrized form. F2(a) (no dynamics for a real Grassmann field) holds on every frame.	Only the symmetrized Lagrangian; the four facts (a)–(f) as assertions: section 3.3 (LAGRANGIAN (a)–(f)), 3.2.3.

ID	severity, kind	the finding	verdict and the skeptic's correction	adopted, and where it stands now
QK-5	major, better for- mulation	J has concrete content: $\sigma_{16} = J\beta$, $J = -i\Omega T_{16}[0]$; with $\Psi^\dagger := \Psi^\dagger J$ the author's conjugate is the observer's Dirac adjoint; fable is four 4D Dirac flavours with a flavour metric of signature (2,2) replaced by 1; legitimate for fermions, not for bosons.	Partly confirmed. Not verified: “J gives reflection positivity” — dropped. The bosonic mechanism is not “a ghost”: for a commuting field H has 8 negative-frequency modes per k and is unbounded below; making it bounded below turns 4 modes into negative-norm modes. The J^* -structure differs from the classical reality structure on J -odd observables, so the classical theory is the limit of the J -even sector only. Missed point: the prescription $b_u^\dagger := \eta_u b_u^\dagger$ is basis-dependent (a Krein boost inside a (4,4) eigenspace gives $\ J_k - J\ = 4.45$ and a non-Hermitian s); the adjoint must be $\Psi^\dagger J$, mode-independent.	Sections 3.5.5, 3.5.6, 3.5.13 (J , KREIN MODES [control]); two symbols for the classical conjugation and the Hilbert adjoint.
QK-6	minor, wrong	The symmetry count: the group commuting with J is $\text{Spin}(4,1) \times \text{Spin}(3)$ (13 generators); the 15 lost are 12 boosts and 3 rotations (the design had called them boosts); chirality $T_{16}[8]$ is lost; $p = \bar{\Psi} T_{16}[8] \Psi$ is anti-Hermitian; G is $\text{Spin}(4,3)$ -invariant (21).	Partly confirmed. Misleading sub-claim: the chirality subspaces are orthogonal in the Hilbert product ($T_{16}[8]$ is Hermitian); what is true is that J is chirality-odd, so each chirality subspace is G -null and the J -vacuum is not chirality-graded. ip is J -Hermitian but classically imaginary, so V depends on s only.	Section 3.5.9, 3.5.7.
QK-7	minor, in- complete	The four candidate matrices are only two ($C_+ T_8 = \sigma_{16}$); the invariant forms are $\text{span}\{\sigma_{16}, \sigma_{16} T_8\}$, both symmetric and chirality-diagonal; Majorana has one quartic scalar and a non-dynamical commuting shadow; complex Weyl spinors do not propagate either; the complex spinor has two scalar bilinears and 6 quartic invariants.	Confirmed (full rank table S S , A S , A A , S A , S S , A S , A A , S A , S S for ranks 0–8; invariant forms of dimension 2; 0 singlets in $\Lambda^2(16)$).	“The minimal, not the unique, field”: sections 3.2.1–3.2.8 (REFINEMENT (a), (b)).
QK-8	minor, hidden as- sumption	When $\sqrt{g}$ depends on x_4 , Ψ is not a canonical Heisenberg field; quantize $\chi = (\sqrt{ g }/H)^{1/2} \Psi$.	Confirmed.	Section 3.5.16; on the dynamical frames, section 4.4.6.
QK-9	minor, hidden as- sumption	Hermiticity needs a J -compatible boundary condition in x_0 : the bag condition $T_{16}[0] \Psi = \pm \Psi$.	Partly confirmed. Overstated: “vanishes at both ends” — the wall is at finite proper distance ($z \approx 3Hx_0^2$), the operator is regular there and needs a condition; at $z \rightarrow \infty$ it is limit-point and none can be imposed.	One-particle space $L^2(dz)$; the condition at the wall only: section 3.5.12.
QK-10	minor, wrong	$j^\mu = i\bar{\Psi} \gamma^\mu \Psi$ has the wrong sign and normalization to be the number current.	Partly confirmed: the calculation is right, “wrong” is overstated — j^μ is a legitimate conserved current that needs its normalization stated.	$n^\mu := -(i/H) \bar{\Psi} \gamma^\mu \Psi$, $n^4 = \Psi^\dagger \Psi / H \geq 0$; a positive-frequency J -mode carries j -charge -1 : section 3.4.10.

ID	severity, kind	the finding	verdict and the skeptic's correction	adopted, and where it stands now
QK- 11	minor, better for- mulation	$V(s)$ needs no Weyl ordering in the Hamiltonian (spectral calculus of one Hermitian operator); the ordering appears only in the Heisenberg equation, as an insertion average.	Partly confirmed: the design's Weyl-symmetrized product is already identical to the insertion average (Fock check to 1.1×10^{-16}); the real addition is that $V(:s:)$ needs no ordering.	Section 3.4.9.

What the quantization lens confirmed as right (11 items): the full Clifford symmetry table; the invariant forms; $G = -i\sigma_{16}\gamma^4 = J$ Hermitian with $J^2 = 1$ and signature (8,8); the eightfold $\pm\omega$ modes for admissible momenta; $\omega^2 = k_s^2 - k_h^2 + m^2$; the J commutation relations; Γ_μ real with $\sigma_{16}\Gamma_\mu$ antisymmetric; the real commuting shadow equals Part VI's **cfLhatFable**; the commutant $M_4(\mathbb{C})$ (four flavours); the Hermiticity of $H_{\text{free}} + \lambda s^2$; the rescaling $\sqrt{|g|} = \sec(6Hx_0)$.

ID	severity, kind	the finding	verdict and the skeptic's correction	adopted, and where it stands now
----	-------------------	-------------	--------------------------------------	-------------------------------------

5.3 The field-equation, energy–momentum-tensor and spin-connection lens (EMT)

ID	severity, kind	the finding	verdict and the skeptic's correction	adopted, and where it stands now
EMT-1	major, implementation risk	The design gave the wrong reason for the covariant derivative in $T_{\mu\nu}$: varying a partial-derivative Lagrangian gives T_{par} , not conserved, differing from T_{cov} in 21 off-diagonal components; the spin-connection variation is identically zero ($\delta\omega_{[cab]} = 0$ for symmetric tetrad variations).	Confirmed (own first-order symmetric tetrad perturbation, all 36 functions generic). Addition: on the canonical frame $T_{\text{cov}} = T_{\text{par}}$ on the diagonal and on (0, 4); (5, 6) vanishes identically; the control needs a non-vacuous pair such as (1, 4), (0, 1), (1, 5), (5, 4), (0, 5). Do not run the symbolic divergence of T_{par} (it ran more than 9 minutes).	Hilbert = T_{cov} off shell, the variation piece zero, Hilbert $\neq T_{\text{par}}$ at a non-vacuous pair: sections 3.6.1, 3.8.12.5 (EMT).
EMT-2	major, wrong	“The connection commutes with J ” holds only for $\gamma^\mu\Gamma_\mu$ and $\Gamma_0, \Gamma_4; \Gamma_5, \Gamma_6, \Gamma_7$ anticommute with J .	Partly confirmed. Part of the finding wrong: in any Hamiltonian Γ_h enters only as $\sigma_{16}\gamma^h\Gamma_h$, and $[J, \gamma^h\Gamma_h] = 0$ for each h separately; the J -odd objects are the bare Γ_h in T_{cov} (0, h), (i , h), (4, h) and the hidden momenta. The expectation argument is right and needs J -eigenmodes.	The full J table, per μ : sections 3.5.8, 3.8.12.6, 4.4.7 (J TABLE).
EMT-3	minor, wrong	“On shell $\hat{L} = sV' - V$ (bilinear part vanishes)” is false: the kinetic bilinear equals $sV'(s)$.	Confirmed.	Section 3.6.4.
EMT-4	minor, incomplete	The on-shell components were missing: trace $7sV' - 8V$; $\rho = V + K_h$; $T^i_i = T^h_h = sV' - V$; $T^0_0 = sV' - V + K_h$; K_h symmetrized, 0 on the rescaled x_4 -only solutions.	Confirmed. Caveats added: calling T^h_h a pressure along the hidden timelike directions is a naming convention; $K_h = 0$ holds on the x_4 -only family, which solves the equation only if $V''s' = 0$.	Section 3.6.4 (EMT ON SHELL).
EMT-5	minor, hidden assumption	Part VI’s drop-out mechanism does not carry over to complex fields; in the symmetrized Lagrangian the connection drops iff the matrix $\{\gamma^\mu, \Gamma_\mu\} = \frac{1}{2}\omega_{[cab]}\gamma^{cab}$ vanishes.	Confirmed.	Sections 3.3.3, 3.8.12.2.
EMT-6	minor, better formulation	With the symmetrized Lagrangian the Hamiltonian density contains no connection; it re-emerges in the equation of motion as $\gamma^\mu\Gamma_\mu = (1/2)(1/\sqrt{g})\partial_\mu(\sqrt{g}\gamma^\mu)$.	Confirmed. Addition: the unsymmetrized and symmetrized integrated Hamiltonians differ only by the surface term $(1/2H)[\text{csc}(6Hx_0)\Psi T_{16}[0]\Psi]$ at the wall, which vanishes exactly when the wall condition kills the normal current.	Sections 3.5.11, 3.8.12.4.
EMT-7	minor, incomplete	On the dynamical frames $\Gamma_0 \neq 0$ when $C' \neq 0$; the complete non-zero ω lists must be asserted.	Confirmed (own routine reproduces GammaSpinCanonical ; the warped and Bianchi-I lists; $\Gamma_4 = 0$).	Section 4.4.2 (the complete lists).

ID	severity, kind	the finding	verdict and the skeptic's correction	adopted, and where it stands now
EMT- 8	minor, hidden as- sumption	The commuting-placeholder conservation proof holds for Grassmann fields only because the on-shell divergence is linear in $\bar{\Psi}$ -left bilinears with coefficients that are functions of the even composites s and ds .	Confirmed (derivation checked step by step).	Section 3.6.3.
EMT- 9	minor, better for- mulation	The exact 8+8 form with its conjugate rows; s is chirality-diagonal, so the axial phase leaves $V(s)$ invariant but not the kinetic term: the only U(1) is the vector one.	Confirmed (the internal-symmetry Lie algebra is 1-dimensional).	Sections 3.4.3, 3.4.6, 3.4.11 (SYMMETRY).
EMT- 10	minor, better for- mulation	The rescaling theorem in frame-independent form: $\gamma^\mu \Gamma_\mu = -(1/2)T_\mu \gamma^\mu$, T_μ the Weitzenböck torsion vector; on the warped frame $\Psi = \sqrt{\sin}(A^3 B^3 C)^{-1/2} \chi$ and $\{\chi, \chi^\dagger\} = H \cot(6Hx_0) G \delta^7$, independent of x_4 .	Confirmed. Wording: the χ anticommutator is x_4 -independent and the dilution is carried by the prefactor; the χ dynamics stay time-dependent through the frame.	Sections 3.8.12.3, 4.4.5, 4.4.6.
EMT- 11	minor, hidden as- sumption	For nonlinear V the operator equation is $\gamma^\mu D_\mu \Psi = H :V'(s)\Psi$;, differing from the naive product by Hartree/Fock contractions.	Partly confirmed. Imprecise: the contraction terms are not all c-number multiples of Ψ ; the J-vacuum sea matrix has a scalar part (divergent, renormalizes V') and a γ^4 part (the sea charge, a constant chemical-potential shift, removed by charge-symmetric ordering or a phase, not by renormalizing V).	Section 3.4.9.

What the skeptic of this lens found **missed**: the x_4 -only family $\Psi = \sqrt{\sin} \Psi'(x_4)$ solves the field equation only if $V''s' = 0$ (for the complex field s' is conserved, so null data $s' = 0$ give x_4 -only solutions for **any** V , with $\rho = V(0)$) — section 3.4.8; normal ordering on an x_4 -dependent background is ambiguous, and “ $\langle :T: \rangle = 0$ in the vacuum” holds only for static backgrounds or adiabatically — section 3.7.3; the witness pairs for T_{cov} against T_{par} are frame-specific; and the per- μ identity $\{\gamma^\mu, \Gamma_\mu\} = 0$ (not only the sum) is what gives the individual pressures — section 3.6.4.

5.4 The Einstein and cosmology lens (E)

ID	severity, kind	the finding	verdict and the skeptic's correction	adopted, and where it stands now
E1	major, wrong	$W = m_0\sigma + (\lambda/2)\sigma^2$ is not frozen-hidden-compatible; the condition holds only for $W = c\sigma^2$, and then the gap is nonzero only while $\lambda g k_F^2/(4\pi^2 v) > 1$. The reviewer wrote the driver as $F = \sigma W' - 2W$.	Partly confirmed. Sign corrected: at self-consistency $F = \rho - 3P_{\text{obs}} + 2P_{\text{hid}} = 2W - \sigma W'$ (the reviewer's own number, $F = +m_0\sigma$, agrees with this sign); with V_0 , $F = 2V_0 + m_0\sigma + \dots$. Stronger than stated: the nontrivial root is never the ground state (at $k_F = 1, 0.6, 0.41$: $\rho = 0.2837, 0.01632, 0.0028633$ against the massless $0.1013, 0.01313, 0.0028631$; roots $m = 3.9785, 1.9816, 0.7479, 0.3468$ at $k_F = 1, 0.8, 0.6, 0.5$, none at 0.40 , $k_{F,c} = 0.40558$).	A frozen-compatible fable is radiation: sections 4.5.6, 4.5.7 (FROZEN).
E2	critical, implemen- tation risk	The Kohn–Sham densities are per observed 3-volume; the 8D source must be per 7-volume ($\sigma_7 = \sigma_{\text{KS}}/v$, gap $m = W'(\sigma_7)$), otherwise conservation fails; $d\rho = \omega_F dn$ holds only at fixed v .	Confirmed (conservation residual -2.5×10^{-11} per 7-volume against 5.4×10^{-3} per 3-volume).	Per-7-volume everywhere, a unit test that the per-3 version fails: sections 4.5.1, 4.5.5, 6.4.2 (per_3_volume_mean_field_violates_8d_conservation).
E3	critical, in- complete	In fable8d the variation of G is catastrophic (12 orders of magnitude); any 8D vacuum energy and any dust drive the hidden sheet; a bare 8D Λ is not a 4D cosmological constant. Make fable4d (with a zero-energy stabilizer) the physical model.	Partly confirmed. The 7D-isotropic attractor is not universal : with baryons plus a frozen-compatible vacuum ($P_{\text{hid}} = -2\rho$, which must obey $\rho' = (3H_B + H_C)\rho$) the sheet re-freezes, $\dot{G}/G \approx 1.5 \times 10^{-14}/\text{yr}$ (within lunar laser ranging) while $\Delta G/G$ since BBN still fails. The stabilizer $P_{\text{stab}} = -F_{\text{total}}/2$ violates the null energy condition along x_0 whenever dust is present ($\rho + P_C = -\rho_{\text{dust}}/2$): a Lagrange multiplier, not a derived stress.	fable4d as the physical model, fable8d as the no-go, P_{stab} printed on every run: sections 4.7.1, 4.7.10.
E4	major, ill-posed	The backward fable8d run from today is neither anchored nor stable (runs to the 8D Kasner point, the constraint drifts).	Confirmed ($H_B/H_A \rightarrow -0.2929 = -(1 - 1/\sqrt{2})$; constraint residual -3.7×10^5 at $a = 10^{-8}$ with LSODA).	The solver refuses it (section 4.7.11).
E5	major, better for- mulation	Replace the circular x0-mismatch no-go by three splitting-free theorems: (a) energy, (b) anisotropy, (c) momentum.	Partly confirmed. (b) as stated (“ $a_4'' = 0$ for any fable state”) holds only for c-number configurations; the Kohn–Sham gas has $P_{\text{obs}} - P_{\text{hid}} = P_{\text{KS}}/v > 0$ and allows $a_4'' < 0$.	(a), (c) as theorems; (b) as $a_4'' = -\kappa(P_{\text{obs}} - P_{\text{hid}})/(2H^2)$: sections 4.2.4–4.2.7.
E6	major, hidden as- sumption	F4 “derives” the volume-preserving a_4 structure only under $C = 1$, $T^4_0 = 0$ and separability; with C dynamical, $H_C = (H_A + H_B)/2$ and positive-energy expansion is allowed.	Confirmed.	“Singled out, not derived”: section 4.3.2.

ID	severity, kind	the finding	verdict and the skeptic's correction	adopted, and where it stands now
E7	minor, incomplete	The decay rates per component; the asymptotic background values; the validity condition; the ghost stiff fluid. The reviewer gave the residual driver as $F_{\text{bg}} = -132H^2 \cot^2 / (\kappa C^2)$.	Partly confirmed. $F_{\text{bg}} = -132$ is wrong (it assumed $P_C = P_{\text{hid}}$); the correct drivers are $\kappa(P_i - T/6) _{\text{bg}} = -12H^2 \cot^2 / C^2$ for A, B and $-72H^2 \cot^2 / C^2$ for C , and only if T_{bg} is not included as a source.	Section 4.3.3, 4.2.8.
E8	major, hidden assumption	$\kappa_4 = \kappa_8/V_h$ needs a finite hidden volume: x_0 has infinite proper length and compact timelike $x_5..x_7$ give closed timelike curves; zero modes along x_0 need a Kaluza–Klein gap; $\sum \Omega \neq 1$ in fable8d .	Confirmed.	The assumptions stated: sections 4.7.1, 4.7.2.
E9	major, ill-posed	The shooting was misposed (two conditions, one unknown in fable8d ; uniqueness needs monotonicity; a branch jump breaks conservation); $W = m_0\sigma$ forces Einstein–de Sitter.	Partly confirmed. “Hot or warm below about 20 eV” had no criterion (use $a_{\text{nr}} < a_{\text{eq}}$ for not hot, $a_{\text{nr}} < 10^{-6}$ for cold-like). Missed simplification: fable4d is algebraic in a , so the shooting is one root-find and $t(a)$ a quadrature — an exact cross-check.	Sections 4.7.1, 4.7.4.
E10	major, better formulation	Four problems with the w definitions; report w_f , the observer-inferred $w_{\text{DE,inf}}$ (with a stated constant subtraction) and both w_{DM} and $w_{\text{DM,eff}}$; the DM/DE split is a convention.	Confirmed.	Section 4.5.10.
E11	minor, wrong	$\Omega_{r0} = 9.15 \times 10^{-5}$ is wrong; at $h = 0.674$, $T_0 = 2.7255 \text{ K}$, $N_{\text{eff}} = 3.046$ it is 9.21×10^{-5} .	Confirmed (9.20961×10^{-5}).	Computed in the code: section 4.7.1, 6.4.3.
E12	minor, hidden assumption	The pre-universe does not define the “beginning of the present universe”; a_i must be chosen on physical grounds and shown irrelevant.	Partly confirmed. Two details wrong: the g_{*s} correction lowers $T(a_i)$ ($Ta = T_0(3.909/g_{*s})^{1/3}$), it does not raise it to “about 0.5 GeV”; the $\rho_r a^4$ factor at $a_i = 10^{-12}$ is 0.46–0.71, not 0.386 (that needs $T > 100 \text{ GeV}$); “any $a_i \leq 10^{-8}$ ” is too loose (that is after BBN).	$a_i \leq \min(10^{-10}, 0.01a_{\text{nr}}(m))$, insensitivity shown, $g_*(T)$ neglected: section 4.7.2.

What the skeptics of this lens found **missed**: the no-sea functional has a **maximum** at the physical root (so a global-minimum rule picks a collapsed state; $E = 0.16123, 0.17007, 0.17028, 0.17012, 0.16350, 0.14277$ at $m = 0.5, 0.9, 1.0, 1.1, 2, 10$ for $W = m_0\sigma$, $k_F = 1$) — section 4.6.1; the spin connection of the warped frame has $\Gamma_{x_0} = +\frac{1}{2} \tan(6Hx_0) C'(x_4) T_{16}[0] \cdot T_{16}[4] \neq 0$ and $[J, \Gamma_h] \neq 0$ — section 4.4.2, 4.4.7; a frozen-compatible vacuum must carry $\rho' = (3H_B + H_C)\rho$, and holding it constant violates the constraint (residual 2.6×10^{-2}); the stabilizer’s NEC violation; and that “ $\dot{G}/G$ today” and “ $\Delta G/G$ since BBN” test different things and must both be reported — section 4.7.10.

5.5 The density-functional and numerics lens (DFT)

ID	severity, kind	the finding	verdict	adopted, and where it stands now
DFT- 1	critical, wrong	“Ground state = global minimum of $E_n[\sigma]$ ” is wrong: T_s is concave, the mass-term root is the maximum , the infimum is the variational collapse $E \rightarrow - m_0 n$. Use the stationary (minimax) point, and for attractive W the convex Walecka functional ($W = \sigma - 10\sigma^2$, $k_F = 1$: $m^* = 0.211249$, $\Omega(m^*) = \rho = 0.121193$).	Confirmed.	Sections 4.6.1, 3.7.4 (Kohn-Sham), 6.4.2 (walecka_functional_is_minimized_at_the_gap_root, ks_functional_stationary_point_is_a_maximum_for_the_mass_term).
DFT- 2	major, wrong	Uniqueness holds for $W''gk_F^2/(4\pi^2V) < 1$, not for any monotone W' ; the repulsive quadratic has three roots, the lowest-energy one of opposite sign.	Confirmed.	Section 4.5.3; root_counts_three_for_strong_repulsion_one_for_concave_w.
DFT- 3	major, wrong	The frozen-hidden condition gives $W = c\sigma^2$ exactly; no massive dark-matter-like fable is frozen-compatible.	Confirmed.	Section 4.5.7.
DFT- 4	critical, hidden as- sumption	For mass-varying W the one-loop Dirac-sea energy $\Delta E_{\text{vac}}(m)$ is not absorbable and is enormous.	Partly confirmed. Overstated: it can be absorbed formally by reading W as the fully renormalized effective potential — which is a fine-tuning; the sea also adds $\sigma_{\text{vac}} = d\Delta E_{\text{vac}}/dm$. The runs use that reading and print $\Delta E_{\text{vac}}(m(t)) - \Delta E_{\text{vac}}(m_{\text{today}})$. (A figure slip in this verdict is corrected in section 5.8.)	Sections 3.7.3, 4.5.11; the column vac_over_rhoc.
DFT- 5	major, wrong	The classical limit is the non-relativistic limit (Pauli forbids a coherent zero mode), with the expansions and the sign rule that removes Part VI’s $Ms > 0$ branch.	Confirmed.	Sections 3.7.7, 4.5.8.
DFT- 6	major, in- complete	The missed theorem $\rho + P_{\text{obs}} = \omega_F n \geq 0$: the quantized fable cannot cross $w = -1$; the Part VI crossing is excluded by $0 < \sigma < \min(n, s_1)$.	Partly confirmed, plus missed: $w_f \geq -1$ holds, but the observer’s $w_{\text{DE,eff}}$ can be phantom and cross -1 when m_{eff} grows (the Das-Corasaniti-Khoury mechanism): power (0.3, 0.05), $\nu = 1/2$, $k_{F0} = 0.1$ gives $w_{\text{DE,eff}} = -3.39$ at $a = 0.7$, -1.21 at 0.9 , crossing near 1.	Sections 3.7.9, 4.5.9, 4.8.1.
DFT- 7	major, wrong	“Bound states below $\sqrt{m^2 + k^2 e^{2a_4}}$ for $k \neq 0$ ” is false in general: with the same-sign MIT wall there is no level at $K = 3$; the first appears near $K_\infty \approx 6.71H$; the opposite-sign wall has an edge band $\omega \approx \mp 0.98K$. The reviewer gave $n = 0$ and $n = 1$ at $K = 80$ as 70.98531 and 79.37150.	Partly confirmed. Labelling corrected: levels must be counted in the full 16-component problem, whose spectrum is $\omega \rightarrow -\omega$ symmetric; at $K = 80$ the positive levels are 70.98531 ($n = 0$), 78.14425 ($n = 1$, from the other irrep) and 79.37150 ($n = 2$).	Sections 4.6.3–4.6.5 (see also section 5.8).

ID	severity, kind	the finding	verdict	adopted, and where it stands now
DFT-8	major, better for- mulation	In this real-Clifford convention MIT is $T_{16}[0]\Psi = \varepsilon\Psi$; zero normal current alone does not single it out; only the MIT pair is Lorentz-covariant; the explicit 2×2 ODE.	Partly confirmed, refined: the covariant, J -preserving, self-adjoint walls are $P = T_{16}[0]Q$ with Q any Hermitian involution in the 8-dimensional commutant of $T_{16}[0..4]$; flavour symmetry gives $Q = \pm 1$; without J -preservation there are more $(T_0 \exp(i\theta T_0 T_5))$.	Section 4.6.3; the wall-state derivation selects $Q = \pm 1$ by flavour symmetry and charge conjugation.
DFT-9	major, ill-posed	The Δ SCF construction as written is ill-posed (moving one quantum changes σ by $O(1/\text{area})$); define it per transverse area with finite fractions; for the mass term the theory is free.	Confirmed.	Section 4.6.6.
DFT-10	major, hidden as- sumption	The gap equation must use the 8D density; couplings run as 4D couplings ($\lambda_4 = \lambda_8/V_{\text{hid}}$).	Confirmed.	Section 4.5.1.
DFT-11	major, in- complete	A degenerate dark-matter fable carries extra radiation: $\Delta N_{\text{eff}} = 6.7, 0.31, 0.014$ at 1, 10, 100 eV ($\propto m^{-4/3}$), so $m \gtrsim 15$ eV; Ω_{r0} must be computed.	Confirmed.	Section 4.7.9.
DFT-12	major, implemen- tation risk	The log closed forms cancel catastrophically for $x = k_F/ m \ll 1$; use series for $x < 0.25$, the asinh forms above, never $\varepsilon - m\sigma$ by subtraction.	Confirmed.	Sections 3.7.5, 4.5.2; <code>fermi_integrals_match_quadrature</code> .
DFT-13	major, implemen- tation risk	The numerical pitfalls: 22 decades of t ; brackets per potential; the poles of $w_{\text{DE,eff}}$; matching points for the wall states (a fixed $z_m = 0.3$ lost the level -70.985 at $K = -80$; a uniform ω grid missed the shallow level 8.05384 at $K = 8$).	Confirmed.	Sections 4.6.4, 4.7.1.
DFT-14	minor, wrong	$E_x/E_H \rightarrow 1/g$ only for $k_F \ll m$; it is $O(1)$ or larger for $k_F \gtrsim m$.	Confirmed.	Section 4.5.12.
DFT-15	minor, in- complete	The pair continuum starts at $\omega_F + m $ (at $q = k_F$), at $2\omega_F$ for $q = 0$; particle-hole excitations are gapless.	Confirmed.	Section 4.6.2.
DFT-16	minor, hidden as- sumption	The parameter domains of power , expdamp , quadratic .	Confirmed.	Section 4.5.3.

What the DFT skeptic found **missed**: the apparent phantom of $w_{\text{DE,eff}}$ (above); the free-streaming bound for a degenerate dark-matter fable ($v_{\text{rms}0} = \sqrt{3/5}k_{F0}/m = 4.5, 2.6$ and 0.21 km/s at 10, 15 and 100 eV against $0.0094\text{--}0.0044$ km/s for Lyman- α -allowed thermal warm dark matter of $3\text{--}5.3$ keV, requiring $m \approx 1.0\text{--}1.8$ keV, far stronger than ΔN_{eff} ; Tremaine–Gunn to be quoted) — section 4.7.9; the wall-condition family $P = T_{16}[0]Q$; the 16-component labelling of the wall levels; the gap solver failing in the non-relativistic limit ($x \lesssim 10^{-7}$: solve in $\delta = 1 - \sigma/n$ with the series for $n - \sigma_{\text{KS}}$); and the **expdamp** bracket $(0, \min(n, s_1))$, not $(-n, n)$ ($\exp(-\sigma/s_1)$ overflowed near $\sigma = -n$).

5.6 What the review changed in the answers to [1] and [2]

Before the review the design expected a mechanism for a time-varying dark-energy w from the condensate and from the classical phantom crossing. After it, answer [1] had to separate three things — no true phantom ($w_f \geq -1$), an apparent phantom and a crossing of -1 in the observer’s inferred dark energy, and the fine-tuning of the Dirac-sea energy — and to rest on the stabilized model, with its null-energy-condition-violating stabilizer stated, and with `fable8d` reported as a no-go. Answer [2] had to add the free-streaming and phase-space bounds, which move the allowed mass to about a keV, where the change of w_{DM} from $1/3$ to 0 happens long before recombination. Both answers, as finally computed, are section 4.8.

5.7 Corrections to Effort A that the review produced

They are listed with their places in section 3.9.4 (the table “The design review, and every correction it made to Effort A”).

5.8 Corrections found after the review, by the implementation and the checks

These are the errors found later — in the review’s own records, in the code, and in earlier pages — each with how it was found and where it was fixed.

1. **The vacuum-energy slip of 10^4 .** The DFT skeptic’s verdict on DFT-4 gave the variation $gm^4/(64\pi^2)$ of the Dirac-sea energy as $3.44 \times 10^{-4}, 3.44, 3.44 \times 10^8, 3.44 \times 10^{12} \rho_c$ at $m = 0.01, 0.1, 1, 10 \text{ eV}$. The first two are off by 10^4 : the correct values are 3.44 and 3.44×10^4 (the quantity scales as m^4 : $g/(64\pi^2) = 0.012665$, $(m/E_c)^4 = 271.9$ at $m = 0.01 \text{ eV}$). The addendum was corrected in place (“corrected: an earlier line here had `3.44e-4` and `3.44` for the first two”), and the correction is stated in section 3.7.3. Notebook 05 prints the exact threshold, `g M^4/(64 pi^2) exceeds rho_c0 for M > (64 pi^2/g)^(1/4)` `E_c = 7.341 meV [nb05 §5.8]`. (This is a different thing from the solver’s own rounding noise in the same quantity, item 4 below.)
2. **The reviewer’s sign of F .** Finding E1 wrote the hidden-sheet driver of the Kohn–Sham fluid as $F = \sigma W' - 2W$. The skeptic’s computation gives $F = \rho - 3P_{\text{obs}} + 2P_{\text{hid}} = 2W - \sigma W'$ ($m_0\sigma$ for the mass term, $2.1539 = m_0\sigma + 2V_0$ at $(m_0 = 1, V_0 = 0.3)$, and 5×10^{-13} , -1.7×10^{-18} for the pure quadratic); the notebook asserts that `sign (F == 2W - sigma W')`, section 4.5.6). Separately, E7’s residual background driver $F_{\text{bg}} = -132H^2 \cot^2 / (\kappa C^2)$ was wrong and replaced by -12 (for A, B) and -72 (for C) in units of $H^2 \cot^2 / C^2$ (section 4.3.3).
3. **The labelling of the first excited wall level (X2).** The review gave $n = 1$ at $K = 80$ as `79.37150`. The wall-state solver, counting the levels exactly in the full 16-component problem, found $n = 1$ at `78.144249332` (irrep `+i`) and `79.371495779` as $n = 2$, and it also refuted the review’s “a second positive level only near $K_\infty \approx 80$ ”: the second level appears at $K_c(2) = 33.91895851H$, almost independent of m (`33.88–34.29` for $m/H = 0.25 \dots 8$) [WG 7, items 10, 11] — section 4.6.5. The same report confirms, refines or refutes 25 claims of the review and of the orchestrator: claim 10 is refuted in part (its $n = 1$) and claim 11 refuted; 6, 14, 16, 17 and 25 are refined; 3 is confirmed with a basis caveat and 15 confirmed and generalized; 19 is confirmed in part (the level exists; the failure of a uniform grid was not re-tested); 20 was not re-tested and 21 not needed by the method; the other thirteen are confirmed [WG 7].
4. **The six solver defects of 2bc936d**, exposed when notebooks 05 and 06 first ran the solver on every case (22:53 PDT) [commit 2bc936d]: 1. *The negative-energy refusal* now is detected inside the right-hand side on every internal CVODE step, and its onset bisected to adjacent doubles: `lorentz` refuses with one line at $a = 0.4677689478$ whatever `--points`

is. Before, the onset moved with the grid, and `--points 11` died with a CVODE `mxstep` error. A minimum step, `CVodeSetMinStep(64 eps max|x|)`, keeps singular approaches fast; successful outputs are byte-identical with it. 2. *The first-order transition* (repulsive quadratic) is also detected inside the right-hand side and bisected, at $a = 0.9962806544$ for any grid; `scipy` agrees to 8×10^{-12} . The message names both branches and the missing Maxwell construction. The gap-root scan now finds the extremum between merging roots, so the root count changes exactly at the fold (before, it lost the pair 1.4×10^{-7} in N early). 3. *The fable8d shooting*. Trial runs no longer use the refusal checks as objective values, and any $|H_A(1) - 1| > 10^{-6}$ is refused instead of reported as converged. For `lorentz` the old code “converged” to $V_0 = 7.5 \times 10^{13}$ with $H_A(1) = 3.3 \times 10^6$. 4. *vac_over_rho_c* (the renormalized Dirac-sea energy over ρ_{c0}) is now evaluated stably in $u = (m - M)/M$: a series for $|u| \leq 0.75$, which starts at order $(m - M)^5$, and the closed form with `log1p` otherwise. It matches 120-digit references to 1.8×10^{-15} . The old closed form left rounding noise of $1.6 \times 10^4 - 5 \times 10^4 \rho_{c0}$ at 1 keV. 5. *gap_residual* is now measured against the natural scale of W' ; it is at most 3.3×10^{-16} in every phase, including $m \rightarrow 0$. 6. *One $\hbar$* (item 7 below).

The tests went from $21 + 8$ to $34/34$ (23 unit + 11 integration), and the `scipy` cross-check still passes (largest difference 1.1×10^{-8}).

5. **The T_{16} JSON that a fresh clone lacked** [commit aca5102]. The wall-state solver’s input `waveguide_T16.json` (the notebook’s $T_{16}[0..8]$, σ_{16} and the block basis U) used to be written into the git-ignored `fermion/dev/`, so a fresh clone without Mathematica could not run `waveguide.py validate` or notebook 07. It is now written next to the solver and committed; the regenerated file is byte-identical, and the derivation log still reports 94 checks, 94 PASS. `waveguide.py validate` no longer returns early when the file is missing (it records one FAIL each for V7 and V8, naming the command that rewrites the file, and still runs its other checks) and exits with status 1 whenever any check fails. Its module docstring, which still cited “81/81 PASS” from an earlier version of the derivation, now cites the committed log’s 94 checks, 94 PASS, 0 FAIL.
6. **The wording of an earlier provenance page** [commit b7ea18d]. The earlier provenance page on the frame field and the canonical spin connection said that the canonical connection “rotates each spatial frame axis against the hidden-space direction and boosts it against time” for all six directions it excites. That holds for the observed directions $j = 1, 2, 3$. For the deflating directions $k = 5, 6, 7$ the roles are exchanged, because $x_5..x_7$ are timelike like x_4 : the $(0, k)$ plane is a **boost** and the $(4, k)$ plane a **rotation**. Only the prose was wrong; the component table was always right. The fact-check of the spin-connection section of the fermion-fable pages found it; the correct statement is in section 3.8.6.
7. **The inconsistent $\hbar$** [commit 2bc936d; the constants of the solver]. The solver used to compute H_0 in eV with the truncated constant $\hbar = 6.582119569 \times 10^{-16}$ eVs, and the reduced Planck mass with $\hbar = 1.054571817 \times 10^{-34}$ Js; the two differ by 6.1×10^{-10} . `HBAR_EV_S` is now `HBAR_J_S/EV_J`, one CODATA-2018 value for both (the same as the Mathematica reference script). This moved E_c from $2.46261317798907812 \times 10^{-3}$ to $2.46261317732986325 \times 10^{-3}$ eV (-2.7×10^{-10}) and Ω_{r0} from $9.20960545742757834 \times 10^{-5}$ to $9.20960546728882807 \times 10^{-5}$ ($+1.1 \times 10^{-9}$); the comments no longer call the truncated constants “exact”; `fermion/crosscheck.py` uses the same convention; notebooks 05–07 assert that the unit system recomputed in Python agrees with the solver’s to 10^{-9} .
8. **The quantization reviewer’s number (QK-1)**: the above-threshold frequencies $\pm 1.705872i$ were wrong; $\pm 1.19896i$ ($= \sqrt{4 - 2.5625i}$) is right, and notebook 05 prints it [nb05 §3.3].
9. **The direction of the g_{*s} correction (E12)**: lower, not higher, temperatures in the past.

10. **The fact-check of the page of Effort A** corrected 7 errors in its markdown and LaTeX before it was committed [commit fee5603]; the corrected text is what section 3 reproduces.
11. **The wall-state solver's own bugs**, found by its checks and fixed before its commit: a Magnus grid too coarse near the wall (errors 6×10^{-5} rad; fixed by geometric grading to $z = 0.05/H$), a trapezoid normalization (10^{-5} ; replaced by Simpson), a wrong identification of the edge branch at $K \geq 80$, a grid mismatch in the band-gap step, and shell escaping in two patches; the first Mathematica check printed `NDSolveValue::precw` (fixed by 20 and 10 extra digits of input) [WG 8].
12. **The generator of the notebooks of Effort 0** said four things wrongly; they were corrected (section 2.5).

6 Effort C: the commands that solve the coupled system, and test, verify, execute and display the calculations

This section is the working manual of the whole solution. It gives **every command**, in full, in a fenced block; for each one what it does and why, what it needs first, how long it takes, what it prints (quoted from the real logs and outputs, with the file named), and how to tell success from failure. Section 6.2 gives the order in which to run them. All commands run in **Git Bash on Windows 11**, where they were run; they are plain POSIX shell and run the same way on Linux and macOS with `.venv/bin/python` in place of `.venv/Scripts/python.exe` and without the `.exe` suffix. Where a PowerShell twin exists, it is given too.

6.1 The toolchain, and how to set it up

What is needed, and the versions that produced every result on this page [the tools' own `--version` output on the development machine, 2026-09-25; the first line of the run logs]:

tool	used for	version	check
Git (Git for Windows, with Git Bash and Perl)	the repository; the shell; Perl for <code>latexmk</code>	<code>git version 2.51.2.windows.1</code>	<code>git --version</code>
Wolfram Language / <code>wolframscript</code>	the notebook (Parts I–VIII), the TeX export, the Mathematica reference runs, the wall-problem derivation and its check	Wolfram Language 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026); WolframScript 1.14.0 for Microsoft Windows (64-bit)	<code>wolframscript -version</code>
Rust (<code>cargo</code> , <code>rustc</code> , MSVC toolchain on Windows)	the solvers <code>fable_cosmo</code> and <code>fable_fermion</code>	<code>cargo 1.91.1 (ea2d97820 2025-10-10)</code> , <code>rustc 1.91.1 (ed61e7d7e 2025-11-07)</code>	<code>cargo --version</code>
the engine	pure-Rust SUNDIALS 7.8.0 CVODE (<code>sundials_rs</code>), sparse-cloned from the <code>rustSolveIt</code> repository of the platform	<code>rustSolveIt_Win11_SUNDIALS_7_8_0 @ a8fdff4</code> ; the binaries report <code>sundials_rs 7.8.0 (pure Rust)</code> , CVODE BDF	<code>setup.sh</code> prints <code>== engine: <url> @ <commit></code>
Python 3.10 or newer, in the virtual environment <code>fable-cosmology/.venv</code>	the notebooks, the cross-check, the wall-state solver, the notebook builder	Python 3.14.5; numpy 2.5.3, scipy 1.18.1, matplotlib 3.11.2, nbconvert 7.17.1, nbclient 0.11.0, nbformat 5.11.1, ipykernel 7.3.0	<code>python --version</code>

tool	used for	version	check
MiKTeX (TeX Live on Linux/macOS) with latexmk	the paper of Effort 0; the LaTeX twins of the provenance pages	MiKTeX-pdfTeX 4.27 (MiKTeX 26.5); Latexmk, John Collins, 9 March 2026. Version 4.88; MiKTeX installed at C:/Program Files/MiKTeX/miktex/bin/x64	pdflatex --version, latexmk -v

How to install them, per platform: **Windows 11** — Git for Windows (<https://git-scm.com/download/win>); Rust with `rustup-init.exe` from <https://rustup.rs> (it uses the MSVC toolchain, which needs the “Desktop development with C++” workload of the Visual Studio Build Tools; rustup offers to install it); Python from <https://www.python.org/downloads/> with “Add python.exe to PATH” ticked; MiKTeX from <https://miktex.org/download>, allowed to install missing packages on the fly; Perl for latexmk is in Git Bash (in PowerShell run `$env:Path += ";C:\Program Files\Git\usr\bin"` or install Strawberry Perl). **macOS (Apple silicon)** — `xcode-select --install`; Rust with `curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh`; Python 3.10+; MacTeX (which contains latexmk and Perl). **Linux (x86-64, Debian/Ubuntu)** — `sudo apt install build-essential git`; Rust as on macOS; `sudo apt install python3 python3-venv`; `sudo apt install latexmk lmodern texlive-latex-recommended texlive-latex-extra texlive-fonts-recommended`. Wolfram Mathematica (developed with 15.0.1) is needed only for the Mathematica steps (sections 6.3, 6.6.1, 6.6.4); everything else runs without it, because the notebook’s log, the reference CSVs, the derivation’s log and its export `waveguide_T16.json` are committed. An internet connection is needed once, for the setup.

Get the repository:

```
git clone https://github.com/once-ere/Pre-Universe_with_Claude.git
cd Pre-Universe_with_Claude
```

The repository carries a `.gitattributes` with `* -text`, so git never rewrites line endings and every file checks out with the bytes that were committed. Every command below starts from the repository root; `cd "$(git rev-parse --show-toplevel)"` takes you there from anywhere inside it.

Set up, once (Git Bash, macOS, Linux; PowerShell twin below):

```
bash fable-cosmology/setup.sh
```

```
powershell -ExecutionPolicy Bypass -File fable-cosmology\setup.ps1
```

- **What it does** [`file: fable-cosmology/setup.sh`]. (1) Creates `fable-cosmology/.venv` unless it exists — with `python3` when that runs, else `python` — upgrades pip, installs `fable-cosmology/requirements.txt` (numpy, scipy, matplotlib, jupyter, nbconvert, nbclient, ipykernel), imports the packages and prints `python ok: numpy <v> scipy <v> matplotlib <v>`. (2) Picks the engine repository for the platform (`uname -s: Linux*` → https://github.com/once-ere/rustSolveIt_linux_SUNDIALS_7_8_0.git, `Darwin*` → https://github.com/once-ere/rustSolveIt_macos-silicon_SUNDIALS_7_8_0.git, `MINGW*|MSYS*|CYGWIN*` → https://github.com/once-ere/rustSolveIt_Win11_SUNDIALS_7_8_0.git) and, unless `rust/vendor/rustSolveIt/sundials_rs/crates/cvode_rs` exists, makes a sparse clone (`git clone --depth 1 --filter=blob:none --sparse <url> rust/vendor/rustSolveIt`, then `git sparse-checkout set sundials_rs`), and prints `== engine: <url> @ <commit>`. (3) For each of the two solver crates, `rust/fable_cosmo` and `rust/fable_fermion`, runs `cargo build --release` inside the crate’s own folder and smoke-tests the binary with `--version`. It ends with `== setup complete`.

- **Why this way.** Both crates depend on the engine by path (`../vendor/rustSolveIt/sundials_rs/crates/sundials_core` and `.../cvscode_rs`), exactly as `rustSolveIt`'s own `planet_Mercury/mercury_rs` does, and each pins `-C target-feature=+fma` in its own `cargo/config.toml` on x86-64, because the engine's deterministic math library uses `f64::mul_add` and requires it. Cargo reads that file from the directory it is invoked in, so **each solver must be built from inside its own folder**, as the script does.
- **How long.** On the development machine the whole setup from nothing (pip, the engine clone over the network, the builds) took 80 s in a fresh clone; the first build of `fable_cosmo` took `Finished release profile [optimized] target(s) in 5.04s` (section 2.3).
- **What it prints** in a fresh clone (recorded for `fable_cosmo` on 2026-09-24, before the second crate existed; the scratch path is shortened to ...):

```

== creating .venv
== installing Python requirements into .venv
WARNING: Cache entry deserialization failed, entry ignored
WARNING: Cache entry deserialization failed, entry ignored
WARNING: Cache entry deserialization failed, entry ignored
python ok: numpy 2.5.3 scipy 1.18.1 matplotlib 3.11.2
== cloning the engine (sparse: sundials_rs only) from
↳ https://github.com/once-ere/rustSolveIt_Win11_SUNDIALS_7_8_0.git
Cloning into 'rust/vendor/rustSolveIt'...
== engine: https://github.com/once-ere/rustSolveIt_Win11_SUNDIALS_7_8_0.git @ a8fdff4
== building rust/fable_cosmo (release)
  Compiling sundials_core v7.8.0
  ↳ (...\\fc\\fable-cosmology\\rust\\vendor\\rustSolveIt\\sundials_rs\\crates\\sundials_core)
  Compiling cvscode_rs v7.8.0
  ↳ (...\\fc\\fable-cosmology\\rust\\vendor\\rustSolveIt\\sundials_rs\\crates\\cvscode_rs)
  Compiling fable_cosmo v0.1.0 (...\\fc\\fable-cosmology\\rust\\fable_cosmo)
    Finished `release` profile [optimized] target(s) in 5.04s
fable_cosmo 0.1.0, sundials_rs 7.8.0 (pure Rust), CVODE BDF
== setup complete

```

With both crates the script prints, after that, `== building rust/fable_fermion (release)`, cargo's lines, and `fable_fermion 0.1.0, sundials_rs 7.8.0 (pure Rust), CVODE BDF` before `== setup complete`. The three pip warnings come from pip's local download cache and are harmless. The setup's output in the final fresh clone is recorded in section 7. - **Success and failure.** Success is the two `--version` lines and `== setup complete`, with exit code 0 (`setup.sh` runs under `set -euo pipefail` and stops at the first failing command; `setup.ps1` checks `$LASTEXITCODE` after each build and each `--version` and throws). Typical failures and their causes: `python: command not found` (neither `python3` nor `python` works: install Python or create the environment by hand with `python3 -m venv fable-cosmology/.venv`); `error: linker 'link.exe' not found` or MSVC errors (the C++ build tools are missing); path errors naming `vendor/rustSolveIt` (the engine clone is incomplete: delete `fable-cosmology/rust/vendor/rustSolveIt` and run the setup again); an error about the target feature `fma` (cargo was run from another directory). - **Idempotent.** Re-running it changes nothing that is already in place.

The same three steps by hand (replace `linux` in the URL by `Win11` or `macos-silicon`):

```

cd "$(git rev-parse --show-toplevel)/fable-cosmology"
python3 -m venv .venv # or: python -m venv .venv
.venv/bin/python -m pip install -r requirements.txt # Windows Git Bash:
↳ .venv/Scripts/python.exe
mkdir -p rust/vendor
git clone --depth 1 --filter=blob:none --sparse
↳ https://github.com/once-ere/rustSolveIt_linux_SUNDIALS_7_8_0.git rust/vendor/rustSolveIt
( cd rust/vendor/rustSolveIt && git sparse-checkout set sundials_rs )
( cd rust/fable_cosmo && cargo build --release )

```

```
rust/fable_cosmo/target/release/fable_cosmo --version
( cd rust/fable_fermion && cargo build --release )
rust/fable_fermion/target/release/fable_fermion --version
```

6.2 The order in which to run everything

step	command (section)	needs	time on the development machine	success looks like
1	<code>bash fable-cosmology/setup.sh</code> (6.1)	Git, Rust, Python, network	80 s from nothing	<code>== setup complete</code>
2	<code>cargo test --release</code> in both crates (6.4.2)	step 1	about 45 s (<code>fable_fermion</code>), under 1 s (<code>fable_cosmo</code>)	test result: ok. 23 passed, 11 passed; 7 passed
3	<code>fable_fermion --constants</code> , one <code>gap</code> solve (6.4.3, 6.4.4)	step 1	instant	the unit block; three roots
4	<code>python fermion/crosscheck.py --jobs 6</code> (6.5)	step 1	about 10 s	CROSSCHECK PASSED: every difference < 1e-06
5	<code>python fermion/waveguide.py validate</code> (6.6.2)	step 1	31–45 s	VALIDATION: 64 of 64 PASS, exit code 0
6	execute notebooks 05, 06, 07 and check them (6.7)	steps 1 (and 5 for 07)	7 s, 93 s, 558 s	7/7 notebooks pass all eight requirements
7	or steps 2 and 6 for all seven notebooks, plus the paper: <code>bash fable-cosmology/run_all.sh</code> (6.8)	step 1	12 min 18 s	<code>== all done</code>
8	<code>python build_tools.py</code> , then <code>verify_nb.wls</code> (6.3.1, 6.3.2)	Python; Mathematica	0.1 s; 4 s	byte-identical notebook; input cells that FAIL to parse: {}
9	<code>wolframscript -file run_from_nb.wls</code> (6.3.4)	Mathematica; for the full 644, step 6 first (it writes the Rust CSVs Part VIII compares with)	208.715 s	Assertions run: 644 passed: 644 failed: 0, cells w/ msgs : 0
10	the TeX export, the reference runs, the wall-problem derivation and its check (6.3.6, 6.3.7, 6.6.1, 6.6.4)	Mathematica	a notebook run; not recorded; 163–171 s; 40 s	14 files written; 701 rows each; 94 checks, 94 PASS, 0 FAIL; 11 checks, 11 PASS, 0 FAIL
11	<code>bash provenance-latex/build_all.sh</code> (6.9)	MiKTeX or TeX Live	—	<code>== all three PDFs built</code>
12	<code>commit</code> , <code>push</code> , <code>git ls-remote</code> , fresh clone (6.10, 7)	everything above	—	the local and remote hashes equal; every check reproduced

The Mathematica steps (8–10) and the Python/Rust steps (1–7) are independent, with one exception: Part VIII’s comparison with the Rust solver runs only when notebook 06 has written `results/nb06_fable4d_mass30eV.csv` and `results/nb06_fable4d_power.csv` (otherwise it is skipped, not failed, and the run reports 642/642 instead of 644/644).

6.3 The Mathematica notebook: build, check, map, run, verify, export, reference runs

The notebook `claude-fable/claude-fable_Einstein-Rosen-2-Planes.nb` holds all the symbolic mathematics: Parts I–V (the author’s pre-universe, its algebra, the canonical frame and spin connection, the bridges), Part VI (Effort 0’s classical fields), Part VII (Effort A) and Part VIII (Effort B’s symbolic side). Every claim is an assertion `cfAssert[label, test]`, which prints `PASS label` or `FAIL label`.

6.3.1 Build the notebook from its manifests

The notebook is never edited by hand. `build_tools.py` reads every `claude-fable/cells_part*.wl` in order and writes both the notebook and `run_all.wls` (the same Input cells as a script).

```
cd "$(git rev-parse --show-toplevel)/claude-fable"
python build_tools.py          # use python3 where python is absent
```

It prints (recorded on 2026-09-24, with the eight committed manifests, in a scratch copy of the directory so that nothing in the repository was touched; it ran in 0.115 s; the last line gives the absolute path of the directory it ran in):

```
manifest files : ['cells_part1.wl', 'cells_part2.wl', 'cells_part3.wl', 'cells_part4.wl',
↪ 'cells_part5.wl', 'cells_part6.wl', 'cells_part7.wl', 'cells_part8.wl']
cells parsed   : 384 {'Title': 8, 'Subtitle': 1, 'Subsubtitle': 1, 'Text': 123, 'Section':
↪ 34, 'Input': 217}
run_all.wls    : 217 Input cells
notebook       : 384 cells -> <repo>\claude-fable\claude-fable_Einstein-Rosen-2-Planes.nb
```

Success: the rebuilt `claude-fable_Einstein-Rosen-2-Planes.nb` and `run_all.wls` are **byte-identical** to the committed ones (`cmp` reports no difference; `git status` shows nothing). With only the seven manifests of Parts I–VII present — the state in which Part VII was run — it prints `cells parsed : 340 {'Title': 7, 'Subtitle': 1, 'Subsubtitle': 1, 'Text': 103, 'Section': 30, 'Input': 198}` and `run_all.wls : 198 Input cells`. The notebook has not changed since `bc04ed1`.

6.3.2 Check the notebook without evaluating it

```
wolframscript -file "$(git rev-parse --show-toplevel)/claude-fable/verify_nb.wls"
```

It imports the `.nb`, counts its cells by style, extracts the Input cells, and checks that every one of them parses. About 4 s (4.118 s when timed). It prints `[file: claude-fable/verify_nb_part8.log, and identically when re-run on 2026-09-24]`:

```
file bytes: 545408
Head: Notebook
cells: 384
style tally: {{Title, 8}, {Subtitle, 1}, {Subsubtitle, 1}, {Text, 123}, {Section, 34},
↪ {Input, 217}}
input cells with plain-string BoxData: 217
total input characters: 347187
input cells that FAIL to parse: {}
first cell: InputForm[(* --- provenance banner, as in the original notebook
↪ -----]
last cell : InputForm[cfAssert["PART VIII [control]: a4 is STILL UNDEFINED -- nothing in Part
↪ VIII gave it a val]
```

Success: input cells that FAIL to parse: `{}`. For the Part VII notebook it printed (`claude-fable/verify_nb_part7.log`) `file bytes: 440320, cells: 340, the tally with 198 Input cells, total input characters: 279955, input cells that FAIL to parse: {}, and the last cell (* --- final tally of every assertion made in this notebook, Parts I–VII ....`

6.3.3 Print the section-to-cell map

```
wolframscript -file "$(git rev-parse --show-toplevel)/claude-fable/nb_section_map.wls" >  
↪ "$(git rev-parse --show-toplevel)/claude-fable/nb_section_map.log"
```

It reads the .nb without evaluating anything and prints, in document order, each Part title and each Section with the range of its Input cells (3.340 s when timed). Its output, `claude-fable/nb_section_map.log`, is the table of contents of the notebook and the key to every “Section n” on this page:

```
== claude-fable_Einstein-Rosen-2-Planes  
0. What this notebook does, and how it is organized | Input cells none  
1. Session setup | Input cells 1-6  
2. Coordinates and the flat 4+4 Minkowski metric eta | Input cells 7-12  
3. SO(4): self-dual and anti-self-dual antisymmetric 4x4 matrices | Input cells 13-15  
4. The real 8x8 Clifford generators tau, their conjugates, and the spinor metric sigma |  
↪ Input cells 16-21  
5. The 16x16 Dirac matrices T16, the chirality operator, sigma16 and the so(4,4) generators  
↪ | Input cells 22-29  
6. A complete basis of the 16x16 matrix algebra | Input cells 30-35  
7. A complete basis of the 8x8 matrix algebra, and the induced flat metric | Input cells  
↪ 36-38  
8. The 4x4 Dirac matrices, and a complete basis of the 4x4 matrix algebra | Input cells  
↪ 39-44  
9. Cartan triality: the constant bridge between the vector and spinor realizations,  
and the split-octonion structure constants | Input cells 45-52  
== PART II -- The wave function of the un-universe, its field equations, and the 3  
↪ generations  
10. Housekeeping: where this notebook lives, and the two helper packages | Input cells  
↪ 53-55  
11. The wave function of the un-universe, and its Lagrangian | Input cells 56-60  
12. The Euler-Lagrange equations, the coupling pattern, and the (z,t) chart | Input cells  
↪ 61-66  
13. Decoupling into four blocks of four, and the closed-form solutions | Input cells 67-78  
14. Bilinears, the two solution branches, and M6 = 3 generations of Einstein-Rosen 2-planes  
↪ | Input cells 79-90  
== PART III -- The curved 4+4 spacetime, its frame field, and the canonical spin connection  
15. The local flat 4+4 Minkowski coordinate system at every spacetime point | Input cells  
↪ 91-99  
16. The generalized Christoffel symbols, and the canonical spin connection | Input cells  
↪ 100-110  
17. The gauge-covariant derivative of the 16-component spinor | Input cells 111-117  
== PART IV -- Three bridges between curved and flat indices: two gauge transformations, and  
↪ one new connection  
18. Bridge 1 -- a locally boosted frame: the same connection in a local gauge | Input  
↪ cells 118-125  
19. Bridge 2 -- the null (light-cone) frame: the same connection in a constant gauge, and  
↪ the spinor metric sigma | Input cells 126-130  
20. Bridge 3 -- the triality frame with totally antisymmetric split-octonion torsion |  
↪ Input cells 131-139  
== PART V -- The fable-5.1 bridge: the connection in which the frame itself is parallel  
21. The fable-5.1 bridge -- the Weitzenboeck (teleparallel) connection of the canonical  
↪ frame | Input cells 140-148  
22. Master comparison of the five spin connections | Input cells 149-152  
== PART VI -- fableScalar and fable: two fields on the pre-universe, and the equation of  
↪ state of dark energy  
23. fableScalar -- a scalar field on the pre-universe, its energy-momentum tensor, and  
↪ wScalar | Input cells 153-160  
24. fable -- the 16-component spinor field, its energy-momentum tensor, and w | Input  
↪ cells 161-169  
25. Reading the two mechanisms side by side | Input cells 170-172  
== PART VII -- fermion fable: the complex 16-spinor, its canonical quantization in 4+4  
↪ dimensions, its field equations in the primordial gravitational field, and its  
↪ energy-momentum tensor operator
```

26. The refinement: why fermion fable is a complex 16-spinor | Input cells 173-179
 27. The Lagrangian, the field equations in the primordial gravitational field, and the
 ↪ canonical spin connection | Input cells 180-184
 28. Canonical quantization in 4+4 dimensions: the Dirac bracket, the Krein structure J, and
 ↪ the admissible sector | Input cells 185-191
 29. The energy-momentum tensor operator of fermion fable: definition, conservation, and rho,
 ↪ P and w | Input cells 192-198
 == PART VIII -- fable as a source of the Einstein equations of the primordial gravitational
 ↪ field
 30. The Einstein tensor of the canonical metric, and what it demands of its source | Input
 ↪ cells 199-203
 31. The generalized warped frame, its asymptotic region, and the 8-dimensional Bianchi-I
 ↪ limit | Input cells 204-207
 32. The canonical spin connection of the interacting system | Input cells 208-210
 33. The coupled equations of fable and the primordial field, and their solution from the
 ↪ radiation era to today | Input cells 211-217
 total Input cells: 217

Where each Section of Parts VII and VIII is discussed on this page:

notebook Section	Input cells	assertions	seconds (Part VII log; Part VIII log)	this page
26	173-179	38	1.957	3.2
27	180-184	36	2.291	3.3, 3.4, 3.8
28	185-191	56	12.311	3.5
29	192-198	40	17.874	3.6, 3.7
30	199-203	28	4.119	4.2
31	204-207	21	12.672	4.3
32	208-210	26	4.694	4.4
33	211-217	39 (41 in the final run)	2.526	4.5, 4.7

6.3.4 Run the notebook end to end, straight out of the .nb

```
cd "$(git rev-parse --show-toplevel)/claude-fable"
wolframscript -file run_from_nb.wls > run_fermion_fable_final.log 2>&1
git checkout -- claude-fable_Einstein-Rosen-2-Planes-eLa.mx
↪ claude-fable_Einstein-Rosen-2-Planes-eLazt.mx
```

- **What it does.** `run_from_nb.wls` imports the `.nb`, takes its Input cells in order, and evaluates each one as a single unit (a multi-line cell becomes one `CompoundExpression`, exactly as the front end evaluates it), through the harness `runner_header.wl`. The harness prints `CELL n t=... s` after each cell, with `MESSAGES: ...` appended if the cell raised any message, prints every `PASS` or `FAIL` label, and ends with the run summary and the assertion counts. The script is self-locating (`$InputFileName`) and finds the notebook from any directory.
- **Why from inside `claude-fable/`.** Run headless, the notebook cannot ask for its own file name, so Section 10 looks for the author's helper packages `ConvertMapleToMathematicaV2.wl` and `EtoExp.wl` in the working directory (and in `Pre-Universe_14SEP26-77/` below it, and one level up); they sit in `claude-fable/`. The reference CSVs of Parts VI and VIII are found as `claude-fable/..fable-cosmology/reference/...` (the log prints these paths), and Part VIII's Rust CSVs as `claude-fable/..fable-cosmology/results/...`
- **The `.mx` restore.** Section 12 of the notebook writes `claude-fable_Einstein-Rosen-2-Planes-eLa.mx` and `...-eLazt.mx` with `DumpSave`, whose bytes are not reproducible from run to run (their content is: the wall-state report records that loading the committed and the rewritten files gives `eLa === eLa` and `eLazt === eLazt` [WG 1]). The two files are restored to the committed version after every run, as the second command does.

- **How long.** 208.715 s for the 217 cells of the final run (total seconds : 208.715); the earlier runs took 232.146 s (Parts I–VII, 198 cells) and 233.289 s (Parts I–VIII, 217 cells). The slowest cell is Part II’s cell 80, the bilinears of the author’s closed-form solution (69.797 s in the final run).
- **What it prints.** First evaluating 217 Input cells straight out of the .nb, then the provenance banner of the original notebook (CopyRight (C) 2022, Patrick L. Nash, under the General Public License., Please cite this work, and this web page, if you use it., Refactored for claude-fable_Einstein-Rosen-2-Planes.nb; Wolfram Language 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)), then every cell’s line, PASS labels and notes. The Part VII portion is quoted in full in section 3.9.2. Part VIII’s last cell prints the Mathematica solution of the two reference cases (section 4.7.12), then the tallies. The final run ends [file: claude-fable/run_fermion_fable_final.log]:

```

identities accepted on numerical evidence alone : 0    (stage 3 returned True)
non-vanishing witnesses                        : 46    (cfNonZeroWitnessQ found a probe point
↪ at which every entry is a
                                     number and one of them is non-zero: a
                                     ↪ complete proof of non-vanishing)

PASS no identity in this notebook was accepted on numerical evidence alone
Assertions run: 644   passed: 644   failed: 0
CELL 217   t=0.000 s
===== RUN SUMMARY =====
cells evaluated : 217
total seconds   : 208.715
cells w/ msgs   : 0
---- slowest cells ----
cell 80  69.797 s
cell 88  22.871 s
cell 204  9.939 s
cell 191  8.828 s
cell 86   8.339 s
cell 87   7.361 s
cell 137  7.359 s
cell 121  4.840 s
cell 196  3.848 s
cell 160  3.824 s
cell 107  3.789 s
cell 193  3.062 s
===== ASSERTIONS =====
assertions run : 644
passed         : 644
FAILED         : 0
=====

```

- **Success and failure.** Success is assertions run : 644, passed : 644, FAILED : 0, cells w/ msgs : 0, and identities accepted on numerical evidence alone : 0. A failed assertion prints FAIL label and is counted in FAILED; a cell that raises any message gets MESSAGES: ... on its CELL line and is counted in cells w/ msgs. The run also prints four EXPECTED MESSAGE(S) lines (three in Part II, one in Part V: messages that the original notebook’s time budgets or its underdetermined Solve calls may raise, announced in advance) and one NOTE in Part III (that a4 is an undefined scalar function inherited from the original notebook); none of them is a message raised by a cell. If the Rust CSVs are absent, Section 33 prints twice NOTE Rust CSV not found; that comparison was skipped, not failed, and the run ends at 642/642 — that is how claude-fable/run_fermion_fable_part8.log ends (total seconds : 233.289), because it was made before notebook 06 wrote them.

The three committed logs of the notebook of 2026-09-24/25, and what separates them:

log	notebook	Input cells	assertions	seconds	witnesses	commit
claude-fable/run_fermion_fable_part7.log	Parts I–VII	198	528/528	232.146	38	4fdfc40
claude-fable/run_fermion_fable_part8.log	Parts I–VIII, Rust CSVs absent	217	642/642	233.289	46	bc04ed1
claude-fable/run_fermion_fable_final.log	Parts I–VIII, Rust CSVs present	217	644/644	208.715	46	1671155

The first two end with a line RUN-DONE that the wrapper which ran them appended; the final one has none. All three were produced by the same command with the output redirected to the named log.

6.3.5 Check a log

```
cd "$(git rev-parse --show-toplevel)/claude-fable"
grep -c '^ PASS' run_fermion_fable_final.log      # 644
grep -c '^ PASS' run_fermion_fable_part8.log     # 642
grep -c '^ PASS' run_fermion_fable_part7.log     # 528
grep -n '^ FAIL' run_fermion_fable_final.log     # no output: no assertion failed
grep -c 'MESSAGES: ' run_fermion_fable_final.log # 0: no cell raised a message
grep -E 'identities accepted|non-vanishing witnesses' run_fermion_fable_final.log | tail -2
# identities accepted on numerical evidence alone : 0 (stage 3 returned True)
# non-vanishing witnesses : 46 (cfNonZeroWitnessQ found a probe
↪ point at which every entry is a
sed -n '/RUN SUMMARY/, $p' run_fermion_fable_final.log # the summary quoted in 6.3.4
diff <(grep '^ PASS' run_fermion_fable_part8.log) <(grep '^ PASS'
↪ run_fermion_fable_final.log)
# 641a642,643
# > PASS fable4d RUNS [fidelity]: the Rust solver's mass30eV run agrees with this cell
↪ to 1e-6 in every column, at every one of the rows its grid covers
# > PASS fable4d RUNS [fidelity]: the Rust solver's power run agrees with this cell to
↪ 1e-6 in every column, at every one of the rows its grid covers
```

The counts are the results on the committed logs (checked for this page). A plain `grep FAIL` also finds the summary line `FAILED : 0` and two `PASS` labels whose text contains the word “FAILS” (control assertions that show a wrong sign failing, lines 298 and 921 of both the Part VIII log and the final log); `grep -n '^ FAIL'` finds only real failures.

6.3.6 Export the field equations, the anticommutator, J and the equations of state to TeX and text

```
wolframscript -file "$(git rev-parse
↪ --show-toplevel)/claude-fable/export_fermion_fable_tex.wls"
```

It is self-locating and sets its working directory to `claude-fable/`, so it runs from anywhere. It evaluates every Input cell of the notebook with the output suppressed (so it takes about as long as a run; its duration was not recorded), prints the assertion count, stops with exit code 1 if the Part VII objects are missing, and writes fourteen files (seven `.tex` and seven `.txt`) into `provenance-latex/generated/`: the 16 component field equations, the 16 adjoint equations, the 8+8 form, the rescaled equations, the anticommutator $\{\Psi_a, \Psi_b^\dagger\} = H \cos(6Hx_0) J_{ab} \delta^7$ with its matrix, J, beta and $\sigma_{16} = J\beta$, and the Kohn–Sham closed forms (section 3.9.6 lists them with the notebook objects they come from). The script does no mathematics of its own beyond formatting. It was run on 2026-09-24 at 20:41 PDT on the 198-cell notebook and printed:

```
evaluating 198 Input cells of C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\claude-fa
↪ ble\claude-fable_Einstein-Rosen-2-Planes.nb (output suppressed)
assertions: 528 failed: 0 cells with messages: 0
```

```
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\ferm_
↪ ion_fable_field_equations.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\ferm_
↪ ion_fable_adjoint_equations.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\ferm_
↪ ion_fable_field_equations_8plus8.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\ferm_
↪ ion_fable_field_equations_rescaled.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\ferm_
↪ ion_fable_anticommutator.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\ferm_
↪ ion_fable_krein_J.tex / .txt
wrote C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated\ferm_
↪ ion_fable_eos.tex / .txt
done: 14 files in
↪ C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\provenance-latex\generated
```

(That console output was captured outside the repository; the fourteen files were committed in 4fd9c40.) On today's 217-cell notebook the first two lines read `evaluating 217 Input cells ... and assertions: ...`; the Part VII objects it exports are the same. **Success:** done: 14 files, and the .txt files unchanged (`git status` shows nothing for them); the .tex files carry the date of the run in their header comment, so a re-run on another day changes that one line. Like every evaluation of the notebook it rewrites the two .mx files, which are then restored as in 6.3.4. The .txt files are quoted verbatim in sections 3.4, 3.5 and 3.7; a test document that \inputs all seven .tex fragments (with `amsmath`) compiled with pdfLaTeX to 9 pages with no errors, warnings or overfull boxes.

6.3.7 The Mathematica reference runs of the coupled system

```
wolframscript -file "$(git rev-parse
↪ --show-toplevel)/fable-cosmology/reference/make_reference_fermion.wls"
```

- **What it does.** An independent Mathematica solution of two `fable4d` cases, written next to the script as `mathematica_fable4d_mass30eV.csv` and `mathematica_fable4d_power.csv`: 701 values of N from $\ln 10^{-10}$ to 0, columns N , a , t , H_A , ρ_f , P_{obs_f} , w_f , m_{eff} , σ , k_F/m ; every row is exact algebra at 40 significant digits (the power-law gap by Brent's method at 45 digits), the age by `NDSolve` at working precision 30, with the stable series of the Fermi-sea integrals for $k_F/|m| < 1/4$ [file: its header]. Self-locating.
- **What it prints** [file: its `Print` statements]: the units ($E_c = \rho_{c0}^{(1/4)} = \dots$ eV; $1/H_0 = \dots$ yr, the three Ω s), the agreement of the series with the closed forms at $k_F/m = 1/5, 1/100$, the parameters of each case (`mass30eV`: $m_0 = \dots$ $E_c = 30$ eV; $kF0 = \dots$ $E_c = \dots$ eV; $V0 = \Omega_{\text{Lambda}} = \dots$; `power`: $\nu = 1/2$, $m_{\text{today}} = 100$ eV, $\epsilon_{\text{today}} = 0.55$; $kF0 = \dots$ E_c ; $\lambda = \dots$), the timings and checks of each case ($|H_A(1) - 1|$, $w_f(a_i)$, the largest gap residual, $m_{\text{eff}}(a_i)$), and wrote `mathematica_fable4d_mass30eV.csv` and `mathematica_fable4d_power.csv` (701 rows each). Its console output and its duration were not captured in a repository log.
- **How it is checked.** By Part VIII: `fable4d RUNS [fidelity]`: the reference CSVs have the expected header and 701 rows and `fable4d RUNS [fidelity]`: at every tenth row ALL TEN columns agree with `fable-cosmology/reference/mathematica_fable4d_mass30eV.csv` and `_power.csv` to $1e-10$ (relative; P_{obs_f} relative to ρ_f ; N and w_f absolute) (the log prints max difference per column $\{0., 0., 0., 0., 0., 0., 0., 0., 0., 0.\}$ for both), and by notebook 06's three-way comparison (section 4.7.12). **Success:** both CSVs byte-identical to the committed ones after a re-run (`git status` shows nothing).

The classical references of Effort 0 are made the same way, by `wolframscript -file fable-cosmology/reference/make_reference.wls` (427 s on its first run of 2026-09-24, 366 s after the fix of 028b379), which prints exactly two lines (section 2.2).

6.4 The Rust solver `fable_fermion`: build, test, and every run

The solver of the coupled (quantized fermion fable, 8-dimensional primordial gravitational field) system, on the pure-Rust SUNDIALS 7.8.0 CVODE engine. Its models, equations, units and normalization are section 4.7.1; this section is how to drive it.

6.4.1 Build it

```
cd "$(git rev-parse --show-toplevel)/fable-cosmology/rust/fable_fermion"
cargo build --release          # -> target/release/fable_fermion(.exe)
./target/release/fable_fermion --version
```

It prints `fable_fermion 0.1.0, sundials_rs 7.8.0 (pure Rust), CVODE BDF` (the first line of notebooks 05, 06 and 07). Build it from inside its folder: the crate's `.cargo/config.toml` pins `rustflags = ["-C", "target-feature=+fma"]` on x86-64, which its comment states is **required** by the vendored engine (its deterministic math library uses `f64::mul_add`), and cargo reads that file only from the directory it is invoked in. `setup.sh` does the same (section 6.1).

file	tests
------	-------

6.4.2 Test it

```
cd "$(git rev-parse --show-toplevel)/fable-cosmology/rust/fable_fermion"
cargo test --release
```

It runs the 23 unit tests of `src/` and the 11 end-to-end tests of `tests/cosmology.rs`; each end-to-end test runs the real binary, writes a CSV and reads it back. About 45 s. The history of the count: 21 + 8 at the finished solver (c5892bd, “re-run independently”), **34/34 (23 unit + 11 integration)** after the six defects were fixed (2bc936d). The 34 tests [listed from the source files]:

file	tests
src/constants. rs	unit_system_matches_the_standard_values
src/cvode_ driver.rs	exponential_decay_forward_and_backward
src/numerics. rs	brent_finds_simple_roots, gauss_kronrod_is_exact_on_smooth_functions
src/potentials. rs	derivatives_match_finite_differences, signs_that_the_bracket_proofs_use
src/kohn_sham. rs	fermi_integrals_match_quadrature, series_and_closed_forms_are_continuous_at_the_switch_points, nonrelativistic_and_ultrarelativistic_limits, per_3_volume_mean_field_violates_8d_conservation, thermodynamic_identities_at_the_self_consistent_point, derivative_along_the_trajectory_matches_finite_differences, expdamp_pins_sigma_below_s1_and_mass_goes_to_zero_at_high_density, multiple_gap_roots_are_found_and_the_lowest_energy_one_is_taken, quadrature_at_the_review_points_and_switch_continuity, nonrelativistic_classical_limit_of_the_mean_field, no_phantom_crossing_rho_plus_p_obs_is_nonnegative, root_counts_three_for_strong_repulsion_one_for_concave_w, walecka_functional_is_minimized_at_the_gap_root, vacuum_energy_vanishes_to_fifth_order_at_the_reference_mass, vacuum_energy_matches_high_precision_references_near_and_far_from_the_reference_mass, gap_residual_is_at_round_off_in_massless_phases, ks_functional_stationary_point_is_a_maximum_for_the_mass_term
tests/ cosmology.rs	version_string, covariant_conservation_along_real_fable8d_runs, constraint_residual_stays_small_on_forward_fable8d_runs, frozen_hidden_theorem, fable8d_with_hidden_frozen_by_hand_reproduces_fable4d, backward_runs, physics_checks_along_runs, negative_energy_refusal_is_found_inside_the_rhs_independently_of_the_output_grid, first_order_transition_is_found_inside_the_rhs_independently_of_the_output_grid, gap_residual_and_vacuum_energy_at_round_off_in_1_kev_runs, insensitivity_to_a_start

What they check, in words: the unit system against standard values; CVODE forward and backward on an exponential; Brent and Gauss–Kronrod; the potentials’ derivatives, U and F against finite differences and the signs the bracket proofs use; the Fermi-sea integrals against Gauss–Kronrod quadrature to 10^{-12} for $k_F/|m|$ from 10^{-8} to 10^6 and both signs of m ; the continuity of the series and closed forms at the switch points; the non- and ultra-relativistic limits; $\varepsilon - 3P = m\sigma$, $\varepsilon + P = \omega_F n$, $dp/dn_8 = \omega_F$ at the gap; the derivatives along a trajectory; the classical limit of the mean field; $w_f \geq -1$ for every potential; the root counts (three for the repulsive quadratic, one for every concave W); the lowest-energy selection; the `expdamp` pinning $\sigma_8 < s_1$; the maximum of the no-sea functional at the mass-term root; the minimum of the

Walecka functional; the fifth-order zero of the Dirac-sea energy; the violation of 8D conservation by a per-3-volume mean field; the Dirac-sea energy against high-precision references; the gap residual at round-off in massless phases; and, on real runs, covariant conservation along `fable8d`, the constraint residual, the frozen-hidden theorem (radiation keeps $H_B = 0$; dust drives it), `fable8d --freeze-hidden = fable4d`, backward runs (refused for `fable8d`, equal to forward for `fable4d`), physics checks, insensitivity to `a_start`, the two refusals located inside the right-hand side independently of the output grid, and the gap residual and the Dirac-sea energy at round-off in 1 keV runs.

What it prints. For each test binary running `N tests`, one `test <name> ... ok` line per test, and a `test result: line. run_all.sh` keeps only the `running/test result` lines; on 2026-09-25 they were [quoted from the recorded run of `run_all.sh`]:

```
running 23 tests
test result: ok. 23 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in
↳ 38.31s
running 0 tests
test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s
running 11 tests
test result: ok. 11 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.93s
running 0 tests
test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s
```

(the four test binaries are the library, the command-line binary, the end-to-end tests and the documentation tests). **Success:** 23 passed and 11 passed, 0 failed, exit code 0. A failing test prints `test <name> ... FAILED`, a panicked at line with the assertion that failed, `test result: FAILED`, and exits non-zero (and `run_all.sh`, `run_all.ps1` then stop).

The classical solver of Effort 0 is tested the same way from `fable-cosmology/rust/fable_cosmo`: 7 tests (`scalar_derivatives_match_finite_differences`, `spinor_derivatives_match_finite_differences_and_w`, `cpl_closed_form_matches_the_integral`, `rho_plus_p_is_twice_the_kinetic_energy`, `friedmann_closure_today`, `x0_grid_rejects_the_singular_ends`, `exponential_decay_to_tolerance`), ending `test result: ok. 7 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s`.

6.4.3 The unit system

```
cd "$(git rev-parse --show-toplevel)/fable-cosmology"
F=rust/fable_fermion/target/release/fable_fermion      # fable_fermion.exe on Windows
$F --constants
```

It prints the unit system the solver computes from CODATA 2018 and IAU constants ($\hbar = c = k_B = 1$; time in $1/H_0$; energy densities per 7-volume in `rho_c0`; masses and momenta in $E_c = \rho_{c0}^{1/4}$) [nb05 §5, verbatim]:

```
# fable_fermion unit system (hbar = c = k_B = 1)
h                = 0.674
H0               = 2.1842852411e-18 1/s = 6.8930799924e-11 1/yr = 1.4377226622e-33 eV
1/H0            = 1.4507302992e10 yr
M_pl (reduced, from G) = 2.4353234593e27 eV = 2.4353234593e18 GeV
rho_c0 = 3 H0^2 M_pl^2 = 3.6777719498e-11 eV^4
E_c = rho_c0^(1/4)    = 2.4626131773e-3 eV (unit of m, kF; sigma, n in E_c^3; W, rho in
↳ E_c^4 = rho_c0)
1 eV             = 4.0607270732e2 E_c
T_CMB            = 2.7255 K = 2.3486541806e-4 eV; T_nu0 = 1.6763891605e-4 eV
Omega_gamma0     = 5.4437728013e-5 (Omega_gamma0 h^2 = 2.4729753331e-5)
Omega_nu(1 species)0 = 1.2363206389e-5
Omega_r0 (N_eff=3.046) = 9.2096054673e-5 (Omega_r0 h^2 = 4.1837027333e-5)
Omega_b0 = 0.02237/h^2 = 4.9243191364e-2
(g_*(T) is not followed: radiation is Omega_r0 A^-4 at all times, which misstates rho_r A^4
↳ by up to a factor ~0.39 before e+e- annihilation and the QCD transition)
```

```

g (fable states per momentum) = 8
a_BBN (T = 1 MeV)           = 1.6763891605e-10   (= T_nu0 / 1 MeV)
a_rec (z = 1090)             = 9.1659028414e-4

```

Every notebook of Effort B recomputes this unit system in Python from CODATA and asserts agreement: the unit system recomputed in Python from CODATA agrees with the solver's to $1e-9$ (E_c, Omega_r0, Omega_nu1, Omega_b0) [nb05 §5, nb06 §5, nb07 §5]. If a notebook fails there, the binary predates the hbar correction of 2bc936d (section 5.8, item 7): rebuild it.

6.4.4 One gap solve

```
$F gap --form quadratic --param m0=0.05 --param lam=20 --kf 1
```

It solves the gap equation $m = W'(\sigma_{KS}(m, k_F)/v)$ once at k_F (and v , default 1) for the potential $W = m_0\sigma + (\lambda/2)\sigma^2$, prints every root with `m`, `sigma8`, `rho8`, `P_obs`, `P_hid`, `gap_residual` (16 significant digits) and the selected root. For this case (the repulsive quadratic, which has three roots) it prints [nb05 §5.5, verbatim]:

```

# potential Quadratic { v0: 0.0, m0: 0.05, lam: 20.0 }, kF = 1e0, v = 1e0, n8 =
↳ 1.3509491152311703e-1, plan = Scan { lo: -2.651898230462341, hi: 2.7518982304623405 }
m,sigma8,rho8,P_obs,P_hid,gap_residual
-2.535353916022531e0,-1.292676958011266e-1,1.909782056219713e-1,1.772147779888931e-1,1.67101
↳ 3717773260e-1,8.506e-17
-1.640336879334347e-2,-3.320168439667173e-3,1.012381942988485e-1,3.387489103006236e-2,1.1023
↳ 51846776195e-4,-1.898e-17
2.644093986040535e0,1.297046993020268e-1,2.039272771420881e-1,1.779694075325236e-1,1.6823309
↳ 02102918e-1,0.000e0
# selected (Lowest): m = -1.640336879334347e-2, rho8 = 1.012381942988485e-1, roots = 3

```

Success: the roots agree with the independent Python solution (notebook 05 prints the three roots of both codes; they agree to 1.3×10^{-15} in `rho`) and the lowest-`rho` root is selected. Notebook 05 also solves the mass term ($m_0 = 1$, so $k_F = x$) at the fifteen $x = k_F/m$ of its accuracy table, with the arguments Python's `repr` gives:

```

for x in 1e-08 1e-06 0.0001 0.01 0.2 0.25 0.3 0.6 1.0 3.0 10.0 99.0 101.0 10000.0 1000000.0;
do
  $F gap --form mass --param m0=1 --kf $x
done

```

(the rows `sigma8`, `rho8`, `P_obs` of the mass term are `sigma_KS`, `eps_KS`, `P_KS`; they agree with the independent Python forms to 3.11×10^{-14} : largest relative difference Rust - Python over these 15 points: $3.11e-14$ [nb05 §5.1]).

6.4.5 The command-line contract

The usage the binary prints (to standard error, with exit code 2) on a malformed command line [file: `fable-cosmology/rust/fable_fermion/src/main.rs`]:

```

usage: fable_fermion fable4d|fable8d --potential
↳ mass|lambda-mass|power|expdamp|lorentz|quadratic|explicit [--form NAME] [--param
↳ KEY=VALUE ...] [--direction forward|backward] [--a-start A] [--points K] [--out FILE.csv]
↳ [--rtol R] [--atol A] [--method bdf|adams] [--branch lowest|positive|negative]
↳ [--freeze-hidden] [--no-shoot] [--no-fable] [--omega-b X] [--omega-r X]
  fable_fermion gap --form NAME --param ... --kf KF [--v V]
  fable_fermion --constants | --version

```

option	meaning	default
fable4d, fable8d	the model: stabilized (physical) or unstabilized (the no-go)	required
gap	solve the gap equation once at k_F (and v) for the potential --form: every root with m , σ_8 , ρ_8 , P_{obs} , P_{hid} , gap_residual , then the selected root	—
--potential NAME	mass, lambda-mass, power, expdamp, lorentz, quadratic, explicit	mass
--form NAME	the form of W for --potential explicit, --no-shoot and gap	the potential's name
--param KEY= VALUE	one parameter; repeat for more; a mass may be given in eV as KEY_ev or in E_c as KEY	see below
--direction	forward (from a_{start} to 1) or backward (from today; fable4d only)	forward
--a-start A	the first scale factor, in $(0, 1)$	10^{-12}
--points K	output rows, equally spaced in $N = \ln A$, first and last included (at least 2)	2001
--out FILE.csv	write the CSV there; without it the CSV goes to standard output	standard output
--rtol R, --atol A	CVODE's tolerances	10^{-10} , 10^{-12}
--method bdf adams	CVODE's family	bdf
--branch	which gap root when there are several: the lowest energy density, the largest positive or the most negative mass	lowest
--freeze-hidden	fable8d with the stabilizer (H_A evolved by its equation)	off
--no-shoot	take the potential's parameters and k_{F0} (or k_{F0_ev}) as given	shoot
--no-fable	radiation and baryons only (the controls of the no-go runs)	with the fable
--omega-b X, --omega-r X	override $\Omega_{\text{b}0}$, $\Omega_{\text{r}0}$	the computed values
--kf KF, --v V	gap only: the Fermi momentum (in E_c) and the hidden volume factor	—, 1

The potentials and their parameters ($W(\sigma)$ in E_c^4 ; the “split” potentials impose the dark matter today: the quasiparticles carry $\Omega_{\text{qp}0} = \omega_{\text{dm}}$ at the mass m_{today} , so k_{F0} follows from $\varepsilon_{\text{KS}}(m_{\text{today}}, k_{F0}) = \omega_{\text{dm}}$):

--potential	$W(\sigma)$	--param keys (default)	what is shot so that the budget closes today
mass (the author's term)	$m_0\sigma$	$m0_ev$ (1)	$\ln k_{F0}$ (the fable is all the non-baryonic matter: Einstein–de Sitter)
lambda-mass power	$V_0 + m_0\sigma$ $m_0\sigma + \lambda\sigma^\nu$	$m0_ev$ (1), ω_{dm} (0.265) $m_{\text{today_ev}}$ (1), ν (0.236), ω_{dm} (0.265)	V_0 , a bare cosmological constant $U_t = (1 - \nu)\lambda\sigma_t^\nu$, the condensate today; then $\lambda = U_t/((1 - \nu)\sigma_t^\nu)$, $m_0 = m_{\text{today}} - \nu\lambda\sigma_t^{\nu-1}$ (may be negative)
expdamp	$V_0 + m_0\sigma e^{-\sigma/s_1}$	$m_{\text{today_ev}}$ (1), x_t (1/2.21), ω_{dm} (0.265)	V_0 ; $s_1 = \sigma_t/x_t$, $m_0 = m_{\text{today}} e^{x_t}/(1 - x_t)$
lorentz	$V_0 + m_0\sigma/(1 + (\sigma/s_1)^2)$	$m_{\text{today_ev}}$ (1), x_t (1/1.5), ω_{dm} (0.265)	V_0 ; $s_1 = \sigma_t/x_t$, $m_0 = m_{\text{today}}(1 + x_t^2)/(1 - x_t^2)$
quadratic	$V_0 + m_0\sigma + (\lambda/2)\sigma^2$	$m_{\text{today_ev}}$ (1), g_q (−0.5), ω_{dm} (0.265)	V_0 ; $\lambda = g_q m_{\text{today}}/\sigma_t$, $m_0 = m_{\text{today}}(1 - g_q)$
explicit	the --form's W	$m0$ or $m0_ev$, $v0$, lam , ν , $s1$	$\ln k_{F0}$, after checking that $\rho_f(A = 1; k_{F0})$ is monotonic

Parameter domains: **power** needs $\lambda > 0$ and $0 < \nu < 1$; **expdamp** $m_0 > 0$, $s_1 > 0$ ($0 < x_t < 1$); **lorentz** $s_1 > 0$. The gap root is provably unique for **mass**, **lambda-mass**, **power**, **expdamp**,

attractive `quadratic` ($\lambda \leq 0$) and for `lorentz` while $n_8 \leq s_1$; otherwise all roots are found by a scan and the lowest-energy one is taken.

The CSV. A run writes `#` comment lines — the version; a machine-readable `# params:` line with every normalized parameter (`model`, `direction`, `spec`, `a_start`, `omega_r0`, `omega_b0`, `freeze`, `branch`, `lnB_i`, `lnC_i`, `e_c_ev`, `omega_nu1`, `rtol`, `atol`, `kf0`, `potential` and the potential's `m0`, `v0`, `lam`, `nu`, `s1`, as `key=value`); `# spec::` one line defining each group of columns — then a header and one row per output point, every number as `{:.14e}` (15 significant digits). The 47 columns: `N`, `a`, `z`, `t`, `B`, `C`, `H_A`, `H_B`, `H_C`, `constraint_residual`, `rho_r`, `rho_b`, `rho_f`, `P_obs_f`, `P_hid_f`, `n_f`, `sigma`, `m_eff`, `kF_over_m`, `w_f`, `rho_qp`, `w_qp`, `rho_U`, `w_DE_eff`, `Omega_r`, `Omega_b`, `Omega_f`, `q_dec`, `G_ratio`, `cs2_adiabatic`, `N_eff_extra`, `P_stab`, `Omega_sum`, `rho_DE_inf`, `w_DE_inf`, `w_DM_intrinsic`, `w_DM_eff`, `Q`, `rho_DE_eff`, `F_hidden`, `dm_dN`, `m_eff_eV`, `kF_eV`, `gap_roots`, `branch`, `gap_residual`, `vac_over_rhoc`. Their meanings:

column	meaning
<code>N</code> , <code>a</code> , <code>z</code>	$\ln A$, A , $1/A - 1$
<code>t</code>	the age in $1/H_0$ ($t(a_{\text{start}}) = 1/(2H_A(a_{\text{start}}))$, then integrated)
<code>B</code> , <code>C</code>	the hidden scale factors (1 today; 1 throughout in <code>fable4d</code>)
<code>H_A</code> , <code>H_B</code> , <code>H_C</code>	the expansion rates, in H_0
<code>constraint_residual</code>	$(S - 3\hat{\rho})/(3H_A^2)$, S the pair sum of the constraint
<code>rho_r</code> , <code>rho_b</code> , <code>rho_f</code>	energy densities per 7-volume, in <code>rho_c0</code>
<code>P_obs_f</code> , <code>P_hid_f</code>	the fable's pressures along the observed and the hidden directions ($P_{x_0} = P_{\text{hid}}$)
<code>n_f</code> , <code>sigma</code>	the fable's number density and scalar density <code>sigma8</code> per 7-volume, in E_c^3
<code>m_eff</code> , <code>kF_over_m</code>	the gap mass in E_c ; $k_F/ m_{\text{eff}} $ (<code>inf</code> when $m_{\text{eff}} = 0$)
<code>w_f</code>	P_{obs_f}/ρ_f , the fable's own equation of state
<code>rho_qp</code> , <code>w_qp</code>	the quasiparticles: $\varepsilon_{\text{KS}}/v$ and $P_{\text{KS}}/\varepsilon_{\text{KS}}$
<code>rho_U</code>	the condensate $W - \sigma_8 W'$ (<code>P_hid_f</code> = $-\rho_U$: an 8-dimensional vacuum energy)
<code>w_DE_eff</code> , <code>rho_DE_eff</code>	$P_{\text{obs}_f}/\rho_{\text{DE_eff}}$, $\rho_{\text{DE_eff}} = \rho_f - \rho_{\text{dust0}} A^{-3}$ $v(1)/v$, $\rho_{\text{dust0}} = m_{\text{eff}}(1)n_f(1)$
<code>Omega_r</code> , <code>Omega_b</code> , <code>Omega_f</code> , <code>Omega_sum</code>	ρ_X/H_A^2 ; $\Omega_{\text{sum}} = \hat{\rho}/H_A^2$ (1 in <code>fable4d</code>)
<code>q_dec</code>	$-1 - (dH_A/dt)/H_A^2$
<code>G_ratio</code>	$1/v = G_4(t)/G_4(\text{today})$
<code>cs2_adiabatic</code>	$(dP_{\text{obs}_f}/dN)/(d\rho_f/dN)$ along the solution (analytic, with the differentiated gap equation)
<code>N_eff_extra</code>	$\rho_f/\rho_{\text{nu}}(1 \text{ species})$
<code>P_stab</code>	$-F/2$ in <code>fable4d</code> and <code>--freeze-hidden</code> ; 0 in <code>fable8d</code>
<code>rho_DE_inf</code> , <code>w_DE_inf</code>	the dark energy an observer infers: $H_A^2 - \Omega_{r0} A^{-4} - \Omega_{b0} A^{-3} - \rho_{\text{dust0}} A^{-3}$ (cold dark matter subtracted with today's mass $m_{\text{eff}}(1)$), and $-1 - (1/3) d \ln \rho_{\text{DE_inf}}/d \ln A$
<code>w_DM_intrinsic</code> , <code>w_DM_eff</code> , <code>Q</code>	$P_{\text{KS}}/\varepsilon_{\text{KS}}$; $w_{\text{DM}} - \sigma_{\text{KS}}(dm/dN)/(3\varepsilon_{\text{KS}})$; $\sigma_8 dm/dt$, the energy the condensate hands to the quasiparticles
<code>F_hidden</code> , <code>dm_dN</code>	$\rho - 3P_{\text{obs}} + 2P_{\text{hid}}$ (all sources); dm_{eff}/dN
<code>m_eff_eV</code> , <code>kF_eV</code>	the mass and the Fermi momentum in eV
<code>gap_roots</code> , <code>branch</code> , <code>gap_residual</code>	the number of gap roots found; the sign of <code>m_eff</code> ; $(m - W'(\sigma_8))/\max(m , S_{W'}(\sigma_8))$, $S_{W'}$ the round-off scale of W'
<code>vac_over_rhoc</code>	$\Delta E_{\text{vac}}(m_{\text{eff}}; M = m_{\text{eff}}(1))/v$ in <code>rho_c0</code> : the renormalized Dirac-sea energy (relativistic Hartree) that the no-sea functional drops; reported, not in the dynamics

Read the CSV skipping the `#` lines (numpy's `genfromtxt(..., names=True)` would take the first comment for the header).

Standard error carries the run's log, one line each: `# normalization: ...` (what was shot, and to what value), `# ground-state check: ...` (split potentials), `# fable today: ...`, `# result: ...` (H_A , B , C today, q_0 , the age, the largest constraint and gap residuals), `# a_nr ...` (with its class: hot, warm or cold-like), `# the fable is ultra-relativistic at a_start ...`, `# Delta N_eff ...` at nucleosynthesis and recombination, `# stabilizer: ...` (fable4d), `# unstabilized fable8d` (the no-go test): `... (fable8d: the G ratio, H_B/H_A today, $d \ln G/dt$ against $10^{-13}/\text{yr}$), # t(A=1) cross-check: ... (CVODE against Gauss-Kronrod quadrature, fable4d), # the DM/DE split ... is a convention, a # stats: line (CVODE steps, right-hand-side evaluations, Newton iterations, error-test failures, Jacobian evaluations and wall time, summed over every integration of the run, shooting included), and the version.`

Exit codes (checked against the binary): 0 success; 1 a solver or physics error with a one-line reason starting `fable_fermion`: (the refusals of 6.4.8, CVODE failures, an unknown gap form, an unwritable output file); 2 a usage error (an unknown model, option or potential name, a non-numeric value, `--points` below 2, `--no-shoot` without `kf0`), with the usage lines printed.

6.4.6 One complete run and its log

The author's mass term at 30 eV, as notebook 05 ran it and printed its whole log [nb05 §5.7, verbatim]:

```
$ fable_fermion fable4d --potential mass --param m0_ev=30 --points 1001 --out
↪ results/nb05_fable4d_mass30.csv
# normalization: closure today (A = B = C = 1): rho_r + rho_b + rho_f = 1; shooting
↪ variable [ln kF0 (the fable number density)] = -2.485515249891274e0; rho_f(A=1; kF0)
↪ strictly increasing on the Lowest branch over ln kF0 in [-40, 40] (1601 grid points, 0
↪ without a gap solution); unique solution ln kF0 = -2.485515249891274e0
# fable today (A = 1, v = 1): potential Mass { m0: 12182.181219597018 }, kF0 =
↪ 8.3282632088e-2 E_c = 2.050929e-4 eV, m_eff = 1.2182181220e4 E_c = 3.000000e1 eV,
↪ rho_qp = 9.5066471258e-1, U = 0.0000000000e0, rho_f = 9.5066471258e-1, w_f =
↪ 9.3473556871e-12, gap roots = 1
# result: H_A(A=1) = 1.000000000000000e0, B(1) = 1.000000000000000e0, C(1) =
↪ 1.000000000000000e0, q0 = 5.000460e-1, age t0 = 6.6660643059e-1/H0, max |constraint
↪ residual| = 4.303e-16, max |gap residual| = 0.0e0, rows with several gap roots = 0
# a_nr (kF = |m_eff|) = 6.837070e-6 (z_nr = 1.4626e5), a_eq = 9.2105e-5: WARM / not hot
↪ (a_nr between 1e-6 and a_eq)
# the fable is ultra-relativistic at a_start: kF/|m_eff| = 6.836e6
# Delta N_eff (fable energy density in units of one massless neutrino species): at BBN (a =
↪ 1.6764e-10, T = 1 MeV) = 3.942638e-1, at recombination (a = 9.1659e-4) = 7.048553e1
# stabilizer: P_stab = -F_hidden/2 ranges over [-2.4622e34, -4.9995e-1] (today
↪ -4.999540e-1); it is a Lagrange multiplier with zero energy density and violates the
↪ null energy condition along x0 wherever dust is present (rho + P_C = -rho_dust/2)
# t(A=1) cross-check: CVODE 6.666064305910978e-1 vs Gauss-Kronrod quadrature
↪ 6.666064296244345e-1 (rel. diff 1.45e-9, quadrature error estimate 2.4e-14)
# NOTE: W = m0 sigma alone (no V0): the closure forces Omega_f0 = 1 - Omega_b0 - Omega_r0 =
↪ 0.950665: an Einstein-de Sitter-like universe (q0 = 0.5000, +1/2 for pure dust)
# the DM/DE split (rho_qp = eps/v vs rho_U = W - sigma W') is a convention: the condensate
↪ has P = -rho_U in all seven spatial directions, i.e. it is an 8D vacuum energy
# stats: model=fable4d direction=Forward potential=mass steps=373 rhs_evals=459
↪ nonlin_iters=456 err_test_fails=18 jac_evals=7 wall=0.010s (summed over every CVODE
↪ integration of the run, shooting included)
# fable_fermion 0.1.0, sundials_rs 7.8.0 (pure Rust), CVODE BDF
```

How to read it: the shooting found $\ln k_{F0}$ uniquely (the energy density today is strictly increasing in it); today $H_A = B = C = 1$ exactly, the constraint holds to 4.303×10^{-16} ; the fable became non-relativistic at $a_{nr} = 6.84 \times 10^{-6}$; it adds 0.394 neutrino species at nucleosynthesis; the stabilizer is negative throughout; CVODE's age agrees with an independent quadrature to 1.45×10^{-9} ; and the run took 373 steps and 10 ms. A `lambda-mass` run at 30 eV logs, among its lines, `# result: H_A(A=1) = 1.000000000000000e0, B(1) = 1.000000000000000e0, C(1)`

= 1.0000000000000000e0, $q_0 = -5.284510e-1$, age $t_0 = 9.5124651250e-1/H_0$, max |constraint residual| = $4.291e-16$, max |gap residual| = $0.0e0$, rows with several gap roots = 0 and # t(A=1) cross-check: CVODE $9.512465124959659e-1$ vs Gauss-Kronrod quadrature $9.512465131954438e-1$ (rel. diff $7.35e-10$, quadrature error estimate $2.4e-14$).

6.4.7 Every run behind the results of this page

Run from fable-cosmology/; each \$ fable_fermion below is \$F of 6.4.3 (the notebooks show the command with the output path relative to fable-cosmology/ and run it with the absolute path). They are listed in the order the committed notebooks ran them, by notebook section, each followed by the line it printed — the # stats: line, or the refusal — exactly as the committed notebooks recorded them (aca5102). Every run takes 3 to 42 ms of CVODE wall time. The fable4d runs take 373–458 CVODE steps; the unstabilized fable8d runs with a fable take 34762–40808, because they integrate the shear of the hidden sheet from a_i on and shoot on two unknowns (the radiation-plus-baryons control takes 2109, the radiation-only control, whose hidden sheet stays frozen, 7, and --freeze-hidden 395).

```
# notebook 05 §5.7: On the pre-universe: the charge is conserved per coordinate volume, the
↳ gas cools from 1/3 to 0
$ fable_fermion fable4d --potential mass --param m0_ev=30 --points 1001 --out
↳ results/nb05_fable4d_mass30.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=373 rhs_evals=459
↳ nonlin_iters=456 err_test_fails=18 jac_evals=7 wall=0.010s (summed over every CVODE
↳ integration of the run, shooting included)

# notebook 05 §5.8: The Dirac-sea energy of a mass-varying fable
$ fable_fermion fable4d --potential power --param nu=0.236 --param m_today_ev=30 --points
↳ 1001 --out results/nb05_fable4d_power0.236_m30.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=390 rhs_evals=482
↳ nonlin_iters=479 err_test_fails=20 jac_evals=7 wall=0.010s (summed over every CVODE
↳ integration of the run, shooting included)

# notebook 06 §5.1: The interval, and why the starting point does not matter
$ fable_fermion fable4d --potential lambda-mass --param m0_ev=30 --a-start 1e-12 --points 1001
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=430
↳ rhs_evals=542 nonlin_iters=539 err_test_fails=25 jac_evals=8 wall=0.010s (summed over
↳ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential lambda-mass --param m0_ev=30 --a-start 1e-11 --points 1001
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=433
↳ rhs_evals=549 nonlin_iters=546 err_test_fails=27 jac_evals=8 wall=0.010s (summed over
↳ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential lambda-mass --param m0_ev=30 --a-start 1e-10 --points 1001
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=438
↳ rhs_evals=525 nonlin_iters=522 err_test_fails=20 jac_evals=8 wall=0.010s (summed over
↳ every CVODE integration of the run, shooting included)
$ fable_fermion fable8d --potential lambda-mass --param m0_ev=30 --a-start 1e-12 --points 1001
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=37544
↳ rhs_evals=42447 nonlin_iters=42339 err_test_fails=654 jac_evals=647 wall=0.040s (summed
↳ over every CVODE integration of the run, shooting included)
$ fable_fermion fable8d --potential lambda-mass --param m0_ev=30 --a-start 1e-11 --points 1001
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=36759
↳ rhs_evals=41619 nonlin_iters=41511 err_test_fails=618 jac_evals=634 wall=0.039s (summed
↳ over every CVODE integration of the run, shooting included)
$ fable_fermion fable8d --potential lambda-mass --param m0_ev=30 --a-start 1e-10 --points 1001
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=34762
↳ rhs_evals=39367 nonlin_iters=39262 err_test_fails=576 jac_evals=593 wall=0.037s (summed
↳ over every CVODE integration of the run, shooting included)

# notebook 06 §5.2: (a) The author's mass term alone: an Einstein-de Sitter universe with a
↳ fable that cools
```

```

$ fable_fermion fable4d --potential mass --param m0_ev=1 --points 1001 --out
↪ results/nb06_mass_m1_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=392 rhs_evals=480
↪ nonlin_iters=477 err_test_fails=20 jac_evals=7 wall=0.010s (summed over every CVODE
↪ integration of the run, shooting included)
$ fable_fermion fable4d --potential mass --param m0_ev=30 --points 1001 --out
↪ results/nb06_mass_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=373 rhs_evals=459
↪ nonlin_iters=456 err_test_fails=18 jac_evals=7 wall=0.010s (summed over every CVODE
↪ integration of the run, shooting included)
$ fable_fermion fable4d --potential mass --param m0_ev=100 --points 1001 --out
↪ results/nb06_mass_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=387 rhs_evals=467
↪ nonlin_iters=464 err_test_fails=16 jac_evals=7 wall=0.010s (summed over every CVODE
↪ integration of the run, shooting included)
$ fable_fermion fable4d --potential mass --param m0_ev=1000 --points 1001 --out
↪ results/nb06_mass_m1000_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=377 rhs_evals=463
↪ nonlin_iters=460 err_test_fails=19 jac_evals=7 wall=0.010s (summed over every CVODE
↪ integration of the run, shooting included)
$ fable_fermion fable4d --potential mass --param m0_ev=2000 --points 1001 --out
↪ results/nb06_mass_m2000_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=mass steps=377 rhs_evals=467
↪ nonlin_iters=464 err_test_fails=18 jac_evals=7 wall=0.010s (summed over every CVODE
↪ integration of the run, shooting included)

# notebook 06 §5.3: (b) A mass term plus a bare cosmological constant:  $\Lambda$ CDM with fable dark
↪ matter
$ fable_fermion fable4d --potential lambda-mass --param m0_ev=30 --points 1001 --out
↪ results/nb06_lm_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=430
↪ rhs_evals=542 nonlin_iters=539 err_test_fails=25 jac_evals=8 wall=0.010s (summed over
↪ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential lambda-mass --param m0_ev=100 --points 1001 --out
↪ results/nb06_lm_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=418
↪ rhs_evals=515 nonlin_iters=512 err_test_fails=21 jac_evals=8 wall=0.010s (summed over
↪ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential lambda-mass --param m0_ev=1000 --points 1001 --out
↪ results/nb06_lm_m1000_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=407
↪ rhs_evals=509 nonlin_iters=506 err_test_fails=21 jac_evals=8 wall=0.009s (summed over
↪ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential lambda-mass --param m0_ev=2000 --points 1001 --out
↪ results/nb06_lm_m2000_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=425
↪ rhs_evals=527 nonlin_iters=524 err_test_fails=21 jac_evals=8 wall=0.010s (summed over
↪ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential lambda-mass --param m0_ev=30 --param omega_dm=0.265
↪ --a-start 1e-10 --points 701 --out results/nb06_fable4d_mass30eV.csv
# prints: # stats: model=fable4d direction=Forward potential=lambda-mass steps=429
↪ rhs_evals=548 nonlin_iters=545 err_test_fails=29 jac_evals=8 wall=0.008s (summed over
↪ every CVODE integration of the run, shooting included)

# notebook 06 §5.4: (c) Potentials with which the fable supplies the dark energy itself
$ fable_fermion fable4d --potential power --param nu=0.236 --param m_today_ev=30 --points
↪ 1001 --out results/nb06_power_0.236_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=390 rhs_evals=482
↪ nonlin_iters=479 err_test_fails=20 jac_evals=7 wall=0.011s (summed over every CVODE
↪ integration of the run, shooting included)
$ fable_fermion fable4d --potential power --param nu=0.236 --param m_today_ev=100 --points
↪ 1001 --out results/nb06_power_0.236_m100_4d.csv

```

```

# prints: # stats: model=fable4d direction=Forward potential=power steps=401 rhs_evals=505
↳ nonlin_iters=502 err_test_fails=23 jac_evals=7 wall=0.010s (summed over every CVODE
↳ integration of the run, shooting included)
$ fable_fermion fable4d --potential power --param nu=0.5 --param m_today_ev=30 --points 1001
↳ --out results/nb06_power_0.5_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=430 rhs_evals=551
↳ nonlin_iters=548 err_test_fails=26 jac_evals=8 wall=0.021s (summed over every CVODE
↳ integration of the run, shooting included)
$ fable_fermion fable4d --potential power --param nu=0.5 --param m_today_ev=100 --points 1001
↳ --out results/nb06_power_0.5_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=433 rhs_evals=545
↳ nonlin_iters=542 err_test_fails=24 jac_evals=8 wall=0.027s (summed over every CVODE
↳ integration of the run, shooting included)
$ fable_fermion fable4d --potential expdamp --param xt=0.4524886877828054 --param
↳ m_today_ev=30 --points 1001 --out results/nb06_expdamp_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=expdamp steps=423
↳ rhs_evals=527 nonlin_iters=524 err_test_fails=21 jac_evals=8 wall=0.019s (summed over
↳ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential expdamp --param xt=0.4524886877828054 --param
↳ m_today_ev=100 --points 1001 --out results/nb06_expdamp_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=expdamp steps=434
↳ rhs_evals=542 nonlin_iters=539 err_test_fails=23 jac_evals=8 wall=0.019s (summed over
↳ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential quadratic --param gq=-0.5 --param m_today_ev=30 --points
↳ 1001 --out results/nb06_quadratic_m30_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=quadratic steps=451
↳ rhs_evals=562 nonlin_iters=559 err_test_fails=23 jac_evals=8 wall=0.012s (summed over
↳ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential quadratic --param gq=-0.5 --param m_today_ev=100 --points
↳ 1001 --out results/nb06_quadratic_m100_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=quadratic steps=451
↳ rhs_evals=558 nonlin_iters=555 err_test_fails=22 jac_evals=8 wall=0.011s (summed over
↳ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential power --param m_today_ev=100 --param nu=0.5 --param
↳ omega_dm=0.55 --a-start 1e-10 --points 701 --out results/nb06_fable4d_power.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=379 rhs_evals=483
↳ nonlin_iters=480 err_test_fails=24 jac_evals=7 wall=0.008s (summed over every CVODE
↳ integration of the run, shooting included)

# notebook 06 §5.7: How heavy must a fable that is all of the dark matter be?  $\Delta N_{\text{eff}}$ , free
↳ streaming, phase space
$ fable_fermion fable4d --potential power --param nu=0.236 --param m_today_ev=1 --points 1001
↳ --out results/nb06_power_0.236_m1_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=397 rhs_evals=487
↳ nonlin_iters=484 err_test_fails=18 jac_evals=7 wall=0.011s (summed over every CVODE
↳ integration of the run, shooting included)
$ fable_fermion fable4d --potential power --param nu=0.5 --param m_today_ev=1 --points 1001
↳ --out results/nb06_power_0.5_m1_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=power steps=429 rhs_evals=553
↳ nonlin_iters=550 err_test_fails=28 jac_evals=8 wall=0.019s (summed over every CVODE
↳ integration of the run, shooting included)
$ fable_fermion fable4d --potential expdamp --param xt=0.4524886877828054 --param
↳ m_today_ev=1 --points 1001 --out results/nb06_expdamp_m1_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=expdamp steps=428
↳ rhs_evals=560 nonlin_iters=557 err_test_fails=32 jac_evals=8 wall=0.018s (summed over
↳ every CVODE integration of the run, shooting included)
$ fable_fermion fable4d --potential quadratic --param gq=-0.5 --param m_today_ev=1 --points
↳ 1001 --out results/nb06_quadratic_m1_4d.csv
# prints: # stats: model=fable4d direction=Forward potential=quadratic steps=444
↳ rhs_evals=576 nonlin_iters=573 err_test_fails=30 jac_evals=8 wall=0.011s (summed over
↳ every CVODE integration of the run, shooting included)

# notebook 06 §5.8: (d) Without the stabilizer: the hidden sheet moves and the Newton
↳ constant varies

```

```

$ fable_fermion fable8d --potential mass --param m0_ev=1 --points 1001 --out
↳ results/nb06_mass_m1_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=mass steps=37019
↳ rhs_evals=41186 nonlin_iters=41081 err_test_fails=499 jac_evals=634 wall=0.039s (summed
↳ over every CVODE integration of the run, shooting included)
$ fable_fermion fable8d --potential mass --param m0_ev=100 --points 1001 --out
↳ results/nb06_mass_m100_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=mass steps=40808
↳ rhs_evals=45860 nonlin_iters=45758 err_test_fails=474 jac_evals=702 wall=0.042s (summed
↳ over every CVODE integration of the run, shooting included)
$ fable_fermion fable8d --potential lambda-mass --param m0_ev=30 --points 1001 --out
↳ results/nb06_lm_m30_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=37544
↳ rhs_evals=42447 nonlin_iters=42339 err_test_fails=654 jac_evals=647 wall=0.039s (summed
↳ over every CVODE integration of the run, shooting included)
$ fable_fermion fable8d --potential lambda-mass --param m0_ev=1000 --points 1001 --out
↳ results/nb06_lm_m1000_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=38558
↳ rhs_evals=42762 nonlin_iters=42657 err_test_fails=595 jac_evals=669 wall=0.040s (summed
↳ over every CVODE integration of the run, shooting included)
$ fable_fermion fable8d --no-fable --omega-b 0 --points 1001 --out
↳ results/nb06_radiation_only_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=none steps=7 rhs_evals=13
↳ nonlin_iters=7 err_test_fails=0 jac_evals=2 wall=0.008s (summed over every CVODE
↳ integration of the run, shooting included)
$ fable_fermion fable8d --no-fable --points 1001 --out results/nb06_radiation_baryons_8d.csv
# prints: # stats: model=fable8d direction=Forward potential=none steps=2109 rhs_evals=2379
↳ nonlin_iters=2370 err_test_fails=37 jac_evals=38 wall=0.010s (summed over every CVODE
↳ integration of the run, shooting included)
$ fable_fermion fable8d --potential lambda-mass --param m0_ev=30 --freeze-hidden --points 1001
# prints: # stats: model=fable8d direction=Forward potential=lambda-mass steps=395
↳ rhs_evals=475 nonlin_iters=472 err_test_fails=15 jac_evals=7 wall=0.011s (summed over
↳ every CVODE integration of the run, shooting included)

# notebook 06 §5.9: The cases the solver refuses, and why
$ fable_fermion fable4d --potential lorentz --points 1001
# prints: exit code 1: fable_fermion: rho_hat = H_A^2 <= 0 from N = -0.7597808064 (a =
↳ 0.4677689478) on: the Kohn-Sham ground state has negative energy there (m_eff =
↳ -3.802078e-2, sigma8 = -9.789374e-4, rho_f = -4.830421e-1 (U = -4.847891e-1) against rho_r
↳ + rho_b = 4.830421e-1); H_A^2 = rho_hat is impossible (the onset located past the last
↳ accepted CVODE step and bisected)
$ fable_fermion fable4d --potential quadratic --param gq=0.5 --points 1001
# prints: exit code 1: fable_fermion: gap-branch jump at N = -0.0037262796 (a =
↳ 0.9962806544): a first-order transition of the Kohn-Sham ground state, the lowest-energy
↳ gap root jumps from the branch m_eff = -1.518260e0 (rho_f = 6.856759e-1) to the branch
↳ m_eff = 4.083552e2 (rho_f = 9.536521e-1); the right-hand side is discontinuous there, and
↳ energy conservation across it would need the Maxwell construction, which this solver does
↳ not implement (the transition detected inside the right-hand side and bisected in N to
↳ adjacent doubles)
$ fable_fermion fable8d --potential mass --direction backward
# prints: exit code 1: fable_fermion: fable8d runs forward only (design review E4):
↳ backward in time the shear modes grow as A^-3 v^-1 relative to H_A ~ A^-2, the solution
↳ runs to the 8D Kasner point H_B/H_A = -0.2929 and the constraint drifts to 0(1); use
↳ fable4d --direction backward

# notebook 07 §5.18: The cosmological runs follow the homogeneous Kohn-Sham ground state at
↳ every epoch
$ fable_fermion fable4d --potential power --param nu=0.236 --param m_today_ev=30 --points 201
# prints: # stats: model=fable4d direction=Forward potential=power steps=387 rhs_evals=488
↳ nonlin_iters=485 err_test_fails=22 jac_evals=7 wall=0.003s (summed over every CVODE
↳ integration of the run, shooting included)
$ fable_fermion fable4d --potential quadratic --param gq=-0.5 --param m_today_ev=30 --points
↳ 201

```

```
# prints: # stats: model=fable4d direction=Forward potential=quadratic steps=458
↳ rhs_evals=570 nonlin_iters=567 err_test_fails=26 jac_evals=8 wall=0.003s (summed over
↳ every CVMODE integration of the run, shooting included)
```

6.4.8 The scans, and the refusals

The 90 scan runs of section 4.7.8 (notebook 06, §5.6; the CSV goes to standard output, 401 rows; each run is re-computed by the independent implementation as it runs; their `# stats:` lines are not printed):

```
# notebook 06 §5.6: Parameter scans: where does a dark-energy fable stay massive through the
↳ matter era?
$ fable_fermion fable4d --potential power --param nu=0.05 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.1 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.15 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.2 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.25 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.3 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.35 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.4 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.45 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.5 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.55 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.6 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential power --param nu=0.65 --param m_today_ev=1000 --points 401
$ fable_fermion fable4d --potential quadratic --param gq=-0.95 --param m_today_ev=1000
↳ --points 401
$ fable_fermion fable4d --potential quadratic --param gq=-0.85 --param m_today_ev=1000
↳ --points 401
$ fable_fermion fable4d --potential quadratic --param gq=-0.75 --param m_today_ev=1000
↳ --points 401
$ fable_fermion fable4d --potential quadratic --param gq=-0.65 --param m_today_ev=1000
↳ --points 401
$ fable_fermion fable4d --potential quadratic --param gq=-0.55 --param m_today_ev=1000
↳ --points 401
$ fable_fermion fable4d --potential quadratic --param gq=-0.45 --param m_today_ev=1000
↳ --points 401
$ fable_fermion fable4d --potential quadratic --param gq=-0.35 --param m_today_ev=1000
↳ --points 401
$ fable_fermion fable4d --potential quadratic --param gq=-0.25 --param m_today_ev=1000
↳ --points 401
$ fable_fermion fable4d --potential quadratic --param gq=-0.15 --param m_today_ev=1000
↳ --points 401
$ fable_fermion fable4d --potential quadratic --param gq=-0.05 --param m_today_ev=1000
↳ --points 401
$ fable_fermion fable4d --potential expdamp --param xt=0.1 --param m_today_ev=1000 --points
↳ 401
$ fable_fermion fable4d --potential expdamp --param xt=0.2 --param m_today_ev=1000 --points
↳ 401
$ fable_fermion fable4d --potential expdamp --param xt=0.3 --param m_today_ev=1000 --points
↳ 401
$ fable_fermion fable4d --potential expdamp --param xt=0.4 --param m_today_ev=1000 --points
↳ 401
$ fable_fermion fable4d --potential expdamp --param xt=0.5 --param m_today_ev=1000 --points
↳ 401
$ fable_fermion fable4d --potential expdamp --param xt=0.6 --param m_today_ev=1000 --points
↳ 401
$ fable_fermion fable4d --potential expdamp --param xt=0.7 --param m_today_ev=1000 --points
↳ 401
$ fable_fermion fable4d --potential expdamp --param xt=0.8 --param m_today_ev=1000 --points
↳ 401
$ fable_fermion fable4d --potential expdamp --param xt=0.9 --param m_today_ev=1000 --points
↳ 401
```

[illegible]

```

$ fable_fermion fable4d --potential expdamp --param xt=0.6 --param m_today_ev=100 --points 401
$ fable_fermion fable4d --potential expdamp --param xt=0.7 --param m_today_ev=100 --points 401
$ fable_fermion fable4d --potential expdamp --param xt=0.8 --param m_today_ev=100 --points 401
$ fable_fermion fable4d --potential expdamp --param xt=0.9 --param m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.265 --param nu=0.3 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.35 --param nu=0.3 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.45 --param nu=0.3 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.55 --param nu=0.3 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.65 --param nu=0.3 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.8 --param nu=0.3 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.265 --param nu=0.5 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.35 --param nu=0.5 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.45 --param nu=0.5 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.5 --param nu=0.5 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.55 --param nu=0.5 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.65 --param nu=0.5 --param
↪ m_today_ev=100 --points 401
$ fable_fermion fable4d --potential power --param omega_dm=0.8 --param nu=0.5 --param
↪ m_today_ev=100 --points 401

```

The refusals. Three runs must fail, and the notebook checks that they do, with exit code 1 and the expected reason (all three refused with exit code 1 and the expected reason [nb06 §5.9]); their messages are in the block of 6.4.7 and discussed in section 4.7.11. They are the solver saying no where the physics or the numerics would be meaningless: `lorentz` (the Kohn–Sham ground state has negative energy from $a = 0.4677689478$ on, so $H_A^2 = \hat{\rho}$ is impossible); the repulsive quadratic (a first-order transition at $a = 0.9962806544$, where energy conservation would need the Maxwell construction, which the solver does not implement); and `fable8d --direction backward` (the shear modes grow backward and the solution runs to the 8D Kasner point). The first two are located inside the right-hand side and bisected to adjacent doubles, so the message and the onset do not depend on `--points`.

6.4.9 The classical solver `fable_cosmo` of Effort 0: its command line and every run

The notebooks 01–04 drive the solver of the classical fields, `fable-cosmology/rust/fable_cosmo` (built and tested exactly as in 6.4.1–6.4.2, from inside its own folder; `fable_cosmo 0.1.0`, `sundials_rs 7.8.0` (pure Rust), `CVODE BDF`). Its usage [file: `fable-cosmology/rust/fable_cosmo/src/main.rs`]:

```

usage: fable_cosmo <scalar-flrw|scalar-cpl|spinor-flrw|scalar-pre|scalar-pre-x0> [--potential
↪ NAME] [--param KEY=VALUE ...] [--no X0] [--n1 X1] [--points K] [--out FILE.csv]
↪ [--no-normalize] [--grid N] [--x0min A] [--x0max B] [--profile-out FILE.csv] [--rtol R]
↪ [--atol A] [--method bdf|adams]
    fable_cosmo --version

```

The models are Model A (`scalar-flrw`: `fableScalar` in the standard 4-dimensional reference cosmology), Model A' (`scalar-cpl`: the same as a test field on the Unite CPL background), Model B (`spinor-flrw`: the classical spinor fable in the reference cosmology), Model C (`scalar-pre`: `fableScalar` on the 8-dimensional pre-universe itself) and Model D (`scalar-pre-x0`, with

--grid, --x0min, --x0max and --profile-out for its dependence on x0); --n0, --n1 set the range of the independent variable, --no-normalize takes the potential's scale as given. Exit code 1 is a solver or physics error (and also an unknown model or potential name), 2 a malformed command line. Every run the committed notebooks 01–04 made, with the line it printed, run from fable-cosmology/ [the committed notebooks 01–04, verbatim]:

```
# notebook 01 §5.1: The exponential potential,  $V = V_0 e^{-\lambda\phi}$ , for  $\lambda = 0.5, 1, 1.5, 2$  and 3
$ fable_cosmo scalar-flrw --potential exp --param lambda=0.5 --out
↪ results/nb01_exp_lambda0.5.csv
# prints: # stats: model=scalar-flrw steps=384 rhs_evals=424 nonlin_iters=421
↪ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=1 --out results/nb01_exp_lambda1.csv
# prints: # stats: model=scalar-flrw steps=385 rhs_evals=434 nonlin_iters=431
↪ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.5 --out
↪ results/nb01_exp_lambda1.5.csv
# prints: # stats: model=scalar-flrw steps=399 rhs_evals=452 nonlin_iters=449
↪ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=2 --out results/nb01_exp_lambda2.csv
# prints: # stats: model=scalar-flrw steps=433 rhs_evals=498 nonlin_iters=495
↪ err_test_fails=8
$ fable_cosmo scalar-flrw --potential exp --param lambda=3 --out results/nb01_exp_lambda3.csv
# prints: exit code 1: fable_cosmo: normalisation cannot bracket Omega_phi = 0.699916:
↪ Omega(10-6 V) = 0.0000011107972084367645, Omega(106 V) = 0.3375115141492524; choose the
↪ scale by hand and pass --no-normalize
$ fable_cosmo scalar-flrw --potential exp --param lambda=3 --no-normalize --param v0=1e6
↪ --out results/nb01_exp_lambda3_scaling.csv
# prints: # stats: model=scalar-flrw steps=662 rhs_evals=697 nonlin_iters=696
↪ err_test_fails=4

# notebook 01 §5.2: The inverse power potential,  $V = V_0 \phi^{-\alpha}$ ,  $\alpha = 1$ : is the tracker
↪ joined?
$ fable_cosmo scalar-flrw --potential invpower --param alpha=1 --out
↪ results/nb01_invpower_alpha1.csv
# prints: # stats: model=scalar-flrw steps=430 rhs_evals=493 nonlin_iters=490
↪ err_test_fails=9
$ fable_cosmo scalar-flrw --potential invpower --param alpha=1 --n0 -25 --points 2501 --out
↪ results/nb01_invpower_alpha1_n0-25.csv
# prints: # stats: model=scalar-flrw steps=517 rhs_evals=619 nonlin_iters=616
↪ err_test_fails=19

# notebook 01 §5.3: The pseudo-Nambu-Goldstone potential, the hilltop, and the constant (the
↪ control)
$ fable_cosmo scalar-flrw --potential pngb --param f=1 --out results/nb01_pngb_f1.csv
# prints: # stats: model=scalar-flrw steps=312 rhs_evals=357 nonlin_iters=354
↪ err_test_fails=5
$ fable_cosmo scalar-flrw --potential pngb --param f=0.5 --param phi_i=0.3 --out
↪ results/nb01_pngb_f0.5.csv
# prints: # stats: model=scalar-flrw steps=313 rhs_evals=356 nonlin_iters=353
↪ err_test_fails=3
$ fable_cosmo scalar-flrw --potential hilltop --param mu=3 --out results/nb01_hilltop_mu3.csv
# prints: # stats: model=scalar-flrw steps=305 rhs_evals=352 nonlin_iters=349
↪ err_test_fails=4
$ fable_cosmo scalar-flrw --potential const --out results/nb01_const.csv
# prints: # stats: model=scalar-flrw steps=306 rhs_evals=370 nonlin_iters=367
↪ err_test_fails=13

# notebook 01 §5.4: Model A': the exponential potential as a test field on the Unite CPL
↪ background
$ fable_cosmo scalar-cpl --potential exp --param lambda=1 --out
↪ results/nb01_cpl_exp_lambda1.csv
# prints: # stats: model=scalar-cpl steps=384 rhs_evals=436 nonlin_iters=433
↪ err_test_fails=6
```

```

$ fable_cosmo scalar-cpl --potential exp --param lambda=2 --out
↳ results/nb01_cpl_exp_lambda2.csv
# prints: exit code 1: fable_cosmo: normalisation cannot bracket rho_phi(a=1) = 2.099748 (=
↳ 3 Omega_DE): rho_phi(10^-6 V) = 0.00000099999995795872962, rho_phi(10^-6 V) =
↳ 1.0248972437664063; choose the scale by hand and pass --no-normalize
$ fable_cosmo scalar-cpl --potential exp --param lambda=2 --no-normalize --param v0=10 --out
↳ results/nb01_cpl_exp_lambda2_hand.csv
# prints: # stats: model=scalar-cpl steps=459 rhs_evals=517 nonlin_iters=514
↳ err_test_fails=7

# notebook 01 §5.9: A finer scan in  $\lambda$ , to answer "which  $\lambda$ " precisely
$ fable_cosmo scalar-flrw --potential exp --param lambda=0.5
# prints: # stats: model=scalar-flrw steps=384 rhs_evals=424 nonlin_iters=421
↳ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=0.6
# prints: # stats: model=scalar-flrw steps=366 rhs_evals=415 nonlin_iters=412
↳ err_test_fails=6
$ fable_cosmo scalar-flrw --potential exp --param lambda=0.7
# prints: # stats: model=scalar-flrw steps=365 rhs_evals=410 nonlin_iters=407
↳ err_test_fails=4
$ fable_cosmo scalar-flrw --potential exp --param lambda=0.8
# prints: # stats: model=scalar-flrw steps=368 rhs_evals=414 nonlin_iters=411
↳ err_test_fails=4
$ fable_cosmo scalar-flrw --potential exp --param lambda=0.9
# prints: # stats: model=scalar-flrw steps=372 rhs_evals=422 nonlin_iters=419
↳ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.0
# prints: # stats: model=scalar-flrw steps=385 rhs_evals=434 nonlin_iters=431
↳ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.1
# prints: # stats: model=scalar-flrw steps=388 rhs_evals=437 nonlin_iters=434
↳ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.2
# prints: # stats: model=scalar-flrw steps=391 rhs_evals=440 nonlin_iters=437
↳ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.3
# prints: # stats: model=scalar-flrw steps=411 rhs_evals=461 nonlin_iters=458
↳ err_test_fails=6
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.4
# prints: # stats: model=scalar-flrw steps=421 rhs_evals=466 nonlin_iters=463
↳ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.5
# prints: # stats: model=scalar-flrw steps=399 rhs_evals=452 nonlin_iters=449
↳ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.6
# prints: # stats: model=scalar-flrw steps=389 rhs_evals=440 nonlin_iters=437
↳ err_test_fails=4
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.7
# prints: # stats: model=scalar-flrw steps=435 rhs_evals=483 nonlin_iters=480
↳ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.8
# prints: # stats: model=scalar-flrw steps=409 rhs_evals=465 nonlin_iters=462
↳ err_test_fails=7
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.9
# prints: # stats: model=scalar-flrw steps=436 rhs_evals=492 nonlin_iters=489
↳ err_test_fails=7
$ fable_cosmo scalar-flrw --potential exp --param lambda=2.0
# prints: # stats: model=scalar-flrw steps=433 rhs_evals=498 nonlin_iters=495
↳ err_test_fails=8

# notebook 02 §5.1: The mass term: dust, exactly
$ fable_cosmo spinor-flrw --potential mass --out results/nb02_mass.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↳ err_test_fails=3

```

```

# notebook 02 §5.2: `lambda-mass`: dust plus a bare cosmological constant
$ fable_cosmo spinor-flrw --potential lambda-mass --param v0=0.7 --param m=0.3 --out
↪ results/nb02_lambda-mass.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3

# notebook 02 §5.3: Power laws: `V = m s + λ s^n`, n = 0.236, 0.5, 0.9
$ fable_cosmo spinor-flrw --potential power --param n=0.236 --out
↪ results/nb02_power_n0.236.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential power --param n=0.5 --out results/nb02_power_n0.5.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential power --param n=0.9 --out results/nb02_power_n0.9.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential power --param n=0.236 --n1 6 --points 1301 --out
↪ results/nb02_power_n0.236_future.csv
# prints: # stats: model=spinor-flrw steps=1257 rhs_evals=1390 nonlin_iters=1387
↪ err_test_fails=10

# notebook 02 §5.4: The potentials that cross the phantom divide: `lorentz`, `expdamp`,
↪ `hilltop`
$ fable_cosmo spinor-flrw --potential lorentz --param v0=1 --param m=1 --param s1=1.5 --out
↪ results/nb02_lorentz.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential expdamp --param v0=1.566 --param m=0.839 --param
↪ s1=2.21 --no-normalize --out results/nb02_expdamp_ref.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential expdamp --param v0=1.566 --param m=0.839 --param s1=1.5
↪ --out results/nb02_expdamp_s1.5.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential expdamp --param v0=1.566 --param m=0.839 --param s1=3.0
↪ --out results/nb02_expdamp_s3.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential hilltop --param v0=1 --param mu=0.05 --param sstar=1.5
↪ --out results/nb02_hilltop_default_start.csv
# prints: exit code 1: fable_cosmo: V(s) = -184913370713936540 is not positive at the start
↪ (s = 1.3188e9 at N0 = -7): rho_Psi = V(s) must be positive; this potential is only valid
↪ at later times -- start at a larger --n0 (the Lorentz potential is positive everywhere)
$ fable_cosmo spinor-flrw --potential hilltop --param v0=1 --param mu=0.05 --param sstar=1.5
↪ --n0 -0.5 --out results/nb02_hilltop.csv
# prints: # stats: model=spinor-flrw steps=122 rhs_evals=138 nonlin_iters=135
↪ err_test_fails=2

# notebook 03 §5.1: Model C: the quadratic potential, `V = ½ m² ϕ²`, m = 1, ϕ_i = 1, to x4 =
↪ 100
$ fable_cosmo scalar-pre --potential quadratic --param m=1 --param phi_i=1 --n1 100 --points
↪ 2001 --out results/nb03_modelC_quadratic.csv
# prints: # stats: model=scalar-pre steps=965 rhs_evals=1163 nonlin_iters=1160
↪ err_test_fails=40

# notebook 03 §5.2: Model C: the quartic potential, `V = ¼ λ ϕ⁴`, λ = 1, and the constant
↪ potential
$ fable_cosmo scalar-pre --potential quartic --param lambda=1 --n1 100 --points 2001 --out
↪ results/nb03_modelC_quartic.csv
# prints: # stats: model=scalar-pre steps=2974 rhs_evals=3832 nonlin_iters=3829
↪ err_test_fails=223

```

```

$ fable_cosmo scalar-pre --potential const --n1 100 --points 2001 --out
↪ results/nb03_modelC_const.csv
# prints: # stats: model=scalar-pre steps=6 rhs_evals=9 nonlin_iters=6 err_test_fails=0

# notebook 03 §5.3: Cross-check of Model C with scipy (DOP853, rtol = 1e-11)
$ fable_cosmo scalar-pre --potential quartic --param lambda=1 --n1 100 --points 2001 --rtol
↪ 1e-12 --atol 1e-14 --out results/nb03_modelC_quartic_tight.csv
# prints: # stats: model=scalar-pre steps=4247 rhs_evals=5187 nonlin_iters=5185
↪ err_test_fails=242

# notebook 03 §5.4: Model D: a profile along the hidden coordinate,  $\varepsilon = 0.05$  and  $\varepsilon = 0.5$ 
$ fable_cosmo scalar-pre-x0 --potential quadratic --param m=1 --param eps=0.05 --grid 61 --n1
↪ 30 --points 301 --profile-out results/nb03_profile_eps0.05.csv --out
↪ results/nb03_modelD_eps0.05.csv
# prints: # stats: model=scalar-pre-x0 steps=294095 rhs_evals=321641 nonlin_iters=321639
↪ err_test_fails=2864
$ fable_cosmo scalar-pre-x0 --potential quadratic --param m=1 --param eps=0.5 --grid 61 --n1
↪ 30 --points 301 --profile-out results/nb03_profile_eps0.5.csv --out
↪ results/nb03_modelD_eps0.5.csv
# prints: # stats: model=scalar-pre-x0 steps=329282 rhs_evals=368139 nonlin_iters=368138
↪ err_test_fails=5712

# notebook 04 §5.1: The best cases, re-run
$ fable_cosmo scalar-flrw --potential exp --param lambda=1 --out results/nb04_exp_lambda1.csv
# prints: # stats: model=scalar-flrw steps=385 rhs_evals=434 nonlin_iters=431
↪ err_test_fails=5
$ fable_cosmo scalar-flrw --potential exp --param lambda=1.5 --out
↪ results/nb04_exp_lambda1.5.csv
# prints: # stats: model=scalar-flrw steps=399 rhs_evals=452 nonlin_iters=449
↪ err_test_fails=5
$ fable_cosmo spinor-flrw --potential expdamp --param v0=1.566 --param m=0.839 --param
↪ s1=2.21 --no-normalize --out results/nb04_expdamp_ref.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential expdamp --param v0=1.566 --param m=0.839 --param s1=3.0
↪ --out results/nb04_expdamp_s3.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential lambda-mass --param v0=0.7 --param m=0.3 --out
↪ results/nb04_lambda-mass.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential power --param n=0.236 --out
↪ results/nb04_power_n0.236.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3
$ fable_cosmo spinor-flrw --potential mass --out results/nb04_mass.csv
# prints: # stats: model=spinor-flrw steps=811 rhs_evals=869 nonlin_iters=866
↪ err_test_fails=3

```

The three refusals are deliberate: an exponential potential with $\lambda = 3$ cannot reach the dark-energy fraction today (the scaling solution caps it at $3/\lambda^2$), nor can $\lambda = 2$ on the CPL background, so each is re-run with the scale chosen by hand (`--no-normalize`); and a potential that is negative at the start is refused with the start to be moved later.

6.5 The independent Python implementation, and comparing results

```

cd "$(git rev-parse --show-toplevel)"
fable-cosmology/.venv/Scripts/python.exe fable-cosmology/fermion/crosscheck.py --jobs 6 #
↪ .venv/bin/python on Linux/macOS

```

- **What it does** [file: its docstring]. An independent implementation of `fable4d` and `fable8d` that shares no code with the Rust crate: it recomputes the unit system from

CODATA (one CODATA-2018 `hbar`, as the solver since 2bc936d); evaluates every Fermi-sea integral by adaptive quadrature (`scipy.integrate.quad`, `epsrel = 1e-12`, split at $k = |m|$), not the closed forms; solves every gap by `brentq` on the documented brackets; integrates `fable8d` with DOP853 at `rtol = 1e-11` (and `fable4d` algebraically). It runs 13 cases with the solver (mass at 1 and 100 eV, `lambda-mass` at 30 eV, power $\nu = 0.236$, `expdamp`, quadratic $g_q = -0.5$, each as `fable4d` and `fable8d`, and `lambda-mass 30 eV fable8d --freeze-hidden`), regenerating their CSVs into the ignored `fable-cosmology/fermion/dev/`, reads each CSV's `# params:` line, re-computes the run on the same grid, and compares $\max |\Delta w_f|$, $\max |\Delta H_A|/H_A$, $\max |\Delta B|/B$, $\max |\Delta G_{\text{ratio}}|/G_{\text{ratio}}$; it also checks, independently of the Rust shooting, that $H_A(A=1) = 1$ and $B(1) = C(1) = 1$. `--jobs N` is the number of worker processes (default 4).

- **What it prints.** A `units:` line (`Omega_r0` and `E_c` of Python and Rust), one row per case (`case`, `max|dw_f|`, `max|dH_A|/H`, `max|dB|/B`, `max|dG|/G`, `H_A(1)py`, `B(1)py`, `nfev`, `secs`), and `CROSSCHECK PASSED:` every difference $< 1e-06$.
- **How long.** About 10 s.
- **Success and failure.** Success is `CROSSCHECK PASSED` and exit code 0. Any difference above 10^{-6} , or $H_A(1)$, $B(1)$ not 1, prints `<-- FAIL` on that row and `CROSSCHECK FAILED: some difference exceeds 1e-6` (or `H_A(1), B(1) != 1`), and exits with code 1. The largest difference on 2026-09-24 was 1.1×10^{-8} ($|dG|/G$ of the `lambda-mass 30 eV fable8d` case), recorded in `c5892bd` (below 10^{-8} then) and `2bc936d` (1.1×10^{-8} after the fixes).

Notebook 06 applies the same independent functions to every run it makes (independent implementation imported from `fermion/crosscheck.py`, `crosscheck.units()` agrees with the solver's unit system to $1e-9$ [nb06 §5]) and prints, per run, the differences and the conservation residual of the independently computed fluid, ending largest difference over the 29 kept runs: $1.08e-08$; over the 90 scan runs: $4.43e-09$; largest conservation residual of the independent fluid: $1.24e-07$ [nb06 §5.10] (section 4.7.12).

Comparing a re-run with the committed results. Every CSV and PNG under `fable-cosmology/results/` is committed as evidence. After a re-run,

```
cd "$(git rev-parse --show-toplevel)"
git diff --stat -- fable-cosmology/results      # lists every result file whose bytes
↪ changed
git status --short -- fable-cosmology/results
```

lists what changed; on every recorded re-run it listed **nothing** (all 59 result files of notebooks 01–04 byte-identical on 2026-09-24 and 2026-09-25; the result files of notebooks 05, 06 and 07 written byte-identically by two successive runs: 63 of 05 and 06, 27 of 07). For a CSV that did change, compare it column by column with this script (save it as `compare_csv.py` and run it from the repository root with the file's path, for example `fable-cosmology/.venv/Scripts/python.exe compare_csv.py fable-cosmology/results/nb06_lm_m30_4d.csv`):

```
import csv, io, math, subprocess, sys
path = sys.argv[1].replace('\\', '/')      # a path relative to the repository root
keep = lambda rows: [r for r in rows if r and not r[0].startswith('#')]  # fable_fermion
↪ CSVs start with '#' lines
old = keep(csv.reader(io.StringIO(subprocess.run(['git', 'show', f'HEAD:{path}'],
↪ capture_output=True, text=True, check=True).stdout)))
new = keep(csv.reader(open(path, newline='')))
if len(old) != len(new):
    print(f'row counts differ: committed {len(old)}, now {len(new)}')
```

```

def num(x):
    try:
        v = float(x)
    except ValueError:
        return None
    return None if math.isnan(v) else v

for j, name in enumerate(new[0]):
    pairs = [(num(a[j]), num(b[j])) for a, b in zip(old[1:], new[1:])]
    d = [abs(x - y) for x, y in pairs if x is not None and y is not None]
    print(f'{name:16s} ' + (f'max |difference| = {max(d):.3e}' if d else '(not numeric)'))

```

It prints one line per column, `max |difference| = ...` (text columns are `(not numeric)`).

6.6 The wall-state solver, its derivation and its Mathematica checks

The discrete states of the fermion fable along the hidden coordinate `x0` (section 4.6). Run from `fable-cosmology/fermion/`, with the Python of the virtual environment:

```

cd "$(git rev-parse --show-toplevel)/fable-cosmology/fermion"
PY=../.venv/Scripts/python.exe          # ../.venv/bin/python on Linux/macOS

```

6.6.1 The derivation (Mathematica)

```
wolframscript -file waveguide_derivation.wls > waveguide_derivation.log
```

- **What it does.** Loads Input cells 1–117 of the notebook (with `DumpSave` blocked, so nothing is written into `claude-fable/`), and derives the wall problem from the notebook's own objects (`T16`, `sigma16`, `gammaCurvedCanonical`, `GammaSpinCanonical`, `cfDcov16`, `gCanonical`, `RicciScalarCanonical`, `eta4488`): the rescaling and the proper distance, the reduction of the 16 components to eight 2×2 blocks, the wall conditions, the symmetries and degeneracies, the endpoints and the exact solutions (section 4.6.3). Section 8 of the script (re)writes the committed `fermion/waveguide_T16.json` (the notebook's `T16[0..8]`, `sigma16` and the block basis `U`, 29 KB), byte-identically.
- **How long.** 162.8 s in the committed run (total time 162.8 s, loading 125. s); 171.4 s in the first committed run (18a2501).
- **What it prints** [file: `fable-cosmology/fermion/waveguide_derivation.log`]: a header (`waveguide_derivation.wls 2026-09-25 00:06:38`, the Mathematica version, the notebook's path), [load] `DumpSave` was blocked while loading: nothing was written into `claude-fable/`, [load] notebook Input cells: 217; evaluated cells 1..117 in 125. s, [load] cells that raised messages while loading (incidental to this script): {}, [load] the notebook's own assertions in cells 1..117: 142 run, 142 passed; not passed: {}, then the 94 PASS lines of its eight sections, and at the end

```

=====
waveguide_derivation.wls: 94 checks, 94 PASS, 0 FAIL
total time 162.8 s
=====

```

- **Success:** 94 checks, 94 PASS, 0 FAIL, the notebook's own 142 assertions passed while loading, and `waveguide_T16.json` unchanged (`git status` shows nothing). Notebook 07 re-reads this log and asserts it (derivation log: 94 PASS lines, 0 FAIL lines; summary line: "94 checks, 94 PASS, 0 FAIL" [nb07 §5.3]).

6.6.2 The solver's validation

```
$PY waveguide.py validate
```

- **What it does.** 64 checks: the analytic cases (constant K with MIT $-$: $\omega = -K$; the closed form for any th ; no level for constant K with MIT $+$; the flat band $\omega = m$ for $g(0) = 0$, absent in the other irrep), the Magnus propagator against DOP853, Hellmann–Feynman ($d\omega/dm = \int(f^2 - g^2)$, $d\omega/dK$), the full 16×16 system with the notebook's T16 (every level of multiplicity exactly 4, a non-level of multiplicity 0, the level located independently), the edge-band slope against its closed form $(m/6H)B(m/6H, 13/12)$, and the block reduction by U . It reads `waveguide_T16.json`.
- **How long.** [done in 31.1 s] in notebook 07's run; 45 s in the report's development run.
- **What it prints.** `waveguide.py validate` (python 3.14.5, numpy 2.5.3), one PASS/FAIL line per check with its numbers, VALIDATION: 64 of 64 PASS and [done in ... s]; notebook 07 §5.3 prints the whole output. Among its lines [nb07 §5.3, verbatim]:

```
PASS V1 constant K=0.3, MIT- (f=-g): single level w = -K [w = -0.2999999999999999, err =
→ 6.11e-16]
PASS V4 true profile K=80.0, g(0)=0: w = m is a level [|w - m| = 4.55e-13; all levels
→ [-80.00428754, -78.51089497, 1.0, 78.51089497, 80.00428754]]
PASS V5 Magnus vs DOP853 mismatch Delta(w), K=80.0, th=-0.785 [max diff 7.50e-09 rad]
PASS V6 level w=+70.98530723 (K=80.0, th=-0.785): norm, wall condition, Hellmann-Feynman
→ dw/dm = Int(f^2-g^2), dw/dK = Int 2fg s6 [norm-1=-1.8e-10, bc=0.0e+00,
→ dw/dm=0.24150470 vs 0.24150473, dw/dK=0.82441698 vs 0.82441698]
PASS V7 16x16 locates the level independently (K=80.0): w16 = 79.3714959054 vs 2x2 block
→ 79.3714957791 [|diff| = 1.3e-07, multiplicity 4, sv_min 1.5e-08]
PASS V9 opposite-sign edge band slope at K -> 0, m = 1.0: omega/K = -(m/6H) B(m/6H, 13/12)
→ [omega/K at K=1e-4: -0.980822728, exact slope -0.980822733]
PASS V8 numeric: U^dag T0(m - iK T1 + i w T4) U = diag of the 8 block matrices (s = +1 for
→ w3=-i, -1 for w3=+i)
VALIDATION: 64 of 64 PASS
[done in 31.1 s]
```

- **Success and failure.** VALIDATION: 64 of 64 PASS and exit status 0. Any failed check makes the exit status 1 and prints `waveguide.py: N validation check(s) FAILED` to standard error. If `waveguide_T16.json` is missing, V7 and V8 each record one FAIL naming the command that rewrites the file (6.6.1), the other checks still run, and the exit status is 1 (section 5.8, item 5).

6.6.3 The levels, thresholds, dispersion, edge band, wall family, convergence, wavefunctions, reference table

```
$PY waveguide.py levels --K 80 --m 1 --wall same # 1.4 s
$PY waveguide.py threshold # 13 s (9.5 s in notebook 07)
$PY waveguide.py figures # dispersion + edge + theta +
→ wavefunctions, ~25 s (24.6 s in notebook 07)
$PY waveguide.py dispersion
$PY waveguide.py edge
$PY waveguide.py theta
$PY waveguide.py wavefunctions
$PY waveguide.py convergence # 18 s
$PY waveguide.py reference # 7 s: the level table read by
→ waveguide_check.wls
$PY waveguide.py all # everything except the exploratory dft
→ scenarios
$PY waveguide.py -h # every command and option
```

The subcommands [file: the module and its argument parser]: `levels` prints all levels of both irreps at `--K` (the transverse momentum $K_\infty = k e^{a_4}$, in units of H ; default 80), `--m` (the

bulk mass m/H ; default 1), `--wall same|opposite` (the MIT wall sign relative to m), or `--theta th` (a uniform wall angle, overriding `--wall`); `threshold` prints $K_c(1)$, $K_c(2)$ against m/H ; `dispersion` the levels $\omega_n(K)$ for $K = 7..100$ and the opposite-sign spectrum; `edge` the opposite-sign edge band; `theta` the uniform self-adjoint family at $K = 3$ and 80 ; `convergence` the dependence of a level on the matching point, the tolerances, `z_a`, the `kappa` grid and the integrator; `wavefunctions` the orbitals at $K = 80$; `figures` the four figure tables at once; `reference` the level table for the Mathematica check; `--quick` limits `dft` to one Δ SCF fraction ($x = 0.05$ instead of $0.01, 0.05, 0.10, 0.20$) [file: `dft_run`]. Tables go to `fermion/dev/` with the prefix `waveguide_` (a folder the repository ignores; notebook 07 writes the final tables and figures into `results/nb07_*`).

`levels --K 80 --m 1 --wall same` prints, for both irreps, `irrep -i: -79.987308552047, -78.144249332428, +70.985307229908, +79.371495779094; irrep +i: -79.371495779091, -70.985307229895, +78.144249332427, +79.987308552047` — the $\omega \rightarrow -\omega$ symmetry between the irreps to 10^{-11} [WG 1]. `threshold` prints the table of section 4.6.5 (notebook 07 prints it again, THRESHOLDS of the same-sign MIT wall ..., with $K_c(1) = 6.70624065$ and $K_c(2) = 33.91895851$ at $m/H = 1$ [nb07 §5.6]); `figures` prints the dispersion table (DISPERSION `omega_n(K)`, same-sign (MIT+) wall, $m/H = 1.0$, $H = 1$: ... [nb07 §5.7]). **Success** for these is exit status 0 and the tables; they are checked by notebook 07, which asserts what they print, and by the Mathematica check (6.6.4).

6.6.4 The independent Mathematica check

```
$PY waveguide.py reference # writes
↪ fermion/dev/waveguide_levels_reference.csv
cd "$(git rev-parse --show-toplevel)"
wolframscript -file fable-cosmology/fermion/waveguide_check.wls
```

- **What it does.** Recomputes the levels in a different language: `NDSolve` at 30 digits, the normalized Wronskian with `FindRoot`, `NullSpace` for the asymptotic data; the 16-component multiplicities with `NDSolve` at $K = 3, 8$ and, at $K = 80$ (where `NDSolve` on the 16×8 matrix system did not finish within 30 minutes), an independently written Magnus/`MatrixExp` propagator; the thresholds by an independently written Prüfer `NDSolve` at 30 digits. It reads the committed `waveguide_T16.json` and the Python level table, which is not committed — so in a fresh clone write it first with `waveguide.py reference`.
- **How long.** 40 s [WG 1].
- **What it prints.** Summary: 11 checks, 11 PASS, 0 FAIL, no messages, with its level table (section 4.6.4), agreement with the Python levels to at most 4.68×10^{-11} , and the thresholds $K_c(1) = 6.70624065181$ and $K_c(2) = 33.9189585083$ [WG 5]. Notebook 07 quotes the four $K = 80$ Mathematica levels and asserts `max |notebook - Mathematica| = 4.7e-11` [nb07 §5.7].
- **Success:** 11 checks, 11 PASS, 0 FAIL, no messages.

6.6.5 The density-functional runs

```
$PY waveguide.py dft # 9.3 min (522.4 s in notebook 07's background
↪ run)
$PY waveguide.py dft --scenario bulk # about a second (0.3 s): is homogeneous
↪ matter self-bound?
$PY waveguide.py dft --scenario strong # 10.9 min; exploratory (lam = -100), does not
↪ converge
$PY waveguide.py dft --scenario convergence # 12.1 min
```

- **What they do.** The local-density Kohn–Sham self-consistency for $W = m_0\sigma + (\lambda/2)\sigma^2$ along z (section 4.6.6): [DFT-a] the author’s mass term, which is free and exact; [DFT-b]

the same-sign wall, $\lambda = -0.01$, $N_s = 100$, its variational check and band gaps, and the Δ SCF first excited states at $x = 0.01, 0.05, 0.1, 0.2$; [DFT-c] the opposite-sign wall at $\lambda = -0.01$ and -10 , $N_s = 0.05$. `--scenario` selects `all` (default), `free`, `same`, `opposite`, `strong`, `bulk` or `convergence`.

- **What dft prints:** every iteration (`it` `n` `max|m_new - m|` `=` ... `E/A` `=` ... `mu` `=` ... `N(check)` `=` ... `z_c` `=` ...), then per scenario the converged state, its stability statement, the Walecka variational check and the band structure, ending [done in 522.4 s]. Notebook 07 starts it in the background and prints its whole output in its §5.17 [nb07 §5.17]; for example the same-sign ground state converges in nine iterations, it 9 `max|m_new - m|` `=` 1.537e-10 `E/A` `=` 937.2277360325 `mu` `=` 11.6958993131 `N(check)` `=` 100.00000000 `z_c` `=` 0.92318, and its variational check reads `d Omega/ds` `=` -48.305804: NOT stationary.
- **What --scenario bulk prints** [nb07 §5.16, verbatim]:

```
[DFT bulk] homogeneous 4D matter (z, x1..x3), g = 8 per 4-momentum, W = m0 sigma + (lam/2)
↪ sigma^2, m0 = 1:
n = kF^4/(4 pi^2), sigma = (m/pi^2) Int_0^kF k^3/w, eps = (1/pi^2) Int_0^kF k^3 w, gap m =
↪ m0 + lam sigma, E = eps - (lam/2) sigma^2
lam = -0.01: min over kF in [0.01, 10] of (E/n - m0) = +0.000033 at kF = 0.0100, m* =
↪ 1.0000, n = 2.533e-10 -> not self-bound (minimum at the lowest density)
lam = -1.00: min over kF in [0.01, 10] of (E/n - m0) = +0.000033 at kF = 0.0100, m* =
↪ 1.0000, n = 2.533e-10 -> not self-bound (minimum at the lowest density)
lam = -10.00: min over kF in [0.01, 10] of (E/n - m0) = +0.000033 at kF = 0.0100, m* =
↪ 1.0000, n = 2.533e-10 -> not self-bound (minimum at the lowest density)
lam = -30.00: min over kF in [0.01, 10] of (E/n - m0) = +0.000033 at kF = 0.0100, m* =
↪ 1.0000, n = 2.533e-10 -> not self-bound (minimum at the lowest density)
lam = -60.00: min over kF in [0.01, 10] of (E/n - m0) = +0.000033 at kF = 0.0100, m* =
↪ 1.0000, n = 2.533e-10 -> not self-bound (minimum at the lowest density)
lam = -80.00: min over kF in [0.01, 10] of (E/n - m0) = -0.038261 at kF = 0.8775, m* =
↪ 0.3965, n = 0.01502 -> SELF-BOUND
lam = -100.00: min over kF in [0.01, 10] of (E/n - m0) = -0.072213 at kF = 0.8575, m* =
↪ 0.3550, n = 0.01369 -> SELF-BOUND
lam = -200.00: min over kF in [0.01, 10] of (E/n - m0) = -0.175598 at kF = 0.7819, m* =
↪ 0.2568, n = 0.009468 -> SELF-BOUND
[done in 0.3 s]
```

- **Success:** the converged scenarios print `converged=True` with the particle number reproduced (`N(check)` `=` 100.00000000); dft ends with exit code 0 (`waveguide.py dft` finished with exit code 0, 554 s after it was started [nb07 §5.17]). The strong-coupling scenario is exploratory and is **expected not to converge** (section 4.6.6).

The development logs of all these commands (`dev/waveguide_run_*.log`) are in the ignored `fermion/dev/`; the committed report `fable-cosmology/fermion/waveguide_REPORT.txt` quotes them, and notebook 07 re-runs and asserts the numbers that matter.

6.7 The Jupyter notebooks: generate, execute, check, and how they display the results

Seven notebooks drive the solvers, display the results and check them: 01–04 for Effort 0, 05–07 for Effort B. They are **generated**, never edited by hand, executed headlessly, and checked.

6.7.1 Generate them

```
cd "$(git rev-parse --show-toplevel)/fable-cosmology"
.venv/Scripts/python.exe notebooks/_build/nbgen.py          # all seven, unexecuted
.venv/Scripts/python.exe notebooks/_build/nbgen.py 05 06 07 # only these
```

`nbgen.py` is the source of truth: it writes `notebooks/ON_*.ipynb` (nbformat 4, unexecuted) from its cell manifests, and refuses (with an assertion) a code cell that is not preceded by a markdown cell of at least 80 characters. It prints one line per notebook, wrote `<name>.ipynb: <n> cells, <m> code cells` [file: its build and `build_fermion` functions]; for the committed notebooks that is 38 cells (16 code) for 05, 45 (20) for 06 and 73 (34) for 07, and 38 (16), 35 (15), 29 (12), 25 (10) for 01–04 [counted from the committed notebooks]. Generating overwrites the executed notebooks with unexecuted ones, so execute them next.

6.7.2 Execute them, and check them

```
cd "$(git rev-parse --show-toplevel)/fable-cosmology"
PY=.venv/Scripts/python.exe          # .venv/bin/python on Linux/macOS
for nb in notebooks/05_fermion_fable_quantum_eos.ipynb
↪ notebooks/06_fable_primordial_gravity_8d.ipynb notebooks/07_fable_dft_states.ipynb; do
    "$PY" -m nbconvert --to notebook --execute --inplace --ExecutePreprocessor.timeout=1800 \
        --ExecutePreprocessor.kernel_name=python3 "$nb"
done
"$PY" notebooks/_build/nbcheck.py
```

- **What it does.** Each notebook is executed headlessly by the Jupyter kernel of the virtual environment, and its outputs (text, tables, figures) are written back **into the .ipynb**. The notebooks locate the solver binary, read the unit system from it (and assert it against CODATA), run every case, write their CSV tables and PNG figures under `results/`, and **assert every claim they print**: a failed assertion stops the execution with an `AssertionError` and `nbconvert` exits non-zero. Notebook 07 also runs `fermion/waveguide.py` and starts its full-resolution dft run in the background (`$ python fermion/waveguide.py dft` started in the background (pid 28636); its output goes to `fermion/dev/nb07_waveguide_dft_full.log`), which its §5.17 waits for. The per-cell timeout is 1800 s.
- **How long.** From the committed notebooks' cell timestamps (the kernel's time from the first cell's start to the last cell's end): 05 **6.9 s**, 06 **92.8 s**, 07 **557.6 s** (and 01–04: 3.3 s, 2.5 s, 31.4 s, 1.8 s). 07 is the one notebook allowed a cell of more than a minute (the wait for the background run).
- **The two interactive cells.** The last two cells of every notebook ask for a name and open a save dialog; headless, they detect the batch run and print `Headless execution: no prompt`; the notebook keeps its name `05_fermion_fable_quantum_eos` and `Headless execution: the save dialog is skipped`; the notebook stays where it is. (likewise for 06 and 07).
- **The checker.** `nbcheck.py` checks every notebook against eight requirements: **R1** it explains how to launch a notebook from a terminal (and how to rebuild its own solver: 05–07 must name the crate `rust/fable_fermion`, and 07 also `fermion/waveguide.py`); **R2** it never sends the reader to another notebook; **R3** every code cell is preceded by explanatory markdown; **R4** it asks the user to name the notebook; **R5** it opens a graphical save dialog, with a fallback; **R6** it describes the field, its Lagrangian and energy–momentum tensor, the equations of motion and the first-order system actually handed to SUNDIALS, with a glossary of every word of the common list (and, for 05–07, of the fermion list: complex 16-spinor, Grassmann numbers, Dirac conjugate, anticommutator, Krein space, fundamental

symmetry J, Fock space, Dirac sea, Kohn–Sham, gap equation, Fermi momentum, degenerate gas, `DeltaN_eff`, stabilizing stress, null energy condition, Bianchi-I, hidden sheet, Newton constant variation; for 07, of the wall list: Hohenberg–Kohn theorem, Kohn–Sham reference system, exchange–correlation energy, local-density approximation, self-consistent field, Δ SCF, Walecka functional, minimax, particle–hole excitation, pair continuum, wall, proper distance, transverse momentum, self-adjoint extension, MIT condition, irrep, Prüfer angle, threshold, edge state, band, surface density, Fermi level); **R7** it is valid nbformat-4, every non-interactive code cell was executed, no cell raised an error, and its metadata pairs it with exactly one solver (`metadata.fable_fermion.model`: `["fable4d", "gap"]` for 05, `["fable4d", "fable8d"]` for 06, `["gap", "fable4d", "waveguide"]` for 07); **R8** it owns its results (a results section naming files under `results/`) and reads one back with plain Python (the `csv` module), independently of the solver. It prints a **FAIL** `<notebook>` block with the failed requirements for each failing notebook, and the count.

- **What it prints, and success.** With all seven executed, on 2026-09-25: 7/7 notebooks pass all eight requirements, exit code 0 (commit `aca5102`: “All seven notebooks pass all eight requirements (nbcheck 7/7)”). Earlier states printed 4/4 ... (before the fermion notebooks) and 6/6 ... (at 23:04 PDT on 2026-09-24, before notebook 07 was executed; 0.11 s).
- **After a run** `git status` shows the notebooks as modified: the execution timestamps of every cell change on every run, and the kernel may split a cell’s printed output into chunks differently; their sources, execution counts, printed text, embedded images and results do not change. Notebook 07 also prints wall-clock times and the process id of its background run, which change from run to run. In a clone at another path, the one line that prints the absolute `results` directory differs too. `git checkout -- fable-cosmology/notebooks` restores the committed notebooks.

6.7.3 How the notebooks display the results

Every notebook has the same eight-part structure (1 how to open it from a terminal; 2 the words used; 3 the field, its Lagrangian and energy–momentum tensor; 4 the equations of motion and the first-order system handed to SUNDIALS; 5 running the solver, section by section; 6 this notebook’s results, read back with plain Python; 7 the conclusions — “What the numbers say” in 05, “Answers” in 06, “What these states mean for the cosmology” in 07; 8 name and save). Their prose states no numerical result: every number is printed by a cell, and the closing summary is **composed from the computed values** and displayed as Markdown output, with an assertion behind each qualitative claim. They display in four ways:

- **Printed tables** in the cell output: the solver’s command line (`$ fable_fermion ...`) and its `# stats`: line for every run, then the table the section computes (for example notebook 06’s table of the mass runs, `run kF0 [eV] a_nr a_i ok dNeff BBN dNeff rec 3P/rho_nu rec q0 t0 [Gyr] Omega_f0`, or notebook 07’s level table at $K = 80$ with each level’s irrep, `kappa`, binding and 16-component multiplicity).
- **Figures**, written as PNG under `results/` and embedded inline in the executed notebook (7 in 05, 8 in 06, 9 in 07):

figure	what it shows
<code>nb05_ks_accuracy.png</code>	“Kohn-Sham Fermi-sea integrals: closed forms versus stable forms”: the relative error of each against quadrature over $10^{-8} \leq x = k_F/ m \leq 10^6$, with the series/closed switch at $x = 0.25$
<code>nb05_w_ks.png</code>	“The degenerate fable gas: from radiation (1/3) to dust (0)”: $w_{KS}(x) = P_{KS}/\varepsilon_{KS}$ with its limits $x^2/5$ and $(1/3)(1 - 2/x^2)$

figure	what it shows
nb05_classical_vs_quantum.png	quantum (Kohn–Sham, lowest-energy gap root) against classical (Part VI, $sV'/V - 1$) w for the five potentials against the density: no phantom crossing after quantization
nb05_gap_roots.png	the gap function $m - W'(\sigma_{KS}(m))$ of the repulsive quadratic (0.05, 20) and its roots; the ground-state m against kF , which jumps (a first-order transition)
nb05_minimax.png	the no-sea functional $E_n[\sigma]/n$ of $W = m_0\sigma$ ($m_0 = 2$, $k_F = 1.5$): the gap root is a maximum, the collapse value $- m_0 $; the Walecka functional of $W = \sigma - 10\sigma^2$ ($k_F = 1$): the gap root is its minimum
nb05_preuniverse_cooling.png	“On the pre-universe the quantum fable gas cools from $w = 1/3$ to $w = 0$ ”: $w_{KS}(a_{nr}/a)$ against a/a_{nr} ($a = e^{-a_4}$), the Rust run’s w_{qp} , and the frozen classical $w = 0$
nb05_vacuum_energy.png	“Renormalized Dirac-sea energy of a mass-varying fable ($g = 8$)”: $ \Delta E_{vac}(m; M) /\rho_{c0}$ against m/M for $M = 1, 30, 100, 1000$ eV, with ρ_{c0} marked
nb06_a_mass_wdm.png	runs (a): $w_{DM} = P_{KS}/\varepsilon_{KS}$ against a : the degenerate fable cools from $1/3$ to 0
nb06_b_lambda_mass.png	runs (b): w_f and w_{DM} , and $w_{DE,inf} + 1$ on $0.3 \leq a \leq 1$: the inferred dark energy is a cosmological constant
nb06_c_de_potentials.png	runs (c): w_f (never below -1), the observer’s $w_{DE,inf}$ (clipped to $[-3, 1]$, poles and crossings marked), m_{eff}/m_{today} , and c_s^2
nb06_c_exchange_vacuum_stabilizer.png	runs (c): w_{DM} and $w_{DM,eff}$, the exchange Q , the Dirac-sea energy relative to the total density, and P_{stab}/ρ
nb06_scan.png, nb06_scan_cpl.png	the scans: the smallest m_{eff}/m_{today} over $10^{-3} \leq a \leq 0.3$ and the smallest c_s^2 against the scanned parameter; the CPL points of $w_{DE,inf}$ and of w_f
nb06_dm_mass_bounds.png	free streaming: v_{rms} today against the fable mass, with the warm-dark-matter velocities
nb06_d_nogo_8d.png	runs (d): $G_4(a)/G_4(today) = 1/v$, H_B/H_A and H_C/H_A (the approach to the 7-dimensional isotropic attractor), and the constraint residual
nb07_homogeneous.png	the homogeneous Kohn–Sham ground state: the no-sea functional of $W = 2\sigma$ ($k_F = 1.5$, a maximum), the Walecka functional of $W = \sigma - 10\sigma^2$ ($k_F = 1$, the minimum), and the repulsive quadratic (one root, then three)
nb07_excitations.png	the particle-hole and pair excitation thresholds of the homogeneous gas ($m = 1$, $k_F = 1.5$) against the total momentum Q
nb07_thresholds.png	“same-sign MIT wall: where the wall levels appear”: $K_c(1)$ (irrep $-i$) and $K_c(2)$ (irrep $+i$) against m/H
nb07_dispersion.png	the positive levels $\omega_n(K)$ of the same-sign wall ($m = H$, each 4-fold) with the continuum edge $R = \sqrt{m^2 + K^2}$, and their binding $R - \omega_n$
nb07_edge_band.png	the opposite-sign wall: $ \omega_{edge} /K$ against the closed-form slope $(m/6H)B(m/6H, 13/12)$, and the edge level against the continuum edge and the bulk mass (“massless fermions on the wall below K^* ”)
nb07_theta.png	“ $K = 3$, $m = H$: levels of the self-adjoint wall family” against θ/π , both irreps, with the continuum edges and the MIT walls marked
nb07_wavefunctions.png	the orbitals $f(z)$, $g(z)$ of $n = 0$ and $n = 1$ at $K = 80$ against the proper distance from the wall
nb07_ks_profiles.png	the Kohn–Sham profiles $\sigma(z)$, $m(z)$, $n(z)$ of the same-sign ground state, of the Δ SCF state with $x = 0.10$, and of the opposite-sign edge-band ground state ($\lambda = -10$)
nb07_strong.png	strong attraction ($\lambda = -100$): the residual of the non-converging iteration, the layer centre drifting away from the wall, and the Walecka functional of the shifted well against its centre (“the wall repels the layer”)

The figures of notebook 07. The fifteen figures of notebooks 05 and 06 are shown in section 4, next to the results they display. The nine figures of notebook 07 follow, exactly as the notebook

wrote them under `fable-cosmology/results/`; each caption is the description of the table above.

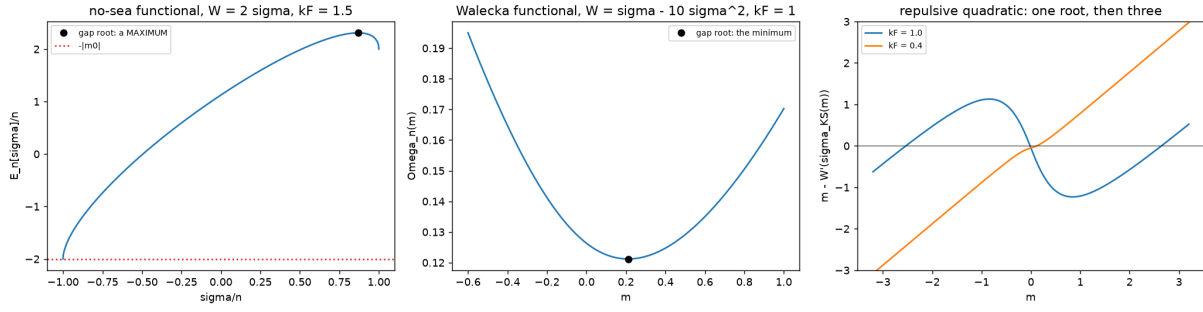


Figure 16: `nb07_homogeneous.png`: the homogeneous Kohn–Sham ground state: the no-sea functional of $W = 2\sigma$ ($k_F = 1.5$, a maximum), the Walecka functional of $W = \sigma - 10\sigma^2$ ($k_F = 1$, the minimum), and the repulsive quadratic (one root, then three).

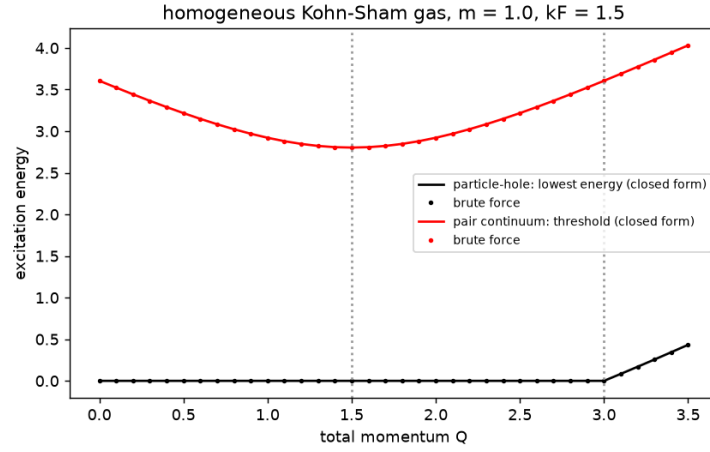


Figure 17: `nb07_excitations.png`: the particle–hole and pair excitation thresholds of the homogeneous gas ($m = 1$, $k_F = 1.5$) against the total momentum Q .

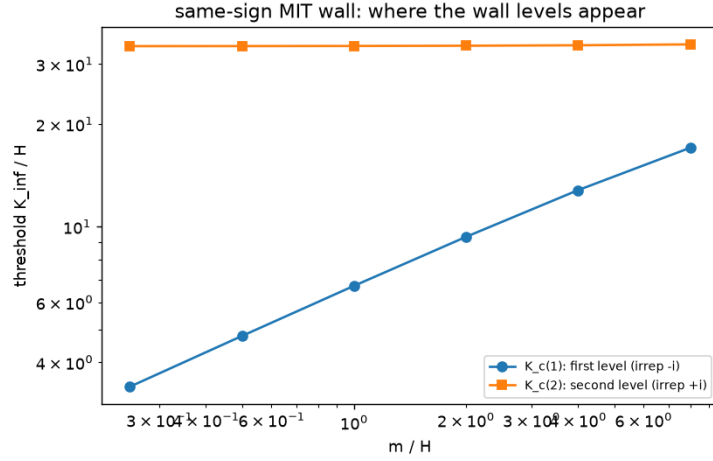


Figure 18: nb07_thresholds.png: “same-sign MIT wall: where the wall levels appear”: $K_c(1)$ (irrep $-i$) and $K_c(2)$ (irrep $+i$) against m/H .

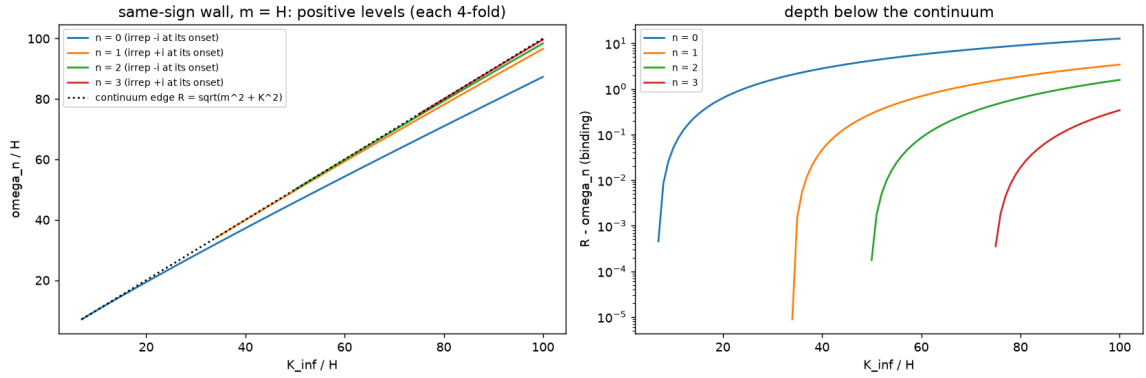


Figure 19: nb07_dispersion.png: the positive levels $\omega_n(K)$ of the same-sign wall ($m = H$, each 4-fold) with the continuum edge $R = \sqrt{m^2 + K^2}$, and their binding $R - \omega_n$.

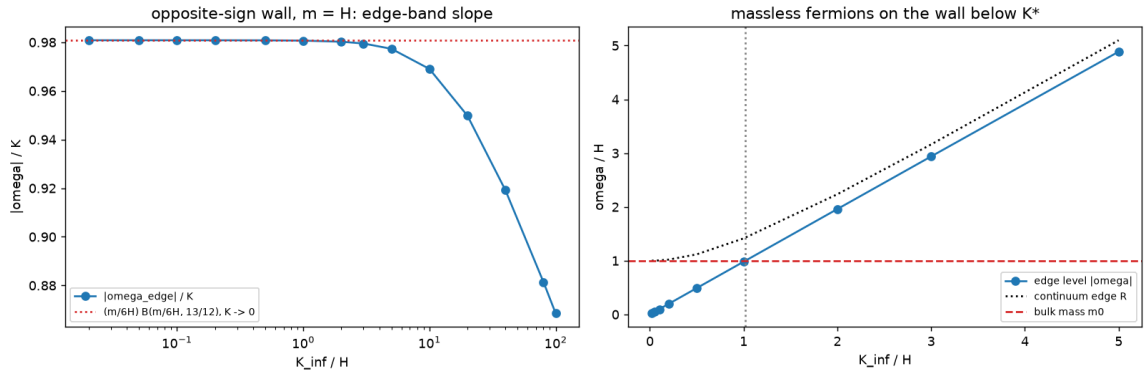


Figure 20: nb07_edge_band.png: the opposite-sign wall: $|\omega_{\text{edge}}|/K$ against the closed-form slope $(m/6H) B(m/6H, 13/12)$, and the edge level against the continuum edge and the bulk mass (“massless fermions on the wall below K^* ”).

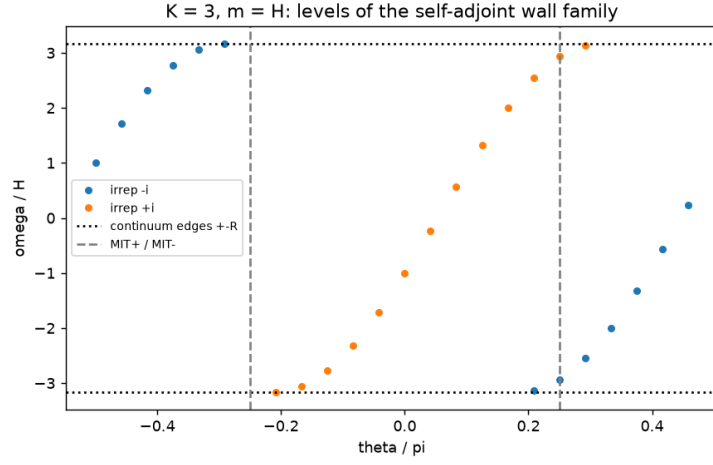


Figure 21: nb07_theta.png: “ $K = 3, m = H$: levels of the self-adjoint wall family” against θ/π , both irreps, with the continuum edges and the MIT walls marked.

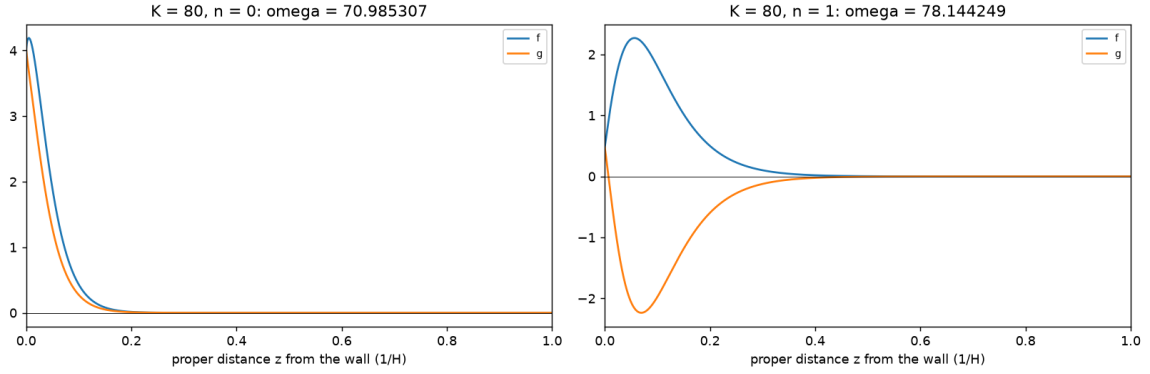


Figure 22: nb07_wavefunctions.png: the orbitals $f(z)$, $g(z)$ of $n = 0$ and $n = 1$ at $K = 80$ against the proper distance from the wall.

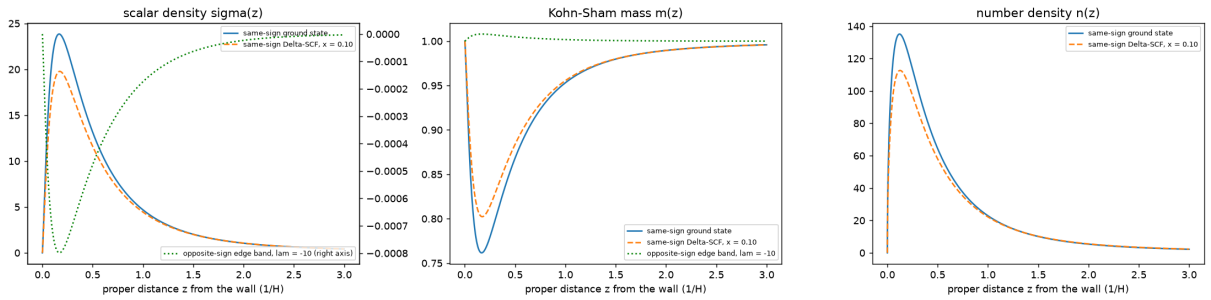


Figure 23: nb07_ks_profiles.png: the Kohn-Sham profiles $\sigma(z)$, $m(z)$, $n(z)$ of the same-sign ground state, of the Δ SCF state with $x = 0.10$, and of the opposite-sign edge-band ground state ($\lambda = -10$).

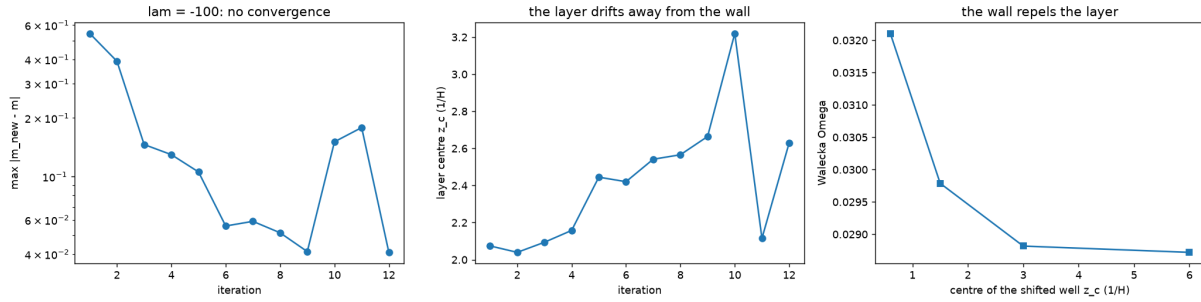


Figure 24: nb07_strong.png: strong attraction ($\lambda = -100$): the residual of the non-converging iteration, the layer centre drifting away from the wall, and the Walecka functional of the shifted well against its centre (“the wall repels the layer”).

- **Result files:** every table the notebook computes is also written as CSV under `results/` (nb05_* 11 files, nb06_* 37, nb07_* 18), and §6 of each notebook reads one back with plain Python and prints it (for example nb06_runs.csv: 29 rows, 34 columns and a line per run with `a_nr`, `Delta N_eff(BBN)` and the CPL fit [nb06 §6]).
- **The composed summary:** notebook 05 prints ten numbered statements, notebook 06 its answers to [1] and [2], notebook 07 nine numbered statements, each ending with a line that every statement is asserted (every statement of this notebook is backed by the numbers above, every claim of the answers is asserted above, every statement of the summary is asserted in the cells above). They are restated in sections 4.5–4.8.

Troubleshooting (what a failing notebook prints, and why): The solver binary ... does not exist (the setup did not finish: run it again); an `AssertionError` naming `omega_r0` in §5 of 05 or 06 (the binary predates the `hbar` correction: rebuild it); the Mathematica reference ... is missing in 06 (restore `reference/mathematica_fable4d_*.csv` with git checkout, or regenerate them, 6.3.7); notebook 07 stopping in its §5.3 with ... `fermion/waveguide_T16.json` is missing (restore it with `git checkout -- fable-cosmology/fermion/waveguide_T16.json`, or rewrite it, 6.6.1); Kernel ... not found or No such kernel named `python3` (run `nbconvert` from the `.venv`, which contains `ipykernel`); the Windows warnings `RuntimeWarning: Proactor event loop does not implement add_reader ...` and `[IPKernelApp] WARNING | Kernel is running over TCP without encryption` are warnings of the `zmq` and `Jupyter` libraries, not errors.

6.8 Reproduce everything with one command

```
bash "$(git rev-parse --show-toplevel)/fable-cosmology/run_all.sh"
```

```
powershell -ExecutionPolicy Bypass -File fable-cosmology\run_all.ps1 # make Perl visible
↪ first (6.1)
```

- **What it does** [file: `fable-cosmology/run_all.sh`], after `setup.sh`, from any directory, under `set -euo pipefail` (it stops at the first failure): == 1. solver tests — `cargo test --release` in `rust/fable_cosmo` (showing its last three lines) and in `rust/fable_fermion` (showing its running, test result, FAILED and panicked lines); == 2. notebooks — every `notebooks/0*.ipynb` in order, 01 to 07, executed in place with the `nbconvert` command of 6.7.2; == 3. notebook requirements — `nbcheck.py`; == 4. the paper — `latexmk -pdf -interaction=nonstopmode -halt-on-error fable_cosmology.tex` in `fable-cosmology/latex/` (its output into `latex/latexmk.log`), then the PDF’s `ls -la` line; and == all done. `run_all.ps1` does the same, shows all of

`cargo test`'s output, and throws on a failing `cargo test` of either solver (it checks `$LASTEXITCODE`), a failing notebook, the checker or the paper build.

- **How long.** 12 min 18 s on 2026-09-25 with the seven notebooks (of which notebook 07 about 9 min and the `fable_fermion` tests about 45 s; the committed execution of notebook 06 took 92.8 s); 3 min 35 s on 2026-09-24 before notebook 07 existed; 51 s (61 s in a fresh clone) before the fermion-fable work.
- **What it printed** on 2026-09-25 (Windows 11, Git Bash, the seven notebooks, 12 min 18 s), verbatim:

```
== 1. solver tests
rust/fable_cosmo

test result: ok. 7 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s

rust/fable_fermion
running 23 tests
test result: ok. 23 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in
↳ 38.31s
running 0 tests
test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s
running 11 tests
test result: ok. 11 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.93s
running 0 tests
test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s
== 2. notebooks
executing notebooks/01_fableScalar_quintessence.ipynb
[NbConvertApp] Converting notebook notebooks/01_fableScalar_quintessence.ipynb to notebook
C:\Users\nsh\Documents\8-dim\Pre-Universe_14SEP26-77\fable-cosmology\.venv\Lib\site-packages\
↳ \zmq\future.py:718: RuntimeWarning: Proactor event loop does not implement add_reader
↳ family of methods required for zmq. Registering an additional selector thread for
↳ add_reader support via tornado. Use
↳ `asyncio.set_event_loop_policy(WindowsSelectorEventLoopPolicy())` to avoid this warning.
self._get_loop()
[IPKernelApp] WARNING | Kernel is running over TCP without encryption. All communication
↳ (including code and outputs) is sent in plain text and is susceptible to eavesdropping.
↳ Use IPC transport or launch with kernel manager-provisioned CurveZMQ keys to enable
↳ transport encryption.
[NbConvertApp] Writing 475134 bytes to notebooks\01_fableScalar_quintessence.ipynb
executing notebooks/02_fable_spinor.ipynb
[NbConvertApp] Converting notebook notebooks/02_fable_spinor.ipynb to notebook
(the same two warning lines)
[NbConvertApp] Writing 532820 bytes to notebooks\02_fable_spinor.ipynb
executing notebooks/03_pre-universe_dynamics.ipynb
[NbConvertApp] Converting notebook notebooks/03_pre-universe_dynamics.ipynb to notebook
(the same two warning lines)
[NbConvertApp] Writing 1090050 bytes to notebooks\03_pre-universe_dynamics.ipynb
executing notebooks/04_dark_matter_dark_energy.ipynb
[NbConvertApp] Converting notebook notebooks/04_dark_matter_dark_energy.ipynb to notebook
(the same two warning lines)
[NbConvertApp] Writing 318320 bytes to notebooks\04_dark_matter_dark_energy.ipynb
executing notebooks/05_fermion_fable_quantum_eos.ipynb
[NbConvertApp] Converting notebook notebooks/05_fermion_fable_quantum_eos.ipynb to notebook
(the same two warning lines)
[NbConvertApp] Writing 734434 bytes to notebooks\05_fermion_fable_quantum_eos.ipynb
executing notebooks/06_fable_primordial_gravity_8d.ipynb
[NbConvertApp] Converting notebook notebooks/06_fable_primordial_gravity_8d.ipynb to notebook
(the same two warning lines)
[NbConvertApp] Writing 1412296 bytes to notebooks\06_fable_primordial_gravity_8d.ipynb
executing notebooks/07_fable_dft_states.ipynb
[NbConvertApp] Converting notebook notebooks/07_fable_dft_states.ipynb to notebook
(the same two warning lines)
[NbConvertApp] Writing 905839 bytes to notebooks\07_fable_dft_states.ipynb
```

```

== 3. notebook requirements

7/7 notebooks pass all eight requirements
== 4. the paper
-rw-r--r-- 1 nsh 197121 2835044 Sep 24 19:40 fable_cosmology.pdf
== all done

```

The only abbreviation is (the same two warning lines), which stands for the `zmq RuntimeError` (two lines) and the `[IPKernelApp] WARNING`, repeated verbatim as after notebook 01; they are warnings of the `zmq` and `Jupyter` libraries on Windows, not of the notebooks, and they change no output. The empty line after `rust/fable_cosmo` is the first of the three lines that `cargo test ... | tail -n 3` keeps. The paper step lists the PDF; `latexmk`'s own output is in `latex/latexmk.log`, which ends with `Latexmk: All targets (fable_cosmology.pdf) are up-to-date when nothing changed (after three pdflatex passes the first time).` - **Success:** `== all done` with exit code 0, `7/7 notebooks pass all eight requirements`, and `git diff --stat -- fable-cosmology/results` empty (6.5).

6.9 The LaTeX build

```

bash "$(git rev-parse --show-toplevel)/provenance-latex/build_all.sh"

powershell -ExecutionPolicy Bypass -File provenance-latex\build_all.ps1

```

- **What it does** [file: `provenance-latex/build_all.sh`]. For each of the three LaTeX twins of the provenance pages of 2026-09-24 — `PROVENANCE-14-FERMION-FABLE-CANONICAL-QUANTIZATION`, `PROVENANCE-15-FERMION-FABLE-AND-THE-PRIMORDIAL-GRAVITATIONAL-FIELD` and `PROVENANCE-16-THE-COMPLETE-SOLUTION-AND-ITS-COMMANDS` (the twin of this page) — it runs `latexmk -pdf -interaction=nonstopmode -halt-on-error <doc>.tex` (as many `pdflatex` passes as the cross-references need), with the output in `provenance-latex/<doc>.latexmk.log`; it stops on the first failure, printing `FAILED`; see `provenance-latex/<doc>.latexmk.log` and `<doc>.log`; it warns (`WARNING: undefined references in <doc>.log`) if the `.log` contains “undefined” or “Rerun to get”; and it lists each PDF. Every document begins with `\input{preamble}` (`provenance-latex/preamble.tex`: `pdfLaTeX`, Latin Modern, A4, 2.5 cm margins, and macros for the notebook's objects, with separate symbols for the classical conjugation `\conjug` and the Hilbert adjoint `\hadj`); the generated fragments of `provenance-latex/generated/` are `\input` where the equations are quoted. `build_all.ps1` does the same in PowerShell and throws on a failing `latexmk`. MiKTeX's `latexmk` needs Perl on the PATH; Git for Windows ships one (`$env:PATH += ";C:\Program Files\Git\usr\bin"` in PowerShell).
- **What it prints:** `== <doc>` and an `ls -la` line per document, and finally `== all three PDFs built`.
- **Success:** exit code 0 and that last line; the `.log` of each document free of errors and undefined references, and no overfull box worse than 10pt (`grep -n -E "^(?!|undefined|Overfull" <doc>.log`). The twin of the page of Effort A was built this way when it was committed: 110 pages, no errors, no undefined references, no overfull box above 5.5pt [commit `fee5603`]. The output of the build of all three twins on the final pushed state is recorded with the fresh-clone verification (section 7).

The paper of Effort 0 is built the same way from `fable-cosmology/latex`:

```

cd "$(git rev-parse --show-toplevel)/fable-cosmology/latex"
latexmk -pdf -interaction=nonstopmode -halt-on-error fable_cosmology.tex

```

A clean build ends with `Output` written on `fable_cosmology.pdf` (24 pages, ...) in `fable_cosmology.log`, with no errors, no undefined references and no overfull boxes; `latexmk`

-c removes the by-products (which git ignores). Its 16 figures are byte-identical copies of the notebooks' PNGs in latex/figures/.

6.10 Git: commit, push, check the remote, clone afresh

Commit and push (every commit of the work was made this way; the notebook's two .mx files are restored first, 6.3.4):

```
cd "$(git rev-parse --show-toplevel)"
git status --short                # what changed
git add <the files of the step>   # named files, never a blanket add
git commit -m "<what the step adds or fixes>" -m "<details>" -m "Co-Authored-By: Claude Opus
↪ 5.5 <noreply@anthropic.com>"
git push origin main
```

origin is https://github.com/once-ere/Pre-Universe_with_Claude.git; the remote named upstream (the author's other repository) is never pushed to. Since pull request #1 (section 2.2) every commit went directly to main.

Check that the remote has what was pushed:

```
cd "$(git rev-parse --show-toplevel)"
git fetch origin
git rev-parse HEAD origin/main    # both must print the same hash
git ls-remote origin refs/heads/main # the hash GitHub serves for main
git status --short                # what is not yet committed
```

When this page was written (2026-09-25), git ls-remote origin refs/heads/main printed

```
16711556effcf920b28a606f4c365149f126dd4e    refs/heads/main
```

which is the local main (git rev-parse HEAD printed the same hash): 1671155, "Record the final notebook run: 217 Input cells, 644/644 assertions, 0 messages". Earlier checks printed bc04ed16096e134857111ae16df5220d4ff18435 (about 21:28 PDT on 2026-09-24) and 2bc936d0dfd6f6c1243214fc6b1b9947f5bde6e (23:26 PDT). The final commits and the ls-remote line of the final push are recorded in section 7.4.

Clone afresh and verify (<scratch> is any empty directory outside the repository; section 7 records the results):

```
cd <scratch>
git clone https://github.com/once-ere/Pre-Universe_with_Claude.git fresh
cd fresh
git log --oneline -1                # the hash of the pushed main
cd claude-fable
cp claude-fable_Einstein-Rosen-2-Planes.nb committed.nb
python build_tools.py                # the four lines of 6.3.1
cmp claude-fable_Einstein-Rosen-2-Planes.nb committed.nb && echo "notebook rebuilds byte for
↪ byte"
wolframscript -file verify_nb.wls    # 6.3.2
cd ../fable-cosmology
bash setup.sh                        # 6.1
bash run_all.sh                      # 6.8: both crates' tests, notebooks
↪ 01-07, nbcheck, the paper
git diff --stat -- results           # empty: every result file reproduced
↪ byte for byte
.venv/Scripts/python.exe fermion/crosscheck.py --jobs 6
(cd fermion && ../venv/Scripts/python.exe waveguide.py validate)
cd ../claude-fable
wolframscript -file run_from_nb.wls > fresh_run.log 2>&1    # after run_all.sh, so the Rust
↪ CSVs exist
sed -n '/RUN SUMMARY/, $p' fresh_run.log    # 217 cells, 644/644, cells w/ msgs 0
```

```
diff <(grep '^ PASS' fresh_run.log) <(grep '^ PASS' run_fermion_fable_final.log) && echo
↪ "every label identical"
git checkout -- claude-fable_Einstein-Rosen-2-Planes-eLa.mx
↪ claude-fable_Einstein-Rosen-2-Planes-eLazt.mx
cd .. && bash provenance-latex/build_all.sh # 6.9
```

The delivered copy of the notebook outside the repository, `C:/Users/nsh/Documents/8-dim/claude-fable_Einstein-Rosen-2-Planes.nb`, is refreshed from `claude-fable/` whenever the notebook changes, next to the author's two helper packages. When this page was written it was 545408 bytes and byte-identical to the repository's notebook (`cmp` reported no difference); when the page of Effort A was written it was still the 2026-09-16 version (273270 bytes).

7 Verification of the repository: the push, and a fresh clone of it

7.1 What was pushed, and what was not yet, when this page was written

The commits of the work (`git log --date=iso, PDT`; every one pushed to `origin/main` of https://github.com/once-ere/Pre-Universe_with_Claude.git):

commit	time	effort	what it contains
e34d454	2026-09-24 18:52:36	0	the 2026-09-16 work as it stood: Part VI (172 Input cells, 358/358), <code>fable_cosmo</code> , notebooks 01–04, results, references, setup and run scripts (section 2.1)
028b379	19:24:32	0	the reference generator runs without underflow messages; pull request #1, merged 19:32:10 (section 2.2)
2f6d322	19:57:29	0	the gaps of 2026-09-16 closed: student instructions, the paper, the provenance page of that work, seven design corrections, <code>setup.sh</code> , <code>run_all.ps1</code> ; a fresh local clone ran <code>setup.sh</code> and <code>run_all.sh</code> end to end (section 2.3)
8459c57	20:02:04	B	a work-in-progress snapshot of the solver crate <code>fable_fermion</code>
e130337	20:14:25	B	a second work-in-progress snapshot of that crate
0a789e2	20:17:18	C	the LaTeX build tree of the three provenance pages: <code>provenance-latex/preamble.tex</code> , <code>build_all.sh</code> , <code>build_all.ps1</code> ; <code>.gitignore</code> names each document's <code>.log</code> and <code>.latexmk.log</code> ; a minimal test document compiled with no warnings
d0f2a99	20:18:53	C	<code>build_all.sh</code> tracked without the executable bit (mode 100644), like the repository's other scripts
c5892bd	20:38:49	B	the finished solver <code>fable_fermion</code> (21 + 8 tests), the independent <code>crosscheck.py</code> (every run below 10^{-8}), <code>report.py</code>
4fd4c40	20:47:28	A	Part VII of the notebook (198 Input cells, 528/528, 0 cells with messages), its log, the checker's log, the TeX export and its fourteen generated files
b7ea18d	20:48:42	A	the wording correction of an earlier provenance page (section 5.8, item 6)
bc04ed1	21:24:04	B	Part VIII of the notebook (217 Input cells, 642/642), its log, and the Mathematica reference runs of the coupled system
18a2501	22:13:42	B	the wall-state solver, its derivation (94/94), its Mathematica check (11/11), its report
fee5603	22:31:20	A, C	the page of Effort A (draft) and its LaTeX twin and PDF (110 pages)
2bc936d	23:12:51	B	six solver defects fixed, 34/34 tests (section 5.8, item 4)
aca5102	2026-09-25 00:23:46	B, 0	notebooks 05–07 executed, their 90 result files, the integration (nbcheck 7/7, <code>run_all.sh</code> passing), the fermion sections of the student instructions, <code>waveguide_T16.json</code> committed, the <code>nbgen.py</code> corrections of notebooks 01–04
1671155	00:28:54	B	the final full run of the notebook, <code>claude-fable/run_fermion_fable_final.log</code> (217 Input cells, 644/644, 0 messages, 208.715 s, 46 witnesses)

When this page was written, `git ls-remote origin refs/heads/main` printed `16711556effcf920b28a606f4c365149f126dd4e refs/heads/main`, the local `main`. Everything that exists as code, notebook, log, result file or generated file is in that pushed history. **Not yet**

committed at that moment: this page, the page of Effort B and the LaTeX twins and PDFs of the pages of Efforts B and C, and an update of the repository’s front page (its table of the notebook’s Parts, the final counts, the table of provenance pages). None of the numbers on this page is taken from an uncommitted file: every one comes from the committed notebooks, result files, logs, reports and commit messages named next to it. They are committed together at the end of the work, and that final state is what the fresh clone verifies.

7.2 What the fresh-clone verification consists of

The final pushed `main` is cloned into an empty directory outside the repository, and everything is rebuilt and re-run there from nothing — a new virtual environment, a new engine clone from GitHub, new builds — with the commands of section 6.10, in this order, each result compared with the committed record:

step	command	compared with
1	<code>git clone https://github.com/once-ere/Pre-Universe_with_Claude.git fresh; git log --oneline -1</code>	the hash <code>git ls-remote</code> reports for <code>main</code>
2	<code>python build_tools.py in claud-fable/; cmp against the committed notebook</code>	byte-identical <code>.nb</code> and <code>run_all.wls</code> (6.3.1)
3	<code>wolframscript -file verify_nb.wls</code>	the output of 6.3.2 (input cells that FAIL to parse: <code>{}</code>)
4	<code>bash fable-cosmology/setup.sh</code>	<code>== setup complete</code> , both <code>--version</code> lines (6.1)
5	<code>bash fable-cosmology/run_all.sh</code>	test result: <code>ok. 7 passed; 23 passed, 11 passed</code> ; the seven notebooks executed; 7/7 notebooks pass all eight requirements; the paper built; <code>== all done</code> (6.8)
6	<code>git diff --stat -- fable-cosmology/results</code>	empty: all 149 result files (59 of notebooks 01–04, 63 of 05 and 06, 27 of 07) reproduced byte for byte; any CSV that differs compared column by column (6.5)
7	<code>python fermion/crosscheck.py --jobs 6; python fermion/waveguide.py validate</code>	CROSSCHECK PASSED: every difference < 1e-06; VALIDATION: 64 of 64 PASS
8	<code>wolframscript -file run_from_nb.wls</code> (after step 5, so the Rust CSVs exist)	cells evaluated : 217, assertions run : 644, passed : 644, FAILED : 0, cells w/ msgs : 0; every PASS label identical to <code>claud-fable/run_fermion_fable_final.log</code> (6.3.5)
9	<code>bash provenance-latex/build_all.sh</code>	<code>== all three PDFs built</code> , no errors, no undefined references, no overfull box worse than 10pt (6.9)
10	<code>git status --short</code> after restoring the two <code>.mx</code> files	only the re-executed notebooks (timestamps) and nothing else

7.3 The results of the fresh-clone verification

{{FRESH-CLONE}}

7.4 The final push

{{FINAL-PUSH}}

8 What remains open

This is the honest list: what the work did not do, what it assumed without deriving, and where its answers stop. Each item names the section where it is discussed.

8.1 Physics that was not done

1. **The Maxwell construction.** For the repulsive quadratic potential ($g_q > 0$) the Kohn–Sham ground state jumps between gap branches at $a = 0.9962806544$ — a first-order transition. Energy conservation across it needs the Maxwell (coexistence) construction, which the solver does not implement; it **refuses** the run instead (sections 4.7.11, 6.4.8). The physics of that transition — coexisting phases, latent heat, bubbles — is not examined.
2. **Exchange and correlation.** E_{xc} is neglected everywhere (the Hartree mean field). That is exact for the author’s mass term, which is free, but for a nonlinear W the Fock exchange of the contact interaction is $1/g$ of the Hartree term only for $k_F \ll m$, and of order one or larger for $k_F \gtrsim m$, the early-universe regime. The ratio of the interaction energy to $3P_{KS}$, which alone controls the neglect there, was **not computed** in the runs (section 4.5.12).
3. **The vacuum energy is fine-tuned, not solved.** For a constant mass the Dirac-sea energy is a constant, absorbed into V_0 as a bare cosmological constant: the cosmological-constant problem is not addressed. For a mass-varying potential the runs read W as the fully renormalized effective potential, which absorbs the sea energy only by fine-tuning: its variation along the runs is 1.31×10^{13} to 2.87×10^{16} times the total density at $a \geq 0.3$ (at least 3.79×10^9 over the scans). The other option — including $\Delta E_{vac}(m)$ in ρ and in the gap equation — was not run (sections 3.7.3, 4.5.11, 4.8.1).
4. **The origin of the stabilizer is unknown.** Every result of Effort B rests on the stabilized model `fable4d`, whose hidden sheet is held by $P_{stab} = -F/2$: a Lagrange multiplier with zero energy density, not a stress derived from any field, which **violates the null energy condition** along x_0 wherever dust is present ($\rho + P_C = -\rho_{dust}/2$; negative on every output row of every stabilized run). Without it (`fable8d`) the Newton constant varies by 6.5×10^8 to 1.9×10^{17} and the model is excluded. What physical mechanism could stabilize the hidden sheet is not known (sections 4.7.1, 4.7.10).
5. **The strong-coupling wall state.** At $\lambda = -100$ homogeneous matter is self-bound, but the Kohn–Sham iteration at the same-sign wall does **not converge** (residual 4.1×10^{-2} after 12 iterations; the layer drifts from $z_c = 2.07$ to 2.63), and the wall repels the layer. A position-constrained iteration, to map $E(z_c)$ self-consistently, was not done; only a rigid-shift upper bound was computed (section 4.6.6).
6. **The same-sign wall’s self-consistent state is not a ground state.** At $\lambda = -0.01$, $N_s = 100$ the iteration converges, but $\mu - m_0 = 10.696 > 0$ (bulk states lie below the Fermi level) and the Walecka functional is not stationary ($d\Omega/ds = -48.3$). A grand-canonical treatment with a bulk fable density was not done (section 4.6.6).
7. **Perturbations were not examined.** Everything is homogeneous background cosmology. Every massive mass-varying case has a negative adiabatic sound speed (c_s^2 down to -0.611 for `power nu = 0.236`, -27 for $\nu = 0.5$, -21.8 for `expdamp`, -47.1 for `quadratic`, -0.364 in Part VIII’s case), which signals an instability of perturbations that is stated but not followed; the free-streaming bound ($m \approx 1.1\text{--}1.8\text{ keV}$) is a velocity-matching estimate, not a

transfer-function computation; structure formation, the CMB and lensing are not computed (sections 4.7.6, 4.7.9).

8. **Renormalized $\langle T \rangle$ on a time-dependent background.** $\langle 0|:T:|0 \rangle = 0$ holds only for a static background or adiabatically; a conserved renormalized expectation on an x_4 -dependent background needs adiabatic subtraction or point-splitting, which was not carried out (section 3.7.3).
9. **Reflection positivity** of the J -quantization was claimed in the design, not verified, and dropped (section 5.2, QK-5).
10. **Other wall conditions.** Only the flavour-symmetric, charge-conjugation-invariant MIT walls $Q = \pm 1$ were solved; the mixed walls $P = T_{16}[0]Q$ and the non- J -preserving covariant walls were described, not solved (section 4.6.3).

8.2 Assumptions made, not derived

1. **a_4 is free.** It is never given a value; nothing in the work determines it. The wall states are computed at a frozen a_4 (the adiabatic approximation, with $K_\infty = k e^{a_4}$), and the cosmology uses the asymptotic region of the generalized frame, not a_4 (sections 3.1.11, 4.3.3, 4.6.3).
2. **The present universe is the asymptotic Bianchi-I region** of the generalized warped frame, valid when $24(H/H_A)^2 \cot^2 / C^2 \ll 1$; the frozen and dust-driven solutions solve the warped equations only up to $O(e^{-12Hz})$ terms, including an x_0 -momentum flux (sections 4.3.3, 4.3.4).
3. **The (0,4) equation singles out, but does not derive, the author's volume-preserving structure** (with $C = 1$ and $T^4_0 = 0$; with C dynamical, positive-energy expansion is allowed) (section 4.3.2).
4. **The admissible theory is a truncation** to fields independent of $x_5 \dots x_7$ with a finite hidden coordinate volume V_{hid} ; compactifying the timelike $x_5 \dots x_7$ produces closed timelike curves; x_0 is taken as compactified at large z with a Kaluza–Klein gap $1/L_0 \gg T_{\text{max}}$, so that all matter is in zero modes (sections 3.5.4, 4.7.1).
5. **The beginning of the present universe** is not fixed by the pre-universe: $a_i \leq \min(10^{-10}, 0.01 a_{\text{nr}}(m))$ is chosen on physical grounds (before nucleosynthesis, fable ultra-relativistic) and shown not to matter; $g_*(T)$ is not followed (a factor 0.46–0.83 on $\rho_r a^4$ at $a \approx 10^{-12}$) (section 4.7.2).
6. **Semiclassical gravity** with the Kohn–Sham expectation value as the source; the split of the fable into dark matter and dark energy is a **convention** (the condensate ρ_U is an 8-dimensional vacuum energy) (sections 4.2.1, 4.5.10).

8.3 Where the answers stop

- [1] A time-varying dark-energy equation of state exists formally in the stabilized model with a mass-varying potential, but in no run is it viable: with a bare Λ the inferred dark energy is a cosmological constant; with a mass-varying potential the fable's own w_f never goes below -1 , while the observer-inferred dark energy has a pole and is phantom or crosses -1 (apparent, from the dark-sector energy exchange); the fable is massless through the matter era unless it is a power law with a positive bare mass ($\nu < 0.279$ at the observed split), and every massive case is adiabatically unstable; the neglected Dirac-sea energy is enormous; the stabilizer violates the null energy condition; without it G varies far beyond the bounds (section 4.8.1). Whether any modification (a derived stabilizer, the sea energy included, perturbations) could make it viable is open.

- [2] A time-varying dark-matter equation of state exists intrinsically ($1/3$ to 0 around $a_{\text{nr}} \propto m^{-4/3}$), but for the masses that free streaming and phase space allow (about 1 keV and above) the change happens around $a_{\text{nr}} \approx 4 \times 10^{-8}$ to 2×10^{-8} , long before recombination: indistinguishable from cold dark matter in the late universe (section 4.8.2).

8.4 Records that are incomplete

- The console output and the duration of the Mathematica reference script of the coupled system (`make_reference_fermion.wls`) and the duration of the TeX export were not captured in a repository log; their products are checked by Part VIII and by notebook 06 instead (sections 6.3.6, 6.3.7).
- Two claims of the review about the wall-state numerics (a fixed matching point losing a level, a uniform frequency grid missing a shallow level) were not re-tested, because the method used does not have those weaknesses (section 5.8, item 3).
- The development logs of the wall-state solver live in the ignored `fable-cosmology/fermion/dev/`; the committed report quotes them, and notebook 07 re-runs what matters.

8.5 What is still to be done when this page is written

- Section 1, the complete set of ideas of all efforts (identical on the three pages of 2026-09-24), which is filled when every effort's results are in.
- The LaTeX twins and PDFs of the pages of Efforts B and C, built with `build_all.sh` (section 6.9).
- The final commit and push, recorded in section 7.4, and the fresh-clone verification of the pushed repository, recorded in section 7.3.